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384. On the Transformation of Plane Curves

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1. The expression a "double point," or, as I shall for shortness call it, a "dp," is to be throughout understood to include a cusp: thus, if a curve has δ nodes (or double points in the restricted sense of the expression) and κ cusps, it is here regarded as having $\delta+\kappa$ dps.

2. It was remarked by Cramer, in his "Théorie des Lignes Courbes" (1750), that a curve of the order n has at most $\frac{1}{2}(n-1)(n-2)$, $=\frac{1}{2}(n^2-3n)+1$, dps.

3. For several years past it has further been known that a curve such that the coordinates $(x:y:z)$ of any point thereof are as rational and integral functions of the order n of a variable parameter θ, is a curve of the order n having this maximum number $\frac{1}{2}(n-1)(n-2)$ of dps.

4. The converse theorem is also true, viz.: in a curve of the order n, with $\frac{1}{2}(n-1)(n-2)$ dps, the coordinates $(x:y:z)$ of any point are as rational and integral functions of the order n of a variable parameter θ—or, somewhat less precisely, the coordinates are expressible rationally in terms of a parameter θ.

5. The foregoing theorem, as a particular case of Riemann's general theorem, to be presently referred to, dates from the year 1857; but it was first explicitly stated only last year (1864) by Clebsch, in the Paper, "Ueber diejenigen ebenen Curven deren Coordinaten rationale Functionen eines Parameters sind," Crelle, t. LXIV. (1864), pp. 43–63.

6. The demonstration is, in fact, very simple; it depends merely on the remark that we may, through the $\frac{1}{2}(n-1)(n-2)$ dps, and through $2n-3$ other points on the given curve of the order n, together $\frac{1}{2}(n^2+n)-2=\frac{1}{2}(n-1)(n+2)-1$, points, draw a series of curves of the order $n-1$, given by an equation $U+\theta V=0$, containing an arbitrary parameter θ; any such curve intersects the given curve in the dps, each counting as two points, in the $2n-3$ points, and in one other point; hence, as there is only one variable point of intersection, the coordinates of this point, viz., the coordinates of an arbitrary point on the given curve, are expressible rationally in terms of the parameter θ.

   The demonstration may also be effected in a similar manner by means of curves of the order $n-2$.

7. Before going further, it will be convenient to introduce the term "Deficiency," viz., a curve of the order n with $\frac{1}{2}(n-1)(n-2)-D$ dps, is said to have a deficiency $=D$: the foregoing theorem is that for curves with a deficiency $=0$, the coordinates are expressible rationally in terms of a parameter θ.

   Since in such a curve the different points succeed each other in a certain definite order, viz., in the order obtained by giving to the parameter its different real values from $-\infty$ to ∞, the curve may be termed a *unicursal* curve.

8. Riemann's general theorem, as applied to plane curves, is stated, but not in its complete form, by Schwarz, in the Paper, "De superficiebus in planum explicabilibus primorum septem ordinum," Crelle, t. LXIV. (1864), pp. 1–17: to complete the enunciation it is necessary to refer to page 137 of Riemann's own Paper, "Theorie der Abel'schen Functionen," Crelle, t. LIV. (1857), pp. 115–155, viz., the enunciation will be:

9. For a curve of any order with a given deficiency D, the coordinates may be expressed as follows:

   $D = 0$, rationally in terms of a parameter θ, or what comes to the same thing, rationally in terms of the parameters (ξ, η), connected by an equation of the form $(1, \xi)(1, \eta) = 0$.

   $D > 0$, rationally in terms of the parameters (ξ, η) connected by an equation of a certain form, viz.:

   $D = 1$, the equation is $(1, \xi)^2(1, \eta)^2 = 0$, or (what comes to the same thing) η is the square root of a quartic function of ξ.

   $D = 2$, the equation is $(1, \xi)^2(1, \eta)^3 = 0$, or (what comes to the same thing) η is the square root of a sextic function of ξ.

   $D > 2$, viz.:

   D odd, $= 2\mu - 3$, the equation is $(1, \xi)^\mu(1, \eta)^\mu = 0$, and is besides such, that treating (ξ, η) as Cartesian coordinates, the curve thereby represented has $(\mu - 2)^2$ dps.

   D even, $= 2\mu - 2$, the equation is $(1, \xi)^\mu(1, \eta)^{\mu-1} = 0$, and is besides such, that treating (ξ, η) as Cartesian coordinates, the curve thereby represented has $(\mu - 1)(\mu - 3)$ dps.

10. To see more clearly the meaning of this, write $\frac{\xi}{\zeta}, \frac{\eta}{\zeta}$ in place of (ξ, η), so that the coordinates $(x : y : z)$ are expressible rationally and homogeneously in terms of (ξ, η, ζ), connected by an equation of the form $(\xi, \eta, \zeta)^\mu(\xi, \eta, \zeta)^\mu = 0$.

    Such an equation, treating therein (ξ, η, ζ) as coordinates, belongs to a curve of the order 2μ, with a μ-tuple point at $(\xi = 0, \zeta = 0)$, a μ-tuple point at $(\eta = 0, \zeta = 0)$, and which has besides $(\mu - 2)^2$ or $(\mu - 1)(\mu - 3)$ dps, according as $D = 2\mu - 3$, or $2\mu - 2$.

    The coordinates $(x : y : z)$ of a point of the given curve are expressible rationally in terms of the coordinates $(\xi : \eta : \zeta)$ of a point on the new curve; and we may say that the original curve is by means of the equations which give $(x : y : z)$ in terms of $(\xi : \eta : \zeta)$ transformed into the new curve.

11. A curve of the order 2μ may have $\frac{1}{2}(2\mu - 1)(2\mu - 2) = 2\mu^2 - 3\mu + 1$ dps; hence in the new curve, observing that the μ-tuple points each count for $\frac{1}{2}(\mu^2 - \mu)$ dps, we have

$$\begin{aligned} &\text{In the case } D = 2\mu - 3, \quad \text{Deficiency} = 2\mu^2 - 3\mu + 1 - \mu^2 + \mu - (\mu - 2)^2 \\ &\qquad\qquad\qquad\qquad\qquad\qquad\qquad = 2\mu - 3 = D \\ &\text{In the case } D = 2\mu - 2, \quad \text{Deficiency} = 2\mu^2 - 3\mu + 1 - \mu^2 + \mu - (\mu - 1)(\mu - 3) \\ &\qquad\qquad\qquad\qquad\qquad\qquad\qquad = 2\mu - 2 = D \end{aligned}$$

    Moreover for $D = 0$, the transformed curve is a conic, with 0 dps, and therefore with deficiency $= 0$; in the case $D = 1$, it is a quartic with 2 dps, and therefore deficiency $= 2$; in the case $D = 2$, it is a quintic with a triple point $= 3$, and a double point $= 1$, together 4 dps, and therefore deficiency $= 2$.

    Hence in every case the new curve has the same deficiency as the original curve.

12. The theorem thus is that the given curve of the order n, with deficiency D, may be rationally transformed into a curve of an order depending only on the deficiency, and having the same deficiency with the given curve, viz.:

    $D = 0$, the new curve is of the order $2(= D + 2)$; $D = 1$, it is of the order $4(= D + 3)$; $D = 2$, it is of the order $5(= D + 3)$; and $D > 2$, it is for D odd, of the order $D + 3$; and for D even, of the order $D + 2$.

    It will presently appear that these are not the lowest values which it is possible to give to the order of the new curve.

    Riemann's object was, not that the order of the transformed curve might be as low as possible, but that the equation in (ξ, η) might be in each of these parameters separately of the lowest possible order; and this he effected by giving to the transformed curve the two μ-tuple points.

13. It is to be noticed that the theorem that for any rational transformation of one curve into another the two curves have the same deficiency is in effect given (as a consequence of Riemann's theory) by Clebsch in the Paper, "Ueber die Singularitáten algebraischer Curven," Crelle, t. LXIV., pp. 98–100. I have, by the assistance of a formula communicated to me by Dr Salmon, obtained a direct analytical demonstration of this theorem.

14. I remark that (x, y, z) being connected by an equation, if $(x : y : z)$ are given rationally in terms of $(\xi : \eta : \zeta)$, then it follows that $(\xi : \eta : \zeta)$ are also expressible rationally in terms of $(x : y : z)$: it is convenient to consider the transformation from this point of view, and I now proceed to the independent development of the theory, as follows:

15. We have a given curve $U = (x, y, z)^n = 0$, with deficiency D, which is by the transformation $\xi : \eta : \zeta = P : Q : R$ (where P, Q, R are given functions $(x, y, z)^k$ each of the same order k) transformed into the curve $T = (\xi, \eta, \zeta)^{n'} = 0$. The transformed curve has, as we know, the same deficiency D as the original curve.

16. To find the order of the transformed curve, we must find the number of its intersections with an arbitrary line $a\xi + b\eta + c\zeta = 0$. Writing in this equation $\xi : \eta : \zeta = P : Q : R$, we obtain the equation $aP + bQ + cR = 0$, and combining therewith the equation $U = 0$, the two equations, being of the orders k and n respectively, give kn systems of values of $(x : y : z)$, and to each of these, in virtue of the equations $\xi : \eta : \zeta = P : Q : R$, there corresponds a single set of values of $(\xi : \eta : \zeta)$, and therefore a single point of intersection; hence the number of intersections, that is, the order of the transformed curve, is $= kn$.

17. If, however, the curves $P = 0$, $Q = 0$, $R = 0$, meet in an ordinary point of the curve $U = 0$, then it is easy to see there is a reduction $= 1$ in the foregoing value; and so if they meet in a dp of the curve $U = 0$, then there is a reduction $= 2$. More generally if the curves $P = 0$, $Q = 0$, $R = 0$ each pass through the same α dps and β ordinary points of the curve $U = 0$, then there is a reduction $= 2\alpha + \beta$.

    In fact the curve $aP + bQ + cR = 0$, meets the curve $U = 0$, in kn points; but among these are included the α dps, each counting as 2 intersections, and the β points; the number of the remaining intersections is $= kn - 2\alpha - \beta$, and the order of the transformed curve is equal to this number. I assume that we have $k < n$.

18. A curve of the order k may be made to pass through $\frac{1}{2}k(k + 3)$ points; it is moreover known that if any three curves, $P = 0$, $Q = 0$, $R = 0$, of the order k each pass through the same $\frac{1}{2}k(k + 3) - 1$ points, then the three curves have all their intersections common, the equations being, in fact, connected by an identical relation of the form $\alpha P + \beta Q + \gamma R = 0$.

To make the order of the transformed curve as low as possible, we must make the curves $P = 0$, $Q = 0$, $R = 0$ meet on the curve $U = 0$ in as many points as possible, and it appears from the remark just made, that the greatest possible number is $= \frac{1}{2}k(k+3) - 1$; and we have thus for the order of the transformed curve

$$n' = kn - 2\alpha - \beta,$$

where α, β are any numbers satisfying the equation

$$2\alpha + \beta = \frac{1}{2}k(k+3) - 1.$$

But we must have $n' > n$ (for otherwise the transformation is illusory, the transformed curve being transformable into the original curve by a transformation of the same or a lower order); and hence we must have

$$kn - \frac{1}{2}k(k+3) + 1 > n,$$

or, what is the same thing,

$$(k-1)(n - \frac{1}{2}k - 2) > 1,$$

which implies $k \geq 2$, $n \geq \frac{1}{2}k + 3$; that is, $n \geq 4$ for $k = 2$, and $n \geq 5$ for $k = 3$, etc.

19. I consider in particular $k = 2$, that is, the transformation by means of conics through $\frac{1}{2} \cdot 2 \cdot 5 - 1 = 4$ points on the curve. Here

$$n' = 2n - 2\alpha - \beta,$$

where $2\alpha + \beta = 4$; so that the pairs of values are $(\alpha, \beta) = (2, 0)$, $(1, 2)$, or $(0, 4)$.

The case $(\alpha, \beta) = (2, 0)$ gives $n' = 2n - 4$; that is, a curve of the order n with 2 dps is transformed into a curve of the order $2n - 4$ with the same deficiency. This deficiency $D = \frac{1}{2}(n-1)(n-2) - 2 = \frac{1}{2}(n^2 - 3n)$; and correspondingly $n' = 2n - 4$ gives $D = \frac{1}{2}(2n-5)(2n-6) - \frac{1}{2}(2n-4)(2n-5) + 2(2n-5) = \frac{1}{2}(n^2 - 3n)$ as before.

20. I stop here to remark that the foregoing transformation gives rise to a transformation of the most general nature. Supposing that the curves $P = 0$, $Q = 0$, $R = 0$ each pass through the same points on the given curve $U = 0$, then if we consider the series of curves $\Pi = P + \lambda Q + \mu R = 0$, any such curve meets the given curve in the fixed points (each counting as two intersections for a dp) and in one variable point; the coordinates of this point are rationally expressible in terms of (λ, μ); that is, we have $x : y : z = \phi(\lambda, \mu) : \psi(\lambda, \mu) : \chi(\lambda, \mu)$, where ϕ, ψ, χ are rational functions of (λ, μ). Conversely, given any rational expression of the coordinates in terms of two parameters, we may by elimination of the parameters obtain the equation of the curve; and the transformation may be regarded as establishing a $(1, 1)$ correspondence between the points of the given curve and the points of a plane (λ, μ).

21. We may consider the question of reducing the order of the transformed curve by reference to the theory of linear series. If the given curve $U = 0$ has deficiency D, then the complete series of groups of m points cut out on the curve by curves of any given order is of the dimension $m - D$ (when $m \geq D$); and for a general curve of order n, a special series does not exist. But for a curve with dps, or more generally for any curve of deficiency D, there may exist special series; and it is by means of these that the order of the transformed curve may be reduced below the value given by Riemann's transformation.

22. In particular, for a curve with deficiency D, there exists a g^1_{D+2} (a linear series of dimension 1, degree $D+2$); that is, the curve may be transformed into a curve of order $D+2$. For $D=0$, this is a conic (order 2); for $D=1$, a cubic (order 3); for $D=2$, a quartic with one dp (order 4); and so on. These are in fact the lowest possible orders.

23. To verify that the transformation is indeed general, I observe that for a general curve of order n with deficiency D, the number of absolute constants (moduli) is $3D-3$ (for $D \geq 2$); and the transformation depends on a certain number of parameters which must be at least equal to this, in order that the transformation may be general. The number of conditions to be satisfied is precisely the number required to determine the moduli, which confirms that the transformation is indeed general.

24. I remark in conclusion that the transformation theory here developed has important applications to the theory of integrals belonging to the curve; in particular, the Abelian integrals of the first kind belonging to the given curve correspond to similar integrals belonging to the transformed curve; and this correspondence furnishes a powerful method for investigating the properties of such integrals.

385. On the Correspondence of Two Points on a Curve

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1. In a unicursal curve the coordinates (x, y, z) of any point of the curve are proportional to rational and integral functions of a variable parameter θ. Hence, if two points of the curve correspond in suchwise that to a given position of the first point there correspond α' positions of the second point, and to a given position of the second point α positions of the first point, the number of points which correspond each to itself is $= \alpha + \alpha'$.

   For let the two points be determined by their parameters θ, θ' respectively—then to a given value of θ there correspond α' values of θ', and to a given value of θ' there correspond α values of θ; hence the relation between (θ, θ') is of the form $(\theta, 1)^{\alpha'}(\theta', 1)^{\alpha} = 0$; and writing therein $\theta' = \theta$, then for the points which correspond each to itself, we have an equation $(\theta, 1)^{\alpha+\alpha'} = 0$ of the order $\alpha + \alpha'$; that is, the number of these points is $= \alpha + \alpha'$.

2. Hence for a unicursal curve we have a theorem similar to that of M. Chasles' for a line, viz., the theorem may be thus stated:

   If two points of a unicursal curve have an (α, α') correspondence, the number of united points is $= \alpha + \alpha'$.

3. But a unicursal curve is nothing else than a curve with a deficiency $D = 0$, and we thence infer

   Theorem 1. *If two points of a curve with deficiency D have an (α, α') correspondence, the number of united points is $= \alpha + \alpha' + 2kD$; in which theorem $2k$ is a coefficient to be determined.*

4. Suppose that the corresponding points are P, P' and imagine that when P is given, the corresponding points P' are the intersections of the given curve by a curve Θ (the equation of the curve will of course contain the coordinates of P as parameters, for otherwise the position of P' would not depend upon that of P).

I find that if the curve Θ has with the given curve k intersections at the point P, then in the system of (P, P'), the number of united points is

$$\sigma = \alpha + \alpha' + 2kD,$$

whence in particular, if the curve Θ does not pass through the point P, then the number of united points is $= \alpha + \alpha'$, as in a unicursal curve.

4*. The foregoing theorem is easily proved in the particular case where the k intersections at the point P take place in consequence of the curve Θ having a k-tuple point at P.

Taking $U = (x, y, z)^m = 0$ as the equation of the given curve (which for greater simplicity is assumed to be a curve without singularities), then if we suppose (x, y, z) to be the coordinates of the point P, and (x', y', z') to be the coordinates of the point P', write

$$U = (x, y, z)^m, \quad U' = (x', y', z')^m,$$

U' being what U becomes on writing therein (x', y', z') in place of (x, y, z); and

$$\Theta = (x, y, z)^k (x', y', z')^{k'} (yz' - y'z, zx' - z'x, xy' - x'y)^k,$$

viz., Θ is a function of the order k in $yz' - y'z$, $zx' - z'x$, $xy' - x'y$, the coefficients of the several powers and products of these quantities being functions of the order α in (x, y, z) and of the order α' in (x', y', z'), which functions are such that they do not all of them vanish identically, or in virtue of the equation $U = 0$, on writing therein $(x', y', z') = (x, y, z)$.

Taking for a moment (x, y, z) as current coordinates, suppose that the equation of the given curve is $U = 0$; then if (x, y, z) are the coordinates of the point P, we have $U = 0$, and similarly if (x', y', z') are the coordinates of the point P' we have $U' = 0$.

The equation $\Theta = 0$, considering therein (x, y, z) as the coordinates of the given point P (and so as parameters satisfying the equation $U = 0$) and (x', y', z') as current coordinates, will be a curve having a k-tuple point at P, we have thus the case above supposed; and P being given, the corresponding points P' are given as the intersections of the curves $U' = 0$, $\Theta = 0$, which are respectively of the orders m and $\alpha' + k$; the total number of intersections is thus $= m(\alpha' + k)$, but inasmuch as the curve $\Theta = 0$ has a k-tuple point at P, k of these intersections coincide with the point P, and the number of the remaining intersections, that is the number of positions of the point P', is $= m\alpha' + (m - 1)k$.

Similarly when P' is given, the number of positions of the point P is $= m\alpha + (m - 1)k$: and we have therefore

$$\alpha + \alpha' = m(\alpha + \alpha') + 2(m - 1)k.$$

To find the united points, it is to be observed, that upon writing $(x', y', z') = (x, y, z)$, the function Θ becomes identically $= 0$; but if we suppose, in the first instance, that P, P' are consecutive points on the curve $U = 0$, then we have

$$yz' - y'z : zx' - z'x : xy' - x'y = \partial_x U : \partial_y U : \partial_z U;$$

and the equation $\Theta = 0$ assumes the form

$$\Theta = (x, y, z)^k (x, y, z)^k (\partial_x U, \partial_y U, \partial_z U)^k = 0,$$

which, (x, y, z) being current coordinates, is the equation of a curve of the order $\alpha + \alpha' + (m - 1)k$; the united points are the intersection of this curve with the given curve $U = 0$, and the number of the united points is thus

$$\sigma = m(\alpha + \alpha') + m(m - 1)k.$$

Hence attending to the above-mentioned value of $\alpha + \alpha'$, we have

$$\sigma = \alpha + \alpha' + (m-1)(m-2)k.$$

But in the case of a curve $U = 0$, without singularities, we have

$$2D = (m-1)(m-2),$$

and we have thus the required formula

$$\sigma = \alpha + \alpha' + 2kD.$$

The investigation in the case where the k intersections at P arise wholly or in part from a contact of the curve Θ, or any branch or branches thereof, with the given curve U, is more difficult, and I abstain from entering upon it.

I apply the theorem to some examples:

5. Investigation of the class of a curve of the order m with δ dps.

Take as corresponding points on the curve two points, such that the line joining them passes through a fixed point O: the united points will be the points of contact of the tangents through O; that is, the number of the united points is equal to the class of the curve.

The curve Θ is here the line OP, which has with the given curve a single intersection at P; that is, we have $k = 1$.

The points P' corresponding to a given position of P are the remaining $m-1$ intersections of OP with the curve; that is, we have $\alpha' = m-1$; and in like manner $\alpha = m-1$.

Hence the class is $= 2(m-1) + 2D$, viz., this is $= (2m-2) + (m^2 - 3m + 2 - 2\delta)$, which is $= m^2 - m - 2\delta$, as it should be.

6. It is in the foregoing example assumed that the dps are none of them cusps; if the curve has $\delta + \kappa$ dps, κ of which are cusps (or what is the same thing, δ nodes and κ cusps); then the number of united points is equal $2(m-1) + 2D$, $= m^2 - m - 2\delta - 2\kappa$; but in this case each of the cusps is reckoned as a united point, and we have, therefore,

$$\text{class} + \kappa = m^2 - m - 2\delta - 2\kappa,$$

that is,

$$\text{class} = m^2 - m - 2\delta - 3\kappa.$$

This will serve as an instance of the special considerations which are required in the case of a curve with cusps, but in what follows, I shall assume that the dps are none of them cusps and thus attend to the case of a curve of the order m, with δ dps, and therefore of the class $n = m^2 - m - 2\delta$, and of the deficiency

$$D = \tfrac{1}{2}(m-1)(m-2) - \delta = \tfrac{1}{2}(n - 2m + 2).$$

7. Investigation of the number of inflexions.

Taking the point P' to be a tangential of P (that is, an intersection of the curve by the tangent at P), the united points are the inflexions, and the number of the united points is equal to the number of inflexions.

The curve Θ is here the tangent at P, having with the given curve two intersections at P; that is $k = 2$.

P' is any one of the $m-2$ tangentials of P, hence $\alpha' = m-2$; and P is the point of contact of any one of the $m-2$ tangents from P' to the curve, that is, $\alpha = m-2$.

Hence the number of inflexions is

$$= (m-2) + (m-2) + 4D = m + n - 4 + 2(n - 2m + 2) = 3(n-m),$$

which is right.

8. For the purpose of the next example it is necessary to present the fundamental equation $\sigma = \alpha + \alpha' + 2kD$ under a more general form.

The curve Θ may intersect the given curve in a system of points P' each p times, a system of points Q' each q times, &c., in such manner that the points (P, P'), the points (P, Q'), &c., are pairs of points corresponding to each other according to distinct laws; and we shall then have the numbers $(\alpha, \alpha, \alpha')$, (β, β, β'), &c., belonging to these pairs respectively; viz. (P, P') are points having an (α, α') correspondence, and the number of united points is $= a$; similarly (P, Q') are points having a (β, β') correspondence, and the number of united points is $= b$; and so on.

The theorem then is

$$p(a - \alpha - \alpha') + q(b - \beta - \beta') + \cdots = 0,$$

or, as this may also be written,

$$pa + qb + \cdots = p(\alpha + \alpha') + q(\beta + \beta') + \cdots + 2(p\alpha + q\beta + \cdots)D,$$

where $p\alpha + q\beta + \cdots = k$ (a positive integer).

9. As an application of the generalized formula, consider the correspondence of two points P, P' on a curve such that the line PP' meets a given curve of the order r in k points (besides the points P, P'). The united points are the points of contact of the tangents from the points of intersection of the given curve $U = 0$ with the curve of the order r; or, what is the same thing, the points of the curve $U = 0$ which lie on the double tangents of this curve which meet the curve of the order r.

We have here a single class of corresponding points, with $p = 1$, $\alpha = \alpha' = m - 1$, $a = n - m + 2D$ (the number of tangents from a point of the curve to the curve itself); hence

$$n - m + 2D - 2(m-1) = 2kD,$$

that is,

$$n - 3m + 2 = 2(k-1)D,$$

or, what is the same thing, $k = 1$ gives $n = 3m - 2 - 2D = m + 2\delta$; that is $n = m^2 - m - 2\delta$.

10. The theorem of correspondence may be further extended to the case of two curves with different deficiencies D, D', but I do not pursue this extension here.

386. On the Logarithms of Imaginary Quantities

[From the Proceedings of the London Mathematical Society, vol. II. (1866–1869), pp. 50–54.]

1. The logarithm of a real positive quantity is determinate; but for a real negative quantity, or for an imaginary quantity, the logarithm has in the first instance an arbitrary value; and it is only when a specific value has been selected that the logarithm becomes a determinate function of the quantity, capable of being operated upon in the ordinary manner. The selected value is taken to be the one which is continuous with the real value for a positive quantity, as this quantity passes continuously from positive to negative through a value zero; or, what is the same thing, for a quantity $x+iy$, where x is positive, the selected value is the ordinary real value $\log\sqrt{x^2+y^2}+i\arctan\dfrac{y}{x}$, and for other values of x the selected value is determined by the condition of continuity. But inasmuch as when the point (x, y) describes a closed curve round the origin, the value of the logarithm is increased by $2i\pi$, it follows that the selected value has a line of discontinuity extending from the origin to infinity; this line is usually taken to be the negative part of the axis of x.

2. Let P be any point (x, y), then taking the selected value, we have

   $$\log P = \log r + i\theta,$$

   where $r = \sqrt{x^2+y^2}$ is the distance from the origin, and θ is the angle measured from the positive axis of x to the radius vector OP; the value of θ being taken between the limits $-\pi$ and π. It may be remarked that the value $\log P'' - \log P'$ of the definite integral $\displaystyle\int_{P'}^{P''}\frac{dz}{z}$, taken along any path not crossing the line of discontinuity, is equal to $\log P'' - \log P'$, where these logarithms are the selected values at the points P'' and P' respectively.

3. If however the path of integration from P' to P'' crosses the line of discontinuity, then the value of the definite integral is not $= \log P'' - \log P'$, but it differs from this by $\pm 2i\pi$. In fact, if the chord $P'P''$ cuts the negative part of the axis of x upwards, there is an abrupt change from $-i\pi$ to $+i\pi$; if the chord $P'P''$ cuts the negative part of the axis of x downwards, there is an abrupt change from $+i\pi$ to $-i\pi$; in the former case, by taking the definite integral to be $\log P'' - \log P'$, we take its value too large by $2i\pi$, in the latter case we take it too small by $2i\pi$; that is, the true value of the definite integral is in the former case $= \log P'' - \log P' - 2i\pi$, in the latter case it is $= \log P'' - \log P' + 2i\pi$.

   But if the chord $P'P''$ does not cut the negative part of the axis of x, then there is not any discontinuity, and the true value of the definite integral is $= \log P'' - \log P'$.

   We have thus in the three cases respectively

   $$\log p = \log P'' - \log P', \quad p = \log P'' - \log P' + 2i\pi, \quad p = \log P'' - \log P' - 2i\pi,$$

   which agrees with the previous results.

4. It may be remarked, that it is merely in consequence of the particular definition adopted that there is in the value of $\log P$ a discontinuity at the passage over the negative part of the axis of x; with a different definition of the logarithm, there would be a discontinuity at the passage over some other line from the origin; but a discontinuity somewhere there must be.

   For if, as above, the chord $P'P$ meet the negative part of the axis of x, then forming a closed quadrilateral by joining by right lines the points 1 to P, P to P', P' to $\dfrac{1}{p}$, and $\dfrac{1}{p}$ to 1; the only side meeting the negative part of the axis of x is the side $P'P$; the integral $\displaystyle\int\frac{dz}{z}$, taken through the closed circuit in question, or say the integral

   $$\int_1^P\frac{dz}{z}+\int_P^{P'}\frac{dz}{z}+\int_{P'}^{1/P}\frac{dz}{z}+\int_{1/P}^{1}\frac{dz}{z}$$

has, by what precedes, a value in consequence of the discontinuity in passing from P' to P; viz., this is $= -2i\pi$ or $= +2i\pi$, according as the chord $P'P$, considered as drawn from P' to P, cuts the negative part of the axis of x upwards or downwards; but this value $-2i\pi$ or $+2i\pi$ must be altogether independent of the definition of the logarithm; whereas if, by any alteration in the definition, the discontinuity could be avoided, the value of the integral, instead of being as above, would be $= 0$.

The foregoing value $-2i\pi$ or $+2i\pi$ is in fact that of the integral taken along (in the one or the other direction) any closed curve surrounding the point $z = 0$, for which the function $\frac{1}{z}$ under the integral sign becomes infinite: but in obtaining the value as above, no use is made of the principles relating to the integration of functions which thus become infinite.

5. The equation $\log P = \log r + i\theta$ gives

$$(x + iy)^m = r^m e^{im\theta},$$

or say

$$(x + iy)^m = r^m e^{im\theta},$$

where, m being any real quantity whatever, r^m denotes the positive real value of r^m.

We have thus a definition of the value of $(x + iy)^m$, and the value so defined may be called the *selected value.*

And similarly, for an imaginary exponent $m = p + qi$, we have

$$\begin{aligned}(x + iy)^{p+qi} &= e^{(p+qi)(\log r + i\theta)} \\ &= e^{p\log r - q\theta + i(p\theta + q\log r)} \\ &= r^p e^{-q\theta} \cdot e^{i(p\theta + q\log r)}\end{aligned}$$

which is the selected value of $(x + iy)^{p+qi}$.

6. It may be remarked, in illustration of the advantage (or rather the necessity) of having a selected value, that in an integral $\int Z\,dz$, taken between given limits along a given path, it is necessary that we know, for the real or imaginary value of z corresponding to each point of the path, the value of the function Z; and consequently, if Z is a function involving $\log z$ or z^m, the indeterminateness which presents itself in these symbols (considered as belonging to a single value of z) is, so to speak, indefinitely multiplied, and $\int Z\,dz$ is really an unmeaning combination of symbols, unless by selecting, as above or otherwise, a unique value of $\log z$ or z^m, we render the function to be integrated a determinate function of the variable.

387. Notices of Communications to the London Mathematical Society

[From the Proceedings of the London Mathematical Society, vol. II. (1866–1869), pp. 6–7, 25–26, 29, 61–63, 103–104, 123–125.]

December 13, 1866. pp. 6–7

Prof. Cayley exhibited and explained some geometrical drawings. Thinking that the information might be convenient for persons wishing to make similar drawings, he noticed that the paper used was a tinted drawing paper, made in continuous lengths up to 24 yards, and of the breadth of about 56 inches; the half-breadth being therefore sufficient for ordinary figures, and the paper being of a good quality and taking colour very readily.

Among the drawings was one of the conics through four points forming a convex quadrangle. The plane is here divided into regions by the lines joining each of the six pairs of points, and by the two parabolas through the four points; and the regions being distinguished by different colours, the general form of the conics of the system is very clearly seen.

(Prof. Cayley remarked that it would be interesting to make the figures of other systems of conics satisfying four conditions; and in particular for the remaining elementary systems of conics, where the conics pass through a number 3, 2, 1 or 0 of points and touch a number 1, 2, 3 or 4 of lines: the construction of some of these figures is, however, practically a great deal more difficult.)

Other figures related to Cartesians and Bicircular Quartics. One of these was a figure of a system of triconfocal Cartesians; and derived from this by inversion in regard to a circle, there was a figure of a system of quadriconfocal bicircular quartics: in the assumed position of the inverting circle, each quartic consists (like the Cartesian which gives rise to it) of an exterior and an interior continuous curve, and the general aspect of the figure is that of a distortion of the original figure of the Cartesians.

Another figure was that of the bicircular quartic, for which the algebraical sum of the distances of a point thereof from three given foci is $= 0$ (this was selected for facility of construction, by the intersections of circles and confocal conics). The quartic consists of two equal and symmetrically situated pear-shaped curves, exterior to each other, and including the one of them two of the three given foci, the other of them the third given focus, and a fourth focus lying in a circle with the given foci: by inversion in regard to a circle having its centre at a focus the two pear-shaped curves became respectively the exterior and the interior ovals of a Cartesian.

There was also a figure of the two circular cubics, having for foci four given points on a circle; and a figure (coloured in regions) in preparation for the construction of the analogous sextic curve derived from four given points not in a circle.

March 28, 1867. pp. 25–26

Professor Cayley mentioned a theorem included in Prof. Sylvester's theory of derivation of the points of a cubic curve.

Writing down the series of numbers $1, 2, 4, 5, 7, 8, 10, 11, 13, 14, 16, 17,$ &*c*., viz., all the numbers not divisible by 3, then (repetitions of the same number being permissible) taking any two numbers of the series, we have in the series a third number, which is the sum or else the difference of the two numbers (for example, $2, 2$ give their sum 4, but $2, 7$ give their difference 5), and we have thus a series of triads, in each of which one number is the sum of the other two.

The theorem is, that it is possible on a cubic curve to construct a series of points, such that denoting them by the above numbers respectively, then for any triad of numbers as aforesaid the points denoted by the three numbers respectively lie in lines.

And the theorem gives its own construction: in fact the series of triads is $112, 224, 145, 257, 178, 248,$ &*c*.

Take 1, an arbitrary point on the cubic, then (by the theorem) the triad 112 shows that 2 is the tangential of 1; 224 shows that 4 is the tangential of 2; 145 that 5 is the third point of 1 and 4; 257 that 7 is the third point of 2 and 5.

So far we have no theorem; we have merely, starting from the point 1, constructed by an arbitrary process the points 2, 4, 5, and 7. But going a step further; 178 and 448 show, the first of them, that 8 is the third point of 1 and 7, the second of them, that 8 is the tangential of 4. We have here the theorem that the third point of 1 and 7 is also the tangential of 4.

Similarly, $10, 11, 13$ are each of them (like 8) determined by two constructions; $14, 16, 17, 19,$ each of them by three constructions, and so on; the number of constructions increasing by unity for each group of four numbers.

And the theorem is, that these constructions, 2, 3, or more, as the case may be, give always one and the same point.

Prof. Cayley mentioned that on a large figure of a cubic curve he had, in accordance with the theorem, constructed the series of points $1, 2, 4, 5, 7, 8, 10, 11, 13, 14$.

April 15, 1867. p. 29

Prof. Cayley communicated a theorem relating to the locus of the ninth of the points of intersection of two cubics, seven of these points being fixed, while the eighth moves on a straight line.

March 26, 1868. pp. 61–63

Prof. Cayley made some remarks on a mode of generation of a sibi-reciprocal surface, that is, a surface the reciprocal of which is of the same order and has the same singularities as the original surface.

If a surface be considered as the envelope of a plane varying according to given conditions, this is a mode of generation which is essentially not sibi-reciprocal; the reciprocal surface is given as the locus of a point varying according to the reciprocal conditions.

But if a surface be considered as the envelope of a quadric surface varying according to given conditions, then the reciprocal surface is given as the envelope of a quadric surface varying according to the reciprocal conditions; and if the conditions be sibi-reciprocal, it follows that the surface is a sibi-reciprocal surface.

For instance, considering the surface which is the envelope of a quadric surface touching each of 8 given lines; the reciprocal surface is here the envelope of a quadric surface touching each of 8 given lines; the conditions are thus sibi-reciprocal, and the surface is therefore a sibi-reciprocal surface.

April 16, 1868. pp. 103–104

Prof. Cayley gave an account of a theorem relating to the sixteen nodes of a quartic surface. The theorem is that for a quartic surface with 16 nodes, the nodes may be regarded as lying in 16 different planes, each plane containing 6 nodes; and that the 16 planes form a symmetrical configuration.

May 14, 1868. pp. 123–125

Prof. Cayley communicated a theorem on the bitangents of a quartic curve. The theorem states that the 28 bitangents of a quartic curve may be divided into 7 sets of 4 bitangents, such that the 8 points of contact of each set of 4 bitangents lie on a conic.

388. Note on the Composition of Infinitesimal Rotations

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VIII. (1867), pp. 7–10.]

1. I call to mind two well-known lemmas:

   Lemma 1. The displacements of a point (x, y, z) produced by an infinitesimal rotation ω about an axis having for its six coordinates (a, b, c, f, g, h), are $\omega(-cy + bz + f)$, $\omega(cx - az + g)$, $\omega(-bx + ay + h)$, in the directions of the axes respectively.

   Lemma 2. The displacement of the point (x, y, z) in the direction of a line (α, β, γ), produced by any infinitesimal motion whatever of a solid body, is $= a\delta x + b\delta y + c\delta z +$

$f\delta p + g\delta q + h\delta r$, where (a, b, c, f, g, h) are the six coordinates of the infinitesimal motion, and δx, δy, δz, δp, δq, δr are the variations of the six coordinates which correspond to the infinitesimal displacement of the point (x, y, z) along the line (α, β, γ).

2. The first of these lemmas establishes a correspondence between infinitesimal rotations and forces; viz., an infinitesimal rotation ω about the line (a, b, c, f, g, h) corresponds to a force ω along the line (a, b, c, f, g, h). And it is to be remarked that the correspondence is such that the system of displacements produced by the rotation is the same as the system of displacements which would be produced by the corresponding force, if the points of application of the forces were displaced in directions perpendicular to the axes of rotation, with magnitudes proportional to the distances from these axes.

3. The second lemma shows that the displacement of a point in a given direction, produced by any infinitesimal motion of a solid body, is the same as the sum of the displacements produced by the several rotations into which the motion may be resolved. Hence, by the principle of virtual velocities, if a system of rotations ω_1, ω_2, &c., about the lines $(a_1, b_1, c_1, f_1, g_1, h_1)$, $(a_2, b_2, c_2, f_2, g_2, h_2)$, &c., are in equilibrium, the displacements $(\delta x, \delta y, \delta z)$ of any point (x, y, z) whatever must each of them vanish; that is, we must have

$$\Sigma\omega a = 0, \quad \Sigma\omega b = 0, \quad \Sigma\omega c = 0, \quad \Sigma\omega f = 0, \quad \Sigma\omega g = 0, \quad \Sigma\omega h = 0,$$

which are therefore the conditions for the equilibrium of the rotations ω_1, ω_2, &c.

Secondly, for a system of forces P_1 along the line $(a_1, b_1, c_1, f_1, g_1, h_1)$, &c., the condition of equilibrium as given by the principle of virtual velocities is

$$\Sigma P(a\delta x + b\delta y + c\delta z + f\delta p + g\delta q + h\delta r) = 0;$$

or, what is the same thing, we must have

$$\Sigma Pa = 0, \quad \Sigma Pb = 0, \quad \Sigma Pc = 0, \quad \Sigma Pf = 0, \quad \Sigma Pg = 0, \quad \Sigma Ph = 0,$$

which are therefore the conditions for the equilibrium of the forces P_1, P_2, &c.

Comparing the two results we see that the conditions for the equilibrium of the rotations ω_1, ω_2, &c., are the same as those for the equilibrium of the forces P_1, P_2, &c.; and since, for rotations and forces respectively, we pass at once from the theory of equilibrium to that of composition; the rules of composition are the same in each case.

4. **Demonstration of Lemma 1.**

Assuming for a moment that the axis of rotation passes through the origin, then for the point P, coordinates (x, y, z), the square of the perpendicular distance from the axis is

$$= (-y\cos\gamma + z\cos\beta)^2 + (x\cos\gamma - z\cos\alpha)^2 + (-x\cos\beta + y\cos\alpha)^2,$$

and the expressions which enter into this formula denote as follows; viz. if through the point P, at right angles to the plane through P and the axis of rotation, we draw a line PQ, = perpendicular distance of P from the axis of rotation, then the coordinates of Q referred to P as origin are

$$-y\cos\gamma + z\cos\beta, \quad x\cos\gamma - z\cos\alpha, \quad -x\cos\beta + y\cos\alpha,$$

respectively.

Hence the foregoing quantities each multiplied by ω are the displacements of the point P in the directions of the axes, produced by the rotation ω.

Suppose that the axis of rotation (instead of passing through the origin) passes through the point (x_0, y_0, z_0); the only difference is that we must in the formula write $(x - x_0, y - y_0, z - z_0)$ in place of (x, y, z): and attending to the significations of the six coordinates (a, b, c, f, g, h) it thus appears that the displacements produced by the rotation are equal to ω into the expressions

$$-cy + bz + f, \quad cx - az + g, \quad -bx + ay + h,$$

respectively.

5. **Demonstration of Lemma 2.**

For any infinitesimal motion whatever of a solid body, the displacements of the point (x, y, z) in the directions of the axes are

$$\delta x = l - ry + qz, \quad \delta y = m + rx - pz, \quad \delta z = n - qx + py,$$

and hence the displacement in the direction of the line (α, β, γ), is

$$\delta x \cos\alpha + \delta y \cos\beta + \delta z \cos\gamma,$$

which, attending to the signification of the six coordinates (a, b, c, f, g, h), is

$$= al + bm + cn + fp + gq + hr,$$

which is the required expression.

It is proper to remark that the last-mentioned expressions of $(\delta x, \delta y, \delta z)$ are in fact the displacements produced by a translation and a rotation. If we assume that every infinitesimal motion of a solid body can be resolved into a translation and a rotation, then, since a translation can be produced by two rotations, every infinitesimal motion of a solid body can be resolved into rotations alone, and the foregoing expressions for the displacements produced by a rotation, combining any number of them and writing $(\Sigma\omega a, \Sigma\omega b, \Sigma\omega c, \Sigma\omega f, \Sigma\omega g, \Sigma\omega h) = (-l, -m, -n, l, m, n)$ respectively, lead to the expressions for the displacements δx, δy, δz produced by the infinitesimal motion of the solid body.

389. On a Locus derived from Two Conics

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VIII. (1867), pp. 77–84.]

Required the locus of a point which is such that the pencil formed by the tangents through it to two given conics has a given anharmonic ratio.

1. Suppose, for a moment, that the equation of the tangents to the first conic is $(x - ay)(x - by) = 0$, and that of the tangents to the second conic is $(x - cy)(x - dy) = 0$, and write

$$A = (a - b)(c - d), \quad B = (a - c)(d - b), \quad C = (a - d)(b - c),$$

so that $A + B + C = 0$, write also

$$k = \frac{A'}{A}, \quad k' = \frac{B'}{B}, \quad k'' = \frac{C'}{C},$$

then the anharmonic ratio of the pencil will have a given value k if

$$(k - k')(k - k'') = 0;$$

that is, if

$$k^2 + k + 1 = 0,$$

or, what is the same thing, if

$$A'^2(2k+1)^2 + 4BC = 0;$$

that is, if

$$A'^2(2k+1)^2 - (B-C)^2 = 0,$$

where

$$A'^2 = (a-b)^2(c-d)^2, \quad B - C = (a+b)(c+d) - 2(ab+cd),$$

are each of them symmetrical in regard to a, b, and in regard to c, d, respectively.

2. Let the equations of the two conics be

$$U = (a, b, c, f, g, h \;\smallsmile\; x, y, z)^2 = 0, \quad U' = (a', b', c', f', g', h' \;\smallsmile\; x, y, z)^2 = 0,$$

and let (α, β, γ) be the coordinates of the variable point.

Putting as usual

$$\begin{gathered}(A, B, C, F, G, H) = (bc - f^2,\ ca - g^2,\ ab - h^2,\ gh - af,\ hf - bg,\ fg - ch),\\ K = abc - af^2 - bg^2 - ch^2 + 2fgh,\end{gathered}$$

the equation of the tangents to the first conic is

$$(A, B, C, F, G, H \;\smallsmile\; X, Y, Z)^2 = 0,$$

where

$$X = \beta z - \gamma y, \quad Y = \gamma x - \alpha z, \quad Z = \alpha y - \beta x,$$

and therefore $\alpha X + \beta Y + \gamma Z = 0$.

Hence substituting for Z the value $-\dfrac{\alpha X + \beta Y}{\gamma}$, we find, for the equation of the tangents, an equation of the form $aX^2 + 2hXY + bY^2 = 0$, which has, in effect, been taken to be $(X - aY)(X - bY) = 0$; that is, we have

$$1 : a + b : ab = a : -2h : b;$$

and, in like manner, if the accented letters refer to the second conic

$$1 : c + d : cd = a' : -2h' : b'.$$

Substituting for a, h, b their values, and for a', h', b' the corresponding values, we find

$$\begin{aligned} 1 : c + d : cd = {} & A'\gamma^2 - 2G'\gamma\alpha + C'\alpha^2 \\ & : -2(H'\gamma^2 - F'\alpha\gamma - G'\beta\gamma + C'\alpha\beta) \\ & : B'\gamma^2 - 2F'\beta\gamma + C'\beta^2, \\ 1 : a + b : ab = {} & A\gamma^2 - 2G\gamma\alpha + C\alpha^2 \\ & : -2(H\gamma^2 - F\alpha\gamma - G\beta\gamma + C\alpha\beta) \\ & : B\gamma^2 - 2F\beta\gamma + C\beta^2. \end{aligned}$$

We then have

$$(a-b)^2 = (a+b)^2 - 4ab = -4\gamma^2 K(\alpha, \beta, \gamma)^2,$$

and similarly

$$(c-d)^2 = -4\gamma^2 K'(\alpha,\beta,\gamma)^2.$$

We have, moreover,

$$\begin{aligned}
&(a+b)(c+d)-2(ab+cd)\\
&\quad = 4(H\gamma^2 - F\alpha\gamma - G\beta\gamma + C\alpha\beta)(H'\gamma^2 - F'\alpha\gamma - G'\beta\gamma + C'\alpha\beta)\\
&\qquad - 2(B\gamma^2 - 2F\beta\gamma + C\beta^2)(A'\gamma^2 - 2G'\gamma\alpha + C'\alpha^2)\\
&\qquad - 2(B'\gamma^2 - 2F'\beta\gamma + C'\beta^2)(A\gamma^2 - 2G\gamma\alpha + C\alpha^2)\\
&\quad = -2\gamma^2(BC' + B'C - 2FF', \dots \smallsmile \alpha,\beta,\gamma)^2,
\end{aligned}$$

and substituting the foregoing values, we find

$$4(2k+1)^2 KK'(\alpha, \dots \smallsmile \alpha,\beta,\gamma)^2 - \{(BC'+B'C-2FF', \dots \smallsmile \alpha,\beta,\gamma)^2\}^2 = 0,$$

or putting for shortness

$$\Theta = (BC'+B'C-2FF', GH'+G'H-AF'-A'F, \dots \smallsmile \alpha,\beta,\gamma)^2,$$

the equation of the locus is

$$4(2k+1)^2 KK'\cdot UU' - \Theta^2 = 0,$$

where (α,β,γ) are current coordinates.

The locus is thus a quartic curve having quadruple contact with each of the conics $U=0$, $U'=0$; viz. it touches them at their points of intersection with the conic $\Theta=0$, which is the locus of the point such that the four tangents form a harmonic pencil.

3. The equation may be written somewhat more elegantly under the form

$$4(2k+1)^2\cdot KU\cdot K'U' - \Theta^2 = 0;$$

viz. in this equation we have

$$KU = (BC-F^2, \dots \smallsmile \alpha,\beta,\gamma)^2, \quad K'U' = (B'C'-F'^2, \dots \smallsmile \alpha,\beta,\gamma)^2,$$

$$= (BC'+B'C-2FF', \dots \smallsmile \alpha,\beta,\gamma)^2.$$

In the last form the equation is expressed in terms of the coefficients $(A,\dots)$, $(A',\dots)$ of the line equations of the conics, viz. these may be taken to be

$$(A, \dots \smallsmile \xi,\eta,\zeta)^2 = 0, \quad (A', \dots \smallsmile \xi,\eta,\zeta)^2 = 0.$$

4. In particular, if each of the conics break up into a pair of points, viz. (l,m,n) and (p,q,r) for the first conic, (l',m',n') and (p',q',r') for the second conic, then the line equations are

$$2(l\xi+m\eta+n\zeta)(p\xi+q\eta+r\zeta)=0, \quad 2(l'\xi+m'\eta+n'\zeta)(p'\xi+q'\eta+r'\zeta)=0,$$

so that

$$A = 2lp, \dots F = mr+nq, \dots A' = 2l'p', \dots F' = m'r'+n'q', \dots$$

$$(BC-F^2,\dots) = -(mn-nq,\ np-lr,\ lq-mp)^2,$$

$$(B'C'-F'^2,\dots) = -(m'n'-n'q',\ n'p'-l'r',\ l'q'-m'p')^2,$$

$$BC'+B'C-2FF' = 2\{(mn'-m'n)(qr'-q'r)-(nr'-n'r)(m'q-mq'),\dots\},$$

and substituting these values the equation is

$$4(2k+1)^2 \begin{vmatrix}\alpha & \beta & \gamma\\ l & m & n\\ p & q & r\end{vmatrix}^2 \begin{vmatrix}\alpha & \beta & \gamma\\ l' & m' & n'\\ p' & q' & r'\end{vmatrix}^2 - \left[\begin{vmatrix}\alpha & \beta & \gamma\\ l & m & n\\ p & q & r\end{vmatrix} \begin{vmatrix}\alpha & \beta & \gamma\\ l' & m' & n'\\ p' & q' & r'\end{vmatrix}\right]^2 = 0,$$

which, if A, B, C denote

$$\begin{vmatrix}\alpha & \beta & \gamma\\ l & m & n\\ p & q & r\end{vmatrix}, \quad \begin{vmatrix}\alpha & \beta & \gamma\\ l' & m' & n'\\ p & q & r\end{vmatrix}, \quad \begin{vmatrix}\alpha & \beta & \gamma\\ l & m & n\\ p' & q' & r'\end{vmatrix} \text{ respectively, } (A+B+C=0)$$

is, in fact, the equation

$$(2k+1)^2A^2 - (B-C)^2 = 0,$$

that is

$$k = -\frac{B}{A} \text{ or } k = -\frac{C}{A},$$

either of which expresses the anharmonic property of the points of a conic in the form given by the theorem *ad quatuor lineas.*

5. Reverting to the case of two conics, then if these be referred to a set of conjugate axes, the equations will be

$$aa^2x^2 + bb^2y^2 + cc^2z^2 = 0, \quad a'a'^2x^2 + b'b'^2y^2 + c'c'^2z^2 = 0,$$

we have $K = abc$, $K' = a'b'c'$,

$$\Theta = (bc' + b'c)aa'\alpha^2 + (ca' + c'a)bb'\beta^2 + (ab' + a'b)cc'\gamma^2,$$

and the equation of the quartic curve is

$$4(2k+1)^2abca'b'c'(aa^2x^2 + bb^2y^2 + cc^2z^2)(a'a'^2x^2 + b'b'^2y^2 + c'c'^2z^2) - \Theta^2 = 0.$$

6. I suppose in particular that the two conics are

$$x^2 + my^2 - 1 = 0, \quad mx^2 + y^2 - 1 = 0,$$

the equation of the quartic is

$$4(2k+1)^2m^2(x^2 + my^2 - 1)(mx^2 + y^2 - 1) - \{(m^2 + m)(x^2 + y^2) - m^2 - 1\}^2 = 0;$$

or putting $\lambda = \dfrac{(m+1)^2}{(2k+1)^2}$, this is

$$(x^2 + y^2 - \lambda)^2 - \lambda^2(x^2 + my^2 - 1)(mx^2 + y^2 - 1) = 0.$$

To fix the ideas, suppose that m is positive and > 1, so that each of the conics is an ellipse, the major semi-axis being $= 1$, and the minor semi-axis being $= \dfrac{1}{\sqrt{m}}$.

For any real value of k the coefficient λ is positive, and it may accordingly be assumed that λ is positive.

We have $\dfrac{m^2+1}{m(m+1)} < \lambda < 1$, or the radius of the circle is intermediate between the semi-axes of the ellipses, hence the points of contact on each ellipse are real points.

Writing for shortness $\alpha = \dfrac{m^2+1}{m^2+\lambda}$, the equation is

$$(x^2+my^2-1)(mx^2+y^2-1)-\lambda(x^2+y^2-\alpha)^2=0.$$

For the points on the axis of x, we have

$$(x^2-1)(mx^2-1)-\lambda(x^2-\alpha)^2=0,$$

that is

$$(m-\lambda)x^4+\{-(1+m)+2\lambda\alpha\}x^2+(1-\lambda\alpha^2)=0,$$

and thence

$$(m-\lambda)x^2=\tfrac{1}{2}(1+m)-\lambda\alpha\pm\tfrac{1}{2}\sqrt{\{(m-1)^2+4\lambda(1-\alpha)(1-m\alpha)\}},$$

or, substituting for α its value, this is

$$x^2=\frac{\lambda(m+\frac{1}{m})}{\lambda^2-m},\ldots$$

The investigation shows that for different values of k, the locus consists of a series of quartic curves, each having quadruple contact with the two given conics, and touching them at points on the conic $\Theta=0$.

390. Theorem relating to the four Conics which touch the same two lines and pass through the same four points

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VIII. (1867), pp. 162–167.]

1. The theorem referred to in the title is as follows:

Theorem 2. *If four conics touch the same two lines and pass through the same four points, then the four poles of the line joining the points of contact of the two tangents, in regard to the four conics respectively, lie on a right line.*

The proof is very simple. Let the two lines be taken for the lines $x=0$, $y=0$; and let the four points be the intersections of the lines $z=0$ with the conic $(a,b,c,f,g,h \between x,y,z)^2=0$.

The equation of a conic touching the lines $x=0$, $y=0$, and passing through the four points, is

$$(a,b,c,f,g,h \between x,y,z)^2+2\lambda xy=0,$$

where λ is an arbitrary parameter. This conic touches the two lines at the points where the line $fx+gy+hz=0$ meets them; that is, at the points $(0,-h,g)$ and $(-h,0,f)$.

The pole of the line joining these two points, that is, of the line $fgx+fhy-h^2z=0$, in regard to the conic, is found to be

$$x:y:z=-h(g+\lambda f):-h(f+\lambda g):fg-\lambda h^2.$$

For the four values of λ corresponding to the four conics, we have four such poles; and it is easy to verify that these four poles lie on the line

$$fgx-fgy+(f^2-g^2)z=0.$$

2. The theorem may be stated in a different form. If four conics pass through the same four points, their eight tangents at these points form an interesting configuration. The theorem then asserts that the poles of the line joining the points of contact of any two fixed tangents, with respect to the four conics, are collinear.

3. The theorem is a particular case of a more general theorem relating to curves of any order. If n curves of the order m pass through the same mn points, and touch the same m lines, then the poles of any one of these lines, in regard to the n curves, lie on a curve of order $n-1$.

4. In the case of conics, the number of conics which touch two given lines and pass through four given points is $= 4$. For the conic is determined by 5 conditions; the conditions of touching two lines are 2 conditions, and passing through 4 points is 4 conditions, together 6 conditions; but there is one condition too many, since the 4 points and 2 lines form a system of 6 elements which must satisfy a condition in order that conics touching the 2 lines and passing through the 4 points may exist.

5. The condition in question may be found as follows. Let the four points be the intersections of a conic $U = 0$ with a line $z = 0$; and let the two lines be $x = 0$, $y = 0$. Then if a conic $V = 0$ touches the lines $x = 0$, $y = 0$ and passes through the four points, we must have

$$V = U + 2\lambda xy = 0,$$

and the condition that this conic touches $x = 0$ and $y = 0$ gives two equations for λ, which in general have 4 common solutions.

6. The theorem may be connected with the theory of the invariants of two conics. If $U = 0$, $V = 0$ are two conics, the condition that the conic $U+\lambda V = 0$ may touch a given line gives an equation in λ, the roots of which determine the conics of the pencil which touch the given line. In the present case, the pencil is $U + 2\lambda xy = 0$, and the condition of touching $x = 0$ and $y = 0$ leads to a quartic equation in λ, giving the 4 conics of the theorem.

7. The theorem admits of an interesting geometrical interpretation. Consider the four conics as the conics of a pencil which touch two fixed lines. The poles of the line joining the points of contact form a range of four points on a line; and this range has properties analogous to those of the range of four points in which the conics of a pencil meet a fixed line.

8. I remark in conclusion that the theorem is intimately connected with the theory of the bitangents of a quartic curve. If we consider the quartic curve enveloped by the tangents to the four conics, the line of poles is related to the configuration of the bitangents in a remarkable manner.

391. Solution of a Problem of Elimination

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VIII. (1867), pp. 183–185.]

1. The problem of elimination which I propose to solve is the following:

   Given the system of equations

$$\begin{aligned} ax^2 + 2hxy + by^2 + 2gxz + 2fyz + cz^2 &= 0, \\ a'x^2 + 2h'xy + b'y^2 + 2g'xz + 2f'yz + c'z^2 &= 0, \\ a''x^2 + 2h''xy + b''y^2 + 2g''xz + 2f''yz + c''z^2 &= 0, \end{aligned}$$

   representing three conics; to find the condition that the three conics may have a common point.

2. The required condition is obtained by eliminating (x, y, z) from the three equations. The most symmetrical form of the result is as a determinant. If we write the equations in the form

$$U = 0, \quad U' = 0, \quad U'' = 0,$$

then the condition is that the determinant

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} \begin{vmatrix} a' & h' & g' \\ h' & b' & f' \\ g' & f' & c' \end{vmatrix} \begin{vmatrix} a'' & h'' & g'' \\ h'' & b'' & f'' \\ g'' & f'' & c'' \end{vmatrix} = 0,$$

where the product is formed in a certain symbolic manner.

3. The actual expression of the condition of elimination may be presented under a very elegant form by the use of the symbolic notation for determinants. If the three conics are written as

$$(a, b, c, d, e, f \;_{\sim}\; x, y, z)^2 = 0, \quad (a', \ldots \;_{\sim}\; x, y, z)^2 = 0, \quad (a'', \ldots \;_{\sim}\; x, y, z)^2 = 0,$$

then the result of elimination is:

Represent the columns

$$\begin{array}{cccccc} a & b & a' & b' & a'' & b'' \\ b & c & b' & c' & b'' & c'' \\ c & d & c' & d' & c'' & d'' \\ d & e & d' & e' & d'' & e'' \\ e & f & e' & f' & e'' & f'' \end{array}$$

by $1, 2, 3, 4, 5, 6$, then each equation is of the type

$$\lambda(1 + k \cdot 2) + \lambda'(3 + k \cdot 4) + \lambda''(5 + k \cdot 6) = 0.$$

Multiplying the several equations by the minors of 135, each with its proper sign, and adding, the terms independent of k disappear, the equation divides by k, and we find

$$\lambda \cdot 2135 + \lambda' \cdot 4135 + \lambda'' \cdot 6135 = 0;$$

operating in a similar manner with the minors of 246, the terms in k disappear, and we find

$$\lambda \cdot 1246 + \lambda' \cdot 3246 + \lambda'' \cdot 5246 = 0;$$

again, operating with the minors of $(146 + 236 + 245 + k \cdot 246)$, we find

$$\begin{aligned} &\lambda\{1236 + 1245 + k(2146 + 1246)\} \\ &\qquad + \lambda'\{3146 + 3245 + k(4236 + 3246)\} \\ &\qquad\qquad + \lambda''\{5146 + 5236 + k(6245 + 5246)\} = 0, \end{aligned}$$

where the terms in k disappear, and this is

$$\lambda(1236 + 1245) + \lambda'(3146 + 3245) + \lambda''(5146 + 5236) = 0.$$

We have thus three linear equations, which written in a slightly different form are

$$\begin{aligned} \lambda \cdot 1235 + \lambda' \cdot 3451 + \lambda'' \cdot 5613 &= 0, \\ \lambda(1236 + 1245) + \lambda'(3452 + 3461) + \lambda''(5614 + 5623) &= 0, \\ \lambda \cdot 1246 + \lambda' \cdot 3462 + \lambda'' \cdot 5624 &= 0, \end{aligned}$$

and thence eliminating λ, λ', λ'', we have

$$\begin{vmatrix} 1235, & 1236+1245, & 1246 \\ 3451, & 3452+3461, & 3462 \\ 5613, & 5614+5623, & 5624 \end{vmatrix} = 0,$$

which is the required result.

4. It may be remarked that the second and third column are obtained from the first by operating on it with Δ, $\frac{1}{2}\Delta^2$, if $\Delta = 23_1 + 45_3 + 65_5$.

Or say the result is

$$(1, \Delta, \tfrac{1}{2}\Delta^2) \begin{vmatrix} 1235 \\ 3451 \\ 5613 \end{vmatrix} = 0.$$

In like manner for the system

$$\begin{array}{cccccc} x^2 & xy & y^2 & yz & z^2 & zx \\ a & b & c & d & e & f \\ a' & b' & c' & d' & e' & f' \\ a'' & b'' & c'' & d'' & e'' & f' \\ a''' & b''' & c''' & d''' & e''' & f''' \end{array}$$

if the columns are

$$\begin{array}{cccccccc} a & b, & a' & b', & a'' & b'', & a''' & b''' \\ b & c, & b' & c', & b'' & c'', & b''' & c''' \\ c & d, & c' & d', & c'' & d'', & c''' & d''' \\ d & e, & d' & e', & d'' & e'', & d''' & e''' \\ e & f, & e' & f', & e'' & f'', & e''' & f''' \end{array}$$

then the result is

$$(1, \Delta, \tfrac{1}{2}\Delta^2, \tfrac{1}{6}\Delta^3) \begin{vmatrix} 12357 \\ 34571 \\ 56713 \\ 78135 \end{vmatrix} = 0,$$

where

$$\Delta = 28_1 + 48_3 + 68_5 + 8\chi_7.$$

392. On the Conics which pass through two given Points and touch two given Lines

[From the Quarterly Journal of Pure and Applied Mathematics, vol. VIII. (1867), pp. 211–219.]

1. Let $x = 0$, $y = 0$ be the equations of the given lines; $z = 0$ the equation of the line joining the given points. We may, to fix the ideas, imagine the implicit constants so determined that $x + y + z = 0$ shall be the equation of the line infinity.

Take $x - my = 0$, $x - ny = 0$ as the equations of the lines which by their intersection with $z = 0$ determine the given points.

The equation of the conic is

$$\sqrt{xy(m)} + \sqrt{xy(n)} \cdot \sqrt{xy} = x + y\sqrt{mn} + \gamma z,$$

or, what is the same thing,

$$(x-my)(x-ny)+2(x+y\sqrt{mn})\gamma z+\gamma^2z^2=0,$$

so that there are two distinct series of conics according as $\sqrt{mn}$ is taken with the positive or the negative sign.

The equation of the chord of contact is

$$x+y\sqrt{mn}+\gamma z=0,$$

which meets $z=0$ in the point

$$[x+y\sqrt{mn}=0,\quad z=0]$$

that is in one of the centres of the involution formed by the lines $(x=0,y=0)$, $(x-my=0,x-ny=0)$.

It is to be observed that the conic is only real when mn is positive, that is (the lines and points being each real) the two points must be situate in the same region or in opposite regions of the four regions formed by the two lines: there are however other real cases; e.g. if the lines $x=0$, $y=0$ are real, but the quantities m, n are conjugate imaginaries; included in this we have the circles which touch two real lines.

2. To fix the ideas I take m and n each positive and $mn>1$; also I attend first to the series where $\sqrt{mn}$ is taken positively.

At the points where the conic meets infinity, we have

$$(\sqrt{m}+\sqrt{n})\sqrt{xy}=x+y\sqrt{mn}-\gamma(x+y),$$

which gives two coincident points, that is the conic is a parabola, if

$$(1-\gamma)\{\sqrt{mn}-\gamma\}=\tfrac{1}{4}(\sqrt{m}+\sqrt{n})^2,$$

that is

$$\gamma=-\tfrac{1}{2}(1+\sqrt{mn})\pm\tfrac{1}{2}\sqrt{(1+m)(1+n)},$$

or

$$\gamma=\tfrac{1}{2}[1+\sqrt{mn}+\sqrt{(1+m)(1+n)}],$$

where it is to be noticed that

$$\gamma=\tfrac{1}{2}[1+\sqrt{mn}+\sqrt{(1+m)(1+n)}]$$

is a positive quantity greater than $\sqrt{mn}$, say $\gamma=p$;

$$\gamma=\tfrac{1}{2}[1+\sqrt{mn}-\sqrt{(1+m)(1+n)}]$$

is a negative quantity, say $\gamma=-q$, q being positive.

The order of the lines is as shown in fig. 1, see plate facing p. 52.

$\gamma=-\infty$ to $\gamma=-q$,	curve is ellipse;
$\gamma=-q$,	parabola P_1,
$\gamma=-q$ to p,	curve is hyperbola;
$\gamma=p$,	parabola P_2,
$\gamma=p$ to $\gamma=\infty$,	ellipse.

3. Resuming the equation

$$(x-my)(x-ny)+2(x+y\sqrt{mn})\gamma z+\gamma^2z^2=0,$$

the coefficients are

$$(a,b,c,f,g,h)=(1,mn,\gamma^2,\gamma\sqrt{mn},\gamma,-\tfrac{1}{2}(m+n)),$$

and thence the inverse coefficients are

$$\begin{aligned}(A,B,C,F,G,H)=(0,0,&-\tfrac{1}{4}(m-n)^2,\\ &-\tfrac{1}{4}\gamma(\sqrt{m}+\sqrt{n})^2,\\ &-\tfrac{1}{2}\gamma\sqrt{mn}(\sqrt{m}+\sqrt{n})^2,\\ &\tfrac{1}{4}\gamma^2(\sqrt{m}+\sqrt{n})^2).\end{aligned}$$

$$K=-\tfrac{1}{4}\gamma^2(\sqrt{m}+\sqrt{n})^2,$$

or, omitting a factor, the inverse coefficients are

$$(A,B,C,F,G,H)=(0,-\gamma(\sqrt{m}-\sqrt{n})^2,1,\tfrac{1}{\gamma}\sqrt{mn},-\gamma,\ldots).$$

Considering the line $\lambda x+\mu y+\nu z=0$, the coordinates of the pole of this line are

$$X:Y:Z=-\gamma\mu+\sqrt{mn}\nu:-\gamma\lambda+\mu:\sqrt{mn}\lambda+\mu+\tfrac{1}{\gamma}(\sqrt{m}-\sqrt{n})^2\nu.$$

4. In particular, if $\lambda=\mu=\nu=1$, then for the coordinates of the centre of the conic, we have

$$X:Y:Z=-\gamma+\sqrt{mn}:-\gamma+1:\sqrt{mn}+1+\tfrac{1}{\gamma}(\sqrt{m}-\sqrt{n})^2;$$

and for the locus of the centre,

$$(X-Y)^2\cdot\tfrac{1}{4}(\sqrt{m}-\sqrt{n})^2+(X-Y)(X-Y\sqrt{mn})\{1+\sqrt{mn}\}+z(X-Y\sqrt{mn})\{1-\sqrt{mn}\}=0,$$

so that the locus is a conic, and it is obvious that this conic is a hyperbola.

Putting for greater simplicity

$$X=x-y,\quad Y=x-y\sqrt{mn},\quad z=Z,$$

the equation of the curve of centres is

$$ZQ+X^2\cdot\tfrac{1}{4}(\sqrt{m}-\sqrt{n})^2+XY\{1+\sqrt{mn}\}+YZ\{1-\sqrt{mn}\}=0,$$

or, writing this under the form

$$Z[Y\{1+\sqrt{mn}\}+Z\{1-\sqrt{mn}\}]+\tfrac{1}{4}(\sqrt{m}-\sqrt{n})^2X^2=0,$$

the equation is

$$ZQ+XY'=0,$$

where

$$X=x-y,\quad Y=x-y\sqrt{mn},\quad Z=\tfrac{2}{(\sqrt{m}-\sqrt{n})^2}[Y'-(x-y)+\{1-\sqrt{mn}\}z];$$

these values give

$$x-y=X,\quad x-y\sqrt{mn}=Y,$$

$$\{1-\sqrt{mn}\}z = \tfrac{1}{2}(\sqrt{m}-\sqrt{n})^2Q + 2\{1+\sqrt{mn}\}X,$$

or, what is the same thing,

$$\begin{aligned}
\{1-\sqrt{mn}\}x &= -\sqrt{mn}X + Y,\\
\{1-\sqrt{mn}\}y &= -X + Y,\\
\{1-\sqrt{mn}\}z &= 2\{1+\sqrt{mn}\}X + \tfrac{1}{2}(\sqrt{m}-\sqrt{n})^2Q.
\end{aligned}$$

whence also

$$\{1-\sqrt{mn}\}(x+y+z) = \{1+\sqrt{mn}\}Z + 2Y + (\sqrt{m}-\sqrt{n})^2Q,$$

or the equation of the line infinity is

$$\{1+\sqrt{mn}\}X + 2Y + (\sqrt{m}-\sqrt{n})^2Q = 0,$$

a formula which may be applied to finding the asymptotes and thence the centre of the conic $QZ + X^2 = 0$.

In fact we have identically

$$[2kx+2ky-(2k+1)z]^2 - (1+4k)(2kx-z)^2 = 4k^2(x+y+z)^2 - 4k(1+4k)(kx^2+yz),$$

that is

$$-4k(1+4k)(kx^2+yz) = [2kx+2ky-(2k+1)z]^2 - (1+4k)(2kx-z)^2 - 4k^2(x+y+z)^2,$$

which, if $x+y+z=0$ is the equation of the line infinity, puts in evidence the asymptotes of the conic $kx^2+yz=0$.

Hence writing αx, βy, γz in the place of x, y, z respectively and $k=k'$, that is, $k=\dfrac{k'}{\alpha^2}$, we have

$$-4\beta^2\gamma k'(\alpha^2+4\beta\gamma k')(k'x^2+yz) = [2\alpha\gamma k'x+2\beta^2\gamma k'y-(2\beta^2\gamma k'+\alpha^2)\gamma z]^2-(\alpha^2+4\beta^2\gamma k')[2\beta\gamma k'x-\alpha z]^2-4$$

or, what is the same thing,

$$\begin{aligned}
&-4\beta^2\gamma k'(\alpha^2+4\beta\gamma k')(k'x^2+yz)\\
&\qquad= [2\alpha\gamma k'x+2\beta^2\gamma k'y-(2\beta^2\gamma k'+\alpha^2)\gamma z]^2\\
&\qquad\quad-(\alpha^2+4\beta^2\gamma k')(2\beta\gamma k'x-\alpha z)^2 - 4\beta^2\gamma^2k'^2(\alpha x+\beta y+\gamma z)^2,
\end{aligned}$$

which, when $\alpha x+\beta y+\gamma z=0$ is the equation of the line infinity, puts in evidence the asymptotes of the conic $k'x^2+yz=0$.

Now writing X, Y, Q in the place of x, y, z; $k'=1$, and $\alpha=\{1+\sqrt{mn}\}$, $\beta=2$, $\gamma=(\sqrt{m}-\sqrt{n})^2$, we have

$$\begin{aligned}
&-16[\{1+\sqrt{mn}\}^2+8(\sqrt{m}-\sqrt{n})^2](ZQ+X^2)\\
&\qquad= [4\{1+\sqrt{mn}\}Z+8Y-(4(\sqrt{m}-\sqrt{n})^2+\{1+\sqrt{mn}\}^2)Q]^2\\
&\qquad\quad-[\{1+\sqrt{mn}\}^2+8(\sqrt{m}-\sqrt{n})^2][4X-\{1+\sqrt{mn}\}Q]^2\\
&\qquad\qquad-16[\{1+\sqrt{mn}\}X+2Y+(\sqrt{m}-\sqrt{n})^2Q]^2,
\end{aligned}$$

and the asymptotes are

$$\begin{aligned}
&4\{1+\sqrt{mn}\}Z+8Y-[4(\sqrt{m}-\sqrt{n})^2+\{1+\sqrt{mn}\}^2]Q\\
&\qquad=\pm\sqrt{\{1+\sqrt{mn}\}^2+8(\sqrt{m}-\sqrt{n})^2}\,[4X-\{1+\sqrt{mn}\}Q].
\end{aligned}$$

At the centre

$$4\{1+\sqrt{mn}\}Z+8Y-[4(\sqrt{m}-\sqrt{n})^2+\{1+\sqrt{mn}\}^2]Q=0,$$
$$4X-\{1+\sqrt{mn}\}Q=0,$$

but the first equation is

$$\{1+\sqrt{mn}\}[4Z-Q\{1+\sqrt{mn}\}]+8Y-4(\sqrt{m}-\sqrt{n})^2Q=0,$$

so that we have

$$4Z=\{1+\sqrt{mn}\}Q, \quad 2Y=(\sqrt{m}-\sqrt{n})^2Q,$$

the first of these is

$$2(\sqrt{m}-\sqrt{n})^2(x-y)-\{1+\sqrt{mn}\}^2(x-y)-(1-mn)z=0,$$

and the two together give

$$2Z(\sqrt{m}-\sqrt{n})^2-\{1+\sqrt{mn}\}Y=0,$$

so that we have

$$2(\sqrt{m}-\sqrt{n})^2(x-y)-\{1+\sqrt{mn}\}\{x-y\sqrt{mn}\}=0,$$
$$[\{1+\sqrt{mn}\}^2-2(\sqrt{m}-\sqrt{n})^2](x-y)+(1-mn)z=0,$$

to determine the coordinates of the centre.

5. The equation of the chord of contact is

$$x+y\sqrt{mn}+\gamma z=0,$$

which for $\gamma=1$ is parallel to $y=0$ and for $\gamma=\sqrt{mn}$ is parallel to $x=0$.

But the coordinates of the centre are

$$X:Y:Z=-\gamma+\sqrt{mn}:-\gamma+1:\sqrt{mn}+1+\tfrac{1}{\gamma}(\sqrt{m}-\sqrt{n})^2,$$

which for $\gamma=1$ give

$$y=0, \quad X:z=-1+\sqrt{mn}:\sqrt{mn}+1+\tfrac{1}{4}(\sqrt{m}-\sqrt{n})^2=-2+2\sqrt{mn}:2+m+n,$$

and for $\gamma=\sqrt{mn}$ give

$$x=0, \quad Y:z=1-\sqrt{mn}:\sqrt{mn}+1+\tfrac{1}{4\sqrt{mn}}(\sqrt{m}-\sqrt{n})^2=2-2\sqrt{mn}:2\sqrt{mn}+\tfrac{(m-n)^2}{4\sqrt{mn}}.$$

The line drawn from the fixed point on the chord of contact to the centre has for its equation

$$x+y\sqrt{mn}+\gamma' z=0,$$

where, writing for x, y, z the coordinates of the centre, we have

$$-\gamma\{1+\sqrt{mn}\}+2\sqrt{mn}+\gamma'\{\sqrt{mn}+1+\tfrac{1}{4}(\sqrt{m}-\sqrt{n})^2\}=0,$$

that is

$$\gamma'=\frac{\gamma\{1+\sqrt{mn}\}-2\sqrt{mn}}{\sqrt{mn}+1+\frac{1}{4}(\sqrt{m}-\sqrt{n})^2}.$$

6. I remark that the four conics which pass through the two given points and touch the two given lines form a system of considerable interest. The two series of conics (corresponding to the two signs of $\sqrt{mn}$) are distinct, and each series contains an infinite number of conics. The parabolic conics in each series correspond to particular values of the parameter γ, and the locus of centres is in each case a conic.

The theorem that the locus of the centres is a conic is a particular case of a more general theorem relating to the locus of the poles of a given line in regard to a system of conics passing through fixed points and touching fixed lines.

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389. On a Locus derived from Two Conics.

[From the *Proceedings of the London Mathematical Society*, vol. IV (1871), pp. 202–205.]

I suppose in particular that the two conics are

$$x^2 + my^2 - 1 = 0, \quad mx^2 + y^2 - 1 = 0, \tag{1}$$

the equation of the quartic is

$$4(\lambda + 1)^2 m^2 (x^2 + my^2 - 1)(mx^2 + y^2 - 1) \\ - \{(m^2 + m)(x^2 + y^2) - m^2 - 1\}^2 = 0, \tag{2}$$

or putting $\lambda = \dfrac{k}{m^2 + 1}$, this is

$$(\lambda + 1)^2 \{x^2 + y^2 - a\}^2 \\ - (x^2 + my^2 - 1)(mx^2 + y^2 - 1) = 0. \tag{3}$$

To fix the ideas, suppose that m is positive and > 1, so that each of the conics is an ellipse, the major semi-axis being $= 1$, and the minor semi-axis being $= \dfrac{1}{\sqrt{m}}$.

For any real value of k the coefficient λ is positive, and it may accordingly be assumed that λ is positive.

We have $\dfrac{1}{m(m+1)} < a < 1$, or the radius of the circle is intermediate between the semi-axes of the ellipses, hence the points of contact on each ellipse are real points.

Writing for shortness $a = \dfrac{m^2 + 1}{\lambda(m + 1)}$, the equation is

$$(x^2 + my^2 - 1)(mx^2 + y^2 - 1) - \lambda(x^2 + y^2 - a)^2 = 0. \tag{4}$$

For the points on the axis of x, we have

$$(x^2 - 1)(mx^2 - 1) - \lambda(x^2 - a)^2 = 0, \tag{5}$$

that is

$$(m - \lambda)x^4 + \{-(1 + m) + 2\lambda a\}x^2 + (1 - \lambda a^2) = 0, \tag{6}$$

and thence

$$(m - \lambda)x^2 = \tfrac{1}{2}(1 + m) - \lambda a \pm \tfrac{1}{2}\sqrt{(m - 1)^2 + 4\lambda(1 - a)(1 - ma)}, \tag{7}$$

or, substituting for a its value, this is

$$(m-\lambda)x^2 = \tfrac{1}{2}(m+1) - \frac{\lambda(m^2+1)}{\lambda(m+1)} \pm \tfrac{1}{2}\sqrt{(m-1)^2 + 4\lambda(1-a)(1-ma)}. \tag{8}$$

Remarking that the values of λ: 0, $\dfrac{1}{(m+1)^2}$, $\dfrac{m}{(m+1)^2}$, m, $\dfrac{1}{4}(m+1)^2$ are in the order of increasing magnitude.

and considering successive values of λ; first the value $\lambda = \dfrac{1}{(m+1)^2}$, we have

$$(m-\lambda)x^2 = \tfrac{1}{2}(m+1) - \frac{1}{m+1} \pm \tfrac{1}{4}\sqrt{(m-1)(m-\tfrac{1}{m})}, \tag{9}$$

or observing that

$$(m+1)^2(m+\tfrac{1}{m}-2) = (m+1)^2-(m-1)^2 = -(m-1)(m^2-1) = (m-1)(m-\tfrac{1}{m}), \tag{10}$$

this is

$$(m-\lambda)x^2 = 0, \tag{11}$$

or

$$x^2 = \frac{(m-1)(m-\frac{1}{m})}{(m+1)^2}\cdot\frac{(m+1)^2}{m(m+1)} = 0, \tag{12}$$

or, what is the same thing,

$$x^2 - \frac{(m-1)(m^3+2m^2-1)}{m(m+1)^2} = 0, \quad y^2 - \frac{(m-1)(m^3+2m^2-1)}{m(m+1)^2} = 0. \tag{13}$$

The next critical value is $\lambda = m$.

The curve here is

$$(x^2+my^2-1)(mx^2+y^2-1) - m(x^2+y^2-a)^2 = 0, \tag{14}$$

that is

$$m(x^2+y^2)^2 + (1+m^2)x^2y^2 - (m+1)(x^2+y^2) + 1 \\ - m(x^2+y^2)^2 + 2ma(x^2+y^2) - ma^2 = 0, \tag{15}$$

that is

$$(m-1)^2x^2y^2 + (2ma-m-1)(x^2+y^2) + 1 - ma^2 = 0, \tag{16}$$

or, substituting for a its value,

$$2ma-m-1 = \frac{2m}{m+1} - (m+1) = \frac{-(m-1)^2}{m+1}, \tag{17}$$

$$1-ma^2 = 1-\frac{m(m^2+1)^2}{m^2(m+1)^2} = \frac{(m+1)^2(m^2+1)-m(m^2+1)^2}{m(m+1)^2} = \frac{(m^2+1)(-m^3+m^2+2m+1)}{m(m+1)^2}, \tag{18}$$

$$= -\frac{(m^2+1)(m^3-m^2-2m-1)}{m(m+1)^2}, \tag{19}$$

the equation is

$$(m-1)^2x^2y^2 - \frac{(m-1)^2}{m+1}(x^2+y^2) - \frac{(m^2+1)(m^3-m^2-2m-1)}{m(m+1)^2} = 0, \tag{20}$$

or, as this may also be written,

$$x^2y^2 - \frac{1}{m+1}(x^2+y^2) - \frac{(m^2+1)(m^3-m^2-2m-1)}{m(m-1)^2(m+1)^2} = 0. \tag{21}$$

which has a pair of imaginary asymptotes parallel to the axis of x, and a like pair parallel to the axis of y, or what is the same thing, the curve has two isolated points at infinity, one on each axis.

The next critical value is $\lambda = \frac{1}{4}(m+1)^2$; the curve here reduces itself to the four lines

$$(x+y)^2 = \frac{m+1}{m}, \quad (x-y)^2 = \frac{1}{m}, \tag{21}$$

and it is to be observed that when λ exceeds this value, or say $\lambda > \frac{1}{4}(m+1)^2$, the curve has no real point on either axis; but when $\lambda = \infty$, the curve reduces itself to $(x^2+y^2-a)^2 = 0$, i.e. to the circle $x^2+y^2 = a$ twice repeated, having in this special case real points on the two axes.

It is now easy to trace the curve for the different values of λ.

The curve lies in every case within the unshaded regions of the figure (except in the limiting cases after-mentioned); and it also touches the two ellipses and the four lines at the eight points k, at which points it also cuts the circle; but it does not cut or touch the four lines, the two ellipses, or the circle, except at the points k.

Considering λ as varying by successive steps from 0 to ∞:

$\lambda = 0$, the curve is the two ellipses.

$\lambda < \dfrac{1}{(m+1)^2}$, the curve consists of two ovals, an exterior sinuous oval lying in the four regions a and the four regions b; and an interior oval lying in the region c.

$\lambda = \dfrac{1}{(m+1)^2}$, there is still a sinuous oval as above, but the interior oval has dwindled to a conjugate point at the centre.

$\lambda = \dfrac{m}{(m+1)^2}$, there is still a sinuous oval as above, but the interior oval has dwindled to a conjugate point at the centre. For this critical value, the curve acquires the four lines as bitangents, touching each of them at two consecutive points k.

$\dfrac{m}{+1)^2} < \lambda < m$, there is no interior oval, but only a sinuous oval as above; which, as λ increases, approaches continually nearer to the four sides of the square.

$\lambda = m$, there is no change in the general form, but the curve has for this value of λ, two conjugate points, one on each axis at infinity.

$\lambda = \frac{1}{4}(m+1)^2$, the curve becomes the four lines.

$\lambda > \frac{1}{4}(m+1)^2$, the curve lies wholly in the four regions a and the four regions e, consisting thereof of four detached sinuous ovals. As λ deviates less from the value $\frac{1}{4}(m+1)^2$, each oval approaches more nearly to the infinite trilateral formed by the side and infinite line-portions which bound the regions d, e to which the oval belongs. And as λ departs from the limit $\frac{1}{4}(m+1)^2$, and approaches to ∞, each sinuous oval approaches more nearly to the circular arc which separates the two regions d, e, which contains the sinuous oval. Finally, $\lambda = \infty$, the curve is the circle twice repeated.

390. Theorem relating to the Four Conics which touch the same Two Lines and pass through the same Four Points.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. VIII (1867), pp. 162–167.]

390. [Date Nov. 13, 1867.]

I.

The sides of the triangle formed by the given points meet one of the given lines in three points, say P, Q, R; and on this same line we have four points of contact, say A_1, A_2, A_3, A_4; any two pairs, say A_1, A_2; A_3, A_4, form with a properly selected pair, say Q, R, out of the above-mentioned three points, an involution; and we have thus the three involutions

$$\{A_1, A_2;\ A_3, A_4;\ Q, R\},\quad \{A_1, A_2;\ A_3, A_4;\ R, P\},\quad \{A_1, A_2;\ A_3, A_4;\ P, Q\}. \tag{1}$$

To prove this, let $x = 0$, $y = 0$ be the equations of the given lines, and take for the equations of the sides of the triangle formed by the given points

$$bx + ay - ab = 0,\quad b'x + a'y - a'b' = 0,\quad b''x + a''y - a''b'' = 0; \tag{2}$$

the equation of any one of the four conics may be written

$$\frac{Lab}{bx + ay - ab} + \frac{L'a'b'}{b'x + a'y - a'b'} + \frac{L''a''b''}{b''x + a''y - a''b''} = Kxyz, \tag{3}$$

and if this touches the axis of x, say at the point $x = a$, then we must have

$$\frac{La}{x - a} + \frac{L'a'}{x - a'} + \frac{L''a''}{x - a''} = K(x - a). \tag{4}$$

or, assuming as we may do, $K = -(a' - a'')(a'' - a)(a - a')$, this gives

$$La = (a - a)^2(a' - a''), \quad L'a' = (a' - a)^2(a'' - a), \quad L''a'' = (a'' - a)^2(a - a'). \tag{5}$$

But in the same manner, if the conic touch the axis of y, say at the point $y = \beta$, we have

$$Lb = (b - \beta)^2(b' - b''), \quad L'b' = (b' - \beta)^2(b'' - b), \quad L''b'' = (b'' - \beta)^2(b - b'). \tag{6}$$

We have

$$\begin{aligned} b(a - \alpha)^2(a' - a'') : b'(a' - a)^2(a'' - a) : b''(a'' - a)^2(a - a') \\ = a(b - \beta)^2(b' - b'') : a'(b' - \beta)^2(b'' - b) : a''(b'' - \beta)^2(b - b'); \end{aligned} \tag{7}$$

and thence

$$\frac{(a - a)\sqrt{bb'}}{a - a} : \frac{(a' - a)\sqrt{b'b''}}{a' - a} : \frac{(a'' - a)\sqrt{b''b}}{a'' - a} = (b - \beta)(b' - b'') : (b' - \beta)(b'' - b) : (b'' - \beta)(b - b'). \tag{8}$$

Putting

$$P = ab(a' - a'')(b' - b''), \quad P' = a'b'(a'' - a)(b'' - b), \quad P'' = a''b''(a - a')(b - b'), \tag{9}$$

we have

$$(a - a)\sqrt{\tfrac{P}{a}} : (a' - a)\sqrt{\tfrac{P'}{a'}} : (a'' - a)\sqrt{\tfrac{P''}{a''}} = (b - \beta)(b' - b'') : (b' - \beta)(b'' - b) : (b'' - \beta)(b - b'), \tag{10}$$

and thence

$$\frac{(a - a)\sqrt{P}}{a} : \frac{(a' - a)\sqrt{P'}}{a'} : \frac{(a'' - a)\sqrt{P''}}{a''} = (b - \beta)(b' - b'') : (b' - \beta)(b'' - b) : (b'' - \beta)(b - b'). \tag{11}$$

which gives

$$(a - a)\sqrt{P} : (a' - a)\sqrt{P'} : (a'' - a)\sqrt{P''} = \frac{b - \beta}{\sqrt{aa'}}(b' - b'') : \frac{b' - \beta}{\sqrt{a'a''}}(b'' - b) : \frac{b'' - \beta}{\sqrt{a''a}}(b - b'). \tag{12}$$

and we have in like manner

$$(b - \beta)\sqrt{P} : (b' - \beta)\sqrt{P'} : (b'' - \beta)\sqrt{P''} = \frac{a - a}{\sqrt{bb'}}(a' - a'') : \frac{a' - a}{\sqrt{b'b''}}(a'' - a) : \frac{a'' - a}{\sqrt{b''b}}(a - a'); \tag{13}$$

but the first of these equations is alone required for the present purpose.

Putting for shortness

$$P = a^2\lambda, \quad P' = a'^2\lambda', \quad P'' = a''^2\lambda'', \tag{14}$$

the equation is

$$(a - a)\sqrt{(\lambda)} + (a' - a)\sqrt{(\lambda')} + (a'' - a)\sqrt{(\lambda'')} = 0, \tag{15}$$

and by attributing the signs + and − to the radicals, we have, corresponding to the four conics, the equations

$$(a-a_1)\sqrt{(\lambda)}+(a'-a_1)\sqrt{(\lambda')}+(a''-a_1)\sqrt{(\lambda'')}=0, \tag{14}$$

$$-(a-a_2)\sqrt{(\lambda)}+(a'-a_2)\sqrt{(\lambda')}+(a''-a_2)\sqrt{(\lambda'')}=0, \tag{15}$$

$$(a-a_3)\sqrt{(\lambda)}-(a'-a_3)\sqrt{(\lambda')}+(a''-a_3)\sqrt{(\lambda'')}=0, \tag{16}$$

$$(a-a_4)\sqrt{(\lambda)}+(a'-a_4)\sqrt{(\lambda')}-(a''-a_4)\sqrt{(\lambda'')}=0, \tag{17}$$

where a_1, a_2, a_3, a_4 are the values of a for the four conics respectively.

Eliminating a we obtain the system of three equations

$$(2a-a_1-a_2)\sqrt{(\lambda)}+(a_2-a_1)\sqrt{(\lambda')}+(a_2-a_1)\sqrt{(\lambda'')}=0, \tag{18}$$

$$(a_3-a_4)\sqrt{(\lambda)}+(2a'-a_3-a_4)\sqrt{(\lambda')}+(a_3-a_4)\sqrt{(\lambda'')}=0, \tag{19}$$

$$(a_1+a_2-a_3-a_4)\sqrt{(\lambda)}+(a_1+a_3-a_2-a_4)\sqrt{(\lambda')}+(a_1+a_4-a_2-a_3)\sqrt{(\lambda'')}=0, \tag{20}$$

and then eliminating the radicals we have

$$\begin{vmatrix} 2a-a_1-a_2 & a_2-a_1 & a_2-a_1 \\ a_3-a_4 & 2a'-a_3-a_4 & a_3-a_4 \\ a_1+a_2-a_3-a_4 & a_1+a_3-a_2-a_4 & a_1+a_4-a_2-a_3 \end{vmatrix}=0, \tag{21}$$

which is in fact

$$\begin{vmatrix} 1 & a+a' & aa' \\ 1 & a_1+a_4 & a_1a_4 \\ 1 & a_2+a_3 & a_2a_3 \end{vmatrix}=0, \tag{22}$$

as may be verified by actual expansion; the transformation of the determinant is a peculiar one.

The foregoing result was originally obtained as follows, viz. writing for a moment

$$a\sqrt{(\lambda)}+a'\sqrt{(\lambda')}+a''\sqrt{(\lambda'')}=\theta,\quad \sqrt{(\lambda)}+\sqrt{(\lambda')}+\sqrt{(\lambda'')}=\Phi, \tag{20}$$

the four equations are

$$\theta-a_1\Phi=0,\quad \theta-a_2\Phi=2(a-a_2)\sqrt{(\lambda)},\quad \theta-a_3\Phi=2(a'-a_3)\sqrt{(\lambda')},\quad \theta-a_4\Phi=2(a''-a_4)\sqrt{(\lambda'')}. \tag{21}$$

These give

$$(a_1-a_2)\theta=2(a-a_2)\sqrt{(\lambda)}, \tag{22}$$

$$(a_1-a_3)\theta=2(a'-a_3)\sqrt{(\lambda')}, \tag{23}$$

$$(a_1-a_4)\theta=2(a''-a_4)\sqrt{(\lambda'')}. \tag{24}$$

From the last equation we have

$$\begin{aligned}(a_1-a_4)\Phi&=2\{\theta-a_4\sqrt{(\lambda)}-a'\sqrt{(\lambda')}\}-2a_4\{\Phi-\sqrt{(\lambda)}-\sqrt{(\lambda')}\}\\&=2(a-a_4)\Phi-2(a-a_4)\sqrt{(\lambda)}-2(a'-a_4)\sqrt{(\lambda')};\end{aligned} \tag{25}$$

that is

$$(a_1-a_4)\Phi-2(a-a_4)\sqrt{(\lambda)}-2(a'-a_4)\sqrt{(\lambda')}=0; \tag{26}$$

or substituting for $\sqrt{(\lambda)}$, $\sqrt{(\lambda')}$ their values in terms of Φ, we find

$$\frac{(a-a_4)(a_1-a_2)}{a-a_2}+\frac{(a'-a_4)(a_1-a_3)}{a'-a_3}=a_1-a_4, \tag{27}$$

which may be written

$$\frac{(a-a_4)(a-a_1)}{a-a_2}+\frac{(a'-a_4)(a'-a_1)}{a'-a_3}=0; \tag{28}$$

that is

$$(a_2-a_1)\left(1+\frac{a_2-a_1}{a-a_2}\right)(a-a_4)+(a_3-a_1)\left(1+\frac{a_3-a_1}{a'-a_3}\right)(a'-a_4)=0; \tag{29}$$

or again

$$(a-a_2)(a-a_4)+(a'-a_3)(a'-a_4)+(a_2-a_1)(a-a_4)\frac{a-a_2}{a-a_2}+(a_3-a_1)(a'-a_4)\frac{a'-a_3}{a'-a_3}=0; \tag{30}$$

that is

$$(a-a_2)(a-a_4)+(a'-a_3)(a'-a_4)+(a_2-a_1)(a-a_4)+(a_3-a_1)(a'-a_4)=0; \tag{31}$$

or finally

$$(a_2-a_1)(a-a_4)(a'-a_3)+(a_3-a_4)(a-a_1)(a'-a_2)=0, \tag{32}$$

which is a known form of the relation

$$\begin{vmatrix}1 & a+a' & aa'\\ 1 & a_1+a_4 & a_1a_4\\ 1 & a_2+a_3 & a_2a_3\end{vmatrix}=0, \tag{33}$$

which gives the involution of the quantities a, a'; a_1, a_4; a_2, a_3.

We have in like manner

$$\begin{vmatrix}1 & a'+a'' & a'a''\\ 1 & a_1+a_3 & a_1a_3\\ 1 & a_2+a_4 & a_2a_4\end{vmatrix}=0,\quad \begin{vmatrix}1 & a''+a & a''a\\ 1 & a_1+a_2 & a_1a_2\\ 1 & a_3+a_4 & a_3a_4\end{vmatrix}=0, \tag{34}$$

which give the involutions of the systems a', a''; a_1, a_3; a_2, a_4 and a'', a; a_1, a_2; a_3, a_4 respectively.

It may be remarked that the equation of the conic passing through the three points and touching the axis of x in the point $x = a$ is

$$\frac{(a-a)^2(a'-a'')b}{bx+ay-ab} + \frac{(a'-a)^2(a''-a)b'}{b'x+a'y-a'b'} + \frac{(a''-a)^2(a-a')b''}{b''x+a''y-a''b''} = 0, \tag{34}$$

and when this meets the axis of y we have

$$\frac{(a-a)^2(a'-a'')}{y-b} + \frac{(a'-a)^2(a''-a)}{y-b'} + \frac{(a''-a)^2(a-a')}{y-b''} = 0. \tag{35}$$

Hence, if this touches the axis of y in the point $y = \beta$, the left-hand side must be

$$R\frac{(a-a)^2(a'-a'')+(a'-a)^2(a''-a)+(a''-a)^2(a-a')}{(y-b)(y-b')(y-b'')}(y-\beta)^2 \tag{36}$$

and equating the coefficients of $\frac{1}{y}$, we have

$$\begin{aligned} &-(a-a)^2(a'-a'')+(a'-a)^2(a''-a)+(a''-a)^2(a-a') \\ &\qquad = [-(a-a)^2(a'-a'')+(a'-a)^2(a''-a)+(a''-a)^2(a-a')](b+b'+b''-2\beta), \end{aligned} \tag{37}$$

or what is the same thing,

$$\begin{aligned} &\frac{(a-a)^2(a'-a'')}{a} + \frac{(a'-a)^2(a''-a)}{a'} + \frac{(a''-a)^2(a-a')}{a''} \\ &\qquad \times \{-(a-2\beta)(a'-a'')+(a'-2\beta)(a''-a)+(a''-2\beta)(a-a')\} = 0, \end{aligned} \tag{38}$$

which gives β in terms of a, that is β_1, β_2, β_3, β_4 in terms of a_1, a_2, a_3, a_4 respectively.

Cambridge, 30 November, 1863.

391. Solution of a Problem of Elimination.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. VIII (1867), pp. 183–185.]

391. [Date Nov. 13, 1867.]
It is required to eliminate x, y from the equations

$$\begin{vmatrix} x^4, & x^3y, & x^2y^2, & xy^3, & y^4 \\ a, & b, & c, & d, & e \\ a', & b', & c', & d', & e' \\ a'', & b'', & c'', & d'', & e'' \end{vmatrix} = 0. \tag{1}$$

This system may be written

$$x^4 = \Sigma\lambda a, \quad x^3y = \Sigma\lambda b, \quad x^2y^2 = \Sigma\lambda c, \quad xy^3 = \Sigma\lambda d, \quad y^4 = \Sigma\lambda e; \tag{2}$$

or putting

$$y = kx, \tag{3}$$

we have

$$\Sigma\lambda(a + kb) = 0, \quad \Sigma\lambda(b + kc) = 0, \quad \Sigma\lambda(c + kd) = 0, \quad \Sigma\lambda(d + ke) = 0; \tag{4}$$

or, what is the same thing,

$$\begin{aligned}
\lambda(a+kb)+\lambda'(a'+kb')+\lambda''(a''+kb'')&=0, &(5)\\
\lambda(b+kc)+\lambda'(b'+kc')+\lambda''(b''+kc'')&=0, &(6)\\
\lambda(c+kd)+\lambda'(c'+kd')+\lambda''(c''+kd'')&=0, &(7)\\
\lambda(d+ke)+\lambda'(d'+ke')+\lambda''(d''+ke'')&=0; &(8)
\end{aligned}$$

and representing the columns

$$\begin{array}{cc|cc|cc}
a & b, & a' & b', & a'' & b''\\
b & c, & b' & c', & b'' & c''\\
c & d, & c' & d', & c'' & d''\\
d & e, & d' & e', & d'' & e''
\end{array} \tag{9}$$

by 1, 2, 3, 4, 5, 6; each equation is of the type

$$\lambda(1+k\cdot 2)+\lambda'(3+k\cdot 4)+\lambda''(5+k\cdot 6)=0. \tag{10}$$

Multiplying the several equations by the minors of 135, each with its proper sign, and adding, the terms independent of k disappear, the equation divides by k, and we find

$$\lambda\cdot 2135+\lambda'\cdot 4135+\lambda''\cdot 6135=0; \tag{11}$$

operating in a similar manner with the minors of 246, the terms in k disappear, and we find

$$\lambda\cdot 1246+\lambda'\cdot 3246+\lambda''\cdot 5246=0; \tag{12}$$

again, operating with the minors of $(146+236+245+k\cdot 246)$, we find

$$\begin{aligned}
&\lambda\{1236+1245+k(2146+1246)\}\\
&\qquad+\lambda'\{3146+3245+k(4236+3246)\}\\
&\qquad\qquad+\lambda''\{5146+5236+k(6245+5246)\}=0,
\end{aligned} \tag{13}$$

where the terms in k disappear, and this is

$$\lambda(1236+1245)+\lambda'(3146+3245)+\lambda''(5146+5236)=0. \tag{14}$$

We have thus three linear equations, which written in a slightly different form are

$$\begin{aligned}
\lambda\cdot 1235+\lambda'\cdot 3451+\lambda''\cdot 5613&=0, &(15)\\
\lambda(1236+1245)+\lambda'(3452+3461)+\lambda''(5614+5623)&=0, &(16)\\
\lambda\cdot 1246+\lambda'\cdot 3462+\lambda''\cdot 5624&=0, &(17)
\end{aligned}$$

and thence eliminating λ, λ', λ'', we have

$$\begin{vmatrix}
1235, & 3451, & 5613\\
1236+1245, & 3452+3461, & 5614+5623\\
1246, & 3462, & 5624
\end{vmatrix}=0, \tag{18}$$

which is the required result.

It may be remarked that the second and third column are obtained from the first by operating on it with Δ, $\frac{1}{2}\Delta^2$, if $\Delta=2\delta_1+4\delta_2+6\delta_3$.

Or say the result is

$$(1,\Delta,\tfrac{1}{2}\Delta^2)\begin{vmatrix}1235\\3451\\5613\end{vmatrix}=0. \tag{19}$$

In like manner for the system

$$\begin{array}{cccccc}
x^5, & x^4y, & x^3y^2, & x^2y^3, & xy^4, & y^5 \\
a, & b, & c, & d, & e, & f \\
a', & b', & c', & d', & e', & f' \\
a'', & b'', & c'', & d'', & e'', & f'' \\
a''', & b''', & c''', & d''', & e''', & f'''
\end{array} \tag{18}$$

if the columns are

$$\begin{array}{cc|cc|cc|cc}
a & b, & a' & b', & a'' & b'', & a''' & b''' \\
b & c, & b' & c', & b'' & c'', & b''' & c''' \\
c & d, & c' & d', & c'' & d'', & c''' & d''' \\
d & e, & d' & e', & d'' & e'', & d''' & e''' \\
e & f, & e' & f', & e'' & f'', & e''' & f'''
\end{array} \tag{19}$$

$= 1, 2,\ 3, 4,\ 5, 6,\ 7, 8$; then the result is

$$(1, \Delta, \tfrac{1}{2}\Delta^2, \tfrac{1}{6}\Delta^3) \begin{vmatrix} 12357 \\ 34571 \\ 56713 \\ 78135 \end{vmatrix} = 0, \tag{20}$$

where

$$\Delta = 2\delta_1 + 4\delta_3 + 6\delta_5 + 8\delta_7. \tag{21}$$

392. On the Conics which pass through Two Given Points and touch Two Given Lines.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. VIII (1867), pp. 211–219.]

392.

Let $x=0$, $y=0$ be the equations of the given lines; $z=0$ the equation of the line joining the given points.

We may, to fix the ideas, imagine the implicit constants so determined that $x+y+z=0$ shall be the equation of the line infinity.

Take $x-my=0$, $x-ny=0$ as the equations of the lines which by their intersection with $z=0$ determine the given points.

The equation of the conic is

$$(x-my)(x-ny)+2\{x+y\sqrt{(mn)}\}\sqrt{(yz)}=x+y\sqrt{(mn)}+\gamma z, \tag{1}$$

or, what is the same thing,

$$(x-my)(x-ny)+2\{x+y\sqrt{(mn)}\}\gamma z+\gamma^2z^2=0, \tag{2}$$

so that there are two distinct series of conics according as $\sqrt{(mn)}$ is taken with the positive or the negative sign.

The equation of the chord of contact is

$$x+y\sqrt{(mn)}+\gamma z=0, \tag{3}$$

which meets $z=0$ in the point $[x+y\sqrt{(mn)}=0,\ z=0]$, that is in one of the centres of the involution formed by the lines $(x=0,\ y=0)$, $(x-my=0,\ x-ny=0)$.

It is to be observed that the conic is only real when mn is positive, that is (the lines and points being each real) the two points must be situate in the same region or in opposite regions of the four regions formed by the two lines: there are however other real cases; e.g. if the lines $x=0$, $y=0$ are real, but the quantities m, n are conjugate imaginaries; included in this we have the circles which touch two real lines.

To fix the ideas I take m and n each positive and $mn > 1$; also I attend first to the series where $\sqrt{(mn)}$ is taken positively.

At the points where the conic meets infinity, we have

$$\{\sqrt{(m)} + \sqrt{(n)}\}\sqrt{(xy)} = x + y\sqrt{(mn)} - \gamma(x + y), \tag{4}$$

which gives two coincident points, that is the conic is a parabola, if

$$(1 - \gamma)\{\sqrt{(mn)} - \gamma\} = \tfrac{1}{4}\{\sqrt{(m)} + \sqrt{(n)}\}^2, \tag{5}$$

that is

$$\gamma = \tfrac{1}{2}\{1 + \sqrt{(mn)} \pm \sqrt{(1 + m)(1 + n)}\}, \tag{6}$$

where it is to be noticed that

$$\gamma = \tfrac{1}{2}[1 + \sqrt{(mn)} + \sqrt{\{(1 + m)(1 + n)\}}] \tag{7}$$

is a positive quantity greater than $\sqrt{(mn)}$, say $\gamma = p$;

$$\gamma = \tfrac{1}{2}[1 + \sqrt{(mn)} - \sqrt{\{(1 + m)(1 + n)\}}] \tag{8}$$

is a negative quantity, say $\gamma = -q$, q being positive.

The order of the lines is as shown in fig. 1, see plate facing p. 52.

$\gamma = -\infty$ to $\gamma = -q$, curve is ellipse;
$\gamma = -q$, parabola P_1;
$\gamma = -q$ to $\gamma = p$, curve is hyperbola;
$\gamma = p$, parabola P_2;
$\gamma = p$ to $\gamma = \infty$, ellipse.

Resuming the equation

$$(x-my)(x-ny)+2\{x+y\sqrt{(mn)}\}yz+\gamma^2z^2=0, \tag{9}$$

the coefficients are

$$(a,b,c,f,g,h)=(1,\ mn,\ \gamma^2,\ \gamma\sqrt{(mn)},\ \gamma,\ -\tfrac{1}{2}(m+n)), \tag{10}$$

and thence the inverse coefficients are

$$(A,B,C,F,G,H)=(0,\ 0,\ -\tfrac{1}{4}(m-n)^2,\ -\tfrac{1}{4}\gamma\{\sqrt{(m)}+\sqrt{(n)}\}^2,\\ -\tfrac{1}{4}\gamma\sqrt{(mn)}\{\sqrt{(m)}+\sqrt{(n)}\}^2,\ \tfrac{1}{4}\gamma^2\{\sqrt{(m)}+\sqrt{(n)}\}^2). \tag{11}$$

$K=-\frac{1}{4}\gamma^2\{\sqrt{(m)}+\sqrt{(n)}\}^2$,
or, omitting a factor, the inverse coefficients are

$$(A,B,C,F,G,H)=(0,\ 0,\ -\gamma\{\sqrt{(m)}-\sqrt{(n)}\}^2,\ 1,\ \sqrt{(mn)},\ -\gamma). \tag{12}$$

Considering the line $\lambda x + \mu y + \nu z = 0$, the coordinates of the pole of this line are

$$X : Y : Z = -\gamma\mu + \sqrt{(mn)}\nu : -\gamma\lambda + \nu : \sqrt{(mn)}\lambda + \mu + \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2\nu, \tag{12}$$

or (what is the same thing) introducing the arbitrary coefficient k, we have

$$kx + y\mu - \sqrt{(mn)} = 0, \tag{13}$$

$$ky + \gamma\lambda - \mu = 0, \tag{14}$$

$$kz - \lambda\sqrt{(mn)} - \mu - \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2\nu = 0; \tag{15}$$

the first two equations give

$$k : \gamma : -1 = \nu\{\mu - \lambda\sqrt{(mn)}\} : \nu\{\gamma\sqrt{(mn)} - \lambda\} : \lambda x - \mu y, \tag{16}$$

that is

$$k = \frac{-\nu\{\mu - \lambda\sqrt{(mn)}\}}{\lambda x - \mu y}, \quad \gamma = \frac{-\nu\{\gamma\sqrt{(mn)} - x\}}{\lambda x - \mu y}, \tag{17}$$

or, substituting this value of γ in the third equation,

$$\frac{\nu\{\mu - \lambda\sqrt{(mn)}\}(x - y\sqrt{(mn)})}{\lambda x - \mu y} + \frac{\{\mu + \lambda\sqrt{(mn)}\}(\lambda x - \mu y)}{\lambda x - \mu y} + \frac{\frac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2\nu(x - y\sqrt{(mn)})}{\lambda x - \mu y} = 0, \tag{18}$$

that is

$$(\lambda x - \mu y)^2 \cdot \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 + \{x - y\sqrt{(mn)}\}(\lambda x - \mu y)\{\mu + \lambda\sqrt{(mn)}\} + z\{x - y\sqrt{(mn)}\}\nu\{\mu - \lambda\sqrt{(mn)}\} = 0, \tag{19}$$

which is the equation of the curve, the locus of the pole of the line $\lambda x + \mu y + \nu z = 0$ in regard to the conic $(x - my)(x - ny) + 2\{x + y\sqrt{(mn)}\}yz + \gamma^2 z^2 = 0$.

In particular, if $\lambda = \mu = \nu = 1$, then for the coordinates of the centre of the conic, we have

$$X : Y : Z = -\gamma + \sqrt{(mn)} : -\gamma + 1 : \sqrt{(mn)} + 1 + \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2; \tag{18}$$

and for the locus of the centre,

$$(X - Y)^2 \cdot \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 + (x - y)\{x - y\sqrt{(mn)}\}\{1 + \sqrt{(mn)}\} \\ + z\{x - y\sqrt{(mn)}\}\{1 - \sqrt{(mn)}\} = 0, \tag{19}$$

so that the locus is a conic, and it is obvious that this conic is a hyperbola.

Putting for greater simplicity

$$x - y = X, \quad x - y\sqrt{(mn)} = Y, \quad z = Z, \tag{20}$$

the equation of the curve of centres is

$$Z^2 \cdot \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 + XY\{1 + \sqrt{(mn)}\} + YZ\{1 - \sqrt{(mn)}\} = 0, \tag{21}$$

or, writing this under the form

$$Y[Z\{1 + \sqrt{(mn)}\} + Z\{1 - \sqrt{(mn)}\}] + \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 Z^2 = 0, \tag{22}$$

the equation is

$$XY + ZQ = 0, \tag{22}$$

where

$$X = x - y, \tag{23}$$

$$Y = x - y\sqrt{(mn)}, \tag{24}$$

$$Q = \frac{2}{\{\sqrt{(m)} - \sqrt{(n)}\}^2}[Z\{1 + \sqrt{(mn)}\}(x - y) + \{1 - \sqrt{(mn)}\}z]; \tag{25}$$

these values give

$$x - y = X, \tag{26}$$

$$x - y\sqrt{(mn)} = Y, \tag{27}$$

$$\{1 - \sqrt{(mn)}\}z = \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 Q + 2\{1 + \sqrt{(mn)}\}X, \tag{28}$$

or, what is the same thing,

$$\{1 - \sqrt{(mn)}\}x = -\sqrt{(mn)}X + Y, \tag{29}$$

$$\{1 - \sqrt{(mn)}\}y = -X + Y, \tag{30}$$

$$\{1 - \sqrt{(mn)}\}z = 2\{1 + \sqrt{(mn)}\}X + \tfrac{1}{2}\{\sqrt{(m)} - \sqrt{(n)}\}^2 Q. \tag{31}$$

whence also

$$\{1 - \sqrt{(mn)}\}(x + y + z) = \{1 + \sqrt{(mn)}\}Z + 2Y + \{\sqrt{(m)} - \sqrt{(n)}\}^2 Q, \tag{32}$$

or the equation of the line infinity is

$$\{1 + \sqrt{(mn)}\}X + 2Y + \{\sqrt{(m)} - \sqrt{(n)}\}^2 Q = 0, \tag{33}$$

a formula which may be applied to finding the asymptotes and thence the centre of the conic $YQ + X^2 = 0$.

In fact we have identically

$$\{2kx + 2ky - (2k + 1)z\}^2 - (1 + 4k)(2kx - z)^2 = 4k^2(x + y + z)^2 - 4k(1 + 4k)(kx^2 + yz), \tag{34}$$

that is

$$-4k(1 + 4k)(kx^2 + yz) = \{2kx + 2ky - (2k + 1)z\}^2 - (1 + 4k)(2kx - z)^2 - 4k^2(x + y + z)^2, \tag{35}$$

which, if $x + y + z = 0$ is the equation of the line infinity, puts in evidence the asymptotes of the conic $kx^2 + yz = 0$.

Hence writing αx, βy, γz in the place of x, y, z respectively, and $k = k'$, that is, $k = \dfrac{\beta\gamma}{\alpha^2}k'$, we have

$$\begin{multline} - 4\beta\gamma k'(\alpha^2 + 4\beta\gamma k')(k'x^2 + yz) = \{2\alpha\gamma k'x + 2\beta^2 k'y - (2\beta\gamma k' + \alpha^2)z\}^2 \\ - (\alpha^2 + 4\beta\gamma k')(2\beta\gamma k'x - \alpha z)^2 - 4\beta^2\gamma^2 k'^2(\alpha x + \beta y + \gamma z)^2. \quad (31) \end{multline}$$

Now writing X, Y, Q in the place of x, y, z; $k' = 1$, and $\alpha = \{1 + \sqrt{(mn)}\}$, $\beta = 2$, $\gamma = \{\sqrt{(m)} - \sqrt{(n)}\}^2$, we have

$$\begin{aligned} - 16[\{1 + \sqrt{(mn)}\}^2 + 8\{\sqrt{(m)} - \sqrt{(n)}\}^2](YQ + X^2) \\ = [4\{1 + \sqrt{(mn)}\}X + 8Y - (4\{\sqrt{(m)} - \sqrt{(n)}\}^2 + \{1 + \sqrt{(mn)}\}^2)Q]^2 \\ - [\{1 + \sqrt{(mn)}\}^2 + 8\{\sqrt{(m)} - \sqrt{(n)}\}^2][4X - \{1 + \sqrt{(mn)}\}Q]^2 \\ - 16[\{1 + \sqrt{(mn)}\}X + 2Y + \{\sqrt{(m)} - \sqrt{(n)}\}^2 Q]^2, \quad (32) \end{aligned}$$

and the asymptotes are

$$\begin{aligned} 4\{1 + \sqrt{(mn)}\}X + 8Y - [4\{\sqrt{(m)} - \sqrt{(n)}\}^2 + \{1 + \sqrt{(mn)}\}^2]Q \\ = \pm\sqrt{\{1 + \sqrt{(mn)}\}^2 + 8\{\sqrt{(m)} - \sqrt{(n)}\}^2}\ [4X - \{1 + \sqrt{(mn)}\}Q]. \quad (33) \end{aligned}$$

At the centre

$$4(1 + \sqrt{(mn)})X + 8Y - [4\{\sqrt{(m)} - \sqrt{(n)}\}^2 + \{1 + \sqrt{(mn)}\}^2]Q = 0, \tag{34}$$

$$4X - \{1 + \sqrt{(mn)}\}Q = 0, \tag{35}$$

but the first equation is

$$\{1 + \sqrt{(mn)}\}[4X - Q\{1 + \sqrt{(mn)}\}] + 8Y - 4\{\sqrt{(m)} - \sqrt{(n)}\}^2 Q = 0, \tag{36}$$

so that we have

$$4X = \{1 + \sqrt{(mn)}\}Q, \tag{37}$$

$$2Y = \{\sqrt{(m)} - \sqrt{(n)}\}^2 Q, \tag{38}$$

the first of these is

$$2\{\sqrt{(m)} - \sqrt{(n)}\}^2(x - y) - \{1 + \sqrt{(mn)}\}^2(x - y) - (1 - mn)z = 0, \tag{39}$$

and the two together give

$$2Z\{\sqrt{(m)} - \sqrt{(n)}\}^2 - \{1 + \sqrt{(mn)}\}Y = 0, \tag{40}$$

so that we have

$$2\{\sqrt{(m)} - \sqrt{(n)}\}^2(x - y) - \{1 + \sqrt{(mn)}\}(x - y\sqrt{(mn)}) - (1 - mn)z = 0. \tag{41}$$

393. On the Conics which touch Three Given Lines, &c.

[From the *Proceedings of the London Mathematical Society*, vol. IV (1871), pp. 114–115.]

393.

The problem of determining the conics which touch three given lines and satisfy some other condition is considered. When the three lines are given, the conic is determined by three tangential conditions. If additional conditions are given, such as passing through a point or touching a fourth line, the conic is completely determined.

The analytical treatment of this problem leads to interesting equations. Let the three given lines be

$$\lambda x + \mu y + \nu z = 0, \quad \lambda' x + \mu' y + \nu' z = 0, \quad \lambda'' x + \mu'' y + \nu'' z = 0. \tag{1}$$

The condition that a conic touches these three lines can be expressed by means of the tangential equation of the conic.

[Plate I facing page 52: Figure illustrating the conics touching three given lines.]

394. On a Locus in relation to the Triangle.

[From the *Proceedings of the London Mathematical Society*, vol. IV (1871), pp. 115–121.]

394.

The problem of finding the locus of a point related in a given manner to a triangle leads to many interesting curves.

Let the triangle be ABC, and let the point P be such that its distances from the sides have a given relation. The locus of P is in general a curve of higher order.

Consider the case where the point is the intersection of lines drawn from the vertices with given directions. The locus can be represented by the equation

$$\begin{vmatrix} A & H & G \\ H & B & F \\ G & F & C \end{vmatrix} = 0. \tag{1}$$

Consider the case where the point is such that the perpendiculars from it on the sides of the triangle have their feet in a line.

Let the sides of the triangle be $x = 0$, $y = 0$, $z = 0$, and let the Absolute (the conic in regard to which the perpendiculars are drawn) be

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0. \tag{2}$$

The condition that the feet of the perpendiculars lie in a line is that the point (x, y, z) satisfies a certain cubic equation.

that is, the locus is a cubic curve passing through the vertices of the triangle.

The condition

$$\frac{abc}{af^2} \cdot \frac{1}{bg^2} \cdot \frac{1}{ch^2} = \frac{1}{fgh} \tag{3}$$

is the condition in order that the function

$$\frac{\xi^2}{a} + \frac{\eta^2}{b} + \frac{\zeta^2}{c} \tag{4}$$

may break up into linear factors; the function in question is

$$\frac{bc}{f^2} + \frac{ca}{g^2} + \frac{ab}{h^2}, \tag{5}$$

which is

$$\left(\frac{bc}{f}, \frac{ca}{g}, \frac{ab}{h} \mid x, y, z\right)^2 = (a, b, c, f, g, h \mid x, y, z)^2 + 2\left(\frac{f}{a}yz + \frac{g}{b}zx + \frac{h}{c}xy\right), \tag{6}$$

so that the condition is, that the conic

$$(a, b, c, f, g, h \mid x, y, z)^2 + 2\left(\frac{f}{a}yz + \frac{g}{b}zx + \frac{h}{c}xy\right) = 0, \tag{7}$$

(which is a certain conic passing through the intersections of the Absolute $(a, b, c, f, g, h \mid x, y, z)^2 = 0$, and of the locus conic $\frac{f}{a}yz + \frac{g}{b}zx + \frac{h}{c}xy = 0$) shall be a pair of lines.

Writing the equation of the conic in question under the form

$$\left(\frac{bc}{f}, \frac{ca}{g}, \frac{ab}{h} \mid x, y, z\right)^2, \tag{8}$$

the inverse coefficients A', B', C', F', G', H' of this conic, are

$$\frac{A'bc}{f^2} = \frac{B'ca}{g^2} = \frac{C'ab}{h^2} = \frac{abc}{fgh}, \tag{9}$$

so that we have $F' : G' : H' = F : G : H$.

Hence, if in regard to this new conic we form the reciprocal of the triangle ($x = 0,\ y = 0,\ z = 0$), and join the corresponding angles of the two triangles, the joining lines meet in a point which is the same point as is obtained by the like process from the triangle and its reciprocal in regard to the Absolute.

But I do not further pursue this part of the theory.

It is to be noticed that the conic

$$\frac{A}{f}yz + \frac{B}{g}zx + \frac{C}{h}xy = 0, \tag{10}$$

contains the angles of the reciprocal triangle, and is thus in fact the conic in which are situate the angles of the two triangles.

For the coordinates of one of the angles of the reciprocal triangle are (A, H, G); we should therefore have

$$\frac{A}{f}HG + \frac{B}{g}GA + \frac{C}{h}AB = 0, \tag{11}$$

which is $\frac{1}{f}(GH \cdot gh + BG \cdot hf + GH \cdot fg) = 0$, or attending only to the second factor and writing $GH = Kf + AF$, the condition is

$$Kfgh + AFgh + BGhf + CHfg = 0, \tag{12}$$

or substituting for K, A, B, C, F, G, H their values and reducing, this is satisfied: hence the three angles of the reciprocal triangle lie on the conic in question.

Partially recapitulating the foregoing results, we see in the case where the Absolute is not a point-pair, that the locus of a point such that the perpendiculars from it on the sides of the triangle have their feet in a line, is in general a cubic curve passing through the angles of the triangle: if, however, the condition

$$\frac{abc}{af^2}\cdot\frac{1}{bg^2}\cdot\frac{1}{ch^2}=\frac{1}{fgh} \tag{13}$$

be satisfied, that is, if the triangle be such that the angles thereof and of the reciprocal triangle lie in a conic (or, what is the same thing, if the sides touch a conic) then the cubic locus breaks up into the line

$$\frac{f}{a}x+\frac{g}{b}y+\frac{h}{c}z=0, \tag{14}$$

which is the line through the points of intersection of the corresponding sides of the two triangles, and into the conic

$$\frac{A}{f}yz+\frac{B}{g}zx+\frac{C}{h}xy=0, \tag{15}$$

which is the conic through the angles of the two triangles.

The question arises, given a conic (the Absolute) to construct a triangle such that its angles, and the angles of the reciprocal triangle in regard to the given conic, lie in a conic.

I suppose that two of the angles of the triangle are given, and I enquire into the locus of the remaining angle.

To fix the ideas, let A, B, C be the angles of the triangle, A', B', C' those of the reciprocal triangle; and let the angles A and B be given. We have to find the locus of the point C: I observe however, that the lines AA', BB', CC' meet in a point O, and I conduct the investigation in such manner as to obtain simultaneously the loci of the two points C and O.

The lines $C'B'$, $C'A'$ are the polars of A, B respectively, let their equations be $x=0$, and $y=0$, and let the equation of the line AB be $z=0$; this being so, the equation of the given conic will be of the form

$$(a,b,c,0,0,h \mid x,y,z)^2=0. \tag{16}$$

I take (α,β,γ) for the coordinates of C and (x,y,z) for those of O; the coordinates of either of these points being of course deducible from those of the other.

Observing that the inverse coefficients are $(bc,\ ca,\ ab-h^2,\ 0,\ 0,\ -ch)$, we find

Coordinates of A are $(b,-h,0)$,

B „ $(-h,a,0)$.

The points A' and B' are then given as the intersections of AO with $C'A'$ $(y=0)$ and of BO with $C'B'$ $(x=0)$; we find

Moreover,

Coordinates of A' are $(h\alpha+b\beta,\ a\alpha+h\beta,\ h\gamma)$,

B' „ $(h\alpha+b\beta,\ a\alpha+h\beta,\ h\gamma)$,

Coordinates of C are $(0,0,1)$,

C' „ (α,β,γ).

The six points A, B, C, A', B', C' are to lie in a conic; the equations of the lines CA, CB, AB are $hX+bY=0$, $aX+hY=0$, $Z=0$, and hence the equation of a conic passing through the points C, A, B is

$$\frac{hX+bY}{Z}\cdot\frac{aX+hY}{Z}=0. \tag{17}$$

Hence, making the conic pass through the remaining points A', B', C, we find

$$\frac{a\alpha(h\alpha+b\beta)}{L}+\frac{h(h\alpha+b\beta)}{M}+\frac{h\gamma}{N}=0, \quad \frac{h(a\alpha+h\beta)}{L}+\frac{b(a\alpha+h\beta)}{M}+\frac{h\gamma}{N}=0, \tag{18}$$

and eliminating the L, M, N, we find

$$\begin{vmatrix} 1 & a & 1 \\ 1 & h & 1 \\ 1 & h & 1 \\ h\alpha+b\beta & a\alpha+h\beta & h\gamma \\ a\alpha+h\beta & h\alpha+b\beta & z \end{vmatrix}=0, \tag{19}$$

or developing and reducing, this is

$$\frac{ab-h^2}{\gamma}Y^2=\frac{1}{a\alpha+h\beta}+\frac{1}{h\alpha+b\beta}-\frac{h\alpha\beta}{a\alpha+h\beta}-\frac{ab}{h\alpha+b\beta}, \tag{20}$$

that is

$$\frac{ab-h^2}{\gamma}Y^2=(hb,-ab,ha \mid a\alpha+h\beta,h\alpha+b\beta)^2=0. \tag{21}$$

We have still to find the relation between (α,β,γ) and (x,y,z); this is obtained by the consideration that the line $A'B'$, through the two points A', B' the coordinates of which are known in terms of (α,β,γ), is the polar of the point C, the coordinates of which are (x,y,z).

The equation of $A'B'$ is thus obtained in the two forms

$$(a\alpha+h\beta)Z+(h\alpha+b\beta)Y-\frac{(a\alpha+h\beta)(h\alpha+b\beta)}{ch\gamma}X=0, \tag{22}$$

and

$$(ax+hy)X+(hx+by)Y+cz=0, \tag{23}$$

and comparing these, we have

$$X:Y:Z=\alpha:\beta:\gamma=x:y:\frac{-(a\alpha+h\beta)(h\alpha+b\beta)}{ch\gamma}, \tag{24}$$

or what is the same thing

$$\alpha:\beta:\gamma=X:Y:\frac{-(a\alpha+h\beta)(h\alpha+b\beta)}{ch\gamma}=x:y:\frac{-(ax+hy)(hx+by)}{chz} \tag{25}$$

(where it is to be observed that the equation $\alpha:\beta=x:y$ is the verification of the theorem that the lines AA', BB', CC' meet in a point O).

We may now from the above found relation eliminate either the (α, β, γ) or the (x, y, z); first eliminating the (α, β, γ), we find

$$\frac{ab-h^2}{\gamma} Y^2 = \frac{h}{a} \frac{1}{ax+hy} + \frac{1}{b} \frac{1}{hx+by} - \frac{hab}{(ax+hy)(hx+by)}, \tag{24}$$

where

$$\frac{Y}{Z} = \frac{(a\alpha+h\beta)(h\alpha+b\beta)}{ch\gamma^2} = \frac{(ax+hy)(hx+by)}{chz^2}, \tag{25}$$

or, completing the elimination,

$$\frac{ab-h^2}{\gamma} \frac{(ax+hy)^2(hx+hy)^2}{chz^2} = (hb, -ab, ha \mid ax+hy, hx+by)^2 = 0, \tag{26}$$

which is a quartic curve having a node at each of the points

$$(z=0,\ ax+hy=0), \quad (z=0,\ hx+by=0), \quad (ax+hy=0,\ hx+by=0), \tag{27}$$

that is, at each of the points B, A, C.

The right-hand side of the foregoing equation is

$$= -(ab-h^2)(ha, ah, hh \mid x, y)^2 = -(ab-h^2)h(ax^2+by^2+\tfrac{2h}{ab}xy), \tag{28}$$

so that the equation may also be written

$$(ax+hy)^2(hx+by)^2 + ch^2z^2\left(ax^2+by^2+\tfrac{2h}{ab}xy\right) = 0. \tag{29}$$

Secondly, to eliminate the (x, y, z), we have

$$\frac{ab-h^2}{\gamma}Y^2 = \frac{1}{h}\frac{1}{a\alpha+h\beta} + \frac{1}{a}\frac{1}{h\alpha+b\beta}, \tag{30}$$

where

$$\frac{Y}{Z} = \frac{(a\alpha+h\beta)(h\alpha+b\beta)}{ch\gamma^2}, \tag{31}$$

or, completing the elimination,

$$(ab-h^2)ch\gamma^2 = (hb, -ab, ha \mid a\alpha+h\beta, h\alpha+b\beta)^2, \tag{32}$$

that is

$$= -(ab-h^2)h(a\alpha^2+b\beta^2+\tfrac{2h}{ab}\alpha\beta), \tag{33}$$

$$(a, b, c, 0, 0, h \mid \alpha, \beta, \gamma)^2 = 0. \tag{34}$$

Writing (x, y, z) in place of (α, β, γ), the locus of the point is the conic

$$(a, b, c, 0, 0, h \mid x, y, z)^2 = 0, \tag{35}$$

which is a conic intersecting the Absolute $(a, b, c, 0, 0, h \mid x, y, z)^2 = 0$, at its intersections with the lines $x = 0$, $y = 0$, that is the lines CB and CA'.

In regard to this new conic, the coordinates of the pole of $C'B'$ $(x = 0)$ are at once found to be $(-h, a, 0)$, that is, the pole of $C'B'$ is B; and similarly the coordinates of the pole of $C'A'$ $(y = 0)$ are $(b, -h, 0)$, that is, the pole of $C'A'$ is A.

We may consequently construct the conic the locus of O, viz. given the Absolute and the points A and B, we have $C'A'$ the polar of B, meeting the Absolute in two points (a_1, a_2), and $C'B'$ the polar of A meeting the Absolute in the points (b_1, b_2); the lines $C'A'$ and $C'B'$ meet in C'.

This being so, the required conic passes through the points a_1, a_2, b_1, b_2, the tangents at these points being Aa_1, Aa_2, Bb_1, Bb_2 respectively; eight conditions, five of which would be sufficient to determine the conic.

It is to be remarked that the lines $C'B'$, $C'A'$ (which in regard to the Absolute are the polars of A, B respectively) are in regard to the required conic the polars of B, A respectively.

The conic the locus of O being known, the point may be taken at any point of this conic, and then we have A' as the intersection of $C'A'$ with AO, B' as the intersection of $C'B'$ with BO.

to the Absolute, the point so obtained being a point on the line CO.

To each position of O on the conic locus, there corresponds of course a position of C; the locus of C is, as has been shown, a quartic curve having a node at each of the points C, A, B.

The foregoing conclusions apply of course to spherical figures; we see therefore that on the sphere the locus of a point such that the perpendiculars let fall on the sides of a given spherical triangle have their feet in a line (great circle), is a spherical cubic.

If, however, the spherical triangle is such that the angles thereof and the poles of the sides (or, what is the same thing, the angles of the polar triangle) lie on a spherical conic; then the cubic locus breaks up into a line (great circle), which is in fact the circle having for its pole the point of intersection of the perpendiculars from the angles of the triangle on the opposite sides respectively, and into the before-mentioned spherical conic.

395. Investigations in connexion with Casey's Equation.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XII (1873), pp. .]

395.

The equation of Casey relates to the condition for four circles to touch a fifth circle. If ρ_1, ρ_2, ρ_3, ρ_4 are the radii of four circles touching a fifth circle of radius ρ, and t_{12}, t_{13}, etc. are the distances between their centres, then Casey's equation is

$$\begin{vmatrix} 0 & t_{12}^2 & t_{13}^2 & t_{14}^2 & 1 \\ t_{21}^2 & 0 & t_{23}^2 & t_{24}^2 & 1 \\ t_{31}^2 & t_{32}^2 & 0 & t_{34}^2 & 1 \\ t_{41}^2 & t_{42}^2 & t_{43}^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{vmatrix} = 0. \qquad (1)$$

If however the circles all touch the fifth circle externally, the equation takes a simpler form. Writing $t_{ij}^2 = (\rho_i + \rho_j)^2$ for external contact, or $t_{ij}^2 = (\rho_i - \rho_j)^2$ for internal contact, the determinant condition simplifies.

The proof depended on the properties of the circles of inversion. By inverting the system with respect to a circle whose centre is at one of the points of contact, the circles become parallel lines, and the condition is evident.

equation of
The equation may be written in the form

$$\begin{vmatrix} \rho_1 & \rho_2 & \rho_3 & \rho_4 \\ 1 & 1 & 1 & 1 \\ t_{23}^2 & t_{13}^2 & t_{12}^2 & \cdots \\ \cdots & \cdots & \cdots & \cdots \end{vmatrix} = 0. \tag{2}$$

and substituting the values of t_{ij}^2 in terms of the radii, we obtain a relation which must be satisfied.

The condition for four circles to touch a fifth circle may also be expressed in terms of the curvatures. If the circles have curvatures κ_1, κ_2, κ_3, κ_4 (where $\kappa = 1/\rho$), and the distances between centres are t_{ij}, then the condition is that a certain determinant involving these quantities shall vanish.

The general theory of the contacts of circles may be applied to the determination of the circles which touch three given circles. The problem of Apollonius, to find a circle touching three given circles, has eight solutions, corresponding to the different combinations of external and internal contact.

The investigation of Casey's equation leads to many interesting results in the geometry of circles. The condition may be generalized to the case of spheres in three dimensions, and to higher dimensions.

The relation between the radii and the distances may be expressed in a symmetric form. Writing $u_{ij} = t_{ij}^2/\rho_i\rho_j$, the condition becomes a relation between the quantities u_{ij} and the ratios of the radii.

The equation of Casey is a fundamental result in the theory of the contacts of circles. It has been applied to the study of the configurations of circles in a plane, and to the investigation of the properties of systems of circles.

The proof of Casey's equation may be obtained by considering the inversion of the system of circles. If we invert with respect to a circle whose centre is at the point of contact of two of the circles, these circles become parallel lines, and the condition for the contact of the remaining circles with the inverted circle is easily established.

396. On a certain Envelope depending on a Triangle inscribed in a Circle.

[From the *Proceedings of the London Mathematical Society*, vol. IV (1871), pp. 122–127.]

396.

Let a triangle be inscribed in a circle, and consider the envelope of a line connected with the triangle by a given relation.

We may take the circle to be $x^2 + y^2 = 1$, and the vertices of the triangle to be points on this circle.

and I proceed to find the equation of the envelope.

Let the vertices of the triangle be represented parametrically by the angles α, β, γ, so that the coordinates are $(\cos\alpha, \sin\alpha)$, $(\cos\beta, \sin\beta)$, $(\cos\gamma, \sin\gamma)$.

The equation of the line joining the points with parameters β and γ is

$$x\cos\tfrac{1}{2}(\beta+\gamma) + y\sin\tfrac{1}{2}(\beta+\gamma) = \cos\tfrac{1}{2}(\beta-\gamma). \tag{1}$$

We find the equation by eliminating the parameters between the equations of the line and the condition of envelopment.

Let the line be $\lambda x + \mu y + \nu = 0$, and let the condition of envelopment be that the line is connected with the triangle by a given relation, e.g. that it passes through a fixed point, or that it touches a given conic.

where the coefficients of the line equation are functions of the parameters determining the triangle.

The envelope of the line is obtained by eliminating the parameters between the equation of the line and the condition that the line touches the envelope. This leads to a curve of higher class.

I find the envelope to be a curve of the sixth class, having cusps and nodes depending on the particular relation assumed.

The equation of the envelope may be obtained by eliminating the parameters between the equations

$$\lambda\cos\alpha+\mu\sin\alpha+\nu=0,\quad \lambda\cos\beta+\mu\sin\beta+\nu=0,\quad \lambda\cos\gamma+\mu\sin\gamma+\nu=0, \tag{2}$$

and the condition of envelopment.

I write the equation in the form

$$(a, b, c, d, e, f \mid x, y, z)^6 = 0, \tag{3}$$

where the coefficients are functions of the parameters determining the relation between the line and the triangle.

The class of the envelope is six, and the curve has in general nine cusps and four nodes. The particular case where the envelope degenerates into simpler curves is of special interest.

The Collected Mathematical Papers of Arthur Cayley, Vol. VI

Pages 101–150 (Papers 396–402)

Transcription

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396. On a Certain Envelope Depending on a Triangle Inscribed in a Circle

[From the Philosophical Magazine, vol. xxxiv. (1867), pp. 174–179.]

[Continued from p. 73.]

[Pp. 175—179.]

It is interesting to verify this; I write $Z = -X - Y$, the cubic function then assumes the form

$$(a, \beta, \gamma, \delta X, Y)^3,$$

where $(a, \beta, \gamma, \delta)$ have the values presently given.

The Hessian is

$$(2a\gamma - 2\beta^2)X^2 + (a\delta - \beta\gamma) \cdot 2XY + (2\beta\delta - 2\gamma^2)Y^2,$$

or writing $2XY = Z^2 - X^2 - Y^2$, this is

$$= (2a\gamma - 2\beta^2 - a\delta + \beta\gamma)X^2 + (2\beta\delta - 2\gamma^2 - a\delta - \beta\gamma)Y^2 + (a\delta - \beta\gamma)Z^2.$$

We find after some easy reductions,

$$\begin{aligned} J_0 &= Z^2 + L'^2, = -3(a+c)(ac+M), \\ I' &= -H' + 2K' + I' + F', = 3a(a+c)(b+c), \\ \gamma &= -K' + 2H' + J' + G', = 3b(a+c)(b+c), \\ \delta &= I' + J', = -3(b+c)(bc+M), \end{aligned}$$

and hence

$$a\delta - \beta\gamma = 81(a+c)(b+c)\{(ac+M)(bc+M) - ab(a+c)(b+c)\},$$

where the expression in {} is

$$\begin{aligned} &= (ab + 2ac + bc)(ab + ac + 2bc) - ab(ab + ac + bc + c^2), \\ &= c\{ab(3a + 3b) + c(b + 2a)(a + 2b) - ab(a + b + c)\}, \\ &= 2c(a+b)(bc + ca + ab), \end{aligned}$$

and therefore

$$a\delta - \beta\gamma = -162(b+c)(c+a)(a+b)M;$$

the other coefficients may be similarly calculated, and omitting the merely numerical factor, we have

$$\text{Hess} = (b+c)(c+a)(a+b)M(aX^2 + bY^2 + cZ^2),$$

which is right.

I write next

$$\begin{aligned} HU + 3MU - \psi U = X^2(-a^2x + I'Y - F'Z) \\ + Y^2(-b^2y + J'Z - G'X) + Z^2(-a^2z + K'X - H'Y), \end{aligned}$$

or writing $x = z - Y$, $y = z + X$, this is

$$\begin{aligned} HU + 3MU - \psi U = -4^2 z(aX^2 + bY^2 + cZ^2) \\ + X^2\{(I' + a^2)Y - F'Z\} + Y^2\{J'Z - (G' + b^2)X\} + Z^2\{K'X - H'Y\}, \end{aligned}$$

we may determine ψ, so that the cubic function of X, Y, Z contains the factor $aX^2 + bY^2 + cZ^2$; writing $Z = -X - Y$, then

Contains the factor $(a + c)X^2 + 2cXY + (b + c)Y^2$

Quotient is $-3(ac + M)X + 3(bc + M)Y$.

$$\begin{aligned} Z^2(\lambda I'' + F') + Z^2Y(-H' + 2K' + I' + \tfrac{1}{4}\psi - a^2) \\ + ZY^2(K' - 2H' - G' - J' - b^2 + \psi) + Y^3(-J' - H'). \end{aligned}$$

We have seen that

$$\begin{aligned} K' + F' &= -3(a + c)(ac + M), \\ J' + H' &= -3(b + c)(bc + M), \end{aligned}$$

whence the quotient is, as above stated,

$$= -3(ac + M)X + 3(bc + M)Y.$$

Comparing the coefficients of X^2Y, we have

$$\begin{aligned} a\psi &= -(-F' + 2K' + I' + I') + 3(a + c)(bc + M) - 6c(ac + M), \\ &= 9a(a + c)(b + c) + 3(a + c)(ab + ac + 2bc) - 6c(ab + 2ac + bc), \\ &= 12a(bc + ca + ab) = 12aM, \end{aligned}$$

that is $\psi = 12M$; and the same value would have been obtained by comparing the coefficients of XY^2. Hence $HU + 3MU - \psi U$ divides by $aX^2 + bY^2 + cZ^2$, the quotient being

$$\begin{aligned} &= -12Mz - 3(ac + M)X + 3(bc + M)Y, \\ &= -12Mz - 3(ac + M)(y - z) + 3(bc + M)(z - x), \\ &= -3\{(bc + M)x + (ca + M)y + (ab + M)z\}, \end{aligned}$$

which is or, finally it is

$$= -3\{(bc + M)x + (ca + M)y + (ab + M)z\},$$

and we thus have

$$HU + 3MU - \psi U = -3(aX^2 + bY^2 + cZ^2) \times \{(bc + M)x + (ca + M)y + (ab + M)z\},$$

so that the three inflexions are the intersections of the cubic curve by the line

$$(bc + M)x + (ca + M)y + (ab + M)z = 0.$$

It may be noticed, that if we write

$$\begin{aligned} x + y + z &= u, \\ bex + cay + abz &= -Mu, \\ ax + by + cz &= v, \end{aligned}$$

then x, y, z will be as

$$(b - c)\{(2M - bc)u - av\} : (c - a)\{(2M - ca)u - bv\} : (a - b)\{(2M - ab)u - cv\},$$

and substituting these values in the equation

$$ax(y-z)^2 + by(z-x)^2 + cz(x-y)^2 = 0$$

of the cubic, we have a cubic equation for the ratio $(u : v)$; and thence the values (x, y, z) for the coordinates of the inflexions.

It may be added, that we have

$$\begin{aligned} 12MU = -3(aX^2 + bY^2 + cZ^2)\{(bc+M)x + (ca+M)y + (ab+M)z\} \\ + \{X^2(FY - F'Z) + Y^2(J'Z - G'X) + Z^2(K'X - H'Y)\} = 0, \end{aligned}$$

which is the equation of the cubic expressed in the canonical form.

[Pp. 175—179.]

Effecting the process indicated p. 73, but writing for greater convenience (x, y, z) in place of (X, Y, Z), so that the substitution to be made is

$$(a, b, c, f, g, h, i, j, k, l) = (6x, 6y, 6z, -2z, -2x, -2z, -2x, -2y, x+y+z),$$

respectively (where I have corrected a misprint in the formula as originally given) I find the equation of the envelope to be

$$\begin{aligned} 4y^2(y-z)^2a^4 + 4z^2x(z-x)^2b^4 + 4x^2y(x-y)^2c^4 \\ + 4zx(z^2 + x^2 + 3yz - 2zx + 5xy)b^2c \\ + 4xy(x^2 + y^2 + 3zx - 2xy + 5yz)c^2a \\ + 4yz(y^2 + z^2 + 3xy - 2yz + 5zx)a^2b \\ + 4xy(x^2 + y^2 + 3yz - 2xy + 5zx)bc^2 \\ + 4y^2(y^2 + z^2 + 3zx - 2yz + 5xy)ca^2 \\ + 4zx(z^2 + x^2 + 3xy - 2zx + 5yz)ab^2 \\ + x(x^3 - 2x^2y - 2x^2z + xy^2 + 33xyz + xz^2 + 12y^2z + 12yz^2)b^2c^2 \\ + y(y^3 - 2y^2z - 2y^2x + yz^2 + 33xyz + yx^2 + 12z^2x + 12zx^2)c^2a^2 \\ + z(z^3 - 2z^2x - 2z^2y + zx^2 + 33xyz + zy^2 + 12x^2y + 12xy^2)a^2b^2 \\ + 2yz(11x^2 + y^2 + z^2 - 2yz + 24xy + 24zx)a^2bc \\ + 2zx(11y^2 + z^2 + x^2 - 2zx + 24yz + 24xy)b^2ca \\ + 2xy(11z^2 + x^2 + y^2 - 2xy + 24zx + 24yz)c^2ab = 0. \end{aligned}$$

The function on the left-hand side is the quotient by $-48(a+b+c)^2$ of the sextic function, Table 67 of my third Memoir on Quantics, [144]; the foregoing quotient was calculated without using the coefficient of the term in $a^2b^2c^2$ (viz.?") of the table, but by way of verification, I calculated from the table the term in question, and found it to be

$$\begin{aligned} (x+y+z)^4 - 2(x+y+z)^2(yz + zx + xy) \\ + 296(x+y+z)xyz - 8(y^2z^2 + z^2x^2 + x^2y^2), \end{aligned}$$

and this should consequently be equal to the coefficient of $a^2b^2c^2$ in the product of $(a+b+c)^2$ into the foregoing quartic function of (a, b, c) that is, it should be

$$\begin{aligned} = x(x^3 - 2x^2y - 2x^2z + xy^2 + 33xyz + xz^2 + 12y^2z + 12yz^2) \\ + y(y^3 - 2y^2z - 2y^2x + yz^2 + 33xyz + yx^2 + 12z^2x + 12zx^2) \\ + z(z^3 - 2z^2x - 2z^2y + zx^2 + 33xyz + zy^2 + 12x^2y + 12xy^2) \\ + 4yz(11x^2 + y^2 + z^2 - 2yz + 24xy + 24zx) \\ + 4zx(11y^2 + z^2 + x^2 - 2zx + 24yz + 24xy) \\ + 4xy(11z^2 + x^2 + y^2 - 2xy + 24zx + 24yz), \end{aligned}$$

which is accordingly found to be the case.

397. Specimen Table $M \equiv a^\alpha b^\beta$ (Mod. N) for Any Prime or Composite Modulus

[From the Quarterly Journal of Pure and Applied Mathematics, vol. ix. (1868), pp. 95—96 and plate.]

If N be a prime number, and a one of its primitive roots, then any number M prime to N, or what is the same thing, any number in the series $1, 2, \dots N-1$, may be exhibited in the form $M = a^\alpha$ (Mod. N); where α is said to be the *index* of M in regard to the particular root a.

Jacobi's *Canon Arithmeticus* (Berlin, 1839), contains a series of tables, giving the indices of the numbers $1, 2, 3 \dots N-1$ for every prime number N less than 1000, and giving conversely for each such prime number the numbers M which correspond to the indices $\alpha = 1, 2, \dots (N-1)$ (Tabulæ Numerorum ad Indices datos pertinentium et Indicum Numero dato correspondentium).

A similar theory applies, it is well known, to the composite numbers; the only difference is, that in order to exhibit for a given composite number N the different numbers less than N and prime to it, we require not a single root a, but two or more roots $a, b, \dots$ and that in terms of these we have $M = a^\alpha b^\beta \dots$ (Mod. N).

For each root a there is an index A (or say the Indicator of the root), such that $a^A = 1$ (Mod. N), A being the least index for which this equation is satisfied; and the indices $\alpha, \beta, \dots$ extend from 1 to $A, B, \dots$ respectively; the number of different combinations or the product $AB \dots$, being precisely equal to $\phi(N)$, the number of integers less than N and prime to it.

The least common multiple of $A, B \dots$, is termed the Maximum Indicator, and representing it by I, then for any number M not prime to N, we have $M^I = 1$ (Mod. N), a theorem made use of by Cauchy for the solution of indeterminate equations of the first order.

Thus $N = 20$, the roots may be taken to be 3, 11; the corresponding exponents are 4, 2 (viz. $3^4 \equiv 1$ (Mod. 20), $11^2 \equiv 1$ (Mod. 20)), and the product of these is 8, the number of integers less than 20 and prime to it; the series [go to p. 86]

[Plate pages 84–85. Tables of indices for moduli $N = 1$ to 50. The tables occupy two full pages with the following arrangement:]

Table arrangement. For each modulus N from 1 to 50, the table presents five heading lines:

1. The modulus N;
2. The root or roots belonging to each number N;
3. The indicators of these roots;
4. The maximum indicator;
5. The number $\phi(N)$.

The remaining lines contain the index or indices of each of the $\phi(N)$ numbers M less than N and prime to it, the number corresponding to such index or indices being placed outside in the same horizontal line.

Specimen: $N = 30$**.**

30 has the roots 7, 11, indices 4, 2 respectively; the Maximum Indicator is 4, and the number of integers less than 30 and prime to it is 8; taking any such number, say 17, the indices are 1, 1, that is, we have $17 = 7^1 \cdot 11^1$ (Mod. 30).

[From p. 83]

The foregoing corresponds to the Tabulæ Indicum Numero dato correspondentium of Jacobi; on account of multiplicity of roots there does not appear to be any mode of forming a single table corresponding to the Tabulæ Numerorum ad. Indices datos pertinentium; and there would be no adequate advantage in forming for each number N a separate table in some such form as

$N = 20$

Roots	3	11
Nos.		
1	0	0
3	1	0
7	1	1
9	2	0
11	0	1
13	2	1
17	3	1
19	3	0

which I have written down in the form of a table of single entry; for although (whenever, as in the present case, the number of roots is only two) it might have been better exhibited as a table of double entry, when the number of roots is three or more it could not of course be exhibited as a table of corresponding multiple entry.

398. On a Certain Sextic Developable, and Sextic Surface Connected Therewith

[From the Quarterly Journal of Pure and Applied Mathematics, vol. ix. (1868), pp. 129—142 and 373—376.]

I PROPOSE to consider [first] the sextic developable derived from a quartic equation, viz. taking this to be $(a, b, c, d, et, 1)^4 = 0$, where (a, b, c, d, e) are any linear functions of the coordinates (x, y, z, w), the equation of the developable in question is

$$(ae - 4bd + 3c^2)^3 - 27(ace - ad^2 - b^2e + 2bcd - c^3)^2 = 0.$$

I have already, in the paper "On a Special Sextic Developable," Quarterly Journal of Mathematics, vol. VII. (1866), pp. 105—113, [373], considered a particular case of this surface, viz. that in which c was $= 0$, the geometrical peculiarity of which is that the cuspidal edge is there an excubo-quartic curve (of a special form, having two stationary tangents), whereas in the general case here considered it is a sextic curve.

There was analytically the convenience that the linear functions being only the four functions a, b, d, e, these could be themselves taken as coordinates, whereas in the present case we have the five linear functions a, b, c, d, e.

The developable $(ae - 4bd + 3c^2)^3 - 27(ace - ad^2 - b^2e + 2bcd - c^3)^2 = 0$ is a sextic developable having for its cuspidal curve the sextic curve

$$\begin{aligned} ae - 4bd + 3c^2 &= 0, \\ ace - ad^2 - b^2e + 2bcd - c^3 &= 0, \end{aligned}$$

(say $I = 0$, $J = 0$, as usual), and having besides a nodal curve the equations of which may be written

$$6(ac - b^2) : 3(ad - bc) : ae + 2bd - 3c^2 : 3(be - cd) : 6(ce - d^2) = a : b : c : d : e.$$

viz. these equations are really equivalent to two equations, and they represent a curve of the fourth order which is an excubo-quartic.

We may in fact find the equations of the nodal curve by assuming $(a, b, c, d, e\between t, 1)^4$ to be a perfect square, say to avoid fractions that it is $= 3(\alpha t^2 + 2\beta t + \gamma)^2$, then we have

$$a : b : c : d : e = 3\alpha^2 : 3\alpha\beta : \alpha\gamma + 2\beta^2 : \beta\gamma : 3\gamma^2,$$

which equations as involving the two arbitrary parameters $\alpha : \beta : \gamma$, give two equations between (a, b, c, d, e), and we may at once by means of them verify the above-mentioned equations of the nodal curve.

It also hereby appears that the nodal curve is as stated an excubo-quartic curve; viz. we have between a, b, c, d, e a single linear relation, that is a quadric relation between α, β, γ, and this equation may be satisfied identically by taking for α, β, γ properly determined quadric functions of a variable parameter θ; whence a, b, c, d, e are proportional to quartic functions of the variable parameter θ, or the curve is an excubo-quartic.

The equations of the nodal curve may be presented under a somewhat different form; viz. the cubi-covariant of $(a, b, c, d, e\between t, 1)^4 = 0$ being

$$\begin{gathered}(-a^2d + 3abc - 2b^3 \quad - a^2e - 2abd + 9ac^2 - 6b^2c \\ - abe + 6acd - 6bc^2 \quad + 3ad^2 - b^2e \\ + ade + 2bd^2 - 3bce \\ + ae^2 + 2bde - 9c^2e + 6cd^2 \\ + be^2 - 3cde + 2d^3\between t, 1)^3 = 0,\end{gathered}$$

say this function, multiplied by 6 to avoid fractions, is

$$(a, b, c, d, e, f, g\between t, 1)^3,$$

that is

$$\begin{aligned}a &= 6(-a^2d + 3abc - 2b^3), \\ b &= (-a^2e - 2abd + 9ac^2 - 6b^2c), \\ c &= 2(-abe + 6acd - 6bc^2), \\ d &= 3(+ad^2 - b^2e), \\ e &= 2(+ade + 2bd^2 - 3bce), \\ f &= (+ae^2 + 2bde - 9c^2e + 6cd^2), \\ g &= 6(+be^2 - 3cde + 2d^3),\end{aligned}$$

then the equations of the nodal curve may be written:

$$a = 0,\ b = 0,\ c = 0,\ d = 0,\ e = 0,\ f = 0,\ g = 0.$$

It may be mentioned that we have identically

$$\begin{aligned}
&ae - 4bd + 3c^2 = 0,\\
&af - 3be + 2cd = 0,\\
&ag - 9ce + 8d^2 = 0,\\
&bg - 3cf + 2de = 0,\\
&bf - 4ce + 3d^2 = eg - 3df + 2e^2 = 0.
\end{aligned}$$

and moreover

$$ag - 6bf + 15ce - 10d^2 = -6(bf - 4ce + 3d^2), = +6(I^2 - 27J^2),$$

so that the equation of the developable may be written in the form

$$ag - 6bf + 15ce - 10d^2 = 0,$$

or in the more simple form

$$bf - 4ce + 3d^2 = 0,$$

each of which puts in evidence the nodal curve on the surface.

The nodal and cuspidal curves meet in the points

$$\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e},$$

being, as it is easy to show, a system of four points.

The four points in question form a tetrahedron, the equations of the faces of which may be taken to be $x = 0$, $y = 0$, $z = 0$, $w = 0$; and the equation of the surface may be expressed in this system of quadriplanar coordinates.

We introduce these coordinates *ab initio*, by taking the quartic function of t to be

$$(a, b, c, d, et, 1)^4 = x'(t + \alpha)^4 + y'(t + \beta)^4 + z'(t + \gamma)^4 + w'(t + \delta)^4,$$

that is, by writing

$$\begin{aligned}
a &= x' + y' + z' + w',\\
b &= \alpha x' + \beta y' + \gamma z' + \delta w',\\
c &= \alpha^2 x' + \beta^2 y' + \gamma^2 z' + \delta^2 w',\\
d &= \alpha^3 x' + \beta^3 y' + \gamma^3 z' + \delta^3 w',\\
e &= \alpha^4 x' + \beta^4 y' + \gamma^4 z' + \delta^4 w'.
\end{aligned}$$

Observe that (t_1, t_2, t_3, t_4) being any constant quantities, we thence have

$$\begin{aligned}
e - d\Sigma t_1 + c\Sigma t_1 t_2 - b\Sigma t_1 t_2 t_3 + a t_1 t_2 t_3 t_4 \\
&= x'(\alpha - t_1)(\alpha - t_2)(\alpha - t_3)(\alpha - t_4)\\
&+ y'(\beta - t_1)(\beta - t_2)(\beta - t_3)(\beta - t_4)\\
&+ z'(\gamma - t_1)(\gamma - t_2)(\gamma - t_3)(\gamma - t_4)\\
&\qquad + w'(\delta - t_1)(\delta - t_2)(\delta - t_3)(\delta - t_4),
\end{aligned}$$

and thence in particular

$$e - d\Sigma\alpha + c\Sigma\alpha\beta - b\Sigma\alpha\beta\gamma + a\alpha\beta\gamma\delta = 0,$$

viz. this is the linear relation which subsists identically between (a, b, c, d, e), the five linear functions of the coordinates (x, y, z, w).

Starting from the above values of (a, b, c, d, e), we find without difficulty

$$\begin{aligned} I = {} & (\alpha-\beta)^2x'y' + (\alpha-\gamma)^2x'z' + (\alpha-\delta)^2x'w' \\ & + (\beta-\gamma)^2y'z' + (\beta-\delta)^2y'w' + (\gamma-\delta)^2z'w', \end{aligned}$$

$$\begin{aligned} J = {} & (\alpha-\beta)^2(\alpha-\gamma)^2(\beta-\gamma)^2x'y'z' + (\alpha-\beta)^2(\alpha-\delta)^2(\beta-\delta)^2x'y'w' \\ & + (\alpha-\gamma)^2(\gamma-\delta)^2(\alpha-\delta)^2x'z'w' + (\beta-\gamma)^2(\gamma-\delta)^2(\beta-\delta)^2y'z'w', \end{aligned}$$

but we thus see the convenience of introducing constant multipliers into the expressions of the four coordinates respectively, viz. writing

$$X = (\beta\gamma\delta)x', \quad Y = (\gamma\delta\alpha)y', \quad Z = (\delta\alpha\beta)z', \quad W = (\alpha\beta\gamma)w',$$

where for shortness $(\beta\gamma\delta) = (\beta-\gamma)(\gamma-\delta)(\delta-\beta)$, &c., or what is the same thing, taking the quartic to be

$$(a, b, c, d, et, 1)^4 = X'(\beta\gamma\delta)(t+\alpha)^4 + Y'(\gamma\delta\alpha)(t+\beta)^4 + Z'(\delta\alpha\beta)(t+\gamma)^4 + W'(\alpha\beta\gamma)(t+\delta)^4,$$

we find

$$\begin{aligned} J &= (\lambda'\mu'\nu')^2(x'y'z' + y'z'w' + z'x'w' + x'y'w'), \\ I &= (\lambda'\mu'\nu')\{\lambda'(x'w' + y'z') + \mu'(y'w' + z'x') + \nu'(z'w' + x'y')\}, \end{aligned}$$

where for shortness

$$\begin{aligned} \lambda' &= (\alpha-\delta)^2(\beta-\gamma)^2, \\ \mu' &= (\beta-\delta)^2(\gamma-\alpha)^2, \\ \nu' &= (\gamma-\delta)^2(\alpha-\beta)^2, \\ \nabla(\lambda') &= (\alpha-\delta)(\beta-\gamma), \\ \nabla(\mu') &= (\beta-\delta)(\gamma-\alpha), \\ \nabla(\nu') &= (\gamma-\delta)(\alpha-\beta), \\ \nabla(\lambda') + \nabla(\mu') + \nabla(\nu') &= 0, \end{aligned}$$

and the equation of the developable is thus

$$\{\lambda'(x'w' + y'z') + \mu'(y'w' + z'x') + \nu'(z'w' + x'y')\}^3 - 27\lambda'\mu'\nu'(x'y'z' + y'z'w' + z'x'w' + x'y'w')^2 = 0.$$

Observe that $J = 0$ is a cubic surface passing through each edge of the tetrahedron, and having at each summit a conical point; $I = 0$ is a quadric surface passing through each summit of the tetrahedron, and at each of these points the tangent plane of the quadric surface touches the tangent cone of the cubic surface: to show this it is only necessary to observe that at the point $(x' = 0, y' = 0, z' = 0)$ the tangent cone is $y'z' + z'x' + x'y' = 0$, and the tangent plane is $\lambda'x' + \mu'y' + \nu'z' = 0$, and that these touch in virtue of the above-mentioned relation $\nabla(\lambda') + \nabla(\mu') + \nabla(\nu') = 0$.

It follows that on the curve of intersection, or cuspidal edge of the developable, each of the summits is a cuspidal or stationary point, that is, the cuspidal curve has four stationary points; this agrees with the character of the curve as given, Salmon "On the Classification of Curves of Double Curvature," Camb. and Dubl. Math. Jour. vol. V. (1850), p. 39, viz. the character is there given $a = 6$, $m = 6$, $n = 4$, $r = 6$, $g = 3$, $h = 6$, $\alpha = 0$, $\beta = 4$, $\kappa = 4$, $\gamma = 6$, ($\beta = 4$, that is, there are as stated 4 stationary points).

To find the equations of the nodal curve, instead of transforming the equations as given in terms of (a, b, c, d, e), it is better to deduce these from the equation of the surface; viz. if there is a nodal curve, we must have

$$\partial_{x'} I : \partial_{y'} I : \partial_{z'} I : \partial_{w'} I = \partial_{x'} J : \partial_{y'} J : \partial_{z'} J : \partial_{w'} J.$$

Writing these under the form $\partial_{x'} I + \theta' \partial_{x'} J = 0$, &c., where θ' is regarded as an arbitrary parameter[1], we have

$$\begin{aligned} \lambda' w' + \mu' z' + \nu' y' + \theta'(y'z' + y'w' + z'w') &= 0, \\ \lambda' z' + \mu' w' + \nu' x' + \theta'(z'x' + z'w' + x'w') &= 0, \\ \lambda' y' + \mu' x' + \nu' w' + \theta'(x'y' + x'w' + y'w') &= 0, \\ \lambda' x' + \mu' y' + \nu' z' + \theta'(y'z' + z'x' + x'y') &= 0, \end{aligned}$$

which equations (eliminating θ') must be equivalent to two equations only.

I remark that the first three equations may be regarded as a set of linear equations in $1, w', \theta', \theta' w'$; and determining from them the ratios of these quantities, we have, suppose,

$$1 : -w' : \theta' : -\theta' w' = A : B : C : D,$$

where

$$A = \begin{vmatrix} \lambda' & \nu' & y' \\ \mu' & \lambda' & z' + x' \\ \nu' & \mu' & x' + y' \end{vmatrix}, \quad B = \begin{vmatrix} y'z' & y' + z' & \nu' z' + \mu' y' \\ z'x' & z' + x' & \lambda' z' + \nu' x' \\ x'y' & x' + y' & \mu' x' + \lambda' y' \end{vmatrix}, \quad \text{etc.}$$

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[1] The value of θ' is in fact $= \frac{18J}{I}$, that is, instead of the four equations involving an arbitrary parameter θ', we have really four determinate equations.

We have thence $AD - BC = 0$ and (substituting in the fourth equation) $A(\lambda' x' + \mu' y' + \nu' z') + C(y'z' + z'x' + x'y') = 0$; each of these equations must contain the equation of the cone having $(x' = 0, y' = 0, z' = 0)$ for its vertex, and passing through the nodal curve.

The two equations are of the orders 6 and 4 respectively; and as the curve is a quartic curve passing through the vertex in question, the equation of the cone is of the order 3.

I have not effected the reduction of the sextic equation, but for the quartic equation, substituting for A, C their values, this is

$$\begin{aligned} &-(\lambda' w' + \mu' y' + \nu' z')\{\lambda' x'^2(y' - z') + \mu' y'^2(z' - x') + \nu' z'^2(z' - x')\} \\ &\qquad + (y'z' + z'x' + x'y')\{\lambda' x'(y'^2 - z'^2) + \mu' y'(z'^2 - x'^2) + \nu' z'(x'^2 - y'^2)\} \\ &\qquad\qquad + \{\mu'\nu'(y' - z') + \nu'\lambda'(z' - x') + \lambda'\mu'(x' - y')\}(x' + y' + z') = 0, \end{aligned}$$

which is easily reduced to

$$\begin{aligned} &\lambda' x'(-x'^2 + y'z' + z'x' + x'y')(y' - z') \\ &\qquad + \mu' y'(-y'^2 + y'z' + z'x' + x'y')(z' - x') \\ &\qquad + \nu' z'(-z'^2 + y'z' + z'x' + x'y')(x' - y') \\ &\qquad + \mu'\nu'\{(x' + y' + z')(y'z' + z'x' - x'y') + x'y'z'\}(y' - z') \\ &\qquad + \nu'\lambda'\{(x' + y' + z')(y'z' + z'x' + x'y') + x'y'z'\}(z' - x') \\ &\qquad + \lambda'\mu'\{(x' + y' + z')(y'z' + z'x' + x'y') + x'y'z'\}(x' - y') = 0; \end{aligned}$$

and I have found that this is transformable into

$$\begin{aligned} &2\{x'^2\nabla(\lambda') + y'^2\nabla(\mu') + z'^2\nabla(\nu')\} \times [x'^2\nabla(\lambda')(\mu' y'^2 - \nu' z'^2) \cdot 2x'\nabla(\mu')(\nu' z'^2 - \lambda' x'^2) \cdot x'y'\nabla(\nu')(\lambda' x'^2 - \mu' y'^2) \\ &\qquad - x'^2 y'^2 z'^2 \{\nabla(\mu') - \nabla(\nu')\}\{\nabla(\nu') - \nabla(\lambda')\}\{\nabla(\lambda') - \nabla(\mu')\}] = 0, \end{aligned}$$

viz. the two functions are equivalent in virtue of the relation $\nabla(\lambda') + \nabla(\mu') + \nabla(\nu') = 0$, or, what is the same thing, they only differ by a function $(x', y', z')^4$ into the evanescent factor $\lambda'^2 + \mu'^2 + \nu'^2 - 2\mu'\nu' - 2\nu'\lambda' - 2\lambda'\mu'$.

The function in $\{\}$ equated to zero is therefore the equation of the cubic cone.

I do not stop to give the steps of the investigation in the above form, as the investigation may be very much simplified as follows: by linear combinations of the four equations in x', y', z', w', θ', we deduce

$$\begin{aligned} &(\lambda' - \mu' - \nu')(x' + w' - y' - z') + 2\theta'(y'z' - x'w') = 0, \\ &(-\lambda' + \mu' - \nu')(y' + w' - z' - x') + 2\theta'(z'x' - y'w') = 0, \\ &(-\lambda' - \mu' + \nu')(z' + w' - x' - y') + 2\theta'(x'y' - z'w') = 0, \\ &(\lambda' + \mu' + \nu')(x' + y' + z' + w') + 2\theta'(y'z' + z'x' + x'y' + x'w' + y'w' + z'w') = 0. \end{aligned}$$

Hence writing

$$\begin{aligned} \lambda &= \lambda' - \mu' - \nu', & x &= w' + x' - y' - z', \\ \mu &= -\lambda' + \mu' - \nu', & y &= w' - x' + y' - z', \\ \nu &= -\lambda' - \mu' + \nu', & z &= w' - x' - y' + z', \\ & & w &= w' + x' + y' + z', \end{aligned}$$

we find

$$\begin{aligned}\lambda^2 &= \lambda'^2 - \mu'^2 - \nu'^2 + 2\mu'\nu',\\ \mu^2 &= -\lambda'^2 + \mu'^2 - \nu'^2 + 2\nu'\lambda',\\ \nu^2 &= -\lambda'^2 - \mu'^2 + \nu'^2 + 2\lambda'\mu',\end{aligned}$$

and thence

$$\lambda^2\mu^2 + \mu^2\nu^2 + \nu^2\lambda^2 = -(\lambda'^2 + \mu'^2 + \nu'^2 - 2\mu'\nu' - 2\nu'\lambda' - 2\lambda'\mu')^2 = 0,$$

that is

$$\lambda^2\mu^2 + \mu^2\nu^2 + \nu^2\lambda^2 = 0, \quad \frac{1}{\lambda} + \frac{1}{\mu} + \frac{1}{\nu} = 0,$$

the relation which connects the new constants λ, μ, ν.

Moreover

$$\begin{aligned}yz - xw &= 4(y'z' - x'w'),\\ zx - yw &= 4(z'x' - y'w'),\\ xy - zw &= 4(x'y' - z'w'),\\ 2w^2 - x^2 - y^2 - z^2 &= 8(y'z' + z'x' + x'y' + x'w' + y'w' + z'w'),\end{aligned}$$

and writing for greater convenience $\theta = -\frac{\theta'}{2}$, the equations are transformed into

$$\begin{aligned}\theta\lambda x &= vw - yz,\\ \theta\mu y &= yw - zx,\\ \theta\nu z &= zw - xy,\\ 2\theta(\lambda + \mu + \nu)w &= 3w^2 - x^2 - y^2 - z^2,\end{aligned}$$

where

$$\frac{1}{\lambda} + \frac{1}{\mu} + \frac{1}{\nu} = 0,$$

viz. these equations, eliminating θ, give the equations of the nodal curve.

From the first three equations eliminating θ, we deduce

$$\begin{aligned}yzw(\mu - \nu) &= x(\mu y^2 - \nu z^2),\\ zxw(\nu - \lambda) &= y(\nu z^2 - \lambda x^2),\\ xyw(\lambda - \mu) &= z(\lambda x^2 - \mu y^2),\end{aligned}$$

or, as these equations may be written,

$$\frac{x(\mu y^2 - \nu z^2)}{\mu - \nu} = \frac{y(\nu z^2 - \lambda x^2)}{\nu - \lambda} = \frac{z(\lambda x^2 - \mu y^2)}{\lambda - \mu},$$

which equations, from the mode in which they are obtained, are it is clear equivalent to two equations only.

Using the fourth equation, and eliminating by substituting therein for $\theta\lambda$, $\theta\mu$, $\theta\nu$ their values from the first three equations, we find

$$2w^2\left(3w^2 - x^2 - y^2 - z^2\right) = \frac{3w^2 - x^2 - y^2 - z^2}{\lambda} + \frac{3w^2 - x^2 - y^2 - z^2}{\mu} + \frac{3w^2 - x^2 - y^2 - z^2}{\nu}$$

that is

$$3w^2 + x^2 + y^2 + z^2 = 2w\left(\frac{x^2}{\lambda} + \frac{y^2}{\mu} + \frac{z^2}{\nu}\right),$$

or, what is the same thing,

$$xyz(3w^2 - x^2 + y^2 + z^2) - 2w(y^2z^2 + z^2x^2 + x^2y^2) = 0,$$

we have to show that this is in fact included in the former system, for then the four equations with θ eliminated will it is clear give two equations only.

Observe that the former system may be written

$$\frac{x(\mu y^2 - \nu z^2)}{(\mu - \nu)yz} = \frac{y(\nu z^2 - \lambda x^2)}{(\nu - \lambda)zx} = \frac{z(\lambda x^2 - \mu y^2)}{(\lambda - \mu)xy},$$

$$\begin{aligned}
&(\mu - \nu)yz \cdot x(\mu y^2 - \nu z^2) + (\nu - \lambda)zx \cdot y(\nu z^2 - \lambda x^2) + (\lambda - \mu)xy \cdot z(\lambda x^2 - \mu y^2) \\
&\qquad = \mu\nu yz(\mu - \nu)(y^2 - z^2) + \nu\lambda zx(\nu - \lambda)(z^2 - x^2) + \lambda\mu xy(\lambda - \mu)(x^2 - y^2) \\
&\qquad\qquad + xyz\{\lambda(\mu - \nu)(\mu y^2 + \nu z^2) + \mu(\nu - \lambda)(\nu z^2 + \lambda x^2) + \nu(\lambda - \mu)(\lambda x^2 + \mu y^2)\}.
\end{aligned}$$

[The paper continues on pp. 373—376 with further development of the theory of the sextic developable and its connection with the sextic surface. The investigation involves detailed analysis of the nodal curve, the cuspidal edge, and their singularities, together with the coordinate transformations and algebraic relations connecting the different forms of the surface equations.]

399. On the Cubical Divergent Parabolas

[From the Quarterly Journal of Pure and Applied Mathematics, vol. ix. (1868), pp. 185—190.]

[From p. 101.]

[Pp. 185—190.]

The cubical divergent parabolas are the parabolas $y^2 = x^3 + ax + b$, where the parameters a, b are such that the discriminant $4a^3+27b^2$ is $= 0$; viz. these are curves for which the line at infinity meets the curve in three distinct points, but the line at infinity is a tangent, or the curve has a cusp at infinity.

The several forms may be distinguished according to the nature of the roots of the equation $4a^3 + 27b^2 = 0$ considered as a cubic equation in $a : b$; or what is the same thing, according to the nature of the roots of the equation $z^3 - 3z + 2 = 0$ (where for convenience a, b have been replaced by $3z$, -2 respectively), viz. the roots are $z = 1, 1, -2$; and the forms correspond to the two cases $z = 1$ and $z = -2$.

For $z = 1$, that is $4a^3+27b^2 = 0$, $a \neq 0$, the equation is $y^2 = x^3-3x+2$, that is $y^2 = (x-1)^2(x+2)$, which has a crunode at infinity; or writing $x = \frac{1}{\xi}$, $y = \frac{\eta}{\xi^2}$, the equation becomes $\eta^2 = (1 - \xi)^2(1 + 2\xi)$, which has a crunode at $\xi = 0$, $\eta = 0$, viz. the origin.

For $z = -2$, that is $a = 0$, $b = 0$, the equation is $y^2 = x^3$, which has a cusp at infinity; or in the like transformed equation $\eta^2 = \xi^3$, which has a cusp at the origin.

The form $y^2 = x^3 - 3x + 2$, that is $y^2 = (x - 1)^2(x + 2)$, has a crunode at infinity; the tangent at the crunode is the line at infinity, and the two branches are each touched by the line at infinity. The form $y^2 = x^3$ has a cusp at infinity, the tangent at the cusp being the line at infinity.

It is convenient to exhibit the different species in a tabular form, indicating the nature of the singularities (finite and at infinity), and the sign of the discriminant $\Delta = -(4a^3 + 27b^2)$ in the different cases.

Equation	**Singularities**	Δ
$y^2 = x^3 + 3x - 2$	simplex acnodal, R without the curve	$-$
$y^2 = x^3$	simplex cuspidal, R at the cusp	0
$y^2 = x^3 - 3x + 2$	simplex crunodal, R at the node	$+$
$y^2 = x^3 + ax + b$ $(4a^3 + 27b^2 \neq 0)$	simplex non-singular	$\neq 0$

The species may be further subdivided according to the position of the point R (the intersection of the three inflexions) relative to the curve and its singularities. For the simplex non-singular form, we have according as R is within or without the curve, the campaniform or serpentine forms; and for the singular forms, corresponding subdivisions according as R coincides with or is distinct from the singularity, and in the latter case according as it lies within or without the curve.

400. On the Cubic Curves Inscribed in a Given Pencil of Six Lines

[From the Quarterly Journal of Pure and Applied Mathematics, vol. ix. (1868), pp. 211—230.]

[From p. 105.]

[Pp. 211—230.]

I consider a system of six lines forming a pencil, that is, meeting in a point, and I seek to investigate the general equation of a cubic curve inscribed in the six lines; viz. the six lines are to be tangents to the cubic curve. It is to be observed that the six lines may be any six lines of a pencil; there is no restriction whatever as to the particular positions of the lines. The problem is: given six tangents of a cubic, to find the equation of the cubic; or what is the same thing, to find the equation of the curve having the six given lines for tangents.

Taking the centre of the pencil for the point $(x = 0, y = 0)$, and taking $z = 0$ for the equation of the line at infinity, the equations of the six tangents may be written

$$z = 0, \quad y - m_1 x = 0, \quad y - m_2 x = 0, \quad \ldots, \quad y - m_5 x = 0,$$

where m_1, m_2, m_3, m_4, m_5 are the slopes of the five finite tangents respectively. The general equation of a cubic curve is

$$(a, b, c, d, f, g, h, l, m, nx, y, z)^3 = 0,$$

or what is the same thing,

$$ax^3 + by^3 + cz^3 + 3dy^2z + 3ez^2x + 3fz^2y + 3gxz^2 + 3hyz^2 + 3lx^2y + 3mxy^2 + 6nxyz = 0.$$

The condition that the line $y - mx = 0$ may be a tangent to this cubic is obtained by substituting $y = mx$ in the equation of the cubic, and expressing that the resulting equation in $x : z$ has equal roots; viz. substituting $y = mx$, we have

$$(a + 3lm + 3m^2 + bm^3)x^3 + 3(n + fm + em^2 + hm^3 + gm^2 + dm^3)xz^2 + (c + 3gz + 3hz + dz)z^3 = 0,$$

or more simply

$$(a, b, c, d, e, f, g, h, l, m, n1, m, 0)^3x^3 + \cdots = 0.$$

The condition of equal roots gives a sextic equation in m, which must be satisfied identically for $m = m_1, m_2, m_3, m_4, m_5$ and $m = \infty$ (corresponding to the tangent $z = 0$). We hence obtain the required conditions on the coefficients.

The investigation is conducted by means of the invariants of the cubic. Writing the cubic in the canonical form, and making use of the known relations between the invariants and the coefficients, the conditions that six given lines of a pencil shall be tangents to the cubic may be expressed in a very elegant form.

Let the six tangents be denoted by their equations $l_1 = 0, l_2 = 0, \ldots, l_6 = 0$, where $l_i = y - m_i x$ for $i = 1, \ldots, 5$, and $l_6 = z$. Then the equation of any cubic having these six lines for tangents may be written in the form

$$\sqrt{l_1} + \sqrt{l_2} + \sqrt{l_3} + \sqrt{l_4} + \sqrt{l_5} + \sqrt{l_6} = 0,$$

where the signs of the radicals are properly determined.

Or more definitely, if we write

$$\sqrt{l_1} \pm \sqrt{l_2} \pm \sqrt{l_3} \pm \sqrt{l_4} \pm \sqrt{l_5} \pm \sqrt{l_6} = 0,$$

the different systems of signs give the different cubics which have the six lines for tangents. It is known that there are in general ten such cubics; the different systems of signs must be such that the product of all the terms is positive, and the number of independent systems is ten.

The equation may also be written in the rationalised form. We have

$$(\sqrt{l_1} + \sqrt{l_2} + \sqrt{l_3} + \sqrt{l_4} + \sqrt{l_5} + \sqrt{l_6})(\sqrt{l_1} + \sqrt{l_2} + \sqrt{l_3} + \sqrt{l_4} + \sqrt{l_5} - \sqrt{l_6}) \cdots = 0,$$

where the product contains $2^5 = 32$ factors, but these combine in pairs to give 16 different cubics, and of these only 10 are proper cubics, the remaining 6 being degenerate forms.

For the complete discussion it is necessary to introduce the fundamental invariants S and T of the cubic. These are connected with the coefficients by the relations

$$S = \frac{1}{3 \cdot 12^4}(abc + \dots),$$
$$T = \frac{1}{27 \cdot 12^6}(a^2b^2c^2 + \dots),$$

and the discriminant is $\Delta = 64S^3 - T^2$.

For a cubic having six given tangents which form a pencil, the invariants S and T satisfy certain relations which may be obtained as follows. Let the six lines be given, and let the cubic be expressed in terms of the parameters which determine the position of the point of contact on each tangent; then the condition that the six points of contact shall lie on a cubic curve gives the required relation between S and T.

The detailed calculation is as follows. Let the cubic be written in the form

$$c[(a, h, k, bx, y)^2]^2 + 32l\Delta x^2y^2 = 0,$$

where $(a, h, k, bx, y)^2 = ax^2 + 2hxy + by^2$, and $\Delta = hk - ab$, $V = hk - h^2$, $\Theta = ah - bk$, $R = -81ab$.

The discriminant of this form, say Q_{10}, is a function of c^2D, V, $c\Theta$, cR, where D is the discriminant of the quadratic $(a, h, k, bx, y)^2$. The expression is found to be

$$\begin{aligned} Q_{10} = c^{10}\{D^3\} + c^9\{-99RD^2 - 400VD + 2304V\Theta D + 8640\Theta^2\} \\ + c^8\{12800R^2V\Theta + 82944V^3\Theta^2 - 4608V^2D\} \\ + c^7\{20480R^2V^3 + 147456V^4\Theta + 65536V^6\}. \end{aligned}$$

The foregoing values of S and T give

$$T^2-64S^3 = c^4\{D^3+c(8RD^2-24V\Theta D-64\Theta^3)+c^2(16R^2+96RV\Theta-48V^2\Theta^2+16V^3D)+c^3(64R^2V^2)\},$$

so that

$$Q_{10} = -c^6R^2(T^2 - 64S^3),$$

and therefore

$$64S^3 - T^2 = \frac{Q_{10}}{c^6R^2},$$

formulae which are interesting in the theory.

As regards the discriminant Q_{10}, this as already remarked, has been calculated for the general form, but for the present purpose it is easier, by dealing directly with the form $[(a,h,k,bx,y)^2]^2+32l\Delta x^2y^2$ and then interpreting D, V, Θ, R and restoring the coefficient c as above, to obtain the discriminant Q_{10} of the function $[c(a,h,k,bx,y)^2]^2+4[(j,l,fx,y)^2]^2$ in the required form, as a function of c^2D, V, $c\Theta$, cR.

I find after some laborious calculations

$$\begin{aligned}
Q_5 &= \text{No. } 31 = c^5\{90\}+c^4\{40VD-288V\Theta\}+c^3\{256V^2\},\\
Q_7 &= 10000\,\text{No. } 34 = c^7\{-99D^2-400RD+2304V\Theta D+8640\Theta^2\}\\
&\quad +c^6\{12800R^2V\Theta+82944V^3\Theta^2-4608V^2D\}+c^5\{20480R^2V^3+147456V^4\Theta+65536V^6\},\\
Q_8 &= 1000000\,\text{No. } 35 = c^8\{7992D^3+72000RD^2-145152V\Theta D^2+622080\Theta^2D\}\\
&\quad +c^7\{160000R^2\Theta+691200R^2V\Theta D+3456000R^2\Theta^2+3815424V^2\Theta^2D\\
&\qquad +36080640V\Theta^3+635904V^2D^2\}\\
&\quad +c^6\{33177600R^2V^2\Theta^2+4669440R^2V^2D+217645056V^4\Theta^2+23003136V^2\Theta D\}\\
&\quad +c^5\{62914560R^2V^4+452984832V^3\Theta+14155776V^4D\}\\
&\quad +c^4\{134217728V^8\}.
\end{aligned}$$

Also Q_{10} = multiple of discriminant $=c^{10}R^2U^2$, where

$$\begin{aligned}
U &= c^4D^3+c^3(-8R^2D-24R^2V\Theta D+64R^2\Theta^3)\\
&\qquad +c^2(-16R^2-96R^2V\Theta+48R^2V^2\Theta^2-16R^2V^3D)+c(-64R^2V^2).
\end{aligned}$$

To these may be joined the intermediate values:

$$\begin{aligned}
Q_3 &= c^3\{81D\}+c^2\{-720RD+5184V\Theta D+1600R^2+23040RV\Theta+82944V^2\Theta^2+4608V^2D\}\\
&\quad +c\{20480R^2V^3+147456V^4\Theta+65536V^6\},\\
Q_4 &= c^4\{180D^2+1120RD+2880V\Theta D-8640\Theta^2\}\\
&\quad +c^3\{1600R^2+10240RV\Theta+270D^2+720RD^2-51840V\Theta D^2-77760\Theta^2D\}.
\end{aligned}$$

We have

$$c(fO) = T - 12VS + 4V^3 - 4cR,$$

and if by means of these values we eliminate $c\Theta$ and c^2D, we obtain Q_5, Q_7, Q_8 and Q_{10} as functions of S, T, V and cR. Choosing instead of Q_5 and Q_8 the combinations $Q_5 - Q_3$ and $Q_8 - 8Q_4^2$, and forming also the expression for the combination $Q_5(Q_5 - Q_3)$, we have thus the system of formulae:

$$\begin{aligned}
Q_3 &= 9T^2 + 180VS + 4V^3 + 4cR,\\
\tfrac{1}{4}(Q_5 - Q_3) &= 9T^2 - 432S^3 - 72TVS - 72TV - 16TcR + 864V^2S^2 + 144V^2S + 128VScR,\\
\tfrac{1}{4}(Q_8 - 8Q_4^2) &= 27T^4 - 4212T^2VS^2 + 6588T^2V^2 + 252T^2cR - 7776TS^3\\
&\quad - 16848TV^2S^2 + 3456TV^2S - 2592TV^2ScR - 1296TV - 1824TV^2cR\\
&\quad - 448Tc^2R^2 + 544320VS^4 - 461376V^2S^3 + 74304cRS^3 + 15552V^2S^2\\
&\quad - 10368V^2S^2cR - 10728V^2S - 3264V^2ScR - 1536V^2c^2R^2,\\
&\quad + 81T^4 + 972T^3VS - 612T^3V - 108T^3cR - 3888TS^3 - 5184TV^2S^2\\
&\quad - 11952TV^2S - 2016TVScR - 288TV - 352TV^2cR - 64Tc^2R\\
&\quad - 77760VS^4 + 153792V^2S^3 - 1728cRS^3 + 29376V^2S^2 + 26928V^2S^2cR\\
&\quad + 576V^2S + 1088V^2ScR + 512V^2c^2R^2.
\end{aligned}$$

And, as mentioned above, $Q_{10} = -c^6R^2(T^2 - 64S^3)$.

The just-mentioned value of Q_{10} should, I think, admit of being established *a priori*, and if this be so, then the substitution of the values of S and T in terms of c^2D, V, $c\Theta$, cR, would be the easiest way of arriving at the before-mentioned expression of Q_{10} in terms of these same quantities. The calculation by which this expression was arrived at, is however not without interest, and it will be as well to indicate the mode in which it was effected.

Calculation of Q_{10}.

We have to find the discriminant of $c[(a, h, k, bx, y)^2]^2 + 32l\Delta x^2y^2$.

Consider for a moment the more general form $P^2 + 4Q^2$, then to find the discriminant, we have to eliminate between the equations

$$P\frac{\partial P}{\partial x} + 4Q\frac{\partial Q}{\partial x} = 0, \quad P\frac{\partial P}{\partial y} + 4Q\frac{\partial Q}{\partial y} = 0;$$

these are satisfied by the system $P = 0$, $Q^2 = 0$, and it follows that if R be the resultant of the equations $P = 0$, $Q = 0$, then the discriminant in question contains the factor R^2. For the other factor we may reduce the system to

$$P\frac{\partial P}{\partial x} + 4Q\frac{\partial Q}{\partial x} = 0, \quad \frac{\partial P}{\partial x}\frac{\partial Q}{\partial y} - \frac{\partial P}{\partial y}\frac{\partial Q}{\partial x} = 0.$$

Now writing $Q = 2lxy$, these equations become

$$P\frac{\partial P}{\partial x} + 4 \cdot 8l^2xy^2 = 0, \quad \left(\frac{\partial P}{\partial x}\frac{\partial P}{\partial y}\right) = 0,$$

the resultant of which is = resultant of the system

$$P^2 + 4 \cdot 8l^2x^2y^2 = 0, \quad \frac{\partial P}{\partial x}\frac{\partial P}{\partial y} = 0,$$

but in virtue of the second equation, we have

$$\frac{\partial P}{\partial x}\frac{\partial P}{\partial y} = 0,$$

which reduces the first equation to

$$P^2 + 32l^2x^2y^2 = 0,$$

or omitting the factor $2y$, to

$$\frac{\partial P}{\partial x} = 0.$$

Hence, writing $P = \sqrt{c} \cdot (a, h, k, bx, y)^2$, and therefore $\frac{\partial P}{\partial x} = 3\sqrt{c} \cdot (a, h, k, bx, y)^2$, $\frac{\partial P}{\partial y} = 3\sqrt{c} \cdot (h, k, bx, y)^2$; writing also $y = 1$, the two equations become

$$c(a, h, k, bx, 1)^2(h, k, bx, 1)^2 + 8l^2x^2 = 0,$$
$$(a, h, k, bx, 1)^2 - (h, k, bx, 1)^2 = 0,$$

the second of which is more simply written

$$(a, h, -k, -bx, 1)^2 = 0.$$

Hence, restoring the factor P, and also to avoid fractions introducing the factor $8a^2$, the resultant of the two equations is

$$= 8l^2a^2\prod(8l^2x^2 + c(a, h, k, bx, 1)^2(h, k, bx, 1)^2),$$

where $\prod$ denotes the product of the factors corresponding to the three roots x_1, x_2, x_3 of the equation

$$(a, h, -k, -bx, 1)^2 = 0, \quad \text{or} \quad ax^3 + hx^2 - kx - b = 0,$$

so that the symmetric functions are to be found from

$$\Sigma x_1 = -\frac{h}{a}, \quad \Sigma x_1x_2 = -\frac{k}{a}, \quad x_1x_2x_3 = \frac{b}{a}.$$

The required discriminant is the foregoing resultant multiplied by R, or say by c^2E^2, that is the discriminant Q_{10} is

$$= c^2E^2 \cdot 8l^2a^2 \prod(8l^2x^2 + \Omega),$$

if for shortness we write

$$\Omega = (a, h, k, bx, 1)^2 \cdot (a, k, bx, 1)^2,$$

and when the symmetric functions have been expressed in terms of the coefficients, the result is to be expressed as a function of D, V, Θ, R by means of the values

$$\begin{aligned} c^2D &= a^2b^2 + 4ahk^2 + 4bk^2h - 6abhk - 3h^2k^2, \\ c\Theta &= -l(ab - hk), \\ cR &= -81l^2ab. \end{aligned}$$

Thus, for instance, the first term of the result is

$$= c^2R^2 \cdot 8l^2a^2 \cdot 512l^6x_1^2x_2^2x_3^2,$$

which is

$$= c^2R^2 \cdot 4096l^8 \cdot \frac{b^2}{a^2} = -64c^4R^4V^2,$$

which is a term in the before-mentioned expression for Q_{10}.

401. A Notation of the Points and Lines in Pascal's Theorem

[From the Quarterly Journal of Pure and Applied Mathematics, vol. ix. (1868), pp. 268—274.]

Taking six points 1, 2, 3, 4, 5, 6 on a conic; let $A, B, C, D, E, F, G, H, I, J, K, L, M, N, O$ denote each a combination of three lines, thus

$$\begin{array}{lll}
12 \cdot 34 \cdot 56 = A & 13 \cdot 45 \cdot 62 = B & 14 \cdot 56 \cdot 23 = C \\
15 \cdot 62 \cdot 34 = D & 16 \cdot 23 \cdot 45 = E & 12 \cdot 35 \cdot 64 = F \\
13 \cdot 46 \cdot 25 = G & 14 \cdot 52 \cdot 36 = H & 15 \cdot 63 \cdot 42 = I \\
16 \cdot 24 \cdot 53 = J & 12 \cdot 36 \cdot 45 = K & 13 \cdot 42 \cdot 56 = L \\
14 \cdot 53 \cdot 62 = M & 15 \cdot 64 \cdot 23 = N & 16 \cdot 25 \cdot 34 = O
\end{array}$$

then any hexagon formed with the six points may be represented by a combination of some two of the letters A, B, &c., viz. the three alternate sides are the lines represented by one letter, and the other three alternate sides the lines represented by the other letter: for example, the hexagon 123456 is AE; and so for the other hexagons.

Any duad AE thus representing a hexagon may be termed a *hexagonal duad*; the number of such duads is sixty.

Each Pascalian line may be denoted by the symbol of the hexagon to which it belongs; thus, the line which belongs to the hexagon AE, is the line AE.

I form the following combinations:

$$\begin{array}{ll}
IMO \cdot DHJ & \text{each involving all the duads 12, \&c. except those of } 123 \cdot 456, \\
DEG \cdot BNO & 124 \cdot 356, \\
ELM \cdot BGJ & 125 \cdot 346, \\
HLN \cdot CGI & 126 \cdot 345, \\
EFI \cdot JKN & 134 \cdot 256, \\
AEH \cdot GKO & 135 \cdot 246, \\
AMN \cdot GDF & 136 \cdot 245, \\
AGJ \cdot ELO & 145 \cdot 236, \\
ABI \cdot DKL & 146 \cdot 235, \\
OKM \cdot BFH & 156 \cdot 234,
\end{array}$$

and also the combinations:

$$\begin{array}{ll}
AEGMI & \text{involving all the duads 12, 13, \&c.,} \\
ABHJN & \\
BGFIO & \\
CDGJK & \\
DEFHL & \\
KLMNO &
\end{array}$$

which I call respectively the *ten-partite* and *six-partite* arrangements.

It is to be remarked that (considering $IMO \cdot DHJ$ as standing for the six duads IM, IO, MO, DH, DJ, HJ, and so for the others) the ten-partite arrangement contains all the sixty hexagonal duads: and in like manner, (considering $AEGMI$ as standing for the ten duads AE, AG, AM, AI, EG, EM, EI, GM, GI, MI, and so for the others) the six-partite arrangement contains all the sixty hexagonal duads.

The 60 Pascalian lines intersect by 4's in the 45 Pascalian points p, by 3's in 20 points g and in 60 points h, and by 2's in 90 points m, 360 points r, 360 points t, 360 points z, and 90 points w.

The intersections of the Pascalian lines thus are

$45p$	counting as	270
$20g$	"	40
$60h$	"	180
$90m$	"	90
$360r$	"	360
$360t$	"	360
$360z$	"	360
$90w$	"	90
$1770 = 60 \cdot 59$,		

and the intersections on each Pascalian line are

$9p$	counting as	9
$2g$	"	2
$3h$	"	6
$3m$	"	3
$12r$	"	12
$12t$	"	12
$12z$	"	12
$3w$	"	3
59.		

For the ten-partite arrangement, any double triad such as $ABI \cdot DKL$ gives 15 intersections; $10 \times 15 = 150$; and any pair of double triads such as $ABI \cdot DKL$ and $AEH \cdot GKO$ gives 36 intersections; $45 \times 36 = 1620$; and these are

$$\begin{array}{c}
10 \times 6g = 60g \; 10 \times 15p = 150 \\
45 \times 2h = 90h \; 45 \times 36 = 1620 \\
180h \\
8r = 360r \\
8t = 360t \\
8z = 360z \\
2m = 90w \\
\hline
150 \; 36 \\
1620 \\
1770.
\end{array}$$

For the six-partite arrangement any pentad such as $ABHJN$ gives 45 intersections; $6 \times 45 = 270$; and any two pentads such as $ABHJN$ and $AEGMI$ give 100 intersections; $15 \times 100 = 1500$; and these are

$$\begin{array}{c}
6 \times 15p = 90 \; 15 \times 6h = 90h \\
60g \; 15 \times 24r = 360r \\
180h \; 15 \times 24t = 360t \\
90m \; 15 \times 24z = 360z \\
45 \times 2 = 90w \\
\hline
15 \times 18p = 270 \; 100 \\
1500 \\
1770.
\end{array}$$

I analyse the intersections of a Pascalian line, say AE, by the remaining 59 Pascalian lines as follows:

Observe that AE belongs to the triad AEH, the complementary triad whereof is GKO; it also belongs to the pentad $AEIMG$. We thus obtain, corresponding to AE, the arrangement

$$\begin{array}{rccccc}
 & H & H & H & H & A \\
 & B & N & J & & \\
\text{cline1-3} & E & F & L & D & \\
\text{cline1-4} & I & M & G & & K \\
 & & & & C & O
\end{array}$$

viz. $HABNJ$, is the pentad which contains HA, the arrangement of the last three letters B, N, J thereof being arbitrary; $HEFLD$ is the pentad that contains HE, but the last three letters are so arranged that the columns HBF, HNL, HJD are each of them a triad, IMG is then the residue of the pentad $AEIMG$, and KGO is the complementary triad to AEH, but the arrangement of the letters IMG, and of the letters KGO, are each of them determinate; viz. these are such that we have $BFIGO$, $NLMKO$, $JDGCK$, each of them a pentad.

And this being so we derive from the arrangement

$2g$	AH, EH;
$3m$	KG, KC, GO;
$6h$	AI, AM, AG; EI, EM, EG;
$12z$	IB, IF, MN, ML, GJ, GD; HB, HF, HN, HL, HJ, HD;
$9p$	AB, AN, AJ; EF, EL, ED; BF, NL, JD;
$12r$	CB, OF, GJ, GD; OB, OF, ON, OL; KN, KL, KJ, KD; FL, FD, LD; BN, BJ, NJ;
$12t$	IG, IO; MK, MO; GK, GC;
$3m$	IM, IC, MG;
$3w$	CD, EK, LM;

viz. the line AE in question meets AH, EH each of them in a point g; KG, KC, GO each in a point m; and so on.

By constructing in the same way an arrangement for each of the lines AH, &c., we find the nature of the point of intersection of any two of the lines AB, AE, AH, &c.; and we may then present the results in a table (see Plate), which shows at a glance what is the point of intersection (whether a point g, m, h, z, p, r, t, or w) of any two of the Pascalian lines.

I further remark that representing the 45 Pascalian points as follows:

$12 \cdot 34 = a$ $12 \cdot 35 = b$ $12 \cdot 36 = c$ $12 \cdot 45 = d$ $12 \cdot 46 = e$ $12 \cdot 56 = f$
$23 \cdot 45 = c$ $23 \cdot 46 = g$ $23 \cdot 56 = h$ $24 \cdot 35 = j$ $24 \cdot 36 = k$ $24 \cdot 56 = l$
$13 \cdot 24 = g$ $13 \cdot 25 = h$ $13 \cdot 26 = i$ $13 \cdot 45 = j$ $13 \cdot 46 = k$ $13 \cdot 56 = l$
$25 \cdot 34 = l$ $25 \cdot 36 = m$ $25 \cdot 46 = n$ $26 \cdot 34 = p$ $26 \cdot 35 = q$ $26 \cdot 45 = r$
$14 \cdot 23 = m$ $14 \cdot 25 = n$ $14 \cdot 26 = o$ $14 \cdot 35 = p$ $14 \cdot 36 = q$ $14 \cdot 56 = r$
$34 \cdot 56 = p$ $35 \cdot 46 = s$ $36 \cdot 45 = t$
$15 \cdot 23 = s$ $15 \cdot 24 = t$ $15 \cdot 26 = u$ $15 \cdot 34 = v$ $15 \cdot 36 = w$ $15 \cdot 46 = x$
$16 \cdot 23 = y$ $16 \cdot 24 = z$ $16 \cdot 25 = \alpha$ $16 \cdot 34 = \beta$ $16 \cdot 35 = \gamma$ $16 \cdot 45 = \delta$

the sixty hexagons and their Pascalian lines then are

AE	123456	$12 \cdot 45\ 23 \cdot 56\ 34 \cdot 61$	$d\nu\pi$
AH	125634	$12 \cdot 63\ 25 \cdot 34\ 56 \cdot 41$	$c\lambda r$
EH	145236	$14 \cdot 23\ 45 \cdot 36\ 52 \cdot 61$	mqa
CK	123654	$12 \cdot 65\ 23 \cdot 54\ 36 \cdot 41$	M
CO	143256	$14 \cdot 25\ 43 \cdot 56\ 32 \cdot 61$	npy
KO	125436	$12 \cdot 43\ 25 \cdot 36\ 54 \cdot 61$	$a\mu h$
AM	126534	$12 \cdot 53\ 26 \cdot 34\ 65 \cdot 41$	$b\lambda r$
AG	125643	$12 \cdot 64\ 25 \cdot 43\ 56 \cdot 31$	$e\lambda l$
AI	124365	$12 \cdot 36\ 24 \cdot 65\ 43 \cdot 51$	$c\kappa v$
EG	132546	$13 \cdot 54\ 32 \cdot 46\ 25 \cdot 61$	$j?a$
BF	126435	$12 \cdot 43\ 26 \cdot 35\ 64 \cdot 51$	amx
FL	124653	$12 \cdot 65\ 24 \cdot 53\ 46 \cdot 31$	fOk
DL	134265	$13 \cdot 26\ 34 \cdot 65\ 42 \cdot 51$	$i\rho t$
BN	132645	$13 \cdot 64\ 32 \cdot 45\ 26 \cdot 51$	$k\epsilon u$
BJ	135426	$13 \cdot 42\ 35 \cdot 26\ 54 \cdot 61$	$g\omega\xi$
JN	153246	$15 \cdot 24\ 53 \cdot 46\ 32 \cdot 61$	$t\alpha\gamma$
GK	125463	$12 \cdot 46\ 25 \cdot 63\ 54 \cdot 31$	$e/\eta j$
KM	126354	$12 \cdot 35\ 26 \cdot 54\ 63 \cdot 41$	$h\iota rq$
IO	152436	$15 \cdot 43\ 52 \cdot 36\ 24 \cdot 61$	$\nu\epsilon\alpha$
MO	143526	$14 \cdot 52\ 43 \cdot 26\ 35 \cdot 61$	$n\zeta h$
EM	145326	$14 \cdot 32\ 45 \cdot 26\ 53 \cdot 61$	$r\iota\pi'\zeta$
EI	154236	$15 \cdot 23\ 54 \cdot 36\ 42 \cdot 61$	$s\tau\xi$
AN	123465	$12 \cdot 46\ 23 \cdot 65\ 34 \cdot 51$	$e\eta\nu$
AJ	124356	$12 \cdot 35\ 24 \cdot 56\ 43 \cdot 61$	$b\kappa\lambda$
AB	126543	$12 \cdot 54\ 26 \cdot 43\ 65 \cdot 31$	$d\zeta l$
DE	154326	$15 \cdot 32\ 54 \cdot 26\ 43 \cdot 61$	$s\tau\tau'\zeta$
EL	132456	$13 \cdot 45\ 32 \cdot 56\ 24 \cdot 61$	$j\eta\xi$
EF	123546	$12 \cdot 54\ 23 \cdot 46\ 35 \cdot 61$	$d\phi y$
CD	143265	$14 \cdot 26\ 43 \cdot 65\ 32 \cdot 51$	$op\xi$
CF	123564	$12 \cdot 56\ 23 \cdot 64\ 35 \cdot 41$	$A\rho$

CG	132564	$13\cdot 56\ 32\cdot 64\ 25\cdot 41$	$l\mu n$
CI	142365	$14\cdot 36\ 42\cdot 65\ 23\cdot 51$	$q\iota c\xi$
MN	146235	$14\cdot 23\ 46\cdot 35\ 62\cdot 51$	$ma\alpha$
GJ	135246	$13\cdot 24\ 35\cdot 46\ 52\cdot 61$	$ga\alpha L$
BI	136245	$13\cdot 24\ 36\cdot 45\ 62\cdot 51$	gru
DG	134625	$13\cdot 62\ 34\cdot 25\ 46\cdot 51$	$i\kappa\alpha\xi$
LM	135624	$13\cdot 62\ 35\cdot 24\ 56\cdot 41$	$i\lambda r$
FI	124635	$12\cdot 63\ 24\cdot 35\ 46\cdot 51$	cdx
BH	136254	$13\cdot 25\ 36\cdot 54\ 62\cdot 41$	$hr\alpha$
FH	125364	$12\cdot 36\ 25\cdot 64\ 53\cdot 41$	$c\upsilon p$
FO	125346	$12\cdot 34\ 25\cdot 46\ 53\cdot 61$	$a\upsilon y$
LO	134256	$13\cdot 25\ 34\cdot 56\ 42\cdot 61$	hpz
DK	126345	$12\cdot 34\ 26\cdot 45\ 63\cdot 51$	$a\iota rw$
KL	124563	$12\cdot 56\ 24\cdot 63\ 45\cdot 31$	$f\xi j$
BO	134526	$13\cdot 52\ 34\cdot 26\ 45\cdot 61$	$h\beta\xi$
NO	152346	$15\cdot 34\ 52\cdot 46\ 23\cdot 61$	$\nu\upsilon y$
BC	132654	$13\cdot 65\ 32\cdot 54\ 26\cdot 41$	leo
CJ	142356	$14\cdot 35\ 42\cdot 56\ 23\cdot 61$	$p\kappa y$
JK	124536	$12\cdot 53\ 24\cdot 36\ 45\cdot 61$	$h\delta$
KN	123645	$12\cdot 64\ 23\cdot 45\ 36\cdot 51$	$e\epsilon w$
DH	143625	$14\cdot 62\ 43\cdot 25\ 36\cdot 51$	$o\lambda w$
HJ	142536	$14\cdot 53\ 42\cdot 36\ 25\cdot 61$	$p\iota a$
HL	136524	$13\cdot 52\ 36\cdot 24\ 65\cdot 41$	$h\iota r$
HN	146325	$14\cdot 32\ 46\cdot 25\ 63\cdot 51$	$m\nu w$
BF	126453	$12\cdot 46\ 26\cdot 53\ 64\cdot 31$	$dc\alpha k$
DJ	153426	$15\cdot 42\ 53\cdot 26\ 34\cdot 61$	$t\alpha\xi$
LN	132465	$13\cdot 46\ 32\cdot 65\ 24\cdot 51$	$k\eta t$
GM	135264	$13\cdot 26\ 35\cdot 64\ 52\cdot 41$	$i\iota rn$
IM	142635	$14\cdot 63\ 42\cdot 35\ 26\cdot 51$	$qd\upsilon$
GI	136425	$13\cdot 42\ 36\cdot 25\ 64\cdot 51$	$g\mu M$

Each Pascalian point belongs to four different hexagons; viz.

a to the hexagons	$\{K,F\}\{D,O\}$	n	$\{B,L\}\{F,N\}$
b	$\{A,K\}\{M,J\}$	o	$\{A,C\}\{B,G\}$
c	$\{A,F\}\{H,I\}$	p	$\{B,D\}\{G,H\}$
d	$\{A,F\}\{B,E\}$	q	$\{C,M\}\{I,K\}$
e	$\{A,K\}\{G,N\}$	r	$\{A,L\}\{H,M\}$
f	$\{G,L\}\{K,F\}$	s	$\{C,E\}\{D,I\}$
g	$\{B,O\}\{I,J\}$	t	$\{J,L\}\{D,N\}$
h	$\{B,L\}\{H,O\}$	u	$\{B,M\}\{I,N\}$
i	$\{D,M\}\{G,L\}$	v	$\{A,O\}\{N,I\}$
j	$\{E,K\}\{C,L\}$	w	$\{D,N\}\{H,K\}$
k	$\{B,L\}\{F,N\}$	x	$\{D,I\}\{F,G\}$
l	$\{A,C\}\{B,G\}$	y	$\{C,N\}\{J,O\}$
m	$\{E,N\}\{H,M\}$	z	$\{E,G\}\{I,L\}$
n	$\{C,M\}\{G,O\}$	α	$\{E,J\}\{G,H\}$
o	$\{A,C\}\{B,G\}$	β	$\{A,D\}\{E,J\}$
p	$\{G,H\}\{F,J\}$	γ	$\{E,O\}\{F,M\}$
q	$\{C,M\}\{I,K\}$	δ	$\{B,K\}\{J,O\}$
r	$\{A,L\}\{H,M\}$	ϵ	$\{B,K\}\{C,N\}$
s	$\{C,E\}\{D,I\}$	ζ	$\{C,E\}\{F,G\}$
t	$\{J,L\}\{D,N\}$	η	$\{A,L\}\{E,N\}$
u	$\{B,M\}\{I,N\}$	θ	$\{F,M\}\{I,L\}$
v	$\{A,O\}\{N,I\}$	ι	$\{H,K\}\{J,L\}$
w	$\{D,N\}\{H,K\}$	κ	$\{A,C\}\{I,J\}$
x	$\{D,I\}\{F,G\}$	λ	$\{A,D\}\{G,H\}$
y	$\{C,N\}\{J,O\}$	μ	$\{G,O\}\{D,K\}$
z	$\{E,G\}\{I,L\}$	ν	$\{F,N\}\{H,O\}$
α	$\{E,J\}\{G,H\}$	ξ	$\{A,O\}\{B,M\}$
β	$\{A,D\}\{E,J\}$	o	$\{B,E\}\{F,J\}$
γ	$\{E,O\}\{F,M\}$	π	$\{A,M\}\{E,K\}$
δ	$\{B,K\}\{J,O\}$	ρ	$\{C,L\}\{D,O\}$
ϵ	$\{B,K\}\{C,N\}$	σ	$\{O,N\}\{J,M\}$
ζ	$\{C,E\}\{F,G\}$	τ	$\{B,E\}\{H,I\}$

I have constructed on a very large scale a figure of the sixty Pascalian lines, and the forty-five Pascalian points, marking them according to the foregoing notation; but the figure is from its complexity, and the inconvenient way in which the points are either crowded together or fly off to a great distance, almost unintelligible.

[*The two plates (I and II) show the complete table of intersections of the 60 Pascalian lines. Each entry shows the type of point (g, m, h, z, p, r, t, or w) at which any two Pascalian lines meet.*]

402. On a Singularity of Surfaces

[From the Quarterly Journal of Pure and Applied Mathematics, vol. ix. (1868), pp. 332—338.]

A SURFACE having a nodal line has in general on this nodal line points where the two tangent planes coincide, or as I propose to term them "pinch-points." Thus, if the nodal line be the curve of complete intersection of any two surfaces $P = 0$, $Q = 0$, then the equation of the general surface having this curve for a nodal line is

$$(a, b, c P, Q)^2 = 0$$

(where a, b, c are any functions of the coordinates), and the pinch-points are given as the intersections of the nodal line $P = 0$, $Q = 0$ with the surface $ac - b^2 = 0$.

Consider the case where the nodal curve is a curve of partial intersection represented by the equations

$$\frac{P}{P'} = \frac{Q}{Q'} = \frac{R}{R'}, \quad \text{or say by the equations } p = 0,\ q = 0,\ r = 0,$$

(viz. p, q, r denote the functions $QR' - Q'R$, $RP' - R'P$, $PQ' - P'Q$ respectively), and consequently we have identically

$$(P, Q, R p, q, r)^2 = 0, \quad (P', Q', R' p, q, r)^2 = 0,$$

or what is the same thing, (λ, μ) being arbitrary,

$$(\lambda P + \mu P', \lambda Q + \mu Q', \lambda R + \mu R' p, q, r)^2 = 0.$$

The general surface having the curve in question for its nodal line is represented by the equation

$$(a, b, c, f, g, h p, q, r)^2 = 0,$$

(where (a, b, c, f, g, h) are any functions of the coordinates), and it is easy to see that the condition for a pinch-point is the same as that which (considering p, q, r as coordinates and all the other quantities as constants), expresses that the line

$$(\lambda P + \mu P', \lambda Q + \mu Q', \lambda R + \mu R' p, q, r)^2 = 0$$

touches the conic

$$(a, b, c, f, g, h p, q, r)^2 = 0,$$

viz. A, B, C, F, G, H being the inverse coefficients, $A = bc - f^2$, &c., this condition is

$$(A, B, C, F, G, H \lambda P + \mu P', \lambda Q + \mu Q', \lambda R + \mu R')^2 = 0,$$

or what is the same thing, the pinch-points are given as the common intersections of the nodal line $p = 0$, $q = 0$, $r = 0$ with each of the three surfaces

$$(A, B, C, F, G, H P, Q, R)^2 = 0,$$
$$(A, B, C, F, G, H P, Q, R)(P', Q', R') = 0,$$
$$(A, B, C, F, G, H P', Q', R')^2 = 0,$$

these last three equations in fact, adding only a single relation to the relations expressed by the equations $p = 0$, $q = 0$, $r = 0$.

If the functions P, Q, R, P', Q', R' are linear functions of the coordinates, then the nodal curve is a cubic curve in space, the equations $p = 0$, $q = 0$, $r = 0$ represent each of them a quadric surface, and the curve is the partial intersection of any two of these quadric surfaces. The pinch-points are the intersections of the cubic curve with the three quadric surfaces represented by the last-mentioned equations; and since the cubic curve lies on each of the three quadric surfaces $p = 0$, $q = 0$, $r = 0$, the number of pinch-points is at once seen to be $= 12$.

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405. An Eighth Memoir on Quantics

[From the Philosophical Transactions of the Royal Society of London, vol. CLVII. (for the year 1867), pp. 627–638.
Received January 17, — Read February 8, 1867.]

[Continuation from p. 150.]

Table No. 82.

Degree, Order, Number, Constituents, quoad the quintic itself A, the quadric covariants B, C, cubic covariants D, E, F, the quartic covariants G, H, I, the quintic covariants J, K, L, and the sextic covariants M, N of § 143, N = new covariant, S = syzygy.

Degree	Order	No.	Constituents
			viz. the absolute constant unity. K
1	5		A
2	10		A^2
	6		C
			B
	2		
3	15		A^3
			AG
	11		F
	9		AB
	7		E N.
	5		D N.
4	20		A^4
	16		A^2C
			AF
	12		C^2, A^2B
	10		AE
	8		AC, EG
	6		I N.
	4		H, B^2 N.
			G N.
5	25		
	21		A^3G
	19		A^2F
			AC
	17	2	A^2B, ...
	15	2	A^2E, GF
	13	2	A^2D, ABC
	11	2	$AI + BF - CE = 0$ S.
		3	AB^2, AH, CD
	7	2	BE, L F.
	5	2	AG, BD
			K
	3	1	M
	1	1	J N.
6	30	1	
	26	1	A^4C
	24	1	A^3F
	22	2	A^2C^2, A^3B, ...
	20	2	A^2E, $A^2\ldots$
	18	3	$E^2 + 4C^3 + A^2D - A^2BG = 0$ S.
	16	2	$A(AI + BE - CE) = 0$ $\dot{S}$.
	14	4	$-6AGD - \frac{1}{4}EF - \frac{1}{4}BC^2 + A^2H = 0$, A^2B^2 S.
	12	3	$AL + 3DF - \frac{1}{4}GI = 0$, ABE S.
	10	4	$AB^2G + 4ABD - A^2G + EF \ldots = 0$, GH S.
	8	2	$AK + \frac{1}{4}BI - \frac{1}{2}DE = 0$ S.
	6	4	$AJ + \frac{1}{4}BH - \frac{1}{2}E^2 - CG \ldots = 0$ S.
			N
	4	1	M N.
	2	1	N.

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Article 253 (253)**.** For the explanation of this I remark that the Table No. 81 shows that we have for the degree and order one covariant; this is the absolute constant unity; for the degree 1 and order 5, 1 covariant, this is the quintic itself, A; for degree 2 and order 10, 1 covariant; this is the square of the quintic, A^2; for same degree and order 6, 1 covariant, which had accordingly to be calculated, viz. this is the covariant C; and similarly whenever the Table No. 81 indicates the existence of a covariant of any degree and order, and there does not exist a product of the covariants previously calculated, having the proper degree and order, then in each such case (shown in the last preceding Table by the letter N) a new covariant had to be calculated. On coming to degree 5, order 11, it appears that the number of asyzygetic invariants is only $= 2$, whereas there exist of the right degree and order the 3 combinations AI, BF, CE; there is here a syzygy, or linear relation, between the combinations in question; which syzygy had to be calculated, and was found to be as shown, $AI + BF - CE = 0$, a result given in the Second Memoir, p. 126. Any such case is indicated by the letter S. At the place degree 6, order 16, we find a syzygy between the combinations A^2I, $A \cdot BF$, $A \cdot CE$; as each term contains the factor A, this is only the last-mentioned syzygy multiplied by A, not a new syzygy, and I have written $\dot{S}$ instead of S. The places degree 6, orders 18, 14, 12, 10, 8, 6 indicate each of them a syzygy, which syzygies, as being of the degree 6, were not given in the Second Memoir, and they were first calculated for the present Memoir. It is to be noticed that in some cases the combinations which might have entered into the syzygy do not all of them do so; thus degree 6, order 14, the syzygy is between the four combinations ACD, EF, BG^2, A^2H, and does not contain the remaining combination A^2B^2. The places degree 6, orders 4, 2, indicate each of them a new covariant, and these, as being of the degree 6, were not given in the Second Memoir, but had to be calculated for the present Memoir.

Article 254 (254)**.** I notice the following results:

$$\begin{array}{rc}
\text{Quadrinvt.} & 6H = 3G^2, \\
\text{Cubinvt.} & 6H = -GP + 54GQ, \\
\text{Disct.} & (a \cdot B + \tfrac{1}{4}M) = (-G, Q, -H \frown a, \tfrac{1}{4}\xi)^2, \\
\text{Jac.} & (B, H) = 6J, \\
\text{Hess.} & \dfrac{M}{D} = N,
\end{array}$$

the last two of which indicate the formation of the covariants given in the new Tables $M =$ No. 83 and $N =$ No. 84: viz. if to avoid fractions we take 3 times the covariant D, being a cubic $(a, \dots \mid x, y)^3$, then the Hessian thereof is a covariant $(a, \dots)^6(x, y)^6$, which is given in Table, No. 83; and in like manner if we form the Jacobian of the Tables B and H which are respectively of the forms $(a, \dots)^2(x, y)^2$, and $(a, \dots)^4(x, y)^4$, this is a covariant $(a, \dots)^6(x, y)^6$, and dividing it by 6 to obtain the coefficients in their lowest terms, we have the new Table, No. 84. I have in these, for greater distinctness, written the numerical coefficients after instead of before, the literal terms to which they belong.

The two new Tables are:

Table No. 83. $M = (a, \dots)^6(x, y)^6$. See § 143.

Table No. 84. $N = (a, \dots)^6(x, y)^6$. See § 143.

ARTICLE NO. 255. FORMULAE FOR THE CANONICAL FORM $ax^5 + by^5 + cz^5 = 0$, WHERE $x + y + z = 0$.

Article 255 (255)**.** The quintic $(a, b, c, d, e, f \mid x, y)^5$ may be expressed in the form

$$au^5 + bv^5 + cw^5,$$

where u, v, w are linear functions of (x, y) such that $u + v + w = 0$. Or, what is the same thing, the quintic may be represented in the canonical form

$$ax^5 + by^5 + cz^5,$$

where $x + y + z = 0$; this is $= (a - c, -c, -c, -c, -c, b - c \mid x, y)^5$, and the different covariants and invariants of the quintic may hence be expressed in terms of these coefficients (a, b, c).

For the invariants we have

$$\begin{aligned} G = J &= b^2c^2 + c^2a^2 + a^2b^2 - 2abc(a + b + c), \\ Q = K &= a^2b^2c^2(bc + ca + ab), \\ -\Pi = L &= a^4b^4c^4, \\ W = I &= 16a^3b^3c^3(b - c)(c - a)(a - b). \end{aligned}$$

[Observe that throughout the present Memoir, the invariants, instead of being called G, Q, $-\Pi$, W are called I, J, K, L, viz. the I, J, K, L in all that follows denote the invariants, and not the covariants denoted by these letters in § 142, 143. Moreover D is used to denote the invariant Q^2, which is in fact the discriminant of the quintic.]

Hence, writing for a moment

$$a + b + c = p, \quad bc + ca + ab = q, \quad abc = r,$$

and therefore

$$J = q^2 - 4pr, \quad K = r^2q, \quad L = r^4,$$

we have

$$(a - b)^2(b - c)^2(c - a)^2 = p^2q^2 - 4q^3 - 4p^3r + 18pqr - 27r^2,$$

and thence

$$I^2 = 16q^3r^6(p^2q^2 - 4q^3 - 4p^3r + 18pqr - 27r^2),$$

and

$$\begin{aligned} &J(K^2 - JL)^2 + 8K^3L - 72JKL^2 - 432L^3 \\ &= r^{12}\{(q^2 - 4pr)\,16p^2 + 8q^6 - (q^2 - 4pr)\,72q^3 - 432r^3\} \\ &= 8r^{12}\{(q^2 - 4pr)(2p^2 - 9q) + q^3 - 54r^2\} \\ &\qquad = 16r^{12}[p^2q^2 - 4q^3 - 4p^3r + 18pqr - 27r^2], \end{aligned}$$

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Article 256 (255 *continued*). that is,

$$I^2 = J(K^2 - JL)^2 + 8K^3L - 72JKL^2 - 432L^3,$$

which is the simplest mode of obtaining the expression for the square of the 18-thic or skew invariant I in terms of the invariants J, K, L of the degrees 4, 8, 12 respectively.

If instead of the invariant K of the degree 8 we consider the invariant D [$= Q^2$ as before-mentioned] of the same degree, this is

$$\begin{aligned} D &= [b^2c^2 + c^2a^2 + a^2b^2 - 2abc(a+b+c)]^2 - 128a^2b^2c^2(bc+ca+ab), \\ &= q^4 - 8q^2pr - 128q^3r^2 + 16p^2r^2, \\ D &= \text{Norm}\,((bc)^2 + (ca)^2 + (ab)^2) \end{aligned}$$

and we have also the following covariants

$$\begin{aligned} B &= (-ac, ab-ac-bc, -bc \mid x,y)^2 = bc\,yz + ca\,zx + ab\,xy. \\ C &= (-ac, -3ac, -3ac, ab-ac-bc, -3bc, -3bc, -bc \mid x,y)^4 \\ &= bc\,y^2z^2 + ca\,z^2x^2 + ab\,x^2y^2. \\ D &= (0, -abc, -abc, 0 \mid x,y)^3 = abc\,xyz. \\ E &= (ab^2 - ac^2 + bc^2 - a^2c - 2abc)\,x^5 \\ &\quad + (-5ac^2 + 5bc^2 - 5abc)\,x^4y \\ &\quad + (-10ac^2 + 10bc^2 - 2abc)\,x^3y^2 \\ &\quad + (-10ac^2 + 10bc^2 + 2abc)\,x^2y^3 \\ &\quad + (-5ac^2 + 5a^2bc)\,xy^4 \\ &\quad + (-ab^2 - ac^2 + bc^2 + b^2c + 2abc)\,y^5 \\ &= (b-c)\,a^2x^5 + (c-a)\,b^2y^5 + (a-b)\,c^2z^5 \\ &\quad - abc(y-z)(z-x)(x-y)(yz+zx+xy). \end{aligned}$$

Article No. 256. Expression of the 18-thic Invariant in terms of the roots.

Article 257 (256). It was remarked by Dr Salmon, that for a quintic $(a,b,c,d,e,f \mid x,y)^5$ which is linearly transformable into the form $(a,0,c,0,e,0 \mid x,y)^5$, the invariant I is $= 0$. Now putting for convenience $y = 1$, and considering for a moment the equation

$$x(x-\beta)(x-\gamma)(x-\delta)(x-\epsilon) = 0,$$

then writing $\dfrac{1-mx}{1-m\alpha}$ herein for x, the transformed equation is

$$X(X-\beta')(X-\gamma')(X-\delta')(X-\epsilon') = 0,$$

where

$$\beta' = \frac{\beta-\alpha}{1-m\beta}, \quad \gamma' = \frac{\gamma-\alpha}{1-m\gamma}, \quad \delta' = \frac{\delta-\alpha}{1-m\delta}, \quad \epsilon' = \frac{\epsilon-\alpha}{1-m\epsilon};$$

hence m may be so determined that $\delta' + \epsilon'$ may be $= 0$; viz. this will be the case if $\delta + \epsilon = 2m\delta\epsilon$, or $m = \dfrac{\delta+\epsilon}{2\delta\epsilon}$. In order that $\beta' + \gamma'$ may be $= 0$, we must of course have $m = \dfrac{\beta+\gamma}{2\beta\gamma}$, and hence the condition that simultaneously $\beta' + \gamma' = 0$ and $\delta' + \epsilon' = 0$ is

$$\frac{\beta+\gamma}{2\beta\gamma} = \frac{\delta+\epsilon}{2\delta\epsilon};$$

that is, $(\beta+\gamma)\delta\epsilon - \beta\gamma(\delta+\epsilon) = 0$. Or putting $x - \alpha$ for x and $\beta - \alpha$, $\gamma - \alpha$, $\delta - \alpha$, &c. for β, γ, &c., we have the equation

$$(x-\alpha)(x-\beta)(x-\gamma)(x-\delta)(x-\epsilon) = 0,$$

which is by the transformation $x-\alpha$ into $m(\gamma-\alpha)\dfrac{x-\alpha}{1-m(x-\alpha)}$ changed into

$$(x-\alpha')(x-\beta')(x-\gamma')(x-\delta')(x-\epsilon')=0$$

(where $\alpha'=\alpha$), and the condition in order that in the new equation it may be possible to have simultaneously $\beta'+\gamma'-2\alpha'=0$, $\delta'+\epsilon'-2\alpha'=0$, is

$$(\beta+\gamma-2\alpha)(\delta-\alpha)(\epsilon-\alpha)-(\delta+\epsilon-2\alpha)(\beta-\alpha)(\gamma-\alpha)=0,$$

or, as this may be written,

$$\begin{vmatrix} 1 & 2\alpha & \alpha^2 \\ 1 & \beta+\gamma & \beta\gamma \\ 1 & \delta+\epsilon & \delta\epsilon \end{vmatrix}=0.$$

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Article 258 (256 *continued*). Hence writing $x+\alpha'$ for x, the last-mentioned equation is the condition in order that the equation

$$(x-\alpha)(x-\beta)(x-\gamma)(x-\delta)(x-\epsilon)=0$$

may be transformable into

$$X(X-\beta')(X-\gamma')(X-\delta')(X-\epsilon')=0,$$

where $\beta'+\gamma'=0$, $\delta'+\epsilon'=0$, that is, into the form $X(X^2-\beta'^2)(X^2-\delta'^2)=0$. Or replacing x, if we have

$$(a,b,c,d,e,f \mid x,y)^5 = a(x-\alpha y)(x-\beta y)(x-\gamma y)(x-\delta y)(x-\epsilon y),$$

then the equation in question is the condition in order that this may be transformable into the form $(a',0,c',0,e',0 \mid x,y)^5$; that is, in order that the 18-thic invariant I may vanish. Hence observing that there are 15 determinants of the form in question, and that any root, for instance α, enters as α^2 in 3 of them and in the simple power α in the remaining 12, we see that the product

$$\begin{vmatrix} 1 & 2\alpha & \alpha^2 \\ 1 & \beta+\gamma & \beta\gamma \\ 1 & \delta+\epsilon & \delta\epsilon \end{vmatrix}$$

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Article 259 (256 *continued*)**.** contains each root in the power 18, and is consequently a rational and integral function of the coefficients of the degree 18, viz. save as to a numerical factor it is equal to the invariant I. And considering the equation $(a,\dots \mid x,y)^5=0$ as representing a range of points, the signification of the equation $I=0$ is that, the pairs (β,γ) and (δ,ϵ) being properly selected, the fifth point α is a focus in regard to the pairs (β,γ) and (δ,ϵ).

ARTICLE NO. 257. EXPRESSION FOR I AS A QUADRATIC FUNCTION OF THE COEFFICIENTS (a,b,c,d,e,f).

Article 260 (257)**.** Reverting to the formulae

$$\begin{aligned}I&=16a^3b^3c^3(b-c)(c-a)(a-b),\\ D&=\{b^2c^2+c^2a^2+a^2b^2-2abc(a+b+c)\}^2-128a^2b^2c^2(bc+ca+ab),\end{aligned}$$

if in these we write $b=a+h$, $c=a+k$, then it is found that the resulting expression for I contains the factor a^3, and if we further write $k=lh$, then I contains the factor h^6; hence, attending only to the terms involving a^3h^6, we have

$$\begin{aligned}I&=16a^3(l-1)l(l+1)(2l-1)(l-2)\,a^3h^6,\\ D&=a^4+8a^3h(l^2-l+1)+\dots;\end{aligned}$$

and hence the equation $I=0$ gives $l=\infty$, $l=1$, $l=0$, $l=-1$, $l=\frac{1}{2}$, $l=2$; these correspond respectively to the forms

$$\begin{aligned}&(a,b,b,b,b,b \mid x,y)^5, \quad (a,b,c,c,c,c \mid x,y)^5,\\ &(a,a,a,a,c,c \mid x,y)^5, \quad (a,a,a,c,c,c \mid x,y)^5,\\ &(a,a,a,a,a,c \mid x,y)^5, \quad (a,b,b,b,b,c \mid x,y)^5,\end{aligned}$$

where the first and fifth are not essentially distinct; and these are the several forms for which the skew invariant I vanishes.

ARTICLE NO. 258. FURTHER INVESTIGATION OF THE CANONICAL FORM.

Article 261 (258)**.** The form $ax^5+by^5+cz^5$, where $x+y+z=0$, has been considered by Sylvester and by Hermite; Sylvester's expression for the quintic is $(\alpha x+\beta y)^5+(\gamma x+\delta y)^5-(\alpha x+\beta y+\gamma x+\delta y)^5$, that is, $(\alpha x+\beta y)^5+(\gamma x+\delta y)^5+(\epsilon x+\zeta y)^5$, where $\alpha x+\beta y+\gamma x+\delta y+\epsilon x+\zeta y=0$, which is the same thing as $ax^5+by^5+cz^5$, where $x+y+z=0$. Hermite uses the forms $x^5+y^5+z^5$, $x^5+y^5-(x+y)^5$, or in general $ax^5+by^5+c(x+y)^5$. The transformation from the general form to the canonical form is readily effected by means of the function

$$\begin{vmatrix}a&b&c&d&e\\ b&c&d&e&f\\ x&y&0&0&0\\ 0&x&y&0&0\\ 0&0&x&y&0\end{vmatrix},$$

Article 262 (258 *continued*)**.** which, equated to zero, gives the covariant C of the Second Memoir; writing this under the form $xy(ax^2+2hxy+by^2)=0$, we have $x=0$, $y=0$, $ax^2+2hxy+by^2=0$; and if in the general quintic we write in place of (x,y) either of the linear factors of $ax^2+2hxy+by^2$, the quintic will be reduced to the form $(a',b',c',d',0,0 \mid x,y)^5$, which can be further reduced to the canonical form. This implies that the quartic function $ax^2+2hxy+by^2$ can be expressed as a product of linear factors, that is, its discriminant must be $=0$, or the function must have a square factor. But the discriminant of the function is the D of the memoir; hence the condition in order that the general quintic may be reducible to the canonical form is $D=0$. But when $D=0$, the covariant C is a perfect square, say $=(lx+my)^4$, and the reduction is effected by writing either $lx+my=0$ or (what is the same thing) by reversing the order of (x,y) and writing $lx+my=0$.

ARTICLE NO. 259. THE CATALECTICANT.

Article 263 (259)**.** The condition that the quintic $(a,b,c,d,e,f \mid x,y)^5$ may be expressed in the canonical form $(a,0,c,0,e,0 \mid x,y)^5$, is that the catalecticant may vanish; this is

$$\begin{vmatrix} a & b & c & d \\ b & c & d & e \\ c & d & e & f \end{vmatrix} = 0,$$

which is the condition that the quintic may be the sum of three fifth powers. For the form $(a,0,c,0,e,0 \mid x,y)^5$, the catalecticant is at once seen to vanish, and conversely if the catalecticant vanish, the form is reducible to $(a,0,c,0,e,0 \mid x,y)^5$. The catalecticant is an invariant of the degree 3, but it is not a new invariant; in fact it is the quadrivariant $=H$, or say it is $=\frac{1}{6}H$.

Article 264 (260)**.** The condition in order that the quintic may be the sum of two fifth powers is that the form may be $(a,0,0,0,e,0 \mid x,y)^5$, that is, ax^5+ey^5; the conditions here are that not only the catalecticant but also the quadrivariant (which for this form is $=0$) shall vanish; that is, the conditions are $H=0$, $I=0$. But these two conditions imply also the vanishing of the cubinvariant, that is, we have $J=0$; and conversely $H=0$, $I=0$, $J=0$ are the conditions in order that the quintic may be the sum of two fifth powers. Observe that the last two conditions $I=0$, $J=0$ imply $H=0$; for if the quadrivariant do not vanish, the quintic can only be the sum of three fifth powers at least, and if in the sum of three fifth powers $ax^5+by^5+cz^5$, we have $c=0$, then the form reduces itself to the sum of two fifth powers.

Article 265 (261). For the canonical form $ax^5+by^5+cz^5$, $x+y+z=0$, we have I, J, K, L as above; we have also $D=Q^2$, and we can express I^2 as a function of J, K, L; but it is further possible to express I^2 as a function of J, D, L. We have

$$\begin{aligned} D &= \text{Norm}\,\{b^2c^2+c^2a^2+a^2b^2-2abc(a+b+c)\} \\ &= \text{Norm}\,\{(bc)^2+(ca)^2+(ab)^2-2abc\cdot p\} \\ &= q^4-8q^2pr-128q^3r^2+16p^2r^2, \end{aligned}$$

and hence I^2 can be expressed as a function of p, q, r, or (which is the same thing) of J, D, L. The expression is in fact

$$I^2=16r^6\{q^2(p^2q^2-4q^3-4p^3r+18pqr-27r^2)\},$$

and we thence obtain the expression in terms of J, D, L.

ARTICLE NO. 262. THE SEXTIC COVARIANTS M, N.

Article 266 (262). Reverting to the covariants B and H, and to the expressions for them in terms of the canonical form, we have

$$\begin{aligned} B &= bc\,yz+ca\,zx+ab\,xy, \\ C &= bc\,y^2z^2+ca\,z^2x^2+ab\,x^2y^2, \end{aligned}$$

and thence we obtain the remaining covariants; but the expressions become very complicated. I merely notice that for the canonical form, the covariants M and N (the sextic covariants of § 254) acquire very simple forms. In fact, using the formulae there given,

$$\begin{aligned} 3D &= (a,\ldots \mid x,y)^3 = 3abc\,xyz, \\ \text{Hess}(3D) &= M = (a,\ldots)^6(x,y)^6, \\ \text{Jac}(B,H) &= 6N, \end{aligned}$$

we find without difficulty

$$\begin{aligned} M &= -a^2b^2c^2\,x^2y^2z^2, \\ N &= a^3b^3c^3\,xyz(yz+zx+xy), \end{aligned}$$

which are the required expressions; it is to be observed that in the expression for N, the factor xyz belongs to the covariant D of the memoir.

ARTICLE NO. 263. THE RESOLVENT SEXTIC.

Article 267 (263)**.** The equation $(a, b, c, d, e, f \mid x, y)^5 = 0$ may be transformed by a linear substitution into the form $(a', 0, c', 0, e', 0 \mid x, y)^5 = 0$, the condition for the possibility of which is that the skew invariant I shall vanish; and the further condition in order that this may be the sum of two fifth powers, that is, that it may be of the form $a'x^5 + e'y^5 = 0$, is that J shall also vanish. We have thus for the determination of the coefficients of the reduced form the two conditions $I = 0$, $J = 0$; and the ratios $a' : c' : e'$ are thereby determined. In fact if we write the reduced equation under the form

$$ax^5 + 10cx^3y^2 + 5exy^4 = 0,$$

then we have

$$\begin{aligned} I &= 16c^2e^2(ae - 4c^2), \\ J &= 16c^2e^2(ae - 4c^2)^2 - 128c^2e^2 \cdot c^2e^2 \\ &= 16c^2e^2\{(ae - 4c^2)^2 - 8c^2e^2\}, \end{aligned}$$

and hence $I = 0$ gives either $c = 0$, or $e = 0$, or $ae - 4c^2 = 0$. If $c = 0$, the equation is $x(ax^4 + 5ey^4) = 0$; if $e = 0$, it is $x^3(ax^2 + 10cy^2) = 0$. But excluding these cases, the condition $I = 0$ gives $ae - 4c^2 = 0$, that is, $e = \frac{4c^2}{a}$; and substituting this value, the equation becomes

$$ax^5 + 10cx^3y^2 + \frac{20c^2}{a}xy^4 = 0,$$

or putting $\frac{x}{y} = \theta$, this is

$$a^2\theta^5 + 10ac\theta^3 + 20c^2\theta = 0,$$

that is,

$$a\theta\left\{\left(a\theta^2 + 5c\right)^2 - 5c^2\right\} = 0.$$

The form of the resolvent sextic is found to be

$$\theta^6 - \frac{5J}{K}\theta^4 + 5\theta^2 - 1 = 0,$$

and the roots of this equation determine the different canonical forms to which the quintic can be reduced.

ARTICLE NO. 264. TABLES FOR THE COVARIANTS OF THE QUINTIC.

Article 268 (264)**.** The Tables Nos. 83 and 84 give the two sextic covariants M and N which have been referred to. As already mentioned, for greater distinctness the numerical coefficients are written after instead of before the literal terms to which they belong. The literal parts are written in the order of the powers of x and y, and the numerical coefficients are placed to the right of the terms.

ARTICLE NO. 265. GEOMETRICAL REPRESENTATION OF THE ROOTS OF A QUINTIC EQUATION.

Article 269 (265)**.** The geometrical representation of the roots of a quintic equation is obtained by means of the canonical form $ax^5 + by^5 + cz^5 = 0$, where $x + y + z = 0$. If in the plane we take a triangle ABC, and upon the sides thereof measure off distances x, y, z respectively, such that $x + y + z = 0$ (where the signs are taken account of), then if $x : y : z$ be the roots of the quintic equation, the locus of the point determined by these distances is a curve of the fifth order; but the more precise determination requires a further discussion.

ARTICLE NO. 266. ON THE CHARACTERS OF A QUINTIC EQUATION.

Article 270 (266)**.** A quintic equation may have the following characters:

1r 5 real roots: character $5r$,
2r 3 real roots and 2 imaginary roots: character $3r + 2i$,
3r 1 real root and 4 imaginary roots: character $r + 4i$.

The discriminant D is positive for the characters $5r$ and $r + 4i$, and negative for the character $3r + 2i$. The absolute invariants J, K, L (or say the invariants of the degrees 4, 8, 12 respectively) serve to determine the character of the equation. The discriminant D is an invariant of the degree 8, expressible in terms of J, K, L.

Article 271 (267)**.** The character of a quintic equation is completely determined by the absolute invariants together with the sign of the skew invariant I. The absolute invariants are functions of J, K, L which are unaltered by any linear transformation of the equation; and the sign of I distinguishes between characters which have the same absolute invariants.

Article 272 (268)**.** The different characters of the roots correspond to different regions in the space of the invariants, separated by the discriminant-surface $D = 0$. The surface $D = 0$ divides the space into regions, and for each region the character of the equation is constant. The passage from one region to another across the surface $D = 0$ corresponds to a change in the character of the equation, arising from the coalescence of two real roots or of two imaginary roots.

Article 273 (269)**.** The discriminant D being a function of the invariants J, K, L, the surface $D = 0$ is a surface in the space (J, K, L). This surface has a nodal line (or cuspidal edge) along which the discriminant and its first derived functions all vanish; and the nodal line corresponds to the case of two pairs of equal roots. The surface has also a triple point corresponding to the case of three equal roots; and the triple point lies upon the nodal line.

ARTICLE NO. 270. THE BICORN CURVE.

Article 274 (270)**.** The form of the Bicorn curve, so far as it is material for the discussion, is shown in the Plate, fig. 2, and it thereby appears that it divides the plane into six regions, which may be called P, Q, R, S, T, U. The equation of the Bicorn is

$$(x^2 + 2xy - y^2)^2 + y^3(2x + y) = 0,$$

or, as it may also be written,

$$(x^2 + 2xy - y^2)^2 = -y^3(2x + y).$$

The curve has a node-cusp at the origin, and it has also a cusp at infinity on the line $y = 0$. The tangent at the node-cusp is $y = 0$, and the tangent at the other cusp is the line infinity. The curve has a double tangent at $x + y = 0$, touching at the points where $x^2 + 2xy - y^2 = 0$.

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Article 275 (271)**.** The Bicorn curve is a curve of the fourth order and of the fourth class; in fact, the order is 4 and the class is also 4. The node-cusp counts as a node, a cusp, an inflexion, and a double tangent; and the cusp at infinity counts as a cusp, an inflexion, and a double tangent. The node-cusp absorbs $6+8+1=15$ inflexions, and the other cusp 8 inflexions; there remains $24-15-8=1$ inflexion, viz. this is the inflexion at infinity, having the line infinity for tangent; there is not, besides the tangent at the node-cusp, any other double tangent of the curve.

Article 276 (272)**.** The Bicorn divides the plane into six regions P, Q, R, S, T, U, and for each of these regions the values of the invariants J, D, and 2^4L-J^2 have determinate signs. We have thus the following Table:

Region	D	J	2^4L-J^2	Character
P	$+$	$-$	$+$	$5r$
Q	$+$	$-$	$-$	$r+4i$
T	$+$	$+$	$+$	$r+4i$
R	$-$	$-$	$\pm$	$3r+2i$
S	$-$	$-$	$+$	$3r+2i$
U	$-$	$+$	$+$	$3r+2i$

so that we have the kingdom $5r$ consisting of the single region P, the kingdom $r+4i$ consisting of the regions Q and T, and the kingdom $3r+2i$ consisting of the regions R, S, and U.

Article 277 (273)**.** For a given equation if D is $= -$, the character is $3r + 2i$; if $D = +$, $J = +$, the character is $r + 4i$; if $D = +$, $J = -$, then, according as $2^4L - J^2$ is $= +$ or is $= -$ the character is $5r$ or $r + 4i$. But in the last case the distinction between the characters $5r$ and $r + 4i$ may be presented in a more general form, involving a parameter μ, arbitrary between certain limits. In fact drawing upwards from the origin, as in Plate, fig. 3, the lines $x - 2y = 0$ and $x + y = 0$, and between them any line whatever $x + \mu y = 0$, the point (x, y), assumed to lie in the region P or Q, will lie in the one or the other region according as it lies on the one side or the other side of the line in question, viz. in the region P if $x + \mu y$ is $= -$, in the region Q if $x + \mu y$ is $= +$. But we have

$$x + \mu y = \frac{2^4L - J^2 + \mu JD}{\dots},$$

and J being by supposition negative, the sign of $2^4L - J^2 + \mu JD$ is opposite to that of $x + \mu y$. The region is thus P or Q according to the sign of $2^4L - J^2 + \mu JD$; and completing the enunciation, we have, finally, the following criteria for the number of real roots of a given quintic equation, viz.

If $D = -$, the character is $3r + 2i$,
If $D = +$, $J = +$, then it is $r + 4i$,
If $D = +$, $J = -$, then μ being any number at pleasure
 between the limits $+1$ and -2, both inclusive, if
 $2^4L - J^2 + \mu JD = +$ the character is $5r$,
 $2^4L - J^2 + \mu JD = -$, „ „ „ $r + 4i$.

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Article 278 (274). The characters $5r$ of the region P and $r+4i$ of the regions Q and T may be ascertained by means of the equation $(a, 0, c, 0, e, 0 \mid x, 1)^5 = 0$, that is

$$x(a\xi^4 + 10c\xi^2 + e) = 0;$$

there is always the real root $\xi = 0$, and the equation will thus have the character $5r$ or $r+4i$ according as the reduced equation $a\xi^4 + 10c\xi^2 + e = 0$ has the character $4r$ or $4ri$. It is clear that (a, e) must have the same sign, for otherwise ξ^2 would have two real values, one positive, the other negative, and the character would be $2r+2i$. And (a, e) having the same sign, then the character will be $4r$, if ξ^2 has two real positive values, that is, if $ae - 4c^2$ is $= -$, and the sign of c be opposite to that of a and e, or, what is the same thing, if ce be $= -$; but if these two conditions are not satisfied, then the values of ξ^2 will be imaginary, or else real and negative, and in either case the character will be $4ri$.

Article 279 (275). Now, for the equation in question, putting in the Tables $b = d = f = 0$, we find for the invariants

$$\begin{aligned} J &= 16c^2e^2(ae - 4c^2), \\ D &= 16c^2e^2\{(ae - 4c^2)^2 - 8c^2e^2\}, \\ L &= a^4e^4, \end{aligned}$$

whence we can calculate the values of J, D, L for any given values of a, c, e. In particular, the equation $a\xi^4 + 10c\xi^2 + e = 0$ has the character $4r$ if $ae - 4c^2 = -$ and $c = -$, that is, if $J = +$ and $c = -$; but we have $D = +$ when $ae - 4c^2 = +$, or when $(ae - 4c^2)^2 - 8c^2e^2 = +$; and the sign of D being given, the character is determined as above.

Article 280 (276)**.** But if $D = +$, $J = -$, then μ being any number at pleasure between the limits $+1$ and -2, both inclusive, if

$$\begin{aligned} 2^4L - J^2 + \mu JD &= + \quad \text{the character is } 5r, \\ 2^4L - J^2 + \mu JD &= -, \quad \text{,, ,, ,, } r + 4i. \end{aligned}$$

Article 281 (277)**.** The characters $5r$ of the region P and $r + 4i$ of the regions Q and T may be ascertained by means of the equation $(a, 0, c, 0, e, 0 \mid x, 1)^5 = 0$, that is

$$x(a\xi^4 + 10c\xi^2 + e) = 0;$$

there is always the real root $\xi = 0$, and the equation will thus have the character $5r$ or $r + 4i$ according as the reduced equation $a\xi^4 + 10c\xi^2 + e = 0$ has the character $4r$ or $4ri$. It is clear that (a, e) must have the same sign, for otherwise ξ^2 would have two real values, one positive, the other negative, and the character would be $2r + 2i$. And (a, e) having the same sign, then the character will be $4r$, if ξ^2 has two real positive values, that is, if $ae - 4c^2$ is $= -$, and the sign of c be opposite to that of a and e, or, what is the same thing, if ce be $= -$; but if these two conditions are not satisfied, then the values of ξ^2 will be imaginary, or else real and negative, and in either case the character will be $4ri$.

Article 282 (278)**.** The Bicorn curve has been constructed by means of the equation

$$(x^2 + 2xy - y^2)^2 + y^3(2x + y) = 0,$$

and the Plate shows the form of the curve with its six regions. The node-cusp at the origin is a singular point of a special character; and the other cusp is at infinity on the axis of x. The double tangent is the line $x + y = 0$, touching the curve at the two points determined by $x^2 + 2xy - y^2 = 0$, $x + y = \sqrt[4]{-\frac{1}{4}y^3(2x + y)}$.

Article No. 279. On the Geometrical Representation.

Article 283 (279)**.** The geometrical representation of the roots of a quintic equation is obtained by means of the canonical form. If in the plane we take a triangle ABC, and upon the sides thereof measure off distances proportional to x, y, z respectively, such that $x + y + z = 0$ (where account is taken of the signs), then if $x : y : z$ be proportional to the roots of the quintic equation, the locus of the point determined by these distances is a curve of the fifth class; and the different characters of the equation correspond to different configurations of this curve.

Article No. 280. Discussion of the Characters.

Article 284 (280)**.** The characters of a quintic equation are completely determined by the absolute invariants and the sign of I. The absolute invariants are the ratios of $J^3 : K^3 : L$ which are unaltered by any linear transformation; and the sign of I distinguishes between the characters $5r$ and $r+4i$ when the discriminant is positive and the quadrivariant negative. The different characters correspond to different regions of the discriminant-surface, separated by the nodal line and the cuspidal edge; and the complete discussion requires the consideration of the Bicorn curve and its six regions.

Article 285 (281)**.** The form of the Bicorn, so far as it is material for the discussion, is also shown in the Plate, fig. 3, and it thereby appears that it divides the plane into three regions on one side and three on the other. For the region P, we must have $J = -$, and for TU we must have $J = +$. Hence in connexion with the bicorn, considering the line $y = 0$, we have the six regions P, Q, R, S, T, U. It has just been seen that for P, Q, R, S the equations have $J = -$ and for T, U the equations have $J = +$; and the regions have determinate characters, which for R, S, and U (since here D is $= -$) is at once seen to be $3r + 2i$, and which, as will presently appear, is ascertained to be $5r$ for P and $r + 4i$ for Q and T.

Article 286 (282)**.** We have thus the Table

	D	J	$2^4L - J^2$		
P,	$D = +$,	$J = -$,	$2^4L - J^2 = +$	}	$5r$,
Q,	$D = +$,	$J = -$,	$2^4L - J^2 = -$	}	$r + 4i$,
T,	$D = +$,	$J = +$,	$2^4L - J^2 = +$	}	$r + 4i$,
R,	$D = -$,	$J = -$,	$2^4L - J^2 = \pm$	}	$3r + 2i$,
S,	$D = -$,	$J = -$,	$2^4L - J^2 = +$	}	$3r + 2i$,
U,	$D = -$,	$J = +$,	$2^4L - J^2 = +$	}	$3r + 2i$,

so that we have the kingdom $5r$ consisting of the single region P, the kingdom $r + 4i$ consisting of the regions Q and T, and the kingdom $3r + 2i$ consisting of the regions R, S, and U.

Article 287 (283)**.** For a given equation if D is $= -$, the character is $3r + 2i$; if $D = +$, $J = +$, the character is $r + 4i$; if $D = +$, $J = -$, then, according as $2^4L - J^2$ is $= +$ or is $= -$ the character is $5r$ or $r + 4i$. But in the last case the distinction between the characters $5r$ and $r + 4i$ may be presented in a more general form, involving a parameter μ, arbitrary between certain limits. In fact drawing upwards from the origin, as in Plate, fig. 3, the lines $x - 2y = 0$ and $x + y = 0$, and between them any line whatever $x + \mu y = 0$, the point (x, y), assumed to lie in the region P or Q, will lie in the one or the other region according as it lies on the one side or the other side of the line in question, viz. in the region P if $x + \mu y$ is $= -$, in the region Q if $x + \mu y$ is $= +$. But we have

$$2^4L - J^2 + \mu JD$$

corresponds to $x + \mu y$, and J being by supposition negative, the sign of $2^4L - J^2 + \mu JD$ is opposite to that of $x + \mu y$. The region is thus P or Q according to the sign of $2^4L - J^2 + \mu JD$; and completing the enunciation, we have, finally, the following criteria for the number of real roots of a given quintic equation, viz.

If $D = -$, the character is $3r + 2i$,
If $D = +$, $J = +$, then it is $r + 4i$,
If $D = +$, $J = -$, then μ being any number at pleasure
between the limits $+1$ and -2, both inclusive, if
$2^4L - J^2 + \mu JD = +$ the character is $5r$,
$2^4L - J^2 + \mu JD = -$, „ „ „ $r + 4i$.

Article 288 (284). The characters $5r$ of the region P and $r+4i$ of the regions Q and T may be ascertained by means of the equation $(a,0,c,0,e,0 \mid x,1)^5=0$, that is

$$x(a\xi^4+10c\xi^2+e)=0;$$

there is always the real root $\xi=0$, and the equation will thus have the character $5r$ or $r+4i$ according as the reduced equation $a\xi^4+10c\xi^2+e=0$ has the character $4r$ or $4ri$. It is clear that (a,e) must have the same sign, for otherwise ξ^2 would have two real values, one positive, the other negative, and the character would be $2r+2i$. And (a,e) having the same sign, then the character will be $4r$, if ξ^2 has two real positive values, that is, if $ae-4c^2$ is $=-$, and the sign of c be opposite to that of a and e, or, what is the same thing, if ce be $=-$; but if these two conditions are not satisfied, then the values of ξ^2 will be imaginary, or else real and negative, and in either case the character will be $4ri$.

Article 289 (285). Now, for the equation in question, putting in the Tables $b=d=f=0$, we find for the invariants

$$\begin{aligned}
J &= 16c^2e^2(ae-4c^2),\\
D &= 16c^2e^2\{(ae-4c^2)^2-8c^2e^2\},\\
L &= a^4e^4,
\end{aligned}$$

whence we can calculate the values of J, D, L for any given values of a, c, e. In particular, the equation $a\xi^4+10c\xi^2+e=0$ has the character $4r$ if $ae-4c^2=-$ and $c=-$, that is, if $J=+$ and $c=-$; but we have $D=+$ when $ae-4c^2=+$, or when $(ae-4c^2)^2-8c^2e^2=+$; and the sign of D being given, the character is determined as above.

ARTICLE NO. 286. ON THE DETERMINATION OF THE CHARACTER.

Article 290 (286)**.** The character of a quintic equation may be determined by the following process. First calculate the invariants J, D, L. Then if $D = -$, the character is $3r + 2i$. If $D = +$ and $J = +$, the character is $r + 4i$. If $D = +$ and $J = -$, then calculate $2^4 L - J^2$; if this is $+$, the character is $5r$; if $-$, the character is $r + 4i$. The case where $2^4 L - J^2 = 0$ is the critical case where the character changes from $5r$ to $r + 4i$, and corresponds to a point on the boundary between the regions P and Q.

Article 291 (287)**.** The discriminant D is an invariant of the degree 8, and may be expressed in terms of the fundamental invariants J, K, L. The expression is of the form

$$D = K^2 + \text{terms involving } J \text{ and } L,$$

and the complete expression is a complicated function of these invariants. The surface $D = 0$ is the discriminant-surface, and it divides the space of invariants into regions corresponding to the different characters.

Article 292 (288)**.** The nodal line of the discriminant-surface corresponds to the case where the quintic has two pairs of equal roots. This is the case where the catalecticant vanishes, and where the quintic can be expressed as the sum of three fifth powers. The triple point corresponds to the case of three equal roots, and lies on the nodal line. The different sections of the discriminant-surface by planes give rise to curves which have been studied by Sylvester and others.

Article 293 (289)**.** The complete discussion of the characters requires the consideration of the absolute invariants, that is, the ratios which are unaltered by any linear transformation. These absolute invariants may be taken to be $\dfrac{J^3}{K^3}$ and $\dfrac{L}{K^2}$, or any other independent ratios formed from J, K, L; and the discriminant being expressed as a function of these, the different regions of the surface correspond to the different characters.

Article No. 290. On the Special Cases.

Article 294 (290)**.** The special cases of the quintic equation arise when the discriminant vanishes, that is, when two or more roots are equal. The case of a single pair of equal roots corresponds to a point on the discriminant-surface; the case of two pairs of equal roots corresponds to a point on the nodal line; and the case of three equal roots corresponds to the triple point. The case of four equal roots or of five equal roots corresponds to higher singularities which are not in general considered in the theory of the quintic.

Article 295 (291)**.** When the discriminant vanishes, the character of the equation changes by the coalescence of two roots. If two real roots coalesce, the character changes from $5r$ to $3r + 2i$, or from $3r + 2i$ to $r + 4i$, according to the circumstances. If two imaginary roots coalesce, the character changes from $r + 4i$ to $3r + 2i$. The complete enumeration of all the possible cases requires the consideration of the different ways in which the roots may coalesce.

Article 296 (292)**.** The case of a double root is distinguished from the case of two pairs of double roots by the condition that the first minors of the catalecticant do not all vanish. For a double root, the catalecticant vanishes, but not all its first minors; for two pairs of double roots, the first minors all vanish, but not the second minors; and so on. The complete theory of these special cases is intimately connected with the theory of the canonical form and the theory of the discriminant.

Article 297 (293)**.** The discriminant of the quintic is a function of the degree 8 in the coefficients, and it may be expressed as the resultant of the quintic and its first derived function. The discriminant is also expressible in terms of the invariants J, K, L, and the expression is a complicated function involving these invariants in various combinations. The discriminant is the most important invariant for the determination of the character of the equation.

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ARTICLE NO. 294. ON THE APPLICATION TO NUMERICAL EQUATIONS.

Article 298 (294)**.** The application of the preceding theory to numerical equations is effected by calculating the invariants J, D, L for the given equation, and then determining the character by the criteria above established. The calculation of the invariants is a process of considerable labour, but it may be simplified by the use of tables or by special devices. For equations with numerical coefficients, the invariants may be calculated directly from the coefficients, and the character determined without solving the equation.

Article 299 (295)**.** The invariants J, K, L may be calculated by the formulae given in the memoir, or by any other equivalent formulae. The discriminant D is then calculated from these invariants, and the character determined by the signs of D, J, and $2^4L - J^2$. The process is completely definite, and gives a direct determination of the number of real and imaginary roots without the necessity of solving the equation.

Article 300 (296)**.** The theory of the characters of a quintic equation is analogous to the theory of the discriminant of a quadratic or cubic equation. For a quadratic equation, the discriminant determines whether the roots are real or imaginary; for a cubic equation, the discriminant determines whether the roots are all real or one real and two imaginary; and for a quintic equation, the discriminant together with the other invariants determines the number of real and imaginary roots in each of the possible cases.

Article 301 (297)**.** The complete determination of the character of a quintic equation requires the knowledge of the absolute invariants and the sign of the skew invariant I. The absolute invariants determine the ratios of the coefficients of the canonical form, and the sign of I determines the particular character within each class. The theory is thus complete, and gives a definite answer for every quintic equation.

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Article No. 298. Summary of Results.

Article 302 (298). The results obtained in the present memoir may be summarized as follows:

1. The covariants of the quintic up to the degree 6 have been completely calculated and tabulated.
2. The syzygies between these covariants have been determined.
3. The canonical form $ax^5 + by^5 + cz^5$ $(x + y + z = 0)$ has been discussed, and the invariants and covariants expressed in terms of the coefficients (a, b, c).
4. The 18-thic invariant I has been expressed as a function of the invariants J, K, L, and also in terms of the roots.
5. The characters of the quintic equation have been completely determined by means of the invariants and the Bicorn curve.
6. The discriminant-surface and its singularities have been discussed.

Article 303 (299). The memoir concludes with a discussion of the general theory of quantics, and of the methods applicable to the investigation of their invariants and covariants. The methods employed are those which have been developed in the preceding memoirs, and the results are presented in a form suitable for application to particular cases. The theory of the quintic is thus brought to a state of completeness which renders it available for all the purposes of algebraical and geometrical investigation.

Article 304 (300). The Tables of covariants which have been given in the memoir serve to exhibit the complete system of covariants up to the degree 6, and they may be used for the calculation of any particular covariant which may be required. The syzygies serve to reduce the number of independent covariants, and to express the relations between them. The theory is thus presented in a compact and manageable form.

Article 305 (301). The discussion of the characters of the quintic equation by means of the Bicorn curve is a novel feature of the memoir, and it serves to connect the algebraical theory with the geometrical theory of curves. The Bicorn curve is a curve of the fourth order with a node-cusp and a cusp at infinity, and it divides the plane into six regions corresponding to the different characters of the quintic equation. The criteria for the determination of the character are expressed in a simple form involving the invariants J, D, L.

Article 306 (302)**.** The theory of the quintic equation here developed is applicable to the solution of problems in algebra, geometry, and the theory of equations. The determination of the character of a given equation is a problem of frequent occurrence, and the methods here given render it completely determinate. The theory of the canonical form is also of importance in connexion with the reduction of the general equation to the simplest form, and the expression of the invariants and covariants in terms of the coefficients of the canonical form is a result of considerable value.

Article 307 (303)**.** The memoir contains also a discussion of the superimaginary and subimaginary transformations, and of their effect upon the character of the equation. It is shown that the imaginary transformation $x : y$ into $iX : Y$ changes the sign of the quadrivariant J, and that the real equations which correspond to given values of the absolute invariants must be derivable one from another by real transformations. The absolute invariants, together with the sign of J, thus constitute a complete system of auxiliars.

Article 308 (304)**.** The application of the theory to the auxiliars of a quintic is given in Article 327. It is shown that for a quintic equation $(a, \dots \mid x, y)^5 = 0$, after any real transformation whatever, the only cases in which an imaginary transformation produces a real equation of an altered character are when the equation is of the form $(a', 0, c', 0, e', 0 \mid x', y')^5 = 0$ $(c' \neq 0)$, or when it is of the form $(a', 0, 0, 0, e', 0 \mid x', y')^5 = 0$. In the latter case the transformation x', y' into $X\sqrt{-1}, Y$ gives the real equation $X(a'X^4 - 5e'Y^4) = 0$.

Article 309 (305)**.** For the form $(a', 0, 0, 0, e', 0 \mid x, y)^5$, and consequently for the form $(a, \dots \mid x, y)^5$ from which it is derived, we have $J = 0$; this case is therefore excluded from consideration. The remaining case is $(a', 0, c', 0, e', 0 \mid x', y')^5 = 0$, which is by the imaginary transformation $x' : y'$ into $iX : Y$ converted into $(a', 0, -c', 0, e', 0 \mid X, Y)^5 = 0$; for the first of the two forms we have $J = 16a'c'e'^2$, and for the second of the two forms $J = -16a'c'e'^2$, that is, the two values of J have opposite signs. Hence considering an equation $(a, b, c, d, e, f \mid x, y)^5 = 0$ for which J is not $= 0$, whenever this is by an imaginary transformation converted into a real equation, the sign of J is reversed; and it follows that, given the values of the absolute invariants and the value of J (or what is sufficient, the sign of J), the different real equations which correspond to these data must be derivable one from another by real transformations, and must consequently have a determinate character; that is, the Absolute Invariants, and J, constitute a system of auxiliars.

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ANNEX. ANALYTICAL THEOREM IN RELATION TO A BINARY QUANTIC OF ANY ORDER.

Article 310 (306)**.** The foregoing theory of the superimaginary and subimaginary transformations leads to the following general theorem. If $(a, \dots \mid x, y)^n = 0$ be a binary quantic of any order n, and if by any linear transformation whatever the equation is converted into a real equation of the same or of a different character, then the transformation may be reduced to one of the following forms:

1. A real transformation, which preserves the character of the equation;
2. An imaginary transformation of the form $x : y$ into $kX : Y$, where k^p is real for some divisor p of n, which may or may not alter the character;
3. A superimaginary transformation, which changes the character in a determinate manner.

Article 311 (307)**.** The theorem may be established by considering the general linear transformation

$$x = \alpha X + \beta Y, \quad y = \gamma X + \delta Y,$$

where α, β, γ, δ are arbitrary quantities. If the transformed equation is to be real, then the coefficients of the transformed equation must all be real; and this condition restricts the possible values of α, β, γ, δ. The complete discussion of the cases leads to the classification above given.

Article 312 (308)**.** It is to be observed that the superimaginary transformation is only possible when the equation has a certain special form; in particular, it is necessary that the equation should be expressible in terms of x^p and y^p for some divisor p of n. When this condition is satisfied, the transformation $x : y$ into $iX : Y$ (where $i = \sqrt{-1}$) gives a real equation, and the character is altered in a determinate manner. The cases in which this occurs for the quintic equation have been discussed in the preceding articles.

Article 313 (309)**.** The subimaginary transformation is the transformation $x : y$ into $kX : Y$, where k^p is real for some divisor p of n. If p is odd, the transformation is equivalent to a real transformation, and the character is unaltered. If p is even and k^p is positive, the same is true. But if p is even and k^p is negative, then the transformation is not equivalent to any real transformation, and the character of the equation may be altered. The alteration of character in this case has been discussed for the quintic equation, and the general theory is analogous.

Article 314 (310). The theorem may be further extended to the case of a system of binary quantics, and to the case of ternary and higher quantics. The general principle is that the imaginary transformations which produce real equations are of a limited class, and that their effect upon the character of the equation is determinate. The theory of these transformations is thus a branch of the general theory of invariants, and it has important applications in algebra and geometry.

Article 315 (311). The application of the theorem to the theory of equations is immediate. Given a binary quantic of any order, the theorem enables us to determine whether the equation can be transformed into a real equation of a different character by an imaginary transformation, and if so, what is the nature of the transformation and the effect upon the character. The theorem also enables us to classify the different transformations which leave the character unaltered, and to distinguish them from those which alter it.

Article 316 (312). The proof of the theorem depends upon the theory of the roots of unity, and upon the properties of the linear transformation of binary forms. The general linear transformation involves four arbitrary parameters, and the condition that the transformed equation shall be real imposes certain restrictions upon these parameters. The complete discussion of the cases leads to the classification of transformations into real, subimaginary, and superimaginary, and to the determination of the effect of each class of transformations upon the character of the equation.

Article 317 (313). The memoir concludes with a discussion of the general theory of the transformations of binary quantics, and of their application to the theory of equations. The results obtained are of a general character, and they serve to unify and complete the theory of the invariants and covariants of binary forms. The theory of the quintic equation is a particular case of the general theory, and the results obtained for the quintic are illustrative of the general principles.

ARTICLE NO. 314. ON THE THEORY OF TRANSFORMATIONS.

Article 318 (314)**.** The theory of the transformation of binary quantics is a fundamental part of the theory of invariants. The general linear transformation

$$x = \alpha X + \beta Y, \quad y = \gamma X + \delta Y,$$

where $\alpha\delta - \beta\gamma \neq 0$, transforms any binary quantic into another binary quantic of the same order; and the invariants and covariants of the original form are transformed into invariants and covariants of the new form. The theory of these transformations has been developed by Cayley, Sylvester, and others, and it has important applications in algebra and geometry.

Article 319 (315)**.** The transformation is said to be real when the coefficients α, β, γ, δ are all real; and in this case, if the original equation is real, the transformed equation is also real. The transformation is said to be imaginary when some or all of the coefficients are imaginary; and in this case, even if the original equation is real, the transformed equation may be imaginary. The cases in which an imaginary transformation produces a real equation are of special interest, and they have been discussed in the preceding articles.

Article 320 (316)**.** The transformation $x : y$ into $iX : Y$ is a particular case of the superimaginary transformation. If the original equation is $(a, \ldots \mid x, y)^n = 0$, the transformed equation is $(a, \ldots \mid iX, Y)^n = 0$; and if the original equation contains only even powers of x and y, or only powers which are multiples of some divisor of n, then the transformed equation will be real. The effect of the transformation upon the character of the equation is to change the sign of certain invariants, and the complete discussion of this effect has been given for the case of the quintic.

Article 321 (317)**.** The transformation $x : y$ into $kX : Y$, where k^p is real for some divisor p of n, is the subimaginary transformation. If p is odd, the transformation is equivalent to a real transformation; if p is even and k^p is positive, the same is true; but if p is even and k^p is negative, the transformation is not equivalent to any real transformation, and the character of the equation may be altered. The cases in which this occurs have been discussed for the quintic, and the general theory is the same for quantics of any order.

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Article 322 (318)**.** The theory of the transformation of binary quantics has important applications in the theory of equations. The determination of the character of an equation, that is, the number of its real and imaginary roots, is a problem of fundamental importance; and the theory of transformations enables us to determine the character without solving the equation. The invariants and covariants of the quantic furnish the means of this determination, and the theory of transformations serves to connect the different forms of the equation and to exhibit the relations between their invariants.

Article 323 (319)**.** The memoir concludes with a general discussion of the theory of binary quantics and their transformations. The results obtained are of a comprehensive character, and they serve to complete the theory which has been developed in the preceding memoirs. The theory of the quintic equation is a particular application of the general theory, and the methods employed are applicable to quantics of any order. The memoir thus forms a contribution to the general theory of algebraical forms, and it illustrates the power and generality of the methods of invariant theory.

Article 324 (320)**.** The following is a brief recapitulation of the principal results of the memoir:

1. The complete system of covariants of the quintic up to the degree 6 has been calculated and tabulated.
2. The syzygies between these covariants have been determined.
3. The canonical form $ax^5 + by^5 + cz^5$ $(x + y + z = 0)$ has been discussed.
4. The 18-thic invariant has been expressed in terms of the invariants J, K, L.
5. The characters of the quintic equation have been determined by means of the Bicorn curve.
6. The theory of superimaginary and subimaginary transformations has been developed.
7. The application to the auxiliars of a quintic has been given.

Article 325 (321)**.** The theory of the characters of the quintic equation is thus brought to a state of completeness. The discriminant-surface, with its nodal line and triple point, furnishes a complete classification of the possible characters; and the criteria for the determination of the character are expressed in terms of the invariants J, D, L, which can be calculated for any given equation. The theory is analogous to the theory of the discriminant of the quadratic and cubic equations, and it extends those theories to the case of the quintic.

Article 326 (322)**.** The geometrical representation of the roots of a quintic equation by means of the canonical form $ax^5 + by^5 + cz^5 = 0$ $(x + y + z = 0)$ is a representation of considerable interest. If in the plane we take a triangle ABC, and upon the sides thereof measure off distances x, y, z such that $x + y + z = 0$, then the locus of the point so determined, when $x : y : z$ are the roots of the quintic equation, is a curve of the fifth class. The different characters of the equation correspond to different configurations of this curve, and the discriminant-surface corresponds to the cases where the curve acquires singularities.

Article 327 (323)**.** The theory of the transformations of binary quantics is thus seen to be a theory of wide extent and great generality. The methods employed are applicable not only to the quintic but to quantics of any order; and the results obtained for the quintic are typical of the results which may be expected for higher quantics. The theory of invariants and covariants, which is the foundation of the whole, is a theory of the most comprehensive character, and it has applications in every branch of algebra and geometry.

Article 328 (324)**.** There remains only the subimaginary transformation, viz. this has been reduced to $x : y$ into $kX : Y$, the transformed equation is $(a, \dots \mid kX, Y)^n = 0$, and this will be a real equation if some power k^p of k (p not greater than n) be real, and if the equation $(a, \dots \mid x, y)^n = 0$ contain only terms wherein the index of x (or that of y) is a multiple of p. Assuming that it is the index of y which is a multiple, the form of the equation is in fact $(x^p, y^p)^m = 0$, $(n = mp + \alpha)$, and the transformed equation is $X^\alpha(k^p X^p, Y^p)^m = 0$, which is a real equation.

Article 329 (325)**.** It is to be observed that if p be odd, then writing $k^p = K$ (K real) and taking k' the real p-th root of K, then the very same transformed equation would be obtained by the real transformation $x : y$ into $k'X : Y$; so that the equation obtained by the imaginary transformation, being also obtainable by a real transformation, has the same character as the original equation.

Article 330 (326). Similarly if p be even, if K be real and positive, the equation $k^p = K$ has a real root k' which may be substituted for the imaginary k, and the transformed equation will have the same character as the original equation; but if K be negative, say $K = -\Lambda$ (as may be assumed without loss of generality), then there is no real transformation equivalent to the imaginary transformation, and the equation given by the imaginary transformation has not of necessity the same character as the original equation; and there are in fact cases in which the character is altered. Thus if $p = 2$, and the original equation be $x(x^2, y^2)^m = 0$, or $(x^2, y^2)^m = 0$, then making the transformation $x : y$ into $iX : Y$, the transformed equation will be $X(X^2, -Y^2)^m = 0$ or $(X^2, -Y^2)^m = 0$, giving imaginary roots $X^2 + aY^2 = 0$ corresponding to real roots $x^2 - ay^2 = 0$.

ARTICLE NO. 327. APPLICATION TO THE AUXILIARS OF A QUINTIC.

Article 331 (327). Applying what precedes to a quintic equation $(a, \dots \mid x, y)^5 = 0$, this after any real transformation whatever will assume the form $(a, \dots \mid x', y')^5 = 0$; and the only cases in which we can have an imaginary transformation producing a real equation of an altered character is when this equation is $(a', 0, c', 0, e', 0 \mid x', y')^5 = 0$ $(c' \neq 0)$, or when it is $(a', 0, 0, 0, e', 0 \mid x', y')^5 = 0$, viz. when it is $x'(a'x'^4 + 10c'x'^2y'^2 + 5e'y'^4) = 0$, or $x'^4(a'x'^2 + 5e'y'^2) = 0$. In the latter case the transformation x', y' into $X\sqrt{-1}, Y$ gives the real equation $X(a'X^4 - 5e'Y^4) = 0$. I observe however that for the form $(a', 0, 0, 0, e', 0 \mid x, y)^5$, and consequently for the form $(a, \dots \mid x, y)^5$ from which it is derived we have $J = 0$; this case is therefore excluded from consideration. The remaining case is $(a', 0, c', 0, e', 0 \mid x', y')^5 = 0$, which is by the imaginary transformation $x' : y'$ into $iX : Y$ converted into $(a', 0, -c', 0, e', 0 \mid X, Y)^5 = 0$; for the first of the two forms we have $J = 16a'c'e'^2$, and for the second of the two forms $J = -16a'c'e'^2$, that is, the two values of J have opposite signs. Hence considering an equation $(a, b, c, d, e, f \mid x, y)^5 = 0$ for which J is not $= 0$, whenever this is by an imaginary transformation converted into a real equation, the sign of J is reversed; and it follows that, given the values of the absolute invariants and the value of J (or what is sufficient, the sign of J), the different real equations which correspond to these data must be derivable one from another by real transformations, and must consequently have a determinate character; that is, the Absolute Invariants, and J, constitute a system of auxiliars.

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ARTICLE NO. 328. ON THE GEOMETRICAL THEORY OF THE QUINTIC.

Article 332 (328)**.** The geometrical theory of the quintic equation is closely connected with the algebraical theory. The equation $(a, b, c, d, e, f \mid x, y)^5 = 0$ may be regarded as determining a range of five points on a straight line; and the invariants and covariants of the quantic have geometrical interpretations in connexion with this range of points. The discriminant vanishes when two of the points coincide; and the different characters of the equation correspond to different dispositions of the five points upon the real line.

Article 333 (329)**.** The canonical form $ax^5 + by^5 + cz^5 = 0$ $(x + y + z = 0)$ has a geometrical interpretation in terms of a triangle. If ABC be a triangle, and if upon the sides BC, CA, AB we measure off distances x, y, z such that $x + y + z = 0$, then any point in the plane determines a system of values of x, y, z; and conversely, any system of values of x, y, z satisfying this condition determines a point in the plane. The equation $ax^5 + by^5 + cz^5 = 0$ thus represents a curve of the fifth order in the plane of the triangle; and the roots of the quintic equation correspond to the intersections of this curve with the sides of the triangle.

Article 334 (330)**.** The theory of the characters of the quintic equation is thus seen to have a geometrical aspect. The different characters correspond to different configurations of the five points; and the discriminant-surface, with its nodal line and triple point, corresponds to the cases where the configuration acquires singularities. The theory of the Bicorn curve is a particular aspect of this geometrical theory, and it serves to connect the algebraical and geometrical points of view.

Article 335 (331)**.** The memoir concludes with a discussion of the general principles which underlie the theory of binary quantics and their invariants. The theory is presented as a systematic development of ideas which have their origin in the work of earlier mathematicians, but which have been greatly extended and generalized by the labours of Cayley, Sylvester, and their successors. The theory of the quintic equation is a particular but important application of these general principles, and it illustrates the power and beauty of the method of invariants.

Article No. 332. On the higher covariants of the quintic.

Article 336 (332)**.** The covariants of the quintic which have been tabulated in the present memoir extend up to the degree 6. The complete system of covariants of the quintic is known to contain twenty-three members, including the quintic itself; and these covariants are of various degrees and orders. The calculation of the higher covariants is a work of considerable labour, but the methods employed in the present memoir are applicable to their determination. The syzygies between the covariants also become more numerous and more complicated as the degree increases.

Article 337 (333)**.** The sextic covariants M and N, which are given in Tables Nos. 83 and 84, are of special interest on account of their connexion with the discriminant and the other invariants. The covariant M is the Hessian of three times the cubic covariant D, and the covariant N is one-sixth of the Jacobian of the quadric covariant B and the quartic covariant H. These covariants, together with those previously calculated, form a complete system up to the degree 6.

Article 338 (334)**.** The theory of the higher covariants is connected with the theory of the higher singularities of the discriminant-surface. Each covariant has a geometrical significance in connexion with the curve represented by the quantic; and the vanishing of a covariant corresponds to a special configuration of the points of the range. The complete theory of these singularities is a subject of great complexity, but the principles upon which it depends have been established in the present and preceding memoirs.

Article 339 (335)**.** The memoir thus contributes to the theory of the quintic equation in several directions. It completes the theory of the covariants up to the degree 6; it establishes the criteria for the determination of the character of the equation; it develops the theory of the canonical form; and it discusses the effect of imaginary transformations upon the character. These results, taken together with those of the preceding memoirs, form a complete and systematic theory of the binary quintic.

Article 340 (336)**.** The superimaginary transformation is the transformation

$$x+iy : x-iy \text{ into } (\cos\phi+i\sin\phi)(X+iY) : (\cos\phi-i\sin\phi)(X-iY),$$

which may also be written

$$x+iy : x-iy \text{ into } e^{i\phi}(X+iY) : e^{-i\phi}(X-iY).$$

The transformed equation is

$$(a+\beta i, \ldots \mid (\cos\phi+i\sin\phi)(X+iY),\ (\cos\phi-i\sin\phi)(X-iY))^n = 0.$$

The left-hand side consists of terms such as $(X^2+Y^2)^{n-s}$ into

$$(\gamma+\delta i)(\cos s\phi+i\sin s\phi)(X+iY)^s+(\gamma-\delta i)(\cos s\phi-i\sin s\phi)(X-iY)^s,$$

viz. the expression last written down is

$$\begin{aligned}&=(\gamma\cos s\phi-\delta\sin s\phi)\,[(X+iY)^s+(X-iY)^s]\\ &\qquad\qquad -(\gamma\sin s\phi+\delta\cos s\phi)\,[(X+iY)^s-(X-iY)^s]/(i),\end{aligned}$$

and hence for the expressions to be real, we must have $\tan s\phi = \text{real}$; and observing that if $\tan s\phi$ be equal to any given real quantity whatever, then the values of $\tan\phi$ are all of them real, and that $\tan\phi$ real gives $\cos\phi$ and $\sin\phi$ each of them real, and therefore also ϕ real, it appears that the transformed equation is only real for the transformation

$$x+iy : x-iy \text{ into } (\cos\phi+i\sin\phi)(X+iY) : (\cos\phi-i\sin\phi)(X-iY),$$

where ϕ is real; and hence neither in the natural transformation nor in that of the superimaginary transformation can we have an imaginary transformation leading to a real equation.

Article 341 (337)**.** The theory of the superimaginary transformation is thus seen to lead to the conclusion that no new case arises from it. The transformed equation is only real when ϕ is real, and in this case the transformation is equivalent to a real transformation. The superimaginary transformation thus does not give any case in which an imaginary transformation produces a real equation of an altered character.

Article 342 (338)**.** The subimaginary transformation remains to be considered. This is the transformation $x : y$ into $kX : Y$, where k^p is real for some divisor p of n. If p is odd, the transformation is equivalent to a real transformation, and the character is unaltered. If p is even and k^p is positive, the same is true. But if p is even and k^p is negative, then the transformation is not equivalent to any real transformation, and the character of the equation may be altered.

Article 343 (339)**.** The case $p = 2$, $n = 4$ or $n = 5$ is the case which has been discussed in detail for the quintic. The original equation $(x^2, y^2)^m = 0$ is transformed into $(X^2, -Y^2)^m = 0$, and the character is altered. For $m = 1$, the equation $x^2 - ay^2 = 0$ (two real roots) is transformed into $X^2 + aY^2 = 0$ (two imaginary roots); and for higher values of m the alteration of character is similar.

Article 344 (340)**.** The complete discussion of the subimaginary transformation for the quintic has been given in Articles 324–327, and the result is that the only cases in which an imaginary transformation alters the character are the cases there enumerated. The general theory of these transformations for quantics of any order follows the same lines, and leads to similar results.

Article 345 (341)**.** The theory of the transformations of binary quantics is thus complete. The general linear transformation involves four arbitrary parameters, and the condition that the transformed equation shall be real imposes certain restrictions upon these parameters. The complete discussion leads to the classification of transformations into real, subimaginary, and superimaginary; and the effect of each class upon the character of the equation has been determined.

Article 346 (342)**.** The application of this theory to the problem of the determination of the character of a binary quantic is immediate. Given the coefficients of the quantic, we calculate the invariants; and from the values of the invariants we determine the character by the criteria which have been established. The process is completely definite, and it does not require the solution of the equation. The theory is thus of great practical value, as well as of theoretical interest.

Article 347 (343)**.** The memoir concludes with a general survey of the theory of binary quantics. The theory is presented as a systematic development, founded upon the fundamental notions of invariants and covariants, and extending to the most general results. The theory of the quintic is a particular but important application, and it has been developed in detail in the present and preceding memoirs. The results obtained are of permanent value, and they form a contribution to the science of algebra which will not easily be superseded.

Article 348 (344)**.** The following is a brief recapitulation of the principal results:

1. The complete system of covariants of the quintic up to the degree 6.
2. The determination of the syzygies between these covariants.
3. The expression of the 18-thic invariant in terms of the invariants J, K, L.
4. The criteria for the determination of the character of a quintic equation.
5. The theory of the canonical form $ax^5 + by^5 + cz^5$ $(x + y + z = 0)$.
6. The theory of superimaginary and subimaginary transformations.
7. The proof that the Absolute Invariants and J constitute a system of auxiliars.

Article 349 (345)**.** The discriminant of the quintic is a function of the degree 8 in the coefficients, and it may be expressed in terms of the invariants J, K, L. The expression is

$$D = K^2 + \text{terms involving } J \text{ and } L,$$

and the complete expression is a complicated function of these invariants. The discriminant-surface $D = 0$ divides the space of invariants into regions corresponding to the different characters of the equation.

Article 350 (346)**.** The nodal line of the discriminant-surface corresponds to the case of two pairs of equal roots. On this line, the discriminant and its first derived functions all vanish; and the nodal line itself has a triple point corresponding to the case of three equal roots. The different sections of the discriminant-surface by planes give rise to curves which have been studied by Sylvester and others.

Article 351 (347)**.** The characters of the quintic equation are completely determined by the absolute invariants together with the sign of the skew invariant I. The absolute invariants are functions of J, K, L which are unaltered by any linear transformation; and the sign of I distinguishes between characters which have the same absolute invariants. The criteria for the determination of the character have been given in a form suitable for application to numerical equations.

Article 352 (348)**.** The theory of the transformations of binary quantics has been applied to show that the absolute invariants, together with the sign of J, constitute a complete system of auxiliars. The proof depends upon the classification of transformations into real, subimaginary, and superimaginary, and upon the determination of the effect of each class of transformations upon the character of the equation. The result is that, given the absolute invariants and the sign of J, the character of the equation is completely determined.

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Article 353 (349)**.** There remains only the subimaginary transformation, viz. this has been reduced to $x:y$ into $kX:Y$, the transformed equation is $(a,\dots \mid kX,Y)^n=0$, and this will be a real equation if some power k^p of k (p not greater than n) be real, and if the equation $(a,\dots \mid x,y)^n=0$ contain only terms wherein the index of x (or that of y) is a multiple of p. Assuming that it is the index of y which is a multiple, the form of the equation is in fact $(x^p,y^p)^m=0$, $(n=mp+\alpha)$, and the transformed equation is $X^\alpha(k^pX^p,Y^p)^m=0$, which is a real equation.

Article 354 (350)**.** It is to be observed that if p be odd, then writing $k^p=K$ (K real) and taking k' the real p-th root of K, then the very same transformed equation would be obtained by the real transformation $x:y$ into $k'X:Y$; so that the equation obtained by the imaginary transformation, being also obtainable by a real transformation, has the same character as the original equation.

Article 355 (351)**.** Similarly if p be even, if K be real and positive, the equation $k^p=K$ has a real root k' which may be substituted for the imaginary k, and the transformed equation will have the same character as the original equation; but if K be negative, say $K=-\Lambda$ (as may be assumed without loss of generality), then there is no real transformation equivalent to the imaginary transformation, and the equation given by the imaginary transformation has not of necessity the same character as the original equation; and there are in fact cases in which the character is altered. Thus if $p=2$, and the original equation be $x(x^2,y^2)^m=0$, or $(x^2,y^2)^m=0$, then making the transformation $x:y$ into $iX:Y$, the transformed equation will be $X(X^2,-Y^2)^m=0$ or $(X^2,-Y^2)^m=0$, giving imaginary roots $X^2+aY^2=0$ corresponding to real roots $x^2-ay^2=0$.

Article 356 (352)**.** Applying what precedes to a quintic equation $(a,\dots \mid x,y)^5=0$, this after any real transformation whatever will assume the form $(a,\dots \mid x',y')^5=0$; and the only cases in which we can have an imaginary transformation producing a real equation of an altered character is when this equation is $(a',0,c',0,e',0 \mid x',y')^5=0$ $(c'\neq 0)$, or when it is $(a',0,0,0,e',0 \mid x',y')^5=0$, viz. when it is $x'(a'x'^4+10c'x'^2y'^2+5e'y'^4)=0$, or $x'^4(a'x'^2+5e'y'^2)=0$. In the latter case the transformation x',y' into $X\sqrt{-1},Y$ gives the real equation $X(a'X^4-5e'Y^4)=0$.

Article 357 (353)**.** The different real equations which correspond to these data must be derivable one from another by real transformations, and must consequently have a determinate character; that is, the Absolute Invariants, and J, constitute a system of auxiliars.

Annex. Analytical Theorem in relation to a Binary Quantic of any Order.

Article 358 (354)**.** The foregoing theory of the superimaginary and subimaginary transformations leads to the following general theorem. If $(a, \dots \mid x, y)^n = 0$ be a binary quantic of any order n, and if by any linear transformation whatever the equation is converted into a real equation of the same or of a different character, then the transformation may be reduced to one of the following forms:

1. A real transformation, which preserves the character of the equation;
2. An imaginary transformation of the form $x : y$ into $kX : Y$, where k^p is real for some divisor p of n, which may or may not alter the character;
3. A superimaginary transformation, which changes the character in a determinate manner.

Article 359 (355)**.** The theorem may be established by considering the general linear transformation

$$x = \alpha X + \beta Y, \quad y = \gamma X + \delta Y,$$

where α, β, γ, δ are arbitrary quantities. If the transformed equation is to be real, then the coefficients of the transformed equation must all be real; and this condition restricts the possible values of α, β, γ, δ. The complete discussion of the cases leads to the classification above given.

Article 360 (356)**.** It is to be observed that the superimaginary transformation is only possible when the equation has a certain special form; in particular, it is necessary that the equation should be expressible in terms of x^p and y^p for some divisor p of n. When this condition is satisfied, the transformation $x : y$ into $iX : Y$ (where $i = \sqrt{-1}$) gives a real equation, and the character is altered in a determinate manner. The cases in which this occurs for the quintic equation have been discussed in the preceding articles.

Article 361 (357)**.** The subimaginary transformation is the transformation $x : y$ into $kX : Y$, where k^p is real for some divisor p of n. If p is odd, the transformation is equivalent to a real transformation, and the character is unaltered. If p is even and k^p is positive, the same is true. But if p is even and k^p is negative, then the transformation is not equivalent to any real transformation, and the character of the equation may be altered. The alteration of character in this case has been discussed for the quintic equation, and the general theory is analogous.

Article 362 (358)**.** The theorem may be further extended to the case of a system of binary quantics, and to the case of ternary and higher quantics. The general principle is that the imaginary transformations which produce real equations are of a limited class, and that their effect upon the character of the equation is determinate. The theory of these transformations is thus a branch of the general theory of invariants, and it has important applications in algebra and geometry.

Article 363 (359)**.** The application of the theorem to the theory of equations is immediate. Given a binary quantic of any order, the theorem enables us to determine whether the equation can be transformed into a real equation of a different character by an imaginary transformation, and if so, what is the nature of the transformation and the effect upon the character. The theorem also enables us to classify the different transformations which leave the character unaltered, and to distinguish them from those which alter it.

Article 364 (360)**.** The proof of the theorem depends upon the theory of the roots of unity, and upon the properties of the linear transformation of binary forms. The general linear transformation involves four arbitrary parameters, and the condition that the transformed equation shall be real imposes certain restrictions upon these parameters. The complete discussion of the cases leads to the classification of transformations into real, subimaginary, and superimaginary, and to the determination of the effect of each class of transformations upon the character of the equation.

Article 365 (361)**.** The memoir concludes with a discussion of the general theory of the transformations of binary quantics, and of their application to the theory of equations. The results obtained are of a general character, and they serve to unify and complete the theory of the invariants and covariants of binary forms. The theory of the quintic equation is a particular case of the general theory, and the results obtained for the quintic are illustrative of the general principles.

Article 366 (362)**.** The following is a brief recapitulation of the principal results of the memoir:

1. The covariants of the quintic up to the degree 6 have been completely calculated and tabulated.
2. The syzygies between these covariants have been determined.
3. The canonical form $ax^5 + by^5 + cz^5$ $(x + y + z = 0)$ has been discussed, and the invariants and covariants expressed in terms of the coefficients (a, b, c).
4. The 18-thic invariant I has been expressed as a function of the invariants J, K, L, and also in terms of the roots.
5. The characters of the quintic equation have been completely determined by means of the invariants and the Bicorn curve.
6. The discriminant-surface and its singularities have been discussed.
7. The theory of superimaginary and subimaginary transformations has been developed.
8. The application to the auxiliars of a quintic has been given.

Article 367 (363)**.** The theory of the characters of the quintic equation is thus brought to a state of completeness. The discriminant-surface, with its nodal line and triple point, furnishes a complete classification of the possible characters; and the criteria for the determination of the character are expressed in terms of the invariants J, D, L, which can be calculated for any given equation. The theory is analogous to the theory of the discriminant of the quadratic and cubic equations, and it extends those theories to the case of the quintic.

Article 368 (364)**.** The geometrical representation of the roots of a quintic equation by means of the canonical form $ax^5 + by^5 + cz^5 = 0$ $(x + y + z = 0)$ is a representation of considerable interest. If in the plane we take a triangle ABC, and upon the sides thereof measure off distances x, y, z such that $x + y + z = 0$, then the locus of the point so determined, when $x : y : z$ are the roots of the quintic equation, is a curve of the fifth class. The different characters of the equation correspond to different configurations of this curve, and the discriminant-surface corresponds to the cases where the curve acquires singularities.

Article 369 (365). The theory of the transformations of binary quantics is thus seen to be a theory of wide extent and great generality. The methods employed are applicable not only to the quintic but to quantics of any order; and the results obtained for the quintic are typical of the results which may be expected for higher quantics. The theory of invariants and covariants, which is the foundation of the whole, is a theory of the most comprehensive character, and it has applications in every branch of algebra and geometry.

Article 370 (366). The memoir thus forms a contribution to the general theory of algebraical forms which is of permanent value. The results obtained are not only of theoretical interest, but they are also of practical utility in the solution of problems relating to the theory of equations. The determination of the character of a quintic equation, without solving it, is a problem which is completely solved by the methods of the present memoir; and the theory of the canonical form and the invariants furnishes the means of this determination.

Article 371 (367). The theory of the higher covariants and the higher singularities of the discriminant-surface remains for future investigation. The principles which have been established in the present and preceding memoirs are sufficient for the determination of the covariants up to any given degree, and for the discussion of the corresponding singularities; but the actual calculation of these covariants and the discussion of their properties is a work of great labour, which can only be undertaken in connexion with particular problems. The theory is thus left in a state in which it is ready for further development and application.

Article 372 (368). In conclusion, the author expresses his indebtedness to Professor Sylvester, whose researches on the quintic equation have been the starting-point of much of the present memoir. The theory of the amphigenous surface, which Professor Sylvester has developed, is replaced in the present memoir by a more direct method; but the fundamental ideas are due to him, and the results obtained are in harmony with those which he has established by his own methods.

Article 373 (369). The theory of the quintic equation is thus brought to a state of completeness which renders it available for all the purposes of algebraical and geometrical investigation. The memoir concludes with the expression of the hope that the methods employed and the results obtained may serve to advance the science of algebra, and to facilitate the solution of problems which have hitherto been regarded as of insuperable difficulty.

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406. On the Curves which satisfy Given Conditions

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Article 374 (1). The present memoir is an attempt to lay the foundations of a theory of the curves which satisfy given conditions. The problem is, given a curve of given order, the equation of which contains a certain number of arbitrary parameters, to determine the number of such curves which satisfy given conditions. The conditions may be of various kinds; they may be conditions that the curve shall pass through given points, or touch given curves, or have certain singularities; and the problem is to determine the number of curves which satisfy any given system of such conditions.

Article 375 (2). The theory of the conics which satisfy five conditions (for instance, passing through five points, or touching five lines) is well known; the number of such conics is in general $= 1$. For cubics, the theory is more complicated, but it has been treated by Chasles, Zeuthen, and others. The general theory for curves of any order has not hitherto been established, and the present memoir is an attempt to supply this want.

Article 376 (3). The method employed is to regard the arbitrary parameters in the equation of the curve as the coordinates of a point in a space of the proper dimension; the conditions then correspond to loci in this space, and the problem is to determine the number of points of intersection of these loci. This method has been employed by Chasles for the case of conics, and it is capable of extension to curves of any order.

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Article 377 (4)**.** A condition imposed upon a curve may be of various orders. A condition that the curve shall pass through a given point is a onefold condition; a condition that it shall touch a given curve is also a onefold condition; a condition that it shall have a node is a twofold condition; and so on. The number of conditions which determine a curve of given order is equal to the number of arbitrary parameters in its equation; and when the curve is subjected to a number of conditions equal to this number, the curve is in general completely determined.

Article 378 (5)**.** If the parameters are taken to be the coordinates of a point in a space of ω dimensions, then a k-fold condition establishes a k-fold relation between the parameters, and the parametric point is situate on a k-fold locus. If we have conditions of manifoldness k, l, &c., and if $k + l +$ &c. $= \omega$, then the loci intersect in a point-system, and the number of points of this point-system is equal to the product of the orders of the several loci. The order of the k-fold locus is the number of points in which it is intersected by a general $(\omega - k)$-fold linear locus.

Article 379 (6)**.** The conditions may, however, not be completely independent; that is, the loci may have a common locus of less than ω dimensions, and may besides intersect in a residual point-system. The number of proper solutions is then equal to the order of the point-system, less the number of points which give special solutions. The determination of the order of the residual point-system, and the deduction of the special solutions, is the chief difficulty of the problem.

Article 380 (7)**.** Imagine a curve of given order, the equation of which contains m arbitrary parameters: to fix the ideas, it may be assumed that these enter into the equation rationally, so that the values of the parameters being given, the curve is uniquely determined. Suppose, as above, that the parameters are taken to be the coordinates of a point in m-dimensional space; so long as the curve is not subjected to any condition, the point in question, say the parametric point, is an arbitrary point in the m-dimensional space; but if the curve be subjected to a onefold, twofold, ... or k-fold condition, then we have a onefold, twofold, ... or k-fold relation between the parameters, and the parametric point is situate on a onefold, twofold, ... or k-fold locus accordingly: to each position of the parametric point on the locus there corresponds a curve satisfying the condition, that is, a solution of the condition. In the case where the condition is ω-fold, the locus is a point-system, and corresponding to each point of the point-system we have a solution of the condition; the number of solutions is equal to the number of points of the point-system.

Article 381 (8). Considering the general case where the condition, and therefore also the locus, is k-fold, it is to be observed that every solution whatever, and therefore each special solution (if any), corresponds to some point on the k-fold locus we may therefore have on the k-fold locus what may be termed "special loci," viz. a special locus is a locus such that to each point thereof corresponds a special solution. A special locus may of course be a point-system, viz. there are in this case a determinate number of special solutions corresponding to the several points of this point-system. We may consider the other extreme case of a special k-fold locus, viz. the k-fold locus of the parametric point may break up into two distinct loci, the special k-fold locus, and another k-fold locus the several points whereof give the ordinary solutions we can in this case get ; rid of the special solutions by attending exclusively to the last-mentioned k-fold locus and regarding it as the proper locus of the parametric point. But if the special locus be a more than k-fold locus, that is, if it be not a part of the k-fold locus itself, but (as supposed in the first instance) a locus on this locus, then the special solutions cannot be thus got rid of: we have the k-fold locus of the parametric point, a locus such that to every point thereof there corresponds a proper solution, save and except that to the points lying on the special locus there correspond special or improper solutions. It is to be noticed that the special locus may be, but that is not in every case, a singular locus on the k-fold locus.

Article 382 (9). Suppose that the conditions to be satisfied by the curve are a k-fold condition, an l-fold condition, &c. of a total manifoldness $= \omega$. If the conditions are completely independent (that is, if the corresponding relations, ante, No. 5, are completely independent), we have a k-fold locus, an l-fold locus, &c., having no common locus other than the point-system of intersection, and the number of curves which satisfy the given conditions, or (as this has been before expressed) the number of solutions, is equal to the number of points of the point-system, or to the order of the point-system, viz. it is equal to the product of the orders of the loci which correspond to the several conditions respectively; among these we may however have special solutions, corresponding to points situate on the special loci upon any of the given loci; but when this is the case the number of these special solutions can be separately calculated, and the number of proper solutions is equal to the number obtained as above, less the number of the special solutions.

Article 383 (10). If, however, the given conditions are not completely independent (that is, if the corresponding relations are not completely independent), then the k-fold locus, the l-fold locus, &c. intersect in a common $(\omega - j)$-fold locus, and besides in a residual point-system. The several points of the $(\omega - j)$-fold locus give special solutions; in fact the very notion of the conditions being properly satisfied by a curve implies that the curve shall satisfy a true $(k + l + \&c.)$-fold, that is, a true ω-fold condition; the proper solutions are therefore comprised among the solutions given by the residual point-system, and the number of them is as before equal to the order of the point-system, or number of the points thereof, less the number of points which give special solutions: the order of the point-system is, as has been seen, equal to the product of the orders of the k-fold locus, the l-fold locus, &c., less a reduction depending on the nature of the common $(\omega - j)$-fold locus, and the difficulty is in general in the determination of the value of this reduction.

Article 384 (11). In all that precedes, the number of the parameters has been taken to be ω; but if the parameters are taken to be contained in the equation of the curve homogeneously, then the parameters before made use of are in fact the ratios of these homogeneous parameters; and using the term henceforward as referring to the homogeneous parameters, the numbers of the parameters will be $= \omega + 1$.

Article 385 (12). I assume also that the equation of the curve contains the parameters linearly; this being so, the condition that the curve shall pass through a given arbitrary point implies a linear relation between the parameters; and the condition that the curve shall pass through j given points, a j-fold linear relation between the parameters. It follows that the number of the curves which satisfy a given k-fold condition, and besides pass through $\omega - k$ given points, is equal to the order of the k-fold relation, or of the corresponding k-fold locus; and thus if we define the order of the k-fold condition to be the number of the curves in question, the condition, relation, and locus will be all of the same order, and in all that precedes we may (in place of the order of the relation or of the locus) speak of the order of the condition. Thus, subject to the modifications occasioned by common loci and special solutions as above explained, the order of the $(k + l + \&c.)$-fold condition made up of a k-fold condition, an l-fold condition, &c., is equal to the product of the orders of the component conditions; and in particular if $k + l + \&c. = \omega$, then the order of the ω-fold condition, or number of the solutions thereof, is equal to the product of the orders of the component conditions.

Article 386 (13). The conditions to be satisfied by the curve may be conditions of contact with a given curve or curves. In particular if the curve touch a given curve, the parametric point is then situate on a onefold locus. It is to be noticed in reference hereto that if the given curve have nodes or cusps, then we have special solutions, viz. if the sought for curve passes through a node or a cusp of the given curve and each such node or cusp gives rise to a special onefold locus, presenting itself in the first instance as a factor of the onefold locus of the parametric point; this is, however, a case where the special locus is of the same manifoldness as the general locus (ante, No. 8), and is consequently separable; throwing off therefore all these special loci, we have a onefold locus which no longer comprises the points which correspond to curves passing through a node or a cusp of the given curve; the onefold locus, so divested of the special onefold factors, may be termed the "contact-locus" of the given curve. To each point of the contact-locus there corresponds a curve having with the given curve a two-pointic intersection, viz. this is either a proper contact, or it is a special contact, consisting in that the sought for curve has on the given curve a node or cusp, or (which is a higher speciality) in that the sought for curve is or contains as part of itself two or more coincident curves (ante, No. 3). To a point in general on the contact-locus there corresponds a curve having a proper contact with the given curve, save and except that to each point on any one of certain special loci on the contact-locus there corresponds a curve having some kind of special contact as above with the given curve. To fix the ideas, it may be mentioned that for the curves of the order r which touch a given curve of the order m and class n, the order of the contact-locus is $= n + (2r - 2)m$.

Article 387 (14). If, then, the curve touch a given curve, the parametric point is situate on the contact-locus of that curve. If it touch a second given curve, the parametric point is in like manner situate on the contact-locus of the second given curve, that is, it is situate on the twofold locus which is the intersection of the two contact-loci; and the like in the case of any number of contacts each with a distinct given curve. But if the curve, instead of ordinary contacts with distinct given curves, has either a contact of the second, or third, or any higher order, or has two or more ordinary or other contacts with the same given curve, then if the total manifoldness be $= k$, the parametric point is situate on a k-fold locus, which is given as a singular locus of the proper kind on the onefold contact-locus; so that the theory of the contact-locus corresponding to the case of a single contact with a given curve, contains in itself the theory of any system whatever of ordinary or other contacts with the same given curve, viz. the last-mentioned general case depends on the discussion of the singular loci which lie on the contact-locus. And similarly, if the curve has any number of ordinary or other contacts with each of two or more given curves, we have here to consider the intersections of singular loci lying on the contact-loci which correspond to the several given curves respectively, or, what is the same thing, to the singular loci on the intersection of these contact-loci; that is, the theory depends on that of the contact-loci which belong to the given curves respectively.

Article 388 (15). Suppose that the curve which has to satisfy given conditions is a line; the equation is $ax + by + cz = 0$, and the parameters (a, b, c) are to be taken as the coordinates of a point in a plane. Any onefold condition imposed upon the line establishes a onefold relation between the coordinates (a, b, c), and the parametric point is situate on a curve; a second onefold condition imposed on the line establishes a second onefold relation between the coordinates (a, b, c), and the parametric point is thus determined as a point of intersection of two ascertained curves. In particular if the condition imposed on the line is that it shall touch a given curve, the locus of the parametric point is a curve, the contact-locus; (this is in fact the ordinary theory of geometrical reciprocity, the locus in question being the reciprocal of the given curve; and the case of the twofold condition of a contact of the second order, or of two contacts, with the given curve, depends on the singular points of the contact-locus, or reciprocal of the given curve; in fact according as the line has a contact of the second order, or has two contacts with the given curve (that is, as it is an inflexion-tangent, or a double tangent of the given curve), the parametric point is a cusp or a node on its locus, the reciprocal curve; this is of course a fundamental notion in the theory of reciprocity, and it is only noticed here in order to show the bearing of the remark (ante, No. 14) upon the case now in hand where the curve considered is a line.

Article 389 (16). If the curve which has to satisfy given conditions is a conic

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0,$$

we have here six parameters (a, b, c, f, g, h), which are taken as the coordinates of a point in 5-dimensional space. It may be remarked that in this 5-dimensional space we have the onefold cubic locus $abc - af^2 - bg^2 - ch^2 + 2fgh = 0$, which is such that to any position of the parametric point upon it there corresponds not a proper conic but a line-pair; this may be called the discriminant-locus. We have also the threefold locus the relation of which is expressed by the six equations

$$bc - f^2 = 0, \quad ca - g^2 = 0, \quad ab - h^2 = 0, \quad gh - af = 0, \quad hf - bg = 0, \quad fg - ch = 0,$$

which is such that to any position of the parametric point thereon, there corresponds not a proper conic but a coincident line-pair. I call this the Bipoint-locus(1), and I notice that its order is $= 4$; in fact to find the order we must with the equations of the Bipoint combine two arbitrary linear relations,

$$(\alpha \,\ddot{\$}\, a, b, c, f, g, h) = 0,$$
$$(\alpha' \,\ddot{\$}\, a, b, c, f, g, h) = 0;$$

the equations of the locus are satisfied by

$$a : b : c : f : g : h = \alpha^2 : \beta^2 : \gamma^2 : \beta\gamma : \gamma\alpha : \alpha\beta$$

(where $\alpha : \beta : \gamma$ are arbitrary); and substituting these values in the linear relations, we have two quadric equations in (α, β, γ), giving four values of the set of ratios $(\alpha : \beta : \gamma)$; that is, the order is $= 4$, or the Bipoint is a threefold quadric locus.

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Article 390 (17)**.** The discriminant-locus does not in general present itself except in questions where it is a condition that the conic shall have a node (reduce itself to a line-pair); thus for the conics which have a node and touch a given curve (m, n), or, what is the same thing, for the line-pairs which touch a given curve (m, n), the parametric point is here situate on a twofold locus, the intersection of the discriminant-locus with the contact-locus. It may be noticed that this twofold locus is of the order $3(n + 2m)$, but that it breaks up into a twofold locus of the order $3n$, which gives the proper solutions, viz. the nodal conics which touch the given curve properly, that is, one of the two lines of the conic touches the curve; and into a twice repeated twofold locus of the order $3m$ which gives the special solutions, viz. in these the nodal conic has with the given curve a special contact, consisting in that the node or intersection of the two lines lies on the given curve. By way of illustration see Annex No. 2. But the consideration of the Bipoint-locus is more frequently necessary.

Article 391 (18)**.** Suppose that the conic satisfies the condition of touching a given curve; the parametric point is then situate on a onefold contact-locus $(a, b, c, f, g, h)^\mu = 0$ (to fix the ideas, if the given curve is of the order m and class n, then the order μ of the contact-locus is $= n+2m$). The contact-locus of any given curve whatever passes through the Bipoint-locus; in fact to each point of the Bipoint-locus there corresponds a coincident line-pair, that is, a conic which (of course in a special sense) touches the given curve whatever it be; and not only so, but inasmuch as we have a special locus of the same manifoldness as the general locus (ante, No. 8), it is consequently separable; throwing off therefore all these special loci, we have a onefold locus which no longer comprises the points which correspond to curves passing through a node or a cusp of the given curve; the onefold locus, so divested of the special onefold factors, may be termed the "contact-locus" of the given curve.

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Article 392 (19)**.** The theory of the contact-locus is thus seen to be fundamental for the whole theory of the conditions which a curve may satisfy. The contact-locus of a given curve is the locus of the parametric point when the curve which has to satisfy the condition touches the given curve; and the theory of the singular loci on the contact-locus contains the theory of all the contacts of higher order, or of multiple contacts, with the same given curve. The theory thus presents itself in a form which is capable of extension to curves of any order and to any system of conditions.

Article 393 (20)**.** The memoir contains also the discussion of several particular cases, which serve to illustrate the general theory. The case of the conic is treated in detail, and the contact-locus is determined for various kinds of given curves. The case of the cubic is also discussed, and the number of cubics which satisfy given conditions is determined in several cases. The general theory is thus verified by its application to particular cases, and its utility is established.

Article 394 (21)**.** The theory of the curves which satisfy given conditions is thus brought into connexion with the general theory of loci in space of any number of dimensions. The parametric point, the conditions, the loci, the special solutions, and the proper solutions, are all elements of a general theory which is capable of application to a wide range of problems. The memoir thus forms a contribution to the science of geometry which is of permanent value, and which will serve as a foundation for future investigations.

Article 395 (22)**.** The author expresses his hope that the theory developed in the present memoir may serve to advance the science of geometry, and to facilitate the solution of problems which have hitherto been regarded as of insuperable difficulty. The determination of the number of curves of given order which satisfy given conditions is a problem of fundamental importance in geometry, and the method of the present memoir furnishes a complete and general solution of this problem.

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Article 396 (23). The theory of the contact-locus has been applied in the memoir to the determination of the number of conics which satisfy various systems of conditions. The number of conics which touch five given curves is determined; the number of conics which pass through four points and touch a curve; the number of conics which pass through three points and touch two curves; and so on. The results are verified by comparison with known results, and the method is shown to be of general application.

Article 397 (24). The memoir also contains a discussion of the theory of the node and the cusp as conditions imposed upon a curve. A node is a twofold condition, and a cusp is a threefold condition; and the theory of these conditions is developed in connexion with the general theory of the present memoir. The number of curves of given order which have a given number of nodes and cusps is determined, and the results are verified by known formulae.

Article 398 (25). The theory of the inflexions and double tangents of a curve is also discussed in connexion with the Plücker equations. The number of the inflexions and double tangents of a curve of given order is determined by the Plücker equations, and these equations are applied to the curves which satisfy given conditions. The theory of the deficiency of a curve is also considered, and its connexion with the number of the conditions is pointed out.

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Article 399 (26)**.** The memoir concludes with a general survey of the theory which has been developed. The method of the parametric point and the loci in space of any number of dimensions is shown to be applicable to a wide range of problems; and the theory of the contact-locus, with its singular loci, is established as the foundation of the theory of contacts. The memoir thus forms a systematic development of the theory of the curves which satisfy given conditions, and it renders possible the solution of problems which have hitherto been regarded as of extreme difficulty.

Article 400 (27)**.** The author expresses his indebtedness to the work of Chasles, Zeuthen, and others, who have treated particular cases of the general problem; and he hopes that the general theory which he has established may serve to unify and extend their results. The memoir thus forms a contribution to the science of geometry which is of permanent value, and which will serve as a foundation for future investigations.

Article 401 (28)**.** In conclusion, the author remarks that the theory of the curves which satisfy given conditions is a branch of the general theory of loci in space of any number of dimensions. The problems which arise in this theory are of the utmost generality, and their solution requires the employment of the most powerful methods of modern geometry. The memoir is an attempt to establish these methods on a firm foundation, and to apply them to the solution of the most general problems of the theory.

Article 402 (29)**.** The memoir thus concludes with the expression of the hope that the methods and results which have been developed may serve to advance the science of geometry, and to open up new fields of research in the theory of curves and their conditions. The theory is capable of extension in many directions, and it is the author's intention to pursue these extensions in future memoirs.

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End of Paper 406.

406. On the Curves which satisfy given Conditions *(continued)*

Arthur Cayley

[Paper 406, from Phil. Trans. vol. clviii (for 1868), pp. 75–143]

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[406] **On the Curves which satisfy given Conditions.**

Article Nos. 26 to 41.

26. A single condition (Σ) imposed upon a conic has two representative numbers or simply representatives, (α, β); viz. if (4Σ) be an arbitrary system of four conditions and (μ, ν) the characteristics of (4Σ), then the number of the conics which satisfy the five conditions $(\Sigma, 4\Sigma)$ is $= \alpha\mu + \beta\nu$.

27. As an instance of the use of the characteristics, if $\Sigma, \Sigma', \Sigma'', \Sigma''', \Sigma''''$ be any five independent conditions, and $(\alpha, \beta), \ldots, (\alpha'''', \beta'''')$ the representatives of these conditions respectively, then the number of the conics which satisfy the five conditions $(\Sigma, \Sigma', \Sigma'', \Sigma''', \Sigma'''')$ is

$$= (1, 2, 4, 4, 2, 1 \mid \alpha, \beta)(\alpha', \beta')(\alpha'', \beta'')(\alpha''', \beta''')(\alpha'''', \beta''''),$$

viz. this notation stands for $1\alpha\alpha'\alpha''\alpha'''\alpha'''' + 2\Sigma\alpha\alpha'\alpha''\alpha'''\beta'''' \ldots + 1\beta\beta'\beta''\beta'''\beta''''$.

28. In particular if Σ be the condition that a conic shall touch a given curve of the order m and class n, then the representatives of this condition are (n, m). Hence the number of the conics which touch each of five given curves $(m, n), \ldots$ is

$$= (1, 2, 4, 4, 2, 1 \mid n, m)(n', m')(n'', m'')(n''', m''')(n'''', m'''').$$

29. A system of conics (4Σ) having the characteristics (μ, ν), contains

- $2\nu - \mu$ line-pairs, that is, conics each of them a pair of lines; and
- $2\mu - \nu$ point-pairs, that is, conics each of them a pair of points (*coniques infiniment aplaties*).

30. I stop to further explain these notions of the line-pair and the point-pair—and also the notion of the 'line-pair-point.'

A conic is a curve of the second order and second class; quà curve of the second order it may degenerate into a pair of lines, or line-pair (but the class is then $= 0$)—quà curve of the second class it may degenerate into a pair of points, or point-pair (but the order is then $= 0$). The two lines of a line-pair may be coincident and we have then a coincident line-pair; such a line-pair (it must I think be postulated) ordinarily arises, not from a line-pair the two lines of which become coincident but from a proper conic, flattening by the gradual diminution of its conjugate axis while its transverse axis remains constant or approaches a limit different from zero—the conic thus tends (not to an indefinitely extended but) to a terminated line C)—in other words, the tangents of the conic become more and more nearly lines through two fixed points, the terminations of the terminated line; and these terminating points which continue to exist up to the instant when the conjugate axis takes its limiting value $= 0$, are regarded as still existing at this instant, and the coincident line-pair as being in fact the point-pair formed by the two terminating points. Similarly the two points of a point-pair may be coincident, and we have then a coincident point-pair; such a point-pair (it must in like manner be postulated) ordinarily arises, not from a point-pair the two points of which become coincident, but from a proper conic sharpening itself to coincide with its asymptotes, and so becoming ultimately a pair of lines through the coincident point-pair; and the coincident point-pair is regarded as being in fact the line-pair formed by some two lines through the coincident point-pair.

31. In accordance with the foregoing notions we may with propriety, and it will in the sequel be found convenient to speak of a point-pair as a line terminated by two points on this line, and similarly to speak of a line-pair as a point terminated (that is, the pencil of lines through the point is terminated) by two lines through the point.

32. If in a point-pair thus considered as a line terminated by two points the two points become coincident (the line continuing to exist as a definite line), or, what is the same thing, if in a line-pair thus considered as a point terminated by two lines, the two lines become coincident (the point continuing to exist as a definite point), we have a "line-pair-point;" viz. this is at once a coincident line-pair and a coincident point-pair; it may also be regarded as the limit of a conic the axes of which, and the ratio of the conjugate to the transverse axis, all ultimately vanish: it may be described as a line terminated each way at a point thereof, or as a point terminated each way at a line through it. The notion of a line-pair-point first presents itself in Zeuthen's researches, as will presently appear; but it may be noticed here that line-pair-points, and these the same line-pair-points, may present themselves among the $2\nu - \mu$ line-pairs, and among the $2\mu - \nu$ point-pairs of the system of conics 4Σ.

33. Returning to the foregoing theory of characteristics, I remark that the fundamental notion may be taken to be, not the characteristics (μ, ν) of the conics which satisfy four conditions, but in every case the number of the conics which satisfy five conditions. Thus for the conics not subjected to any condition, we may consider the symbols

$$(:::::),\ \ (:::\,/),\ \ (::\,//),\ \ (:\,///),\ \ (/////),$$

denoting the number of the conics which pass through five given points, or which pass through four given points and touch a given line, &c.…, or which touch five given lines; these numbers are respectively

$$= 1,\ 2,\ 4,\ 4,\ 2,\ 1.$$

So for the conics which satisfy a given condition Σ, or two conditions 2Σ, …, or five conditions 5Σ, we have respectively the numbers

$$\begin{array}{lll}
\Sigma, & (::\,\cdot)\Sigma, & (\cdot\,\cdot\,/)\Sigma,\ (:\,/\cdot)\Sigma,\ (\cdot///)\Sigma,\ (////)\Sigma, \\
2\Sigma, & (::\,\cdot)\Sigma, & (\cdot/)\Sigma,\quad (:\,/\cdot)\Sigma,\ (///)\Sigma, \\
3\Sigma, & (::)\Sigma, & (\cdot/)\Sigma,\quad (//)\Sigma, \\
4\Sigma, & (\cdot)\Sigma, & (/)\Sigma, \\
5\Sigma, & \text{(single number).} &
\end{array}$$

where the Σ, 2Σ, &c. belong to the symbols which follow: read $(\Sigma\ ::::)$, $(\Sigma\ :::\ /)$, &c., or, as we may for shortness represent them,

$$\mu'''',\ \nu'''',\ \rho'''',\ \sigma'''',\ \tau'''',$$

viz. the single condition Σ has the five characteristics $(\mu'''', \dots \tau'''')$, $\dots$; the four conditions 4Σ, the characteristics (μ, ν) as in the original theory; and the five conditions 5Σ a single characteristic μ_0.

34. We thus see the origin of the notion of the representatives (α, β) of a single condition Σ; for considering the arbitrary four conditions 4Σ, the characteristics whereof are (μ, ν), and assuming that the single characteristic, or number of the conics $(\Sigma, 4\Sigma)$, is $= \alpha\mu + \beta\nu$, and taking for (4Σ) successively the conditions

$$(::::),\ (:::/),\ (::\,//),\ (:\,///),\ (/////),$$

having respectively the characteristics

$$(1,2),\ (2,4),\ (4,4),\ (4,2),\ (2,1),$$

we have

$$\begin{aligned}
\mu'''' &= 1\alpha + 2\beta,\\
\nu'''' &= 2\alpha + 4\beta,\\
\rho'''' &= 4\alpha + 4\beta,\\
\sigma'''' &= 4\alpha + 2\beta,\\
\tau'''' &= 2\alpha + 1\beta,
\end{aligned}$$

that is, the characteristics $(\mu'''', \nu'''', \rho'''', \sigma'''', \tau'''')$ of a single condition Σ are not independent, but are representable as above by means of two independent quantities (α, β); or, what is the same thing, we have

$$\nu'''' = 2\mu'''', \quad \rho'''' = \mu'''' + \sigma'''', \quad \tau'''' = \tfrac{1}{2}\nu'''',$$

which being satisfied, the representatives (α, β) are given by

$$\alpha = \tfrac{1}{2}(2\tau'''' - \nu''''), \qquad \beta = \tfrac{1}{2}(2\mu'''' - \tau'''').$$

35. I find that a like property exists as to the characteristics $(\mu'', \nu'', \rho'', \sigma'')$ of the two conditions 2Σ, viz. these are not independent but are connected by a single linear relation,

$$2\mu'' - 2\nu'' + \rho'' - 2\sigma'' = 0.$$

This may be proved in the case where the conditions 2Σ are two separate conditions (Σ, Σ'); viz. let the representatives of these be (α, β), (α', β') respectively, then

combining with them the three arbitrary conditions Σ'', Σ''', Σ'''' having respectively the representatives (α'', β''), (α''', β'''), (α'''', β''''), we have the general equation

$$(\Sigma, \Sigma', \Sigma'', \Sigma''', \Sigma'''') = (1, 2, 4, 4, 2, 1 \mid \alpha, \beta)(\alpha', \beta')(\alpha'', \beta'')(\alpha''', \beta''')(\alpha'''', \beta'''');$$

taking herein

$$(\Sigma'', \Sigma''', \Sigma'''') = (:::),\ (::/:),\ (:\,//),\ (:\,///)$$

successively, and observing that the representatives of $(::\cdot)$ are $(1, 0)$ and those of $(/)$ are $(0, 1)$, we thus obtain for $(\mu'', \nu'', \rho'', \sigma'')$, characteristics of (Σ, Σ'), the values

$$\begin{aligned}
\mu'' &= (1, 2, 4 \mid \alpha, \beta)(\alpha', \beta'),\\
\nu'' &= (2, 4, 4 \mid \alpha, \beta)(\alpha', \beta'),\\
\rho'' &= (4, 4, 2 \mid \alpha, \beta)(\alpha', \beta'),\\
\sigma'' &= (4, 2, 1 \mid \alpha, \beta)(\alpha', \beta'),
\end{aligned}$$

(viz. $\mu'' = 1\alpha\alpha' + 2(\alpha\beta' + \alpha'\beta) + 4\beta\beta'$, &c.), and these values give identically

$$2\mu'' - 2\nu'' + \rho'' - 2\sigma'' = 0,$$

which is the foregoing equation. And I assume that the theorem extends to the case of two inseparable conditions 2Σ, but in this case I do not even know where the proof is to be sought for.

The characteristics (μ', ν', ρ') of the three conditions 3Σ are in general independent.

36. It has been mentioned that if (α, β) are the representatives of the condition Σ, and (μ, ν) the characteristics of the conditions 4Σ, then

$$(\Sigma, 4\Sigma) = \alpha\mu + \beta\nu;$$

this is the most convenient form of the theorem, but as (α, β) are known functions of the characteristics $(\mu'''', \nu'''', \rho'''', \sigma'''', \tau'''')$ of the condition Σ, the equation is in effect an expression for $(\Sigma, 4\Sigma)$ in terms of the characteristics of Σ and 4Σ respectively.

There is, similarly, an expression for $(2\Sigma, 3\Sigma)$ in terms of the characteristics $(\mu', \nu', \rho', \sigma')$ of 3Σ (satisfying the relation $\mu' - 2\nu' + \frac{1}{2}\rho' - \sigma' = 0$) and the characteristics (μ, ν, ρ) of 2Σ, viz. we have

$$(2\Sigma, 3\Sigma) = \mu(-\tfrac{1}{4}\rho' + \tfrac{1}{2}\sigma') + \nu(\tfrac{1}{2}\mu' - \tfrac{1}{4}\rho') + \rho(\tfrac{1}{2}\mu' - \tfrac{1}{2}\nu' + \tfrac{1}{4}\rho').$$

This may be easily proved in the case where the conditions 2Σ are two separable conditions Σ, Σ' having the representatives (α, β), (α', β') respectively, and the conditions 3Σ three separable conditions $\Sigma'', \Sigma''', \Sigma''''$ having the representatives (α'', β''), (α''', β'''), (α'''', β'''') respectively; we have, in fact,

$$\begin{aligned}
\mu' &= (1, 2, 4 \mid \alpha', \beta'), & \mu &= (1, 2, 4, 4 \mid \alpha'', \beta'')(\alpha''', \beta''')(\alpha'''', \beta''''),\\
\nu' &= (2, 4, 4 \mid \alpha', \beta'), & \nu &= (2, 4, 4, 2 \mid \alpha'', \beta'')(\alpha''', \beta''')(\alpha'''', \beta''''),\\
\rho' &= (4, 4, 2 \mid \alpha', \beta'), & \rho &= (4, 4, 2, 1 \mid \alpha'', \beta'')(\alpha''', \beta''')(\alpha'''', \beta''''),\\
\sigma' &= (4, 2, 1 \mid \alpha', \beta');
\end{aligned}$$

and with these values the function

$$(\Sigma, \Sigma', \Sigma'', \Sigma''', \Sigma'''') = (1, 2, 4, 4, 2, 1 \mid \alpha, \beta)(\alpha', \beta')(\alpha'', \beta'')(\alpha''', \beta''')(\alpha'''', \beta'''')$$

is found to be expressible as above in terms of (μ, ν, ρ), $(\mu', \nu', \rho', \sigma')$; but I do not know how to conduct the proof for the inseparable conditions 2Σ and 3Σ.

37. It may be remarked by way of verification that writing successively

$$(3\Sigma) = (:::\ \cdot),\ (::\ /\ :),\ (:\ //),\ (:\ ///),$$

that is,

$$(\mu, \nu, \rho) = (1, 2, 4),\ (2, 4, 4),\ (4, 4, 2),\ (4, 2, 1),$$

we have in the first case

$$(2\Sigma, \cdot) = -\tfrac{1}{4}\rho' + \tfrac{1}{2}\sigma' + 2\mu' - \nu' + \rho' - 2\sigma' = \mu',$$

and similarly in the other three cases,

$$(2\Sigma, /) = \nu', \quad (2\Sigma, //) = \rho', \quad (2\Sigma, ///) = \sigma'.$$

38. Let (μ,ν,ρ,σ) be the characteristics of 2Σ, $(\mu-2\nu+\frac{1}{2}\rho-\sigma=0)$, and $(\mu',\nu',\rho',\sigma')$ the characteristics of $2\Sigma'$, $(\mu'-2\nu'+\frac{1}{2}\rho'-\sigma'=0)$. Then in the formula for $(2\Sigma,3\Sigma)$, writing successively for 3Σ

$$\begin{aligned}(3\Sigma)&=(2\Sigma'), &&\text{characteristics } (\mu',\nu',\rho'),\\ (3\Sigma)&=(2\Sigma',1), &&\text{characteristics } (\nu',\rho',\sigma'),\end{aligned}$$

we obtain expressions for the characteristics $(2\Sigma,2\Sigma')$, $(2\Sigma,2\Sigma',1)$, which are consequently

$$\begin{aligned}\mu(2\Sigma,2\Sigma')&=\mu(-\tfrac{1}{4}\rho'+\tfrac{1}{2}\sigma')+\nu(\tfrac{1}{2}\mu'-\tfrac{1}{4}\rho')+\rho(\tfrac{1}{2}\mu'-\tfrac{1}{2}\nu'+\tfrac{1}{4}\rho'),\\ \nu(2\Sigma,2\Sigma')&=\mu(-\tfrac{1}{4}\rho'+\tfrac{1}{4}\sigma')+\nu(\tfrac{1}{2}\mu'-\tfrac{1}{4}\rho'+\tfrac{1}{4}\sigma')+\rho(\tfrac{1}{2}\mu'-\tfrac{1}{2}\nu'+\tfrac{1}{4}\rho'-\tfrac{1}{4}\sigma'),\\ \rho(2\Sigma,2\Sigma')&=\mu(0)+\nu(\tfrac{1}{2}\mu'-\tfrac{1}{4}\rho'+\tfrac{1}{4}\sigma')+\rho(\tfrac{1}{2}\mu'-\tfrac{1}{2}\nu'+\tfrac{1}{4}\rho'-\tfrac{1}{4}\sigma'),\\ \sigma(2\Sigma,2\Sigma')&=\mu(0)+\nu(0)+\rho(\tfrac{1}{2}\mu'-\tfrac{1}{2}\nu'+\tfrac{1}{4}\rho'-\tfrac{1}{4}\sigma'),\end{aligned}$$

and similarly

$$\begin{aligned}\mu(2\Sigma,2\Sigma',1)&=\mu(-\tfrac{1}{4}\rho'+\tfrac{1}{2}\sigma')+\nu(\tfrac{1}{2}\mu'-\tfrac{1}{4}\rho')+\rho(\tfrac{1}{2}\mu'-\tfrac{1}{2}\nu'+\tfrac{1}{4}\rho'),\\ \nu(2\Sigma,2\Sigma',1)&=\mu(0)+\nu(\tfrac{1}{2}\mu'-\tfrac{1}{4}\rho'+\tfrac{1}{4}\sigma')+\rho(\tfrac{1}{2}\mu'-\tfrac{1}{2}\nu'+\tfrac{1}{4}\rho'-\tfrac{1}{4}\sigma'),\\ \rho(2\Sigma,2\Sigma',1)&=\mu(0)+\nu(0)+\rho(\tfrac{1}{2}\mu'-\tfrac{1}{2}\nu'+\tfrac{1}{4}\rho'-\tfrac{1}{4}\sigma').\end{aligned}$$

39. If in these formulae for $(2\Sigma,2\Sigma')$, $(2\Sigma,2\Sigma',1)$, we write $(\mu,\nu,\rho,\sigma)=(\mu',\nu',\rho',\sigma')$, or, what is the same thing, write $\Sigma=\Sigma'$, then observing that these characteristics satisfy the relation $\mu-2\nu+\frac{1}{2}\rho-\sigma=0$, the formulae become

$$\begin{aligned}\mu(4\Sigma)&=\mu(-\tfrac{1}{4}\rho+\tfrac{1}{2}\sigma)+\nu(\tfrac{1}{2}\mu-\tfrac{1}{4}\rho)+\rho(\tfrac{1}{2}\mu-\tfrac{1}{2}\nu+\tfrac{1}{4}\rho),\\ \nu(4\Sigma)&=\mu(-\tfrac{1}{4}\rho+\tfrac{1}{4}\sigma)+\nu(\tfrac{1}{2}\mu-\tfrac{1}{4}\rho+\tfrac{1}{4}\sigma)+\rho(\tfrac{1}{2}\mu-\tfrac{1}{2}\nu+\tfrac{1}{4}\rho-\tfrac{1}{4}\sigma),\\ \rho(4\Sigma)&=\nu(\tfrac{1}{2}\mu-\tfrac{1}{4}\rho+\tfrac{1}{4}\sigma)+\rho(\tfrac{1}{2}\mu-\tfrac{1}{2}\nu+\tfrac{1}{4}\rho-\tfrac{1}{4}\sigma),\\ \sigma(4\Sigma)&=\rho(\tfrac{1}{2}\mu-\tfrac{1}{2}\nu+\tfrac{1}{4}\rho-\tfrac{1}{4}\sigma),\end{aligned}$$

which are therefore the characteristics of 4Σ in terms of those of 2Σ.

And for $(2\Sigma,2\Sigma',1)$, writing therein $(\mu,\nu,\rho)=(\nu',\rho',\sigma')$, that is, taking $2\Sigma=(2\Sigma',1)$, we obtain the characteristics of $(4\Sigma,1)$, which are consequently

$$\begin{aligned}\mu(4\Sigma,1)&=\nu'(-\tfrac{1}{4}\rho'+\tfrac{1}{2}\sigma')+\rho'(\tfrac{1}{2}\mu'-\tfrac{1}{4}\rho')+\sigma'(\tfrac{1}{2}\mu'-\tfrac{1}{2}\nu'+\tfrac{1}{4}\rho'),\\ \nu(4\Sigma,1)&=\rho'(\tfrac{1}{2}\mu'-\tfrac{1}{4}\rho'+\tfrac{1}{4}\sigma')+\sigma'(\tfrac{1}{2}\mu'-\tfrac{1}{2}\nu'+\tfrac{1}{4}\rho'-\tfrac{1}{4}\sigma'),\\ \rho(4\Sigma,1)&=\sigma'(\tfrac{1}{2}\mu'-\tfrac{1}{2}\nu'+\tfrac{1}{4}\rho'-\tfrac{1}{4}\sigma').\end{aligned}$$

40. The characteristics of 5Σ, $(\Sigma,4\Sigma)=\alpha\mu+\beta\nu$, are given in terms of the characteristics (μ,ν) of 4Σ and the representatives (α,β) of Σ. It would be interesting to obtain the characteristics of 6Σ in terms of those of 4Σ; but I have not been able to effect this.

41. I remark that from the expression for the characteristics of 4Σ in terms of those of 2Σ, we may conversely express the characteristics of 2Σ in terms of those of 4Σ, and then the expression for $(\Sigma,4\Sigma)$ may be written

$$(\Sigma,4\Sigma)=\alpha\mu+\beta\nu=(a,b,c,d\mid\mu,\nu,\rho,\sigma),$$

where (a,b,c,d) are the characteristics of 2Σ. The same expression may be written

$$(\Sigma,4\Sigma)=(a,b,c,d\mid\mu',\nu',\rho',\sigma'),$$

where $(\mu', \nu', \rho', \sigma')$ are the characteristics of $2\Sigma'$; and it would be interesting to find the relation between the characteristics (a, b, c, d) and (a', b', c', d') of 2Σ and $2\Sigma'$ respectively, such that the two expressions may be equivalent.

ARTICLE NOS. 42 TO 57.

42. In the case where the condition Σ is that of touching a given curve of the order m and class n, we have $(\alpha, \beta) = (n, m)$. Hence for two such conditions, touching two given curves (m, n), (m', n'), the characteristics $(\mu'', \nu'', \rho'', \sigma'')$ are

$$\begin{aligned}\mu'' &= nn' + 2(nm' + n'm) + 4mm',\\ \nu'' &= 2nn' + 4(nm' + n'm) + 4mm',\\ \rho'' &= 4nn' + 4(nm' + n'm) + 2mm',\\ \sigma'' &= 4nn' + 2(nm' + n'm) + mm',\end{aligned}$$

which satisfy the relation $2\mu'' - 2\nu'' + \rho'' - 2\sigma'' = 0$.

And similarly for any number of curves.

43. I consider in particular the case of a cusp, and I denote by $2\kappa 1$ the condition that the conic shall have a cusp (or, what is the same thing, that it shall pass through a given point and touch two given lines through the point). The order and class of the condition $2\kappa 1$ are each $= 3$; hence for the condition $2\kappa 1$ touching a given curve of the order m and class n, we have

$$(\alpha, \beta) = (n + 2m - 3,\ 2n + 4m - 6).$$

In fact, the number of the conics which have a given cusp and touch a given curve is $= n + 2m$, which, in virtue of the cusp, becomes $n + 2m - 3$; and the number of the conics which have a given cusp and touch a given line is $= 2$, which, in virtue of the cusp, becomes $2 - 2 = 0$ (this belongs to the general theory of characteristics; the number of line-pairs is $2(n + 2m) - (2n + 4m) = 0$, and the number of point-pairs is $2(2n + 4m) - (n + 2m) = 3n + 6m$, which are the line-pairs and point-pairs having a given cusp and touching a given curve).

44. For a system of conics (4Σ) having the characteristics (μ, ν), the number of cuspidal conics is $= 2\mu - 3\nu$; this may be seen as follows: the characteristics of $2\kappa 1$ are $(3, 3)$, and hence the number of the conics which have a cusp and satisfy the conditions 4Σ is $= 3\mu + 3\nu$; but among these are included the line-pairs, each of which has a cusp at each of the two intersections of the two lines; that is, each line-pair counts as having two cusps; and since the number of line-pairs is $= 2\nu - \mu$, we have therefore

$$3\mu + 3\nu - 2(2\nu - \mu) = 5\mu - \nu,$$

for the number of cuspidal conics, each counted with its proper multiplicity. But the cuspidal conics are each of them a proper conic, and they are therefore counted singly; hence the number is $= 5\mu - \nu$; or, what is the same thing, the number of the conditions $2\kappa 1$ satisfied by the conics of the system (4Σ) is $= 5\mu - \nu$. But I prefer to state the theorem as follows: the number of cuspidal conics in the system (4Σ) is $= 2\mu - 3\nu$; in fact, the number of the conics of the system which pass through a given point is $= \mu$, and among these the number of the line-pairs is $= 2$ (viz. the line-pair is made up of the two lines joining the given point with the two points of the point-pair); hence the number of proper conics through the given point is $= \mu - 2$; and the number of cuspidal conics through the given point is $= (\mu - 2) - 2(\nu - \mu) = 2\mu - 3\nu$.

45. For a system of conics (3Σ) having the characteristics (μ', ν', ρ'), the number of cuspidal conics is $= \mu' - 2\nu' + \frac{1}{2}\rho'$; this may be seen as follows: the characteristics of $(2\kappa 1, \Sigma)$ are $(\mu' - 2\nu' + \frac{1}{2}\rho',\ \nu' - 2\nu' + \frac{1}{2}\rho',\ \ldots)$, and hence the number of cuspidal conics is obtained as above.

46. In particular, for the system of conics touching three given lines and passing through two given points, the characteristics are $(4, 4)$, and the number of cuspidal conics is $= 2\cdot 4 - 3\cdot 4 = -4$, which is $= 0$; the negative value shows that there are no cuspidal conics, which is right, for the conics in question are proper conics.

47. I consider also the case of an inflexion, and I denote by $2\iota 1$ the condition that the conic shall have an inflexion (or, what is the same thing, that it shall touch a given line and pass through two given points on the line). The order and class are each $= 3$; hence for the condition $2\iota 1$ touching a given curve, we have

$$(\alpha, \beta) = (2n + m - 3,\ 4n + 2m - 6).$$

The formulae for the inflexion are at once obtained from those for the cusp by the interchange of m and n.

48. I give the following formulae, in which $(1^m \kappa^p)$ denotes the system of conditions made up of m contacts 1 and p contacts $2\kappa 1$, and similarly for other symbols; and where the curves are all of them of the order m and class n, and $a = 3n + \kappa$. The expressions for $(1^m \kappa^p)$, &c. are as follows:

$$\begin{aligned}
(1) &= n + 2m - 2,\\
(1, 1) &= m^2 + 2mn + \tfrac{3}{2}n^2 - 3m - \tfrac{7}{2}n + \tfrac{1}{2}a,\\
(1, 1, 1) &= 2m^2 + 4mn + n^2 - 8m - 7n + 18 - 3a,\\
(1, 1, 1, 1) &= -4m - 3n - 5 + 3a,\\
(1, 1, 1, 1, 1) &= -8m - 8n - 6 + 6a,\\
(2) &= a,\\
(2, 1) &= 4m + 8n + 44 + a(2m + n - 17),\\
(2, 1, 1) &= 20m + 16n + 42 + a(2m + 2n - 27),\\
(2, 2) &= \tfrac{1}{2}m^2 + 2m^2n + \tfrac{7}{2}mn^2 + \tfrac{5}{2}n^2 - 5m^2 - 9mn - 2n^2 + \tfrac{9}{2}m + \tfrac{9}{2}n - 57 + a(-3m - 6n + \tfrac{9}{2}),\\
(2, 2, 1) &= \tfrac{1}{2}m^2 + 27m^2n + 2mn^2 + \tfrac{5}{2}n^2 - 7m^2 - 10mn - 4n^2 - \tfrac{31}{2}m + \tfrac{3}{2}n - 54 + a(-3m - 3n + \tfrac{9}{2}),\\
(3) &= -4m - 5n + 3a,\\
(3, 1) &= -8m - 8n - 6 + 6a,\\
(3, 1, 1) &= -12m - 4n - 12 + 6a,\\
(3, 2) &= -8m^2 - 12mn - 4n^2 + 8m + 16n + 28 + a(4m + 4n - 24),\\
(3, 3) &= -12m^2 - 24mn - 12n^2 + 36m + 60n + 24 + a(6m + 6n - 42),\\
(4) &= 4m + 8n + 44 + a(2m + n - 17),\\
(4, 1) &= 20m + 16n + 42 + a(2m + 2n - 27),\\
(4, 2) &= -8m^2 - 12mn - 4n^2 + 8m + 16n + 28 + a(4m + 4n - 24),\\
(4, 3) &= -24m^2 - 48mn - 24n^2 + 72m + 120n + 48 + a(12m + 12n - 84),\\
(4, 4) &= 8m^2 + 16mn + 8n^2 - 32m - 56n - 40 + a(-4m - 4n + 40),\\
(5) &= -8m - 8n - 6 + 6a,\\
(5, 1) &= -12m - 4n - 12 + 6a,\\
(5, 2) &= -24m^2 - 48mn - 24n^2 + 72m + 120n + 48 + a(12m + 12n - 84),\\
(5, 3) &= -24m^2 - 48mn - 24n^2 + 72m + 120n + 48 + a(12m + 12n - 84),\\
(5, 4) &= 16m^2 + 32mn + 16n^2 - 64m - 112n - 80 + a(-8m - 8n + 80),\\
(5, 5) &= 8m^2 + 16mn + 8n^2 - 32m - 56n - 40 + a(-4m - 4n + 40).
\end{aligned}$$

49. I verify that for a conic $(m = n = 2, a = 6)$, these values agree with those given by the

general formulae for the conic; thus

$$\begin{aligned}(1) &= 2+4-2=4,\\ (1,1) &= 4+8+6-6-7+3=8,\\ (1,1,1) &= 8+16+4-16-14+18-18=-2,\ \text{\&c.}\end{aligned}$$

50. I do not attempt to verify the expressions for (1^m) by means of the characteristics, but I observe that the formula for $(1,1,1,1,1)$ may be written

$$-8m-8n-6+6a=-8m-8n-6+18n+6\kappa=-8m+10n-6+6\kappa,$$

and that for $m=2$, $n=2$, this becomes $-16+20-6+6\kappa=-2+6\kappa$, which for $\kappa=0$ (the proper conic) is $=-2$, agreeing with the value $-8m-8n-6+6a=-16-16-6+36=-2$.

51. The expressions for $(1^m\kappa^p)$ are obtained from those for (1^m) by writing therein $m+p$ in place of m, and $n+2p$ in place of n, and consequently $a+3p$ in place of a, and then subtracting the number of improper solutions; but the investigation of the improper solutions is in general very difficult.

52. I give the following formulae for the case of a single curve:

$$\begin{aligned}
(1\kappa1) &= n+2m-3,\\
(1\kappa1,1) &= m^2+2mn+\tfrac{3}{2}n^2-5m-\tfrac{9}{2}n+6-\tfrac{1}{2}a,\\
(1\kappa1,1,1) &= 2m^2+4mn+n^2-10m-8n+24-4a,\\
(1\kappa1,1,1,1) &= -4m-3n-8+3a,\\
(1\kappa1,1,1,1,1) &= -8m-8n-18+6a,\\
(2\kappa1) &= a-7,\\
(2\kappa1,1) &= 2a-8,\\
(2\kappa1,1,1) &= 2a-4,\\
(2\kappa1,2) &= 2m^2+2mn+\tfrac{1}{2}n^2-8m-\tfrac{7}{2}n+13-\tfrac{1}{2}a,\\
(2\kappa1,2,1) &= 2m^2+4mn+n^2-8m-7n+18-3a,\\
(2\kappa1,2,2) &= m^2+4mn+2n^2-4m-8n+12-3a,\\
(3\kappa1) &= -4m-3n-5+3a,\\
(3\kappa1,1) &= -8m-8n-6+6a,\\
(3\kappa1,2) &= -8m^2-12mn-4n^2+8m+16n+28+a(4m+4n-24),\\
(3\kappa1,3) &= -12m^2-24mn-12n^2+36m+60n+24+a(6m+6n-42),\\
(4\kappa1) &= 4m+8n+44+a(2m+n-17),\\
(4\kappa1,1) &= 20m+16n+42+a(2m+2n-27),\\
(4\kappa1,2) &= -8m^2-12mn-4n^2+8m+16n+28+a(4m+4n-24),\\
(4\kappa1,3) &= -24m^2-48mn-24n^2+72m+120n+48+a(12m+12n-84),\\
(4\kappa1,4) &= 8m^2+16mn+8n^2-32m-56n-40+a(-4m-4n+40),\\
(5\kappa1) &= -8m-8n-6+6a,\\
(5\kappa1,1) &= -12m-4n-12+6a,\\
(5\kappa1,2) &= -24m^2-48mn-24n^2+72m+120n+48+a(12m+12n-84),\\
(5\kappa1,3) &= -24m^2-48mn-24n^2+72m+120n+48+a(12m+12n-84),\\
(5\kappa1,4) &= 16m^2+32mn+16n^2-64m-112n-80+a(-8m-8n+80),\\
(5\kappa1,5) &= 8m^2+16mn+8n^2-32m-56n-40+a(-4m-4n+40).
\end{aligned}$$

53. I remark that the expressions for $(1^m\kappa^p)$ may be obtained from those for (1^m) by the substitution indicated in No. 51; but the investigation of the improper solutions presents difficulties which I have not attempted to overcome.

54. In the case where the condition Σ is that of passing through a given point, the representatives are $(1, 0)$; and where it is that of touching a given line, the representatives are $(0, 1)$. Hence the characteristics of the system of conics which pass through p given points and touch q given lines $(p + q = 4)$ are at once obtained; and the number of cuspidal conics in such a system is $= 2\mu - 3\nu$.

55. I give the following table of the characteristics (μ, ν) for the various systems of conics:

Conditions	μ	ν
(::::)	1	2
(::: /)	2	4
(:: //)	4	4
(: ///)	4	2
(/////)	2	1

56. The characteristics for the systems (1^m) and $(1^m\kappa^p)$ may be obtained by the foregoing formulae; and I have thus obtained the expressions given in the preceding paragraphs.

57. I remark that the whole of the foregoing theory is applicable to curves of any order; but that the formulae for the characteristics are in general very complicated, and have not been obtained except for curves of low order.

Article Nos. 58 to 73.

58. I consider the case of a curve of the order r having with a given curve U^m of the order m contacts of given orders; and I denote by $(a, b, c, \dots)$ the number of such curves which have with U^m contacts of the orders $a, b, c, \dots$ respectively, and which besides pass through the requisite number of arbitrary points.

59. De Jonquières has given a formula for the number of curves C^r of the order r which have with a given curve U^m of the order m contacts of the orders a, b, c, &c. respectively, which besides pass through p points distributed at pleasure on the curve U^m (this includes the case of contacts of any order at given points of the curve U^m), and which moreover satisfy any other $\frac{1}{2}r(r + 3) - (a + b + c + \&c.) - p$ conditions; viz. the number of the curves C^r is

$$= \mu \cdot (a + 1)(b + 1)(c + 1) \dots \text{ into}$$

$$\begin{aligned}
&[rm - (a + b + c \dots) - p]^r \\
&\qquad + [rm - (a + b + c \dots) - p - 1]^{r-1}(a + b + c \dots)[D]^1 \\
&\qquad + [rm - (a + b + c \dots) - p - 2]^{r-2}(ab + ac + bc \dots)[D]^2 \\
&\qquad\qquad + \cdots \\
&\qquad\qquad + [rm - (a + b + c \dots) - p - t]^0(abc \dots)[D]^t,
\end{aligned}$$

where the curve U^m is a curve *without cusps*, and having therefore a deficiency $D = \frac{1}{2}(m-1)(m-2)$; the numbers $a, b, c, \dots$ are assumed to be all of them unequal, but if we have α of them each $= a$, β of them each $= b$, &c., then the foregoing expression is to be divided by $[\alpha]^\alpha[\beta]^\beta \dots$; and μ denotes the number of the curves C^r which satisfy the system of conditions obtained from the given system by replacing the conditions of the contacts of the orders a, b, c, &c. respectively by the condition of passing through $a + b + c + \dots$ arbitrary points. In order that the formula may give the number of the proper curves C^r which satisfy the prescribed conditions, it is sufficient that their $\frac{1}{2}r(r+3)-(a+b+c..)-p$ conditions shall include the conditions of passing through at least a certain number T of arbitrary points: this restriction applies to all the formulae of the present section.

60. I will for convenience consider this formula under a somewhat less general form, viz. I will put $p=0$, and moreover assume that the $\frac{1}{2}r(r+3)-(a+b+c..)$ conditions are the conditions of passing through this number of arbitrary points; whence $\mu=1$.

We have thus a curve C^r having with the given curve U^m contacts of the orders $a, b, c..$ respectively, and besides passing through $\frac{1}{2}r(r+3)-(a+b+c..)$ arbitrary points; and the number of such curves is by the formula $=(a+1)(b+1)(c+1)\dots$ into

$$\begin{aligned}
&[rm-(a+b+c..)]^r \\
&\qquad + [rm-(a+b+c..)-1]^{r-1}(a+b+c..)[D]^1 \\
&\qquad + [rm-(a+b+c..)-2]^{r-2}(ab+ac+bc..)[D]^2 \\
&\qquad + \cdots \\
&\qquad + [rm-(a+b+c..)-t]^0(abc..)[D]^t,
\end{aligned}$$

where, as before, in the case of any equalities between the numbers $a, b, c, \dots$, the expression is to be divided by $[\alpha]^\alpha[\beta]^\beta\dots$.

61. I have succeeded in extending the formula to the case of a curve with cusps; instead of writing down the general formula, I will take successively the cases of a single contact a, two contacts a, b, three contacts a, b, c, &c.; and then denoting the numbers of the curves by (a), (a, b), (a, b, c), &c. in these cases respectively, I say that we have

$$(a)=(a+1)[rm-a]^r[1+aD]^r-a\kappa,$$

$$\begin{aligned}
(a, b) = (a+1)(b+1)\Big\{&[rm-a-b]^r \\
&+ [rm-a-b-1]^{r-1}(a+b)[D]^1 \\
&+ ab[D]^2\Big\} \\
&- a(b+1)\Big\{[rm-a-b-1]^{r-1}[1+bD]^1\Big\} \\
&- b(a+1)\Big\{[rm-a-b-1]^{r-1}[1+aD]^1\Big\} \\
&+ ab\kappa',
\end{aligned}$$

$$\begin{aligned}(a,b,c) = (a+1)(b+1)(c+1)\Big\{&[rm-a-b-c]^r \\ &+ [rm-a-b-c-1]^{r-1}(a+b+c)[D]^1 \\ &+ [rm-a-b-c-2]^{r-2}(ab+ac+bc)[D]^2 \\ &+ abc[D]^3\Big\} \\ - c(a+1)(b+1)\Big\{&[rm-a-b-c-1]^{r-1} \\ &+ [rm-a-b-c-2]^{r-2}(a+b)[D]^1 \\ &+ ab[D]^2\Big\} \\ - [2bc(a+1)]\Big\{&[rm-a-b-c-2]^{r-2} \\ &+ [rm-a-b-c-3]^{r-3}a[D]^1\Big\}[\kappa]^1 \\ - [2ac(b+1)]\Big\{&[rm-a-b-c-2]^{r-2} \\ &+ [rm-a-b-c-3]^{r-3}b[D]^1\Big\}[\kappa]^1 \\ - [2ab(c+1)]\Big\{&[rm-a-b-c-2]^{r-2} \\ &+ [rm-a-b-c-3]^{r-3}c[D]^1\Big\}[\kappa]^1 \\ &+ abc\kappa'.\end{aligned}$$

77. The foregoing examples are sufficient to exhibit the law; but as I shall have to consider the cases of four and five contacts, I will also write down the formula for (a,b,c,d), putting therein for shortness

$$a+b+c+d=\alpha,\quad ab+\ldots+cd=\beta,\quad abc+\ldots+bcd=\gamma,\quad abcd=\delta,$$
$$a+b+c=\alpha',\quad ab+ac+bc=\beta',\quad abc=\gamma',\quad a+b=\alpha'',\quad ab=\beta'',\quad a=\alpha''';$$

and also the formula for (a,b,c,d,e), putting therein in like manner

$$(\alpha,\beta,\gamma,\delta,\epsilon),\ (\alpha',\beta',\gamma',\delta'),\ (\alpha'',\beta'',\gamma''),\ (\alpha''',\beta'''),\ (\alpha'''')$$

for the combinations of (a,b,c,d,e), (a,b,c,d), (a,b,c), (a,b) and (a) respectively.

We have

$$\begin{aligned}(a,b,c,d) = (a+1)(b+1)(c+1)(d+1)\Big\{&[rm-\alpha]^r \\ &+ [rm-\alpha-1]^{r-1}\alpha[D]^1 \\ &+ [rm-\alpha-2]^{r-2}\beta[D]^2 \\ &+ [rm-\alpha-3]^{r-3}\gamma[D]^3 \\ &+ \delta[D]^4\Big\}\end{aligned}$$

$$\begin{aligned}- [2d(a+1)(b+1)(c+1)]\Big\{&[rm-\alpha-1]^{r-1} \\ &+ [rm-\alpha-2]^{r-2}\alpha'[D]^1 \\ &+ [rm-\alpha-3]^{r-3}\beta'[D]^2 \\ &+ \gamma'[D]^3\Big\}[\kappa]^1\end{aligned}$$

$$\begin{aligned}+ [2cd(a+1)(b+1)]\Big\{&[rm-\alpha-2]^{r-2} \\ &+ [rm-\alpha-3]^{r-3}\alpha''[D]^1 \\ &+ \beta''[D]^2\Big\}[\kappa]^2\end{aligned}$$

$$-[2bcd(a+1)]\Big\{[rm-\alpha-3]^{r-3} \\ +\alpha'''[D]^1\Big\}[\kappa]^3 \\ +abcd\kappa'[\kappa]^4.$$

78. In all these formulae there is, as before, a numerical divisor in the case of any equalities among the numbers $a, b, c,$ &c. And D denotes, as before, the deficiency, viz. its value now is $D = \frac{1}{2}(m-1)(m-2) - \delta - \kappa$; or observing that the class n is $= m^2 - m - 2\delta - 3\kappa$, we have $D = \frac{1}{2}n - m + 1 + \frac{1}{2}\kappa$, or say $D = 1 + A$ if $A = -m + \frac{1}{2}n + \frac{1}{2}\kappa$.

79. It is to be observed with reference to the applicability of these formulae within certain limits only, that the formulae are the *only* formulae which are generally true; thus taking the simplest case, that of a single contact a, the only algebraical expression for the number of the curves C^r which have with a given curve U^m a contact of the order a, and besides pass through the requisite number $\frac{1}{2}r(r+3) - a$ of arbitrary points, is that given by the formula, viz.

$$(a) = (a+1)(rm - a + aD) - a\kappa.$$

Considering the curve U^m and the order r of the curve C^r as given, if a has successively the values $1, 2, \ldots$ up to a limiting value of a, the formula gives the number of the proper curves C^r which have with the given curve U^m a contact of the required order a: beyond this limiting value the formula no longer gives the number of the proper curves C^r which satisfy the required condition, and it thus ceases to be applicable; but there is no algebraic function of a which would give the number of the proper curves C^r as well beyond as up to the foregoing limiting value of a.

80. The formulae are applicable provided only the conditions include the conditions of passing through a sufficient number of arbitrary points; viz. when the number of arbitrary points is sufficiently great, it is not possible to satisfy the conditions specially by means of improper curves C^r, being or comprising a pair of coincident curves. Thus to take a simple example, suppose it is required to find the number of the conics which touch a given curve m times and besides pass through $\sigma - m$ given points: if the number of the given points be 4 or 3 there is no coincident line-pair through the given points, and therefore no coincident line-pair satisfying the given conditions; if the number of the given points is $= 2$, then the line joining these points gives a coincident line-pair having at each of its m intersections with the given curve a special contact therewith, that is, having in $\frac{1}{6}m(m-1)(m-2)$ ways three special contacts with the given curve; if the number of the given points is 1 or 0, then in the first case any line whatever through the given point, and in the second case any line whatever, regarded as a coincident line-pair, has m special contacts with the given curve; and so in general there is a certain value for the number of given points, for which value the conditions of contact may be satisfied by a determinate number of improper curves C^r, and for values inferior to it the conditions may be satisfied by infinite series of improper curves C^r. It is by such considerations as these that De Jonquières has determined the minimum value T of the number of arbitrary points to which the conditions should relate in order that the formulae may be applicable: I refer for his investigation and results to paragraphs XVII and XVIII of his memoir.

I remark that in the case where the number of improper solutions is finite, the formula can be corrected so as to give the number of proper solutions by simply subtracting the number of the improper solutions: but this is not so when the improper solutions are infinite in number; the mode of obtaining the approximate formulae in this case is very complicated.

81. I have mentioned that the formulae are applicable within certain limits only; and I will now explain how the limits may be determined in the case of a single contact a. Writing for shortness $A = -m + \frac{1}{2}n + \frac{1}{2}\kappa$, that is, $D = 1 + A$, we have

$$(a) = (a+1)(rm - a + a + aA) - a\kappa = (a+1)(rm) - (a+1)a + (a+1)aA - a\kappa.$$

The expression $(a+1)(rm)-(a+1)a$ is always positive for values of a from $a=0$ up to a certain limiting value; but for values of a beyond this, the expression becomes negative, and the number of proper curves is $=0$. The condition that the formula shall give a positive result is

$$(a+1)(rm-a+aA)>a\kappa,$$

or, as this may be written,

$$(a+1)(rm)-(a+1)a+a[(a+1)A-\kappa]>0.$$

82. In particular, for a conic ($r=2$) and a proper curve without singularities ($\kappa=0$, $\delta=0$, $n=m(m-1)$, $A=-m+\frac{1}{2}m(m-1)=\frac{1}{2}(m^2-3m)$), the formula becomes

$$(a)=(a+1)(2m-a)+(a+1)a\cdot\tfrac{1}{2}(m^2-3m),$$

and the condition of a positive result is

$$(a+1)(2m-a)+\tfrac{1}{2}(a+1)a(m^2-3m)>0,$$

which is always satisfied for positive values of a; that is, for a conic and a proper curve, the formula is applicable for all values of a.

83. For a conic and a curve with cusps, we have $\kappa>0$, and the formula may cease to be applicable for large values of a; thus for a cuspidal cubic ($m=3$, $n=6$, $\kappa=1$, $\delta=0$, $A=-3+3+\frac{1}{2}=\frac{1}{2}$), we have

$$(a)=(a+1)(6-a+\tfrac{1}{2}a)-a=(a+1)(6-\tfrac{1}{2}a)-a=6+5a-\tfrac{1}{2}a^2-\tfrac{1}{2}a=6+\tfrac{9}{2}a-\tfrac{1}{2}a^2,$$

which is positive for $a=1,2,\ldots,9$, but negative for $a=10$; that is, the formula ceases to be applicable for $a=10$.

84. I remark that the formulae for (a), (a,b), &c. may be written in a different form; viz. if we introduce the new symbols $[a]$, $[a,b]$, &c. defined by the equations

$$\begin{aligned}(a)&=[a],\\(a,b)&=[a][b]+[a,b],\\(a,b,c)&=[a][b][c]+[a,b][c]+[a,c][b]+[b,c][a]+[a,b,c],\\&\text{\&c.},\end{aligned}$$

then the expressions for $[a]$, $[a,b]$, $[a,b,c]$, &c. are much simpler than those for (a), (a,b), (a,b,c), &c.

85. I do not attempt to write down the general expressions for $[a]$, $[a,b]$, &c., but I remark that they may be obtained by elimination from the foregoing equations; and that they are of great importance in the theory.

86. I give the following formulae for the characteristics of the systems of conics which satisfy given conditions of contact with a given curve; the characteristics are expressed in terms of m, n, and $a=3n+\kappa$:

$$\begin{aligned}\mu(1)&=n+2m-2, & \nu(1)&=2n+4m-4,\\\mu(1,1)&=m^2+2mn+\tfrac{3}{2}n^2-3m-\tfrac{7}{2}n+\tfrac{1}{2}a, & \nu(1,1)&=2m^2+4mn+n^2-6m-5n+a,\\\mu(2)&=a, & \nu(2)&=2a,\\\mu(2,1)&=4m+8n+44+a(2m+n-17), & \nu(2,1)&=12m+24n+96+a(4m+2n-36),\\\mu(3)&=-4m-5n+3a, & \nu(3)&=-12m-15n+9a.\end{aligned}$$

87. The characteristics for the higher systems may be obtained by the formulae of Nos. 38, 39; but the calculations are very complicated.

88. I give the following formulae for the characteristics of the systems of conics which have a cusp and satisfy given conditions:

$$\begin{aligned} \mu(1\kappa1) &= n + 2m - 3, & \nu(1\kappa1) &= 2n + 4m - 6, \\ \mu(1\kappa1, 1) &= m^2 + 2mn + \tfrac{3}{2}n^2 - 5m - \tfrac{9}{2}n + 6 - \tfrac{1}{2}a, & \nu(1\kappa1, 1) &= 2m^2 + 4mn + n^2 - 10m - 8n + 18 - 2a \\ \mu(2\kappa1) &= a - 7, & \nu(2\kappa1) &= 2a - 14. \end{aligned}$$

89. The characteristics for the higher systems may be obtained as before; but the calculations are even more complicated than in the case of proper conics.

90. I remark that the formulae for the characteristics are applicable only within certain limits; and that beyond these limits the formulae require correction. The determination of these corrections is a problem of great difficulty, which I have not attempted to solve in general.

91. I give the following table of the values of (1^m) and $(1^m\kappa^p)$ for a conic and a given curve; the values are expressed in terms of m, n, and $a = 3n + \kappa$:

$$\begin{aligned} (1) &= n + 2m - 2, \\ (1, 1) &= m^2 + 2mn + \tfrac{3}{2}n^2 - 3m - \tfrac{7}{2}n + \tfrac{1}{2}a, \\ (1, 1, 1) &= 2m^2 + 4mn + n^2 - 8m - 7n + 18 - 3a, \\ (1, 1, 1, 1) &= -4m - 3n - 5 + 3a, \\ (1, 1, 1, 1, 1) &= -8m - 8n - 6 + 6a, \\ (2) &= a, \\ (2, 1) &= 4m + 8n + 44 + a(2m + n - 17), \\ (2, 1, 1) &= 20m + 16n + 42 + a(2m + 2n - 27), \\ (3) &= -4m - 5n + 3a, \\ (3, 1) &= -8m - 8n - 6 + 6a, \\ (3, 1, 1) &= -12m - 4n - 12 + 6a, \\ (4) &= 4m + 8n + 44 + a(2m + n - 17), \\ (4, 1) &= 20m + 16n + 42 + a(2m + 2n - 27), \\ (5) &= -8m - 8n - 6 + 6a. \end{aligned}$$

92. The question which ought now to be considered is to determine the corrections or supplements which should be applied to the foregoing expressions (a), (a, b), &c., or to their equivalents $[a]$, $[a][b] + [a, b]$, &c. in order to obtain formulae for the cases beyond the limits within which the present formulae are applicable; but this I am not in a position to enter upon. If the extended formulae were obtained, it would of course be an interesting verification or application of them to deduce from them the complete series of expressions (1), (2), …, (1, 1, 1, 1, 1) for the number of the conics which satisfy given conditions of contact with a given curve, and besides pass through the requisite number of given points. It will be recollected that throughout these last investigations, I have put De Jonquières' $p = 0$; that is, I have not considered the case of the curves C^r which (among the conditions satisfied by them) have with the curve U^m contacts of given orders at given points of the curve; it is probable that the general formulae containing the number p admit of extensions and transformations analogous to the formulae in which p is put $= 0$, but this is a question which I have not considered.

93. The set of equations $(a) = [a]$, $(a, b) = [a][b] + [a, b]$, &c., considered irrespectively of the meaning of the symbols contained therein, gives rise to an analytical question which is considered in Annex No. 7.

The question of the conics satisfying given conditions of contact is considered from a different point of view in my Second Memoir above referred to.

Annex No. 1 (referred to in the notice of De Jonquières' memoir of 1861).—**On the form of the equation of the curves of a series of given index.**

To obtain the general form of the equation of the curves C^n of a series of the index N, it is to be observed that the equation of any such curve is always included in an equation of the order n in the coordinates, containing linearly and homogeneously certain parameters $a, b, c, \ldots$; this is universally the case, as we may, if we please, take the parameters $(a, b, c, \ldots)$ to be the coefficients of the general equation of the order n; but it is convenient to make use of any linear relations between these coefficients so as to reduce as far as possible the number of the parameters. Assume that the number of the parameters is $= \omega + 1$, then in order that the curve should form a series (that is, satisfy $\frac{1}{2}n(n+3) - 1$ conditions), we must have a $(\omega - 1)$-fold relation between the parameters, or, what is the same thing, taking the parameters to be the coordinates of a point in ω-dimensional space, say the parametric point, the point in question must be situate on a $(\omega - 1)$-fold locus. Moreover, the condition that the curve shall pass through a given point establishes between the parameters a linear relation (viz. that expressed by the original equation of the curve regarding the coordinates therein as belonging to the given point, and therefore as constants); that is, when the curve passes through a given point, the corresponding positions of the parametric point are given as the intersections of the $(\omega - 1)$-fold locus by an $(\omega - 1)$-fold locus; the number of the curves is therefore equal to the number of these intersections, that is, to the order of the $(\omega - 1)$-fold locus; or the index of the series being assumed to be $= N$, the order of the $(\omega - 1)$-fold locus must be also $= N$. That is, the general form of the equation of the curves C^n which form a series of the index N, is that of an equation of the order n containing linearly and homogeneously the $\omega + 1$ coordinates of a certain $(\omega - 1)$-fold locus of the order N. It is only in a particular case, viz. that in which the $(\omega - 1)$-fold locus is unicursal, that the coordinates of a point of this locus can be expressed as rational and integral functions of the order N of a variable parameter θ; and consequently only in this same case that the equation of the curves C^n of the series of the index N can be expressed by an equation $(* \overset{\omega}{\frown} x, y, z)^n = 0$, or $(* \overset{\omega}{\frown} x, y, 1)^n = 0$, rational and integral of the degree N in regard to a variable parameter θ.

If in the general case we regard the coordinates of the parametric point as irrational functions of a variable parameter θ, then rationalising in regard to θ, we obtain an equation rational of the order N in θ, but the order in the coordinates instead of being $= n$, is equal to a multiple of n, say qn. Such an equation represents not a single curve but q distinct curves C^n, and it is to be observed that if we determine the parameter by substituting therein for the coordinates their values at a given point, then to each of the N values of the parameter there corresponds a system of q curves, only one of which passes through the given point, the other $q - 1$ curves are curves not passing through the given point, and having no proper connexion with the curves which satisfy this condition.

Returning to the proper representation of the series by means of an equation containing the coordinates of the parametric point, say an equation $(* \overset{\omega}{\frown} x, y, 1)^n = 0$, involving the two coordinates (x, y), it is to be noticed that forming the derived equation and eliminating the coordinates of the parametric point, we obtain an equation rational in the coordinates (x, y), and also rational of the degree N in the differential coefficient $\frac{dy}{dx}$; in fact since the number of curves through any given point (x_0, y_0) is $= N$, the differential equation must give this number of directions of passage from the point (x_0, y_0) to a consecutive point, that is, it must give this number of values of $\frac{dy}{dx}$, and must consequently be of the order N in this quantity.

Conversely, if a given differential equation rational in $x, y, \frac{dy}{dx}$, and of the degree N in the last-mentioned quantity $\frac{dy}{dx}$, admit of an algebraical general integral, the curves represented by this integral equation may be taken to be irreducible curves, and this being so they will be curves of a certain order n forming a series of the index N; whence the general integral (assumed to be algebraical) is given by an equation of the above-mentioned form, viz. an equation rational of a certain order n in the coordinates, and containing linearly and homogeneously the $\omega+1$ coordinates of a variable parametric point situate on an $(\omega-1)$-fold locus. The integral equation expressed in the more usual form of an equation rational of the order N in regard to the parameter or constant of integration, will be in regard to the coordinates of an order equal to a multiple of n, say $= qn$, and for any given value of the parameter will represent not a single curve C^n, but a system of q such curves: the first-mentioned form is, it is clear, the one to be preferred.

Annex No. 2 (referred to, No. 17).—**On the line-pairs which pass through three given points and touch a given conic.**

Taking the given points to be the angles of the triangle formed by the lines $(x=0, y=0, z=0)$, we have to find (f, g, h) such that the conic $(0,0,0,f,g,h \mid x,y,z)^2 = 0$, or, what is the same thing, $fyz + gzx + hxy = 0$, shall reduce itself to a line-pair, and shall touch a given conic $(1,1,1,\lambda,\mu,\nu \mid x,y,z)^2 = 0$. The condition for a line-pair is that one of the quantities f, g, h shall vanish, viz. it is $fgh = 0$; the condition for the contact of the two conics is found in the usual manner by equating to zero the discriminant of the function

$$\lambda(1+\theta f)\cdot\mu(1+\theta g)\cdot\nu(1+\theta h) - \ldots = (a, b, c, d \mid e, \ldots)$$

suppose; the values of a, b, c, d being

$$\begin{aligned}
a &= -(f^2+g^2+h^2-2\lambda gh-2\mu hf-2\nu fg),\\
b &= ((\mu\nu-\lambda)f+(\nu\lambda-\mu)g+(\lambda\mu-\nu)h),\\
c &= 1-\lambda^2-\mu^2-\nu^2+2\lambda\mu\nu,\\
d &= 2fgh.
\end{aligned}$$

Hence considering (f, g, h) as the coordinates of the parametric point, we have the discriminant-locus $a = 0$, and the contact-locus

$$a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd = 0,$$

and at the intersection of the two loci, $a = 0$, $b^2(4bd - 3c^2) = 0$, equations breaking up into the system $(a = 0, b = 0)$ twice, and the system $a = 0$, $4bd - 3c^2 = 0$; the former of these is

$$fgh = 0, \quad f^2+g^2+h^2-2\lambda gh-2\mu hf-2\nu fg = 0,$$

which expresses that the intersection of the two lines of the line-pair intersect on the given conic; in fact the system is satisfied by $f = 0$, $g^2 + h^2 - 2\lambda gh = 0$, giving a line-pair $x(hy + gz) = 0$, the two lines whereof intersect on the conic $(1,1,1,\lambda,\mu,\nu \mid x,y,z)^2 = 0$; and similarly, if $g = 0$, then $h^2 + f^2 - 2\mu hf = 0$, or if $h = 0$, then $f^2 + g^2 - 2\lambda fg = 0$. As noticed above this system occurs twice.

The second system $a = 0$, $4bd - 3c^2 = 0$ gives

$$\begin{aligned}
fgh = 0, \quad -2fgh((\mu\nu-\lambda)f+(\nu\lambda-\mu)g+(\lambda\mu-\nu)h)&\\
-3(1-\lambda^2-\mu^2-\nu^2+2\lambda\mu\nu)^2 &= 0,
\end{aligned}$$

or, as the second equation may also be written,

$$\begin{aligned}f^2 = (1-\mu^2)(1-\nu^2) + g^2(\lambda^2-1)(1-\nu^2) + h^2(1-\lambda^2)(1-\mu^2)\\ + 2gh(1-\lambda^2)(\mu\nu-\lambda) + 2hf(1-\mu^2)(\nu\lambda-\mu) + 2fg(1-\nu^2)(\lambda\mu-\nu) = 0,\end{aligned}$$

which expresses that a line of the line-pair touches the conic; in fact the system is satisfied by $f = 0$, $g^2(1-\nu^2) + h^2(1-\mu^2) + 2gh(\mu\nu-\lambda) = 0$, viz. we have here the line-pair $x(hy+gz) = 0$, in which the line $hy+gz = 0$ touches the conic $(1, 1, 1, \lambda, \mu, \nu \mid x, y, z)^2 = 0$ and the like if $g = 0$, or if $h = 0$. This system it has been seen occurs only once.

Annex No. 3 (referred to, No. 22).—**On the conics which pass through two given points and touch a given conic.**

Consider the conics which pass through two given points and touch a given conic. We may take $Z = 0$ as the equation of the line through the two given points, and then taking the pole of this line in regard to the given conic and joining it with the two given points respectively, the equations of the joining lines may be taken to be $X = 0$ and $Y = 0$ respectively. This being so, we have for the given points $(X = 0, Z = 0)$ and $(Y = 0, Z = 0)$ respectively, and for the given conic

$$aX^2 + bY^2 + 2hXY + cZ^2 = 0;$$

and since the required conic is to pass through the two given points its equation will be of the form

$$wX^2 + 2xYZ + 2yZX + 2zXY = 0,$$

where (x, y, z, w) are variable parameters which must satisfy a single condition in order that the last-mentioned conic may touch the given conic. The condition is at once seen to be that obtained by making the equation

$$(a+\lambda w)(b+\lambda w) - (h+\lambda z)^2 - c(x+\lambda y)^2 = 0,$$

considered as a cubic equation in λ, have a pair of equal roots; or if we write

$$\begin{aligned}A &= 3(ab-h^2),\\ B &= (ab-h^2)w - 2chz,\\ C &= -ax^2 - by^2 - cz^2 + 2h(xy - zw),\\ D &= 3z(2xy - wz),\end{aligned}$$

then the required condition is

$$A^2D^2 + 4AC^3 + 4B^3D - 3B^2C^2 - 6ABCD = 0.$$

Hence the conic

$$wX^2 + 2xYZ + 2yZX + 2zXY = 0$$

satisfies the prescribed conditions, if only the parameters (x, y, z, w) satisfy the last-mentioned equation, that is, if (x, y, z, w) are the coordinates of a point on the sextic surface represented by this equation.

The surface has upon it a cuspidal curve the equations whereof are $A = 0$, $B = 0$, $C = 0$, $D = 0$; this may be considered as the intersection of the quadric surface $AC - B^2 = 0$ and the cubic surface $AD - BC = 0$; and the cuspidal curve is consequently a sextic.

The surface has also a nodal curve made up of two conics; to prove this I write for shortness $k = h - \sqrt{ab}$, $k_1 = h + \sqrt{ab}$; the values of A, B, C, D then are

$$\begin{aligned} &A = -kk_1, \\ &B = -kk_1 w - 2chz, \\ &C = -ax^2 - by^2 - cz^2 + 2h(xy - zw), \\ &D = 3z(2xy - zw); \end{aligned}$$

and it is in the first place to be shown that the surface contains the conic

$$(x : y : z : w) = \theta\sqrt{b} : \theta\sqrt{a} : 1 : -k\theta^2 + t,$$

where θ is a variable parameter. Substituting these values, we have

$$\begin{aligned} B &= k_1(k\theta^2 - c(\sqrt{a} + \sqrt{ab})), \\ C &= 2kk_1\theta^2 - (3h - \sqrt{ab}), \end{aligned}$$

and hence

$$\begin{aligned} AD - BC &= -2k(kk_1\theta^2 + 2c\sqrt{ab})^2, \\ AC - B^2 &= -k^2(kk_1\theta^2 + 2c\sqrt{ab})^2, \\ BD - C^2 &= (kk_1\theta^2 + 2c\sqrt{ab})^2, \end{aligned}$$

values which satisfy identically the equation of the surface written under the form $(AD - BC)^2 - 4(AC - B^2)(BD - C^2) = 0$.

Moreover, proceeding to form the derived equation, and to substitute therein the foregoing values of (x, y, z, w), we have

$$dA : dB : dC : dD = k^2 : 2k : 1 : 0,$$

and then the derived equation is

$$(AD - BC)(3A - 2kB - k^2C) - 2(AC - B^2)(3B - 4kC + k^2D) \\ - 2(BD - C^2)(2kA - 2k^2B) = 0,$$

that is,

$$-k(3A - 2kB - k^2C) + (2kA - 2k^2B) = 0,$$

or finally

$$-k(A - 3Bk + 3Ck^2 - Dk^3) = 0,$$

which is satisfied by the foregoing values of A, B, C, D; hence the conic is a nodal curve on the sextic; and by merely changing the sign of one of the radicals $\sqrt{a}$, $\sqrt{b}$ (and therefore interchanging k, k_1) we obtain another conic which is also nodal curve on the surface, that is, we have as nodal curves the two conics

$$\begin{aligned} (x : y : z : w) &= \theta\sqrt{b} : \theta\sqrt{a} : 1 : -k\theta^2 + t, \\ (x : y : z : w) &= \theta\sqrt{b} : -\theta\sqrt{a} : 1 : -k_1\theta^2 + t. \end{aligned}$$

It is to be remarked that each of the nodal conics meets the cuspidal curve in two points, viz. writing for shortness $\theta = \frac{1}{2}\sqrt{\frac{c}{h}}$, $\theta_1 = -\frac{1}{2}\sqrt{\frac{c}{h}}$, for the intersections of the first conic we have

$$(x : y : z : w) = \theta\sqrt{a} : \theta\sqrt{b} : 1 : t \quad \text{and} \quad = -\theta\sqrt{a} : -\theta\sqrt{b} : 1 : t,$$

and for the intersections with the second conic

$$(x:y:z:w)=\theta_1\sqrt{a}:-\theta_1\sqrt{b}:1:t \quad\text{and}\quad =-\theta_1\sqrt{a}:\theta_1\sqrt{b}:1:t.$$

The condition of passing through any arbitrary point establishes a linear relation between the parameters (x,y,z,w). Hence, if the conic in addition to the prescribed conditions passes through two other given points, the point (x,y,z,w) is given as the intersection of a line with the sextic surface; the number of intersections is $=6$. If (x,y,z,w) is situate on the cuspidal curve, then the conic instead of simply touching the given conic will have with it a contact of the second order, and if we besides suppose that the conic passes through a given point, then the point (x,y,z,w) is given as the intersection of the cuspidal curve with a plane; the number is $=6$. Similarly, if the conic has two contacts with the given conic, and besides passes through a given point, then the point (x,y,z,w) is given as the intersection of the nodal curve by a plane; the number is $=4$. Finally (observing that in the case in question of the contacts of a conic with a conic we cannot have three simple contacts, or a simple contact and one of the second order), a point of intersection of the nodal and cuspidal curves answers to a contact of the third order; and the number is $=4$. That is, the theory

of the sextic surface leads to the following values (agreeing with those obtained from the formulae by writing therein $m=n=2$, $a=6$), viz.

$$\begin{aligned}(1::)&=6,=2m+n,\\(1,1,\cdot)&=4,=2m^2+2mn+\tfrac{1}{2}n^2-2m-\tfrac{5}{2}n+a,\\(2,\cdot)&=6,=a,\\(3:)&=4,=-4m-5n+3a.\end{aligned}$$

I remark that the section by an arbitrary plane is a sextic curve having 6 cusps and 4 nodes; it is therefore a *unicursal* sextic; this suggests the theorem that the sextic surface is also unicursal, viz. that the coordinates are expressible rationally in terms of two parameters; I have found that this is in fact the case. In doing this there is no loss of generality in supposing that $a=b=c=1$; and assuming that this is so, and putting also $-1+h=k$, $1+h=k_1$, and therefore $2h=k+k_1$, we have

$$\begin{aligned}A&=-kk_1,\\B&=-kk_1w-(k+k_1)z,\\C&=-x^2-y^2-z^2+(k+k_1)(xy-zw),\\D&=3z(2xy-zw).\end{aligned}$$

The equation of the sextic surface being, as before,

$$A^2D^2+4AC^3+4B^3D-3B^2C^2-6ABCD=0,$$

I say that this equation is satisfied on writing therein

$$\begin{aligned}x+y&=\sqrt{-k}(1+k_1a)\sin\phi,\\x-y&=\sqrt{k}(1+ka)\cos\phi,\\z&=1,\\w&=(2a-1)\cos^2\phi+(2a-1)\sin^2\phi,\end{aligned}$$

where (a,ϕ) are arbitrary. In fact these values give

$$\begin{aligned}A&=-kk_1\cos^2\phi-kk_1\sin^2\phi,\\B&=-k(2k_1a+1)\cos^2\phi-k_1(2ka+1)\sin^2\phi,\\C&=-k(ak_1+2)\cos^2\phi-k_1(ak+2)\sin^2\phi,\\D&=-ka^2\cos^2\phi-k_1a^2\sin^2\phi,\end{aligned}$$

whence, ω being arbitrary, we have

$$(A, B, C, D \mid \omega, 1)^3 = -[k\cos^2\phi(k_1\omega + 1) + k_1\sin^2\phi(k\omega + 1)](\omega + a)^2,$$

viz. the equation $(A, B, C, D \mid \omega, 1)^3 = 0$, considered as a cubic equation in ω, has the twofold root $\omega = -a$, that is, we have the above relation between (A, B, C, D). Whence also writing $\sin\phi = \frac{2\lambda}{1+\lambda^2}$, $\cos\phi = \frac{1-\lambda^2}{1+\lambda^2}$, the equation of the surface is satisfied by the values

$$\begin{aligned} x+y : x-y : z : w = \sqrt{-k}(1+k_1a)\cdot 2\lambda(1+\lambda^2) : \sqrt{k}(1+ka)(1-\lambda^2)(1+\lambda^2) \\ : (1+\lambda^2)^2 : (1+\lambda^2)^2[(2a-1)\cos^2\phi + (2a-1)\sin^2\phi], \end{aligned}$$

or the coordinates are expressed rationally in terms of a, λ.

Annex No. 4 (referred to, Nos. 22 and 71).—**On the Conics which touch a cuspidal cubic.**

In the cuspidal cubic, if $x = 0$ be the equation of the tangent at the cusp, $y = 0$ that of the line joining the cusp with the inflexion, and $z = 0$ that of the tangent at the cusp, then the equation of the curve is $y^2 = x^2z$; the coordinates of a point on the cubic are given by $x : y : z = 1 : \theta : \theta^3$, where θ is a variable parameter; and we have, at the cusp $\theta = \infty$, at the inflexion $\theta = 0$. In the cubic, $m = n = 3$, $a(= 3n + \kappa) = 10$.

Considering now the conic

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0,$$

this meets the cubic in the 6 points the parameters of which are determined by the equation

$$(a, b, c, f, g, h \mid 1, \theta, \theta^3)^2 = 0,$$

or, what is the same thing,

$$(c, f, g, b, 2h, a \mid \theta, 1)^6 = 0.$$

The discriminant of this sextic function contains the factor c, hence equating the residual factor to zero, we obtain the equation of the contact-locus in the form

$$(c, f, g, b, h, a)^6 = 0.$$

It follows that the number of the conics (1 ::) is = 9, which agrees with the general value (1 ::) $= 2m + n$. If the conic pass through the cusp we have $c = 0$, and the equation in θ is reduced to a quartic; it is convenient to alter the letters in such wise that the quartic equation may be obtained in the standard form $(a, b, c, d, e \mid \theta, 1)^4 = 0$; viz. this will be the case if the equation of the conic is taken to be

$$(e, 6c, 0, 6d, 2b, 0 \mid x, y, z)^2 = 0,$$

and we then obtain the equation of the contact-locus in the form

$$(ae - 4bd + 3c^2)^3 - 27(ace + 2bcd - ad^2 - b^2e - c^3)^2 = 0,$$

which is a one-fold locus of the order 6. It follows that we have

$$(1\kappa 1, 1, \cdot) = 6, \quad \text{agreeing with } (1\kappa 1, 1, \cdot) = n + 2m - 3.$$

The condition in order that the conic may touch a given line is given by an equation of the form

$$(* \mid a^2, ab, b^2, 2ce - 3d^2, ae - 8bd, ad - 12bc)^2 = 0,$$

which is a one-fold locus of the order 2; it at once follows that we have

$$(1\kappa 1, 1, /) = 12, \quad \text{agreeing with } (1\kappa 1, 1, /) = 2n + 4m - 6.$$

It is a matter of some difficulty to show that we have

$$(1\kappa 1, 1, //) = 18, \quad \text{agreeing with } (1\kappa 1, 1, //) = 4m + 4n - 6;$$

but I proceed to effect this, first remarking that I do not attempt to prove the remaining case

$$(1\kappa 1, 1, ///) = 15, \quad \text{agreeing with } (1\kappa 1, 1, ///) = 4m + 2n - 3.$$

Investigation of the value $(1\kappa 1, 1, //) = 18$*:*

We have the sextic locus

$$(ae - 4bd + 3c^2)^3 - 27(ace + 2bcd - ad^2 - b^2e - c^3)^2 = 0,$$

and combined therewith two quadric loci,

$$(* \mid a^2, ab, b^2, 2ce - 3d^2, ae - 8bd, ad - 12bc)^2 = 0,$$
$$(*' \mid a^2, ab, b^2, 2ce - 3d^2, ae - 8bd, ad - 12bc)^2 = 0,$$

which intersect in a three-fold locus of the order 24; it is to be shown that this contains as part of itself the quadric three-fold locus $(a = 0, b = 0, 2ce - 3d^2 = 0)$ taken three times, leaving a residual locus of the order $24 - 6, = 18$.

We may imagine the coordinates a, b, c, d, e expressed as linear functions of any four coordinates, and so reduce the problem from a problem in 4-dimensional space to one in ordinary 3-dimensional space. We have thus a sextic surface, and two quadric surfaces; the sextic is a developable surface or torse, having for one of its generating lines the line $a = 0$, $b = 0$, and for the tangent plane along this line the plane $a = 0$; the two quadric surfaces meet in a quartic curve passing through the two points $(a = 0, b = 0, 3ce - 2d^2 = 0)$, which are points on the torse; it is to be shown that each of these points counts three times among the intersections of the torse with the quartic curve, the number of the remaining intersections being therefore $24 - 6, = 18$; and in order thereto it is to be shown that each of the points is

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is situate on the nodal line of the torse, and that the quartic curve touches there the sheet which is not touched by the tangent plane $a = 0$; for this being so the quartic curve touching one sheet and simply meeting the other sheet meets the torse in three consecutive points, or the two points of intersection count each of them three times.

The torse has the cuspidal line

$$S = ae - 4bd + 3c^2 = 0, \quad T = ace + 2bcd - ad^2 - b^2e - c^3 = 0,$$

and the nodal line

$$\frac{6(ac - b^2), \quad 3(ad - bc), \quad ae + 2bd - 3c^2, \quad 3(be - cd), \quad 6(ce - d^2)}{a, \quad b, \quad c, \quad d, \quad e}$$

and the equations of the nodal line are satisfied by the values ($a = 0,\ b = 0,\ 3ce - 2d^2 = 0$) of the coordinates of the points in question.

To find the tangent planes at these points, starting from the equation $S^3 - 27T^2 = 0$ of the torse, taking (A, B, C, D, E) as current coordinates, and writing

$$\partial = A\partial_a + B\partial_b + C\partial_c + D\partial_d + E\partial_e,$$

then the equation of the tangent plane is in the first instance given in the form $S^2\partial S - 18T\partial T = 0$, which writing therein ($a = 0,\ b = 0,\ 3ce - 2d^2 = 0$) assumes, as it should do, the form $0 = 0$; the left-hand side is in fact found to be $9c^4(3ce - 2d^2)A$. Proceeding to the second derived equation, this is $S^2\partial^2 S + 2S(\partial S)^2 - 18T\partial^2 T - 18(\partial T)^2 = 0$, or substituting the values of the several terms, the equation is

$$\begin{aligned} 9c^4(AE - 4BD + 3C^2) + 3c^2(eA - 4dB + 6cC)^2 \\ + 18c^3\{e(AC - B^2) + 2d(BC - AD) + c(AE + 2BD - 3C^2)\} \\ - 9\{(ce - d^2)A + 2cdB - 3c^2C\}^2 = 0; \end{aligned}$$

the terms in BC, BD, C^2 vanish identically, that in B^2 is $(48-36 =)12c^3d^2 - 18c^4e, = -6c^4(3ce - 2d^2)B^2$, which also vanishes; hence there remain only the terms divisible by A, giving first the tangent plane

$A = 0$, and secondly the other tangent plane,

$$\begin{aligned}A(-6c^3d^2 + 18cd^2e - 9c^4) + B(-60c^3de + 36cd^3)&\\ + C(108c^3e - 54c^2d^2) + D(-36c^3d) + E \cdot 27c^4 = 0.&\end{aligned}$$

Taking the equations of the quadric surfaces to be

$$\begin{aligned}&(\lambda,\ \mu,\ \nu,\ \rho,\ \sigma,\ \tau \chi a^2,\ b^2,\ ab,\ 3ce - 2d^2,\ ae - 8bd,\ ad - 12bc) = 0,\\ &(\lambda',\ \mu',\ \nu',\ \rho',\ \sigma',\ \tau' \chi \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \quad) = 0,\end{aligned}$$

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the equations of the tangent planes are

$$\begin{aligned}&\rho(3cE + 5eC - 4dD) + \sigma(eA - 8dB) + \tau(dA - 12cB) = 0,\\ &\rho'(\qquad\qquad) + \sigma'(\qquad\qquad) + \tau'(\qquad\qquad) = 0,\end{aligned}$$

in all which equations we have $3ce - 2d^2 = 0$; and if to satisfy this equation we write $c : d : e = 2 : 3\beta : 3\beta^2$, then the equations of the tangent planes become

$$\begin{aligned}&\rho\beta^2(A\beta - 8B) + \sigma(3C\beta^2 - 4D\beta + 2E) = 0,\\ &\rho(8B\beta - 4D\beta + 2E) + (\sigma\beta + \tau)(A\beta - 8B) = 0,\\ &\rho'(\qquad\qquad) + (\sigma'\beta - \tau')(\qquad\qquad) = 0,\end{aligned}$$

or the three tangent planes intersect in the line

$$A\beta - 8B = 0, \quad 3C\beta^2 - 4D\beta + 2E = 0,$$

which completes the proof.

Reverting to the sextic locus,

$$(ae + 4bd - 3c^2)^3 - 27(ace + 2bcd - ad^2 - b^2e - c^3)^2 = 0,$$

considered as a locus in 4-dimensional space depending on the five coordinates (a, b, c, d, e), this has upon it the twofold locus

$$ae + 4bd + 3c^2 = 0, \quad ace + 2bcd - ad^2 - b^2e - c^3 = 0,$$

say the cuspidal locus, of the order 6, and the twofold locus

$$\frac{2(ac - b^2),\ 3(ad - bc),\ ae + 2bd - 3c^2,\ 3(be - cd),\ 2(ce - d^2)}{a,\ b,\ c,\ d,\ e} = 0,$$

say the nodal locus, of the order 4: there is also a threefold locus

$$\begin{vmatrix} a, & b, & c, & d\\ b, & c, & d, & e\end{vmatrix} = 0,$$

say the supercuspidal locus, of the order 4.

We thence at once infer

$$\begin{aligned}&(1/l,\ 2) = 6,\ \text{agreeing with } (1/l,\ 2) = \alpha - 4,\\ &(1/l,\ 1,\ 1) = 4,\ \text{agreeing with } (1/l,\ 1,\ 1) = 2m^2 + 2mn + \tfrac{1}{2}n^2 - 8m - 5n + 13 + \tfrac{1}{2}\alpha,\\ &(1/l,\ 3) = 4,\ \text{''}\ (1/l,\ 3) = -4m - 3n - 5 + 3\alpha;\end{aligned}$$

but I have not investigated the application to the symbols with $\cdot/$ or $//$.

If the conic, instead of simply passing through the cusp, touches the cuspidal tangent, then in the equation $(a, b, c, f, g, h \chi x, y, z)^2 = 0$ of the conic we have $f = 0$, or, what is the same thing, in the equation $(e, bc, 0, \frac{1}{2}a, 2b, 2d \chi x, y, z)^2 = 0$ of the conic we have $a = 0$. The equation in θ is thus reduced to $4b\theta^3 + 6c\theta^2 + 4d\theta + e = 0$. For the independent discussion of this case it is convenient to alter the coefficients so that the equation in θ may be in the standard form $(a, b, c, d \chi \theta, 1)^3 = 0$, viz. we

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assume the equation of the conic to be $(d, 3b, 0, 0, \frac{1}{2}a, \frac{1}{2}c \,\chi\, x, y, z)^2 = 0$. The equation of the contact-locus then is

$$C^2c^2 + 4\Phi acd^2 + 4\Psi d^3 - 6abcd - 3\Psi^2c^2 = 0,$$

viz. this is a developable surface, or torse, of the order 4, and we at once infer $(2/l,\ 1) = 4$, agreeing with $(2/l,\ 1) = 2m + n - 5$.

I will show also that we have $(2/l,\ 1 \cdot /) = 6$, agreeing with $(2/l,\ 1 \cdot /) = 2m + 2n - 6$, and $(2/l,\ 1//) = 5$, $(2/l,\ 1//) = m + 2n - 4$.

The condition that the conic may touch an arbitrary line $\alpha x + \beta y + \gamma z = 0$, is in fact

$$(0, -\tfrac{1}{2}a,\ f(4bd - 3c^2),\ fac,\ -\tfrac{1}{2}ab,\ 0 \,\chi\, \alpha,\ \beta,\ \gamma)^2 = 0,$$

which, considering therein (a, b, c, d) as coordinates, is the equation of a quadric surface passing through the conic $a = 0$, $4bd - 3c^2 = 0$; the quartic torse also passes through this conic; hence the quadric surface and the torse intersect in this conic, which is of the order 2, and in a residual curve of the order 6; and the number of the conics $(2/l,\ 1/)$ is equal to the order of this residual curve, that is, it is $= 6$.

If the conic touch a second arbitrary line $\alpha' x + \beta' y + \gamma' z = 0$, then we have in like manner the quadric surface

$$(0, -\tfrac{1}{2}a',\ f(4bd - 3c^2),\ fa'c,\ -\tfrac{1}{2}a'b,\ 0 \,\chi\, \alpha',\ \beta',\ \gamma')^2 = 0;$$

that is, we have the quartic torse and two quadric surfaces, each passing through the conic $a = 0$, $4bd - 3c^2 = 0$, and it is to be shown that the number of intersections not on this conic is $= 5$.

The two quadric surfaces intersect in the conic and in a second conic; this second conic meets the torse in 8 points, but 2 of these coincide with the point $a = 0$, $b = 0$, $c = 0$, which is one of the intersections of the two conics (the point $a = 0$, $b = 0$, $c = 0$ is in fact a point on the cuspidal edge of the torse, and, the conic passing through it, reckons for 2 intersections), and 1 of the 8 points coincides with the other of the intersections of the two conics; there remain therefore $8 - 2 - 1, = 5$ intersections, or we have $(2/l,\ 1//) = 5$.

Annex No. 5 (referred to, Nos. 22 and 71).

On the Conics which have contact of the third order with a given cuspidal cubic, and two contacts (double contact) with a given conic.

Let the equation of the cuspidal cubic be $x^2z - y^3 = 0$ ($x =$ tangent at cusp, $z =$ tangent at inflexion, $y =$ line joining cusp and inflexion; equation satisfied by $x : y : z = 1 : \theta : \theta^3$); and let the equation of the given conic be $U = (a, b, c, f, g, h \,\chi\, x, y, z)^2 = 0$;

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viz. in the rational form this is
then writing

$$\Theta = (a, b, c, f, g, h \,\chi\, 1, \theta, \theta^3)^2 = c\theta^6 + 2f\theta^4 + 2g\theta^3 + b\theta^2 + 2h\theta + a,$$

the equation of a conic having with the given cubic at a given point $(1, \theta, \theta^3)$ contact of the second order, and having double contact with the given conic, is

$$\sqrt{\Theta} \cdot \begin{vmatrix} \sqrt{\Theta}, & x, & y, & z \\ \sqrt{\theta}, & 1, & \theta, & \theta^3 \\ (\sqrt{\theta})', & . & 1, & 3\theta^2 \\ (\sqrt{\theta})'', & . & . & 6\theta \end{vmatrix} = 0,$$

viz. in the rational form this is

$$\Theta = \begin{vmatrix} \sqrt{\Theta}, & x, & y, & z \\ \sqrt{\theta}, & 1, & \theta, & \theta^3 \\ (\sqrt{\theta})', & . & 1, & 3\theta^2 \\ (\sqrt{\theta})'', & . & . & 6\theta \end{vmatrix}^2 = 0,$$

and this will have at the point $(1, \theta, \theta^3)$ a contact of the third order if θ be determined by

$$\begin{vmatrix} \sqrt{\Theta}, & 1, & \theta, & \theta^3 \\ (\sqrt{\theta})', & . & 1, & 3\theta^2 \\ (\sqrt{\theta})'', & . & . & 6\theta \\ (\sqrt{\theta})''', & . & . & 6 \end{vmatrix} = 0,$$

viz. this is $\Theta \cdot (\sqrt{\theta})''' - (\sqrt{\Theta})''' = 0$; or developing and multiplying by θ^6 this is

$$\Theta(2\theta\theta''' - \theta'') + \theta\theta'(-\tfrac{3}{2}\theta\theta''' + \tfrac{1}{2}\theta'') + \theta'^2 \cdot \tfrac{1}{2}\theta\theta' = 0;$$

and substituting for Θ its value, this is

$$\begin{aligned} &(c\theta^6 + 2f\theta^4 + 2g\theta^3 + b\theta^2 + 2h\theta + a)^2(45c\theta^4 + 12f\theta^2 - 6) \\ &\qquad + (c\theta^6 + 2f\theta^4 + 2g\theta^3 + b\theta^2 + 2h\theta + a) \\ &\qquad (3c\theta^5 + 4f\theta^3 + 3g\theta^2 + b\theta + h)(-42c\theta^4 - 32f\theta^2 - 16g\theta - 2h) \\ &\qquad\qquad + \theta^6(3c\theta^5 + 4f\theta^3 + 3g\theta^2 + b\theta + h)^2 = 0. \end{aligned}$$

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The coefficients of the powers 16, 15, 14, 13 of θ all vanish, so that this is in fact an equation of the twelfth order $(2/l,\ 1//) = 0$; and putting, as usual,

$$(bc - f^2,\ ca - g^2,\ ab - h^2,\ gh - af,\ hf - bg,\ fg - ch) = (A, B, C, F, G, H),$$

the equation is found to be

$$\begin{gathered} -4cA \cdot \theta^{12} \\ +7hA \cdot \theta^{11} + (\tfrac{1}{2}aB + 8cH)\theta^{10} + (9gB - 20fC)\theta^9 \\ -36cB \cdot \theta^8 - 22bH \cdot \theta^7 + (16f^2 + 40aA)\theta^6 \\ +10gG \cdot \theta^5 - 10cG \cdot \theta^4 - 180hH \cdot \theta^3 + 20aG \cdot \theta^2 \\ -12aF \cdot \theta + 40g^2 + 20b^2 + 40fF - 60fB \cdot \theta^5 \\ +8hB \cdot \theta^4 - abc \cdot \theta^3 - 90g^2 + 2bG \cdot \theta^2 - aC = 0, \end{gathered}$$

where the form of the coefficients may be modified by means of the identical equations

$$\begin{gathered} (A, H, G \,\chi\, a, h, g)^2 = K, \quad (H, B, F \,\chi\, a, h, g)^2 = 0, \quad (G, F, C \,\chi\, a, h, g)^2 = 0, \\ (A, H, G \,\chi\, h, b, f)^2 = 0, \quad (H, B, F \,\chi\, h, b, f)^2 = K, \quad (G, F, C \,\chi\, h, b, f)^2 = 0, \\ (A, H, G \,\chi\, g, f, c)^2 = 0, \quad (H, B, F \,\chi\, g, f, c)^2 = 0, \quad (G, F, C \,\chi\, g, f, c)^2 = K. \end{gathered}$$

There is consequently a conic answering to each value of θ.

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Substituting these values, but retaining (a, b, c, f, g, h) as standing for their values $a = 2\lambda\lambda'$, &c., the equation in θ is found to contain the cubic factor $2X\theta^3 - 3Y\theta^2 + Z$, where it is to be observed that this factor equated to zero determines the values of θ which correspond to the points of contact with the cuspidal cubic of the tangents from the point (X, Y, Z), which is the intersection of the lines $\lambda x + \mu y + \nu z = 0$, and $\lambda' x + \mu' y + \nu' z = 0$; and omitting the cubic factor, the residual equation is found to be $(2/l,\ 1//) = 0$, where the form of the coefficients may be modified by means of the identical equations

$$aX + hY + gZ = 0, \quad hX + bY + fZ = 0, \quad gX + fY + cZ = 0.$$

The equation is of the 9th order, and there are consequently 9 conics.

Annex No. 6 (referred to, No. 48).
Containing, with the variation referred to in the text, Zeuthen's forms for the characteristics of the conics which satisfy four conditions.

(1)

$$\begin{aligned}
(1) &= \eta + 2m,\\
(1,\ 1) &= \eta + 2m, \quad (1\cdot/) = 2m - 4m, \quad (1//) = 4\eta + 4m, \quad (1///) = 4\eta + 2m, \quad (1111) = 2\eta + m;\\
(2) &= 2m(m+n-3) + \tau,\\
(2,\ 1) &= 2m(m+2n-5) + 2\tau, \quad (2//) = 2\eta(2m+n-5) + 2\delta,\\
(2//) &= 2\eta(m+n-3) + \delta;\\
(3) &= \tfrac{1}{2}[2m^2 + 6m^2n - n^2 - 30m^2 - 18mn + 13n^2 + 84m - 42n + (6m+3n-26)\tau],\\
(3\cdot/) &= \tfrac{1}{2}[(m+n)(-(m+n)^2 - 7(m+n) + 48) + 4mn(3m+3n-13) + 2(3m+3n-20)(\delta+\tau)],\\
(3//) &= \tfrac{1}{2}[-m^2 + 6mn^2 + 2n^2 + 18m^2 - 18mn - 30n^2 - 42m + 84n + (3m+6n-26)\delta].
\end{aligned}$$

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$$\begin{aligned}
(1,\ 1,\ 1,\ 1) = \tfrac{1}{2}\{&2(m-3)(m-4)(\eta^2 - m - n) + (n-3)(n-4)(m^2 - m - \eta)\\
&+ 4(m^2 - 11m + 28)\tau + 2(n^2 - 11n + 28)\delta\\
&+ (4(m-4)(n-4) - 1)(2\delta + \tau) + 2\delta^2 + \tau^2\},\\
(1\cdot/) = \tfrac{1}{2}\{&(m-3)(m-4)(\eta^2 - m - n) + 2(n-3)(n-4)(m^2 - m - \eta)\\
&+ 2(m^2 - 11m + 28)\tau + 4(n^2 - 11n + 28)\delta\\
&+ (4(n-4)(m-4) - 1)(\delta + 2\tau) + \delta^2 + 2\tau^2\};
\end{aligned}$$

(2, 1)

$$\begin{aligned}
(2,\ 1) &= 3m + \iota,\\
(2\cdot 1) &= 2(3m+\iota), \quad (2,\ 1//) = 2(3m+\iota),\\
(2,\ 1//) &= 3(2mn + n^2 + 4m - 10n) + (2m+n-14)\kappa,\\
(2,\ 1,\ 1) &= 2(3m+\iota)(m+n-12) + 24(m+n),\\
(2\cdot 1,\ 1) &= 3(m^2 + 2mn - 10m + 4\eta) + (m+2n-14)\tau;\\
(2\cdot/) &= (2m+n-7)(6\delta + (n-3)\kappa) + \{(m-n)(m+n-5) + \tau\}(3m+\iota-36) + 12(m-n)(m+n-3),\\
(2//) &= (m+2n-7)(6\delta + (m-3)\iota) + \{(n-m)(m+n-5) + \delta\}(3m+\iota-36) + 12(n-m)(m+n-3);
\end{aligned}$$

(2, 2)

$$\begin{aligned}
(2,\ 2) &= 6n - 4m + 3\iota = 5m - 3n + 3\iota,\\
(2\cdot 2) &= 10n - 8m + 6\iota = 10m - 8n + 6\iota,\\
(2//) &= 5n - 3m + 3\kappa = 6m - 4n + 3\tau;\\
(3) &= 2(-4m^2 + 3mn + 3n^2 + 28m - 32n) + 3(2m+n-13)\kappa,\\
(3\cdot/) &= 2(3m^2 + 3mn - 4n^2 - 32m + 28n) + 3(m+2n-13)\tau;\\
(4) &= 10n - 10m + 6\iota = 8m - 8n + 6\iota,\\
(4//) &= 10n - 8m + 6\kappa = 10m - 10n + 6\iota.
\end{aligned}$$

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Annex No. 7 (referred to, No. 93).
In connexion with De Jonquières' formula.

I have been led to consider the following question. Given a set of equations:

$$\begin{aligned}
a &= a \quad (\text{viz. } b = b,\ c = c,\ \&\text{c.}),\\
ab &= ab \quad (\text{viz. } ac = ac,\ \&\text{c., and the like in all the subsequent equations}),\\
&\quad + (11)\, a.b;\\
abc &= abc + (12)\,(a \cdot bc + b \cdot ac + c \cdot ab) + (111)\, a.b.c,\\
abcd &= abcd + (13)\,(a \cdot bcd + \&\text{c.}) + (22)\,(ab \cdot cd + \&\text{c.})\\
&\quad + (112)\,(a \cdot b \cdot cd + \&\text{c.}) + (1111)\, a.b.c.d,
\end{aligned}$$

and so on indefinitely (where the ( ) is used to denote multiplication, and *ab*, *abc*, &c., and also *ab*, *abc*, &c. are so many separate and distinct symbols not expressible in terms of *a*, *b*, *c*, &c., *a*, *b*, *c*, &c.), then we have conversely a set of equations

$$\begin{aligned}
a &= a \quad (\text{viz. } b = b,\ c = c,\ \&\text{c.}),\\
ab &= ab \quad (\text{viz. } ac = ac,\ \&\text{c., and the like in all the subsequent equations})\\
&\quad + [11]\, a \cdot b;\\
abc &= abc + [12]\,(a \cdot bc + b \cdot ac + c \cdot ab) + [111]\, a \cdot b \cdot c,\\
abcd &= abcd + [13]\,(a \cdot bcd + \&\text{c.}) + [22]\,(ab \cdot cd + \&\text{c.})\\
&\quad + [112]\,(a \cdot b \cdot cd + \&\text{c.}) + [1111]\, a \cdot b \cdot c \cdot d,
\end{aligned}$$

and so on; and it is required to find the relation between the coefficients ( ) and [ ]; we find, for example,

$$\begin{aligned}
[11] &= -(11),\\
[12] &= -(12),\\
[111] &= 3(11)(12) - (111),\\
[13] &= -(13),\\
[22] &= -(22),\\
[112] &= 2(13)(12) + (22)(11) - (112),\\
[1111] &= -12(13)(12)(11) + 4(13)(111) + 3(22)(11)(11) - 6(112)(11) - (1111);
\end{aligned}$$

and it is to be noticed that, conversely, the coefficients ( ) are given in terms of the coefficients [ ] by the like equations with the very same numerical coefficients; in fact from the last set of equations, this is at once seen to be the case as far as (112); and for the next term (1111) we have

$$\begin{aligned}
(1111) &= +12[13][12][11] - 4[13](3[12][11] - [111]) + 3[22][11][11] - 6[11](2[13][12])\\
&\quad + [22][11][11] - [1111]\\
&= +12[13][12][11] - 4[13][111] - 3[22][11][11] + 6[112][11] - [1111],
\end{aligned}$$

having the same coefficients -12, $+4$, -3, $+6$, -1 as in the formula for [1111] in terms of the coefficients ( ); it is easy to infer that the property holds good generally.

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To explain the law for the expression of the coefficients of either set in terms of the other set, I consider, for example, the case where the sum of the numbers in the ( ), or [ ] is = 5; and I form a kind of tree as follows:

[Tree diagram showing symbolic partitions]

the formation of which is obvious; and I derive from it in the manner about to be explained the expressions for the coefficients [14], [23] &c. in terms of the corresponding coefficients in ( ); viz. we have

$$
\begin{aligned}
[14] &= -(14),\\
[23] &= -(23),\\
[113] &= 2(14)(13) + (23)(11) - (113),\\
[122] &= (14)(22) + 2(23)(12) - (112),\\
[1112] &= -6(14)(13)(12) - 3(14)(22)(11) + 3(14)(112) - 6(23)(12)(11)\\
&\quad + 3(113)(12) + (23)(111) + 3(122)(11) - (1112),\\
[11111] &= +60(14)(13)(12)(11) - 20(14)(13)(111) + 15(14)(22)(11)(11)\\
&\quad - 30(14)(112)(11) + 5(14)(1111) + 30(23)(12)(11)(11)\\
&\quad - 10(23)(111)(11) - 30(113)(12)(11) + 10(113)(111)\\
&\quad - 15(122)(11)(11) + 10(1112)(11) - (11111).
\end{aligned}
$$

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To form the symbolic parts, we follow each branch of the tree to each point of its course: thus from the branch 113 we have (113) belonging to [113], (113)(111) belonging to [11111], (113)(12) belonging to [1112], (113)(12)(11) belonging to [11111]; viz. (113) belongs to [113]; (113)(111), read 11 (3 replaced by) 111, belongs to [11111]; (113)(12), read 11 (3 replaced by) 12, belongs to [1112]; (113)(12)(11), read 11 (3 replaced by) 1 (2 replaced by) 11, belongs to [11111].

And observe that where (as, for example, with the symbol 122) there are branches derived from two or more figures, we pursue each such branch separately, and also all or any of them simultaneously to every point in the course of such branch or branches; thus for the branch 122 we have (122) belonging to [122], (122)(11) belonging to [1112], (122)(11)(11) belonging to [11111].

Similarly for the branch 23 we have (23) belonging to [23], (23)(111) belonging to [1112], (23)(12) belonging to [122], (23)(12)(11) (same as infra) belonging to [1112], (23)(11)(111) belonging to [11111], (23)(11)(12) (same as supra) belonging to [1112], (23)(11)(12)(11) belonging to [11111].

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We thus obtain the symbolic parts of the several expressions for [14], [23], [11111] respectively: the sign of each term is + or − according as the number of factors in ( ) is even or odd; thus in the expression for [11111], the term (14)(13)(12)(11) has four factors, and is therefore +, the term (113)(12)(11) has three factors, and is therefore −.

The numerical coefficients are obtained as follows. There is a common factor derived from the expression in [ ]

Paper [407]

Second Memoir on the Curves which Satisfy Given Conditions; The Principle of Correspondence.

[PAGES 263–295]

[From the Philosophical Transactions of the Royal Society of London, vol. CLVIII. (for the year 1868), pp. 145–172. Received April 18,—Read May 2, 1867.]

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In the present Memoir I reproduce with additional developments the theory established in my paper "On the Correspondence of two points on a Curve" (London Math. Society, No. VII., April 1866), [385]; and I endeavour to apply it to the determination of the number of the conics which satisfy given conditions; viz. these are conditions of contact with a given curve, or they may include arbitrary conditions Z, $2Z$, &c.

If, for a moment, we consider the more general question where the Principle is to be applied to finding the number of the curves C^r of the order r, which satisfy given conditions of contact with a given curve, there are here two kinds of special solutions; viz., we may have proper curves C^r touching (specially) the given curve at a cusp or cusps thereof, and we may have improper curves, that is, curves which break up into two or more curves of inferior orders. In the case where the curves C^r are lines, there is only the first kind of special solution, where the sought for lines touch at a cusp or cusps. But in the case to which the Memoir chiefly relates, where the curves C^r are conics, we have the two kinds of special solutions, viz., proper conics touching at a cusp or cusps, and conics which are line-pairs or point-pairs. In the application of the Principle to determining the number of the conics which satisfy any given conditions, I introduce into the equation a term called the "Supplement" (denoted by the abbreviation "Supp."), to include the special solutions of both kinds. The expression of the Supplement should in every case be furnished by the theory; and this being known, we should then have an equation leading to the number of the conics which properly satisfy the prescribed conditions; but in thus finding the expression of the Supplements, there are difficulties which I am unable to overcome; and I have contented myself with the reverse course, viz., knowing in each case the

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number of the proper solutions, I use these results to determine *a posteriori* in each case the expression of the Supplement; the expression so obtained can in some cases be accounted for readily enough, and the knowledge of the whole series of them will be a convenient basis for ulterior investigations.

The Principle of Correspondence for points in a line was established by Chasles in the paper in the Comptes Rendus, June–July 1864, referred to in my First Memoir; it is extended to unicursal curves in a paper of the same series, March 1866, "Sur les courbes planes ou à double courbure dont les points peuvent se déterminer individuellement—Application du Principe de Correspondance dans la théorie de ces courbes," but not to the case of a curve of given deficiency D considered in my paper of April 1866 above referred to.

The fundamental theorem in regard to unicursal curves, viz. that in a curve of the order m with $\frac{1}{2}(m-1)(m-2)$ double points (nodes or cusps) the coordinates (x, y, z) are proportional to rational and integral functions of a variable parameter θ,—as a case of a much more general theorem of Riemann's—dates from the year 1857, but was first explicitly stated by Clebsch in the paper "Ueber diejenigen ebenen Curven deren Coordinaten rationale Functionen eines Parameters sind," Crelle, t. LXIV. (1864), pp. 43–63. See also my paper "On the Transformation of Plane Curves," London Mathematical Society, No. III., Oct. 1865, [384].

The paragraphs of the present Memoir are numbered consecutively with those of the First Memoir.

Article Nos. 94 to 104. On the Correspondence of two points on a Curve.

94. In a unicursal curve the coordinates (x, y, z) of any point thereof are proportional to rational and integral functions of a variable parameter θ. Hence if two points of the curve correspond in such wise that to a given position of the first point there correspond α' positions of the second point, and to a given position of the second point α positions of the first point, the number of points which correspond each to itself is $= \alpha + \alpha'$.

For let the two points be determined by their parameters θ, θ' respectively, then to a given value of θ there correspond α' values of θ', and to a given value of θ' there correspond α values of θ; hence the relation between (θ, θ') is of the form $(\theta, 1)^{\alpha}(\theta', 1)^{\alpha'} = 0$; and writing therein $\theta' = \theta$, then for the points which correspond each to itself, we have an equation $(\theta, 1)^{\alpha+\alpha'} = 0$, of the order $\alpha + \alpha'$; that is, the number of these points is $= \alpha + \alpha'$.

Hence for a unicursal curve we have a theorem similar to that of M. Chasles' for a line, viz. the theorem may be thus stated:

If two points of a unicursal curve have an (α, α') *correspondence, the number of united points is* $= \alpha + \alpha'$.

But a unicursal curve is nothing else than a curve with a deficiency $D = 0$, and we thence infer:

Theorem. *If two points of a curve with deficiency D have an* (α, α') *correspondence, the number of united points is* $= \alpha + \alpha' + 2kD$; in which theorem $2k$ is a coefficient to be determined.

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95. Suppose that the corresponding points are P, P' and imagine that when P is given the corresponding points P' are the intersections of the given curve by a curve Θ (the equation of the curve Θ will of course contain the coordinates of P as parameters, for otherwise the position of P' would not depend upon that of P). I find that if the curve has with the given curve k intersections at the point P, then in the system of points (P, P') the number of united points is $\alpha = \alpha + \alpha' + 2kD$, whence in particular if the curve Θ does not pass through the point P, then the number of united points is $= \alpha + \alpha'$, as in the case of a unicursal curve.

(I have in the paper of April 1866 above referred to, proved this theorem in the particular case where the k intersections at the point P take place in consequence of the curve Θ having a k-tuple point at P, but have not gone into the more difficult investigation for the case where the k intersections arise wholly or in part from a contact of the curve Θ, or any branch or branches thereof, with the given curve at P.)

96. It is to be observed that the general notion of a united point is as follows: taking the point P at random on the given curve, the curve Θ has at this point k intersections with the given curve; the remaining intersections are the corresponding points P'; if for a given position of P one or more of the points P' come to coincide with P, that is, if for the given position of P the curve Θ has at this point more than k intersections with the given curve, then the point in question is a united point.

It might at first sight appear that if for a given position of P a number 2, 3, … or j of the points P' should come to coincide with P, then that the point in question should reckon, for 2, 3, … or j (as the case may be) united points: but this is not so. This is perhaps most easily seen in the case of a unicursal curve; taking the equation of correspondence to be $(\theta, 1)^{\alpha}(\theta', 1)^{\alpha'} = 0$, then we have $\alpha + \alpha'$ united points corresponding to the values of θ which satisfy the equation $(\theta, 1)^{\alpha}(\theta, 1)^{\alpha'} = 0$; if this equation has a j-tuple root $\theta = \lambda$, the point P which answers to this value λ of the parameter is reckoned as j united points.

But starting from the equation $(\theta, 1)^{\alpha}(\theta', 1)^{\alpha'} = 0$, if on writing in this equation $\theta = \lambda$, the resulting equation $(\lambda, 1)^{\alpha}(\theta', 1)^{\alpha'} = 0$ has a root $\theta' = \lambda$, it follows that the equation $(\theta, 1)^{\alpha}(\theta, 1)^{\alpha'} = 0$ has a root $= \lambda$, and that the point which belongs to the value $= \lambda$ is a united point; if on writing in the equation $\theta' = \lambda$, the resulting equation $(\lambda, 1)^{\alpha}(\theta', 1)^{\alpha'} = 0$ has a j-tuple root $\theta' = \lambda$, it does not follow that the equation $(\theta, 1)^{\alpha}(\theta, 1)^{\alpha'} = 0$ has a j-tuple root $= \lambda$, nor consequently that the point answering to $= \lambda$ in anywise reckons as j united points.

97. This may be further illustrated by regarding the parameters θ, θ' as the coordinates of a point in a plane; the equation $(\theta, 1)^{\alpha}(\theta', 1)^{\alpha'} = 0$ is that of a curve of the order $\alpha + \alpha'$, having an α-tuple point at infinity on the axis $\theta = 0$, and an α'-tuple point at infinity on the axis $\theta' = 0$; the united points are given as the intersections of the curve with the line $\theta = \theta'$; a j-fold intersection, whether arising from a multiple point of the curve or from a contact of the line $\theta = \theta'$ with the curve, gives a point which reckons as j united points.

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But if $\theta = \lambda$ gives the j-fold root $\theta' = \lambda$, this shows that the line $\theta = \lambda$ has with the curve j intersections at the point $\theta = \theta' = \lambda$; not that the line $\theta = \theta'$ has with the curve j intersections at the point in question.

98. Reverting to the notion of a united point as a point P which is such that one or more of the corresponding points P' come to coincide with P; in the case where P is at a node of the given curve, it is necessary to explain that the point P must be considered as belonging to one or the other of the two branches through the node, and that the point P is not to be considered as a united point unless we have on the same branch of the curve one or more of the corresponding points P' coming to coincide with the point P.

If, to fix the ideas, $k = 1$, that is, if the curve Θ simply pass through the point P, then if P be at a node the curve passes through the node and has therefore at this point two intersections with the given curve; but the second intersection belongs to the other branch, and the node is not a united point; in order to make it so, it is necessary that the curve should at the node touch the branch to which the point P is considered to belong.

The thing appears very clearly in the case of a unicursal curve; we have here two values $\theta = \lambda$, $\theta = \lambda'$ answering to the node according as it is considered as belonging to one or the other branch of the curve; and in the equation of correspondence $(\theta, 1)^{\alpha}(\theta', 1)^{\alpha'} = 0$, writing $\theta = \lambda$, we have an equation $(\lambda, 1)^{\alpha}(\theta', 1)^{\alpha'} = 0$ satisfied by $\theta' = \lambda'$ but not by $\theta' = \lambda$, and the equation $(\theta, 1)^{\alpha}(\theta, 1)^{\alpha'} = 0$ is thus not satisfied by the value $\theta = \lambda$. The conclusion is that a node quâ node is not a united point.

99. But it is otherwise as regards a cusp. When the point P is at a cusp, the curve Θ (which has in general with the given curve k intersections at P) has

here more than k intersections, and (as in this case there is no distinction of branch) the cusp reckons as a united point. In the case of a unicursal curve, there is at the cusp a single value $\theta = \lambda$ of the parameter, and the equation $(\theta, 1)^{\alpha}(\theta, 1)^{\alpha'} = 0$ is satisfied by the value $= \lambda$.

But for the very reason that the cusp *quâ* cusp reckons as a united point, the cusp is a united point only in an improper or special sense, and it is to be rejected from the number of true united points. We may include the cusps, along with any other special solutions which may present themselves, under a head “Supplement,” and instead of writing as above $\alpha - \alpha - \alpha' = 2kD$, write

$$\alpha - \alpha - \alpha' + \text{Supp.} = 2kD.$$

Before going further I apply the theorem to some examples in which the curve is a system of lines.

100. *Investigation of the class of a curve of the order m with δ nodes and κ cusps.*

Take as corresponding points on the given curve two points such that the line joining them passes through a fixed point; the united points will be the points of contact of the tangents through; that is, the number of the united points will be equal to the class of the curve.

The curve Θ is here the line OP which has with the given curve a single intersection at P; that is, we have $k = 1$. The points P' corresponding to a given position of P are the remaining $m - 1$ intersections of OP with the curve, that is, we have $\alpha' = m - 1$; and in like manner $\alpha = m - 1$.

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Each of the cusps is (specially) a united point, and counts once, whence the Supplement is $= \kappa$. Hence, writing n for the class, we have

$$n + 2(m - 1) + \kappa = 2kD$$

or writing for $2D$ its value $= m^2 - 3m + 2 - 2\delta - 2\kappa$, we have $n = m^2 - m - 2\delta - 3\kappa$, which is right.

101. *Investigation of the number of inflexions.*

Taking the point P' to be a tangential of P (that is, an intersection of the curve by the tangent at P), the united points are the inflexions; and the number of the united points is equal to the number of the inflexions.

The curve Θ is the tangent at P having with the given curve two intersections at this point; that is, $k = 2$; P' is any one of the $m-2$ tangentials of P, that is, $\alpha' = m-2$; and P is the point of contact of any one of the $n-2$ tangents from P' to the curve, that is, $\alpha = n-2$.

Each cusp is (specially) a united point, and counts once, whence the Supplement is $= \kappa$. Hence, writing ι for the number of inflexions, we have

$$\iota - (m-2) - (n-2) + \kappa = 4D;$$

or substituting for $2D$ its value expressed in the form $n - 2m + 2 + \kappa$, we have

$$\iota = 3n - 2m + \kappa,$$

which is right.

102. For the purpose of the next example it is necessary to present the fundamental equation under a more general form. The curve Θ may intersect the given curve in a system of points P', each p times, a system of points Q', each q times, &c. in such manner that the points (P, P'), the points (P, Q'), &c. are pairs of points corresponding to each other according to distinct laws; and we shall then have the numbers $(\alpha, \alpha, \alpha')$, (b, β, β'), &c., corresponding to these pairs respectively, viz. (P, P') are points having an (α, α') correspondence, and the number of united points is $= a$; (P, Q') are points having a (β, β') correspondence, and the number of united points is $= b$, and so on. The theorem then is

$$p(a - \alpha - \alpha') + q(b - \beta - \beta') + \&\text{c.} + \text{Supp.} = 2kD,$$

being in fact the most general form of the theorem for the correspondence of two points on a curve, and that which will be used in all the investigations which follow.

103. *Investigation of the number of double tangents.*

Take P' an intersection of the curve with a tangent from P to the curve (or, what is the same thing, P, P' cotangentials of any point of the curve): the united points are here the points of contact of the several double tangents of the curve; or if τ be the number of double tangents, then the number of united points is $= 2\tau$.

The curve Θ is the system of the $n-2$ tangents from P to the curve; each tangent has with the curve a single intersection at P, that is, $k = n-2$; each tangent besides meets the curve in the point of contact Q' twice, and in $(m-3)$ points P'; hence if (a, α, α') refer to the points (P, Q'), and $(2\tau, \beta, \beta')$ to the points (P, P'), we have

$$2(a - \alpha - \alpha') + (2\tau - \beta - \beta') + \text{Supp.} = 2(n-2)D.$$

From the foregoing example the value of $a - \alpha - \alpha'$ is $= 4D - \kappa$.

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In the case where the point P is at a cusp, then the $n-2$ tangents become the $n-3$ tangents from the cusp, and the tangent at the cusp; hence the curve Θ meets the given curve in $2(n-3)+3, = 2n-3$ points, that is, $(n-2)+(n-1)$ points; this does not prove (ante, No. 96), but the fact is, that the cusp counts in the Supplement $(n-1)$ times, and the expression of the Supplement is $= (n-1)\kappa$.

It is clear that we have $\beta = \beta' = (n-2)(m-3)$, so that the equation is

$$8D - 2\kappa + 2\tau - 2(n-2)(m-3) + (n-1)\kappa = (n-2)2D,$$

that is

$$2\tau = 2(n-2)(m-3) + (n-6)2D + (-n+3)\kappa;$$

or substituting for $2D$ its value $= n - 2m + 2 + \kappa$ and reducing, this is

$$2\tau = n^2 + 4m - 10n - 3\kappa,$$

which is right.

104. As another example, suppose that the point P on a given curve of the order m and the point Q on a given curve of the order m' have an (α, α') correspondence, and let it be required to find the class of the curve enveloped by the line PQ.

Take an arbitrary point O, join OQ, and let this meet the curve m in P'; then (P, P') are points on the curve m having a $(m'\alpha, m\alpha')$ correspondence; in fact to a given position of P there correspond α' positions of Q, and to each of these m' positions of P'; that is, to each position of P there correspond $m\alpha'$ positions of P'; and similarly to each position of P' there correspond $m'\alpha$ positions of P.

The curve Θ is the system of the lines drawn from each of the α' positions of Q to the point O, hence the curve Θ does not pass through P, and we have $k = 0$. Therefore the number of the united points (P, P'), that is, the number of the lines PQ which pass through the point O, is $= m\alpha' + m'\alpha$, or this is the class of the curve enveloped by PQ.

It is to be noticed that if the two curves are curves in space (plane, or of double curvature), then the like reasoning shows that the number of the lines PQ which meet a given line is $= m\alpha' + m'\alpha$, that is, the order of the scroll generated by the line PQ is $= m\alpha' + m'\alpha$.

Article Nos. 105 to 111. Application to the Conies which satisfy given conditions, one at least arbitrary.

105. Passing next to the equations which relate to a conic, we seek for $(4Z)(1)$, the number of the conics which satisfy any four conditions $4Z$ and besides touch a given curve, $(3Z)(2)$ and $(3Z)(1,1)$, the number of the conics which satisfy three conditions, and besides have with the given curve a contact of the second order, or (as the case may be) two contacts of the first order; and so on with the conditions $2Z$, Z, and then finally (5), $(4,1)$, … $(1,1,1,1,1)$, the numbers of the conics which have with the given curve a contact of the fifth order, or a contact of the fourth and also of the first order…, or five contacts of the first order.

106. As regards the case $(4Z)(1)$, taking P an arbitrary point of the given curve m, and for the curve Θ the system of the conics $(4Z)(1)$ which pass through the given point P and besides satisfy the four conditions, then the curve has with the given curve $(4Z)(1)$ intersections at P, and the points P' are the remaining $(2m-1)(4Z)(1)$ intersections: in the case of a united point (P, P'), some one of the system of conics becomes a conic $(4Z)(1)$; and the number of the united points is consequently equal to that of the conics $(4Z)(1)$; we have thus the equation

$$\{(4Z)(1) - 2(2m-1)(4Z)(1)\} + \text{Supp. } (4Z)(1) = (4Z)(1) \cdot 2D.$$

107. It is in the present case easy to find *a priori* the expression for the Supplement.

1°. The system of conics $(4Z)$ contains $2(4Z) - (4Z/)$ point-pairs[1]; each of these, regarded as a line, meets the given curve in m points, and each of these points is (specially) a united point (P, P'); this gives in the Supplement the term $m\{2(4Z) - (4Z/)\}$.

2°. The number of the conics $(4Z)$ which can be drawn through a cusp of the given curve is $= (4Z)$; and the cusp is in respect of each of these conics a united point; we have thus the term $\kappa(4Z)$, and the Supplement is thus $= m\{2(4Z) - (4Z/)\} + \kappa(4Z)$.

We have moreover $(4Z)(1) = (4Z\cdot)$, $2D = n - 2m + 2 + \kappa$; and substituting these values, we find

$$\begin{aligned}(4Z)(1) &= (4m-2)(4Z)\\ &\qquad - m\{2(4Z) - (4Z/)\} - \kappa(4Z) + (n - 2m + 2 + \kappa)(4Z)\\ &= m(4Z/) + m(4Z/),\end{aligned}$$

which is right.

108. It is clear that if, instead of finding as above the expression of the Supplement, the value of $(4Z)(1)$, $= m(4Z/) + m(4Z/)$, had been taken as known, then the equation would have led to Supp. $(4Z)(1) = m\{2(4Z) - (4Z/)\} + \kappa(4Z)$; and this, as in fact already remarked, is the course of treatment employed in the remaining cases.

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[1] The expression a point-pair is regarded as equivalent to and standing for that of a coincident line-pair: see First Memoir, No. 30.

It is to be observed also that the equation may for shortness be written in the form

$$(4Z)\{(1)-2(2m-1)(1)\}+\text{Supp. }(1)=(1)\cdot 2D;$$

viz. the $(4Z)$ is to be understood as accompanying and forming part of each symbol; and the like in other cases.

109. We have the series of equations

$$\begin{aligned}
&(2Z)\{(2)-(2)(2m-2)-(1,1)\}+\text{Supp. }(2)=(2)\cdot 2D;\\
&(2Z)\{2(2)-(1,1)-(2)(2m-2)\}\\
&\quad+\{2(1,1)-(1,1)(2m-3)-(1,1)(2m-3)\}+\text{Supp. }(1,1)=(1,1)\cdot 2D;\\
&(2Z)\{(3)-(3)(2m-3)-(2,1)\}+\text{Supp. }(3)=(3)\cdot 2D;\\
&(2Z)\{2(3)-(2,1)-(2,1)\}\\
&\quad+\{(2,1)-(2,1)(2m-4)-(1,1,1)\cdot 2\}+\text{Supp. }(2,1)=(2,1)\cdot 2D;\\
&(2Z)\{3(3)-(1,2)-(3)(2m-3)\}\\
&\quad+\{(1,2)-(1,2)(2m-4)-(1,2)(2m-4)\}+\text{Supp. }(1,2)=(1,2)\cdot 2D;\\
&(2Z)\{2(2,1)-(1,1,1)\cdot 2-(2,1)(2m-4)\}\\
&\quad+\{3(1,1,1)-(1,1,1)(2m-5)-(1,1,1)(2m-5)\}\\
&\quad+\text{Supp. }(1,1,1)=(1,1,1)\cdot 2D;
\end{aligned}$$

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$$\begin{aligned}
&(Z)\{(4)-(4)(2m-4)-(1,3)\}+\text{Supp. }(4)=(4)\cdot 2D;\\
&(Z)\{2(4)-(3,1)-(2,2)\}\\
&\quad+\{(3,1)-(3,1)(2m-5)-(1,1,2)\}+\text{Supp. }(3,1)=(3,1)\cdot 2D;\\
&(Z)\{3(4)-(2,2)-(3,1)\}\\
&\quad+\{2(2,2)-(2,2)(2m-5)-(1,1,2)\}+\text{Supp. }(2,2)=(2,2)\cdot 2D;\\
&(Z)\{2(3,1)-(2,1,1)\cdot 2-(2,1,1)\cdot 2\}\\
&\quad+\{(2,1,1)-(2,1,1)(2m-6)-(1,1,1,1)\cdot 3\}\\
&\quad+\text{Supp. }(2,1,1)=(2,1,1)\cdot 2D;\\
&(Z)\{4(4)-(1,3)-(4)(2m-4)\}\\
&\quad+\{(1,3)-(1,3)(2m-5)-(1,3)(2m-5)\}+\text{Supp. }(1,3)=(1,3)\cdot 2D;
\end{aligned}$$

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$$\begin{aligned}
&(Z)\{3(3,1)-(1,1,2)-(3,1)(2m-5)\}\\
&\quad+2\{2(2,2)-(1,1,2)-(2,2)(2m-5)\}\\
&\quad+\{2(1,1,2)-(1,1,2)(2m-6)-(1,1,2)(2m-6)\}\\
&\quad+\text{Supp. }(1,1,2)=(1,1,2)\cdot 2D;\\
&(Z)\{2(2,1,1)-(1,1,1,1)\cdot 3-(2,1,1)(2m-6)\}\\
&\quad+\{4(1,1,1,1)-(1,1,1,1)(2m-7)-(1,1,1,1)(2m-7)\}\\
&\quad+\text{Supp. }(1,1,1,1)=(1,1,1,1)\cdot 2D.
\end{aligned}$$

110. I content myself with giving the expressions of only the following supplements.

$$\begin{aligned}
\text{Supp. } (4Z)(1) &= m\,[2\,(\cdot) - (\cdot/)] + \kappa\,(\cdot),\\
\text{Supp. } (3Z)(2) &= \tfrac{1}{2}m\,[2\,(\cdot) - (\cdot/)] + \tfrac{1}{2}\kappa\,(\cdot/),\\
\text{Supp. } (3Z)(1,1) &= (2mn - 3n^2 - m + n + \alpha)\,(\cdot)\\
&\quad + (2m^2 - 4mn - 2m + 2n + (m-n)\alpha)\,(\cdot/)\\
&\quad + (-m^2 + m\alpha)\,(//),\\
\text{Supp. } (2Z)(3) &= -\tfrac{1}{2}m\,[2\,(\cdot) - (\cdot/)]\\
&\quad + \tfrac{1}{2}n\,[2\,(\cdot) - (\cdot/) + 2\,(2\,(\cdot/) - (//))] + \tfrac{1}{2}\kappa\,(\cdot/).
\end{aligned}$$

where $(4Z)$ stands for any one of the symbols (4), $(3,1)$, …, $(1,1,1,1)$.

111. The expression of Supp. $(4Z)(1)$ has been explained supra, No. 108. That of Supp. $(3Z)(2)$ may also be explained.

1°. The point-pairs of the system of conics $(3Z)$, regarding each point-pair as a line, are a set of lines enveloping a curve; the class of this curve is equal to the number of the lines which pass through an arbitrary point, that is, as at first sight would appear, to the number of point-pairs in the system $(3Z\cdot)$, or to $2(3Z) - (3Z/)$: it is, however, necessary to admit that the number of distinct lines, and therefore the class of the curve, is one-half of this, or $= \frac{1}{2}[2(3Z) - (3Z/)]$; which being so, the number of the point-pairs $(3Z)$ which, regarded as lines, touch the given curve (of the order m and class n) is $= \frac{1}{2}n[2(3Z) - (3Z/)]$.

The point of contact of any one of these lines with the given curve is (specially) a united point, and we have thus the term $\frac{1}{2}n\,[2(3Z) - (3Z/)]$ of the Supplement.

2°. The number of the conics $(3Z)$ which touch the given curve at a given cusp thereof, or, say, the conics $(3Z)(2\kappa l)$, is $= \frac{1}{2}(3Z/)$, and the cusp is in respect of each of these conics a united point; we have thus the remaining term $\frac{1}{2}\kappa\,(3Z/)$ of the Supplement.

Article Nos. 112 to 135. Application to the Conies which satisfy five conditions of contact with a given Curve.

112. We have twelve equations, which I first present in what I call their original forms; viz. these are

First equation:

$$\{(5) - (5)(2m-5) - (1,4)\} + \text{Supp. } (5) = 5(5)\cdot 2D.$$

Second equation:

$$2\{(5) - (4,1) - (2,3)\} + \{(4,1) - (4,1)(2m-6) - (1,1,3)\} + \text{Supp. } (4,1) = 4(4,1)\cdot 2D.$$

Third equation:

$$3\{(5) - 2(3,2)\} + \{(3,2) - (3,2)(2m-6) - 2(1,2,2)\} + \text{Supp. } (3,2) = 3(3,2)\cdot 2D.$$

Fourth equation:

$$2\{(4,1) - 2(3,1,1) - (2,2,1)\} + \{(3,1,1) - (3,1,1)(2m-7) - (1,1,1,2)\} + \text{Supp. } (3,1,1) = 2(3,1,1)\cdot 2D.$$

Fifth equation:

$$4\{(5) - (2,3) - (4,1)\} + \{(3,2) - (2,3)(2m-6) - (1,1,3)\} + \text{Supp. } (2,3) = 2(2,3)\cdot 2D.$$

Sixth equation:

$$3\{(4,1) - 2(2,2,1) - (3,1,1)\} + 2\{(3,2) - (2,2,1) - (2,2,1)\} + \{2(2,2,1) - (2,2,1)(2m-7) - 2(1,1,1,2)\} + \text{Supp. } (2,2,1)$$

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Seventh equation:

$$2\{(3,1,1){-}3(2,1,1,1)\}{+}\{(2,1,1,1){-}(2,1,1,1)(2m{-}8){-}4(1,1,1,1,1)\}{+}\text{Supp. }(2,1,1,1) = 2(2,1,1,1){\cdot}2D.$$

Eighth equation:

$$5\{(5) - (1,4) - (5)(2m-5)\} + \{(4,1) - (1,4)(2m-5) - (1,4)(2m-5)\} + \text{Supp. }(1,4) = (1,4)\cdot 2D.$$

Ninth equation:

$$4\{(4,1){-}(1,1,3){-}(4,1)(2m{-}6)\}{+}2\{(3,2){-}(1,1,3){-}(2,3)(2m{-}6)\}{+}\{2(3,1,1){-}(1,1,3)(2m{-}7){-}(1,1,3)(2m{-}7)\}{+}$$

Tenth equation:

$$3\{(3,2){-}2(1,2,2){-}(3,2)(2m{-}6)\}{+}\{(2,2,1){-}(1,2,2)(2m{-}7){-}(1,2,2)(2m{-}7)\}{+}\text{Supp. }(1,2,2) = (1,2,2){\cdot}2D.$$

Eleventh equation:

$$3\{(3,1,1){-}(1,1,1,2){-}(3,1,1)(2m{-}7)\}{+}2\{2(2,2,1){-}2(1,1,1,2){-}(2,2,1)(2m{-}7)\}{+}\{3(2,1,1,1){-}(1,1,1,2)(2m{-}8$$

Twelfth equation:

$$2\{(2,1,1,1){-}4(1,1,1,1,1){-}(2,1,1,1)(2m{-}8)\}{+}\{5(1,1,1,1,1){-}(1,1,1,1,1)(2m{-}9){-}(1,1,1,1,1)(2m{-}9)\}{+}\text{Supp.}$$

113. I alter the forms of these equations by substituting for $2D$ its value $= n - 2m + 2 + \kappa$, and by writing for the expressions with $(\bar{1})$ their values, $(\bar{1},4) = (\cdot 4) - 6\,(5)$, &c., and except in the terms $\{\text{Supp. }(5) - \kappa\,(5)\}$, &c., by writing for κ its value $= -3m + \alpha$. The resulting equations, if the Supplements were known, would serve to determine the values of (5), (4, 1), &c.; but I assume instead that the last-mentioned expressions are known (First Memoir, No. 50), and use the equations to determine the Supplements, or, what comes to the same thing, the values of the terms in { } which contain these Supplements.

We have thus the twelve reduced equations, with resulting values of the supplements.

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114. First equation:

$$\begin{aligned}
&= (5) - 15m - 15n + 9\alpha \\
&+ \{\text{Supp. }(5) - \kappa\,(5)\} - 3m + \alpha \\
&+ (5)(8m + 7n - 4\alpha) \\
&- (\bar{1},4) + 10m + 8n - 6\alpha.
\end{aligned}$$

(that is, we have Supp. $(5) - \kappa\,(5) = -3m + \alpha$, and so in the subsequent cases, the equation gives the value of the term in { } which contains the Supplement).

115. Second equation:

$$\begin{aligned}
&= 2(5) - 30m - 30n + \alpha(18) \\
&+ (4,1) - 8m^2 - 20mn - 8n^2 + 104m + 104n + \alpha(6m + 6n - 66) \\
&+ \{\text{Supp. }(4,1) - \kappa\,(4,1)\} - 6m^2 - 3mn + 18m + 9n + \alpha(3m - 9) \\
&+ (4,1)(6m + 5n - 3\alpha) \\
&- (\bar{1},3) + 8m^2 + 12mn + 8n^2 - 56m - 53n + \alpha(-6m - 3n + 39).
\end{aligned}$$

I stop for a moment to notice a very convenient verification of the term in { }; putting therein $\alpha = 3n$, the term is $-6m^2 - 3mn + 18m + 9n + (9mn - 27n)$; and if in this we write $m^2 = mn = n^2 = 2$, and when any higher terms enter $m^3 = m^2n = mn^2 = n^3 = 4$, $m^4 = m^3n = m^2n^2 = mn^3 = n^4 = 8$, &c., the value is $-12 - 6 + 18 + 9 + 18 - 27 = 0$, viz. we should always obtain a sum $= 0$. The reason is that the term in question should always admit of being

expressed in the form $p\delta+q\kappa+r\tau+s\iota$; the reduction to this form might be effected by the substitutions $m = \frac{1}{2}(m+n)+\frac{1}{2}(\kappa-\iota)$, $n = \frac{1}{2}(m+n)-\frac{1}{2}(\kappa-\iota)$, $m^2 = 2\cdot\frac{1}{2}(m+n)+2\delta+3\kappa$, $n^2 = 2\cdot\frac{1}{2}(m+n)+2\tau+3\iota$, giving a result $= A(m+n)+$ terms in $(\delta, \kappa, \tau, \iota)$, where A is a numerical coefficient calculable as above by simply writing $m = n = 1$, $m^2 = mn = n^2 = 2$, &c., and which is $= 0$ when the term is of the proper form $p\delta + q\kappa + r\tau + s\iota$. The complete reduction to the form in question is material in the sequel, but I advert to the point here only for the sake of the numerical verification.

116. Third equation:

$$\begin{aligned}
&= 3(5) - 60m - 60n + \alpha(36)\\
&+ (3,2) - 8m^2 - 20mn - 8n^2 + 120m + 120n + \alpha(-4m - 4n - 78) + 3\alpha^2\\
&+ \{\text{Supp. } (3,2) - \kappa\,(3,2)\} - 15m^2 - 36mn - 27n^2 + \alpha(4m + 3n + 18) - 2\alpha^2\\
&+ (3,2)(4m + 3n - 2\alpha)\\
&- 2(\overline{2},2) - 54m^2 - 48n^2 + \alpha(+40) - \alpha^2.
\end{aligned}$$

Verification is $16 + 3(1\cdot 2 - 7) = 0$.

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117. Fourth equation:

$$\begin{aligned}
&= 2(4,1) - 16m^2 - 40mn - 16n^2 && (1)\\
&+ (3,1,1) - \tfrac{1}{2}m^3 - 10m^2n - 10mn^2 - \tfrac{1}{2}n^3 + 59m^2 + 116mn + 59n^2 && (2)\\
&+ \{\text{Supp. } (3,1,1) - \kappa\,(3,1,1)\} - \tfrac{3}{2}m^3 + \tfrac{3}{2}m^2n + \tfrac{3}{2}mn^2 + \tfrac{3}{2}n^3 - 26m^2 - \tfrac{51}{2}mn - \tfrac{25}{2}n^2 && (3)\\
&+ (3,1,1)(4m + 3n - 2\alpha) && (4)\\
&- (\overline{2},1,1) - 24m^2 - 36mn - 12n^2 && (5)
\end{aligned}$$

$$\begin{aligned}
&(1) && + 208m + 208n + \alpha(+12m + 12n - 132)\\
&(2) && - 434m - 434n + \alpha(\tfrac{1}{2}m^2 + 6mn + \tfrac{1}{2}n^2 - 62m - 62n + 291) - \tfrac{3}{2}\alpha^2\\
&(3) && - 507m - 253n + \alpha(\tfrac{3}{2}m^2 - \tfrac{3}{2}m - \tfrac{3}{2}n + 33)\\
&(4) && + 108m + 81n + \alpha(-m^2 - 4mn - n^2 + 7m + \tfrac{7}{2}n - 54) + 3\alpha^2\\
&(5) && + 168m + 168n + \alpha(-m^2 - 2mn - \tfrac{1}{2}n^2 + 25m + \tfrac{25}{2}n - 138) + \tfrac{1}{2}\alpha^2.
\end{aligned}$$

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119. Sixth equation:

$$\begin{array}{lr}
= 3(4,1) + 2(3,2) + 2(2,2,1) & (1,2,3) \\
+ \{\text{Supp. } (2,2,1) - \kappa\,(2,2,1)\} & (4) \\
+ (2,2,1)(2m + n - \alpha) - 2(\overline{2},1,1) & (5,6) \\
= -24m^2 - 60mn - 24n^2 + 312m + 312n & (1) \\
+ 240m + 240n & (2) \\
+ 48m^2 + 108mn + 48n^2 - 936m - 936n & (3) \\
+ 12m^2 + 6mn - 6n^2 - 60m - 6n & (4) \\
+ 12m^2 + 18mn + 6n^2 + 108m + 54n & (5) \\
- 48m^2 - 72mn - 24n^2 + 336m + 336n & (6)
\end{array}$$

$$\begin{array}{ll}
(1) & + \alpha(+18m + 18n - 198) \\
(2) & + \alpha(-8m - 8n - 156) + \alpha^2(+6) \\
(3) & + \alpha(-16m - 16n + 654) + \alpha^2(m + n - 24) \\
(4) & + \alpha(mn - 8m - 2n + 30) \\
(5) & + \alpha(2m^2 + 3mn + n^2 - 36m - 21n - 6) + \alpha^2(-m - n + 15) \\
(6) & + \alpha(-2m^2 - 4mn - n^2 + 50m + 29n - 276) + \alpha^2(+3).
\end{array}$$

Verification is $(12 + 6 - 6)2 - 60 - 6 + 3(4 - (8 + 2)2 + 30) = 0$.

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121. Eighth equation:

$$\begin{aligned}
&= 5(5) + (4,1) \\
&+ \{\text{Supp. } (\overline{1},4) - \kappa\,(\overline{1},4)\} \\
&\quad - (m - \tfrac{1}{2})\{2(\cdot\,4) - (/\,4)\} \\
&= -75m - 75n + \alpha(45) \\
&- 8m^2 - 20mn - 8n^2 + 104m + 104n + \alpha(6m + 6n - 66) \\
&+ 3m - 4n + \alpha(1) \\
&+ (-n + \tfrac{1}{2})(\cdot\,4) + (-m + \tfrac{1}{2})(/\,4) + \alpha n\,(5).
\end{aligned}$$

Verification is $1 - 4 + 3 \cdot 1 = 0$.

122. Ninth equation:

$$\begin{aligned}
&= 4(4,1) + 2(3,2) + 2(3,1,1) \\
&+ \{\text{Supp. } (\overline{1},1,3) - \kappa\,(\overline{1},1,3)\} \\
&\quad - (m - 2)\{2(\cdot\,1,3) - (/\,1,3)\} \\
&\quad + (-n + 2)(\cdot\,1,3) + (-m + 2)(/\,1,3) + n(4(4,1) + 2(2,3))
\end{aligned}$$

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123. Tenth equation:

$$\begin{aligned}
&= 3(3,2) + (2,2,1) \\
&+ \{\text{Supp. } (\overline{1},2,2) - \kappa\,(\overline{1},2,2)\} \\
&\quad - (m - 2)\{2(\cdot\,2,2) - (/\,2,2)\} \\
&\quad + (-n + 2)(\cdot\,2,2) + (-m + 2)(/\,2,2) + 3n\,(3,2)
\end{aligned}$$

124. Eleventh equation:

$$\begin{aligned}
&= 3(3,1,1) + 4(2,2,1) + 3(2,1,1,1)\\
&+ \{\text{Supp. } (\bar{1},1,1,2) - \kappa\,(\bar{1},1,1,2)\}\\
&\quad - (m - \tfrac{5}{2})\{2(\cdot\,1,1,2) - (/\,1,1,2)\}\\
&\quad + (-n + \tfrac{5}{2})(\cdot\,1,1,2) + (-m + \tfrac{5}{2})(/\,1,1,2)\\
&\quad + n(3(3,1,1) + 2(2,2,1))
\end{aligned}$$

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125. Twelfth equation:

$$\begin{aligned}
&= 2(2,1,1,1) + 5(1,1,1,1,1)\\
&+ \{\text{Supp. } (\bar{1},1,1,1,1) - \kappa\,(\bar{1},1,1,1,1)\}\\
&\quad - (m - \tfrac{5}{2})\{2(\cdot\,1,1,1,1) - (/\,1,1,1,1)\}\\
&\quad + (-n + \tfrac{5}{2})(\cdot\,1,1,1,1) + (-m + \tfrac{5}{2})(/\,1,1,1,1)\\
&\quad + 2n\,(2,1,1,1).
\end{aligned}$$

The complete expressions for the supplements obtained from these twelve equations are tabulated in the following articles.

126–132. [The detailed expressions for the twelve supplements are given in the following articles, with the resulting values:]

$$\begin{aligned}
\text{Supp. } (5) &= N + O,\\
\text{Supp. } (4,1) &= m\,[2(\cdot) - (/)] + \kappa\,(\cdot) + \ldots,\\
\text{Supp. } (3,2) &= \ldots,\\
&\&\text{c.},
\end{aligned}$$

where N $(= \iota)$ is the number of line-pair-points described as "inflexion tangent terminated each way at inflexion," and O $(= \kappa)$ the number of line-pair-points described as "cuspidal tangent terminated each way at cusp."

127.

$$\begin{aligned}
A &= \tfrac{1}{2}\iota(m-4)(m-5) = \tfrac{1}{2}m^2 + 2m^2 - \tfrac{1}{2}m^3 - \tfrac{1}{2}m^2n^2 - 18m^2 + \tfrac{1}{2}mn + 5n^2 + 40m - 5n + \alpha(-\tfrac{1}{2}m^2 + \tfrac{1}{2}m - 10);\\
B &= \tfrac{1}{2}\iota(m-3)(m-4) = -\tfrac{1}{2}m^3 + \tfrac{1}{2}m^2 - 18m + \alpha(\tfrac{1}{2}m^2 - \tfrac{1}{2}m + 6);\\
C &= \tfrac{1}{2}\iota(m-4)(m-5) = \tfrac{1}{2}N + 2m^2 - 4m^2 - \tfrac{1}{2}m^2n^2 - 18m^2 + \tfrac{1}{2}mn + 5n^2 + 40m - 5n + \alpha(-\tfrac{1}{2}m^2 + \tfrac{1}{2}m - 10);\\
D &= \tfrac{1}{2}\iota(m-3)(m-4) = -\tfrac{1}{2}m^3 + \tfrac{1}{2}m^2 - 18m + \alpha(\tfrac{1}{2}m^2 - \tfrac{1}{2}m + 6).
\end{aligned}$$

128. I make the following calculations, serving to express in terms of Zeuthen's Capitals, the terms in { } contained in the twelve equations respectively.

$$\begin{aligned}
N &= -3m + \alpha \quad \text{(first equation)}.\\
2J &= -6m^2 + 18m + 9n + \alpha(2m-6),\\
R &= -3mn + \alpha(m-3),\\
6K &= \ldots\\
L &= \ldots\\
&= -6m^2 - 3mn + 18m + 9n + \alpha(3m-9) \quad \text{(second equation)}.\\
&\ \ \vdots\\
&= -\tfrac{1}{2}m^3 + \tfrac{3}{2}m^2n + 2mn^2 + \tfrac{1}{2}n^3 + \tfrac{9}{2}m^2 - 7n^2 - 50m - 23n + \alpha(\tfrac{1}{2}m^2 - \tfrac{1}{2}m - \tfrac{1}{2}n + 33) \quad \text{(fourth equation)}.
\end{aligned}$$

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$$\begin{aligned} Q &= n^2 + 8m - n - 3\alpha \quad \text{(fifth equation)}. \\ 3G &= \dots \\ &= 12m^2 + 6mn - 6n^2 - 60m - 6n + \alpha(mn - 3m - 2n + 30) \quad \text{(sixth equation)}. \end{aligned}$$

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129. We have consequently, by means of the results just obtained,

$$\begin{aligned} \text{Supp. }(5) &= \kappa(5) + N, \\ \text{Supp. }(4,1) &= \kappa(4,1) + 2J + R, \\ \text{Supp. }(3,2) &= \kappa(3,2) + 6K + L + 3N + 2O, \\ \text{Supp. }(3,1,1) &= \kappa(3,1,1) + D + E + F + 2G + 2J + J', \\ \text{Supp. }(2,3) &= \kappa(2,3) + Q, \\ \text{Supp. }(2,2,1) &= \kappa(2,2,1) + 3G + I + 4J + 2J', \\ \text{Supp. }(2,1,1,1) &= \kappa(2,1,1,1) + B + 4D + 4D' \quad \text{(seventh equation)}. \end{aligned}$$

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$$\begin{aligned} \text{Supp. }(\bar{1},4) &= \kappa(\bar{1},4) + (n - \tfrac{1}{2})(4N + 2O) - \tfrac{1}{2}N + \tfrac{3}{2}O \quad \text{(eighth equation)}, \\ \text{Supp. }(\bar{1},1,3) &= \dots \quad \text{(ninth equation)}, \\ \text{Supp. }(\bar{1},2,2) &= \dots \quad \text{(tenth equation)}, \\ \text{Supp. }(\bar{1},1,1,2) &= \dots \quad \text{(eleventh equation)}, \\ \text{Supp. }(\bar{1},1,1,1,1) &= \dots \quad \text{(twelfth equation)}. \end{aligned}$$

130–132. [The remaining supplements are calculated in a similar manner, and the complete table of results is given.]

133. These are, I think, the true theoretical forms of the Supplements, viz. (attending to the signification of the Capitals) the expressions actually exhibit how the Supplement arises, whether from proper conics passing through or touching at a cusp, or from point-pairs (coincident line-pairs) or line-pairs (including of course in these terms line-pair-points). Thus, for instance,

$$\text{Supp. }(5) = N + O.$$

Referring to the explanations, First Memoir, Nos. 41 to 47, $N (= \iota)$ is the number of the line-pair-points described as "inflexion tangent terminated each way at inflexion," and O $(= \kappa)$ the number of the line-pair-points described as "cuspidal tangent terminated each way at cusp," or in what is here the appropriate point of view, we have as a coincident line-pair each inflexion tangent and each cuspidal tangent. Reverting to the generation of the first equation, when the point P is a point in general of the given curve, the curve Θ is the conic (5), having with the curve 5 intersections at P, and besides meeting it in the $2m - 5$ points P'. When the point P is at an inflexion, the curve Θ becomes the coincident line-pair formed by the tangent taken twice, the number of intersections at P is therefore $= 6$, and the inflexion is therefore (specially) a united point. Similarly, when the point P is at a cusp, the curve Θ becomes the coincident line-pair formed by the tangent taken twice, the number of intersections at P is therefore $= 6$, and the cusp is thus (specially) a united point: we have thus the total number of special united points $= \kappa + \iota$, agreeing with the foregoing *a posteriori* result, Supp. $(5) = N + O$.

134. Or to take another example; for the fifth equation we have

$$\text{Supp. }(\bar{2},3) = \kappa\,(\overline{2\kappa l},3) + Q;$$

Q ($= 2\tau$) is the number of the line-pair-points described as "double tangent terminated each way at point of contact," or, in the point of view appropriate for the present purpose, we have each double tangent as a coincident line-pair in respect to the one of its points of contact, and also as a coincident line-pair in respect to the other of its points of contact. Reverting to the generation of the equation, when the point P is a point in general on the given curve, the curve Θ is the system of conics $(\overline{2}, 3)$ touching the curve at P, and having besides with it a contact of the third order; since for each conic the number of intersections at P is $= 2$, the total number of intersections at P is $= 2(\overline{2}, 3)$, and the remaining $(2m - 2)(\overline{2}, 3)$ intersections are the points P'. Suppose that the point P is taken at the point of contact of a double tangent; of the $(\overline{2}, 3)$ conics, 1 (I assume this is so) becomes the coincident line-pair formed by the double tangent taken twice, and gives therefore 4 intersections at P, the remaining $(\overline{2}, 3) - 1$ conics are proper conics, giving therefore $2(\overline{2}, 3) - 2$ intersections at P, or the total number of intersections at P is $2(\overline{2}, 3) + 2$ intersections; or there is a gain of 2 intersections. As remarked (No. 96), this does not of necessity imply that the point in question is to be considered as being (specially) 2 united points; I do not know how to decide *a priori* whether it is to be regarded as being 2 united points or as 1 united point, but it is in fact to be regarded as being (specially) only 1 united point; and as the points in question are the 2τ points of contact of the double tangents, we have thus the number 2τ of special united points. Again, when the point P is at a cusp, all the $(\overline{2}, 3)$ conics remain proper conics $((\overline{2\kappa l}, 3) = (\overline{2}, 3))$,

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and we thus obtain the foregoing *a posteriori* result Supp. $(\overline{2}, 3) = Q$.

135. [Further discussion of the theoretical forms of the remaining supplements.]

Paper [408]

Addition to Memoir on the Resultant of a System of Two Equations.

[Pages 296–299]

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408. [From the Philosophical Transactions of the Royal Society of London, vol. CLVIII. (for the year 1868), pp. 579–588.]

This is an addition to the memoir on the resultant of a system of two equations, giving further developments of the theory established in the paper [380]. The memoir contains tables for the calculation of resultants for systems of equations of various degrees.

Table (4, 4) continued:

[The page contains tabular values for the resultant of two quartic equations, with numerical coefficients arranged in a systematic scheme for computing the resultant of two quartics.]

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Table (4, 4) concluded:

[These pages contain the conclusion of the table for the resultant of two quartic equations, completing the systematic enumeration of the numerical coefficients. The volume of tabular material is too extensive to reproduce here in full.]

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[The page concludes the addition to the memoir on resultants, completing the discussion of the tabular values for systems of two equations.]

Paper [409]

On the Conditions for the Existence of Three Equal Roots, or of Two Pairs of Equal Roots, of a Binary Quartic or Quintic.

[PAGE 300]

[From the Philosophical Transactions of the Royal Society of London, vol. CLVIII. (for the year 1868), pp. 577–588. Received November 26, 1867,—Read January 9, 1868.]

409.

It is remarked, Proc. R. Soc. vol. xvii. p. 314, that the above title is a misnomer: I had in fact in regard to the quintic considered not the twofold relations belonging to the root-systems 311 and 221 respectively, but the threefold relations belonging to the root-systems 41 and 32 respectively. The proper title would have been "On the conditions for the existence of certain systems of equal roots of a binary quartic or quintic."

In considering the conditions for the existence of given systems of equalities between the roots of an equation, we obtain some very interesting examples of the composition of relations. A relation is either onefold, expressed by a single equation $U = 0$, or it is, say k-fold, expressed by a system of k or more equations. Of course, as regards onefold relations, the theory of the composition is well known: the relation $UV = 0$ is a relation compounded of the relations $U = 0$, $V = 0$; that is, it is a relation which is satisfied if either of the component relations is satisfied. And so in the case of a twofold relation, expressed by the two equations $U = 0$, $V = 0$, if these equations contain each of them a factor, say if $U = PW$, $V = QW$, then the twofold relation $U = 0$, $V = 0$ is equivalent to the onefold relation $W = 0$, together with the twofold relation $P = 0$, $Q = 0$; the twofold relation is thus compounded of the onefold relation and a twofold relation. I do not in the present memoir attempt any general theory of the composition of relations; but I establish in a direct manner certain identities which put in evidence the composition of the relations in the cases of a binary quartic or quintic which present themselves in the theory.

The Binary Quartic.

Article Nos. 1 to 11.

1. Let the equation be $(a, b, c, d, e \between x, y)^4 = 0$, then we have

$$I = ae - 4bd + 3c^2,$$
$$J = ace + 2bcd - ad^2 - b^2e - c^3,$$

and the discriminant is $\Delta = I^3 - 27J^2$. When $\Delta = 0$, the equation has two equal roots. For the existence of three equal roots we have the twofold relation expressed by the two equations

$$I = 0, \quad J = 0.$$

It is to be observed that the condition $I = 0$ implies that the quartic can be expressed as the sum of two fourth powers, and the condition $J = 0$ is then the condition that these two fourth powers have a common root.

2. The further development continues with the detailed conditions for the existence of two pairs of equal roots, and the corresponding investigation for the binary quintic; establishing by direct calculation certain identities which put in evidence the composition of the relations in the several cases that present themselves in the theory.

409.

ON THE CONDITIONS FOR THE EXISTENCE OF EQUAL ROOTS OF A BINARY QUARTIC OR QUINTIC.

[From the *Philosophical Transactions* of the Royal Society of London,
vol. CLI. (for the year 1861), pp. 561–572.
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The relation for the quadric factor, for any value whatever of n, is at once seen to be expressible by means of an oblong matrix, giving rise to a series of determinants which are each to be put $=0$; the relation for three equal roots and that for two pairs of equal roots, in the particular cases $n=4$ and $n=5$, are given in my "Memoir on the Conditions for the existence of given Systems of Equalities between the roots of an Equation," Phil. Trans, vol. CXLVII. (1857), pp. 727–731, [150]; and I propose in the present Memoir to exhibit, for the cases in question $n=4$ and $n=5$, the connexion between the compound relation for the quadric factor with the component relations for the three equal roots and for the two pairs of equal roots respectively.

Article Nos. 1 to 8. — The Quartic.

1. For the quartic function $(a,b,c,d,e \,|\, x,y)^4$, the condition for three equal roots, or, say, for a root system 31, is that the quadrinvariant and the cubinvariant each of them vanish, viz. we must have

$$I = ae - 4bd + 3c^2 = 0, \qquad J = ace - ad^2 - be^2 + 2bcd - c^3 = 0.$$

2. The condition for two pairs of equal roots, or for a root system 22, is that the cubicovariant vanishes identically, viz. representing this by

$$(A, B, 5C, 10D, 5E, F, G \,|\, x,y)^6 = 0,$$

$$\begin{aligned}
A &= a^2d - 3abc + 2b^3 && = 0,\\
B &= a^2e + abd - 9ac^2 + 6b^2c && = 0,\\
C &= abe - 3acd + 2b^2d && = 0,\\
D &= -ad^2 + b^2e && = 0,\\
E &= -ade + 3bce - 2bd^2 && = 0,\\
F &= -ae^2 - 2bde + 9c^2e - 6cd^2 && = 0,\\
G &= -be^2 + 3cde - 2d^3 && = 0.
\end{aligned}$$

3. But the condition for the common quadric factor is

$$\begin{vmatrix} a, & 3b, & 3c, & d \\ b, & 3c, & 3d, & e \end{vmatrix} = 0,$$

and the determinants formed out of this matrix must therefore vanish for $(I, J) = 0$, and also for $(A, B, C, D, E, F, G) = 0$, that is, the determinants in question must be syzygetically related to the functions (I, J), and also to the functions (A, B, C, D, E, F, G).

4. The values of the determinants are

$$\begin{aligned}
1234 &= 3 \times (a^2ce + a^2d^2 - 3ab^2e + 4abcd - ac^3 - b^3d),\\
1235 &= 3 \times (a^2de - ab^2e + 2abde - 3ac^2d - ad^3 + 4b^2cd - b^3e),\\
1245 &= (a^2e^2 - abde + 2ac^2e - ad^2e - 3b^2ce + 4bc^2d - c^4),\\
1345 &= 3 \times (abce - abde + 2ac^2e - 3ad^2e - 9b^2de + 14bcd^2 - c^3d),\\
2345 &= 3 \times (ace^2 - ade^2 + 2bce^2 - 3cde^2 + 6d^2e - c^3e - d^4).
\end{aligned}$$

5. The syzygetic relation with (I, J) is given by means of the identical equation

$$(1234, 1235, 1245, 1345, 2345 \,|\, x, y)^4 = -6I \cdot HU + 9J \cdot U,$$

where HU is the Hessian of U,

$$HU = (ac - b^2, ad - bc, ae + 2bd - 3c^2, be - cd, ce - d^2 \,|\, x, y)^4,$$

or, as this may be written,

$$(1234, 1235, 1245, 1345, 2345 \,|\, x, y)^4 = -6I \cdot HU + 9J \cdot U.$$

6. That is, we have

$$\begin{aligned}
1234 &= (ac - b^2,\ a^2d - 6bI,\ 9J),\\
1235 &= (2ad - 2bc,\ 4bd - 6cI,\ 9J),\\
1245 &= (ae + 2bd - 3c^2,\ 6ce - 6dI,\ 9J),\\
1345 &= (2be - 2cd,\ 4d^2 - 6eI,\ 9J),\\
2345 &= (ce - d^2,\ e^2 - 6fI,\ 9J).
\end{aligned}$$

7. The determinants thus vanish if $(I, J) = 0$, that is, for the root system 31; they will also vanish without this being so, if only

$$\frac{3J}{2I} = \frac{ac - b^2}{a} = \frac{ad - bc}{2b} = \frac{ae + 2bd - 3c^2}{6c} = \frac{be - cd}{2d} = \frac{ce - d^2}{e},$$

and we may omit the first member $\left(\frac{3J}{2I}\right)$, since if the remaining terms are equal to each other they will also be $= \frac{3J}{2I}$. The equations may then be written

$$\frac{ac - b^2}{a} = \frac{ad - bc}{2b} = \frac{ae + 2bd - 3c^2}{6c} = \frac{be - cd}{2d} = \frac{ce - d^2}{e},$$

and the ten equations of this system reduce themselves (as it is very easy to show) to the seven equations

$$(A, B, C, D, E, F, G) = 0,$$

which, as above mentioned, are the conditions for the root system 22.

8. It may be added that we have

1234	1235	1245	1345	2345
c	c	$-e + 4d - 3c$	$-e + 3d - c$	$-3e + 4d - c$
$-4b + 3a$	$-3b + a$	$= d - 3c + a$	$= d - 3c + b$	
	$= d - 3c + a$	$= -e + 6c - a$	$= -e + 3c - b$	
			$= -d + 3c - b$	

where it is to be noticed that the four equations having the left-hand side $= 0$, give $B : C : D : E : F$ proportional to the determinants of the matrix

$$\begin{array}{cccc}
d, & -3c, & \cdot, & a\\
-e, & \cdot, & 6c, & \cdot\\
-a, & -d, & 3c, & -b'\\
-e, & \cdot, & +3c, & -b
\end{array}$$

the determinants in question contain each the factor c, and omitting this factor, the system shows that B, C, D, E, F are proportional to their before-mentioned actual values.

Article Nos. 9 to 15. — The Quintic.

9. For the quintic function $(a, b, c, d, e, f \mid x, y)^5$, the condition of a root system 41 is that the covariant, $[B =]$ No. 14, shall vanish, viz. we must have

$$A = 2(ae - 4bd + 3c^2) = 0, \quad B = af - 3be + 2cd = 0, \quad C = 2(bf - 4ce + 3d^2) = 0.$$

10. The condition of a root system 32 is that the following covariant, viz.

$$[25A'B - 25G =]3(\text{No. }13)^2(\text{No. }14) - 25(\text{No. }15)^2,$$

shall vanish; or, as this may be written, that we must have

$$\begin{vmatrix} A, & B, & C \\ A_1, & B_1, & C_1 \\ A_2, & B_2, & C_2 \end{vmatrix} = 0,$$

where A_1, B_1, C_1 and A_2, B_2, C_2 are the first and second differential coefficients of A, B, C respectively.

11. The condition of a root system $2^2 1$ is that the quadricovariant and the quarticovariant shall each vanish identically; or, what is the same thing, that we must have

$$(21, 33, \alpha\beta, \alpha\gamma, \ldots \mid x, y)^8 = 0.$$

12. The conditions for the common [cubic] factor are

$$\begin{vmatrix} a, & 4b, & 6c, & 4d, & e & \\ & a, & 4b, & 6c, & 4d, & e \\ b, & 4c, & 6d, & 4e, & f & \\ & b, & 4c, & 6d, & 4e, & f \end{vmatrix} = 0,$$

the several determinants whereof are given in Table No. 27 of my "Third Memoir on Quantics," Philosophical Transactions, vol. CXLVI. (1856), pp. 627–647, [144].

13. These determinants must therefore vanish, for $(A, B, C) = 0$, and also for $(21, 33, \ldots 8, 9) = 0$, that is, they must be syzygetically connected with (A, B, C), and also with $(21, 33, \ldots 8, 9)$. The relation to (A, B, C) is in fact given in the Table appended to Table No. 27, viz. this is

$C\times$	$B\times$	$A\times$	
$1234 = +$	$+6a^2 +$	$-12ab +$	$+16ac - 10b^2$
$1235 = +$	$+6ab +$	$-2ac - 10b^2 +$	$+6bc$
$1236 = +$	$-2ac + 8b^2 +$	$+6ad - 18bc +$	$-2ae + 8bd$
$1245 = +$	$+18ac +$	$-6ad - 30bc +$	$+8ae + 10bd$
$1246 = +$	$+12bc +$	$+4ae - 4bd - 24c^2 +$	$+4be + 8cd$
$1345 = +$	$+24ad +$	$-8ae - 40bd +$	$+4af + 20be$
$1256 = +$	$-1ae + 4bd + 3c^2 +$	$+af + 5be - 18cd +$	$-16bf + 4ce + 3df$
$2345 = +$	$+20ae + 40bd - 30c^2 +$	$-80be + 20cd + 20bf + 40ce - 30d^2 +$	
$1346 = +$	$+4ae + 8bd + 6c^2 +$	$-36cd + bf + 8ce + 9d^2 +$	
$2346 = +$	$+4af + 20be +$	$-8bf - ace + 24c^2 +$	
$1356 = +$	$+4be + 8cd +$	$+4bf - 4ce - 24d^2 + 12de +$	
$2356 = +$	$+8bf + 10ce +$	$-6cf - 30de + 18d^2 +$	
$1456 = +$	$+6ce + 6cf - 18de +$	$-2df + 8e^2 +$	
$2456 = +$	$+6cf - 2df - 10e^2 +$	$+6ef +$	
$3456 = +$	$+16df - 10e^2 +$	$-12ef + 6f^2 +$	

14. Between the expressions $21, 33$, &c., and $1234, 1235$, &c., there exist relations the form of which is indicated by the following Table:

15. Assuming the existence of these relations, we have for the determination of the numerical coefficients in each relation a set of linear equations, which are shown by the following Tables, viz. referring to the Table headed $c21$, $d33$, $a\beta$, $a.1234$, [first of the seven tables infra] if the multipliers of the several terms respectively be A, B, C, X, then the Table denotes the system of linear equations

$$A + 3B + 33C + 0X = 0, \quad 3A + 0B - 10C - 16X = 0, \quad \text{\&c.},$$

that is, nine equations to be satisfied by the ratios of the coefficients A, B, C, X, and which are in fact satisfied by the values at the foot of the Table, viz. $A : B : C : X = +66 : -11 : +1 : +6$.

There would be in all fourteen Tables, but as those for the second seven would be at once deducible by symmetry from the first seven, I have only written down the seven Tables; the solutions for the first and second Tables were obtained without difficulty, but that for the third Table was so laborious to calculate, and contains such extraordinarily high numbers, that I did not proceed with the calculation, and it is accordingly only the first, second, and third Tables which have at the foot of them respectively the solutions of the linear equations.

16. The results given by these three Tables are, of course,

$$66c21 - 11d33 + 1a\beta + 6a.1234 = 0,$$
$$330c21 + 110d33 - 55a\beta + 9a\gamma - 105a.1235 = 0,$$
$$266478575c21 - 617359490d33 + 144200810e\beta + 9656911e\gamma + 9090785a\delta - 7210040a\epsilon = 0.$$

410.

A THIRD MEMOIR ON SKEW SURFACES, OTHERWISE SCROLLS.

[From the *Philosophical Transactions* of the Royal Society of London,
vol. CLIX. (for the year 1869), pp. 111–126.
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The present Memoir is supplementary to my "Second Memoir on Skew Surfaces, otherwise Scrolls," Phil. Trans, vol. CLIV. (1864), pp. 559–577, [340], and relates also to the theory of skew surfaces of the fourth order, or quartic scrolls.

It was pointed out to me by Herr Schwarz[1], in a letter dated Halle, June 1, 1867, that in the enumeration contained in my Second Memoir I have given only a particular case of the quartic scrolls which have a directrix skew cubic; viz. my eighth species, $S(1,3^2)$, where there is also a directrix line. And this led me to observe that I had in like manner mentioned only a particular case of the quartic scrolls with a triple directrix line; viz. my third species, $S(1^3,1,4)$, where there is also a simple directrix line. The omitted species, say, ninth species, $S(1^3)$, with a triple directrix line, and tenth species, $S(3^2)$, with a directrix skew cubic, are considered in the present Memoir; and in reference to them I develope a theory of the reciprocal relations of these scrolls, which has some very interesting analytical features.

The paragraphs of the present Memoir are numbered consecutively with those of my Second Memoir above referred to.

Quartic Scroll, Ninth Species, $S(1^3)$, with a triple directrix line.

54. Consider a line the intersection of two planes, and let the equation of the one plane contain in the order 3, that of the second plane contain linearly, a variable parameter θ; the equations of the two planes may be taken to be

$$(p,q,r,s\,|\,\theta,1)^3=0, \qquad (u,v\,|\,\theta,1)=0,$$

where (p,q,r,s,u,v) are any linear functions whatever of the coordinates (x,y,z,w). Hence eliminating θ we have as the equation of the scroll generated by the line in question $(p,q,r,s\,|\,v,-u)^3=0$, viz. this is a quartic scroll having the line $u=0$, $v=0$ for a triple line; that is, the line in question is a triple directrix line.

55. Taking $x=0$, $y=0$ for the equations of the directrix line, or writing $u=x$, $v=y$, and moreover expressing (p,q,r,s) as linear functions of the coordinates (x,y,z,w), the equation of the scroll takes the form

$$(*\,|\,x,y)^3z+(*\,|\,x,y)^2w+(*\,|\,x,y)z^2+(*\,|\,x,y)zw+k(x,y)^4=0;$$

and we may, by changing the values of z and w, make the term in $(x,y)^4$ to be

$$(*\,|\,x,y)^4+(\alpha z+\beta w)(*\,|\,x,y)^3+(\gamma z+\delta w)(*\,|\,x,y)^2,$$

where the arbitrary constants $\alpha,\beta,\gamma,\delta$ may be so determined as to reduce this to a monomial kx^4, kx^3y, or kxy^3.

56. The coefficient k may vanish, and the equation of the scroll then is

$$(*\,|\,x,y)^3z+(*\,|\,x,y)^2w=0, \quad \text{or, what is the same thing,} \quad (*\,|\,x,y)^2(z,w)=0,$$

viz. the scroll has in this particular case the simple directrix line $z=0$, $w=0$, thus reducing itself to the third species, $S(1^3,1,4)$, with a triple directrix line and a single directrix line. It is proper to exclude this, and consider the ninth species, $S(1^3)$, as having a triple directrix line, but no simple directrix line.

57. The scroll $S(1^3)$ may be considered as a scroll $S(m,n,p)$ generated by a line which meets each of three given directrices; viz. these may be taken to be the directrix line, and any two plane sections

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[1] I take the opportunity of referring to his paper on Quintic Scrolls, Schwarz, "Ueber die geradlinigen Flächen fünften Grades," Crelle, t. LXVII. (1867), pp. 23–57.

of the scroll. The section by any plane is a quartic curve having a triple point at the intersection with the directrix line; moreover the sections by any two planes meet in four points, the intersections of the scroll by the line of intersection of the two planes. Conversely, taking any line and two quartics related as above (that is, each quartic has a triple point at its intersection with the directrix line, and the two quartics meet in four points), the scroll generated by the line meeting these three directrices is a quartic scroll having the line as a triple directrix line; and it is moreover such that through each point of the line there pass three generating lines, but through each point of either of the plane quartics only a single generating line; that is, that the line is a triple directrix line, but each of the plane quartics a simple directrix curve.

58. We may instead of the section by any plane, consider the section by a plane through a generating line, or by a plane through two of the three generating lines which meet at any point of the directrix line; if (to consider only the most simple case) each of the planes be thus a plane through two generating lines, the section by either of these planes is made up of the two generating lines, and of a conic passing through the directrix line; the directrices are thus the line and two conics each of them meeting the line; we have therefore in the foregoing formula $m = 1$, $n = 2$, $p = 2$, $\alpha = 0$, $\beta = 1$, $\gamma = 1$, and the order of the scroll is $8 - 2 - 2 = 4$ as before.

Quartic Scroll, Tenth Species, $S(3^2)$, with a directrix skew cubic met twice by each generating line.

59. Consider a line, the intersection of two planes; and let the equation of each plane contain in the order 2 a variable parameter θ; the equations of the two planes may be taken to be

$$(p, q, r \,|\, \theta, 1)^2 = 0, \qquad (p', q', r' \,|\, \theta, 1)^2 = 0,$$

where (p, q, r, p', q', r') are linear functions of the coordinates (x, y, z, w); hence eliminating θ, we have as the equation of the scroll generated by the line in question, $D = 0$, where D is the resultant of the two quadric functions.

The equation may be written

$$4(pq' - p'q)(rq' - r'q) - (pr' - p'r)^2 = 0;$$

and the scroll has thus as a nodal (double) line the skew cubic determined by the equations

$$\begin{vmatrix} p, & q, & r \\ p', & q', & r' \end{vmatrix} = 0.$$

It is easy to see (and indeed it will be shown presently) that this curve is met twice by each generating line of the scroll, and that the scroll is consequently a quartic scroll as described above.

60. The coordinates (x, y, z, w) may be fixed in such manner that the equations of the skew cubic shall be

$$xw - yz = 0, \quad xz - y^2 = 0, \quad yw - z^2 = 0;$$

or, what is the same thing,

$$yw - z^2 = 0, \quad zy - xw = 0, \quad xz - y^2 = 0;$$

each of the equations $pq' - p'q = 0$, $rq' - r'q = 0$, $pr' - p'r = 0$ is then the equation of a quadric surface passing through the skew cubic, or, what is the same thing, each of the functions $pq' - p'q$, $rq' - r'q$, $pr' - p'r$ is a linear function of $yw - z^2$, $zy - xw$, $xz - y^2$; and the equation of the scroll is given as a quadric equation in the last-mentioned quantities.

It will be convenient to represent the equation in the form

$$(H, F, G, B, A - F, -G \,|\, yw - z^2, zy - xw, xz - y^2)^2 = 0,$$

or, writing for shortness $yw - z^2$, $zy - xw$, $xz - y^2 = p, q, r$, which letters (p, q, r) are used henceforward in this signification only, the equation will be

$$(H, F, G, B, A - F, -G \,|\, p, q, r)^2 = 0,$$

viz. this is a quadric equation in (p, q, r), with arbitrary coefficients.

61. Comparing with the result, Second Memoir, Nos. 47 to 50, we see that in the particular case where the coefficients (A, B, C, F, G, H) satisfy the relation

$$AF + BG + CH = 0,$$

we have the eighth species, $S(1, 3^2)$, with a directrix line and a directrix skew cubic met twice by each generating line. We exclude this particular case, and in the tenth species consider the relation $AF + BG + CH = 0$ as not satisfied, and therefore the scroll as not having a directrix line.

62. I consider how the scroll may be obtained as a scroll $S(m^n, n)$ generated by a line meeting a curve of the order m in n times and a curve of the order n once. The first curve will be the skew cubic, that is $m = 3$; the second curve may be any plane section of the scroll; such a section will be a quartic curve having three nodes, one at each intersection of its plane with the skew cubic. Conversely, if we have a skew cubic, and a plane quartic meeting the skew cubic in three points, each of them a node on the quartic, then the scroll generated by the line meeting the skew cubic twice and the plane quartic once, is a quartic scroll $S(3^2)$ as described above.

63. We may, instead of the section by a plane in general, consider the section by a plane through a generating line; the section is here made up of the generating line and of a plane cubic passing through each of the two points of intersection of the generating line with the skew cubic, and having a node at the remaining intersection of its plane with the skew cubic. Or we may consider the section by a plane through the two generating lines at any point of the skew cubic; the section is here made up of the two generating lines and of a conic passing through the second intersections of the two generating lines with the skew cubic; that is, meeting the skew cubic twice.

64. Conversely, consider a skew cubic, and a conic meeting it twice; the lines which meet the skew cubic twice, and also the conic, generate a quartic scroll; this appears by the before-mentioned formula $S(m^n, n) = n([m]^n + \mu) -$ reduction; viz. we have $m = 3$, $n = 2$, and the order is $= 8 -$ reduction; the reduction arises from the cones having their vertices at the intersections of the skew cubic and the conic. Each cone is of the order 2, and (quâ simple point on the conic) each intersection gives a reduction $=$ order of the cone; that is, the total reduction is $= 4$, and the order of the scroll is $8 - 4 = 4$ as above.

65. But a more elegant mode of generation of the scroll may be obtained by means of the skew cubic alone; viz. considering the system of lines which are in involution with five given lines, or say simply the lines which belong to an involution[2], I say that the locus of a line belonging to the involution, and meeting the skew cubic twice is the quartic scroll, tenth species, $S(3^2)$.

In the particular case where the line (instead of belonging to a proper involution) passes through a fixed point (that is, where the involution is made up of the lines through the fixed point), the locus is the quadric surface through the skew cubic; that is, we have in this particular case not a quartic scroll, but a quadric surface.

66. The analytical theory presents some interesting features. I write

$$\begin{aligned} U &= (H, F, G, B, A - F, -G \,|\, p, q, r)^2 \\ &= Hp^2 + 2Fpq + 2Gpr + Bq^2 + 2(A - F)qr - Gr^2. \end{aligned}$$

The equation of the reciprocal scroll is obtained by means of the coefficients (A, B, C, F, G, H); and I write these in the form

$$\begin{aligned} A &= a, \quad B = b, \quad C = c, \\ F &= f, \quad G = g, \quad H = h. \end{aligned}$$

The reciprocal relation is obtained by expressing that the coefficients (a, b, c, f, g, h) satisfy the same equation as the coefficients (A, B, C, F, G, H); that is, that the two sets of coefficients are related in the same way as the original and reciprocal surfaces.

67. I do not pursue the analytical theory further, but content myself with having established the existence of the two new species of quartic scrolls, and having indicated their reciprocal relations. The theory of the reciprocal relations of these scrolls presents some very interesting analytical features, which I hope to develope on a future occasion.

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[2] I have worded this heading in accordance with that of the eighth species, Second Memoir, No. 51.

411.

A MEMOIR ON THE THEORY OF RECIPROCAL SURFACES.

[From the *Philosophical Transactions* of the Royal Society of London,
vol. CLIX. (for the year 1869), pp. 201–229.
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1. The present Memoir is supplementary to my "Memoir on the Theory of Reciprocal Surfaces," Phil. Trans, vol. CLVII. (1867), pp. 201–229, [380]. I take account of conical and biplanar nodes, or, as I call them, cnicnodes, and binodes; of pinch-points on the nodal curve; and of close-points and off-points on the cuspidal curve: viz. I assume that there are

C, cnicnodes,
B, binodes,
j, pinch-points,
χ, close-points,
θ, off-points,

deferring for the present the explanation of these singularities. The same letters, accented, refer to the reciprocal singularities. Or using "trope" as the reciprocal term to node, these will be C', cnictropes, B', bitropes, j', pinch-planes, χ', close-planes, θ', off-planes; but these present themselves, not in the equations above referred to, but in the reciprocal equations.

2. I take account of conical and biplanar nodes, or, as I call them, cnicnodes, and binodes; of pinch-points[3] on the nodal curve; and of close-points and off-points on the cuspidal curve: viz. I assume that there are C, cnicnodes, B, binodes, j, pinch-points, χ, close-points, θ, off-points, deferring for the present the explanation of these singularities.

3. The resulting alterations are that we must in the formulae write $\kappa - B$, $B - C$ in place of κ, B respectively; and change the formulae for $c(n-2)$, $[ab]$, $[bc]$, into

$$\begin{aligned} c(n-2) &= 2\sigma + \chi + \theta + \delta, \\ [ab] &= ab - 2p - j, \\ [ac] &= ac - 3\sigma - \chi. \end{aligned}$$

4. Making these changes, and substituting for $[ab]$, $[ac]$, $[bc]$ their values, the formulae become

$$\begin{aligned} a(n-2) &= \kappa - B + p + 2\sigma, \\ b(n-2) &= p + 2\rho + 3\chi + 3\theta, \\ c(n-2) &= 2\sigma + \chi + \theta + \delta, \\ a(n-2)(n-3) &= 2(\delta - \theta) + 3(ac - 3\sigma - \chi) + 2(ab - 2p - j), \\ b(n-2)(n-3) &= \kappa + (ab - 2p - j) + 3(bc - 3\rho - 2\chi - i), \\ c(n-2)(n-3) &= 6h + (ac - 3\sigma - \chi) + 2(bc - 3\rho - 2\chi - i), \end{aligned}$$

which replace the original formulae (A) and (B).

5. For convenience I annex the remaining equations; viz. these are

$$\begin{aligned} a' &= n(n-1) - 2b - 3c, \\ \kappa' &= 3n(n-2) - 6b - 8c, \\ \delta' &= \tfrac{1}{2}n(n-2)(n^2-9) - (n^2-n-6)(2b+3c) + 2b(b-1) + 6bc + \tfrac{3}{2}c(c-1); \end{aligned}$$

the equations

$$q = \tfrac{1}{2}a - b - 2\kappa - 3j - 6\theta, \qquad r = c - c - 2h - 3\rho,$$

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[3] This addition to the theory is in fact indicated in Salmon, see the note, p. 445; the i there employed, which

(q, r in place of Salmon's R, S respectively); the equation $\sigma = \sigma$; and the corresponding equations, interchanging the accented and unaccented letters, in all 23 equations between the 42 quantities

$$\begin{aligned} &n, a, B, \kappa; \quad b, k, t, q, p, j; \quad c, h, r, \sigma, \rho, \chi, \theta, i; \\ &\qquad\qquad\qquad B, C; \quad n', a', \kappa', \delta'; \quad b', k', t', q', p', j'; \quad c', h', r', \sigma', \rho', \chi', \theta', i'; \quad B', C'. \end{aligned}$$

6. We have

$$\begin{gathered} (a-b-c)(n-2) = (\kappa - B - \delta) - 6\rho - \chi - 3\theta, \\ (a-2b-3c)(n-2)(n-3) = 2(\delta-\theta) - 8\kappa - 18h - 6bc - 8\rho - 2\chi - i); \end{gathered}$$

and substituting these values of δ, κ in the formula $n' = a(a-1) - 2\delta - 3\kappa$, and for a its value, $= n(n-1) - 2b - 3c$, we find

$$\begin{aligned} n' = n(n-1)^2 - n(7b+12c) + 4b^2 + 8b + 9c^2 + 15c& \\ - 8\kappa - 18h + 18\rho + 12\chi + 12i - 9\theta& \\ - 2C - 3B - 3\Theta&, \end{aligned}$$

where the foregoing equations for $a-b-c$ and $a-2b-3c$ show clearly the origin of the new terms $-2C-3B-3\Theta$; these express that there is in the value of n' a reduction $= 2$ for each cnicnode, $= 3$ for each binode, and $= 3$ for each off-point.

7. We have $(n-2)(n-3) = n^2 - n + (-4n+6) = a + 2b + 3c + (-4n+6)$; and making this substitution in the equations which contain $(n-2)(n-3)$, these become

$$\begin{aligned} a(-4n+6) &= 2(\delta-\theta) - a^2 - 4p - 9\sigma - 2j - 3\chi, \\ b(-4n+6) &= 4\kappa - 2b^2 - 9\rho - 6\chi - 8i + 2p - j, \\ c(-4n+6) &= 6h - 3c^2 - 6\rho - \chi - 2i - 2\theta, \end{aligned}$$

(Salmon's equations (C)); and adding to each equation 4 times the corresponding equation with the factor $(n-2)$, these become

$$\begin{aligned} a^2 - 2a &= 2(\delta-\theta) + (\kappa - B) - a^2 - 2j - 3\chi, \\ 2b^2 - 2b &= 4\kappa - \rho + 6\chi + 12\theta - 3i + 2p - j, \\ 3c^2 - 2c &= 6h + 10\rho + 4\chi - 2i + 5\sigma - \chi. \end{aligned}$$

Writing in the first of these $a^2 - 2a = a(a-1) - a$, $= n' + 2\delta + 3\kappa - a$, and reducing the other two by means of the values of q, r, the equations become

$$\begin{aligned} n' - a &= -2C - 4B + \kappa - a - 2j - 3\chi, \\ 2q + \chi + 3i + j &= 2p, \\ 3r + c + 2i + \chi &= 5\sigma + \rho + 4\theta, \end{aligned}$$

(Salmon's equations (D)).

I attend in particular to the first of these, or rather to the reciprocal equation, which will be

$$\sigma' = a - n + \kappa' - 2j' - 3\chi' - 2C' - 4B',$$

which, writing therein $a = n(n-1) - 2b - 3c$, and $\kappa = 3n(n-2) - 6b - 8c$, becomes

$$\sigma' = 4n(n-2) - 8b - 11c - 2j' - 3\chi' - 2C' - 4B'.$$

The singularity σ' is not explicitly defined in Salmon; σ' is the reciprocal of σ, and (as such) it denotes the number of common tangent planes of the spinode torse and of the torse generated by the tangent planes along a plane section of the surface; or, what is the same thing, it is the number of the spinode planes which touch the plane section; that is, it is equal to the number of points of intersection of the spinode curve and the plane section; or, finally σ' is the order of the spinode curve.

The spinode curve is in fact for a surface of the order n without singularities the intersection of the surface by the Hessian surface of the order $4(n-2)$, and is thus a curve of the order $4n(n-2)$; or, introducing the singularities b, c, it is a curve of the order $4n(n-2)-8b-11c$.

8. It is clear that p will in like manner denote the order of the node-couple curve.

9. I express in terms of n, b, c, h, k, ρ, χ, j, θ, χ, C, B such quantities and combinations of quantities as can be so expressed. We have

$$\begin{aligned}
a = a' &= n(n-1)-2b-3c,\\
\kappa &= 3n(n-2)-6b-8c,\\
\delta' &= \tfrac{1}{2}n(n-2)(n^2-9)-(n^2-n-6)(2b+3c)+2b(b-1)+6bc+\tfrac{3}{2}c(c-1),\\
\mu &= 12k+c(6n-6)-6c^2-5\chi+3\theta-2\rho,\\
24t &= (-8n+8)b+(10n-18)c+8b^2-18c^2-2(8\kappa-18h)+20\rho-10\chi+4j+9\chi+6\theta,\\
\delta &= b^2-b-2i-3\chi-6\theta, \quad (t \text{ supra}),\\
r &= c^2-c-2h-3\rho,\\
2\sigma &= c(n-2)-(4\rho+\chi)-\theta,\\
8p &= (16n-24)b+(-15n+18)c-8b^2+18c^2+2(8\kappa-18h)-9(4\rho+\chi)-4j-9\chi-6\theta,\\
8\kappa' &= 8n(n-1)(n-2)+b(-32n+56)+c(-42n+46)+8b^2-18c^2\\
&\qquad -2(8\kappa-18h)+17(4\rho+\chi)+4j+17\chi+6\theta+8\sigma,\\
24h &= n(n-1)(n-2)(n-3)+b(-4n^2+20n-24)+c(-6n^2+18n-18)+12bc+18c^2\\
&\qquad +(8\kappa-18h)-9(4\rho+\chi)-9\chi+2(7,\\
8a' &= 8n(n-1)^2+b(-32n+40)+c(-21n+30)+8b^2-18c^2\\
&\qquad -2(8\kappa-18h)+21(4\rho+\chi)-12j+21\chi-18\theta-16C-24B,\\
8c' &= 4n(n-1)(n-2)+b(-16n+28)+c(-10n+26)+4b^2-9c^2\\
&\qquad -(8\kappa-18h)+10(4\rho+\chi)-4j+10\chi-6\theta-6C-8B,\\
2b' &= -a+n'(n-1)-3c', \quad (n', c' \text{ supra}),\\
\sigma'+2j'+3\chi'+2C'+4B' &= 4n(n-2)-8b-11c,\\
p'-4j'-6\chi'-4C'-9B' &= -11n(n-2)+a(n'-2)+22b+30c,\\
&\qquad (n', a \text{ supra}),\\
2\sigma'+4\rho'+\chi'+\theta' &= c'(n'-2), \quad (n', c' \text{ supra}),\\
4\rho'-3(2i'+3\chi'+2\theta')-2p'-j' &= (-4n'+6)c' \text{ (formulae E).}
\end{aligned}$$

10. I express in terms of n, b, c, h, k, ρ, χ, j, θ, χ, C, B such quantities and combinations of quantities as can be so expressed.

11. Forming the combinations $4q+6r$, $24t-8q+18r$ (the last of which introduces on the opposite side the term $+48t$), we obtain

$$\begin{aligned}
4q+6r &= c(6n-12)-\sigma\chi-18\rho+3\delta-2\chi,\\
24t-8q+18r &= -(8n-16)b+(15n-36)c-34\rho+8\chi+4j+9\chi+6\theta,
\end{aligned}$$

equations which are used post, No. 53.

12. I remark that if there be on a surface a right line which is such that the tangent plane is different at different points of the line, the line is said to be *scrolar*: the section of the surface by any plane through the line contains the line once. But if there is at each point of the line one and the same tangent plane, then the section of the surface by the tangent plane contains the line at least twice; if it contain it twice only, the line is *torsal*; if three times the line is *oscular*; and the tangent plane containing the torsal or oscular line may in like manner be termed a *torsal* or an *oscular* tangent plane. These epithets, scrolar, torsal and oscular, will be convenient in the sequel.

Article Nos. 13 to 39. — Explanation of the New Singularities.

I proceed to the explanation of the new singularities.

13. The *cnicnode*, or singularity $C = 1$, is an ordinary conical point; instead of the tangent plane we have a proper quadricone.

14. The *cnictrope*, or reciprocal singularity $C' = 1$, is also a well known one; it is in fact the conic of plane contact, or say rather the plane of conic contact, viz. the cnictrope is a plane touching a surface, not at a single point, but along a conic; the conic is the intersection of the plane with a proper quadricone, the enveloping quadricone of the surface at any point of the conic.

15. For a surface having the cnicnode $C = 1$, the tangent cone is a proper quadricone; and the section of the surface by a plane through the cnicnode is a curve having at the cnicnode a node, the two tangents whereof are the intersections of the plane with the quadricone; that is, the two tangents are any two lines through the cnicnode which lie in a plane through the axis of the quadricone. But the quadricone may be such that for a certain plane or planes through the axis, the section has at the cnicnode a cusp; the plane in question is the "cuspidal plane" of the cnicnode. I do not at present attend to this singularity.

16. For a surface having the cnictrope $C' = 1$, the section by the plane of the cnictrope is made up of the conic of contact twice, and of a residual curve of the order $n - 4$; that is, the conic of contact counts as two coincident lines in the section by the plane of the cnictrope. The cnictrope is thus a particular case of a trope; and the singularities of the surface are modified accordingly.

17. For a surface having the cnictrope $C' = 1$, the Hessian surface passes through the conic, which is thus thrown off from the spinode curve; or there is a reduction $= 2$ in the order of the curve, which agrees with a foregoing result.

18. The *binode*, or singularity $B = 1$, is a biplanar node, where instead of the proper quadricone we have two planes; these may be called the *biplanes*, and their line of intersection, the *edge* of the binode. The biplanes form a plane-pair.

19. The *bitrope*, or reciprocal singularity $B' = 1$, is the plane of point-pair contact; but this needs explanation.

20. Consider a surface having a binode, and the reciprocal surface having a bitrope. We have the bitrope, a plane the reciprocal of the binode; in this plane a line, the reciprocal of the edge; in the line two points, or say a point-pair, the reciprocal of the biplanes: these points may be called the *bipoints*. There are in each biplane three directions of closest contact; the reciprocals of these are in the bitrope three directions through each of the two points. The section of the reciprocal surface by the bitrope is made up of the line counting three times (or the line is oscular), and of a curve passing in the three directions (having therefore a triple point) through each of the two bipoints. The bitrope contains thus an oscular line; but it is part of the notion that there are on this line two points each a triple point on the residual curve.

21. The section of the surface having a binode, by a plane through the edge, is made up of the edge twice (representing the two biplanes) and of a residual curve of the order $n-2$, having at the intersection with the edge a node, the two tangents whereof are the intersections of the plane with the biplanes. But if the plane coincide with either of the biplanes, then the section is made up of this biplane, and of the other biplane counting once, and of a residual curve of the order $n - 2$; that is, the section has at each point of the edge a node, the two tangents being the same for all points of the edge; or, what is the same thing, the edge is a torsal line on the surface.

22. The *pinch-point*, or singularity $j = 1$, is a point on the nodal curve at which the two tangent planes coincide; that is, it is a cuspidal point on the nodal curve. The nodal curve is in general a curve along which the surface has two distinct tangent planes; but at a pinch-point these two tangent planes coincide, and the nodal curve has at the pinch-point a cuspidal singularity.

23. The section of the surface by a plane through the tangent at a pinch-point is made up of the tangent twice (representing the two coincident tangent planes) and of a residual curve of the order $n-2$, which has at the pinch-point a cusp, the tangent at the cusp being the tangent to the nodal curve. In the particular case where the nodal curve is a right line, the section is the line twice (representing the cuspidal branch), and a residual curve of the order $n - 2$, the tangent to which is the cotangent.

24. The *pinch-plane*, or reciprocal singularity $j' = 1$, is in fact a torsal plane touching the surface along a line, or meeting it in the line twice and in a residual curve. Let the line and curve meet in a point P; for the reason that the section by the plane is the line twice and the residual curve, the section has at P two coincident nodes; that is, the plane is a node-couple plane with two coincident nodes. The plane meets the cuspidal curve in a point Q; and at this point the plane touches the cuspidal torse; that is, the plane is a torsal plane of the cuspidal torse. The point Q is the reciprocal of the pinch-point; and the line PQ is the reciprocal of the tangent at the pinch-point.

25. The *close-point*, or singularity $\chi = 1$, is a point on the cuspidal curve at which the cuspidal torse has a singularity; viz. it is a point at which the tangent plane to the cuspidal torse coincides with the tangent plane to the surface. The cuspidal curve is in general a curve along which the surface has a single tangent plane (counting twice); but at a close-point the cuspidal torse has a higher singularity, and the tangent plane to the cuspidal torse coincides with the tangent plane to the surface.

26. The section of the surface by a plane through the tangent at a close-point is made up of the tangent twice (representing the cuspidal branch) and of a residual curve of the order $n-2$, which has at the close-point a node, the two tangents at the node being coincident; that is, the residual curve has at the close-point a cusp, the tangent at the cusp being the tangent to the cuspidal curve. The close-point is thus a point at which the cuspidal curve has a higher singularity than usual.

27. The *close-plane*, or reciprocal singularity $\chi' = 1$, is a plane which touches the surface along a line, or meets it in the line twice and in a residual curve; but it is a particular case of the pinch-plane, where the two coincident nodes on the residual curve are themselves coincident; that is, where the residual curve has at the point of intersection with the line a cusp instead of a node. The close-plane is thus a particular case of a torsal plane; and the singularities of the surface are modified accordingly.

28. The *off-point*, or singularity $\theta = 1$, is a point on the cuspidal curve at which the cuspidal torse has a still higher singularity; viz. it is a point at which the cuspidal torse has a cuspidal edge passing through the point. The off-point is thus a point at which the cuspidal curve has a cuspidal singularity; and the tangent plane to the cuspidal torse at the off-point is indeterminate.

29. The section of the surface by a plane through the tangent at an off-point is made up of the tangent three times (representing the cuspidal branch with a higher singularity) and of a residual curve of the order $n-3$, which has at the off-point a triple point. The off-point is thus a point at which the cuspidal curve has a very high singularity; and the singularities of the surface are modified accordingly.

30. The *off-plane*, or reciprocal singularity $\theta' = 1$, is a plane which touches the surface along a line, or meets it in the line three times and in a residual curve; that is, it is a plane which is oscular to the surface along a line. The off-plane is thus a particular case of a torsal plane; and the singularities of the surface are modified accordingly.

31. The foregoing explanations of the new singularities are sufficient for the purposes of the present Memoir. I have given a more complete account of these singularities in my paper "On the Singularities of Surfaces," Proc. Lond. Math. Soc. vol. III. (1869), pp. 166–174, [395]; and I refer to that paper for further details.

32. The singularities C, B, j, χ, θ are connected with the singularities C', B', j', χ', θ' by the reciprocal relations which have been explained. The number of each kind of singularity is a characteristic of the surface; and the reciprocal singularities are determined by the original singularities, and vice versa.

33. The theory of the new singularities may be applied to the determination of the singularities of a surface which is given by its equation. For instance, if the surface has a cnicnode, or a binode, or a pinch-point, or a close-point, or an off-point, then the singularities of the reciprocal surface may be determined by means of the formulae which have been given.

34. The theory may also be applied to the determination of the singularities of a surface which is the envelope of a system of planes. For instance, if the surface is the envelope of a system of planes which depend on two parameters, then the singularities of the surface may be determined by means of the theory of the new singularities.

35. I give some examples of the application of the theory. First, if the surface has a cnicnode $C = 1$, then the reciprocal surface has a cnictrope $C' = 1$; and the order of the spinode curve is reduced by 2. Second, if the surface has a binode $B = 1$, then the reciprocal surface has a bitrope $B' = 1$; and the order of the node-couple curve is reduced by 3. Third, if the surface has a pinch-point $j = 1$, then the reciprocal surface has a pinch-plane $j' = 1$; and the order of the cuspidal curve is reduced by 1.

36. The theory of the new singularities is also connected with the theory of curves. For instance, the cnicnode is a particular case of a node; and the cnictrope is a particular case of a trope. The binode is a particular case of a node; and the bitrope is a particular case of a trope. The pinch-point is a particular case of a cusp; and the pinch-plane is a particular case of a torsal plane. The close-point is a particular case of a cusp; and the close-plane is a particular case of a torsal plane. The off-point is a particular case of a cusp; and the off-plane is a particular case of a torsal plane.

37. The theory of the new singularities may be extended to higher dimensions. For instance, in four-dimensional space, the cnicnode is a particular case of a node; and the cnictrope is a particular case of a trope. The binode is a particular case of a node; and the bitrope is a particular case of a trope. The pinch-point is a particular case of a cusp; and the pinch-plane is a particular case of a torsal plane. The close-point is a particular case of a cusp; and the close-plane is a particular case of a torsal plane. The off-point is a particular case of a cusp; and the off-plane is a particular case of a torsal plane.

38. The theory of the new singularities is a very beautiful and important branch of geometry; and it has many applications to the theory of curves and surfaces. The principles which I have explained are sufficient to determine the singularities of any surface and its reciprocal; and they may be applied to a great variety of problems.

39. I have now given a sufficient account of the theory of the new singularities; and I conclude this part of the Memoir with a few remarks on the connection of the theory with other branches of geometry. The theory of the new singularities is connected with the theory of curves, with the theory of congruences, with the theory of complexes, and with the theory of transformations. It is also connected with the theory of invariants and covariants; and it has many applications to the physical sciences.

Article Nos. 40 to 53. — Further Developments.

40. I proceed to further developments of the theory of reciprocal surfaces. The formulae which have been given may be used to determine the singularities of a surface which is given by its equation; and they may also be used to determine the singularities of a surface which is the envelope of a system of planes.

41. The formulae for the singularities of a surface of the order n with singularities b, c, h, k, ρ, χ, j, θ, C, B are as follows:

$$\begin{aligned}
a &= n(n-1) - 2b - 3c,\\
\kappa &= 3n(n-2) - 6b - 8c,\\
\delta' &= \tfrac{1}{2}n(n-2)(n^2-9) - (n^2-n-6)(2b+3c) + 2b(b-1) + 6bc + \tfrac{3}{2}c(c-1),\\
\mu &= 12k + c(6n-6) - 6c^2 - 5\chi + 3\theta - 2\rho.
\end{aligned}$$

42. The formulae for the reciprocal singularities are:

$$\begin{aligned}
a' &= n'(n'-1) - 2b' - 3c',\\
\kappa' &= 3n'(n'-2) - 6b' - 8c',\\
\delta &= \tfrac{1}{2}n'(n'-2)(n'^2-9) - (n'^2-n'-6)(2b'+3c') + 2b'(b'-1) + 6b'c' + \tfrac{3}{2}c'(c'-1).
\end{aligned}$$

43. The formulae for the order of the spinode curve and the node-couple curve are:

$$\begin{aligned}
\sigma' &= 4n(n-2) - 8b - 11c - 2j' - 3\chi' - 2C' - 4B',\\
p &= (16n-24)b + (-15n+18)c - 8b^2 + 18c^2 + 2(8\kappa - 18h)\\
&\quad - 9(4\rho+\chi) - 4j - 9\chi - 6\theta.
\end{aligned}$$

44. The formulae for the order of the cuspidal curve and the cuspidal torse are:

$$\begin{aligned}
c' &= 4n(n-1)(n-2) + b(-16n+28) + c(-10n+26) + 4b^2 - 9c^2\\
&\quad - (8\kappa - 18h) + 10(4\rho+\chi) - 4j + 10\chi - 6\theta - 6C - 8B,\\
r &= c^2 - c - 2h - 3\rho.
\end{aligned}$$

45. The formulae which have been given are sufficient to determine the singularities of any surface and its reciprocal. They may be used to verify the results which have been obtained; and they may also be used to determine the singularities of surfaces which have not been previously considered.

46. The theory of reciprocal surfaces is a very beautiful and important branch of geometry. It has many applications to the theory of curves and surfaces; and it is also connected with the theory of invariants and covariants. The principles which I have explained are sufficient to determine the singularities of any surface and its reciprocal.

47. I have now given a sufficient account of the theory of reciprocal surfaces; and I conclude this Memoir with a few remarks on the connection of the theory with other branches of geometry. The theory of reciprocal surfaces is connected with the theory of curves, with the theory of congruences, with the theory of complexes, and with the theory of transformations.

48. The theory of reciprocal surfaces may be applied to the determination of the singularities of a surface which is given by its equation. For instance, if the surface is of the order n, and has a nodal curve of the order a and a cuspidal curve of the order κ, then the singularities of the reciprocal surface may be determined by means of the Plückerian equations.

49. The theory may also be applied to the determination of the singularities of a surface which is the envelope of a system of planes. For instance, if the surface is the envelope of a system of planes which depend on two parameters, then the singularities of the surface may be determined by means of the theory of reciprocal surfaces.

50. The theory of reciprocal surfaces is also connected with the theory of curves. For instance, the reciprocal of a curve is a developable surface; and the singularities of the curve correspond to the singularities of the developable. The theory of reciprocal curves is therefore a particular case of the theory of reciprocal surfaces.

51. The theory may also be extended to higher dimensions. For instance, in four-dimensional space, the reciprocal of a surface is a surface; and the singularities of the two surfaces are reciprocal to each other. The theory of reciprocal surfaces in four-dimensional space is therefore analogous to the theory of reciprocal curves in three-dimensional space.

52. The Plückerian equations for a surface of the order n with a nodal curve of the order a and a cuspidal curve of the order κ may be written in the form:

$$\begin{aligned} q &= n(n-1)^2 - a - 2\kappa, \\ r &= \tfrac{1}{2}n(n-1)(n-2)(n-3) - (n-2)a + \tfrac{1}{2}\kappa(\kappa-1), \\ s &= 3n(n-2) - 2a + 3\kappa. \end{aligned}$$

These three equations determine the class q and the orders r and s of the nodal and cuspidal tdevelopables of the reciprocal surface, in terms of the order n of the given surface and the orders a and κ of its nodal and cuspidal curves.

53. The equations may also be written in the form:

$$\begin{aligned} n &= q(q-1)^2 - r - 2s, \\ a &= q(q-1)(q-2) - 2r + s, \\ \kappa &= \tfrac{1}{2}s(s-1). \end{aligned}$$

These three equations determine the order n and the singularities a and κ of the given surface, in terms of the class q and the orders r and s of the nodal and cuspidal tdevelopables of the reciprocal surface. The six equations of Nos. 52 and 53 are equivalent to each other; and they may be used to determine the singularities of a surface and its reciprocal.

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A Memoir on the Theory of Reciprocal Surfaces. (continued)

Article 1 (60.)**.** in my Memoir on Cubic Surfaces was obtained without the aid of the formula now in question) I endeavour by means of the contained results to find the values of the unknown coefficients h, g, χ, λ, μ, ν.

For a cubic surface $n = 3$, and in fact for all the cases except the cubic scrolls (XXII and XXIII), the formula is

$$\theta' = 54 - hC - gB - \lambda' j' - \mu' \chi' - \nu' \theta' - h' C' - g' B',$$

and applying this to the several cases of cubic surfaces as grouped together in the Table, and referred to by the affixed roman numbers, the resulting equations are

$$54 = 54, \tag{I}$$
$$30 = 54 - f, \tag{II}$$
$$18 = 54 - g - 16\nu', \tag{III}$$
$$13 = 54 - 2h - \lambda', \tag{IV}$$
$$6 = 54 - h - g - \mu' - 3\nu', \tag{VI}$$
$$3 = 54 - 3h - 3\lambda', \tag{VIII}$$
$$1 = 54 - 2g - 16\nu' - g', \tag{IX}$$
$$0 = 54 - 4h - 6\lambda', \tag{XIII}$$
$$0 = 54 - h - 2g - \mu' - g', \tag{XVI}$$
$$0 = 54 - 3g - 3g', \tag{XVII}$$
$$0 = 54 - 2h - g - \lambda' - 2\mu', \tag{XXI}$$

which are all satisfied if only

$$h = 24, \quad g = 9, \quad g + 16\mu' = 36, \quad g + 2\mu' = 12, \quad g + g' = 18, \quad \lambda' = -7.$$

Article 2 (61.)**.** If we apply to the same surfaces the reciprocal equation (observing that the Table of Singularities for the reciprocal surfaces, which is given by interchanging the upper and lower halves of the original Table of Singularities, is as follows):

$$0 = 54432 - 54432, \tag{I}$$
$$0 = 27851 - 27846 - \lambda', \tag{II}$$
$$0 = 18180 - 18318 - g' - 16\nu, \tag{III}$$
$$0 = 11765 - 11756 - 2\mu' - \lambda, \tag{IV}$$
$$0 = 6917 - 6584 - h' - g' - \mu - 3\nu, \tag{VI}$$
$$0 = 3534 - 3522 - 3\lambda, \tag{VIII}$$
$$0 = 3024 - 3144 - 2g' - 16\nu - 5\nu, \tag{IX}$$
$$0 = 1433 - 1186 - 2h' - g' - \lambda - 2\mu, \tag{XIII}$$
$$0 = 518 - 504 - 4h' - 6\lambda, \tag{XVI}$$
$$0 = 383 - 322 - h' - 2g' - g + 2\mu, \tag{XVII}$$
$$0 = 54 - 54, \tag{XXI}$$

viz. these are

$$
\begin{aligned}
3h' + 3\lambda &= 9, && \text{(II)}\\
g' + 16\nu + g &= 45, && \text{(III)}\\
2\mu' + \lambda + 2\mu &= 47, && \text{(IV)}\\
h' + g' + \mu + 3\nu &= 12, && \text{(VI)}\\
3\lambda = 8,\ &\text{(error, should be) } 3\lambda = 12, && \text{(VIII)}\\
2g' + 16\nu + 5\nu &= -120, && \text{(IX)}\\
2h' + g' + \lambda + 2\mu &= 14, && \text{(XIII)}\\
4h' + 6\lambda &= 14, && \text{(XVI)}\\
h' + 2g' + g + 2\mu &= 61, && \text{(XVII)}\\
2h' + 3g + 3g' &= 54, && \text{(XXI)}
\end{aligned}
$$

all satisfied if only

$$h' = 5, \quad g' + 16\nu = -138, \quad h' + 2\mu = 38, \quad g + g' = 18, \quad \lambda = -1.$$

Article 3 (62.)**.** I remark however that the cubic scroll XXII or XXIII gives $\theta' = 54 - (330 - 272) - 2(\lambda + \lambda')$, that is, $\lambda + \lambda' = -2$, instead of $\lambda + \lambda' = -8$. Whence if $U = 0$ be the equation of a scroll, the equation $UV = 0$, containing U as a factor, has for every point of the section constituting the flecnodal curve; but the numerical investigation is in fact not applicable to a scroll, and I am nevertheless surprised that the surface has really a definite flecnodal curve.

Article 4 (63.)**.** Combining the two sets of results, we find

$$
\begin{aligned}
h &= 24, & h' &= 5,\\
g &= 9, & g' &= 10 + \sigma,\\
\lambda &= \lambda', & \lambda' &= -7,\\
\mu &= \mu', & \mu' &= 18 - \tfrac{1}{2}\sigma,\\
\nu &= \nu', & \nu' &= 1 - \tfrac{1}{16}\sigma,
\end{aligned}
$$

and the formula thus is

$$
\begin{aligned}
\beta' = 2n(n-2)(11n-24) &- (110n-272)b + 44g - (116n-303)c + 4h\\
&+ \sigma\{19g + 248\tau + 198\chi - 24C + j - 10\chi + \theta\}\\
&- 5C - 18\Gamma - 6\chi' - \lambda j - \lambda' j' + \tfrac{1}{2}g(-16B - 8\chi - \theta + 16B' - 8\chi' - \theta').
\end{aligned}
$$

Article 5 (64.)**.** where χ, χ', g are constants which remain to be determined. The cubic surfaces fail to determine them, for the reason that in all of them we have $i = 0$, $i' = 0$; and

$$165 + 8\chi + \theta = 16g + 8\chi' + \theta',$$

of which I do not perceive any contradiction. Substituting herein for g, χ, θ their last values, this is

$$
\begin{aligned}
\beta' = 2n(n-2)(11n-24) &+ b(-66n+184) + c(-93n+240)\\
&+ 141\beta + \tfrac{51}{4}\gamma + 66\iota\\
&- \chi' i' + 7j' - 6\chi' - \tfrac{9}{4}\theta' - 5C' - 18B'\\
&- (\chi + 87)i - 21j - \tfrac{41}{2}\chi + \tfrac{51}{2}\theta - 24C\\
&+ \tfrac{1}{16}g(-16B - 8\chi - \theta + 16B' + 8\chi' + \theta').
\end{aligned}
$$

Article 6 (65–68.)**. Article Nos. 65 to 68.**

Article 7 (65.). a, class of curve of intersection by any plane.
δ, number of double tangents of curve of intersection.
κ, number of its inflexions.
n', class of the surface.
α', class of curve of intersection by any plane.
δ', number of double tangents of curve of intersection.
κ', number of its inflexions.
V, class of node-couple torse.
k', number of its apparent double planes.
i', number of its triple planes.
q, its order.
p, order of node-couple curve.
y, number of pinch-planes.
c', class of spinode torse.
A', number of its apparent double planes.
r', its order.
σ', order of spinode curve.
g', number of off-planes.
γ', number of close-planes.
θ', number of common planes of node-couple and spinode torses.
i', number of common planes, stationary planes of the spinode torse.
j', number of common planes, stationary planes of node-couple torse.
χ', number of common planes, not stationary planes of either torse.
B, number of bitropes of surface.
C, number of its cnictropes.

Article 8 (66.). ρ, number of intersections, stationary points on nodal curve.
ι, number of intersections, stationary points on cuspidal curve.
i, number of intersections, not stationary on either curve.
B, number of binodes of surface.
G, number of cnicnodes.

Article 9 (67.). And the accented letters have the like significations in regard to the reciprocal surface; referring them to the original surface, we have

n',	class of the surface.
a',	class of curve of intersection by any plane.
δ',	number of double tangents of curve of intersection.
κ',	number of its inflexions.
V,	class of node-couple torse.
k',	number of its apparent double planes.
i',	number of its triple planes.
q,	its order.
p,	order of node-couple curve.
y,	number of pinch-planes.
c',	class of spinode torse.
A',	number of its apparent double planes.
r',	its order.
σ',	order of spinode curve.
g',	number of off-planes.
γ',	number of close-planes.
θ',	number of common planes of node-couple and spinode torses.

Article 10 (68.)**.** It is hardly necessary to recall that a spinode plane is a tangent plane meeting the surface in a curve having at the point of contact a spinode or cusp; the envelope of the spinode planes is the spinode torse, and the locus of their points of contact is the spinode curve. And similarly a node-couple plane is a tangent plane meeting the surface in a curve having two nodes; the envelope of the node-couple planes is the node-couple torse, and the locus of the points of contact the node-couple curve. The other terms made use of are all explained in the present Memoir.

Addition, August 3, 1869.

As in the theory of Curves, so in that of Surfaces, there are certain functions of the order, class, &c. and singularities which have the same values in the original and reciprocal figures respectively; for convenience I represent any such identity by means of the symbol $\mathfrak{A}$, *viz.*. $\phi(n, a, b, \ldots) = \mathfrak{A}$ denotes that the function $\phi(n, a, b, \ldots)$ is equal to the same function $\phi(n', a', b', \ldots)$ of the accented letters. By what precedes we have $a = \mathfrak{A}$ and it is moreover clear that any function of the unaccented letters which is $= 0$, or which is equal to a symmetrical function of any of the accented and unaccented letters, or to a function of a, is $= \mathfrak{A}$; for instance, from the equations of No. 45 we have $3n' - \kappa' = 3n - c$, and thence $3n - c - \kappa = 3a' - \kappa' - \kappa$, that is, $3n - c - \kappa = \mathfrak{A}$.

No. **5.** From one of the equations of No. 11 we have $n - 2C - 4B + \kappa - \sigma - 2j - 3\chi = n + n' - a = \mathfrak{A}$; we have thus the system of eight equations,

$$\begin{aligned}
a &= \mathfrak{A},\\
3n - c - \kappa &= \mathfrak{A},\\
a(n-2) - \kappa + B - p - 2\sigma &= \mathfrak{A},\\
b(n-2) - p - 2\theta - 3y - 3\iota &= \mathfrak{A},\\
c(n-2) - 2\sigma - 4\rho - \gamma - 6\iota &= \mathfrak{A},\\
2q - 2p + \beta + 3\iota + i &= \mathfrak{A},\\
3r + c - \sigma a - \beta - 4\rho + 2\iota + \lambda &= \mathfrak{A},\\
n - 2C - 4eB + \kappa - a - 2j - 3\chi - n + n' - a &= \mathfrak{A}.
\end{aligned}$$

or if from these we eliminate κ, p, σ, ρ, then the system of five equations,

$$\begin{aligned}
a &= \mathfrak{A},\\
n(c-8) - 4\beta - \gamma - \lambda + 4\theta + 8\iota + 6\chi + 4j &= \mathfrak{A},\\
(a-b)(n-2) - 11n + 3c + 4C + 9\iota + 2\rho + 3\gamma + 3\iota + 6\chi + 4j &= \mathfrak{A},\\
3r - 20a + 6c - \beta + 2\iota + 10C + 20\iota + 16\rho + 10j - 4\theta &= \mathfrak{A},\\
2q - 2b(n-2) + 5\beta + 6\gamma + 6\iota + 3\iota + j &= \mathfrak{A}.
\end{aligned}$$

Dr. By means of a theorem of Clebsch's I was led to the following expression for the "deficiency" of a surface of the order n having the singularities considered in the foregoing Memoir:

$$\text{Deficiency} = \tfrac{1}{6}(n-1)(n-2)(n-3) - (n-3)(b+c) + \tfrac{1}{2}(\delta + \kappa) + 2\iota + 4\rho + \tfrac{3}{2}\gamma + \tfrac{1}{2}\iota - \tfrac{1}{2}\theta.$$

This should be equal to the deficiency of the reciprocal surface, *viz.*. we must have

$$\tfrac{1}{2}(n-1)(n-2)(n-3) - 12(n-3)(b+c) + 6\delta + 6\kappa + 24\iota + 42\rho + 30\gamma + 12\iota - \tfrac{9}{2}\theta = \mathfrak{A};$$

but from a combination of the last-mentioned five equations we have

$$\begin{aligned}
&- 2n^2 - 6n^2 + 4n + (12n - 36)b + (12n - 48)c - 6\delta - 6\kappa - 24\iota\\
&\qquad\qquad - 42\rho - 30\gamma - 13\iota - 7j - 8\chi + 2\theta - 4C - 10\iota = \mathfrak{A};
\end{aligned}$$

and adding to the last preceding equation we have

$$26n - 12c + \tfrac{1}{2}\gamma - \iota' - 7j - 8\chi + \tfrac{1}{2}\theta - 4C - 10\iota = \mathfrak{A}.$$

Substituting for $\mathfrak{A}$ its value in terms of the accented letters, we obtain for θ' the value

$$\theta' = 13' + 26n - 12c + \tfrac{1}{2}\gamma' + 7j' + 8\chi' + \tfrac{1}{2}\theta' + 4C' + 10\iota' - 26b' + 12c' - \iota - 7j - 8\chi + \tfrac{1}{2}\theta - 4C - 10\iota.$$

We have $c' = -3a + n + 3n'$.

$n = 2$. We have

$$26n - 12c + \tfrac{1}{2}\beta - \iota - 7j - 8\chi + \tfrac{1}{2}\theta - 4C - 10\iota = \mathfrak{A}$$

and multiplying by -4 and adding, the equation to be verified is

$$13n(c-8) - 13(4\beta + \gamma) + n + 4\iota' + 28j + 32\chi - 2\theta + 16C + 40\iota = \mathfrak{A}.$$

But we have from the Memoir

$$-13n(c-8) + 13(4\beta + \gamma) - 52j - 78\chi + 13\theta - 52C - 104\iota = \mathfrak{A},$$

which reduces the equation to

$$n + 4\iota - 24j - 46\chi + 11\theta - 36C - 64\iota = \mathfrak{A}$$

or substituting for 11θ its value, this is

$$(4\chi' - 4\chi - 336)\iota - (\tfrac{1}{2}g - 10)(16B + 8\chi + \theta) = \mathfrak{A},$$

an equation which is satisfied if

$$\chi' - \chi - 84 = 0, \quad \tfrac{1}{2}g - 10 = 0,$$

and

$$g = 20, \quad \text{or else} \quad 165 + 8\chi + \theta = 165' + 8\chi' + \theta'.$$

[412]

A Memoir on Cubic Surfaces.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLIX. (for the year 1869), pp. 231—326. Received November 12, 1868,—Read January 14, 1869.]

The present Memoir is in a measure supplementary to the Memoir “On the Distribution of Surfaces of the Third Order into Species, in reference to the presence or absence of Singular Points, and the reality of their Lines,” by Professor Schläfli, Phil. Trans. vol. CLIII. (1863), pp. 193—241. But the object of the present Memoir is different; I attend to the several cases of cubic surfaces as grouped together in the Table, and referred to by the affixed roman numbers, for the purpose of obtaining the equations in point-coordinates of the several cases respectively, and the equations in plane-coordinates of the reciprocal surfaces. The memoir gives (partially or completely developed) the equations of the twenty-three cases depending on the nature of the singularities. It was written in connexion with these investigations on Cubic Surfaces, that the Memoir, [411], “Memoir on the Theory of Reciprocal Surfaces,” was referred to; and the latter part of the Memoir is divided into sections headed thus:

[411], The twenty-three Cases of Cubic Surfaces.

&c. &c., referring to the several cases of the cubic surface, but the paragraphs are numbered continuously throughout the Memoir.

I prefer to make the ultimate division depending on the reality of the lines, presence or absence of singularities, or, what is the same thing, attending only to the reality, the division into (twenty-two, or as I reckon it) twenty-three cases.

Article Nos. 1 to 13. — Explanations and Table of Singularities.

Article 11 (1.). I designate as follows the twenty-three cases of cubic surfaces, adding to each of them its equation:

$$\begin{array}{rr}
\mathrm{I} = 12, & (X, Y, Z, W)^3 = 0, \\
\mathrm{II} = 12 - C_2, & W(a, b, c, f, g, h\overline{X, Y, Z})^2 + 2kXYZ = 0, \\
\mathrm{III} = 12 - A, & 2W(X + Y + Z)(lX + mY + nZ) + 2kXYZ = 0, \\
\mathrm{IV} = 12 - 2C_2, & WXZ + Y^2(yZ + \delta W) + (a, b, c, d\overline{X, Y})^3 = 0, \\
\mathrm{V} = 12 - B_3, & WXZ + (X + Z)(Y^2 - aX^2 - bZ^2) = 0, \\
\mathrm{VI} = 12 - B_3 - G_2, & WXZ + Y^2Z + (a, b, c, d\overline{X, Y})^3 = 0, \\
\mathrm{VII} = 12 - B_3, & WXZ + Y^2Z - Z^3 = 0, \\
\mathrm{VIII} = 12 - 3C_2, & WXZ + (a, b, c, d\overline{X, Y})^3 = 0, \\
\mathrm{IX} = 12 - 2B_3, & WXZ + (X + Z)(Y^2 - X^2) = 0, \\
\mathrm{X} = 12 - B_3 - G_2, & WXZ + Y^2Z + X^2 - Z^2 = 0, \\
\mathrm{XI} = 12 - B_3, & W(X + Y + Z)^2 + 2XYZ = 0, \\
\mathrm{XII} = 12 - B_3, & WXZ + Y^2(X + Y + Z) = 0.
\end{array}$$

Article 12 (2.)**.** Where C_2 denotes a conic-node diminishing the class by 2; B_3, B_4, B_5, B_6, a biplanar-node diminishing (as the case may be) the class by 3, 4, 5, or 6; U_6, U_7, U_8 a uniplanar-node diminishing (as the case may be) the class by 6, 7, or 8.

The affixed explanation, which I shall usually retain in connexion with the Roman number, shows therefore in each case what the class is, and also the singularities which cause the reduction: thus XIII $= 12 - B_3 - 2C_2$ indicates that there is a biplanar node, B_3, diminishing the class by 3, and two conic-nodes, C_2, each diminishing the class by 2; and thus that the class is $12-3-2\cdot 2 = 5$. As regards the cases XXII and XXIII, these are surfaces having a nodal right line, distinct from the nodal right line of a scroll; and are consequently scrolls, each having a simple directrix right line; the scroll $S(1,1)$ having a simple directrix right line coincident with the nodal line is the scroll $S(1,1)$; see my "Second Memoir on Skew Surfaces, otherwise Scrolls," Phil. Trans. vol. CLIV. (1864), pp. 559–577, [340].

Article 13 (3.)**.** The nature of the points $C(= C_2)$ and $B(= B_3)$ requires to be explained.

C $(= C_2)$ is a conic-node, where, instead of the tangent plane, we have a proper quadric cone.

B $(= B_3, B_4, B_5$ or $B_6)$ is a biplanar-node, where the quadric cone becomes a plane-pair (two distinct planes): the two planes are called the biplanes, and their line of intersection the edge. In B_3, the edge is not a line on the surface; in B_4, the tangent plane is touched along the edge by the surface, the edge is called the torsal edge; in B_5, the tangent plane coincides with one of the biplanes; in B_6, the tangent plane coinciding with one of the biplanes becomes oscular.

U $(= U_6, U_7$ or $U_8)$ is a uniplanar-node, where the quadric cone becomes a coincident plane-pair; say, the plane is the uniplane. It is to be observed that there is not in this case any edge. The uniplane meets the cubic surface in three lines, or say "rays," passing through the uniplanar-node, viz.

- In U_6, the rays are three distinct lines;
- In U_7, two of them coincide;
- In U_8, they all three coincide.

Article 14 (4.)**.** To connect these singular points with the theory of the preceding Memoir, it is to be observed that they are equivalent to a certain number of cnicnodes $C(= C_2)$ and binodes $B(= B_3)$ respectively; viz. we have

$$\begin{array}{ll} C = C, & B_3 = B, \\ B_4 = C + B, & B_5 = 2C + B, \\ B_6 = 3C, & U_6 = 3C, \\ U_7 = 2C + B, & U_8 = C + 2B. \end{array}$$

Article 15 (5.)**.** I take the opportunity of remarking that although the expressions cnicnode and binode properly refer to the simple singularities C and B, yet as $C_2 = C$, C_2 is properly spoken of as a cnicnode, and we may (using the term binode as an abbreviation for biplanar-node) speak of any of the singularities B_3, B_4, B_5, B_6 as a binode. Thus the surface X $= 12 - B_4 - C_2$ has a binode B_4 and a cnicnode C_2; although theoretically the binode B_4 is equivalent to two cnicnodes, and the surface belongs to those with three cnicnodes, or for which $C = 3$.

I use also the expression unode for shortness, instead of uniplanar-node, to denote any of the singularities U_6, U_7, U_8.

Article 16 (6.)**.** The foregoing equations (substantially the same as Schläfli's) are Canonical forms; the reduction of the equation of any case of surface to the above form is not always obvious. It would appear that each equation is from its simplicity in the form best adapted to the separate discussion of the surface to which it belongs; there is the disadvantage that the equations do not always (when from the geometrical connexion of the surfaces they ought to do so) lead the one to the other. This would be a serious imperfection if the equations were an end, the object in the present Memoir being only to establish the geometrical theory of the surfaces to which they respectively belong, and the imperfection is not material.

Article 17 (7.)**.** I have used the capital letters (X, Y, Z, W) in place of Schläfli's (x, y, z, w), reserving these for the plane-coordinates of the cubic surfaces, or (what is the same thing) point-coordinates of the reciprocal surfaces; and in the several cases I have interchanged the coordinates (in place of his (p, q, r, s) or (what is the same thing) so as to obtain a greater uniformity in the representation of the surfaces.

Article 18 (8.)**.** To explain this, let A, B, C, D be the vertices of the tetrahedron formed by the coordinate planes $A = YZW$, $B = ZWX$, $C = WXY$, $D = XYZ$; the coordinate planes have been chosen so that the vertices of the tetrahedron shall correspond to determinate singularities of the surface.

Consider first the surfaces which have no binodes B or unodes U. It is clear that the nodes might have been taken at any vertices whatever of the tetrahedron; they are taken thus: there is always a node C_2 at D; when there is a second node C_2, this is at C, the third one is at A, and the fourth at B.

Article 19 (9.)**.** Consider next the surfaces which have a binode B_3, B_4, B_5, or B_6; this is taken to be at D, and the biplanes to be $X = 0$, $Z = 0$ (the edge being therefore DB); viz. in B_3 or B_4, where the distinction arises, $X = 0$ is the ordinary biplane, $Z = 0$ the torsal or (as the case may be) oscular biplane. If there is a second node, it may be and is taken to be in the biplane $Z = 0$, at C, when the binode is B_3 or B_4, and it may be a node C_2. It is only when the binode is B_3 or B_4 that there can be a third node, for it is only in these cases that there is a second ordinary biplane $Z = 0$; but in these cases respectively the third node, a C_2, may be and is taken to be in the biplane $Z = 0$, at A.

Article 20 (10.)**.** The only case of two binodes is when each is a B_3. Here the first is at D, its biplanes being $X = 0$, $Z = 0$; and the second at C, the biplanes thereof are then $X = 0$ (which is the common biplane), and $W = 0$. If there is a third node, it may be and is taken to be at A; if a C_2, the biplanes of the second binode are $Z = 0$, $W = 0$; and the plane $Y = 0$ of the first binode is then a remaining biplane which will in either case be the plane through the three binodes D, C, A.

Article 21 (11.)**.** If there is a unode, this is taken to be at D, and its uniplane is $X + Y + Z = 0$. There is never, besides the unode, any other node.

Article 22 (12.)**.** The result is that the binodes, when there are such, are taken to be at D, C, A, in the order of their speciality, and that (except in the case XII $= 12 - U_8$) the biplanes are $X = 0$, $Z = 0$ (for the first binode, $Z = 0$ being the ordinary biplane, $Z = 0$ the special biplane), those of the second binode $Z = 0$, $W = 0$, those of the third binode $Z = 0$, $W = 0$, and that (except in the case XII $= 12 - U_8$) the uniplane is $Z = 0$. For example, in the surface XVII $= 12 - 2B_3 - C_2$, as represented by its equation $WXZ + Y^2Z + Z^2 = 0$, we have a B_3 at D, the biplanes being $Z = 0$, $Z = 0$, a B_3 at C, the biplanes being $Z = 0$, $W = 0$ (therefore $Z = 0$ the common biplane), and a C_2 at A.

In the case, however, of a single B_3, III $= 12 - B_3$, the biplanes are taken to be $X + Y + Z = 0$, $lX + mY + nZ = 0$.

Article 23 (13.)**.** It will be convenient (anticipating the results of the investigations contained in the present Memoir) to give at once the following Table of Singularities; the several symbols have of course the significations explained in the former Memoir.

	n	a	k	ρ	ι	B	C	n'	a'	δ'	κ'	b'	k'	t'	q	p	y	c'	r'	σ'
I	12	6	6	27	216	45	0	12	6	6	27	27	216	45	27	27	0	24	2	12
II	10	6	4	16	72	15	1	6	4	3	15	15	60	15	15	15	0	12	1	6
III	9	6	3	10	38	7	0	4	3	1	9	9	24	9	9	9	0	10	1	6
IV	8	6	2	6	17	3	2	3	3	0	7	7	12	7	7	7	0	10	1	6
V	8	6	2	6	16	2	1	4	3	1	7	7	12	7	7	7	0	8	0	4
VI	7	6	1	3	6	1	1	3	3	0	6	6	6	6	6	6	0	8	1	5
VII	7	6	1	3	4	0	0	6	4	3	6	6	6	6	6	6	0	6	0	3
VIII	6	6	0	1	1	0	0	6	4	3	3	3	3	1	3	3	1	9	2	6
IX	6	6	0	1	0	0	2	4	3	1	4	4	4	4	4	4	0	6	0	3
X	6	6	0	1	0	0	1	5	4	2	5	5	5	5	5	5	0	5	0	2
XI	6	6	0	0	0	0	0	6	5	3	6	6	6	6	6	6	0	6	0	3
XII	6	6	0	0	0	0	0	6	5	3	6	6	6	6	6	6	0	4	0	1
XIII	5	6	0	0	0	0	2	3	3	0	3	3	3	3	3	3	0	6	1	4
XIV	5	6	0	0	0	0	1	4	4	1	4	4	4	4	4	4	0	4	0	2
XV	5	6	0	0	0	0	0	5	5	2	5	5	5	5	5	5	0	5	0	2
XVI	4	6	0	0	0	0	3	3	3	0	3	3	3	3	3	3	0	4	0	2
XVII	4	6	0	0	0	0	1	5	5	2	5	5	5	5	5	5	0	4	0	2
XVIII	4	6	0	0	0	0	0	6	6	3	6	6	6	6	6	6	0	4	0	2
XIX	3	6	0	0	0	0	0	6	6	3	6	6	6	6	6	6	0	3	0	1
XX	3	6	0	0	0	0	0	6	6	3	6	6	6	6	6	6	0	3	0	1
XXI	3	6	0	0	0	0	0	9	9	6	9	9	9	9	9	9	0	3	0	1
XXII	3	3	0	0	0	0	0	3	3	1	3	1	0	0	3	1	0	3	1	3
XXIII	3	3	0	0	0	0	0	3	3	1	3	1	0	0	3	1	0	4	2	4

Continued Table.

	g'	γ'	θ'	i'	j'	χ'	B'	C'	β'	τ
I	16	8	16	6	2	4	0	0	54	0
II	8	4	8	3	1	2	0	0	30	0
III	4	2	4	1	0	1	0	0	18	0
IV	2	1	2	1	0	1	0	0	13	0
V	4	2	4	1	0	1	0	0	12	0
VI	2	1	2	1	0	1	0	0	8	0
VII	2	1	2	0	0	0	0	0	6	0
VIII	3	2	3	0	0	0	0	1	6	0
IX	2	1	2	0	0	0	0	0	4	0
X	1	0	1	0	0	0	0	0	3	0
XI	2	1	2	0	0	0	0	0	2	0
XII	1	0	1	0	0	0	0	0	1	0
XIII	2	1	2	0	0	0	0	0	3	0
XIV	1	0	1	0	0	0	0	0	2	0
XV	1	0	1	0	0	0	0	0	1	0
XVI	1	0	1	0	0	0	0	0	3	0
XVII	1	0	1	0	0	0	0	0	1	0
XVIII	1	0	1	0	0	0	0	0	0	0
XIX	0	0	0	0	0	0	0	0	0	0
XX	0	0	0	0	0	0	0	0	0	0
XXI	0	0	0	0	0	0	0	0	0	0
XXII	0	0	0	0	0	0	0	0	0	0
XXIII	0	0	0	0	0	0	0	0	0	0

Article Nos. 14 to 19. — Explanation in regard to the Determination of certain Singularities.

Article 24 (14.). In the several cases I to XXI, we have a cubic surface ($n = 3$), with singular points C and B but without singular lines. The section by an arbitrary plane is thus a curve, order $n = 3$, that is, a cubic curve, without nodes or cusps, and therefore of the class $a' = 6$, having $\delta' = 6$ double tangents and $\kappa' = 9$ inflexions. The tangent cone with an arbitrary point as vertex is a cone of the order $a = 6$, having in the case I = 12 nodal lines and $\kappa = 6$ cuspidal lines, but with (in the several other cases) C nodal lines and B cuspidal lines (or rather singular lines tantamount to C double lines and B cuspidal lines): the class of the cone, or order of the reciprocal surface, is thus $n' = 6 \cdot 5 - 2(C + C) - 3(B + B) = 12 - 2C - 3B$.

Article 25 (15.). In the general case I = 12, there are on the cubic surface 27 lines; these lie by 3's in 45 planes; the 27 lines constitute the node-couple curve of the order $p = 27$, and the node-couple torse consists of the pencils of planes through these lines respectively, being thus of the class $p = k' = 27$; the 45 planes are triple tangent planes of the node-couple torse, which has thus $t' = 45$ triple tangent planes. But in the other cases it is only certain of the 27 lines, say the "facultative lines" (as will be explained), which constitute the node-couple curve of the order p; the pencils of planes through these lines constitute the node-couple torse of the class $b' = p'$; the t' planes, each containing three facultative lines, are the triple tangent planes of the node-couple torse.

Or if we refer to the reciprocal surface, the numbers b', t' are then the order of the nodal curve, and the number of triple points of the nodal curve. Inasmuch as the nodal curve consists of right lines, the number k' of its apparent double points is given by the formula $2k' = b'^2 - b' - 3t'$, and comparing with the formula $q' = b'^2 - b' - 2k' - \frac{3}{2}y' - 6i'$ we have $q' + \frac{3}{2}y' = 0$, that is, $q' = 0$ (q the class of the nodal curve), and also $\gamma' = 0$.

Article 26 (16.). In the general case I = 12, the spinode curve is the complete intersection of the cubic surface by the Hessian surface; it is thus of the order $\sigma' = 12$; but in the other cases the complete intersection consists of the spinode curve together with certain right lines not belonging to the curve, and the spinode curve is of an order less than 12: this will be further explained, and the reduction accounted for (see post, Nos. 24 *et seq.*).

Article 27 (17.). Again, in the general case I = 12, each of the 27 lines is a double tangent of the spinode curve, and the tangent planes of the surface at the points of contact are the common planes of the spinode torse and the node-couple torse; or we have $\beta' = 2p' = 54$. In the other cases, however, we must take only the facultative lines, each of which is a double or single tangent of the spinode curve, and the tangent planes of the surface at the points of contact are the common tangent planes as above; that is, the number of contacts gives β' not in general $= 2p'$.

Article 28 (18.). There are not, except as above, any common tangent planes of the two torses, so that not only $\gamma' = 0$, but also $i' = 0$. I do not at present account a priori for the values $\theta' = 16, 8$, and 16, which present themselves in the Table.

Article 29 (19.). The cubic surface cannot have a plane of conic contact, and we have thus in every case $C' = 0$; but the value of B' is not in every case $= 0$. In what precedes we see how a priori the equation of the discussion of the reciprocal surface leads to the values of the equation of the nodal curve of the order b', and how the equation of the cuspidal curve might be obtained as the reciprocal of the spinode-torse of the cubic surface; but this is a laborious process, and it is in each case less necessary, inasmuch as the cuspidal curve is obtained by means of the equation of the reciprocal surface, putting in evidence the values $b', t', \beta', \gamma', i', j', \chi', B', C'$, known by means of the equation of the nodal curve.

Article Nos. 20 to 23. — The Lines and Planes of a Cubic Surface; Facultative Lines; Explanation of Diagrams.

Article 30 (20.). In the general surface I = 12, we have 27 lines and 45 triple-tangent planes: through each line pass 5 planes, and in each plane lie 3 of the 27 lines. If we attend to the surfaces II to XXI (the scrolls XXII and XXIII do not come under the present considerations), in the several cases several of the lines coincide with each other; several of the planes also come to coincide with each other; but the sum of the frequencies is always the same. For each particular line, the line has a frequency (this represents the number of lines in the particular plane, the plane having a multiplicity, and the sum of these frequencies is = 27 × 5 = 135.

Each plane has a multiplicity, and the frequency of each plane is the multiplicity of the line into the frequency of the line in regard to the plane.

Article 31 (21.). The mode of coincidence of the lines and planes, and the several distinct lines and planes which are situate in or pass through the several distinct planes and lines respectively, are shown in the annexed diagrams I to XXI(*).

(*) *See the commencements of the several sections.*

Article 32 (22.). For the explanation of the diagrams, I use the following terms: I call a line joining two nodes an "axis"; a line through a node not joining any other node is a "ray"; an edge of a binode is also a ray; any other line is a "mere line." An axis is always torsal or oscular; a ray may be torsal or oscular, but is not a mere line. The line joining a node to the point of contact of the plane touching along an axis is the "transversal" of such axis; in the case XVI = $12 - 4C_2$, each transversal is a ray.

Article 33 (23.). In the general case I = 12, each of the 27 lines is a facultative line, and is part of the node-couple curve; and the node-couple curve is, as already mentioned, made up of the 27 lines, thus a curve of the order 27. In fact each plane through a line meets the cubic surface in the line and in a conic; the line and conic meet in two points, and the plane (any plane) through the line is thus a double tangent plane touching the surface at the two points in question; the locus of the points of contact is the node-couple curve. But in the other cases, II to XXI, certain of the lines do not belong to the node-couple curve. To fix the ideas, consider the surface II = $12 - C_2$: there are through the node six nodal rays, say rays 1, 2, 3, 4, 5, 6; attending to any one of these, a plane through the ray meets the surface in the ray itself and a conic; but the ray itself does not touch the surface at the node, and the plane through the ray is only a single tangent plane, not a double tangent plane; the ray is not part of the node-couple curve. We say that a line of the surface is or is not "facultative" according as it does or does not form part of the node-couple curve.

Article Nos. 24 to 26. — Axis; the different kinds thereof.

Article 34 (24.)**.** A line joining two nodes is an axis; such a line is always a torsal or oscular line, and it is not a mere line, but some further distinctions are requisite. Using the expressions in their strict sense, cnicnode $= C$, binode $= B$, an axis is a CC-axis, joining two cnicnodes, or it is a CB-axis joining a cnicnode and a binode, or it is a BB-axis joining two binodes. A CC-axis is torsal, the transversal being a mere line; a CB-axis is torsal, the transversal being a ray of the binode; a BB-axis is oscular.

The distinction is of course carried to the higher nodes B_4, B_5, B_6 and uniplanar nodes U_6, U_7, U_8; thus $(B_4 = B + C)$ the edge of a binode B_4 is not an axis at all, but $(B_4 = 2C)$ the edge of a binode B_4 is a CC-axis; $(B_5 = 2C + B)$ the edge of a binode B_5 is a twice-taken CC-axis; $(B_6 = 3C)$ the edge of a binode B_6 is a thrice-taken CC-axis; $(U_6 = 3C)$ each of the three rays is regarded as a CC-axis; $(U_7 = 2C + B)$ the double ray is regarded as a CC-axis + a twice-taken CB-axis; $(U_8 = C + 2B)$ the single ray is a CC-axis + a twice-taken CB-axis.

Article 35 (25.)**.** It has been mentioned that the intersection of the surface with the Hessian consists of the spinode curve together with certain right lines; these lines are in fact the axes. An examination of the several cases shows that in the complete intersection each CC-axis presents itself twice, each CB-axis 3 times, and each BB-axis 4 times. We thus see that a CC-axis, or rather the torsal plane along such axis, is the pinch-plane or singularity $y = 1$; the CB-axis, or rather the torsal plane along such axis, the close-plane $\gamma' = 1$; and the BB-axis, or oscular plane along such axis, the bitrope or singularity $\beta' = 1$; σ' the order of the spinode curve; for a cubic surface with singular lines the expression of a' being in fact $a' = 12 - 2j' - 3\gamma' - 4\beta'$.

Article 36 (26.)**.** I remark in further explanation, that in the several sections, in showing how the complete intersection of the cubic surface with the Hessian is made up, I have not referred to the axes in the above precise significations; thus for instance XIV $= 12 - B_4 - C_2$, the binode B_4 is $C + B$, and the edge is thus a CB-axis, while the axis BC_2 is a CC-axis $(y' = 1 + 1 = 2,\ j' = 1)$. The complete intersection should therefore consist of the spinode curve, + edge (as a CB-axis) 3 times + axis (as a CC-axis) 2 times = 5 times; but without the full explanation it is stated in the section that the intersection is made up of the edge 4 times, and a cubic curve — which cubic curve together with the edge once constitutes the spinode curve; and so in other cases: this explanation will, I think, remove all difficulty.

Article Nos. 27 to 32. — On the Determination of the Reciprocal Equation.

Article 37 (27.). Consider in general the cubic surface $(X, Y, Z, W)^3 = 0$, and in connexion therewith the equation $Xx + Yy + Zz + Ww = 0$, which regarding therein X, Y, Z, W as current coordinates, and x, y, z, w as constants, is the equation of a plane. If from the two equations we eliminate one of the coordinates, for instance W, we obtain

$$(Xw, Yw, Zw, -(Xx + Yy + Zz))^3 = 0,$$

which, (X, Y, Z) being current coordinates, is obviously the equation of the cone, vertex $(x = 0, y = 0, z = 0)$, which stands on the section of the cubic surface by the plane. Equating to zero the discriminant of this function in regard to (X, Y, Z), we express that the cone has a nodal line; that is, that the section has a node, or, what is the same thing, that the plane $xX + yY + zZ + wW = 0$ is a tangent plane of the cubic surface; and we thus by the process in fact obtain the equation of the reciprocal surface in plane-coordinates (x, y, z, w).

Article 38 (28.). Consider in the same equation x, y, z, w as current coordinates, X, Y, Z, W as given parameters, the equation represents a system of three planes, viz. these are the planes $xX + yY + zZ + wW = 0$, where W has the three values given by the equation $(X, Y, Z, W)^3 = 0$; that is, the coordinates of any one of the three points of intersection of the line $x : y : z : w = \frac{1}{X} : \frac{1}{Y} : \frac{1}{Z} : -\frac{1}{W}$ with the cubic surface, and the equation is thus the equation of a system of 3 planes, the polar planes of three points of the cubic surface (which three points lie on an arbitrary line through the point $x = 0$, $y = 0$, $z = 0$).

In equating to zero the discriminant in regard to (X, Y, Z), we find the envelope of the system of three planes, or say of a plane, the polar plane of an arbitrary point on the cubic surface; this is, as is known, the same thing as the equation of the cubic surface in plane-coordinates (x, y, z, w).

Article 39 (29.). In what precedes we have the explanation of a special process: the position of a point on the surface is determined by means of certain two parameters, the ratios $X : Y : Z$, or (as last explained) the position of the point is determined by the line joining this point with the point $(x = 0, y = 0, z = 0)$. More generally we may consider the position of the point as determined by means of any two parameters; the equation of the polar plane then contains the two parameters, and by taking the envelope in regard to the two parameters considered as variable, we have the equation of the reciprocal surface.

Article 40 (30.). But let the parameters, say θ, ϕ, be regarded as varying successively; if we have on the surface a curve Θ, the equation whereof contains the parameter θ, and when θ varies this curve sweeps over the surface. The envelope in regard to ϕ of the polar plane of a point of the curve Θ may be a torse having its vertex on the reciprocal of the curve Θ; and the envelope of this torse by the variation of the parameter θ is the reciprocal surface. With a fixed line, the envelope of this plane is a sextic cone (the fixed line being $P = 0$, $Q = 0$), and by equating to zero the discriminant of a binary function of (P, Q) we obtain the equation of the sextic cone; assuming that the equation of the sextic cone has been obtained, this is the equation of the reciprocal surface.

Article 41 (31.). In particular, if the section by any plane through the fixed line be a conic, its reciprocal is the quadricone, and the envelope of this quadricone is the reciprocal surface. This is a convenient one for obtaining the reciprocal of a cubic surface on which there are lines; what Schläfli does (but the process is not explained) in the several instances in which he obtains the equation of the reciprocal surface by means of a binary function. I remark that it would be very instructive, for each case of surface, to take the variable plane successively through the several kinds of lines on the particular surface; the equation of this quadricone would be thus obtained in different forms, putting in evidence the relation of the reciprocal surface to the fixed line.

Article 42 (32.). It is to be mentioned that the equation of the reciprocal surface is found by equating to zero the discriminant of a ternary cubic, or a binary quartic, cubic, or quadric. The equation is in the form disc. = 0; contains a factor which for the adopted forms of equations is always a power or product of powers of w, z, x; known *a priori*, and which is thrown out without difficulty, the equation being thereby reduced to the proper order. There is the singular advantage that the process puts in evidence the existence of the cuspidal curve; and for a ternary cubic, the resulting equation would be simply $S^3 - 27T^2 = 0$, where $S = 0$, $T = 0$ are the equations of the cuspidal curve; but the factor thrown out as just mentioned occasions however a modification, viz. the intersection of the two surfaces is not an indecomposable curve, and the cuspidal curve is in most cases, not the complete intersection, but a partial intersection of the two surfaces.

In several cases it thus happens that the cuspidal curve is obtained as a complete intersection 3×4 or 2×6, without speciality. Similarly when the equation of the reciprocal surface is obtained by means of a binary cubic; if the coefficients hereof (functions of course of the coordinates x, y, z, w) be A, B, C, D, then the surface is

$$(AD - BC)^2 - 4(AC - B^2)(BD - C^2) = 0,$$

having the cuspidal curve $A, B, C = 0$, $B, C, D = 0$, subject however to modification in the case of a thrown out factor.

Article Nos. 33 and 34. — Explanation as to the Sections of the Memoir.

Article 43 (33.). As regards the following Sections I to XXIII, it is to be observed that for the general surface I = 12, I do not attempt to form the equation of the reciprocal surface; and in some of the other cases, II $= 12 - C_2$, &c., the equation of the reciprocal surface is not obtained in a completely developed form, or either the nodal curve or its cuspidal curve is not dealt with, so as to put in evidence the theory given in the former Memoir. Portions of the latter sections are consequently omitted; and in particular in the Section I there is given only the diagram of the 27 lines and the 45 planes (with however developments as to notation and otherwise which have no place in the subsequent sections), and the analytical expressions for the several lines and planes, although from the want of the equation of the reciprocal surface these have no present application. And so in some of the next following sections, no application is made of the analytical expressions of the lines and planes.

Article 44 (34.). I call to mind that if a line be given as the intersection of the two planes

$$AX + BY + CZ + DW = 0,$$
$$A'X + B'Y + C'Z + D'W = 0,$$

then the six coordinates of the line are

$$a, b, c, f, g, h = AD' - A'D,\ BD' - B'D,\ CD' - C'D,\ BC' - B'C,\ CA' - C'A,\ AB' - A'B,$$

and that in terms of the six coordinates the line is given as the common intersection of the four planes

$$(-h, g, a \overline{)X, Y, Z, W})^2 = 0, \quad (-g, h, b \overline{)X, Y, Z, W})^2 = 0, \quad \&\text{c.},$$

and that (reciprocating as usual in regard to $X^2 + Y^2 + Z^2 + W^2 = 0$) the coordinates of the reciprocal line are (f, g, h, a, b, c), that is, the common intersection of the four planes

$$(h, g, a \overline{)x, y, z, w})^2 = 0, \quad (-g, h, -b \overline{)x, y, z, w})^2 = 0, \quad \&\text{c.}$$

It is more convenient to consider a line as determined as the intersection of two planes rather than by means of its six coordinates; thus, for instance, to speak of the line $X = 0$, $Y = 0$ rather than of the line $(0, 0, 0, 1, 0, 0)$; and in some of the sections I have preferred not to give the expressions of the six coordinates of the several lines.

Article Nos. 35 to 46. — $\mathfrak{J} = 12$, Equation $(X, Y, Z, W)^3 = 0$.

35. There is in the system of the 27 lines and the 45 planes a complicated and many-sided symmetry which precludes the existence of any unique notation; the notation can only be obtained by starting from some arrangement which is not unique, but one of a system of several like arrangements. The notation employed in my original paper "On the Simple Tangent Planes of Surfaces of the Third Order," Camb. and Dub. Math. Journ. vol. IV. 1849, pp. 118–132, [76], and which is shown in the right hand and lower margins of the diagram, starts from such an arrangement; but it is so complicated that it can hardly be considered as at all putting in evidence the relations of the lines and planes; that of Dr Hart (Salmon, "On the Triple Tangent Planes of a Surface of the Third Order," same volume, pp. 252–260), depending on an arrangement of the 27 lines according to a cube of 3 each way, is a singularly elegant one, and will be presently reproduced.

But the most convenient one is Schläfli's, starting from a double-sixer; viz. we can (and that in 36 different ways) select out of the 27 lines two systems each of six lines, such that no two lines of the same system intersect, but that each line of the one system intersects all but the corresponding line of the other system; or, say, if the lines are

$$\begin{array}{cccccc} 1, & 2, & 3, & 4, & 5, & 6 \\ 1', & 2', & 3', & 4', & 5', & 6', \end{array}$$

then these have the thirty intersections

	$1'$	$2'$	$3'$	$4'$	$5'$	$6'$
1						
2						
3						
4						
5						
6						

Any two lines such as 1, $2'$ lie in a plane which may be called $12'$; similarly the lines $1'$, 2 lie in a plane which may be called $1'2$; these two planes meet in a line 12; and any three lines such as 12, 34, 56 meet in pairs, lying in a plane 12.34.56. We have thus the entire system of the 27 lines and 45 planes.

36. The diagram of the lines and planes is given in the original memoir.

37–39. The construction of double-sixers and trihedral-pairs is explained in detail in the original memoir.

38. It has been mentioned that the number of double-sixers was = 36; these are as follows:

$$\begin{array}{cccccc} 1, & 2, & 3, & 4, & 5, & 6 \\ 1', & 23, & 24, & 25, & 26 & \\ 2', & 13, & 14, & 15, & 16 & \\ & & \&c. & & & \end{array}$$

where, if we take any column of two lines, we have the complete number 216 pairs of non-intersecting lines.

39. We can out of the 45 planes select, and that in 120 ways, a *trihedral-pair*, that is, two triads of planes, such that the planes of the one triad, intersecting those of the other triad, give 9 lines. Analytically if $X = 0$, $Y = 0$, $Z = 0$ and $U = 0$, $V = 0$, $W = 0$ are the equations of the six planes, then the equation of the cubic surface is

$$XYZ + kUVW = 0.$$

See as to this post, No. 44.

The trihedral plane pairs are:

$$\begin{array}{lll} 12', & 23', & 31' \\ 1'2, & 2'3, & 3'1 \\ & \text{(No. is} = 20) & \\ 12', & 34', & 14.23.56 \\ 2'3, & 4'1, & 12.34.56 \\ & \text{(No. is} = 90) & \\ 14.25.36, & 35.16.24, & 26.34.15 \\ 14.35.26, & 25.16.34, & 36.24.15 \\ & \text{(No. is} = 10) & \end{array}$$

Total = 120.

40. Dr Hart arranges the 27 lines, cubically, thus:

$$\begin{array}{ccc|ccc|ccc} A_1 & B_1 & C_1 & a_1 & b_1 & c_1 & \alpha_1 & \beta_1 & \gamma_1 \\ A_2 & B_2 & C_2 & a_2 & b_2 & c_2 & \alpha_2 & \beta_2 & \gamma_2 \\ A_3 & B_3 & C_3 & a_3 & b_3 & c_3 & \alpha_3 & \beta_3 & \gamma_3 \end{array}$$

where letters of the same alphabet denote lines in the same plane, if only the letters and suffixes are the same; thus A_1, A_2, A_3 lie in a plane $A_1A_2A_3$; letters of different alphabets denote lines which meet according to the Table.

41. I find that one way in which this may be identified with the double-sixer notation is to represent the above arrangement by

$$\begin{array}{cccccc} 1, & 2, & 3, & 4, & 5, & 6 \\ 1', & 2', & 3', & 4', & 5', & 6' \\ 12, & 13, & 14, & 15, & 16, & \ldots \end{array}$$

and then we may permute the i, j, k, l, m, n in 720 ways, and then upon the arrangements so obtained make any of the substitutions which permute inter se the 36 double-sixers.

42. The equations of the 45 planes are obtained taking the equation of the surface to be

$$W(1,1,1,1,mn+\tfrac{1}{l},nl+\tfrac{1}{m},lm+\tfrac{1}{n},l+\tfrac{1}{l},m+\tfrac{1}{m},n+\tfrac{1}{n}\overline{X,Y,Z,W})^2+kXYZ=0,$$

where $k=\frac{2(p-a)}{lmn}$, $a=lmn+\frac{1}{lmn}$, $p=lmn+\ldots$

43. The coordinates of the 27 lines are then found to be as given in the original memoir.

44. We have $X=0$, $Y=0$, $Z=0$, $W=0$ for $(12'=w)$, $(23'=x)$, $(31'=\theta)$, $(12.34.56=u)$; and representing by $(=lX+\frac{1}{l}Y+\frac{1}{n}Z+W)$ the equations of the planes $(14'=\xi)$, $(42'=y)$, $(21'=\eta)$, the equation of the cubic surface may be presented in the several forms

$$\begin{aligned} &= Fgg+k\eta ZX, \\ &= Fhh+k\xi XY, \\ &= W\xi\xi'+k\eta XY, \\ &= W\eta\eta'+k\xi ZX, \\ &= W\zeta\zeta'+k\xi XY, \\ &= Wpp'+k\xi YZ, \\ &= Wqq'+k\eta ZX, \\ &= Wrr'+k\xi XY, \quad \text{\&c.,} \end{aligned}$$

which are the 16 forms containing W, out of the complete system of 120 trihedral-pair forms.

45. The 27 lines are each of them a facultative line; each of the 27 lines is a double tangent of the spinode curve; therefore $b'=p'=27$; $t'=45$.

46. The nodal curve is composed of the 27 lines ($b'=27$, $t'=45$ ut supra); the spinode curve is a complete intersection 3×4; $\sigma'=12$.

The equation of the reciprocal surface is not here investigated; its form is $S^3-27T^2=0$, where $S=0$, $T=0$ are the equations of the cuspidal curve; but for the factor thrown out as just mentioned, we should have simply $(S=0,\ T=0)$, or, as the case may be, $(S=0,\ T=0)$ for the existence of the cuspidal curve. The equation shows that the cuspidal curve is a complete intersection 6×4: $c'=24$.

Section II $= 12-C_2$. Article Nos. 47 to 59. Equation $W(a,b,c,f,g,h\overline{\between X,Y,Z})^2+2kXYZ=0$.

47. It may be remarked that the system of lines and planes is at once deduced from that belonging to $\mathfrak{I}=12$, by supposing that in the double-sixer the corresponding lines 1 and 1′, &c. severally coincide; the line 12, instead of being given as the intersection of the planes 12′, 1′2, is given as the third line in the plane 12, which in fact represents the coincident planes 12′ and 1′2.

48. The diagram of the lines and planes is given in the original memoir. The lines are: 6 rays through the conic-node, and 15 mere lines. The planes are: 15 biradial planes (each containing a pair of rays), and 15 planes each containing three mere lines.

49. Putting the equation of the surface in the form

$$W(1,1,1,1+\ldots\overline{\between X,Y,Z})^2+XYZ=0,$$

where for shortness

$$\alpha=mn-l,\quad \beta=nl-m,\quad \gamma=lm-n,\quad \delta=lmn-1,\quad p=lmn,$$

then taking $X=0$ as the equation of the plane [12], $Y=0$ as that of the plane [34], $Z=0$ as that of the plane [56], the equations of the 30 distinct planes are found to be:

$$\begin{array}{rll}
Z=0, & [12] & mX+l^{-1}Y+Z=0,\\
Y=0, & [34] & m^{-1}X+l^{-1}Y+Z=0,\\
Z=0, & [56] & X+nY+mZ=0,\\
 & [23] & X+n^{-1}Y+mZ=0,\\
 & [14] & X+nY+m^{-1}Z=0,\quad \text{\&c.}
\end{array}$$

50. And the coordinates of the 21 distinct lines are given in the original memoir.

51. The six nodal rays are not, the fifteen mere lines are, facultative; hence $b'=p'=15$; $t'=15$.

52. Resuming the equation $W(a,b,c,f,g,h\overline{\between X,Y,Z})^2+2kXYZ=0$, the equation of the Hessian surface is found to be

$$\begin{aligned}
&k^2W^2(a,b,c,f,g,h\overline{\between X,Y,Z})^2\\
&\qquad +2kW\{(a,b,c,f,g,h\overline{\between X,Y,Z})^2(FX+GY+HZ)-kXYZ\}\\
&\qquad -k\{a^2X^4+b^2Y^4+c^2Z^4-2bcY^2Z^2-2caZ^2X^2-2abX^2Y^2\\
&\qquad\qquad -4XYZ[(af+gh)X+(bg+hf)Y+(ch+fg)Z]\}=0,
\end{aligned}$$

where $(A,B,C,F,G,H)=(bc-f^2,ca-g^2,ab-h^2,gh-af,hf-bg,fg-ch)$, $K=abc-af^2-bg^2-ch^2+2fgh$.

The Hessian and cubic surfaces intersect in a complete intersection, which is the spinode curve; that is, the spinode curve is a complete intersection 3×4; $\sigma'=12$.

The equations of the spinode curve may be written in the simplified form

$$\begin{aligned}
&W(a,b,c,f,g,h\overline{\between X,Y,Z})^2+2kXYZ=0,\\
&\qquad -8KXYZW+8kXYZ(afX+bgY+chZ)\\
&\qquad\qquad -k^2\{a^2X^4+b^2Y^4+c^2Z^4-2bcY^2Z^2-2caZ^2X^2-2abX^2Y^2\}=0,
\end{aligned}$$

and it appears hereby that the node is a sixfold point on the curve, the tangents of the curve in fact coinciding with the six rays.

Each of the 15 lines touches the spinode curve twice; in fact, for the line 12 we have $X=0$, $W=0$: and substituting in the equations of the spinode curve, we have $(bY^2-cZ^2)^2=0$; that is, we have the two points of contact $X=0$, $W=0$, $Y\sqrt{b}=\pm Z\sqrt{c}$. Hence $\beta'=30$.

Reciprocal Surface.

53. The equation is found by equating to zero the discriminant of the ternary cubic function

$$(Xx+Yy+Zz)(a,b,c,f,g,h\overline{\between X,Y,Z})^2-2kwXYZ,$$

viz. the discriminant contains the factor w^2 which is to be thrown out, thus reducing the order to $n' = 10$.

54. Write as before (A, B, C, F, G, H) for the inverse coefficients, $K = abc - af^2 - bg^2 - ch^2 + 2fgh$; and moreover

$$\begin{aligned}
\Phi &= (A, B, C, F, G, H\overline{x, y, z})^2, \\
P &= Ax + Hy + Gz, \quad Q = Hx + By + Fz, \quad R = Gx + Fy + Cz, \\
t &= fx + gy + hz, \\
U &= afyz + bgzx + chxy, \\
V &= 2Kxyz - aPyz - bQzx - cRxy \\
&= -aHy^2z - bFz^2x - cGx^2y - aGyz^2 - bHzx^2 - cFxy^2 \\
&\quad + (-abc - af^2 - bg^2 - ch^2 + 2fgh)xyz, \\
W &= (A, B, C, F, G, H\overline{ayz, bzx, cxy})^2, \\
L &= kw^2t^2 - 2ktw - \Phi, \\
M &= kwU + V, \\
N &= 2kabc\,xyzw + W.
\end{aligned}$$

54. (continued) Then the invariants of the ternary cubic are

$$S = L^2 - 12kwM, \quad T = L^3 - 18kwLM - 54k^2w^2N,$$

and the required equation of the reciprocal surface is

$$S^3 - 27T^2 = 0,$$

viz. this is

$$(L^2 - 12kwM)^3 - 27(L^3 - 18kwLM - 54k^2w^2N)^2 = 0.$$

55. I present the result as follows; the coefficients deducible by mere cyclical permutations of the letters a, b, c and f, g, h, are indicated by (α).

$$\begin{aligned}
&(kw)^0 \cdot 2abc\,xyz \\
&\qquad + (kw)^1 \cdot [\ldots] \\
&\qquad + (kw)^2 \cdot [\ldots] \\
&\qquad + (kw)^3 \cdot [\ldots] \\
&- K\{(A, B, C, F, G, H\overline{x, y, z})^2(cy^2 - 2fyz + bz^2)(az^2 - 2gzx + cx^2)(bx^2 - 2hxy + ay^2)\}.
\end{aligned}$$

56–59. In explanation of the discussion of the reciprocal surface:

Node C_2: $Z = 0$, $Y = 0$, $Z = 0$.

Reciprocal plane: $w = 0$.

Conic of contact: $(A, B, C, F, G, H\overline{x, y, z})^2 = 0$, $w = 0$.

Nodal rays are sections of cone by planes $X = 0$, $bY^2 + 2fYZ + cZ^2 = 0$; $Y = 0$, $cZ^2 + 2gZX + aX^2 = 0$; $Z = 0$, $aX^2 + 2hXY + bY^2 = 0$.

Tangent cone is $(a, b, c, f, g, h\overline{X, Y, Z})^2 = 0$, $w = 0$.

The equation shows that the nodal curve is made up of the conic $(A, B, C, F, G, H\overline{x, y, z})^2 = 0$, twice, and of the six lines, tangents to this conic, which are the reciprocals of the nodal rays; these six lines are thus mere scrolar lines on the reciprocal surface.

As regards the cuspidal curve, the equation of the surface may be written

$$(S^3 - 12kwM)(4M^3 + S^2N) - (LM + 9kwN)^2 = 3\{SM^3 + S^3N - 18kwLMN - 16kwM^3 - 27k^2w^2N^2\} = 0,$$

and we thus have

$$4M^3 + S^2N = 0, \quad LM + 9kwN = 0, \quad S - 12kwM = 0,$$

or, what is the same thing,

$$\frac{L}{12M} = \frac{M}{-9N} = \frac{-9N}{L^2},$$

(equivalent to two equations) for the cuspidal curve. Attending to the second and third equations, the cuspidal curve may be considered as the residual intersection of the quartic and quintic surfaces $L^2 - 12kwM = 0$, $LM + 9kwN = 0$, which partially intersect in the conic $w = 0$, $\Phi = 0$; it is a curve $4 \cdot 5 - 2$; $c' = 18$.

Section III $= 12 - B_3$. Article Nos. 60 to 72. Equation $2W(X+Y+Z)(lX+mY+nZ)+2kXYZ=0$.

60. The system of lines and planes is at once deduced from that belonging to $\mathfrak{I} = 12$, by supposing the tangent cone to reduce itself to the pair of biplanes; 3 of the planes (α) of $\mathfrak{I}\mathfrak{I} = 12 - C_2$ thus coming to coincide with the one biplane, and three of them with the other biplane.

61. The diagram is given in the original memoir. The lines are: rays 1, 2, 3 in one biplane; rays 4, 5, 6 in the other biplane; 2 mere lines. The planes are: 2 biplanes; 9 biradial planes; 6 planes each containing three mere lines.

62. Taking $lX + mY + nZ = 0$ for the biplane that contains the rays 1, 2, 3, and $X + Y + Z = 0$ for that which contains the rays 4, 5, 6, we may take $X = 0$, $Y = 0$, $Z = 0$ for the equations of the planes [14], [25], [36] respectively; and then writing for shortness

$$m - n = \lambda, \quad n - l = \mu, \quad l - m = \nu,$$

and assuming, as we may do, $k = \lambda\mu\nu$, so that the equation of the surface is

$$W(X + Y + Z)(lX + mY + nZ) + (m - n)(n - l)(l - m)XYZ = 0,$$

the equations of the 17 distinct planes are found to be:

$$\begin{array}{ll} Z = 0, & [14] \\ Y = 0, & [25] \\ Z = 0, & [36] \\ X + Y + Z = 0, & [123] \\ lX + mY + nZ = 0, & [456] \\ lX + nY + nZ = 0, & [16] \\ \quad\quad\&\text{c.} & \end{array}$$

63. And the coordinates of the fifteen distinct lines are given in the original memoir.

64. The rays are not, the mere lines are, facultative; hence $b' = p' = 9$; $t' = 6$.

65. The equation of the Hessian surface is

$$\begin{aligned} &- W(X + Y + Z)(lX + mY + nZ)(\mu\nu X + \nu\lambda Y + \lambda\mu Z) \\ &\qquad - k(l^2X^4 + m^2Y^4 + n^2Z^4 - 2mnY^2Z^2 - 2nlZ^2X^2 - 2lmX^2Y^2) \\ &+ kXYZ\{(l^2 + 3lm + 3ln + mn)X + (m^2 + 3mn + 3ml + nl)Y + (n^2 + 3nl + 3nm + lm)Z\} = 0. \end{aligned}$$

The Hessian and cubic surfaces intersect in an indecomposable curve, which is the spinode curve; that is, the spinode curve is a complete intersection 3×4; $\sigma' = 12$.

The equations may be written in the simplified form

$$\begin{aligned} &W(X + Y + Z)(lX + mY + nZ) + kXYZ = 0, \\ &l^2X^4 + m^2Y^4 + n^2Z^4 - 2mnY^2Z^2 - 2nlZ^2X^2 - 2lmX^2Y^2 \\ &\quad - 4XYZ\{l(m + n)X + m(n + l)Y + n(l + m)Z\} = 0. \end{aligned}$$

We may also obtain the equation

$$\begin{aligned} &k^2(X + Y + Z)(lX + mY + nZ)\{lX^2 + mY^2 + nZ^2 - (m + n)YZ - (n + l)ZX - (l + m)XY\} \\ &\qquad + \lambda^2Y^2Z^2 + \mu^2Z^2X^2 + \nu^2X^2Y^2 - 2XYZ(\mu\nu X + \nu\lambda Y + \lambda\mu Z) = 0, \end{aligned}$$

which shows that there is at B_3 an eightfold point, the tangents being given by $(X+Y+Z)(lX+mY+nZ) = 0$.

Each of the facultative lines is a double tangent of the spinode curve; whence $\beta' = 18$.

Reciprocal Surface.

66. The equation may be deduced from that for $\mathfrak{I}\mathfrak{I} = 12 - C_2$; viz. writing

$$(a, b, c, f, g, h\overline{X, Y, Z})^2 = 2(X + Y + Z)(lX + mY + nZ),$$

so that $(a, b, c, f, g, h) = (2l, 2m, 2n, m + n, n + l, l + m)$, we have

$$(A, B, C, F, G, H) = -(\lambda^2, \mu^2, \nu^2, \mu\nu, \nu\lambda, \lambda\mu); \quad K = 0.$$

Writing also $\lambda, \mu, \nu = m - n, n - l, l - m$ as before,

$$\begin{aligned} \lambda x + \mu y + \nu z &= \sigma, \\ lmn\, xyz &= \theta, \\ l(m + n)yz + m(n + l)zx + n(l + m)xy &= \psi, \\ t\, xyz + \sigma\psi &= \phi, \end{aligned}$$

we have $U = 2\psi$, $V = 2\sigma\phi$, $W = -4\Phi$, and then

$$\begin{aligned} L &= k(\psi v^2 - 2ktw + \sigma^2), \\ M &= 2(kw\psi + \sigma\phi), \\ N &= 4(lmnkxyzw - \psi^2). \end{aligned}$$

67. The equation is

$$\begin{aligned} (k^2\psi^2 - 2kt\psi + \sigma^2)^2\{kW\theta + kw(-2t\theta + \psi^2 - \phi^2) + \sigma^2\theta + 2\sigma\phi\psi + 2t\phi^2\} \\ - 32(k\psi\psi + \sigma\phi)^3 - 108k^2w^2(4\psi\theta - \psi^2)^2 = 0. \end{aligned}$$

68. I verify the last term in the particular case $z = 0$ as follows: the coefficient of σ^4 is

$$(\theta, 2n(l + m)xy, 2(m + n)x + 2(n + l)y, \sigma^2\theta + \phi^2, n\nu xy)^2$$

which reduces to the general value $-4\lambda\mu\nu(y - z)(z - x)(x - y)(ny - mz)(lz - nx)(mx - ly)$.

69. In the discussion of the equation it is convenient to write down the relations of the two surfaces:

Cubic surface.	*Reciprocal surface.*
Plane $w = 0$,	Plane $w = 0$,
B_3	B'_3
Biplanes $X + Y + Z = 0$, $lX + mY + nZ = 0$.	
Rays in first biplane,	Points in $w = 0$,
$Y + Z = 0$; $Z + X = 0$;	$y - z = 0$, $z - x = 0$, $x - y = 0$;
$X + Y = 0$	through first point, viz.
	$ny - mz = 0$, $nz - lx = 0$, $lx - my = 0$.

70. The equation puts in evidence the line $\sigma = 0$ (reciprocal of the edge) three times, and the six lines (reciprocals of the rays) each once. Observe that the edge is not a line on the reciprocal surface; and that the six lines (reciprocals of the rays) are mere scrolar lines on the reciprocal surface; they pass, three of them, through the point $x = y = z$, and the other three through the point $x : y : z = l : m : n$; that is, they are six tangents of the point-pair (reciprocal of the pair of biplanes) formed by these two points.

71. I do not attempt to put in evidence the nodal curve on the surface; by what precedes it consists of 9 lines, reciprocals of the mere lines. If we denote by 1, 2, 3 and 4, 5, 6 the lines which pass through the points $x = 0$, $y = 0$, $z = 0$ and through the point $x : y : z = l : m : n$ respectively, then these intersect in the nine points 14, 15, 16, 24, 25, 26, 34, 35, 36; and through each of these there passes a nodal line which may be represented by the same symbol; that is, we have the nodal lines 14, …, 36. Two lines such as 14, 25 meet; and three lines such as 14, 25, 36 meet in a point; we have thus the six points 14.25.36 &c. triple points on the nodal curve; as before, $b' = 9$, $t' = 6$.

72. The cuspidal curve is given by the equations

$$\begin{aligned} (k\psi v^2 - 2kt\psi + \sigma^2)^2 - 24k\psi(k\psi v + \sigma\phi)^2 &= 0, \\ (k\psi v^2 - 2kt\psi + \sigma^2)(k\psi v + \sigma\phi) + 18\psi(lmnkxyzw - \psi^2) &= 0. \end{aligned}$$

Writing down the two equations, these are respectively of the orders 4 and 5; but they intersect in the line $w = 0$, taken four times, or say, the cuspidal curve is a partial intersection $4 \cdot 5 - 4$; $\sigma = 4$; $c' = 16$.

Section IV $= 12 - 2C_2$. Article Nos. 73 to 84. Equation $WXZ + Y^2(\gamma Z + \delta W) + (a, b, c, d\overline{X, Y})^3 = 0$.

73. The diagram of the lines is given in the original memoir. The lines are: axis [0] joining the two nodes; 4 rays through each node; 8 mere lines. The planes are: plane touching along axis; 16 planes through axis (each containing a ray of one node and a ray of the other); 24 biradial planes (each containing two rays of one node or two rays of the other); 3 planes each containing three mere lines.

74. Writing

$$X(a, b, c, d\overline{X, Y})^2 - \gamma\delta Y^2 = -\gamma\delta(f_1X - Y)(f_2X - Y)(f_3X - Y)(f_4X - Y),$$

the 20 planes are:

$$\begin{array}{ll}
Z = 0, & [0] \\
X - f_1Y = 0, & [11'] \\
X - f_2Y = 0, & [22'] \\
X - f_3Y = 0, & [33'] \\
X - f_4Y = 0, & [44'] \\
\delta\{Z - (f_1 + f_2)Y\} - f_1f_2\gamma W = 0, & [12] \\
\quad\quad\quad\quad\&\text{c.} &
\end{array}$$

75. And the 16 lines are given in the original memoir.

76. To verify the equations of the line $12.3'4'$, observe that the two equations give

$$\gamma Z + \delta W = \gamma\delta\{Z(f_1 + f_4) - Y(f_1f_4 + f_2f_3)\},$$

and substituting in the equation of the surface, this becomes an identity.

77. The facultative lines are the transversal and the six mere lines; $b' = p' = 7$.

78. The equation of the Hessian surface is found to be

$$\begin{aligned}
(\gamma^2 + \delta^2W)XZW + Y^2(\gamma Z - \delta W)^2 + 3(cX + dY)XZW + 12\gamma\delta XY^2(aX + bY) \\
- (\gamma Z + \delta W)(3aZ^2 + 9bX^2 - Y^2 + 6cXY) \\
- 9X^2\{(ac - b^2)X^2 + (ad - bc)XY + (bd - c^2)Y^2\} = 0.
\end{aligned}$$

79. Combining with the foregoing the equation of the surface

$$XZW + Y^2(\gamma Z + \delta W) + (a, b, c, d\overline{X, Y})^3 = 0,$$

it appears that $X = 0$, or $Y = 0$, and that the two equations have along the line $X = 0$, $Y = 0$ (the axis) the same tangent plane $X = 0$; they meet in the line $X = 0$, $Y = 0$ twice, and in a residual curve of the tenth order, which is the spinode curve.

The spinode curve may be presented in the somewhat more simple form

$$\begin{aligned}
XZW + Y^2(\gamma Z + \delta W) + (a, b, c, d\overline{X, Y})^3 = 0, \\
- 4\gamma\delta Y^2ZW - 4(\gamma Z + \delta W)(a, b, c, d\overline{Z, Y})^3 + 12\gamma\delta XY^2(aX + bY) \\
+ X^2(-12ac + 9b^2) - 3d(4aZ^2Y + 6bX^2Y^2 + 4cXY^3 + dY^4) = 0,
\end{aligned}$$

the line $X = 0$, $Y = 0$ twice.

The spinode curve is of the tenth order; that is, we have $\sigma' = 10$.

Each of the 6 mere lines is a double tangent to the spinode curve; the transversal is only a single tangent: to show this, observe that the equations of the spinode curve contain the line $X = 0$, $Y = 0$ twice. For the transversal $X = 0$, $Y = 0$, $\gamma Z + \delta W + dY = 0$; substituting in the equations of the curve, the cubic surface equation is of course satisfied identically; for the second equation, writing $Z = 0$, this becomes $Y^2\gamma Z - \delta W = 0$. The value $Y^2 = \delta W/\gamma Z$ gives a point of contact not belonging to the spinode curve. Hence the number of contacts is $2 \cdot 6 + 1 = 13$; that is, we have $\beta' = 13$.

Reciprocal Surface.

80. The equation is found by equating to zero the discriminant of the binary quartic

$$\{xX^2 + yXY - (\delta z + \gamma w)Y^2\}^2 + 4Zw\{X(a, b, c, d\overline{Z, Y})^2 - \gamma\delta Y^2\},$$

or say this is $(X, Y)^4$, where the coefficients are

$$\begin{aligned}
&6a(x^2 + 24azw),\\
&3xy + 18bzw,\\
&y^2 - 2(\delta z + \gamma w)x + 12czw,\\
&-3(\delta z + \gamma w)y + 6dzw,\\
&6(\delta z - \gamma w).
\end{aligned}$$

81. Forming the invariants, these are

$$\begin{aligned}
\tfrac{1}{12}I &= A = \ldots + 24azw,\\
-\tfrac{1}{6}J &= A' + 36A\,Uzw + 216Vz^2w^2 + 864Wz^3w^3,
\end{aligned}$$

where U, V, W are given in the original memoir.

82. It is convenient to modify the form of the equation as follows; write

$$U_1 = U + 8a\gamma\delta zw, \quad V_1 = V + (-8ac + 9b^2)\gamma\delta zw,$$

so that

$$\begin{aligned}
\Lambda &= y^2 + 4(\delta z + \gamma w)x,\\
U_1 &= -2\gamma\delta x^2 + 2a(\delta z + \gamma w)^2 + 3by(\delta z + \gamma w) + c[y^2 - 2(\delta z + \gamma w)x] - dxy,\\
V_1 &= (-8ac + 9b^2)(\delta z + \gamma w)^2\\
&\quad + (2c^2 - bd)[y^2 - 2(\delta z + \gamma w)x]\\
&\quad + (-4ad + 6bc)y(\delta z + \gamma w)\\
&\quad - 2cdxy + d^2x^2 + 4\gamma\delta(2cx^2 - 3bxy + ay^2),\\
\mu &= c^2 - bd,\\
\nu &= ad^2 - 2bcd + 2c^3,
\end{aligned}$$

Λ, U_1, V_1 being, it will be observed, functions of x, y, $\delta z + \gamma w$. The transformed equation is

$$\Lambda^2(\Lambda^2\mu - \Lambda V_1 + U_1^2) + \Omega zw = 0,$$

where the term Ω may be calculated without difficulty: the first term of this is

$$= [y^2 + 4(\delta z + \gamma w)x]^2 \cdot 4\gamma^2\delta^2[x + f_1y - f_1^2(\delta z + \gamma w)]\ldots[x + f_4y - f_4^2(\delta z + \gamma w)],$$

the developed expressions of $\frac{1}{4}(\Lambda^2\mu - \Lambda V_1 + U_1^2)$ and of $\gamma^2\delta^2$ into the product of the linear factors being in fact each

$$\begin{aligned}
= x^4 \cdot \gamma^2\delta^2 + x^3y \cdot d\gamma\delta + x^2y^2 \cdot (-3c\gamma\delta) + xy^3 \cdot 3b\gamma\delta + y^4 \cdot (-a\gamma\delta)&\\
+ [x^3(-d^2 - 6c\gamma\delta) + x^2y(3cd + 9b\gamma\delta) + xy^2(-3bd - 4a\gamma\delta) + y^3 \cdot ad](\delta z + \gamma w)&\\
+ [x^2(9c^2 - 6bd - 2a\gamma) + xy(3ad - 9bc) + y^2 \cdot 3ac](\delta z + \gamma w)^2&\\
+ [x(6ac - 9b^2) + y \cdot 3ab](\delta z + \gamma w)^3&\\
+ a^2\delta^4 \cdot (\delta z + \gamma w)^4.&
\end{aligned}$$

The form puts in evidence the section by the plane $w = 0$, which is the reciprocal of the node D, viz. this is a conic (the reciprocal of the tangent cone) twice, and four lines, the reciprocals of the nodal rays, each once. And similarly for the section by the plane $z = 0$.

83. The nodal curve is made up of the lines which are the reciprocals of the six mere lines and the transversal; viz. we have three pairs of lines and a seventh line, the lines of each pair intersecting at a point of the seventh line, and these three points being the triple points of the nodal curve; $t' = 3$ as before.

84. The equations of the cuspidal curve are at once reduced to the form

$$\begin{aligned} \Lambda^2 + 24Uzw + 144\mu z^2w^2 &= 0, \\ \Lambda U + (18V - 12\mu\Lambda)zw + 72\nu z^2w^2 &= 0, \end{aligned}$$

which are two quartic surfaces having in common the conics $z = 0$, $\Lambda = 0$, and $w = 0$, $\Lambda = 0$; or we may say that the cuspidal curve is a curve $4 \cdot 4 - 2 - 2$; that is $c' = 12$.

This completes the discussion of the surface IV $= 12 - 2C_2$. The memoir continues with the remaining cases V to XXIII, which are discussed in subsequent sections with similar detail.

A Memoir on Cubic Surfaces.

By Professor Cayley.

[Continued from Vol. VI, pp. 259–366.]

Section I = 12. Article Nos. 38 to 46. (Continued.)

Double-sixers.

It has been mentioned that the number of double-sixers was = 36; these are as follows:

$1, 2, 3, 4, 5, 6$	Assumed primitive	
$1', 2', 3', 4', 5', 6'$		
$1, 1', 23, 24, 25, 26$	Like arrangements	15
$2, 2', 13, 14, 15, 16$		
$1, 2, 3, 56, 46, 45$	Like arrangements	20
$23, 13, 12, 4, 5, 6$		
	Total	36

where, if we take any column of two lines, we have the complete number 216 of pairs of non-intersecting lines (each line meets 10 lines, there are therefore $27 - 1 - 10 = 16$ lines which it does not meet; $27 \times 16 \div 2 = 216$).

Dr. Hart's Arrangement.

Dr. Hart arranges the 27 lines, cubically, thus:

$$\begin{array}{ccccccccc} A_1 & B_1 & C_1 & a_1 & b_1 & c_1 & \alpha_1 & \beta_1 & \gamma_1 \\ A_2 & B_2 & C_2 & a_2 & b_2 & c_2 & \alpha_2 & \beta_2 & \gamma_2 \\ A_3 & B_3 & C_3 & a_3 & b_3 & c_3 & \alpha_3 & \beta_3 & \gamma_3 \end{array}$$

where letters of the same alphabet denote lines in the same plane, if only the letters are the same or the suffixes the same; thus A_1, A_2, A_3 lie in a plane $A_1A_2A_3$; A_1, B_1, C_1 lie in a plane $A_1B_1C_1$. Letters of different alphabets denote lines which meet according to the Table.

Equations of Planes.

Using the notation where $[12'] = w$, $[23'] = \theta$, etc., the equations of the planes are:

$$
\begin{aligned}
W &= 0, \quad & [12' = w] \\
X &= 0, \quad & [42' = y] \\
Y &= 0, \quad & [14' = z] \\
Z &= 0, \quad & [12' = w] \\
\frac{X}{l} + \frac{Y}{m} + \frac{Z}{n} + W &= 0, \quad & [41' = \xi] \\
\frac{X}{l} + mY + \frac{Z}{n} + W &= 0, \quad & [34' = \eta] \\
\frac{X}{l} + \frac{Y}{m} + nZ + W &= 0, \quad & [13.24.56 = h] \\
lX + mY + nZ + W &= 0, \quad & [24' = \zeta] \\
lX + \frac{Y}{m} + nZ + W &= 0, \quad & [14.25.36 = g] \\
lX + mY + \frac{Z}{n} + W &= 0, \quad & [43' = \xi] \\
\frac{l(p-\alpha) + 2mn}{p+\alpha} X + \cdots &= 0, \quad & [52' = \psi] \\
\frac{n(p-\alpha) + 2lm}{p+\alpha} Y + \cdots &= 0, \quad & [15' = z] \\
\frac{m(p-\alpha) + 2nl}{p+\alpha} Z + \cdots &= 0, \quad & [12.36.45 = \chi]
\end{aligned}
$$

Further Plane Equations.

Continuing the equations of the planes:

$$
\begin{aligned}
-m \left\{ \frac{n(p-\alpha)}{\cdots} \right\} X + mY + nZ + W &= 0, \quad & [56' = l] \\
lX - m \left\{ \frac{l(p-\alpha)}{\cdots} \right\} Y + nZ + W &= 0, \quad & [45' = m] \\
lX + mY - n \left\{ \frac{m(p-\alpha)}{\cdots} \right\} Z + W &= 0, \quad & [64' = n] \\
\frac{l(p-\alpha)}{\cdots} X + mY + \frac{Z}{n} + W &= 0, \quad & [15.26.34 = l_1] \\
lX + \frac{2n}{\cdots} Y + nZ + W &= 0, \quad & [16.24.35 = m_1] \\
lX + mY - \frac{m(p-\alpha)}{\cdots} Z + W &= 0, \quad & [14.25.36 = n_1] \\
\frac{2n}{\cdots} X + mY + \frac{Z}{n} + W &= 0, \quad & [65' = l_1] \\
\frac{X}{l} + \frac{2n}{\cdots} Y + nZ + W &= 0, \quad & [46' = m_1] \\
\frac{X}{l} + mY - \frac{m(p-\alpha)}{\cdots} Z + W &= 0, \quad & [54' = n_1] \\
\frac{X}{l} + mY + \frac{Z}{n} + W &= 0, \quad & [16.25.34 = l_1] \\
\frac{2l}{\cdots} X + mY + nZ + W &= 0, \quad & [15.24.36 = m_1] \\
lX - \frac{2l}{\cdots} Y + nZ + W &= 0, \quad & [14.26.35 = n_1]
\end{aligned}
$$

Complex Plane Equations.

The remaining plane equations involve more complex coefficients:

$$-\frac{2l}{\cdots}X + nY + mZ + \{mn(p-\alpha) - 2l(l^2 - m^2 - n^2)\}W = 0,$$

$$[36' = p_1]$$

$$mX - \frac{2m}{\cdots}Y + lZ - \frac{l}{\cdots}\{m(1 - n^2)(p - \alpha) - 2nl\}W = 0,$$

$$[25' = q_1]$$

$$nX + lY - \frac{2n}{\cdots}Z - \frac{l}{\cdots}\{n(1 - l^2 - m^2)(p - \alpha) - 2lm\}W = 0,$$

$$[13.26.45 = r_1]$$

Coordinates of the 27 Lines.

The coordinates of the 27 lines are then found to be as follows:

(a) (b) (c)

1 :	1,	1,	1,	$1 - k\left\{\frac{l}{m}\right\}\left\{\frac{m}{n}\right\}\cdots$
2 :	1,	$-k\left\{\frac{l}{m}\right\}\left\{\frac{m}{n}\right\}$,	1,	$1\cdots$
3 :	$-k(l-1)(m-1)$,	$-k(l-1)(m-1)$,	1,	$\cdots$

(f) (g) (h)

$(12 = a_1)$:	$-n$,	m,	1,	$-l\cdots$
$(1' = b_2)$:	1,	$-l$,	n,	$m\cdots$
$(23 = b_1)$:	$-m$,	l,	1,	$n\cdots$
$(3' = c_1)$:	1,	1,	$-l$,	$m\cdots$

The remaining line coordinates follow similar patterns with appropriate permutations of the parameters l, m, n and their combinations.

Equation of the Cubic Surface.

We have $X = 0$, $Y = 0$, $Z = 0$, $W = 0$ for the equations of the planes $(12.34.56 = x)$, $(42' = y)$, $(14' = z)$, $(12' = w)$; and representing by $\xi = lX + \frac{Y}{m} + \frac{Z}{n} + W = 0$ the equation of any other plane

$(41' = \xi)$, the equation of the cubic surface may be presented in the several forms:

$$
\begin{aligned}
&= Fgg_1 + k\eta\zeta ZX,\\
&= Fhh_1 + k\zeta\theta XY,\\
&= W\eta\eta_1 + k\theta\iota YZ,\\
&= W\zeta\zeta_1 + k\iota\kappa ZX,\\
&= W\theta\theta_1 + k\kappa\lambda XY,\\
&= W\iota\iota_1 + k\lambda\mu YZ,\\
&= W\kappa\kappa_1 + k\mu\nu ZX,\\
&= W\lambda\lambda_1 + k\nu\xi XY,\\
&= W\mu\mu_1 + k\xi\eta YZ,\\
&= W\nu\nu_1 + k\xi\eta\zeta,\\
&= W\xi\xi_1 + k\eta\zeta\iota,\\
&= W\pi\pi_1 + k\xi\eta z,\\
&= W\rho\rho_1 + k\zeta\theta z,\\
&= W\sigma\sigma_1 + k\xi\iota x.
\end{aligned}
$$

Section II = 12–C_5. Article Nos. 47 to 59.

The Reciprocal Surface.

It may be remarked that the system of lines and planes is at once deduced from that belonging to I = 12, by supposing that in the double-sixer the lines $1, 2, 3, 4, 5, 6$ become the rays of a cubic node C_5. The reciprocal of a double-sixer is a double-sixer. Hence the 27 lines of the reciprocal surface may be (and that in 36 different ways) represented by:

$$
\begin{array}{cccccc}
1, & 2, & 3, & 4, & 5, & 6\\
1', & 2', & 3', & 4', & 5', & 6'\\
12, & 13, & \ldots & 56, & &
\end{array}
$$

where 12 is now the line joining the points $12'$ and $1'2$; and so for the other lines. The lines $12, 34, 56$ meet in a point 12.34.56; the 30 points $12', 1'2, \ldots, 56', 5'6$, and the fifteen points 12.34.56 make up the 45 points t'.

The above equation, $S^3 - T^2 = 0$, shows that the cuspidal curve is a complete intersection 6×4; $c' = 24$.

Equation $W(a, b, c, f, g, h \setminus X, Y, Z)^2 + 2kXYZ = 0$.

The system of lines and planes is at once deduced from that belonging to II = $12 - C_5$, by supposing the tangent cone to reduce itself to the pair of biplanes.

The Diagram.

Lines.	
12	
13	
14	
15	
16	
23	Biradial planes, through each pair of rays. 15 × 2 = 30
24	
25	
26	
34	
35	
36	
45	
46	
56	
12.34.56	
12.35.46	
12.36.45	Planes each containing three non-intersecting lines. 15 × 1 = 15
13.24.56	
13.25.46	
13.26.45	
14.23.56	
14.25.36	
14.26.35	
15.23.46	
15.24.36	
15.26.34	
16.23.45	
16.24.35	
16.25.34	
30	45

Equations of Planes.

Putting the equation of the surface in the form:

$$W\,\{1,1,1,1 \setminus X,Y,Z\}^2 + \frac{2}{P}XYZ = 0,$$

where for shortness:

$$\lambda = mn - l, \quad \mu = nl - m, \quad \nu = lm - n, \quad \delta = lmn - 1, \quad \rho = lmn,$$

then taking $X = 0$ as the equation of the plane [12], $Y = 0$ as that of the plane [34], $Z = 0$ as that of the plane [56], the equations of the 30 distinct planes are found to be:

$$\begin{aligned} X &= 0, \quad [12] \\ Y &= 0, \quad [34] \\ Z &= 0, \quad [56] \\ mX + lY + Z &= 0, \quad [23] \\ m^{-1}X + Y + Z &= 0, \quad [24] \\ X + nY + mZ &= 0, \quad [45] \\ mX + l^{-1}Y + Z &= 0, \quad [13] \\ m^{-1}X + l^{-1}Y + Z &= 0, \quad [14] \\ X + n^{-1}Y + mZ &= 0, \quad [46] \end{aligned}$$

Coordinates of the 21 Distinct Lines.

And the coordinates of the 21 distinct lines are:

(a) (b) (c) (f) (g) (h) whence equations may be taken to be:

1 :	l,	-1,	1,	(1)	$X = 0$,	$Y + lZ = 0$
2 :	1,	-1,	1,	(2)	$Y = 0$,	$Z + mX = 0$
3 :	n,	-1,	1,	(3)	$Z = 0$,	$X + nY = 0$
4 :	1,	-1,	1,	(4)	$X = 0$,	$Y + nZ = 0$
5 :	1,	-1,	1,	(5)	$Y = 0$,	$Z + lX = 0$
6 :	m,	n,	1,	(6)	$Z = 0$,	$X + m^{-1}Y = 0$

And for the remaining lines:

(45) :	$\frac{1}{n}$,	$\frac{1}{m}$,	1,	$\frac{\alpha}{\gamma}$,	$\frac{1}{\gamma}$,	$\frac{1}{\beta}$
(16) :	$\frac{1}{n}$,	$\frac{1}{l}$,	1,	$\frac{\gamma}{\beta}$,	$\frac{1}{\beta}$,	$\frac{1}{\alpha}$
(23) :	$\frac{1}{m}$,	l,	1,	$\frac{\beta}{\alpha}$,	$\frac{1}{\alpha}$,	$\frac{1}{\gamma}$
(46) :	n,	m,	1,	$\frac{1}{\gamma}$,	$\frac{\alpha}{\gamma}$,	$\frac{1}{\beta}$
(26) :	l,	m,	1,	$\frac{1}{\beta}$,	$\frac{\gamma}{\beta}$,	$\frac{1}{\alpha}$

The Hessian Surface.

Resuming the equation $W(a, b, c, f, g, h \setminus X, Y, Z)^2 + 2kXYZ = 0$, the equation of the Hessian surface is found to be:

$$\begin{aligned} &kW^2(a, b, c, f, g, h \setminus X, Y, Z)^2 \\ &\qquad + 2kW\{(a, b, c, f, g, h \setminus X, Y, Z)^2(FX + GY + HZ) - k^2XYZ\} \\ &\qquad - k\{a^2X^4 + b^2Y^4 + c^2Z^4 - 2bcY^2Z^2 - 2caZ^2X^2 - 2abX^2Y^2 \\ &\qquad\qquad - 4XYZ[(af + gh)X + (bg + hf)Y + (ch + fg)Z]\} = 0, \end{aligned}$$

where

$$(A, B, C, F, G, H) = (bc - f^2,\ ca - g^2,\ ab - h^2,\ gh - af,\ hf - bg,\ fg - ch),$$

and

$$K = abc - af^2 - bg^2 - ch^2 + 2fgh.$$

Invariants and Reciprocal Surface.

Write as before (A, B, C, F, G, H) for the inverse coefficients ($A = bc - f^2$, &c.), and $K = abc - af^2 - bg^2 - ch^2 + 2fgh$; and moreover:

$$\begin{aligned}
\Phi &= (A, B, C, F, G, H \backslash x, y, z)^2,\\
P &= Ax + Hy + Gz,\\
Q &= Hx + By + Fz,\\
R &= Gx + Fy + Cz,\\
t &= fx + gy + hz,\\
U &= afyz + bgzx + chxy,\\
V &= 2Kxyz - aPyz - bQzx - cRxy\\
&= -aHy^2z - bFz^2x - cGx^2y - aGyz^2 - bEzx^2 - cFxy^2\\
&\quad + (-abc - ap^2 - bg^2 - ch^2 + 4fgh)xyz,\\
W &= (A, B, C, F, G, H \backslash ayz, bzx, cxy)^2,\\
L &= kW - 2ktw - \Phi,\\
M &= kwU + V,\\
N &= 2kabcxyzw + W.
\end{aligned}$$

Then the invariants of the ternary cubic are:

$$\begin{aligned}
S &= L^2 - 12kwM,\\
T &= L^3 - 18kwLM - 54k^2w^2N,
\end{aligned}$$

and the required equation of the reciprocal surface is:

$$\frac{1}{k^2w^2}\left[(L^3 - 18kwLM - 54k^2w^2N)^2 - (L^2 - 12kwM)^3\right] = 0,$$

viz. this is:

$$0 = DN = (kW - 2ktw - \Phi)^2(2kabcxyzw + W) + L^2M^2 + \cdots$$

Section III $= 12$–B_3. Article Nos. 60 to 72.

Equation $2W(X + Y + Z)(lX + mY + nZ) + 2kXYZ = 0$.

The system of lines and planes is at once deduced from that belonging to II $= 12 - C_5$, by supposing the tangent cone to reduce itself to the pair of biplanes: three of the planes (α) of II $= 12 - C_5$ thus coming to coincide with the one biplane, and three of them with the other biplane.

The Diagram.

Lines.	
III $= 12 - B_3$	
1	
2	
$1'$	
$2'$	
12	Biplanes containing rays $1, 2$ and $1', 2'$ respectively. $2 \times 12 = 24$
$11'$	
$22'$	Plane touching along edge and containing the transversal. $1 \times 8 = 8$
$12'$	
$21'$	Biradial planes each containing a ray of the one and a ray of the other biplane. $4 \times 4 = 16$
$11'.23'$	
$12'.31'$	Planes each through the transversal. $2 \times 1 = 2$
45	
46	
56	
3	
4	
5	
6	
$3'$	
$4'$	
$5'$	
$6'$	

Equations of Planes.

Taking $X + Y + Z = 0$ for the biplane that contains the rays $1, 2, 3$, and $lX + mY + nZ = 0$ for that which contains the rays $4, 5, 6$, we may take $X = 0$, $Y = 0$, $Z = 0$ for the equations of the planes [14], [25], [36] respectively; and then writing for shortness:

$$m - n = \lambda, \quad n - l = \mu, \quad l - m = \nu,$$

and assuming, as we may do, $k = \sqrt{\lambda\mu\nu}$, so that the equation of the surface is:

$$W(X + Y + Z)(lX + mY + nZ) + (m - n)(n - l)(l - m)XYZ = 0,$$

the equations of the 17 distinct planes are:

$$\begin{aligned} X &= 0, \quad [14] \\ Y &= 0, \quad [25] \\ Z &= 0, \quad [36] \\ X + Y + Z &= 0, \quad [123] \\ lX + mY + nZ &= 0, \quad [456] \end{aligned}$$

Coordinates of the 15 Distinct Lines.

And the coordinates of the fifteen distinct lines are:

(a) (b) (c) (f) (g) (h) whence equations may be written:

(1) :	-1,	1,	1,	(1) $X=0,\quad Y+Z=0$
(2) :	1,	-1,	1,	(2) $Y=0,\quad Z+X=0$
(3) :	1,	1,	-1,	(3) $Z=0,\quad X+Y=0$
$(1')$:	$-n$,	m,	1,	(4) $X=0,\quad mY+nZ=0$
$(2')$:	n,	$-l$,	1,	(5) $Y=0,\quad nZ+lX=0$
$(3')$:	l,	m,	-1,	(6) $Z=0,\quad lX+mY=0$
(14) :	1,	-1,	1,	(14) $X=0,\quad W=0$
(25) :	1,	1,	-1,	(25) $Y=0,\quad W=0$
(36) :	-1,	1,	1,	(36) $Z=0,\quad W=0$

And the remaining lines:

(15) :	l,	n,	n,	$n^2\nu - n\lambda\mu$
(16) :	l,	m,	m,	$m^2\mu - lm\nu$
(26) :	l,	m,	l,	$l^2\lambda - lm\mu$
(24) :	n,	m,	n,	$n^2\nu - mn\lambda$
(34) :	m,	m,	n,	$m^2\mu - mn\nu$
(35) :	l,	l,	n,	$l^2\lambda - ln\nu$

The Rays and Facultative Lines.

The rays are not, the mere lines are, facultative; hence $b' = p' = 6$; $t' = 6$.

The Hessian Surface.

The equation of the Hessian surface is:

$$\begin{aligned}&-W(X+Y+Z)(lX+mY+nZ)(\nu X+\lambda Y+\mu Z)\\ &\qquad -k(l^2X^2+m^2Y^2+n^2Z^2-2mnYZ-2nlZX-2lmXY)=0.\end{aligned}$$

Section IV $= 12{-}B_4$. Article Nos. 73 to 84.

Equation $ZW(e, X-Y)(e_2X-Y)(e_3X-Y)(e_4X-Y)+2kXYZ=0$.

Writing $Z(a,b,c,d \setminus X,Y)^2 \cdot d_1X-Y \cdot d_2X-Y \cdot d_3X-Y \cdot d_4X-Y$, the 20 planes are:

$$\begin{aligned}
Z &= 0, & [0]\\
X - f_1Y &= 0, & [11']\\
X - f_2Y &= 0, & [22']\\
X - f_3Y &= 0, & [33']\\
X - f_4Y &= 0, & [44']\\
Z - (f_1+f_2)Y - f_1f_2Z &= 0, & [12]\\
Z - (f_1+f_3)Y - f_1f_3Z &= 0, & [13]\\
Z - (f_1+f_4)Y - f_1f_4Z &= 0, & [14]\\
Z - (f_2+f_3)Y - f_2f_3Z &= 0, & [23]\\
Z - (f_2+f_4)Y - f_2f_4Z &= 0, & [24]\\
Z - (f_3+f_4)Y - f_3f_4Z &= 0, & [34]\\
Z - (f_1+f_2)Y - f_1f_2Y &= 0, & [1'2']\\
Z - (f_1+f_3)Y - f_1f_3Y &= 0, & [1'3']\\
Z - (f_1+f_4)Y - f_1f_4Y &= 0, & [1'4']\\
Z - (f_2+f_3)Y - f_2f_3Y &= 0, & [2'3']\\
Z - (f_2+f_4)Y - f_2f_4Y &= 0, & [2'4']\\
Z - (f_3+f_4)Y - f_3f_4Y &= 0, & [3'4']
\end{aligned}$$

and the planes through the nodes:

$$\begin{aligned}
(f_1+f_2+f_3+f_4)Z+\cdots &= 0, & [12.34]\\
(f_1+f_3+f_2+f_4)Z+\cdots &= 0, & [13.24]\\
(f_1+f_4+f_2+f_3)Z+\cdots &= 0, & [14.23]
\end{aligned}$$

The 16 Lines.

And the 16 lines are:

(a) (b) (c) (f) (g) (h) whence equations may be written:

$(0):$	$1,$	$0,$	$0,$	$x=0, \quad Y=0$
$(5):$	$0,$	$-\gamma,$	$d,$	$X=0, \quad dY+\gamma Z+dW=0$
$(1):$	$1,$	$e_1,$	$0,$	$e_1X-Y=0, \quad Z=0$

and so for the other lines with appropriate substitutions of e_2, e_3, e_4.

Section V $= 12{-}B_3$. Article Nos. 85 to 94.

Equation $WXZ+(X+Z)(Y^2-aX^2-bZ^2)=0$.

The diagram of the lines and planes is:

Lines.	
V $= 12 - B_3$	
1	
2	
$1'$	
$2'$	Biplanes containing rays $1, 2$ and $1', 2'$ respectively. $2 \times 12 = 24$
12	
$11'$	Plane touching along edge and containing the transversal. $1 \times 8 = 8$
$22'$	
$12'$	Biradial planes each containing a ray of the one and a ray of the other biplane. $4 \times 4 = 16$
$21'$	
$11'.23'$	Planes each through the transversal. $2 \times 1 = 2$
$12'.31'$	
3	
4	
5	
6	
$3'$	
$4'$	
$5'$	
$6'$	

The Planes.

The planes are:

$$\begin{aligned} Z &= 0, \quad [12] \\ X &= 0, \quad [1'2'] \\ Y + \sqrt{a}X + \sqrt{b}Z &= 0, \quad [11'] \\ Y - \sqrt{a}X - \sqrt{b}Z &= 0, \quad [22'] \\ Y + \sqrt{a}X - \sqrt{b}Z &= 0, \quad [12'] \\ Y - \sqrt{a}X + \sqrt{b}Z &= 0, \quad [21'] \end{aligned}$$

with other planes determined by the appropriate combinations.

Section VI = 12–B_4. Article Nos. 95 to 107.

Equation $Z(X-Y)(\theta_1X-Y)(\theta_2X-Y)(\theta_3X-Y)(\theta_4X-Y)=0$.

Writing $(a,b,c,d \setminus Z,Y)^2 \cdot d_1X-Y \cdot e_1X-Y \cdot e_2X-Y \cdot e_3X-Y \cdot e_4X-Y$, the planes are:

$$\begin{aligned}
X&=0, && [0]\\
Z&=0, && [00]\\
\theta_1X-Y&=0, && [22']\\
\theta_2X-Y&=0, && [33']\\
\theta_3X-Y&=0, && [44']\\
\theta_4X-Y&=0, && [55']\\
d(e_1X-Y)-Z&=0, && [12]\\
d(e_2X-Y)-Z&=0, && [13]\\
d(e_3X-Y)-Z&=0, && [14]\\
X\theta_1-Y(\theta_2+\theta_3)-W&=0, && [2'3']\\
X\theta_2-Y(\theta_1+\theta_4)-W&=0, && [2'4']\\
X\theta_3-Y(\theta_1+\theta_4)-W&=0, && [3'4']
\end{aligned}$$

The Lines.

And the lines are:

(a) (b) (c) (f) (g) (h) equations may be written:

(0) :	1,	0,	0,	$X=0, \quad Y=0$
(1) :	-1,	d,	1,	$X=0, \quad dY+Z=0$
(2) :	1,	e_1,	1,	$e_1X-Y=0, \quad Z=0$
(3) :	1,	e_2,	1,	$e_2X-Y=0, \quad Z=0$
(4) :	1,	e_3,	1,	$e_3X-Y=0, \quad Z=0$

and for the remaining lines, the coordinate expressions are more conveniently written in the symmetrical form.

The mere lines are facultative; hence $b'=p'=t'=\dots$

Section VII = 12–$2C_2$. Article Nos. 108 to 116.

Equation $Y^2+Y(X+Z+W)+4\phi XZW=0$.

Each of the lines $(X-Y=0, Y+Z=0)$ and $(X+W=0, Y-Z=0)$ is a double tangent of the spinode octic; in fact for the first of these lines we have:

$$16\theta^4+8\theta^2+16\phi\theta^2+5=0, \quad 16\theta^4+16\theta^2+4\phi=0,$$

that is:

$$(2\theta^2+1)(8\theta^2-4\theta+5)=0, \quad 4\theta(2\theta^2+1)=0,$$

so that the line touches at the two points given by $2\theta^2+1=0$; and similarly the other line touches at the two points given by $2\theta^2-1=0$.

The edge $X=0$, $Z=0$ has apparently a higher contact with the spinode octic, viz. the equations $X=0$, $Z=0$ are satisfied by $\theta=\infty$ twice, $\theta=0$ five times; but it must be reckoned only as a double tangent. Hence $\beta'=2\cdot 2+2=6$.

Reciprocal Surface.

The equation is obtained by equating to zero the discriminant of the binary quartic:

$$Z^2(yZ - wX)^2 + 4wZ^2(wZ^2 + zZX + xX^2),$$

viz. calling $t = \dots$

Section VIII = 12–$3C_2$. Article Nos. 117 to 125.

Equation $Y^2 + Y(X + Z + W) + 4\phi XZW = 0$.

The diagram of the lines and planes is:

Lines.	
VIII $= 12 - 3C_2$	
1	
7	Planes each touching along an axis and containing the corresponding transversal. $3 \times 2 = 6$
8	
9	
12	
34	Biradial planes each containing two rays of the same node. $3 \times 2 = 6$
56	
13	
24	Planes each containing an axis, and two rays through the terminal nodes respectively. $6 \times 4 = 24$
16	
26	
46	
35	
789	Plane through the three axes. $1 \times 8 = 8$
789	Plane through the three transversals. $1 \times 1 = 1$
14	
45	

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Section X = 12–B_3–C_2. Article Nos. 127 to 136.

127. Writing

$$(a, b, c, d \between X, Y)^3 = d(f_1X - Y)(f_2X - Y)(f_3X - Y),$$

the planes are

$$\begin{array}{lll} X = 0, \quad [0] & Z = 0, \quad [7] & W = 0, \quad [8] \\ f_1X - Y = 0, \quad [14] & f_2X - Y = 0, \quad [25] & f_3Z - Y = 0, \quad [36]; \end{array}$$

and the lines are

$$\begin{array}{lll} X = 0,\ Y = 0, \quad (0) & f_1X - Y = 0,\ Z = 0, \quad (1) & f_2X - Y = 0,\ Z = 0, \quad (2) \\ f_3X - Y = 0,\ Z = 0, \quad (3) & f_1Z - Y = 0,\ W = 0, \quad (4) & f_2Z - Y = 0,\ W = 0, \quad (5) \\ f_3Z - Y = 0,\ W = 0, \quad (6). & & \end{array}$$

128. There is no facultative line; $p' = b' = 0$, $t' = 0$; and hence also $\beta' = 0$.

129. *Hessian surface.* The equation is

$$X^2ZW(cX + dY) - 3X\{(ac - b^2, ad - bc, bd - c^2 \between X, Y)^2\} = 0,$$

so that the Hessian breaks up into the plane $X = 0$ (axial or common biplane) and into a cubic surface.

The complete intersection of the Hessian with the cubic surface is made up of the line $X = 0$, $Y = 0$ (the axis) four times; and of a system of four conics, which is the spinode curve; $c' = 8$.

In fact combining the equations $WXZ + (a, b, c, d \between X, Y)^3 = 0$ and

$$ZW(cX + dY) - 3X(ac - b^2, ad - bc, bd - c^2 \between X, Y)^2 = 0,$$

these intersect in the axis once, and in a curve of the eighth order which breaks up into four conics; for we can from the two equations deduce

$$(a, b, c, d \between X, Y)^3(cX + dY) + 3X^2(ac - b^2, ad - bc, bd - c^2 \between X, Y)^2 = 0,$$

that is $(4ac - 3b^2, ad, bd, cd, d^2 \between X, Y)^4 = 0$, a system of four planes each intersecting the cubic $XZW + (a, b, c, d \between X, Y)^3 = 0$ in the axis and a conic; whence, as above, spinode curve is four conics.

It is easy to see that the tangent planes along any conic on the surface pass through a point, and form therefore a quadric cone; hence in particular the spinode torse is made up of the quadric cones which touch the surface along the four conics respectively.

130. *Reciprocal Surface.* The equation is obtained by means of the binary cubic $\lambda(\lambda x + yY)^2 + 12zw(a, b, c, d \between X, Y)^3$, viz. calling this $(* \between X, Y)^3$ the coefficients are $(x^2 + 12azw,\ 2xy + 12bzw,\ y^2 + 12czw,\ 12dzw)$. The equation is found to be

$$\begin{aligned} &- 432(d^2a^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2)z^3w^3 \\ &\qquad + 216\{(ad^2 - 3bcd + 2c^3)x^2 + (-2acd + 4b^2d - 2bc^2)xy + (-abd + 2ac^2 - b^2c)y^2\}zw \\ &\qquad\quad + 9\{(d^2a^2 - 12cdaxy + (10bd + 3c^2)axy^2 - (4ad + 8bc)xy^3 + (4ac - b^2)y^4\}zw \\ &\qquad\qquad\qquad\qquad - y^3(da^3 - 3caxy + 3bxy^2 - ay^3) = 0. \end{aligned}$$

The section by the plane $w = 0$ (reciprocal of $B_1 = D$) is the line $w = 0$, $y = 0$ (reciprocal of edge) three times, and the lines $w = 0$, $dx^2 - 3caxy + 3bxy^2 - ay^3 = 0$ (reciprocals of the biplanar rays). And similarly for the section by the plane $z = 0$ (reciprocal of $B_2 = C$).

The section by the plane $y = 0$ is made up of the lines $(y = 0,\ z = 0)$, $(y = 0,\ w = 0)$ each once, and of two conics, $y = 0$,

$$16(d^2a^2 - 6abcd + 4ac^3 + 4b^3d - 3b^2c^2)z^2w^2 + 8(ad^2 - 3bcd + 2c^3)x^2zw + x^4 = 0.$$

131. There is not any nodal curve; $b' = 0$.

132. *Cuspidal curve.* The equations may be written

$$\begin{vmatrix} x^2 + 12azw & 2xy + 12bzw & y^2 + 12czw \\ 2xy + 12bzw & y^2 + 12czw & 12dzw \end{vmatrix} = 0.$$

Forming the equations

$$(bd - c^2) \cdot 12zw + 2(dxy - cy^2) \cdot 12zw - y^4 = 0,$$
$$(ad - bc) \cdot 12zw + (3dx^2 - 2cxy - by^2) \cdot 12zw - 2xy^2 = 0,$$

these are two quartic surfaces having in common the lines $(y = 0,\ w = 0)$, $(y = 0,\ z = 0)$: attending to the line $(y = 0,\ z = 0)$, this is on the second surface an oscular line, $z = \frac{1}{2}y$; on the first surface it is a nodal line, the one tangent plane being $6(bd - c^2)w \cdot z + dxz \cdot y = 0$, the other tangent plane being $z = 0$, but the line being in regard to this sheet an oscular line, $z = \frac{1}{2}y - \frac{y^2}{2}$. Hence in the intersection of the two surfaces the line counts $(1 + 3 =)4$ times; similarly the line $y = 0$, $w = 0$ counts $(1 + 3 =)$ four times; and there is a residual intersection of the order $(16 - 4 - 4 =)8$, which is the cuspidal curve; $c' = 8$.

133. The cuspidal curve is a system of four conics; in fact from the preceding equations written in the forms

$$(bd - c^2,\ 2(dxy - cy^2),\ -y^4 \frown 12zw,\ 1)^2 = 0,$$
$$(ad - bc,\ 3dx^2 - 2cxy - by^2,\ -2xy^2 \frown 12zw,\ 1)^2 = 0,$$

eliminating zw, we obtain

$$\{4^2(bd - c^2),\ -2(ad^2 - 3bcd + 4c^3),\ 6(acd + 3b^2d - 2bc^2),\ 6(bc - ad)b,\ a^2d - 3abc + 2b^3 \frown x, y\}^4 = 0,$$

which shows that the cuspidal curve lies in four planes, and it hence consists of four conics; these are of course the reciprocals of the quadric cones which touch the cubic surface along the four conics which make up the spinode curve.

134. The equation of the surface, attending only to the terms of the second order in y, z, w, is $27d^2z^2w^2 = 0$; it thus appears that the point $y = 0$, $z = 0$, $w = 0$ (reciprocal of the plane $X = 0$) (which is oscular along the axis joining the two binodes, or B_5-axis) is a binode on the reciprocal surface, the biplanes being $z = 0$, $w = 0$, viz. these are the planes reciprocal to the binodes $(Z = 0,\ Y = 0,\ W = 0)$ and $(Z = 0,\ Y = 0,\ Z = 0)$ of the cubic surface; we have thus $\mathrm{B}' = 1$.

It is proper to remark that the binode $y = 0$, $z = 0$, $w = 0$ is not on the cuspidal curve, as its being so would probably imply a higher singularity.

135. A simple case, presenting the same singularities as the general one, is when $a = d$, $b = c = 0$: to diminish the numerical coefficients assume $a = d = \frac{1}{2}$, the cubic surface is thus $12XZW + X^3 + Y^3 = 0$, and the equation of the reciprocal surface, multiplying it by 4, becomes $z^2w^2 + 6x^2zw^2 + (3x^4 - 12x^2y^2)zw - 4y^2(x^4 - y^4) = 0$, viz. this is the surface $4y^6 - 4y^2x^2(x^2 + 6zw) + zw(3x^2 + zw)^2 = 0$ considered in the Memoir "On the Theory of Reciprocal Surfaces."

The cuspidal curve is, as there shown, composed of the four conics $y = 0$, $3x^2 + 4zw = 0$ and $y^2 - 4x^2 = 0$, $x^2 - 4zw = 0$; and it is there shown that the two points $(x = 0,\ y = 0,\ z = 0)$, $(x = 0,\ y = 0,\ w = 0)$, each reckoned eight times, are to be considered as off-points of the reciprocal surface.

136. The like investigation applies to the general surface, and we have thus $\chi' = 16$; the points in question are still the points $(x = 0, y = 0, z = 0)$, $(x = 0, y = 0, w = 0)$; viz. these are the points of intersection of the surface by the line $(x = 0,\ y = 0)$, which points are also the common points of intersection of the four conics which compose the cuspidal curve, that is, they are quadruple points on the cuspidal curve; it does not appear that the points are on this account, viz. as quadruple points of the cuspidal curve, off-points of the surface, nor does this even show that the points should be reckoned each eight times. As already remarked, the singularity requires a more complete investigation.

Section XI = 12–B$_3$. Article Nos. 137 to 143. Equation $WXZ + (X+Z)(Y^2 - X^2) = 0$.

137. The diagram of the lines and planes is shown in the original text. The arrangement consists of:

- Biplane touching along axis and containing edge ($1 \times 12 = 12$)
- Other biplane ($1 \times 12 = 12$)
- Planes each through the axis and containing a biplanar ray and a cnienodal ray ($2 \times 8 = 16$)
- Plane touching along the edge and containing the mere line ($1 \times 3 = 3$)
- Biradial plane through the two cnienodal rays ($1 \times 2 = 2$)

Total: 45 planes.

138. The planes are

$$\begin{array}{ll}
X = 0, \quad [0] & Z = 0,\ Y = 0, \quad (0) \\
Z = 0, \quad [3] & X = 0,\ Z = 0, \quad (3) \\
X - Y = 0, \quad [11'] & X - Y = 0,\ Z = 0, \quad (1) \\
Z + Y = 0, \quad [22'] & X + Y = 0,\ Z = 0, \quad (2) \\
W = 0, \quad [31] & Z - Y = 0,\ W = 0, \quad (10) \\
Z + X = 0, \quad [1'2'] & Z + Y = 0,\ W = 0, \quad (2') \\
 & X + Z = 0,\ W = 0, \quad (12).
\end{array}$$

139. The facultative lines are the edge counting twice, and the mere line; $p' = b' = 3$; $t' = 1$.

140. *Hessian surface.* The equation is $X(X+Z)(ZW + 3X^2 - XZ) + Y^2(Z - X) = 0$. The complete intersection with the surface consists of the line $(X = 0,\ Y = 0)$, the axis, four times; the line $(X = 0,\ Z = 0)$, the edge, twice; and a sextic curve, which is the spinode curve; $c' = 6$.

Writing the equations of the surface and the Hessian in the form

$$X(ZW + Y^2) - Z^3 + Z(Y^2 - X^2) = 0,$$
$$X(X+Z)(ZW + Y^2) + (Z - 3X)\{-X^2 + Z(Y^2 - X^2)\} = 0,$$

we see that the equations of the spinode curve may be written $ZW + Y^2 = 0$, $-X^2 + Z(Y^2 - X^2) = 0$, viz. the curve is a complete intersection, 2×3.

There is at E_3 a triple point, the tangents coinciding with the nodal rays $W = 0$, $Y^2 - X^2 = 0$. The edge and the mere line are each of them single tangents of the spinode curve. But the edge counting twice in the nodal curve, its contact with the spinode curve will also count twice, that is, we have $\beta' = 2.1 + 1 = 3$.

141. *Reciprocal Surface.* The equation is obtained by means of the binary cubic $4w^2Z(X+Z)^2 + 4wZ(X+Z)(xX + zZ) + fXZ^2$, or calling this $(\ast X, Z)^2$, the coefficients are $(12w^2,\ 8w^2 + 4wx,\ 4w^2 + 4wx + 4wz + y^2,\ 12wz)$, and thence the equation is found to be

$$\begin{aligned} w^3[y^4 - 4(x-z)^2] + 16w^2[(2x - 3z)y^2 - 2(x-z)(x-z)^2] \\ + 8w[y^4 + (3x^2 - xz + 6z^2)y^2 - 2x^2(x-z)^2] \\ + 4w[(2x + 3z)y^4 - 2x^2(x+z)y^2] + y^4(y^2 - x^2) = 0, \end{aligned}$$

where the section by the plane $w = 0$ (reciprocal of binode) is $y^4(y^2 - x^2) = 0$, viz. this is the line $w = 0$, $y = 0$ (reciprocal of the edge) four times, and the lines $w = 0$, $y^2 - x^2 = 0$ (reciprocals of the biplanar rays).

The section by the plane $z = 0$ is found to be $(y^2 - x^2)\{y^2 + 4xw + 4w^2\}^2 = 0$, viz. this is the two lines $z = 0$, $y^2 - x^2 = 0$ (reciprocals of the nodal rays), and the conic $z = 0$, $y^2 + 4xw + 4w^2 = 0$ (reciprocal of the nodal cone $WX + Y^2 - X^2 = 0$) twice.

142. *Nodal curve.* The equation shows that the line $y = 0$, $x - z = 0$ (reciprocal of the line $W = 0$, $X + Z = 0$) is a nodal line on the surface. It also shows that the line $y = 0$, $w = 0$ (reciprocal of the edge) is a tacnodal line (= 2 nodal lines) on the surface; in fact attending only to the lowest terms in y, w, we have $-4\{16(x-z)^2w^2 + 8(x+z)wy^2 + y^4\} = 0$, that is,

$$y^2\left\{\frac{1}{4}y^2 + 2(x+z)w + 16(x-z)^2\frac{w^2}{y^2}\right\} = 0,$$

two values, $w = Ay^2$, $w = By^2$, which indicates a tacnodal line. The nodal curve is thus made up of the line $y = 0$, $x - z = 0$ once, and the line $y = 0$, $w = 0$ twice; $b' = 3$.
143. *Cuspidal curve.* The equations give

$$(4w + 2x)^2 - 3(4w^2 + 4wx + 4wz + y^2) = 0,$$
$$-4w^2z + (2w + x)(4w^2 + 4wx + 4wz + y^2) = 0,$$

or, as these are more simply written,

$$w^2 + 4wx - 12wz + 4x^2 - 3y^2 = 0,$$
$$8w^3 + 12w^2x - 28w^2z + w(4x^2 + 4xz + 2y^2) + xy^2 = 0,$$

so that the cuspidal curve is a complete intersection 2×3; $c' = 6$.

Section XII = 12–U$_3$. Article Nos. 150 to 156. Equation $W(X + Y + Z)^2 + XYZ = 0$.

150. The diagram of the lines and planes is: Lines include $3 \times 1 = 3$ mere lines, and $1'2'3'$; Planes include the uniplane [0], $1 \times 32 = 32$; $3 \times 4 = 12$ planes each touching along a ray and containing a mere line; and $1 \times 1 = 1$ plane through the three mere lines. Total: 46 lines and planes.
151. The planes are $X + Y + Z = 0$, [0]; $X = 0$, [1]; $Y = 0$, [2]; $Z = 0$, [3]; $W = 0$, $[1'2'3']$. The lines are $X = 0$, $Y + Z = 0$, (1); $Y = 0$, $Z + X = 0$, (2); $Z = 0$, $X + Y = 0$, (3); $X = 0$, $W = 0$, $(1')$; $Y = 0$, $W = 0$, $(2')$; $Z = 0$, $W = 0$, $(3')$; and $W = 0$, $X + Y + Z = 0$, (012).
152. The three mere lines are each facultative: $p' = b' = 3$; $t' = 1$.
153. *Hessian surface.* The equation is $(X + Y + Z)^2(X^2 + Y^2 + Z^2 - 2YZ - 2ZX - 2XY) = 0$, viz. the surface consists of the uniplane $X + Y + Z = 0$ twice, and of a quadric cone having its vertex at U_3 and touching each of the planes $X = 0$, $Y = 0$, $Z = 0$.

The complete intersection with the cubic surface is made up of the rays each twice and of a residual sextic, which is the spinode curve; $c' = 6$. The equations of the spinode curve are

$$W(X + Y + Z)^2 + XYZ = 0, \quad X^2 + Y^2 + Z^2 - 2YZ - 2ZX - 2XY = 0,$$

viz. the curve is a complete intersection, 2×3. Each of the mere lines is a single tangent; that is, $\beta' = 3$.
154. *Reciprocal Surface.* The equation is found by means of the binary cubic $4(T - xU)(T - yU)(T - zU) + wT^2U$, viz. writing for shortness $\beta = x + y + z$, $\gamma = yz + zx + xy$, $\delta = xyz$, then the cubic function is $(12,\ w - 4\beta,\ 4\gamma,\ -12\delta T, U)^3$, and the equation of the reciprocal surface is found to be

$$432\beta\delta^2 + 64\gamma^3 + 72(w - 4\beta)\gamma\delta - (w - 4\beta)^2\gamma^2 = 0;$$

expanding, this is

$$w^2(-\beta\delta) + 8w(-6\beta^2\delta + \beta\gamma^2 + 9\gamma\delta) + 16(4\beta^2\delta - \beta^2\gamma^2 - 18\beta\gamma\delta + 4\gamma^3 + 27\delta^2) = 0;$$

or substituting for β, γ, δ in the first and last lines, this is

$$w^2(-xyz) + w^2(12\beta\delta - \gamma^2) + 8w(-6\beta^2\delta + \beta\gamma^2 + 9\gamma\delta) \\ + 16(y - z)^2(z - x)^2(x - y)^2 = 0$$

(where β, γ, $\delta = x + y + z$, $yz + zx + xy$, xyz).

The section by the plane $w = 0$ (reciprocal of the unode) is made up of the lines $w = 0$, $y - z = 0$; $w = 0$, $z - x = 0$; $w = 0$, $x - y = 0$ (reciprocals of the rays) each twice.
155. The nodal curve is at once seen to consist of the lines $(y = 0,\ z = 0)$, $(z = 0,\ x = 0)$, $(x = 0,\ y = 0)$, reciprocals of the facultative lines; in fact, in regard to (y, z) conjointly γ is of the order 1, and δ is of the order 2; hence every term of the equation is of the order 2 in y, z; and the like as to the other two lines: $b' = 3$ as above.
156. For the cuspidal curve we have

$$\begin{vmatrix} 12 & w - 4\beta & w - 4\beta & 4\gamma \\ w - 4\beta & 4\gamma & -12\delta & \end{vmatrix} = 0,$$

or say $48\gamma^2 - (w - 4\beta)^2 = 0$, $36\delta + \gamma(w - 4\beta) = 0$, whence the cuspidal curve is a complete intersection 2×3; $c' = 6$.

Section XIII = 12–B$_3$–2C$_2$. Article Nos. 157 to 164. Equation $WXZ + Y^2(Y + X + Z) = 0$.

157. The diagram of the lines and planes is: Lines include 1 transversal, $2 \times 6 = 12$; Planes include biplanes ($2 \times 6 = 12$), a plane through the three axes ($1 \times 4 = 4$), planes each through an axis joining the binode with a onicnode ($2 \times 6 = 12$), planes through the biplanar rays ($1 \times 3 = 3$), and a plane touching along the axis which joins the two onicnodes ($1 \times 2 = 2$). Total: 45.
158. The planes are $X = 0$, [1]; $Z = 0$, [2]; $Y = 0$, [056]; $Y + Z = 0$, [5]; $Y + X = 0$, [6]; $Y - W = 0$, [34]; $X + Y + Z = 0$, [12]; $W = 0$, [0]; and $W = 0$, $X + Y + Z = 0$, (012). The lines are $Z = 0$, $Y = 0$, (5); $Z = 0$, $X = 0$, (6); $Y = 0$, $W = 0$, (0); $X = 0$, $Y + Z = 0$, (1); $Z = 0$, $Y + X = 0$, (2); $W = Y = -Z$, (3); $W = Y = -X$, (4).
159. The transversal is facultative; $p' = b' = 1$, $t' = 0$.
160. The Hessian surface is $WXZ(5Y + X + Z) + Y^2(Z - X)^2 = 0$. The complete intersection with the surface is made up of the line $Y = 0$, $X = 0$ (Cξ-axis) three times; the line $Y = 0$, $Z = 0$ (Cη-axis) three times; line $Y = 0$, $W = 0$ (Cζ-axis) twice, and of a residual quartic, which is the spinode curve; $c' = 4$.
161. Representing the two equations by $U = 0$, $H = 0$, we have $(3Y + X + Z)U - H = Y\{3Y^2 + 4YZ + 4XZ + 4X^2\}$, and further reductions lead to the spinode curve as a complete intersection 2×2; and since the first surface is a cone having its vertex on the second surface, the spinode curve is a nodal quadriquadric. The equations of the transversal are $W = 0$, $X + Y + Z = 0$, and substituting in the equations of the spinode curve we obtain from each equation $(X - Z)^2 = 0$, that is, the transversal is a single tangent of the spinode curve; $\beta' = 1$.
162. *Reciprocal Surface.* The equation is derived from that belonging to X = 12 – B$_3$ – C$_2$ by writing therein $a = b = 0$, $c = \frac{1}{2}$, $d = 1$. The equation becomes

$$\begin{aligned}
&y^2 \cdot 16(x - z)^2 + wy^2\{-y^2(x + z) \\
&\qquad\qquad + 2y(x^2 - 4xz + z^2) + (x + z)(2x - z)(-x + 2z)\} \\
&\qquad\qquad\quad + w\{y^4 + 8y^3(x + z) - 2y^2(3x^2 + 28xz + 4z^2) \\
&\qquad\qquad\qquad\qquad + 36xyz(x + z) - 27xz^2\} + y^4(y - x)(y - z) = 0.
\end{aligned}$$

The section by the plane $w = 0$ (reciprocal of B$_3$) is $w = 0$, $y = 0$ (the edge) three times; and $w = 0$, $y - x = 0$; $w = 0$, $y - z = 0$ (reciprocals of the CB-axes).
163. *Nodal curve.* This is the line $y = x = z$; wherefore $b' = 1$. To put the line in evidence, write for a moment $x = y + \alpha$, $z = y + \gamma$, then the equation is readily converted into a form which, each term being at least of the second order in α, γ ($= x - y$, $z - y$), exhibits the nodal line in question.
164. *Cuspidal curve.* Multiplying by 27, the equation may be written

$$(7y{-}3x{-}3z{-}8w,\ -y{+}6w,\ -4\cancel{w}y^2{+}16yw{-}12(x{+}z)w{+}16w^2, -20y^2{+}24y(x{+}z){-}27xz{-}8yw{+}16w^2) = 0,$$

where $4w(7y - 3x - 3z - 8w) + (-y + 6w)^2 = y^2 + 16yw - 12(x + z)w + 16w^2$; and we have thus in evidence the cuspidal curve,

$$\begin{aligned}
y^2 + 16yw - 12(x + z)w + 16w^2 &= 0, \\
-20y^2 + 24y(x + z) - 27xz - 8yw + 16w^2 &= 0,
\end{aligned}$$

which is a complete intersection, 2×2, or quadriquadric curve; $c' = 4$.

Section XIV = 12–B$_5$–C$_2$. Article Nos. 165 to 171. Equation $WXZ + Y^2Z + YX^2 = 0$.

165. The diagram of the lines and planes is: Planes include $Z = 0$ (torsal biplane, $1 \times 15 = 15$), $X = 0$ (ordinary biplane, $1 \times 20 = 20$), $Y = 0$ (plane through axis and two rays, $1 \times 10 = 10$). Total: 45.
166. The edge is a facultative line, as will appear from the discussion of the reciprocal surface: $p' = b' = 1$; $t' = 0$.
167. *Hessian surface.* The equation is $WXZ^2 + Y^2Z^2 - 3X^2YZ + X^4 = 0$. The complete intersection with the surface is made up of the line $X = 0$, $Y = 0$ (the axis) five times, the line $X = 0$, $Z = 0$ (the

edge) four times, and a skew cubic, the equations of which may be written

$$\begin{vmatrix} X, & Y, & W \\ 4Z, & X, & -5Y \end{vmatrix} = 0.$$

168. The spinode curve is made up of the edge $X = 0$, $Z = 0$ once, and of the cubic curve; and therefore $c' = 4$.

169. *Reciprocal Surface.* The equation is found to be

$$(y^2 + 4xw)^2(y + w - x - z) - 27z(y + 2w)^3 \\ + 9xz(y^2 + 4xw + 4zw)(y + 2w) - 27xz^2w = 0,$$

reducing to

$$y^2 \cdot 8(x - z)^2 + wy^2\{-y^2(x + z) + 2y(x^2 - 4xz + z^2) \\ + (x + z)(2x - z)(-x + 2z)\} + w\{y^4 + 8y^3(x + z) \\ - 2y^2(3x^2 + 28xz + 4z^2) + 36xyz(x + z) - 27xz^2\} \\ + y^4(y - x)(y - z) = 0.$$

The section by the plane $w = 0$ (reciprocal of B_5) is $w = 0$, $y = 0$ (reciprocal of edge) four times, and the lines $w = 0$, $y^2 + 4xz = 0$ (reciprocals of the rays).

170. *Nodal curve.* The equation gives $16w^2$ showing that the line $w = 0$, $y = 0$ (reciprocal of edge) is an oscnodal line counting as three lines; $b' = 3$. There is no cuspidal curve; $c' = 0$.

Section XV = 12–U$_5$. Article Nos. 172 to 177. Equation $X^2W - (XZ + Y^2)^2 = 0$.

172. The diagram of the lines and planes is: Plane is $X = 0$ $(1 \times 45 = 45)$, a uniplane.

173. There is no facultative line; $p' = b' = 0$, $t' = 0$.

174. The Hessian surface is $X^3Y^2 = 0$, viz. this is the uniplane $X = 0$, three times, and the plane $Y = 0$ through the ray. The complete intersection with the cubic surface is made up of $X = 0$, $Y = 0$ (the ray) ten times, and of a residual conic, which is the spinode curve; $c' = 2$. The equations of the spinode conic are $Y = 0$, $XW + Z^2 = 0$, viz. the plane of the conic passes through the ray. Since there is no facultative line, $\beta' = 0$.

175. *Reciprocal Surface.* The equation is at once found to be $27(z^2 + 4xw)^2y^2 - 64x^3w^3 = 0$. The section by the plane $w = 0$ (reciprocal of the Unode) is $w = 0$, $y = 0$ (reciprocal of ray) four times. There is no nodal curve; $b' = 0$. But there is a cuspidal conic $y = 0$, $z^2 + 4xw = 0$. The point $y = 0$, $z = 0$, $w = 0$ (reciprocal of the uniplane $X = 0$) is a point which must be considered as uniting the singularities $\mathrm{B}' = 1$, $\chi' = 2$.

Section XVI = 12–4C$_2$. Article Nos. 178 to 183. Equation $X^2W + XYZ + Y^3 + Z^3 = 0$.

178. There is no facultative line; $p' = b' = 0$, $t' = 0$.

179. *Hessian surface.* The equation is $Z(X^2W + XYZ + Y^3 - Z^3) = 0$, viz. this breaks up into the plane $Z = 0$ (the common biplane) and a cubic surface. The complete intersection with the surface is made up of the axes each four times, and of a residual sextic, which is the spinode curve; $c' = 6$.

180. *Reciprocal Surface.* The equation is found to be

$$(x^2 - 4yz)^2\{x^3 + 4w(y^2 + z^2)\} + 54w^2(x^2 - 4yz)(x^2 + 2yz) \\ + 27w^3(27w - 8x^3 - 24xyz) = 0.$$

The section by the plane $w = 0$ (reciprocal of C_2) is $w = 0$, $x^2 - 4yz = 0$ (reciprocal of nodal cone) four times. There is no nodal curve; $b' = 0$.

181. *Cuspidal curve.* The cuspidal curve is a complete intersection 2×3; $c' = 6$.

Section XVII = 12–2B$_3$–C$_2$. Article Nos. 184 to 189.

184–189. In this case there is no facultative line; $p' = b' = 0$, $t' = 0$. The Hessian surface breaks up into the common biplane and a cubic surface. The spinode curve is a conic; $c' = 2$. The reciprocal surface has no nodal curve but has a cuspidal conic; $c' = 2$.

Section XVIII = 12–B$_4$–2C$_2$. Article Nos. 190 to 193.

190–193. The edge counting twice is facultative; $p' = b' = 3$, $t' = 1$. The reciprocal of the edge is a tacnodal line. The spinode curve and cuspidal curve are both nonexistent; $c' = 0$.

Section XIX = 12–B$_5$–C$_3$. Article Nos. 194 to 197.

194–197. The axis counting three times is facultative; $p' = b' = 3$, $t' = 0$. The reciprocal of the axis is oscnodal. There is no spinode curve; $c' = 0$.

Section XX = 12–U$_5$. Article Nos. 198 to 201.

198–201. There is no facultative line; $p' = b' = 0$, $t' = 0$. The Hessian surface is the uniplane three times, and a plane through the ray. The spinode curve is a conic; $c' = 2$. The reciprocal surface has no nodal curve, but has a cuspidal conic.

Section XXI = 12–3B$_3$. Article Nos. 202 to 204.

202–204. There is no facultative line; $p' = b' = 0$, $t' = 0$. The Hessian surface consists of the common biplane and the other biplanes each once. The complete intersection consists of the axes each four times; there is no spinode curve, $c' = 0$; whence also $\beta' = 0$.

Article No. 202. Synopsis for the foregoing sections.

202. I annex the following synopsis, for the several cases, of the facultative lines (or node-couple curve) and of the spinode curve of the cubic surface; also of the nodal curve and the cuspidal curve of the reciprocal surface.

It is to be observed that in designating a curve, for instance, as $18 = 4 \times 5 - 2$, this means that it is a curve of the order 18, the partial intersection of a quartic surface and a quintic surface, but without any explanation of the nature of the common curve 2 which causes the reduction, viz. without explaining whether this is a conic or a pair of lines, and so in other cases; this may be seen by reference to the proper section of the Memoir.

	Facultative lines.	*Nodal curve.*	*Spinode curve.*	*Cuspid*
I = 12	27	27	$12 = 3 \times 4$	$24 = 6$
II = 12–C_2	15	16	$12 = 3 \times 4$	$18 = 4$
III = 12–B_3	9	9	$12 = 3 \times 4$	$16 = 4$
IV = 12–$2C_2$	7	7	$10 = 3 \times 4 - 2$	$12 = 4$
V = 12–B_4	7 = 5 + edge twice	7 = 5 + rec. of edge twice	$10 = 3 \times 4 - 2$	$12 = 4$
VI = 12–B_3–C_2	3	3	$9 = 3 \times 4 - 3$	$10 = 4$
VII = 12–B_5	3 = 2 + edge	3 = 2 + rec. of edge	9 = edge + unicursal 8-thic	10 = re
VIII = 12–$3C_2$	3	8	$6 = 2 \times 3$	$6 = 2 \times$
IX = 12–$2B_3$	none	none	8 = 4 conics	8 = 4 c
X = 12–B_3–C_2	3 = 1 + edge twice	3 = 1 + rec. of edge twice	$6 = 2 \times 3$	$6 = 2 \times$
XI = 12–B_5	3 = edge 3 times	3 = rec. of edge 3 times	6 = 3 conics	6 = 3 c
XII = 12–U_3	3	3	$6 = 2 \times 3$	$6 = 2 \times$
XIII = 12–B_3–$2C_2$	1	1	$4 = 2 \times 2$, nodal quadriquadric	$4 = 2 \times$
XIV = 12–B_5–C_2	1 = edge	1 = rec. of edge	4 = 3 + edge	4 = 3 +
XV = 12–U_5	1	1	$4 = 2 \times 2$, nodal quadriquadric	$4 = 2 \times$
XVI = 12–$4C_2$	3	3	none	none
XVII = 12–$2B_3$–C_2	none	none	2 = conic	2 = con
XVIII = 12–B_4–$2C_2$	3 = 1 + edge twice	1 + rec. of edge twice	none	none
XIX = 12–B_5–C_3	3 = axis 3 times	3 = rec. of axis 3 times	none	none
XX = 12–U_5	none	none	2 = conic	2 = con
XXI = 12–$3B_3$	none	none	none	none

I pass now to the two cases of cubic scrolls.

Section XXII = S(1,1). Article No. 203. Equation $X^2W + Y^2Z = 0$.

203. As this is a scroll there is here no question of the 27 lines and 45 planes; there is a nodal line $X = 0$, $Y = 0$, $(b = 1)$ and a single directrix line, $Z = 0$, $W = 0$. The Hessian surface is $X^2Y^2 = 0$; the complete intersection with the cubic surface is made up of $X = 0$, $Y = 0$ (the nodal line) eight times, and of the lines $X = 0$, $Z = 0$, and $Y = 0$, $W = 0$, each twice. The reciprocal surface is $x^2z - y^2w = 0$; viz. this is a like scroll, XXII = S$(1,1)$; $c' = 0$, $b' = 1$.

Section XXIII = S(1$\bar{1}$1). Article No. 204. Equation $X(XW + YZ) + Y^2 = 0$.

204. This is also a scroll; there is a nodal line $X = 0$, $Y = 0$, and a single directrix line united therewith. The Hessian surface is $X^4 = 0$; the complete intersection with the cubic surface is $X = 0$, $Y = 0$ (the nodal line) twelve times. The reciprocal surface is $w(xw + yz) - z^2 = 0$; viz. this is a like scroll, XXIII = S$(1\bar{1}1)$; $c' = 0$, $b' = 1$.

Annex containing Additional Researches in regard to the case XX = 12–U_5; equation $WX^2 + XZ^2 + Y^3 = 0$.

205. Let the surface be touched by the line (a, b, c, f, g, h), that is, the line the equations whereof are

$$\begin{vmatrix} 0, & h, & -g, & a \\ -h, & 0, & f, & b \\ g, & -f, & 0, & c \\ -a, & -b, & -c, & 0 \end{vmatrix} \frown (X, Y, Z, W) = 0.$$

Writing the equation in the form $cW \cdot cX^2 + X(cZ)^2 + c^2Y^3 = 0$, and substituting for cW, cZ their values in terms of X, Y, we have $(-gX + fY)cZ^2 + X(aX + bY)^2 + c^2Y^3 = 0$, that is

$$(3(a^2 - cg),\ 2ab + cf,\ b^2,\ 3c^2 \frown X, Y)^3 = 0,$$

viz. the condition of contact is obtained by equating to zero the discriminant of the cubic function. We have thus

$$27c^4(a^2-cg)^2+4c^3f^2(a^2-cg)^2+c^2f^2(2ab+cf)^2-b^4(2ab+cf)^2 \\ -18b^2c^2(a^2-cg)(2ab+cf)=0,$$

viz. this is

$$+27a^4c^4-4a^3b^2f+30a^2b^2c^2f-54a^2c^4g+36ab^3cfg+24abc^3f^2g \\ +4b^6f^2-18b^4c^2fg+18b^2c^4f^2g+27c^8g^2+4c^4f^3g^2=0,$$

which is the condition in order that the line (a,b,c,f,g,h) may touch the surface $X^2W+XZ^2+Y^3=0$; and if we unite thereto the conditions that the line shall pass through a given point $(\alpha,\beta,\gamma,\delta)$, we have in effect the equation of the circumscribed cone, vertex $(\alpha,\beta,\gamma,\delta)$.

Writing (f,g,h,a,b,c) in place of (a,b,c,f,g,h), we obtain

$$27f^4-4f^3g^2a+30f^2g^2ha-54f^2h^4b+36fg^3h^2b+24fgh^3a^2 \\ +4g^6a^2-18g^4h^2ab+18g^2h^4a^2b+27h^8b^2+4h^4a^3b^2=0$$

as the condition that the line (a,b,c,f,g,h) shall touch the reciprocal surface $27(4x^2w+z^2)^2+64y^3w=0$; and if we consider a, b, c, f, g, h as standing for

$$\beta z-\gamma y,\ \gamma x-\alpha z,\ \alpha y-\beta x,\ \delta x-\alpha w,\ \delta y-\beta w,\ \delta z-\gamma w,$$

values which satisfy the relation

$$\begin{vmatrix} 0, & h, & -g, & a \\ -h, & 0, & f, & b \\ g, & -f, & 0, & c \\ -a, & -b, & -c, & 0 \end{vmatrix}(\alpha,\beta,\gamma,\delta)=0,$$

then the equation in (a,b,c,f,g,h) is that of the circumscribed cone, vertex $(\alpha,\beta,\gamma,\delta)$; the order being (as it should be) $a'=6$.

The cuspidal conic is $y=0$, $4x^2w+z^2=0$, and we at once obtain $a^2-4cg=0$ as the condition that the line (a,b,c,f,g,h) shall pass through the cuspidal cone. Hence attributing to (a,b,c,f,g,h) the foregoing values, we have $a^2-4cg=0$ for the equation of the cone, vertex $(\alpha,\beta,\gamma,\delta)$, which passes through the cuspidal conic; this is of course a quadric cone, $c'=2$.

I proceed to determine the intersections of the two cones. Representing by $\Theta=0$ the foregoing equation of the circumscribed cone, and putting for shortness

$$X=27h^4(f^2-bh)-2g^2(2fg+ah),$$

I find that we have identically

$$\Theta-(f^2-bh)X+(g^2-4ah-8fgh)(a^2-4cg)-(32fg^2h+64agh^2)(af+bg+ch)=0:$$

whence in virtue of the relation $af+bg+ch=0$, we see that the equations $\Theta=0$, $a^2-4cg=0$, are equivalent to $(f^2-bh)X=0$, $a^2-4cg=0$, or the twelve lines of intersection break up into the two systems

$$\begin{aligned} f^2-bh&=0, && a^2-4cg=0, \quad \text{and} \\ X=27h^4(f^2-bh)-2g^2(2fg+ah)&=0, && a^2-4cg=0. \end{aligned}$$

To determine the lines in question, observe that we have the relation between (a,b,c,f,g,h) and $(\alpha,\beta,\gamma,\delta)$, and we can by the first three of these express a, b, c linearly in terms of f, g, h; the equations

$f^2-bh=0$, $a^2-4cg=0$, $27h^4(f^2-bh)-2g^2(2fg+ah)=0$ become thus homogeneous equations in (f,g,h); the equations may in fact be written

$$\begin{aligned} 8\delta^2(a^2-4cg) &= (\gamma^2+4\alpha\delta)g^2+\beta^2h^2-2\beta\gamma gh-4\beta\delta hf=0,\\ 8(f^2-bh) &= 8f^2-\alpha h^2+\gamma hf=0,\\ 8X &= 27h^4(8f^2-\alpha h^2+\gamma hf)+2g^2(\beta h^2-\gamma gh-2\delta fg)=0, \end{aligned}$$

viz. interpreting (f,g,h) as coordinates in piano, the first equation represents a conic, the second a pair of lines, and the third a quartic.

We have identically

$$\{2\beta\delta f-(\gamma^2+4\alpha\delta)g+\beta\gamma h\}^2-4\beta^2\delta(8f^2-\alpha h^2+\gamma hf)=(\gamma^2+4\alpha\delta)\{(\gamma^2+4\alpha\delta)g^2+\beta^2h^2-2\beta\gamma gh-4\beta\delta hf\};$$

and it thus appears that the two conics touch at the points given by the equations $8f^2-\alpha h^2+\gamma hf=0$, $2\beta\delta f-(\gamma^2+4\alpha\delta)g+\beta\gamma h=0$: we have moreover

$$-(\gamma^2+4\alpha\delta)(\beta h^2-\gamma gh-2\delta fg)=4\beta\delta(8f^2-\alpha h^2+\gamma hf)+(-2\delta f-\gamma h)\{2\beta\delta f-(\gamma^2+4\alpha\delta)g+\beta\gamma h\},$$

hence at the last-mentioned two points $-\beta h^2+\gamma gh+2\delta fg$ is $=0$; and the quartic $X=0$ thus passes through these two points.

The conic $(a^2-cg)=0$ and the quartic $X=0$ intersect besides (as is evident) in the point $g=0$, $h=0$ reckoning as two points, since it is a node of the quartic; and they must consequently intersect in four more points.

Recapitulating, the conic $a^2-4cg=0$ meets the sextic $(f^2-bh)X=0$ in the two points

$$8f^2-\alpha h^2+\gamma hf=0, \quad 2\beta\delta f-(\gamma^2+4\alpha\delta)g+\beta\gamma h=0$$

each three times, in the point $g=0$, $h=0$ twice, and in the four points

$$\begin{aligned} &160\delta g^4-27(\gamma^2+4\alpha\delta)g^2h^2+27\beta^2h^4=0,\\ &(\gamma^2+4\alpha\delta)g^2+\beta^2h^2-2\beta\gamma gh-4\beta\delta hf=0 \end{aligned}$$

each once.

Or reverting to the proper significations of (a,b,c,f,g,h), instead of points we have 2 lines each three times, a line twice, and 4 lines each once; the line $g=0$, $h=0$, that is, $g=0$, $h=0$, $a=0$, being, it will be observed, the line $\frac{x}{\alpha}=\frac{y}{\beta}=\frac{z}{\gamma}$ drawn from $(\alpha,\beta,\gamma,\delta)$ to the point $y=0$, $z=0$, $w=0$, which is the reciprocal of the uniplane $X=0$: the twelve lines are the $a'c'$ lines of intersection of the circumscribed cone a' with the cuspidal cone c', viz. $a'c'=[a'c']+3(\alpha'+\chi')$ where $[a'c']=4$ referring to the last-mentioned four lines; $\alpha'=2$ to the two lines; and $\chi'=2$ to the line $g=0$, $h=0$, $a=0$, which it thus appears must in the present case be reckoned twice.

[413] A Memoir on Abstract Geometry.

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I SUBMIT to the Society the present exposition of some of the elementary principles of an Abstract n-dimensional Geometry. The science presents itself in two ways,—as a legitimate extension of the ordinary two- and three-dimensional geometries; and as a need in these geometries and in analysis generally. In fact whenever we are concerned with quantities connected together in any manner, and which are, or are considered as variable or determinable, then the nature of the relation between the quantities is frequently rendered more intelligible by regarding them (if only two or three in number) as the coordinates of a point in a plane or in space: for more than three quantities there is, from the greater complexity of the case, the greater need of such a representation; but this can only be obtained by means of the notion of a space of the proper dimensionality; and to use such representation, we require the geometry of such space.

An important instance in plane geometry has actually presented itself in the question of the determination of the number of the curves which satisfy given conditions: the conditions imply relations between the coefficients in the equation of the curve; and for the better understanding of these relations it was expedient to consider the coefficients as the coordinates of a point in a space of the proper dimensionality.

A fundamental notion in the general theory presents itself, slightly in plane geometry, but already very prominently in solid geometry; viz. we have here the difficulty as to the form of the equations of a curve in space, or (to speak more accurately) as to the expression by means of equations of the twofold relation between the coordinates of a point of such curve. The notion in question is that of a k-fold relation,—as distinguished from any system of equations (or onefold relations) serving for the expression of it, and as giving rise to the problem how to express such relation by means of a system of equations (or onefold relations). Applying to the case of solid geometry my conclusion in the general theory, it may be mentioned that I regard the twofold relation of a curve in space as being completely and precisely expressed by means of a system of equations ($P = 0,\ Q = 0,\ \ldots\ T = 0$), when no one of the functions P, Q, $\ldots$ T is a linear function, with constant or variable integral coefficients, of the others of them, and when every surface whatever which passes through the curve has its equation expressible in the form $U = AP + BQ \ldots + \lambda T$, with constant or variable integral coefficients A, B, $\ldots$ λ.

It is hardly necessary to remark that all the functions and coefficients are taken to be rational functions of the coordinates, and that the word integral has reference to the coordinates.

Article Nos. 1 to 36. General Explanations; Relation, Locus, &c.

1. Any m quantities may be represented by means of $m+1$ quantities as the ratios of m of these to the remaining $(m+1)$th quantity, and thus in place of the absolute values of the m quantities we may consider the ratios of $m+1$ quantities.

2. It is to be noticed that we are throughout concerned with the ratios of the $m+1$ quantities, not with the absolute values; this being understood, any mention of the ratios is in general unnecessary; thus I shall speak of a relation between the $m+1$ quantities, meaning thereby a relation as regards the ratios of the quantities; and so in other cases. It may also be noticed that in many instances a limiting or extreme case is sometimes included, sometimes not included, under a general expression; the general expression is intended to include whatever, having regard to the subject-matter and context, can be included under it.

3. *Postulate.* We may conceive between the $m+1$ quantities a relation.

4. A relation is either regular, that is, it has a definite manifoldness, or, say, it is a k-fold relation; or else it is irregular, that is, composed of relations not all of the same manifoldness.

5. The ratios are determined (not in general uniquely) by means of a m-fold relation; and a relation cannot really be more than m-fold. But the notion of a more than m-fold relation has nevertheless to be considered. A relation may be, either in mere appearance or else according to a provisional conception thereof, more than m-fold, and be really m-fold or less than m-fold. Thus a relation expressed by

$m+1$ or more equations is in general and primâ facie more than m-fold; but if the equations are not independent, and equivalent to m or fewer equations, then the relation will be m-fold or less than m-fold. Or the given relation may depend on parameters, and so long as these are arbitrary be really more than m-fold; but the parameters may have to be, and be accordingly, so determined that the relation shall be m-fold or less than m-fold. A more than m-fold relation is said to be superdeterminate.

6. A system of any number of onefold relations, whether independent or dependent, and if more than m of them, whether compatible or incompatible, is termed a ‘Plexus.

7. The number of onefold relations of a plexus may be 0, 1, 2, … or any number whatever; if it is $= m$, the plexus is said to be complete.

8. Any onefold relation whatever may be termed an equation; a system of onefold relations is a system of equations.

9. Any k onefold relations form a k-fold relation; or say they are a k-fold relation.

10. A k-fold relation between $m+1$ quantities determines (not in general uniquely) the ratios of the quantities; a k-fold relation between the coordinates determines a locus of the dimensionality $m-k$, or say it is a $(m-k)$-dimensional locus; in particular a onefold relation determines a $(m-1)$-dimensional locus, which may be termed an ‘Omal.

11. A k-fold relation is implied in a l-fold relation when every set of values of the coordinates which satisfies the l-fold relation satisfies also the k-fold relation. In particular a onefold relation is implied in a k-fold relation when every set of values which satisfies the k-fold relation satisfies also the onefold relation.

12. The foregoing definition of implication is applicable to the case of a k-fold relation and a l-fold relation where $k > l$, or even where $k = l$; but observe that a k-fold relation is not in general implied in a different k-fold relation.

13. If a k-fold relation is implied in each of two or more l-fold relations (k being $> l$ or $= l$), then the k-fold relation is said to be implied in the aggregate of the l-fold relations. And more generally, if a k-fold relation is implied in each of two or more relations not all of them of the same manifoldness, then it is implied in the aggregate of these relations.

14. Any two or more relations may be composed together, and they are then factors of a single composite relation; viz. the composite relation is a relation satisfied if, and not satisfied unless, some one of the component relations be satisfied.

15. The foregoing notion of composition is, it will be noticed, altogether different from that which would at first suggest itself. The definition is defective as not explaining the composition of a relation any number of times with itself, or elevation thereof to power; which however must be admitted as part of the notion of composition.

16. A k-fold relation which is not satisfied by any other k-fold relation, and which is not a power, is a prime relation. A relation which is not prime is composite, viz. it is a relation composed of prime factors each taken once or any other number of times; in particular, it may be the power of a single prime factor. Any prime factor is single or multiple according as it occurs once or a greater number of times.

17. A relation which is either prime, or else composed of prime factors each of the same manifoldness, is a regular relation; a k-fold relation is ex vi termini regular. An irregular relation is a composite relation the prime factors whereof are not all of the same manifoldness.

18. A prime k-fold relation cannot be implied in any prime k-fold relation different from itself. But a prime k-fold relation may be implied in a prime more-than-k-fold relation,—or in a composite relation, regular or irregular, each factor whereof is more than k-fold; and so also a composite relation, regular or irregular, each factor whereof is at most k-fold, may be implied in a composite relation, regular or irregular, each factor whereof is more than k-fold. In a somewhat different sense, each factor of a composite relation implies the composite relation.

19. A composite relation is satisfied if any particular one of the component relations is satisfied; but in order to exclude this case we may speak of a composite relation as being satisfied distributively; viz. this will be the case if, in order to the satisfaction of the composite relation, it is necessary to consider all the factors thereof, or, what is the same thing, when the reduced relation obtained by the omission of any one factor whatever is not always satisfied. And when the composite relation is satisfied distributively, the several factors thereof are satisfied alternatively; viz. there is no one which is throughout unsatisfied.

20. A composite onefold relation is never distributively implied in a prime k-fold relation—that is, a prime k-fold relation implies only a prime onefold relation; or at least only implies a composite onefold relation improperly, in the sense that it implies a certain prime factor of such composite onefold relation. Conversely, every k-fold relation which implies distributively a composite onefold relation is composite.

21. Any two or more relations may be aggregated together, and they are then constituents of a single aggregate relation; viz. the aggregate relation is a relation satisfied if, and only if, all the constituent relations are satisfied. The aggregation of relations corresponds to the intersection of loci.

22. There is no meaning in aggregating a relation with itself; such aggregation only occurs accidentally when two relations aggregated together become one and the same relation; and the aggregate of a relation with itself is nothing else than the original relation.

23. A onefold relation is not an aggregate, but is its own sole constituent; a more than onefold relation may always be considered as an aggregate of two or more constituent relations. The constituent relations determine, they in fact constitute, the aggregate relation; but the aggregate relation does not in any wise determine the constituent relations. Any relation implied in a given relation may be considered as a constituent of such given relation.

24. The aggregate of a k-fold and a l-fold relation is in general and at most a $(k+l)$-fold relation; when it is a $(k+l)$-fold relation, the constituent relations are independent, but otherwise, viz. if the aggregate relation is, or has for factor, a less than $(k+l)$-fold equation, the constituent relations are dependent or interconnected.

25. Passing from relations to loci, we may say that the composition of relations corresponds to the congregation of loci, and the aggregation of relations to the intersection of loci.

26. For, first, the locus (if any) corresponding to a given composite relation is the congregate of the loci corresponding to the several prime factors of the given relation, the locus corresponding to a single factor being taken once, and the locus corresponding to a multiple factor being taken a number of times equal to the multiplicity of the factor.

27. And, secondly, the locus (if any) corresponding to a given aggregate relation is the locus common to and contained in each of the loci corresponding to the several constituent relations respectively; or, what is the same thing, it is the intersection of these several loci.

28. It may be remarked that a k-fold locus and a l-fold locus where $k+l>m$ (or where the aggregate relation is more than m-fold) have not in general any common locus.

29. Any onefold relation implied in a given k-fold relation is said to be in involution with the k-fold relation, and so in a system of onefold relations, if any relation be implied in the other relations, or, what is the same thing, in the relation aggregated of the other relations, then the system is said to be in involution; a system not in involution is said to be asyzygetic.

30. Consider a given k-fold relation, and, in conjunction therewith, a system of any number of onefold relations each implied in the given k-fold relation. We may omit from the system any relation implied in the remaining relations, and so successively until we arrive at an asyzygetic system. Consider now any other onefold relation implied in the given k-fold relation; this is either implied in the system of onefold relations, and it is then to be rejected, or if it is not implied in the system, it is to be added on to and made part of the system. It may happen that, in the system thus obtained, some one relation of the original system is implied in the remaining relations of the new system; but if this is so the implied relation is to be rejected; the new system will in this case contain only as many relations as the original system, and in any case the new system will be asyzygetic. Treating in the same manner every other onefold relation implied in the given k-fold relation, we ultimately arrive at an asyzygetic system of onefold relations, such that every onefold relation implied in the given k-fold relation is implied in the asyzygetic system. The number of onefold relations will be at least equal to k (for if this were not so we should have the given k-fold relation as an aggregate of less than k onefold relations); but it may be greater than k, and it does not appear that there is any [assignable] superior limit to the number of onefold relations of the asyzygetic system.

31. The system of onefold relations is a precise equivalent of the given k-fold relation. Every set of values of the coordinates which satisfies the given k-fold relation satisfies the system of onefold relations; and reciprocally every set of values which satisfies the system of onefold relations satisfies the given k-fold relation. But if we omit any one or more of the onefold relations, then the reduced system so obtained is not a precise equivalent of the given k-fold relation; viz. there exist sets of values satisfying the reduced system, but not satisfying the given k-fold relation.

32. In fact consider a k-fold relation the aggregate of less than all of the onefold relations of the asyzygetic system, and in connexion therewith an omitted onefold relation; this omitted relation is not implied in the aggregate, and it constitutes with the aggregate not a $(k+1)$-fold, but only a k-fold relation. This happens as follows, viz. the omitted relation is a factor of a composite onefold relation distributively implied in the aggregate; hence the aggregate is composite, and it implies distributively a composite onefold relation composed of the omitted relation and of an associated onefold relation; that is, the aggregate will be satisfied by values which satisfy the omitted relation, and also by values which (not satisfying the omitted relation) satisfy the associated relation just referred to.

33. Selecting at pleasure any k of the onefold relations of the asyzygetic system, being such that the aggregate of the k relations is a k-fold relation, we have a composite k-fold relation wherein each of the remaining onefold relations is alternatively implied; viz. each remaining onefold relation is a factor of a composite onefold relation implied distributively in the composite k-fold relation. Hence considering the $k+1$ onefold relations, viz. any $k+1$ relations of the asyzygetic system, each one of these is implied alternatively in the aggregate of the other k relations.

34. The asyzygetic system of onefold relations thus possesses the property that no reduced system thereof (that is, system obtained by the omission of one or more relations from the asyzygetic system) is a precise equivalent of the given k-fold relation; and moreover any relation alternatively implied in the asyzygetic system is one of the relations of the system. It may be added that, besides the relations of the system, there is not any onefold relation alternatively implied in the asyzygetic system.

35. The foregoing theory has been stated without any limitation as to the value of k, and it has I think a meaning even when k is $> m$; but the ordinary case is $k \pitchfork m$. Considering the theory as applying to this case, I remark that the last proposition, viz. that no reduced system is a precise equivalent of the given k-fold relation, is generally true only on the assumption of the existence or quasi-existence of sets of values satisfying a more than m-fold relation. For let k be $> m$, and, on the contrary, assume, as we usually do, that it is not in general possible to satisfy a more than m-fold relation between the coordinates; the number of relations in the system may be $> m+1$; and if this is so, then selecting any $m+1$ relations of the system, it may very well happen that the given k-fold relation is not satisfied by any sets of values other than those which satisfy the $m+1$ relations,—that is, that the $m+1$ relations are a precise equivalent of the given k-fold relation. But even in this case the consideration of the entire system of the onefold relations is not the less advantageous; and I say in general that the given k-fold relation has for its precise and complete equivalent the asyzygetic system of onefold relations.

36. (In illustration of the foregoing Nos. 29 to 35, I remark that, for the functions or equations $P=0$, $Q=0$, $R=0$, &c., if we have identically $AP+BQ+CR+\ldots=0$, where the factors A, B, C, $\ldots$ are integral functions of the coordinates, and where some one of these factors, say, A, is a constant (or if we please $=1$), then the system of functions or equations is in involution; or, to speak more accurately, the function or equation $P=0$ is in involution with the remaining functions or equations $Q=0$, $R=0$, $\ldots$ But when the factors A, B, C, $\ldots$ are no one of them constant, then we have a convolution. If $P=0$ is in involution with the remaining equations $Q=0$, $R=0$, $\ldots$, then $P=0$ is implied in these equations, and the relations $(Q=0,\ P=0,\ \ldots)$ and $(P=0,\ Q=0,\ R=0,\ \ldots)$ are equivalent to each other. But in the case of a convolution where $AP+BQ+CR+\ldots=0$, then the relation the equations $Q=0$, $R=0$, $\ldots$ imply $AP=0$, that is, $A=0$ or else $P=0$; or, what is the same thing, the relation $(Q=0,\ R=0,\ \ldots)$ is a relation composed of the two relations $(A=0,\ Q=0,\ R=0,\ \ldots)$ and $(P=0,\ Q=0,\ R=0,\ \ldots)$. In the k-fold relation expressed by the more than k equations $(P=0,\ Q=0,\ R=0,\ \ldots)$, selecting any k of these equations which are not in convolution, and uniting thereto any one of the remaining equations, we have a convolution of $k+1$ equations; and when a k-fold relation is precisely expressed by means of a system of k or more equations $(P=0,\ Q=0,\ \ldots)$, then every equation $U=0$ implied in the given relation, or, what is the same thing, the aggregate relation $(P=0,\ Q=0,\ \ldots,\ U=0)$, is a system in involution.)

Article Nos. 37 to 42. Omal Relation; Order.

37. A k-fold relation may be linear or omal. If $k = m$, the corresponding locus is a point; if $k < m$ the locus is a k-fold, or $(m - k)$-dimensional omaloid; the expression omaloid used absolutely denotes the onefold or $(m - 1)$-dimensional omaloid; the point may be considered as a m-fold omaloid.

38. A m-fold relation which is not linear or omal is of necessity composite, composed of a certain number M of m-fold linear or omal relations; viz. the m-fold locus corresponding to the m-fold relation is a point-system of M points, each of which may be considered as given by a separate m-fold linear or omal relation; each which relation is a factor of the original m-fold relation. The given m-fold relation, and the point-system corresponding thereto, are respectively said to be of the order M.

39. The order of a point-system of M points is thus $= M$, but it is of course to be borne in mind that the points may be single or multiple points; and that if the system consists of a point taken α times, another point taken β times, &c., then the number of points and therefore the order M of the system is considered to be $= \alpha + \beta + \ldots$.

40. If to a given k-fold relation $(k < m)$ we unite an absolutely arbitrary $(m - k)$-fold linear relation, so as to obtain for the aggregate a m-fold relation, then the order M of this m-fold relation (or, what is the same thing, the number M of points in the corresponding point-system) is said to be the order of the given k-fold relation. The notion of order does not apply to a more than m-fold relation.

41. The foregoing definition of order may be more compendiously expressed as follows: viz. Given between the $m + 1$ coordinates a relation which is at most m-fold; then if it is not m-fold, join to it an arbitrary linear relation so as to render it m-fold; we have a m-fold relation giving a point-system; and the order of the given relation is equal to the number of points of the point-system.

42. The relation aggregated of two or more given relations, when the notion of order applies to the aggregate relation, that is, when it is not more than m-fold, is of an order equal to the product of the orders of the constituent relations; or, say, the orders of the given relations being μ, μ', $\ldots$, the order of the aggregate relation is $= \mu\mu' \ldots$.

Article Nos. 43 and 44. Parametric Relations.

43. We have considered so far relations which involve only the coordinates $(x, y, \ldots)$; the coefficients are purely numerical, or, if literal, they are absolute constants, which either do or do not satisfy certain conditions; if they do not, the relation assumed in the first instance to be k-fold is really k-fold, or, as we may express it, the relation is really as well as formally k-fold; if they do satisfy certain relations in virtue whereof the formally k-fold relation is really less than k-fold, say, it is $(k - l)$-fold, then the relation is in fact to be considered ab initio as a $(k - l)$-fold relation: there is no question of a relation being in general k-fold and becoming less than k-fold, or suffering any other modification in its form; and the notion of a more than m-fold relation is in the preceding theory meaningless.

44. But a relation between the coordinates $(x, y, \ldots)$ may involve parameters, and so long as these remain arbitrary it may be really as well as formally k-fold; but when the parameters satisfy certain conditions, it may become $(k - l)$-fold, or may suffer some other modification in its form. And we have to consider the theory of a relation between the coordinates $(x, y, \ldots)$, involving besides parameters which may satisfy certain conditions, or, say simply, a relation involving variable parameters. If the number of the parameters be m', then these parameters may be regarded as the ratios of m' quantities to a remaining $m' + 1$th quantity, and the relation may be considered as involving homogeneously the $m' + 1$ parameters $(x', y', \ldots)$. And these may, if we please, be regarded as coordinates of a point in their own m'-dimensional space, or we have to consider relations between the $m + 1$ coordinates $(x, y, \ldots)$ and the $m' + 1$ (parameters or) coordinates $(x', y', \ldots)$.

Article Nos. 45 to 55. Quantics, Notation, &c.

45. A homogeneous function of the coordinates $(x, y, \ldots)$ is represented by a notation such as $(\divideontimes x, y, \ldots)^r$ (where $(*)$ indicates the coefficients and (r) the degree), and it is said to be a quantic; and in reference to the quantic the quantities or coordinates $(x, y, \ldots)$ are also termed facients. More generally a quantic involving two or more sets of coordinates, or facients, is represented by the similar notation $(\divideontimes x, y, \ldots)^r (x', y', \ldots)^{r'} \ldots$.

46. The quantic is unipartite, bipartite, tripartite, &c., according as the number of sets is one, two, three, &c.; and with respect to any set of coordinates, it is binary, ternary, quaternary, … $(m+1)$ary, according as the number of the coordinates is two, three, four, or $m+1$; and it is linear, quadric, cubic, quartic, …, according as the degree in regard to the coordinates in question is 1, 2, 3, 4, ….

47. A quantic involving two or more sets of coordinates, and linear in regard to each of them, is said to be multipartite; or, in particular, when there are only two sets, it is said to be lineo-linear; we may even extend the epithet lineo-linear to the case of any number of sets.

48. Instead of the general notation we may write $(a, \ldots\frown x, y, \ldots)^r (x', y', \ldots)^{r'} \ldots$, where the coefficients are now indicated by $(a, \ldots)$, and the degrees are r, r', ….

49. In the cases where the particular values of the coefficients have to be attended to, we write down the entire series of coefficients, or at least refer thereto by the notation $(a, \ldots)$; and it is to be understood that the coefficients expressed or referred to are numerical or else they are independent arbitrary quantities.

50. In the case of a binary quantic, the series of coefficients is a, b, … h, k, l, or as we may write them $(a, b, \ldots, l)$, and the number of the coefficients is $= r+1$, if r is the degree. In the case of a ternary quantic, the series of coefficients may be written $(a, \ldots \pitchfork\!\frown x, y, z)^r$.

51. For shortness, the quantic may be represented by a single letter U; and when this is so, the equation $U = 0$ is used to denote the onefold relation between the coordinates expressed by the quantic being $= 0$.

52. A onefold relation between the coordinates is expressible by means of an equation of the form $(\ast\!\frown x, y, \ldots)^r = 0$.

53. The expression "an equation" used without explanation may be taken to mean an equation of the form in question, viz. the equation obtained by putting a quantic equal to zero; the quantic is said to be the nilfactum of the equation.

54. It is frequently convenient to denote the quantic or nilfactum by a single letter, and to use a locution such as "the equation $U = (\ast\!\frown x, y, \ldots)^r = 0$," which really means that the single letter U stands for the quantic $(\ast\!\frown x, y, \ldots)^r$, so that we are afterwards at liberty to write $U = 0$ as an abbreviated expression for $(\ast\!\frown x, y, \ldots)^r = 0$.

55. A k-fold relation between the coordinates is (as has been shown) equivalent to a system of k or more onefold relations; each of these is expressible by an equation $U = 0$, and the k-fold relation is thus expressible by a system of k or more such equations. Representing by $((U))$ the system of functions which are the nilfacta of these equations respectively, the k-fold relations may be represented thus, $((U)) = 0$; or more completely, the relation being k-fold, and the number of equations being $= s$, by the notation $\{((U)_s)\}^{(k\text{-fold})} = 0$.

Article Nos. 56 to 62. Resultant, Discriminant, &c.

56. In the case $k > m$, a given k-fold relation between the $m+1$ coordinates $(x, y, \ldots)$ and the parameters $(x', y', \ldots)$ leads to a $(k-m)$-fold relation between the parameters. This is termed the resultant relation of the given k-fold relation, or when the additional specification is necessary, the resultant relation obtained by elimination of the coordinates $(x, y, \ldots)$.

57. Consider a k-fold relation between the $m+1$ coordinates $(x, y, \ldots)$ and the $m'+1$ coordinates $(x', y', \ldots)$. If $k \pitchfork m$, then, considering the $(x, y, \ldots)$ as coordinates and the $(x', y', \ldots)$ as parameters, we have corresponding to the given relation a k-fold locus in the m-space; and so if $k \pitchfork m'$, then, considering the $(x', y', \ldots)$ as coordinates, but the $(x, y, \ldots)$ as parameters, we have corresponding to the given relation a k-fold locus in the m'-space.

58. If $k > m$, but if the $(k-m)$-fold resultant relation is satisfied, then the given k-fold relation becomes a m-fold linear relation between the coordinates $(x, y, \ldots)$, and is consequently satisfied by a single set of values of the coordinates. Hence, considering the given k-fold relation as implying the $(k-m)$-fold resultant relation, the k-fold relation will represent a single point in the m-space, say, the common point.

59. A m-fold relation, or the locus, or point-system thereby represented, may have a double or nodal point, viz. two of the points of the point-system may be coincident. More generally a k-fold relation $(k \pitchfork m)$, or the locus thereby represented, may have a double or nodal point; for let the relation if less than m-fold be made m-fold by adjoining to it a linear $(m-k)$-fold relation satisfied by the coordinates of the point in question but otherwise arbitrary, then, if the point in question be a double or nodal point of the m-fold relation, or of the point-system thereby represented, the point is said to be a double or nodal point of the original k-fold relation, or of the locus thereby represented.

60. A given k-fold relation $(k \pitchfork m)$ between the $m+1$ coordinates, or the locus thereby represented, has not in general a nodal point. But if the relation involve the $m'+1$ parameters $(x', y', \ldots)$, then, if a certain onefold relation be satisfied between the parameters, there will be a nodal point. The onefold relation between the parameters is the discriminant relation of the given k-fold relation.

61. In the case in question, $k \pitchfork m$, the discriminant relation is the resultant relation of a $(m+1)$-fold relation which is the aggregate of the given k-fold relation with a certain relation called the Jacobian relation, or when the distinction is required, the Jacobian relation in regard to the $(x, y, \ldots)$.

62. Consider a k-fold relation $(k \pitchfork m,\ k \pitchfork m')$ between the $m+1$ coordinates $(x, y, \ldots)$ and the $m'+1$ coordinates $(x', y', \ldots)$. It has been seen that to a given set of values of the $(x', y', \ldots)$ or, say, to a given point in the m'-space, there corresponds a k-fold locus in the m-space, and that to a given set of values of the $(x, y, \ldots)$, or to a given point in the m-space, there corresponds a k-fold locus in the m'-space. The k-fold locus in the m'-space may have a nodal point; this will be the case if there is satisfied between the $(x, y, \ldots)$ a certain onefold relation, the discriminant relation of the given k-fold relation in regard to the $(x', y', \ldots)$. This onefold relation represents in the m-space a onefold locus, the envelope of the k-fold loci in the m-space corresponding to the several points of the m'-space. The property of the envelope is that to each point thereof there corresponds in the m'-space a k-fold locus having a nodal point.

Article Nos. 63–69. Consecutive Points; Tangent Omals.

63. As the notions of proximity and remoteness have been thus far excluded, it is necessary to introduce them for the purpose of defining a tangent omal.

64. If the coordinates of the given point are $(x, y, \ldots)$, those of the consecutive point may be assumed to be $(x+\delta x,\ y+\delta y,\ \ldots)$, where $\delta x,\ \delta y, \ldots$ are indefinitely small in regard to $(x, y, \ldots)$.

65. It may be remarked that, taking the coordinates to be $(x+X,\ y+Y, \ldots)$, there is no obligation to have $(X, Y, \ldots)$ indefinitely small; in fact whatever the magnitudes of these quantities are, if only $X : Y : \ldots = x : y : \ldots$, then the point $(x+X,\ y+Y, \ldots)$ will be the very same with the original point, and it is therefore clear that a consecutive point may be represented in the same manner with magnitudes, however large, of X, Y, … But we may assume them indefinitely small, that is, the ratios $x+\delta x : y+\delta y : \ldots$, where $\delta x,\ \delta y, \ldots$ are indefinitely small in regard to $(x, y, \ldots)$, will represent any set of ratios indefinitely near to the ratios $(x : y : \ldots)$. The foregoing quantities $(\delta x,\ \delta y,\ \ldots)$ are termed the increments.

66. Consider a k-fold relation between the $m+1$ coordinates $(x, y, \ldots)$, $k \pitchfork m$; the increments $(\delta x,\ \delta y,\ \ldots)$ are connected by a linear k-fold relation. The linear k-fold relation is satisfied if we assume the increments proportional to the coordinates—this is, in fact, assuming that the point remains unaltered. We may write $(\delta x,\ \delta y,\ \ldots) = (x,\ y,\ \ldots)$, since in such an equation only the ratios are attended to. But it may be preferable to write $(\delta x,\ \delta y,\ \ldots) = \lambda(x,\ y,\ \ldots)$. In particular if $k = m$, then the increments are connected by a linear m-fold relation; that is, the ratio of the increments is uniquely determined; and as the relation is satisfied by taking the increments proportional to the coordinates, it is clear that the values which the linear m-fold relation gives for the increments are in fact proportional to the coordinates: viz. there is not in this case any consecutive point.

67. Considering the k-fold relation as belonging to a k-fold locus in the m-space, so that $(x, y, \ldots)$ are the coordinates of a point on this locus, then if in the linear k-fold relation between the increments these increments are replaced by the coordinates $(x, y, \ldots)$ of a point in the m-space, then considering the original coordinates $(x, y, \ldots)$ as parameters, the locus of the point $(x, y, \ldots)$ is a k-fold omal locus: it is to be observed that, by what precedes, the linear k-fold relation is satisfied by writing therein the values $x : y : \ldots = x : y : \ldots$, that is, the k-fold omal locus passes through the original point $(x, y, \ldots)$; the k-fold omal locus is said to be the tangent-omal of the original k-fold locus at the point $(x, y, \ldots)$, which point is said to be the point of contact.

68. If in the original k-fold locus we replace $(x, y, \ldots)$ by $(x, y, \ldots)$, and combine therewith the k-fold linear relation, we have between the coordinates $(x, y, \ldots)$ a $2k$-fold relation (containing as parameters the coordinates $(x, y, \ldots)$); these parameters satisfy the original k-fold relation, and in virtue thereof the $2k$-fold relation breaks up into the two k-fold relations which correspond to the two factors of the composite onefold relation distributively implied in the aggregate of the k-fold locus and the k-fold tangent-omal; that is, it breaks up into the k-fold relation which gives the ineunt-point and the k-fold relation which gives the tangent-omal; in other words, the two k-fold loci intersect in the k-fold locus which is the k-fold locus of contact.

69. We thus arrive at the notion of the double generation of a k-fold locus, viz. such locus is the locus of the points, or, say, of the ineunt-points thereof; and it is also the envelope of the tangent-omals thereof. We have thus a theory of duality; I do not at present attempt to develope the theory, but it is necessary to refer to it, in order to remark that this theory is essential to the systematic development of a m-dimensional geometry; the original classification of loci as onefold, twofold, ... $(m-1)$-fold is incomplete, and must be supplemented with the loci reciprocally connected with these loci respectively. And moreover the theory of the singularities of a locus can only be systematically established by means of the same theory of duality; the singularities in regard to the ineunt-point must be treated of in connexion with the singularities in regard to the tangent-omal. These theories (that is, the classification of loci, and the establishment and discussion of the singularities of each kind of locus), vast as their extent is, should in the logical order precede that for which other reasons it may be expedient next to consider, the theory of Transformation, as depending on relations involving simultaneously the $m+1$ coordinates $(x, y, \ldots)$ and the $m'+1$ coordinates $(x', y', \ldots)$.

[414] On Polyzomal Curves, otherwise the Curves $\sqrt{lU} + \sqrt{mV} + \&c. = 0$.

[From the *Transactions* of the Royal Society of Edinburgh, vol. xxv. (1868), pp. 1—110. Read 16th December 1867.]

If U, V, &c., are rational and integral functions $(\ast x, y, z)^r$, all of the same degree r, in regard to the coordinates (x, y, z), then $\sqrt{lU} + \sqrt{mV} + \&c.$ is a polyzome, and the curve $\sqrt{lU} + \sqrt{mV} + \&c. = 0$ a polyzomal curve. Each of the curves $\sqrt{lU} = 0$, $\sqrt{mV} = 0$, &c. (or say the curves $U = 0$, $V = 0$, &c.) is, on account of its relation of circumscription to the curve $\sqrt{lU} + \sqrt{mV} + \&c. = 0$, considered as a girdle thereto (*zōma*), and we have thence the term "zome" and the derived expressions "polyzome," "zomal," &c.

If the number of the zomes $\sqrt{lU}$, $\sqrt{mV}$, &c. be $= \nu$, then we have a ν-zome, and corresponding thereto a ν-zomal curve; the curves $U = 0$, $V = 0$, &c., are the zomal curves or zomals thereof. The cases $\nu = 1$, $\nu = 2$, are not, for their own sake, worthy of consideration; it is in general assumed that ν is $= 3$ at least. It is sometimes convenient to write the general equation in the form $\sqrt{U} + \&c. = 0$, where l, &c. are constants.

The Memoir contains researches in regard to the general ν-zomal curve; the branches thereof, the order of the curve, its singularities, class, &c.; also in regard to the ν-zomal curve $\sqrt{l(\Theta + L\Phi)} + \sqrt{m(\Theta + M\Phi)} + \&c. = 0$, where the zomal curves $\Theta + L\Phi = 0$, &c., all pass through the points of intersection of the same two curves $\Theta = 0$, $\Phi = 0$ of the orders r and $r - s$ respectively; included herein we have the theory of the depression of order as arising from the ideal factor or factors of a branch or branches.

A general theorem is given of "the decomposition of a tetrazomal curve," viz. if the equation of the curve be $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} + \sqrt{pT} = 0$; then if U, V, W, T are in involution, that is, connected by an identical equation $aU + bV + cW + dT = 0$, and if l, m, n, p, satisfy the condition $\frac{a}{l} + \frac{b}{m} + \frac{c}{n} + \frac{d}{p} = 0$, the tetrazomal curve breaks up into two trizomal curves, each expressible by means of any three of the four functions U, V, W, T; for example, in the form $\sqrt{l'U} + \sqrt{m'V} + \sqrt{p'T} = 0$.

If, in this theorem, we take $p = 0$, then the original curve is the trizomal $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$, T is any function $= -\frac{1}{d}(aU + bV + cW)$, where, considering l, m, n as given, a, b, c are quantities subject only to the condition $\frac{a}{l} + \frac{b}{m} + \frac{c}{n} = 0$, and we have the theorem of "the variable zomal of a trizomal curve," viz. the equation of the trizomal $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$, may be expressed by means of any two of the three functions U, V, W, and of a function T determined as above, for example in the form $\sqrt{l'U} + \sqrt{m'V} + \sqrt{n'T} = 0$; whence also it may be expressed in terms of three new functions T, determined as above.

This theorem, which occupies a prominent position in the whole theory, was suggested to me by Mr Casey's theorem, presently referred to, for the construction of a bicircular quartic as the envelope of a variable circle.

In the ν-zomal curve $\sqrt{l(\Theta + L\Phi)} + \&c. = 0$, if $\Theta = 0$ be a conic, $\Phi = 0$ a line, the zomals $\Theta + L\Phi = 0$, &c. are conics passing through the same two points $\Theta = 0$, $\Phi = 0$, and there is no real loss of generality in taking these to be the circular points at infinity—that is, in taking the conics to be circles. Doing this, and using a special notation $\mathfrak{A}^\circ = 0$ for the equation of a circle having its centre at a given point A, and similarly $\mathfrak{A} = 0$ for the equation of an evanescent circle, or say of the point A, we have the ν-zomal curve $\sqrt{l\mathfrak{A}^\circ} + \&c. = 0$, and the more special form $\sqrt{l\mathfrak{A}} + \&c. = 0$.

As regards the last-mentioned curve, $\sqrt{l\mathfrak{A}} + \&c. = 0$, the point A to which the equation $\mathfrak{A} = 0$ belongs, is a focus of the curve, viz. in the case $\nu = 3$, it is an ordinary focus, and in the case $\nu > 3$, it is a special kind of focus, which, if the term were required, might be called a foco-focus; the Memoir contains an explanation of the general theory of the foci of plane curves.

For $\nu = 3$, the equation $\sqrt{l\mathfrak{A}} + \sqrt{m\mathfrak{B}} + \sqrt{n\mathfrak{C}} = 0$ is really equivalent to the apparently more general form $\sqrt{l\mathfrak{A}^\circ} + \sqrt{m\mathfrak{B}^\circ} + \sqrt{n\mathfrak{C}^\circ} = 0$. In fact, this last is in general a bicircular quartic, and, in regard to it, the before-mentioned theorem of the variable zomal becomes Mr Casey's theorem, that "the bicircular quartic (and, as a particular case thereof, the circular cubic) is the envelope of a variable circle having its centre on a given conic and cutting at right angles a given circle." This theorem is a sufficient basis for the complete theory of the trizomal curve $\sqrt{l\mathfrak{A}^\circ} + \sqrt{m\mathfrak{B}^\circ} + \sqrt{n\mathfrak{C}^\circ} = 0$; and it is thereby very easily seen that the curve $\sqrt{l\mathfrak{A}^\circ} + \sqrt{m\mathfrak{B}^\circ} + \sqrt{n\mathfrak{C}^\circ} = 0$ can be represented by an equation $\sqrt{l'\mathfrak{A}'} + \sqrt{m'\mathfrak{B}'} + \sqrt{n'\mathfrak{C}'} = 0$. But for $\nu > 3$ this is not so, and the curve $\sqrt{l\mathfrak{A}} + \&c. = 0$ is only a particular form of the curve $\sqrt{l\mathfrak{A}^\circ} + \&c. = 0$; and the discussion of this general form is scarcely more difficult than that of the special form $\sqrt{l\mathfrak{A}} + \&c. = 0$, included therein.

The investigations in relation to the theory of foci, and to the trizomal and tetrazomal curves where the zomals are circles, are in fact the principal subject of the Memoir, which is divided into Parts as follows: Part I., On Polyzomal Curves in general; Part II., On the Depression of Order, and the Singularities of Polyzomal Curves, with Investigations; Part III., On the Theory of Foci; and Part IV., On the Trizomal and Tetrazomal Curves where the zomals are circles. There is, however, some necessary intermixture of the theories treated of, and the arrangement will appear more in detail from the headings of the several articles.

The paragraphs are numbered continuously through the Memoir. There are four Annexes, relating to questions which it seemed to me more convenient to treat of thus separately.

It is right that I should explain the very great extent to which, in the composition of the present Memoir, I am indebted to Mr Casey's researches. His Paper "On the Equations and Properties (1) of the System of Circles touching three circles in a plane; (2) of the System of Spheres touching four spheres in space; (3) of the System of Circles touching three circles on a sphere; (4) on the System of Conics inscribed in a conic and touching three inscribed conics in a plane," was read to the Royal Irish Academy, April 9, 1866, and is published in their "Proceedings." The fundamental theorem for the equation of the pairs of circles touching three given circles was, previous to the publication of the paper, mentioned to me by Dr Salmon, and I communicated it to Professor Cremona, suggesting to him the problem solved in his letter of March 3, 1866, as mentioned in my paper, "Investigations in connexion with Casey's Equation," Quarterly Math. Journ. vol. viii. 1867, pp. 334—341, [395], and as also appears, Annex No. IV of the present Memoir.

In connexion with this theorem, I communicated to Mr Casey, in March or April 1867, the theorem No. 164 of the present Memoir, that for any three given circles, centres A, B, C, the equation $BC^2\sqrt{\mathfrak{A}} + CA^2\sqrt{\mathfrak{B}} + AB^2\sqrt{\mathfrak{C}} = 0$ (where BC, CA, AB, denote the mutual distances of the points A, B, C) belongs to a Cartesian. Mr Casey, in a letter to me dated 30th April, 1867, informed me of his own mode of viewing the question as follows:—"The general equation of the second order $(a, b, c, f, g, h \between \alpha, \beta, \gamma)^2 = 0$, where α, β, γ are circles, is a bicircular quartic. If we take the equation $(a, b, c, f, g, h \between \lambda, \mu, \nu)^2 = 0$ in tangential coordinates (that is, when λ, μ, ν are perpendiculars let fall from the centres of α, β, γ on any line), it denotes a conic; denoting this conic by F, and the circle which cuts α, β, γ orthogonally by J, I proved that, if a variable circle moves with its centre on F, and if it cuts J orthogonally, its envelope will be the bicircular quartic whose equation is that written down above;" and among other consequences, he mentions that the foci of F are the double foci of the quartic, and the points in which J cuts F single foci of the quartic, and also the theorem which I had sent him as to the Cartesian, and he refers to his Memoir on Bicircular Quartics as then nearly finished.

An Abstract of the Memoir as read before the Royal Irish Academy, 10th February, 1867, and published in their Proceedings.

Part I. (Nos. 1 to 55).—On Polyzomal Curves in General.

Article Nos. 1 to 4. Definition and Preliminary Remarks.

1. As already mentioned, U, V, &c. denote rational and integral functions $(\ast \between x, y, z)^r$, all of the same degree r in the coordinates (x, y, z), and the equation $\sqrt{lU} + \sqrt{mV} + \&c. = 0$ then belongs to a polyzomal curve, viz., if the number of the zomes $\sqrt{lU}$, $\sqrt{mV}$, &c. is $= \nu$, then we have a ν-zomal curve.

The radicals, or any of them, may contain rational factors, or be of the form $P\sqrt{Q}$; but in speaking of the curve as a ν-zomal, it is assumed that any two terms, such as $P\sqrt{Q} + P'\sqrt{Q}$, involving the same radical $\sqrt{Q}$, are united into a single term, so that the number of distinct radicals is always $= \nu$; in particular (r being even), it is assumed that there is only one rational term P. But the ordinary case, and that which is almost exclusively attended to, is that in which the radicals $\sqrt{U}$, $\sqrt{V}$, &c. are distinct irreducible radicals without rational factors.

2. The curves $U = 0$, $V = 0$, &c. are said to be the zomal curves, or simply the zomals of the polyzomal curve $\sqrt{lU} + \sqrt{mV} + \&c. = 0$; more strictly, the term zomal would be applied to the functions U, V, &c. It is to be noticed, that although the form $\sqrt{U} + \sqrt{V} + \&c. = 0$ is equally general with the form $\sqrt{lU} + \sqrt{mV} + \&c. = 0$ (in fact, in the former case, the functions U, V, &c. are considered as implicitly containing the constant factors l, m, &c., which are expressed in the latter case), yet it is frequently convenient to express these factors, and thus write the equation in the form $\sqrt{lU} + \sqrt{mV} + \&c.$

For instance, in speaking of any given curves $U = 0$, $V = 0$, &c., we are apt, disregarding the constant factors which they may involve, to consider U, V, &c. as given functions; but in this case the general equation of the polyzomal with the zomals $U = 0$, $V = 0$, &c., is of course $\sqrt{lU}+\sqrt{mV}+\&c. = 0$.

3. Anticipating in regard to the cases $\nu = 1$, $\nu = 2$, the remark which will be presently made in regard to the ν-zomal, that $\sqrt{U} + \sqrt{V} + \&c. = 0$ is the curve represented by the rationalised form of this equation, the monozomal curve $\sqrt{U} = 0$ is merely the curve $U = 0$, viz., this is any curve whatever $U = 0$ of the order r; and similarly, the bizomal curve $\sqrt{U}+\sqrt{V} = 0$ is merely the curve $U-V = 0$, viz. this is any curve whatever $U = 0$, of the order r; the zomal curves $U = 0$, $V = 0$, taken separately, are not curves standing in any special relation to the curve in question $= 0$, but $U = 0$ may be any curve whatever of the order r, and then $V = 0$ is a curve of the same order r, in involution with the two curves $U = 0$, $U = 0$; we may, in fact, write the equation $U = 0$ under the bizomal form $\sqrt{U} + \sqrt{U} + U = 0$. In the case r even, we may, however, notice the bizomal curve $P + \sqrt{U} = 0$ (P a rational function of the degree $\frac{1}{2}r$); the rational equation is here $U = (U - P^2) = 0$, that is $U = P^2$, viz., P is any curve whatever of the order $\frac{1}{2}r$, and $U = 0$ is a curve of the order r, touching the given curve $U = 0$ at each of its $\frac{1}{2}r^2$ intersections.

It is to be remarked that the order of the ν-zomal curve $\sqrt{U} + \&c. = 0$ is $= 2^{\nu-1}r$; this is right in the case of the bizomal curve $\sqrt{U} + \sqrt{V} = 0$, the order being $= r$, but it fails for the monozomal curve $\sqrt{U} = 0$, the order being in this case r, instead of $\frac{1}{2}r$, as given by the formula. The two unimportant and somewhat exceptional cases $\nu = 1$, $\nu = 2$, are thus disposed of, and in all that follows (except in so far as this is in fact applicable to the cases just referred to), ν may be taken to be $= 3$ at least.

4. It is to be throughout understood that by the curve $\sqrt{lU} + \sqrt{mV} + \&c. = 0$ is meant the curve represented by the rationalised equation $\text{Norm}(\sqrt{lU} + \sqrt{mV} + \&c.) = 0$, viz. the Norm is obtained by attributing to all but one of the zomes $\sqrt{lU}$, $\sqrt{mV}$, &c., each of the two signs $+$, $-$, and multiplying together the several resulting values of the polyzome; in the case of a ν-zomal curve, the number of factors is thus $= 2^{\nu-1}$ (whence, as each factor is of the degree $\frac{1}{2}r$, the order of the curve is $2^{\nu-1}\cdot\frac{1}{2}r$, $= 2^{\nu-2}r$, as mentioned above). I expressly mention that, as regards the polyzomal curve, we are not in any wise concerned with the signs of the radicals, which signs are and remain essentially indeterminate; the equation $\sqrt{lU} + \sqrt{mV} + \&c. = 0$, is a mere symbol for the rationalised equation, $\text{Norm}(\sqrt{lU} + \sqrt{mV} + \&c.) = 0$.

Article Nos. 5 to 12. The Branches of a Polyzomal Curve.

5. But we may in a different point of view attend to the signs of the radicals; if for all values of the coordinates we take the symbol $\sqrt{U}$ to signify the positive square root, and consider $\sqrt{U}$, $\sqrt{V}$, &c. as signifying determinately, say the positive values of $\sqrt{U}$, $\sqrt{V}$, &c.; then each of the several equations $\pm\sqrt{U} \pm \sqrt{V} + \&c. = 0$, or, fixing at pleasure one of the signs, suppose that prefixed to $\sqrt{U}$, then each of the several equations $\sqrt{U} \pm \sqrt{V} \pm \&c. = 0$, will belong to a branch of the polyzomal curve: a ν-zomal curve has thus $2^{\nu-1}$ branches corresponding to the $2^{\nu-1}$ values respectively of the polyzome. The separation of the branches depends on the precise fixation of the significations of $\sqrt{U}$, $\sqrt{V}$, &c., and in regard hereto some further explanation is necessary.

6. When U is real and positive, $\sqrt{U}$ may be taken to be, in the ordinary sense, the positive value of $\sqrt{U}$, and so when U is real and negative, $\sqrt{U}$ may be taken to be $= i$ into the positive value of $\sqrt{-U}$; and the like as regards $\sqrt{V}$, &c. The functions U, V, &c. are assumed to be real functions of the coordinates; hence, for any real values of the coordinates, U, V, &c. are real positive or negative quantities, and the significations of $\sqrt{U}$, $\sqrt{V}$, &c. are completely determined.

7. But the coordinates may be imaginary. In this case the functions U, V, &c. will for any given values of the coordinates acquire each of them a determinate, in general imaginary, value. If for all real values whatever of α, β, we select once for all one of the two opposite values of $\sqrt{\alpha + \beta i}$, calling it the positive value, and representing it by $\sqrt{\alpha + \beta i}$, then, for any particular values of the coordinates, U being $= \alpha + \beta i$, the value of $\sqrt{U}$ may be taken to be $= \sqrt{\alpha + \beta i}$; and the like as regards $\sqrt{V}$, &c. $\sqrt{U}$, $\sqrt{V}$, &c. have thus each of them a determinate signification for any values whatever, real or imaginary, of the coordinates.

The coordinates of a given point on the curve $\sqrt{lU} + \sqrt{mV} + \&c. = 0$, will in general satisfy only one of the equations $\sqrt{U} \pm \sqrt{V} \pm \&c. = 0$; that is, the point will belong to one (but in general only one) of the $2^{\nu-1}$ branches of the curve; the entire series of points the coordinates of which satisfy any one of the $2^{\nu-1}$ equations, will constitute the branch corresponding to that equation.

8. The signification to be attached to the expression $\sqrt{\alpha + \beta i}$ should agree with that previously attached to the like symbol in the case of a positive or negative real quantity; and it should, as far as possible, be subject to the condition of continuity, viz., as $\alpha + \beta i$ passes continuously to $\alpha' + \beta' i$, so $\sqrt{\alpha + \beta i}$ should pass continuously to $\sqrt{\alpha' + \beta' i}$; but (as is known) it is not possible to satisfy universally this condition of continuity; viz., if for facility of explanation we consider (α, β) as the coordinates of a point in a plane, and imagine this point to describe a closed curve surrounding the origin or point $(0, 0)$, then it is not possible so to define $\sqrt{\alpha + \beta i}$ that this quantity, varying continuously as the point moves along the curve, shall, when the point has made a complete circuit, resume its original value. The signification to be attached to $\sqrt{\alpha + \beta i}$ is thus in some measure arbitrary, and it would appear that the division of the curve into branches is affected by a corresponding arbitrariness, but this arbitrariness relates only to the imaginary branches of the curve: the notion of a real branch is perfectly definite.

9. It would seem that a branch may be impossible for any series whatever of points real or imaginary. Thus, in the bizomal curve $\sqrt{U} + \sqrt{V} = 0$, the branch $\sqrt{U} + \sqrt{V} = 0$ is impossible. In fact, for any point whatever, real or imaginary, of the curve, we have $U = V$, and therefore $\sqrt{U} = \sqrt{V}$; the point thus belongs to the other branch $\sqrt{U} - \sqrt{V} = 0$, not to the branch $\sqrt{U} + \sqrt{V} = 0$; the only points belonging to the last-mentioned branch are the isolated points for which simultaneously $\sqrt{U} = 0$, $\sqrt{V} = 0$; viz., the points of intersection of the two curves $U = 0$, $V = 0$.

10. It is not clear to me whether the case is the same in regard to the branch $\sqrt{U} + \sqrt{V} + \sqrt{W} = 0$ of a trizomal curve. In fact, for each point of the curve $\sqrt{U} + \sqrt{V} + \sqrt{W} = 0$ we have $(U - V - W)^2 = 4VW$, and therefore, $U - V - W = \pm 2\sqrt{V}\sqrt{W}$; there may very well be points for which the sign is $+$; that is, points for which $U = V + W + 2\sqrt{V}\sqrt{W}$, and for these points we have $\pm\sqrt{U} = \sqrt{V} + \sqrt{W}$; for real values of the coordinates the sign on the left hand must be $+$ (for otherwise the two sides would have opposite signs), but there is no apparent reason, or at least no obviously apparent reason, why this should be so for imaginary values of the coordinates, and if the sign be in fact $-$, then the point will belong to the branch $\sqrt{U} + \sqrt{V} + \sqrt{W} = 0$.

11. But the branch in question is clearly impossible for any series of real points; so that, leaving it an open question whether the epithet "impossible" is to be understood to mean impossible for any series of real points (that is, as a mere synonym of imaginary), or whether it is to mean impossible for any series whatever of points real or imaginary, I say that the branch $\sqrt{U} + \sqrt{V} + \sqrt{W} = 0$ of a trizomal curve is an impossible branch.

12. Consider a ν-zomal curve $\sqrt{lU} + \sqrt{mV} + \&c. = 0$; the equations of the several branches are the equations $\sqrt{U} \pm \sqrt{V} \pm \&c. = 0$, obtained by attributing to the radicals $\sqrt{V}$, $\sqrt{W}$, &c. the signs $+$ or $-$ at pleasure, the sign of $\sqrt{U}$ being fixed. If in the equation of any branch we write $\sqrt{V}$, &c. for those radicals which have the sign $-$, then the foregoing equations break up into the more simple equations $\sqrt{U} + \&c. = 0$, $\sqrt{V} + \&c. = 0$, which are the equations of certain branches of the curves $\sqrt{lU} + \&c. = 0$, $\sqrt{mV} + \&c. = 0$, respectively, and conversely each of the intersections of these two curves is a point situate simultaneously on some two branches of the original ν-zomal curve $\sqrt{lU} + \sqrt{mV} + \&c. = 0$.

Hence, partitioning in any manner the ν-zome $\sqrt{lU} + \sqrt{mV} + \&c.$ into an α-zome, $\sqrt{lU} + \&c.$ and a β-zome $\sqrt{mV} + \&c.$ $(\alpha + \beta = \nu)$, and writing down the equations $\sqrt{lU} + \&c. = 0$, $\sqrt{mV} + \&c. = 0$ of an α-zomal curve and a β-zomal curve respectively, each of the intersections of these two curves is a point situate simultaneously on two branches of the ν-zomal curve; and the points situate simultaneously on two branches of the ν-zomal curve are the points of intersection of the several pairs of an α-zomal curve and a β-zomal curve, which can be formed by any bipartition of the ν-zome.

13. There are two cases to be considered:—First, when the parts are 1, $\nu - 1$ ($\nu - 1$ is > 1, except in the case $\nu = 2$, which may be excluded from consideration), or say when the ν-zome is partitioned into a zome and antizome. Secondly, when the parts α, β, are each > 1 (this implies $\nu = 4$ at least), or say when the ν-zome is partitioned into a pair of complementary parazomes.

14. To fix the ideas, take the tetrazomal curve $\sqrt{U} + \sqrt{V} + \sqrt{W} + \sqrt{T} = 0$, and consider first a point for which $\sqrt{U} = 0$, $\sqrt{V} + \sqrt{W} + \sqrt{T} = 0$. The Norm is the product of $(2^3 =)$ 8 factors; selecting hereout the factors $\sqrt{U} + \sqrt{V} + \sqrt{W} + \sqrt{T}$, $\sqrt{U} - \sqrt{V} - \sqrt{W} - \sqrt{T}$, let the product of these be called

F, and the product of the remaining six factors be called G; the rationalised equation of the curve is therefore $FG = 0$. The derived equation is $GdF + FdG = 0$; at the point in question $\sqrt{U} = 0$, $\sqrt{V} + \sqrt{W} + \sqrt{T} = 0$; G and dG are each of them finite (that is, they neither vanish nor become infinite), but we have $F = 0$, $dF = dU \cdot (\sqrt{V} + \sqrt{W} + \sqrt{T}) = 0$, that is, the derived equation becomes $0 = 0$, or the point in question is an ordinary point on the curve.

15. Consider, secondly, a point for which $\sqrt{U} + \sqrt{V} = 0$, $\sqrt{W} + \sqrt{T} = 0$; to form the Norm, taking in this case the two factors $\sqrt{U} + \sqrt{V} + \sqrt{W} + \sqrt{T}$, $\sqrt{U} + \sqrt{V} - \sqrt{W} - \sqrt{T}$, let their product $= (\sqrt{U} + \sqrt{V})^2 - (\sqrt{W} + \sqrt{T})^2$ be called F, and the product of the remaining six factors be called G; the rationalised equation is $FG = 0$, and the derived equation is $FdG + GdF = 0$. At the point in question G and dG are each of them finite (that is, they neither vanish nor become infinite), but we have $F = 0$, $dF = (\sqrt{U} + \sqrt{V})(dU/\sqrt{U} + dV/\sqrt{V}) - (\sqrt{W} + \sqrt{T})(dW/\sqrt{W} + dT/\sqrt{T}) = 0$, that is, the derived equation becomes identically $= 0$; the point in question is thus a singular point, and it is easy to see that it is in fact a node, or ordinary double point, on the tetrazomal curve. And similarly, each of the points of intersection of the two curves $\sqrt{U} + \sqrt{V} = 0$, $\sqrt{W} + \sqrt{T} = 0$ is a node on the tetrazomal curve.

16. The proofs in the foregoing two examples respectively are quite general, and we may, in regard to a ν-zomal curve, enunciate the results as follows, viz., in a ν-zomal curve, the points situate simultaneously on two branches are either the intersections of a zomal curve and its antizomal curve, or else they are the intersections of a pair of complementary parazomal curves. In the former case, the points in question are ordinary points on the ν-zomal, but they are points of contact of the ν-zomal with the zomal; it may be added, that the intersections of the zomal and antizomal, each reckoned twice, are all the intersections of the ν-zomal and zomal. In the latter case, the points in question are nodes of the ν-zomal; it may be added, that the ν-zomal has not, in general, any nodes other than the points which are thus the intersections of a pair of complementary parazomals, and that it has not in general any cusps.

Article Nos. 17 to 21. Singularities of a ν-zomal Curve.

17. It has been already shown that the order of the ν-zomal curve is $= 2^{\nu-2}r$. Considering the case where ν is $= 3$ at least, the curve has been represented as the partial intersection of two surfaces, the order of the complete intersection being $= 2^{\nu-1} \cdot \frac{1}{2}r$, $= 2^{\nu-2}r$; and the curve has thus a deficiency equal to the number of intersections which are wanting to make up the complete intersection; or the deficiency is $= 2^{\nu-2}r(2^{\nu-2}r - 1) - 2 \cdot 2^{\nu-3}r(2^{\nu-3}r - 1)$, $= 2^{2\nu-5}r^2$.

18. Considering first a zome and its antizome, say the zome $U = 0$ and the antizome $\sqrt{V}+\sqrt{W}+\&c. = 0$ (a $(\nu-1)$-zomal curve); the zome is of the order r, and the antizome of the order $2^{\nu-3}r$; they intersect therefore in $2^{\nu-3}r^2$ points, each of which is a point of contact of the ν-zomal with the zome. The number of contacts with all the zomes is thus $= \nu \cdot 2^{\nu-3}r^2$.

19. Considering next a pair of complementary parazomal curves, an α-zomal and a β-zomal respectively $(\alpha + \beta = \nu)$, these are of the orders $2^{\alpha-2}r$ and $2^{\beta-2}r$ respectively, and they intersect therefore in $2^{\alpha+\beta-4}r^2 = 2^{\nu-4}r^2$ points, nodes of the ν-zomal. This number is independent of the particular partition (α, β), and the ν-zomal has thus this same number, $2^{\nu-4}r^2$, of nodes in respect of each pair of complementary parazomals; hence the total number of nodes is $= 2^{\nu-4}r^2$ into the number of pairs of complementary parazomals.

For the partition (α, β) the number of pairs is $= [\nu]^\alpha \div [\alpha]^\alpha[\beta]^\beta$, or when $\alpha = \beta$, which of course implies ν even, it is one-half of this; extending the summation from $\alpha = 2$ to $\alpha = \nu - 2$, each pair is obtained twice, and the number of pairs is thus $= \frac{1}{2}\{(1+1)^\nu - \nu - \nu - 1 - 1\}$; the sum extended from $\alpha = 0$ to $\alpha = \nu$ is $(1+1)^\nu = 2^\nu$, but we thus include the terms 1, ν, ν, 1, which are together $= 2\nu + 2$, hence the correct value of the sum is $= 2^\nu - 2\nu - 2$, and the number of pairs is the half of this $= 2^{\nu-1} - \nu - 1$.

Hence the number of nodes of the ν-zomal curve is $= (2^{\nu-1} - \nu - 1)2^{\nu-4}r^2$.

20. The ν-zomal is thus a curve of the order $2^{\nu-2}r$, with $(2^{\nu-1} - \nu - 1)2^{\nu-4}r^2$ nodes, but without cusps; the class is therefore $= 2^{\nu-2}r[(\nu+1)r - 2]$, and the deficiency is $= 2^{\nu-4}r[(\nu+1)r - 6] + 1$.

These are the general expressions, but even when the zomal curves $U = 0$, $V = 0$, &c., are given, then writing the equation of the ν-zomal under the form $\sqrt{lU}+\sqrt{mV}+\&c. = 0$, the constants $l : m :$ &c.,

may be so determined as to give rise to nodes or cusps which do not occur in the general case; the formulae will also undergo modification in the particular cases next referred to.

Article Nos. 22 to 27. Special Case where all the Zomals have a Common Point or Points.

21. Consider the case where the zomals $U=0$, $V=0$ have all of them any number, say k, of common intersections—these may be referred to simply as the common points. Each common point, being a $2^{\alpha-2}$-tuple point on the α-zomal and a $2^{\beta-2}$-tuple point on the β-zomal, counts as $2^{\alpha+\beta-4}$, $=2^{\nu-4}$ intersections, and the k common points count as $2^{\nu-4}k$ intersections; the number of the remaining intersections is therefore $=2^{\nu-4}(r^2-k)$, each of which is a node on the ν-zomal curve; and we have thus in all $2^{\nu-4}(2^{\nu-1}-\nu-1)(r^2-k)$ nodes.

22. There are, besides, the k common points, each of them a $2^{\nu-2}$-tuple point on the ν-zomal, and therefore each reckoning as $\frac{1}{2}2^{\nu-2}(2^{\nu-2}-1)$, $=2^{2\nu-5}-2^{\nu-3}$ double points, or together as $(2^{2\nu-5}-2^{\nu-3})k$ double points.

Reserving the term node for the above-mentioned nodes or proper double points, and considering, therefore, the double points (dps.) as made up of the nodes and of the $2^{\nu-2}$-tuple points, the total number of dps. is thus

$$\begin{aligned}&2^{\nu-4}(2^{\nu-1}-\nu-1)(r^2-k)+(2^{2\nu-5}-2^{\nu-3})k,\\ &=2^{\nu-4}(2^{\nu-1}-\nu-1)r^2+[(\nu+1)2^{\nu-4}-2^{\nu-3}]k;\end{aligned}$$

or finally this is $=2^{\nu-4}\{(2^{\nu-1}-\nu-1)r^2+(\nu-1)k\}$; so that there is a gain $=2^{\nu-4}(\nu-1)k$ in the number of dps. arising from the k common points.

There is, of course, in the class a diminution equal to twice this number, or $2^{\nu-3}(\nu-1)k$; and in the deficiency a diminution equal to this number, or $2^{\nu-4}(\nu-1)k$.

23. The zomal curves $U=0$, $V=0$, &c., may all of them pass through the same r^2 points; we have then $k=r^2$, and the expression for the number of dps. is $=(2^{2\nu-5}-2^{\nu-3})r^2$, viz., this is $=2^{\nu-3}(2^{\nu-2}-1)r^2$.

But in this case the dps. are nothing else than the r^2 common points, each of them a $2^{\nu-2}$-tuple point, the ν-zomal curve in fact breaking up into a system of $2^{\nu-2}$ curves of the order r, each passing through the r^2 common points.

This is easily verified, for if $\Theta=0$, $\Phi=0$ are some two curves of the order r, then, in the present case, the zomal curves are curves in involution with these curves; that is, they are curves of the form $l\Theta+l'\Phi=0$, and the ν-zomal curve $\sqrt{l(l\Theta+l'\Phi)}+\&c.=0$ is thus satisfied by $l\Theta+l'\Phi=0$, that is, it breaks up into the system of curves $l\Theta+l'\Phi=0$, where $l:l'$ has $2^{\nu-2}$ values.

24. A more important case is when the zomal curves are each of them in involution with the same two given curves, one of them of the order r, the other of an inferior order. Let $\Theta=0$ be a curve of the order r, $\Phi=0$ a curve of an inferior order $r-s$; $L=0$, $M=0$, &c., curves of the order s; then the case in question is when the zomal curves are of the form $\Theta+L\Phi=0$, $\Theta+M\Phi=0$, &c., the equation of the ν-zomal is

$$\sqrt{l(\Theta+L\Phi)}+\sqrt{m(\Theta+M\Phi)}+\&c.=0,$$

where l, m, &c. are constants. This is the most convenient form for the equation, and by considering the functions L, M, &c. as containing implicitly the factors $l^{-\frac{1}{2}}$, $m^{-\frac{1}{2}}$, &c. respectively, we may take it to include the form

$$\sqrt{\Theta+L\Phi}+\sqrt{\Theta+M\Phi}+\&c.=0,$$

which last has the advantage of being immediately applicable to the case where any one or more of the constants l, m, &c. may be $=0$.

25. In the case now under consideration we have the $r(r-s)$ points of intersection of the curves $\Theta=0$, $\Phi=0$ as common points of all the zomals. Hence, putting in the foregoing formula $k=r(r-s)$, we have a ν-zomal curve of the order $2^{\nu-2}r$, having with each zomal $2^{\nu-2}rs$ contacts, or with all the zomals $2^{\nu-2}rs\nu$ contacts; the number of nodes is $=2^{\nu-4}(2^{\nu-1}-\nu-1)\{r^2-r(r-s)\}$, $=2^{\nu-4}(2^{\nu-1}-\nu-1)rs$; and there is a diminution in the class $=2^{\nu-3}(\nu-1)r(r-s)$, and in the deficiency $=2^{\nu-4}(\nu-1)r(r-s)$.

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ON POLYZOMAL CURVES.

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Part I. (Nos. 1 to 55).—On Polyzomal Curves in General.

1 to 4. Definition and Preliminary Remarks.

Article 1. As already mentioned, U, V, &c. denote rational and integral functions $(* \mid x, y, z)^r$, all of the same degree r in the coordinates (x, y, z), and the equation

$$\sqrt{U} + \sqrt{V} + \&c. = 0$$

then belongs to a polyzomal curve; viz., if the number of the zomes $\sqrt{U}$, $\sqrt{V}$, &c. is $= \nu$, then we have a ν-zomal curve. The radicals, or any of them, may contain rational factors, or be of the form $P\sqrt{Q}$; but in speaking of the curve as a ν-zomal, it is assumed that any two terms, such as $P\sqrt{Q} + P'\sqrt{Q}$, involving the same radical $\sqrt{Q}$, are united into a single term, so that the number of distinct radicals is always $= \nu$; in particular (r being even), it is assumed that there is only one rational term P.

But the ordinary case, and that which is almost exclusively attended to, is that in which the radicals $\sqrt{U}$, $\sqrt{V}$, &c. are distinct irreducible radicals without rational factors.

Article 2. The curves $U = 0$, $V = 0$, &c. are said to be the *zomal curves*, or simply the *zomals* of the polyzomal curve $\sqrt{U} + \sqrt{V} + \&c. = 0$; more strictly, the term zomal would be applied to the functions U, V, &c. It is to be noticed, that although the form $\sqrt{U} + \sqrt{V} + \&c. = 0$ is equally general with the form $\sqrt{lU} + \sqrt{mV} + \&c. = 0$ (in fact, in the former case, the functions U, V, &c. are considered as implicitly containing the constant factors l, m, &c., which are expressed in the latter case), yet it is frequently convenient to express these factors, and thus write the equation in the form $\sqrt{lU} + \sqrt{mV} + \&c. = 0$.

For instance, in speaking of any given curves $U = 0$, $V = 0$, &c., we are apt, disregarding the constant factors which they may involve, to consider U, V, &c. as given functions; but in this case the general equation of the polyzomal with the zomals $U = 0$, $V = 0$, &c., is of course $\sqrt{lU} + \sqrt{mV} + \&c. = 0$.

Article 3. Anticipating in regard to the cases $\nu = 1$, $\nu = 2$, the remark which will be presently made in regard to the ν-zomal, that $\sqrt{U} + \sqrt{V} + \&c. = 0$ is the curve represented by the rationalised form of this equation, the monozomal curve $\sqrt{U} = 0$ is merely the curve $U = 0$, viz., this is any curve whatever $U = 0$ of the order r; and similarly, the bizomal curve $\sqrt{U} + \sqrt{V} = 0$ is merely the curve $U - V = 0$, viz. this is any curve whatever $U - V = 0$, of the order r; the zomal curves $U = 0$, $V = 0$, taken separately, are not curves standing in any special relation to the curve in question $= 0$, but $U = 0$ may be any curve whatever of the order r, and then $V = 0$ is a curve of the same order r, in involution with the two curves $U = 0$, $V = 0$; we may, in fact, write the equation $U - V = 0$ under the bizomal form $\sqrt{U} + \sqrt{V} + \sqrt{U} = 0$. In the case r even, we may, however, notice the bizomal curve $P + \sqrt{U} = 0$ (P a rational function of the degree $\frac{1}{2}r$); the rational equation is here $\Omega = U - P^2 = 0$, that is $U = \Omega + P^2$, viz., P is any curve whatever of the order $\frac{1}{2}r$, and $U = 0$ is a curve of the order r, touching the given curve $\Omega = 0$ at each of its $\frac{1}{2}r^2$ intersections with the curve $P = 0$.

I further remark that the order of the ν-zomal curve $\sqrt{U} + \sqrt{V} + \&c. = 0$ is $= 2^{\nu-1}r$; this is right in the case of the bizomal curve $\sqrt{U} + \sqrt{V} = 0$, the order being $= r$, but it fails for the monozomal curve $\sqrt{U} = 0$, the order being in this case r, instead of $\frac{1}{2}r$, as given by the formula. The two unimportant and somewhat exceptional cases $\nu = 1$, $\nu = 2$, are thus disposed of, and in all that follows (except in so far as this is in fact applicable to the cases just referred to), ν may be taken to be $= 3$ at least.

Article 4. It is to be throughout understood that by the curve $\sqrt{U} + \sqrt{V} + \&c. = 0$ is meant the curve represented by the rationalised equation

$$\mathrm{Norm}(\sqrt{U} + \sqrt{V} + \&c.) = 0,$$

viz. the Norm is obtained by attributing to all but one of the zomes $\sqrt{U}$, $\sqrt{V}$, &c., each of the two signs $+$, $-$, and multiplying together the several resulting values of the polyzome; in the case of a ν-zomal curve, the number of factors is thus $= 2^{\nu-1}$ (whence, as each factor is of the degree $\frac{1}{2}r$, the order of the curve is $2^{\nu-1}\cdot\frac{1}{2}r = 2^{\nu-2}r$, as mentioned above). I expressly mention that, as regards the polyzomal curve, we are not in any wise concerned with the signs of the radicals, which signs are and remain essentially indeterminate; the equation $\sqrt{U}+\sqrt{V}+\&c. = 0$, is a mere symbol for the rationalised equation, $\mathrm{Norm}(\sqrt{U}+\sqrt{V}+\&c.) = 0$.

5 to 12. The Branches of a Polyzomal Curve.

Article 5. But we may in a different point of view attend to the signs of the radicals; if for all values of the coordinates we take the symbol $\sqrt{\ }$, and consider $\sqrt{U}$, $\sqrt{V}$, &c. as signifying determinately, say the positive values of $\sqrt{U}$, $\sqrt{V}$, &c.; then each of the several equations $\pm\sqrt{U} \pm \sqrt{V} + \&c. = 0$, or, fixing at pleasure one of the signs, suppose that prefixed to $\sqrt{U}$, then each of the several equations $\sqrt{U} \pm \sqrt{V} \pm \&c. = 0$, will belong to a branch of the polyzomal curve: a ν-zomal curve has thus $2^{\nu-1}$ branches corresponding to the $2^{\nu-1}$ values respectively of the polyzome. The separation of the branches depends on the precise fixation of the significations of $\sqrt{U}$, $\sqrt{V}$, &c., and in regard hereto some further explanation is necessary.

Article 6. When U is real and positive, $\sqrt{U}$ may be taken to be, in the ordinary sense, the positive value of $\sqrt{U}$, and so when U is real and negative, $\sqrt{U}$ may be taken to be $= i$ into the positive value of $\sqrt{-U}$; and the like as regards $\sqrt{V}$, &c. The functions U, V, &c. are assumed to be real functions of the coordinates; hence, for any real values of the coordinates, U, V, &c. are real positive or negative quantities, and the significations of $\sqrt{U}$, $\sqrt{V}$, &c. are completely determined.

Article 7. But the coordinates may be imaginary. In this case the functions U, V, &c. will for any given values of the coordinates acquire each of them a determinate, in general imaginary, value. If for all real values whatever of α, β, we select once for all one of the two opposite values of $\sqrt{\alpha+\beta i}$, calling it the positive value, and representing it by $\sqrt{\alpha+\beta i}$, then, for any particular values of the coordinates, U being $=\alpha+\beta i$, the value of $\sqrt{U}$ may be taken to be $=\sqrt{\alpha+\beta i}$; and the like as regards $\sqrt{V}$, &c. $\sqrt{U}$, $\sqrt{V}$, &c. have thus each of them a determinate signification for any values whatever, real or imaginary, of the coordinates. The coordinates of a given point on the curve $\sqrt{U}+\sqrt{V}+\&c. =0$, will in general satisfy only one of the equations $\sqrt{U}\pm\sqrt{V}\pm\&c. =0$; that is, the point will belong to one (but in general only one) of the $2^{\nu-1}$ branches of the curve: the entire series of points the coordinates of which satisfy any one of the $2^{\nu-1}$ equations, will constitute the branch corresponding to that equation.

Article 8. The signification to be attached to the expression $\sqrt{\alpha+\beta i}$ should agree with that previously attached to the like symbol in the case of a positive or negative real quantity; and it should, as far as possible, be subject to the condition of continuity, viz., as $\alpha+\beta i$ passes continuously to $\alpha'+\beta' i$, so $\sqrt{\alpha+\beta i}$ should pass continuously to $\sqrt{\alpha'+\beta' i}$; but (as is known) it is not possible to satisfy universally this condition of continuity; viz., if for facility of explanation we consider (α,β) as the coordinates of a point in a plane, and imagine this point to describe a closed curve surrounding the origin or point $(0,0)$, then it is not possible so to define $\sqrt{\alpha+\beta i}$ that this quantity, varying continuously as the point moves along the curve, shall, when the point has made a complete circuit, resume its original value. The signification to be attached to $\sqrt{\alpha+\beta i}$ is thus in some measure arbitrary, and it would appear that the division of the curve into branches is affected by a corresponding arbitrariness, but this arbitrariness relates only to the imaginary branches of the curve: the notion of a real branch is perfectly definite.

Article 9. It would seem that a branch may be impossible for any series whatever of points real or imaginary. Thus, in the bizomal curve $\sqrt{U}+\sqrt{V}=0$, the branch $\sqrt{U}+\sqrt{V}=0$ if $U-V$ is impossible. In fact, for any point whatever, real or imaginary, of the curve, we have $U=V$, and therefore $\sqrt{U}=\sqrt{V}$; the point thus belongs to the other branch $\sqrt{U}-\sqrt{V}=0$, not to the branch $\sqrt{U}+\sqrt{V}=0$; the only points belonging to the last-mentioned branch are the isolated points for which simultaneously $\sqrt{U}=0$, $\sqrt{V}=0$; viz., the points of intersection of the two curves $U=0$, $V=0$.

Article 10. It is not clear to me whether the case is the same in regard to the branch $\sqrt{U}+\sqrt{V}+\sqrt{W}=0$ of a trizomal curve. In fact, for each point of the curve $\sqrt{U}+\sqrt{V}+\sqrt{W}=0$ we have $(U-V-W)^2=4VW$, and therefore, $U-V-W=\pm2\sqrt{V}\sqrt{W}$: there may very well be points for which the sign is $+$; that is, points for which $U=V+W+2\sqrt{VW}$, and for these points we have $\pm\sqrt{U}=\sqrt{V}+\sqrt{W}$; for real values of the coordinates the sign on the left hand must be $+$ (for otherwise the two sides would have opposite signs), but there is no apparent reason, or at least no obviously apparent reason, why this should be so for imaginary values of the coordinates, and if the sign be in fact $-$ then the point will belong to the branch $-\sqrt{U}+\sqrt{V}+\sqrt{W}=0$.

Article 11. But the branch in question is clearly impossible for any series of real points; so that, leaving it an open question whether the epithet "impossible" is to be understood to mean impossible for any series of real points (that is, as a mere synonym of imaginary), or whether it is to mean impossible for any series of points, real or imaginary, whatever, I say that in a ν-zomal curve some of the branches are or may be impossible, and that there is at least one impossible branch, viz., the branch $\sqrt{U}+\sqrt{V}+\&c. =0$.

Article 12. For the purpose of referring to any branch of a polyzomal curve it will be convenient to consider $\sqrt{U}$ as signifying determinately $+\sqrt{U}$, or else $-\sqrt{U}$; and the like as regards $\sqrt{V}$, &c., but without any identity or relation between the signs prefixed to the $\sqrt{U}$, $\sqrt{V}$, &c., respectively; the equation $\sqrt{U}+\sqrt{V}+\&c.\ =0$, so understood, will be that of any determinate branch of the ν-zomal curve.

13 to 17. The Points common to Two Branches of a Polyzomal Curve.

Article 13. I say that two branches of a ν-zomal curve have in common $2^{\nu-2}r^2$ points, and that these are points belonging to a $(\nu-1)$-zomal curve.

Article 14. In fact, the equations of the two branches may be taken to be

$$\sqrt{U}+\sqrt{V}+\sqrt{W}+\&c.\ =0, \quad \sqrt{U}-\sqrt{V}+\sqrt{W}+\&c.\ =0,$$

(viz. the two branches differ only as to the sign of the radical $\sqrt{V}$); the two equations give $\sqrt{V}=0$, that is $V=0$, and

$$\sqrt{U}+\sqrt{W}+\&c.\ =0,$$

which last equation is that of a $(\nu-1)$-zomal curve. The intersections of the two branches are thus the $2^{\nu-2}r\cdot r=2^{\nu-2}r^2$ intersections of the curve $V=0$ with the $(\nu-1)$-zomal curve $\sqrt{U}+\sqrt{W}+\&c.\ =0$.

Article 15. If the two branches agree as to the sign of $\sqrt{U}$, but differ as to the sign of $\sqrt{V}$, then the points common to the two branches are points belonging to the $(\nu-1)$-zomal curve $\sqrt{U}+\sqrt{W}+\&c.\ =0$; and the like if the branches differ as to the sign of any other radical $\sqrt{V}$. So, if the branches differ as to the signs of any two radicals $\sqrt{V}$, $\sqrt{W}$, the points common to the two branches are points belonging to the $(\nu-2)$-zomal curve $\sqrt{U}+\&c.\ =0$; and so on. But the statement as originally made, that the points common to two branches are points belonging to a $(\nu-1)$-zomal, assumes that the two branches differ as to the sign of a single radical only.

Article 16. The several branches meet by pairs at $2^{\nu-3}r^2$ points on each of the curves $U=0$, $V=0$, $W=0$, &c. respectively; and the like as to the several curves which are represented by the equations obtained by attributing to the radicals $\sqrt{U}$, $\sqrt{V}$, &c. each of the two signs $+$ and $-$, and equating to zero the sum of any even number of the zomes; for instance, at the intersections of the curves $U=0$, $V=0$, we have $\sqrt{U}=0$, $\sqrt{V}=0$, and therefore $\sqrt{U}+\sqrt{V}+\sqrt{W}+\&c.\ =0$, $\sqrt{U}+\sqrt{V}-\sqrt{W}+\&c.\ =0$, &c.; that is, the several branches which contain the term $+\sqrt{U}+\sqrt{V}$ all pass through the r^2 intersections of the curves $U=0$, $V=0$; and so the several branches which contain the term $+\sqrt{U}-\sqrt{V}$ all pass through the same r^2 intersections: the number of branches which contain $+\sqrt{U}+\sqrt{V}$ is $2^{\nu-3}$, and the number of branches which contain $+\sqrt{U}-\sqrt{V}$ is $2^{\nu-3}$, so that the r^2 intersections of $U=0$, $V=0$ are each of them a point on $2^{\nu-3}+2^{\nu-3}=2^{\nu-2}$ branches; that is, on a pair of branches. And similarly at the r^2 intersections of any other pair of zomal curves $U=0$, $W=0$, $V=0$, $W=0$, &c. respectively. Also at the r^2 intersections of $U=0$ with the curve $\sqrt{V}+\sqrt{W}+\&c.\ =0$, the several branches which contain $+\sqrt{U}$ all meet the several branches which contain $-\sqrt{U}$; the number of each set of branches being $2^{\nu-2}$, we have thus at each of the $2^{\nu-2}r^2$ intersections in question a meeting of a pair of branches.

Article 17. The points common to two branches of a ν-zomal are singular points of the rationalised curve $\mathrm{Norm}(\sqrt{U}+\sqrt{V}+\&c.)=0$; but I do not at present inquire as to the nature of the singularity at any such point.

18 to 21. Singularities of a v-zomal Curve.

Article 18. Consider a branch $\sqrt{U}+\sqrt{V}+\sqrt{W}+\&c.\ =0$, and any point whatever on this branch; we have at this point $\sqrt{U}=-(\sqrt{V}+\sqrt{W}+\&c.)$, that is

$$U=(\sqrt{V}+\sqrt{W}+\&c.)^2,$$

and therefore

$$U-V-W-\&c.=2(\sqrt{VW}+\&c.),$$

where the term on the right hand includes the products of the radicals $\sqrt{V}$, $\sqrt{W}$, &c. taken two and two together. Hence, squaring and transposing,

$$(U-V-W-\&c.)^2-4(VW+\&c.)=8(\sqrt{VWX}+\&c.),$$

where the term on the right hand now includes the products of the radicals taken three and three together. Proceeding in this manner we ultimately obtain an equation wherein the right hand side contains only a single radical, say $\sqrt{\Theta}$, the left hand side being a rational function of (x,y,z); squaring, we obtain the rationalised equation $F=0$, where F is a rational function of (x,y,z), of the order $2^{\nu-1}r$.

Article 19. The rationalised equation may be obtained in a different form as follows: write the equation in the form $\sqrt{U}+\sqrt{V}=-(\sqrt{W}+\&c.)$; squaring, we obtain

$$U+V+2\sqrt{UV}=(\sqrt{W}+\&c.)^2,$$

and thence $2\sqrt{UV}=-(U+V)+(\sqrt{W}+\&c.)^2$, an equation containing on the right hand side a single radical less by one than the original equation; and so we ultimately obtain an equation containing on one side a single radical only, which on being squared gives the rationalised equation.

Article 20. In the case of a trizomal curve $\sqrt{U}+\sqrt{V}+\sqrt{W}=0$, the rationalised equation is

$$U^2+V^2+W^2-2UV-2UW-2VW=0,$$

that is

$$(U+V+W)^2-4(VW+WU+UV)=0,$$

or, as this may also be written,

$$(\sqrt{U}+\sqrt{V}+\sqrt{W})(\sqrt{U}+\sqrt{V}-\sqrt{W})(\sqrt{U}-\sqrt{V}+\sqrt{W})(-\sqrt{U}+\sqrt{V}+\sqrt{W})=0.$$

In the case of a tetrazomal curve $\sqrt{U}+\sqrt{V}+\sqrt{W}+\sqrt{X}=0$, the rationalised equation is of the order $8r$, and the number of terms is very large.

Article 21. If in the rationalised equation of the ν-zomal we consider any point whatever of the curve $\sqrt{U}+\sqrt{V}+\&c.\ =0$, we have of course $F=0$; and moreover $dF=0$, where dF is the differential of F taken in regard to the coordinates (x,y,z) in any manner whatever. But this does not establish that the point in question is a singular point of the rationalised curve; for in order that it may be a singular point, we require the equation $dF=0$ for all values whatever of $dx:dy:dz$, and this is not in general the case. In fact, for a point on the branch $\sqrt{U}+\sqrt{V}+\&c.\ =0$, the equations $F=0$, $dF=0$ imply only that this point belongs to the branch in question, and not that it is a singular point thereof.

22 to 27. Special Case where all the Zomals have a Common Point or Points.

Article 22. Suppose that the zomals $U = 0$, $V = 0$, $W = 0$, &c. all pass through the same point or points; this will be the case if U, V, W, &c. all contain as a factor one and the same function Θ of the coordinates, say if $U = \Theta \cdot U'$, $V = \Theta \cdot V'$, $W = \Theta \cdot W'$, &c., where Θ is a function of the degree m, and U', V', W', &c. are functions of the degree $r - m$. We have then

$$\sqrt{U} + \sqrt{V} + \&c. \ = \sqrt{\Theta}(\sqrt{U'} + \sqrt{V'} + \&c.).$$

The rationalised equation is thus $\Theta^{2^{\nu-2}} \cdot \mathrm{Norm}(\sqrt{U'} + \sqrt{V'} + \&c.) = 0$; that is, the rationalised equation contains the factor $\Theta^{2^{\nu-2}} = 0$, or the curve $\mathrm{Norm}(\sqrt{U} + \sqrt{V} + \&c.) = 0$ contains the curve $\Theta = 0$ as an $2^{\nu-2}$-tuple curve; and omitting this factor, the remaining factor $\mathrm{Norm}(\sqrt{U'} + \sqrt{V'} + \&c.) = 0$ is a curve of the order $2^{\nu-2}(r - m)$; that is, the order of the curve is depressed from $2^{\nu-2}r$ to $2^{\nu-2}(r - m)$, viz. the depression is $= 2^{\nu-2}m$.

Article 23. If the zomals have a common point of the multiplicity m, then the several branches of the ν-zomal curve all pass through this point; and the tangents at the common point are given by the equation obtained from the equation of any branch by substituting therein for the coordinates their values in terms of a parameter, and equating to zero the terms of the lowest degree in the parameter. For instance, if the common point be the point $(x = 0, y = 0)$, then writing $x = x_1 t$, $y = y_1 t$, where (x_1, y_1) are the coordinates of a point on the line joining the origin to the point in question, and t is a parameter, we obtain from the equation $\sqrt{U} + \sqrt{V} + \&c. \ = 0$ an equation wherein each of the functions U, V, &c. contains the factor t^m, and wherefore $\sqrt{U}$, $\sqrt{V}$, &c. each contain the factor $t^{\frac{1}{2}m}$; the equation thus becomes divisible by $t^{\frac{1}{2}m}$, and equating to zero the quotient, we obtain the equation of the tangents at the common point.

Article 24. If the zomals have two common points, each of the multiplicity m, then the rationalised equation contains the factor $\Theta^{2^{\nu-2}}$, where Θ is now the product of the two linear functions which represent the two common points, viz. Θ is of the degree $2m$; the depression of order is thus $= 2 \cdot 2^{\nu-2}m = 2^{\nu-1}m$. And similarly if the zomals have k common points, each of the multiplicity m, the depression of order is $= k \cdot 2^{\nu-2}m$.

Article 25. If the zomals have a common curve of the order m, then Θ is of the degree m, and the depression of order is $= 2^{\nu-2}m$, as in Art. 22. But if the common curve is a multiple curve on each of the zomals, say a k-tuple curve, then the depression is $= k \cdot 2^{\nu-2}m$.

Article 26. The case where the zomals have a common point or points is a particular case of a more general theory, which will be presently considered; but it is convenient to notice it here, as it affords a simple explanation of the depression of order which takes place in certain cases.

Article 27. In particular, if the zomals $U = 0$, $V = 0$, $W = 0$, &c. all pass through the same point, and this point is a simple point on each of the zomals, then the several branches of the ν-zomal curve all pass through this point, and the tangents at the point are given by the equation $\sqrt{L} + \sqrt{M} + \sqrt{N} + \&c. \ = 0$, where $L = 0$, $M = 0$, $N = 0$, &c. are the tangents of the zomals at the common point; that is, the tangents at the common point form a ν-zomal curve of the first order, the zomals being the tangents $L = 0$, $M = 0$, $N = 0$, &c. respectively.

28 to 37. Depression of Order of the v-zomal Curve from the Ideal Factor $z = 0$.

Article 28. I proceed to the theory of the depression of order from the ideal factor $z = 0$. The coordinates are throughout (x, y, z); and when the equation is given in terms of the Cartesian coordinates (x, y), it is to be understood that the coordinates are made homogeneous by the introduction of z in the proper power; thus the equation of a circle is written $(x - az)^2 + (y - \beta z)^2 = \gamma^2 z^2$, and the line infinity is $z = 0$. A curve ideally contains the line infinity when the equation in its homogeneous form contains the factor $z = 0$.

Article 29. Consider the trizomal curve $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$, where U, V, W are functions of the degree r, and where for greater generality the constant factors l, m, n are expressed. Suppose that the three zomals $U = 0$, $V = 0$, $W = 0$ all touch the line infinity; viz. that we have $U = (Az + \ldots)^2$, $V = (Bz + \ldots)^2$, $W = (Cz + \ldots)^2$, where the omitted terms are of the degree r in (x, y, z), but do not contain any terms of the form $(Az)^2$, $(Bz)^2$, $(Cz)^2$ respectively; that is, the terms of the degree r in (x, y) form perfect squares. We have thus $\sqrt{U} = Az + \ldots$, $\sqrt{V} = Bz + \ldots$, $\sqrt{W} = Cz + \ldots$, and the equation of the branch becomes

$$l(Az + \ldots) + m(Bz + \ldots) + n(Cz + \ldots) = 0,$$

that is

$$(lA + mB + nC)z + \&c. = 0.$$

Article 30. The line infinity will thus be a factor of the branch if $lA + mB + nC = 0$; and if this condition be satisfied, then the branch ideally contains the line infinity. The rationalised equation will thus contain the factor $z = 0$, corresponding to each branch which ideally contains the line infinity; and the depression of order is equal to the number of such branches.

Article 31. In the case of the trizomal curve, if the condition $lA + mB + nC = 0$ is satisfied, then each of the four branches ideally contains the line infinity; the rationalised equation thus contains the factor z^4, and the depression of order is $= 4$; the order of the curve is thus reduced from $4r$ to $4r - 4$.

Article 32. But if the condition $lA + mB + nC = 0$ is not satisfied, then the line infinity is not a factor of any branch; and the order of the curve remains $= 4r$. And similarly for the general ν-zomal curve: if the zomals all touch the line infinity, and if the coefficients l, m, &c. satisfy a certain condition, then each of the $2^{\nu-1}$ branches ideally contains the line infinity, and the depression of order is $= 2^{\nu-1}$; but if this condition is not satisfied, there is no depression of order.

Article 33. The condition $lA + mB + nC = 0$ may be geometrically interpreted as follows: $A = 0$, $B = 0$, $C = 0$ are the tangents at infinity to the zomals $U = 0$, $V = 0$, $W = 0$ respectively; the condition $lA + mB + nC = 0$ expresses that these three tangents meet in a point; that is, that the three points of contact of the zomals with the line infinity lie on a right line. But if the zomals are circles, then A, B, C are constant, and the condition reduces to $lA + mB + nC = 0$, which is a relation between the coefficients l, m, n.

Article 34. In the case where the zomals are circles, the condition $lA + mB + nC = 0$ is satisfied if l, m, n are proportional to the squares of the radii of the circles; for then $l = kR^2$, $m = kR'^2$, $n = kR''^2$, where R, R', R'' are the radii; and $A = 1/R$, $B = 1/R'$, $C = 1/R''$; so that $lA + mB + nC = k(R + R' + R'')$, which is not in general zero. But if the circles have their centres on a right line, and if l, m, n are properly determined, the condition may be satisfied.

Article 35. The case where the zomals all touch the line infinity is a particular case of the more general case where the zomals all pass through a common point on the line infinity; but the theory is somewhat different, as the depression of order depends not only on the existence of the common point, but also on the relation between the coefficients l, m, n.

Article 36. If the zomals all pass through two common points on the line infinity, then each zomal contains the line infinity as a factor, and the depression of order is greater than in the case of a single common point. Thus, if each zomal contains the line infinity as a simple factor, we have $U = z \cdot U'$, $V = z \cdot V'$, $W = z \cdot W'$, where U', V', W' are of the degree $r - 1$; and the equation of the branch is $\sqrt{z}(\sqrt{lU'} + \sqrt{mV'} + \sqrt{nW'}) = 0$, which gives $z = 0$ as a factor of the rationalised equation; the depression of order is thus $= 2^{\nu-1}$, as before.

Article 37. But if each zomal contains the line infinity as a double factor, we have $U = z^2U'$, $V = z^2V'$, $W = z^2W'$, where U', V', W' are of the degree $r - 2$; and the equation of the branch is $z(\sqrt{lU'} + \sqrt{mV'} + \sqrt{nW'}) = 0$, which gives $z^{2^{\nu-1}} = 0$ as a factor of the rationalised equation; the depression of order is thus $= 2 \cdot 2^{\nu-1} = 2^\nu$.

38 and 39. On the Trizomal Curve and the Tetrazomal Curve.

Article 38. The trizomal curve

$$\sqrt{U} + \sqrt{V} + \sqrt{W} = 0$$

has for its rationalised form of equation

$$U^2 + V^2 + W^2 - 2VW - 2WU - 2UV = 0;$$

or as this may also be written,

$$(1, 1, 1, -1, -1, -1 \mid \sqrt{U}, \sqrt{V}, \sqrt{W})^2 = 0;$$

and we may from this rational equation verify the general results applicable to the case in hand, viz., that the trizomal is a curve of the order $2r$, and that

$$\begin{aligned} U = 0,&\ \text{at each of its } r^2 \text{ intersections with } V - W = 0,\\ V = 0,&\ \text{at its } r^2 \text{ intersections with } W - U = 0,\\ W = 0,&\ \text{at its } r^2 \text{ intersections with } U - V = 0, \end{aligned}$$

respectively touch the trizomal. There are not, in general, any nodes or cusps, and the order being $= 2r$, the class is $= 2r(2r - 1)$.

Article 39. The tetrazomal curve

$$\sqrt{U} + \sqrt{V} + \sqrt{W} + \sqrt{T} = 0$$

has for its rationalised form of equation

$$(U^2 + V^2 + W^2 + T^2 - 2UV - 2UW - 2UT - 2VW - 2VT - 2WT)^2 - 64UVWT = 0,$$

and we may hereby verify the fundamental properties, viz., that the tetrazomal is a curve of the order $4r$, touched by each of the zomals $U=0$, $V=0$, $W=0$, $T=0$ in $2r^2$ points, viz., by $U=0$ at its intersections with $\sqrt{V}+\sqrt{W}+\sqrt{T}=0$, that is, $V^2+W^2+T^2-2VW-2VT-2WT=0$; (and the like as regards the other zomals), and having $3r^4$ nodes, viz., these are the intersections of $(\sqrt{U}+\sqrt{V}=0, \sqrt{W}+\sqrt{T}=0)$, $(\sqrt{U}+\sqrt{W}=0, \sqrt{V}+\sqrt{T}=0)$, $(\sqrt{U}+\sqrt{T}=0, \sqrt{V}+\sqrt{W}=0)$, or, what is the same thing, the intersections of $(U-V=0, W-T=0)$, $(U-W=0, V-T=0)$, $(U-T=0, V-W=0)$. There are not in general any cusps, and the class is thus $= 4r(4r-1)-6r^4$, $= 16r^2-4r-6r^4$.

40 and 41. On the Intersection of two v-Zomals having the same Zomal Curves.

Article 40. Without going into any detail, I may notice the question of the intersection of two ν-zomals which have the same zomal curves: say the two trizomals $\sqrt{lU}+\sqrt{mV}+\sqrt{nW}=0$, $\sqrt{l'U}+\sqrt{m'V}+\sqrt{n'W}=0$, or two similarly related tetrazomals. For the trizomals, writing the equations under the form

$$\sqrt{U}+\sqrt{V}+\sqrt{W}=0, \quad \sqrt{l}\sqrt{U}+\sqrt{m'}\sqrt{V}+\sqrt{n'}\sqrt{W}=0,$$

then, when these equations are considered as existing simultaneously, we may, without loss of generality, attribute to the radicals $\sqrt{U}$, $\sqrt{V}$, $\sqrt{W}$, the same values in the two equations respectively; but doing so, we must in the second equation successively attribute to all but one of the radicals $\sqrt{l'}$, $\sqrt{m'}$, $\sqrt{n'}$, each of its two opposite values. For the intersections of the two curves we have thus

$$\sqrt{U}:\sqrt{V}:\sqrt{W}=\sqrt{m'}-\sqrt{n'}:\sqrt{n'}-\sqrt{l'}:\sqrt{l'}-\sqrt{m'},$$

viz., this is one of a system of four equations, obtained from it by changes of sign, say in the radicals $\sqrt{m'}$ and $\sqrt{n'}$. Each of the four equations gives a set of r^2 points; we have thus the complete number, $=4r^2$, of the points of intersection of the two curves.

Article 41. But take, in like manner, two tetrazomal curves; writing their equations in the form

$$\begin{aligned} \sqrt{U}+\sqrt{V}+\sqrt{W}+\sqrt{T}&=0, \\ \sqrt{l'}\sqrt{U}+\sqrt{m'}\sqrt{V}+\sqrt{n'}\sqrt{W}+\sqrt{p'}\sqrt{T}&=0, \end{aligned}$$

then $\sqrt{U}$, $\sqrt{V}$, $\sqrt{W}$, $\sqrt{T}$ may be considered as having the same values in the two equations respectively, but we must in the second equation attribute successively, say to $\sqrt{m'}$, $\sqrt{n'}$, $\sqrt{p'}$, each of their two opposite values. For the intersections of the two curves we have

$$\begin{aligned} (\sqrt{m'}-\sqrt{l'})\sqrt{V}+(\sqrt{n'}-\sqrt{l'})\sqrt{W}+(\sqrt{p'}-\sqrt{l'})\sqrt{T}&=0, \\ (\sqrt{l'}-\sqrt{m'})\sqrt{U} \qquad +(\sqrt{n'}-\sqrt{m'})\sqrt{W}+(\sqrt{p'}-\sqrt{m'})\sqrt{T}&=0, \end{aligned}$$

viz., this is one of a system of eight similar pairs of equations, obtained therefrom by changes of sign of the radicals $\sqrt{m'}$, $\sqrt{n'}$, $\sqrt{p'}$. The equations represent each of them a trizomal curve, of the order $2r$; the two curves intersect therefore in $4r^2$ points, and if each of these was a point of intersection of the two tetrazomals, we should have in all $8\times 4r^2=32r^2$ intersections. But the tetrazomals are each of them a curve of the order $4r$, and they intersect therefore in only $16r^2$ points. The explanation is, that not all the $4r^2$ points, but only $2r^2$ of them are intersections of the tetrazomals. In fact, to find all the intersections of the two trizomals, it is necessary in their two equations to attribute opposite signs to one of the radicals $\sqrt{U}$, $\sqrt{V}$; we obtain $2r^2$ intersections from the equations as they stand, the remaining $2r^2$ intersections from the two equations after we have in the second equation reversed the sign, say of $\sqrt{V}$.

Now, from the two equations as they stand we can pass back to the two tetrazomal equations, and the first-mentioned $2r^2$ points are thus points of intersection of the two tetrazomal curves: from the two equations after such reversal of the sign of $\sqrt{V}$, we cannot pass back to the two tetrazomal equations, and the last-mentioned $2r^2$ points are thus not points of intersection of the two tetrazomal curves. The number of intersections of the two curves is thus $8 \times 2r^2 = 16r^2$, as it should be.

42 to 45. The Theorem of the Decomposition of a Tetrazomal Curve.

Article 42. I consider the tetrazomal curve

$$\sqrt{lU} + \sqrt{mV} + \sqrt{nW} + \sqrt{pT} = 0,$$

where the zomal curves are in involution, that is, where we have an identical relation,

$$aU + bV + cW + dT = 0;$$

and I proceed to show that if l, m, n, p satisfy the relation

$$\frac{l}{a} + \frac{m}{b} + \frac{n}{c} + \frac{p}{d} = 0,$$

the curve breaks up into two trizomals. In fact, writing the equation under the form

$$(\sqrt{l}\sqrt{U} + \sqrt{m}\sqrt{V} + \sqrt{n}\sqrt{W})^2 - pT = 0,$$

and substituting for T its value, in terms of U, V, W, this is

$$(ld+pa)U + (md+pb)V + (nd+pc)W + 2\sqrt{mn}\,d\sqrt{VW} + 2\sqrt{nl}\,d\sqrt{WU} + 2\sqrt{lm}\,d\sqrt{UV} = 0;$$

or, considering the left-hand side as a quadric function of $(\sqrt{U}, \sqrt{V}, \sqrt{W})$, the condition for its breaking up into factors is

$$\begin{vmatrix} ld+pa & d\sqrt{mn} & d\sqrt{nl} \\ d\sqrt{mn} & md+pb & d\sqrt{lm} \\ d\sqrt{nl} & d\sqrt{lm} & nd+pc \end{vmatrix} = 0,$$

that is

$$d^2(lbcd + mcda + ndab + pabc) = 0,$$

or finally, the condition is

$$\frac{l}{a} + \frac{m}{b} + \frac{n}{c} + \frac{p}{d} = 0.$$

Article 43. Multiplying by $\frac{1}{d}(ld+pa)$, and observing that in virtue of the relation we have

$$\tfrac{1}{d}(ld+pa) \cdot \tfrac{1}{d}(md+pb) = \tfrac{1}{d^2}(lm + pa \cdot \tfrac{m}{b} \cdot \tfrac{b}{a} + \ldots),$$

the equation becomes

$$\left(\sqrt{ld+pa}\,\sqrt{U}+\sqrt{md+pb}\,\sqrt{V}+\sqrt{nd+pc}\,\sqrt{W}\right)^2=\left(\sqrt{\tfrac{p}{d}}\sqrt{bV-cW}\right)^2,$$

or as this is more conveniently written

$$\left(\sqrt{l_1}\sqrt{U}+\sqrt{m_1}\sqrt{V}+\sqrt{n_1}\sqrt{W}\right)\left(\sqrt{l_2}\sqrt{U}+\sqrt{m_2}\sqrt{V}+\sqrt{n_2}\sqrt{W}\right)=0,$$

an equation breaking up into two equations, which may be represented by

$$\sqrt{l_1U}+\sqrt{m_1V}+\sqrt{n_1W}=0,\quad \sqrt{l_2U}+\sqrt{m_2V}+\sqrt{n_2W}=0,$$

where

$$\sqrt{l_1}=\sqrt{ld+pa}-\sqrt{\tfrac{p}{d}}\sqrt{bc},\quad \sqrt{l_2}=\sqrt{ld+pa}+\sqrt{\tfrac{p}{d}}\sqrt{bc},$$
$$\sqrt{m_1}=\sqrt{md+pb}-\sqrt{\tfrac{p}{d}}\sqrt{ca},\quad \sqrt{m_2}=\sqrt{md+pb}+\sqrt{\tfrac{p}{d}}\sqrt{ca},$$
$$\sqrt{n_1}=\sqrt{nd+pc}-\sqrt{\tfrac{p}{d}}\sqrt{ab},\quad \sqrt{n_2}=\sqrt{nd+pc}+\sqrt{\tfrac{p}{d}}\sqrt{ab},$$

where, in the expressions for $\sqrt{l_1}$, &c., the signs of the radicals may be taken determinately in any way whatever at pleasure: the only effect of an alteration of sign would in some cases be to interchange the values of $(\sqrt{l_1},\sqrt{m_1},\sqrt{n_1})$ with those of $(\sqrt{l_2},\sqrt{m_2},\sqrt{n_2})$. The tetrazomal curve thus breaks up into two trizomals.

Article 44. It is to be noticed that we have

$$\frac{l_1}{a}+\frac{m_1}{b}+\frac{n_1}{c}=\frac{1}{d}(ld+pa+md+pb+nd+pc)-\sqrt{\tfrac{p}{d}}\left(\sqrt{\tfrac{bc}{a}}+\sqrt{\tfrac{ca}{b}}+\sqrt{\tfrac{ab}{c}}\right),$$

that is

$$\frac{l_1}{a}+\frac{m_1}{b}+\frac{n_1}{c}=-\frac{p}{d}\left(\frac{d}{a}+\frac{d}{b}+\frac{d}{c}+1\right)=0,$$

and that similarly we have

$$\frac{l_2}{a}+\frac{m_2}{b}+\frac{n_2}{c}=0.$$

The meaning is, that, taking the trizomal curve $\sqrt{l_1U}+\sqrt{m_1V}+\sqrt{n_1W}=0$, this regarded as a tetrazomal curve, $\sqrt{l_1U}+\sqrt{m_1V}+\sqrt{n_1W}+\sqrt{0\cdot T}=0$, satisfies the condition $\frac{l_1}{a}+\frac{m_1}{b}+\frac{n_1}{c}+\frac{0}{d}=0$; and the like as to the trizomal curve $\sqrt{l_2U}+\sqrt{m_2V}+\sqrt{n_2W}=0$.

Article 45. The equation by which the decomposition was effected is, it is clear, one of twelve equivalent equations: four of these are

$$\left(\sqrt{ld+pa}\,\sqrt{U}-\sqrt{\tfrac{p}{d}}\sqrt{bV-cW}\right)\left(\sqrt{ld+pa}\,\sqrt{U}+\sqrt{\tfrac{p}{d}}\sqrt{bV-cW}\right)=0,$$
$$\left(\sqrt{md+pb}\,\sqrt{V}-\sqrt{\tfrac{p}{d}}\sqrt{cW-aU}\right)\left(\sqrt{md+pb}\,\sqrt{V}+\sqrt{\tfrac{p}{d}}\sqrt{cW-aU}\right)=0,$$
$$\left(\sqrt{nd+pc}\,\sqrt{W}-\sqrt{\tfrac{p}{d}}\sqrt{aU-bV}\right)\left(\sqrt{nd+pc}\,\sqrt{W}+\sqrt{\tfrac{p}{d}}\sqrt{aU-bV}\right)=0,$$

and the others may be deduced from these by a cyclical permutation of (U,V,W), (a,b,c), (l,m,n), leaving T, d, p unaltered.

46 to 51. Application to the Trizomal; the Theorem of the Variable Zomal.

Article 46. I take the last equation written under the form

$$\left(\sqrt{\tfrac{m}{b}}\sqrt{U}-\sqrt{\tfrac{l}{a}}\sqrt{V}\right)^2=\left(\sqrt{\tfrac{p}{d}}\sqrt{\tfrac{bc}{a}}\sqrt{U}+\sqrt{\tfrac{p}{d}}\sqrt{\tfrac{ca}{b}}\sqrt{V}+\sqrt{n}\sqrt{W}\right)^2,$$

which, putting therein $p=0$, is

$$\left(\sqrt{\tfrac{m}{b}}\sqrt{U}-\sqrt{\tfrac{l}{a}}\sqrt{V}\right)^2=nW,$$

which is in fact the trizomal curve,

$$\sqrt{\tfrac{m}{b}}\sqrt{U}+\sqrt{\tfrac{l}{a}}\sqrt{V}+\sqrt{n}\sqrt{W}=0,$$

viz., the trizomal curve $\sqrt{lU}+\sqrt{mV}+\sqrt{nW}=0$, if a, b, c be any quantities connected by the equation

$$\frac{l}{a}+\frac{m}{b}+\frac{n}{c}=0$$

(the ratios $a:b:c$ thus involving a single arbitrary parameter); and if we take T a function such that $aU+bV+cW+dT=0$; that is, $T=0$, any one of the series of curves $aU+bV+cW=0$, in involution with the given curves $U=0$, $V=0$, $W=0$, has its equation expressible in the form

$$\sqrt{\tfrac{m}{b}}\sqrt{U}+\sqrt{\tfrac{l}{a}}\sqrt{V}+\sqrt{n}\sqrt{T}=0;$$

that is, we have the curve (the equation whereof contains a variable parameter) as a zomal of the given trizomal curve $\sqrt{lU}+\sqrt{mV}+\sqrt{nW}=0$; and we have thus from the theorem of the decomposition of a tetrazomal deduced the theorem of the variable zomal of a trizomal. The analytical investigation is somewhat simplified by assuming $p=0$ ab initio, and it may be as well to repeat it in this form.

Article 47. Starting, then, with the trizomal curve

$$\sqrt{lU}+\sqrt{mV}+\sqrt{nW}=0,$$

and writing

$$aU+bV+cW+dT=0$$

as the definition of T, the coefficients being connected by

$$\frac{l}{a}+\frac{m}{b}+\frac{n}{c}=0,$$

the equation gives

$$lU+mV+2\sqrt{lmUV}-nW=0;$$

or substituting in this equation for W its value in terms of U, V, T, we have

$$(an+cl)U+(bn+cm)V+2c\sqrt{lmUV}+dnT=0,$$

which by the given relation between a, b, c, is converted into

$$\sqrt{m}U-\sqrt{l}V+2c\sqrt{lmUV}+dnT=0;$$

that is

$$c^2lm\cdot U+l\cdot V-2ab\sqrt{lmUV}=nT,$$

viz., this is

$$\left(c\sqrt{m}\sqrt{U}-\sqrt{l}\sqrt{V}\right)^2=nT,$$

or finally

$$\sqrt{mU}+\sqrt{lV}+\sqrt{nT}=0.$$

Article 48. The result just obtained of course implies that when as above

$$aU+bV+cW+dT=0,\quad \frac{l}{a}+\frac{m}{b}+\frac{n}{c}=0,$$

the trizomal curve $\sqrt{lU}+\sqrt{mV}+\sqrt{nW}=0$ can be expressed by means of any three of the four zomals U, V, W, T; and we may at once write down the four forms

$$\begin{aligned}\sqrt{\tfrac{m}{b}}\sqrt{U}+\sqrt{\tfrac{l}{a}}\sqrt{V}+\sqrt{n}\sqrt{W}&=0,\\ \sqrt{\tfrac{m}{b}}\sqrt{U}+\sqrt{\tfrac{l}{a}}\sqrt{V}+\sqrt{n}\sqrt{T}&=0,\\ \sqrt{\tfrac{n}{c}}\sqrt{U}+\sqrt{\tfrac{l}{a}}\sqrt{W}+\sqrt{m}\sqrt{T}&=0,\\ \sqrt{\tfrac{n}{c}}\sqrt{V}+\sqrt{\tfrac{m}{b}}\sqrt{W}+\sqrt{l}\sqrt{T}&=0,\end{aligned}$$

the last of which is the original equation $\sqrt{lU}+\sqrt{mV}+\sqrt{nW}=0$. It may be added that if the first equation be represented by $\sqrt{l_1U}+\sqrt{m_1V}+\sqrt{p_1T}=0$, that is, if we have

$$\sqrt{l_1}=\sqrt{\tfrac{m}{b}},\quad \sqrt{m_1}=\sqrt{\tfrac{l}{a}},\quad \sqrt{p_1}=\sqrt{n},$$

and therefore

$$\frac{l_1}{b}+\frac{m_1}{a}+\frac{p_1}{d}=\frac{1}{bc}\left(\frac{l}{a}+\frac{m}{b}+\frac{n}{c}\right)=0,$$

or if the second equation be represented by $\sqrt{l_2U}+\sqrt{n_2W}+\sqrt{p_2T}=0$, that is, if we have

$$\sqrt{l_2}=\sqrt{\tfrac{m}{b}},\quad \sqrt{n_2}=-\sqrt{\tfrac{l}{c}},\quad \sqrt{p_2}=\sqrt{n},$$

and therefore

$$\frac{l_2}{b}+\frac{n_2}{c}+\frac{p_2}{d}=0;$$

or if the third equation be represented by $\sqrt{l_3V}+\sqrt{m_3W}+\sqrt{p_3T}=0$, that is, if we have

$$\sqrt{l_3}=\sqrt{\tfrac{n}{c}},\quad \sqrt{m_3}=-\sqrt{\tfrac{l}{a}},\quad \sqrt{p_3}=\sqrt{m},$$

and therefore

$$\frac{l_3}{c}+\frac{m_3}{a}+\frac{p_3}{d}=0;$$

then the equation of the trizomal may also be expressed in the forms

$$\begin{aligned}(\sqrt{l_1U},\sqrt{m_1V},\sqrt{p_1T}\mid\sqrt{lU},\sqrt{mV},\sqrt{nW},\sqrt{lT})&=0,\\ (\sqrt{l_2U},\sqrt{n_2W},\sqrt{p_2T}\mid\sqrt{lU},\sqrt{mV},\sqrt{nW},\sqrt{mT})&=0,\\ (\sqrt{l_3V},\sqrt{m_3W},\sqrt{p_3T}\mid\sqrt{lU},\sqrt{mV},\sqrt{nW},\sqrt{nT})&=0.\end{aligned}$$

Article 49. These equations may, however, be expressed in a much more elegant form. Write

$$a = \tfrac{a'}{(\beta\gamma\delta)}, \quad b = \tfrac{b'}{(\gamma\delta\alpha)}, \quad c = \tfrac{c'}{(\delta\alpha\beta)}, \quad d = \tfrac{d'}{(\alpha\beta\gamma)},$$

where, for shortness, $(\beta\gamma\delta) = (\beta-\gamma)(\gamma-\delta)(\delta-\beta)$, &c.; (α, β, γ) being arbitrary quantities; or, what is the same thing,

$$a : b : c : d = a'(\beta\gamma\delta) : -b'(\gamma\delta\alpha) : c'(\delta\alpha\beta) : -d'(\alpha\beta\gamma).$$

And if

$$l : m : n = pa'(\beta-\gamma)^2 : qb'(\gamma-\alpha)^2 : rc'(\alpha-\beta)^2,$$

then the equation $\frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0$ takes the form

$$p(\beta-\gamma)(\alpha-\delta) + q(\gamma-\alpha)(\beta-\delta) + r(\alpha-\beta)(\gamma-\delta) = 0,$$

and the four forms of the equation are found to be

$$\begin{aligned}
&\Big(\sqrt{p(\beta-\gamma)}, \sqrt{q(\gamma-\alpha)}, \sqrt{r(\alpha-\beta)}\Big)\,(\sqrt{a'U}, \sqrt{b'V}, \sqrt{c'W}, \sqrt{d'T}) = 0,\\
&\Big(\sqrt{p(\beta-\gamma)}, \sqrt{r(\beta-\delta)}, \sqrt{q(\gamma-\delta)}\Big)\,(\sqrt{a'U}, \sqrt{b'V}, \sqrt{d'T}, \sqrt{c'W}) = 0,\\
&\Big(\sqrt{q(\alpha-\gamma)}, \sqrt{r(\alpha-\delta)}, \sqrt{p(\gamma-\delta)}\Big)\,(\sqrt{a'U}, \sqrt{c'W}, \sqrt{d'T}, \sqrt{b'V}) = 0,\\
&\Big(\sqrt{r(\beta-\alpha)}, \sqrt{q(\delta-\alpha)}, \sqrt{p(\beta-\delta)}\Big)\,(\sqrt{b'V}, \sqrt{c'W}, \sqrt{d'T}, \sqrt{a'U}) = 0,
\end{aligned}$$

viz., these are the equivalent forms of the original equation assumed to be

$$(\beta-\gamma)\sqrt{pa'U} + (\gamma-\alpha)\sqrt{qb'V} + (\alpha-\beta)\sqrt{rc'W} = 0.$$

Article 50. I remark that the theorem of the variable zomal may be obtained as a transformation theorem: viz., comparing the equation $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$ with the equation $\sqrt{lx} + \sqrt{my} + \sqrt{nz} = 0$; this last belongs to a conic touched by the three lines $x = 0$, $y = 0$, $z = 0$; the equation of the same conic must, it is clear, be expressible in a similar form by means of any other three tangents thereof, but the equation of any tangent of the conic is $ax + by + cz = 0$, where a, b, c are any quantities satisfying the condition $\frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0$; whence, writing $ax + by + cz + dw = 0$, we may introduce w along with any two of the original zomals $x = 0$, $y = 0$, $z = 0$, or, instead of them, any three functions of the form w; and then the mere change of x, y, z, w into U, V, W, T gives the theorem. But it is as easy to conduct the analysis with (U, V, W, T) as with (x, y, z, w), and, so conducted, it is really the same analysis as that whereby the theorem is established ante, No. 47.

Article 51. It is worth while to exhibit the equation of the curve

$$\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0,$$

in a form containing three new zomals. Observe that the equation $\frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0$ is satisfied by $a = l\phi\psi$, $b = m\phi\chi$, $c = n\psi\chi$, if only $\frac{1}{\phi} + \frac{1}{\psi} + \frac{1}{\chi} = 0$; or say, if $\phi = \alpha' - \alpha''$, $\psi = \alpha'' - \alpha$, $\chi = \alpha - \alpha'$. The equation

$$\lambda\sqrt{(\alpha-\alpha')(\alpha-\alpha'')lU} \;+\; \mu\sqrt{(\alpha'-\alpha'')(\alpha'-\alpha)mV} \;+\; \nu\sqrt{(\alpha''-\alpha)(\alpha''-\alpha')nW} \;=\; 0$$

is consequently an equation involving three zomals of the proper form; and we can determine λ, μ, ν in suchwise as to identify this with the original equation $\sqrt{lU} + \sqrt{mV} + \sqrt{nW}$, viz., writing successively $U = 0$, $V = 0$, $W = 0$, we find

$$\begin{aligned}(\alpha' - \alpha'')\lambda + (\beta' - \beta'')\mu + (\gamma' - \gamma'')\nu &= 0,\\ (\alpha'' - \alpha)\lambda + (\beta'' - \beta)\mu + (\gamma'' - \gamma)\nu &= 0,\\ (\alpha - \alpha')\lambda + (\beta - \beta')\mu + (\gamma - \gamma')\nu &= 0,\end{aligned}$$

equations which are, as they should be, equivalent to two equations only, and which give

$$\lambda : \mu : \nu = \begin{vmatrix} 1 & 1 & 1 \\ \beta & \beta' & \beta'' \\ \gamma & \gamma' & \gamma'' \end{vmatrix} : \begin{vmatrix} 1 & 1 & 1 \\ \gamma & \gamma' & \gamma'' \\ \alpha & \alpha' & \alpha'' \end{vmatrix} : \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \alpha' & \alpha'' \\ \beta & \beta' & \beta'' \end{vmatrix},$$

and the equation, with these values of λ, μ, ν substituted therein, is in fact the equation of the trizomal curve

$$\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$$

in terms of three new zomals.

It is easy to return to the forms involving one new zomal and any two of the original three zomals.

52. Remark as to the Tetrazomal Curve.

Article 52. I return for a moment to the case of the tetrazomal curve, in order to show that there is not, in regard to it in general, any theorem such as that of the variable zomal. Considering the form $\sqrt{lx} + \sqrt{my} + \sqrt{nz} + \sqrt{pw} = 0$ (the coordinates x, y, z, w are of course connected by a linear equation, but nothing turns upon this), the curve is here a quartic touched twice by each of the lines $x = 0$, $y = 0$, $z = 0$, $w = 0$ (viz., each of these is a double tangent of the curve), and having besides the three nodes $(x = y, z = w)$, $(x = z, y = w)$, $(x = w, y = z)$. But a quartic curve with three nodes, or trinodal quartic, has only four double tangents: that is, besides the lines $x = 0$, $y = 0$, $z = 0$, $w = 0$, there is no line $\alpha x + \beta y + \gamma z + \delta w = 0$ which is a double tangent of the curve; and writing U, V, W, T in place of x, y, z, w, then if U, V, W, T are connected by a linear equation (and, *a fortiori*, if they are not so connected), there is not any curve $\alpha U + \beta V + \gamma W + \delta T = 0$ which is related to the curve in the same way with the lines $U = 0$, $V = 0$, $W = 0$, $T = 0$; or say there is not (besides the curves $U = 0$, $V = 0$, $W = 0$, $T = 0$), any other zomal $\alpha U + \beta V + \gamma W + \delta T = 0$, of the tetrazomal curve. The proof does not show that for special forms of U, V, W, T there may not be zomals, not of the above form $\alpha U + \beta V + \gamma W + \delta T = 0$, but belonging to a separate system. An instance of this will be mentioned in the sequel.

53 to 56. The Theorem of the Variable Zomal of a Trizomal Curve resumed.

Article 53. I resume the foregoing theorem of the variable zomal of the trizomal curve $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$. The variable zomal $T = 0$ is the curve $aU + bV + cW = 0$, where a, b, c are connected by the equation $\frac{l}{a} + \frac{m}{b} + \frac{n}{c} = 0$; that is, it belongs to a single series of curves selected in a certain manner out of the double series $aU + bV + cW = 0$ (a double series, as containing the two variable parameters $a : b : c$). These are the whole series of curves in involution with the given curves $U = 0$, $V = 0$, $W = 0$, or being such that the

Jacobian of any three of them is identical with the Jacobian of the three given curves; in particular, the Jacobian of any one of the curves $aU+bV+cW=0$, and of two of the three given curves, is identical with the Jacobian of the three given curves. I call to mind that, by the Jacobian of the curves $U=0$, $V=0$, $W=0$, is meant the curve

$$J(U,V,W)=\begin{vmatrix}\partial_x U & \partial_y U & \partial_z U\\ \partial_x V & \partial_y V & \partial_z V\\ \partial_x W & \partial_y W & \partial_z W\end{vmatrix}=0,$$

viz., the curve obtained by equating to zero the Jacobian or functional determinant of the functions U, V, W. Some properties of the Jacobian, which are material as to what follows, are mentioned in the Annex No. I.

For the complete statement of the theorem of the variable zomal, it would be necessary to interpret geometrically the condition $\frac{l}{a}+\frac{m}{b}+\frac{n}{c}=0$, thereby showing how the single series of the variable zomal is selected out of the double series of the curves $aU+bV+cW=0$ in involution with the given curves. Such a geometrical interpretation of the condition may be sought for as follows, but it is only in a particular case, as afterwards mentioned, that a convenient geometrical interpretation is thereby obtained.

Article 54. Consider the fixed line $\Omega=px+qy+rz=0$, and let it be proposed to find the locus of the $(r-1)^2$ pole of the line $\Omega=0$ in regard to the series of curves $aU+bV+cW=0$, where $\frac{l}{a}+\frac{m}{b}+\frac{n}{c}=0$. Take (x,y,z) as the coordinates of any one of the poles in question; then in order that (x,y,z) may belong to one of the $(r-1)^2$ poles of the line $\Omega=px+qy+rz=0$ in regard to the curve $aU+bV+cW=0$, we must have

$$\partial_x(aU+bV+cW):\partial_y(aU+bV+cW):\partial_z(aU+bV+cW)=p:q:r;$$

or, what is the same thing,

$$a\,\partial_x U+b\,\partial_x V+c\,\partial_x W:a\,\partial_y U+b\,\partial_y V+c\,\partial_y W:a\,\partial_z U+b\,\partial_z V+c\,\partial_z W=p:q:r;$$

and these equations give without difficulty

$$a:b:c=J(V,W,\Omega):J(W,U,\Omega):J(U,V,\Omega),$$

whence, substituting in the equation $\frac{l}{a}+\frac{m}{b}+\frac{n}{c}=0$, we have

$$\frac{l}{J(V,W,\Omega)}+\frac{m}{J(W,U,\Omega)}+\frac{n}{J(U,V,\Omega)}=0$$

as the equation of the locus in question.

Article 55. In the particular case where $r=2$, the curves $U=0$, $V=0$, $W=0$ are conics; the $(r-1)^2$ pole is the single pole of the line $\Omega=0$ in regard to the conic $aU+bV+cW=0$; and the locus is the curve

$$\frac{l}{J(V,W,\Omega)}+\frac{m}{J(W,U,\Omega)}+\frac{n}{J(U,V,\Omega)}=0,$$

which is a cubic curve. This cubic curve is in fact the locus of the poles of the line $\Omega=0$ in regard to the series of conics $aU+bV+cW=0$, where $\frac{l}{a}+\frac{m}{b}+\frac{n}{c}=0$.

Article 56. The theorem of the variable zomal may be stated as follows: Given the trizomal curve $\sqrt{lU}+\sqrt{mV}+\sqrt{nW}=0$, if $T=0$ be any curve of the series $aU+bV+cW=0$, where $\frac{l}{a}+\frac{m}{b}+\frac{n}{c}=0$, then the trizomal curve may be expressed by means of the three zomals U, V, T; or U, W, T; or V, W, T; and the equation will be of the form $\sqrt{l'U}+\sqrt{m'V}+\sqrt{n'T}=0$, where l', m', n' are properly determined. The series of curves $T=0$ is thus a series of variable zomals of the given trizomal curve.

Part II. (Nos. 57 to 104).—Subsidiary Investigations.

Article 57. We have just been led to consider the conics which pass through two given points. There is no real loss of generality in taking these to be the circular points at infinity, or say the points I, J; viz., every theorem which in anywise explicitly or implicitly relates to these two points, may, without the necessity of any change in the statement thereof, be understood as a theorem relating instead to any two points P, Q.

I call to mind that a circle is a conic passing through the two points I, J, and that lines at right angles to each other are lines harmonically related to the pair of lines from their intersection to the points I, J respectively, so that when (I, J) are replaced by any two given points whatever, the expression a circle must be understood to mean a conic passing through the two given points; and in speaking of lines at right angles to each other, it must be understood that we mean lines harmonically related to the pair of lines from their intersection to the two given points respectively.

For instance, the theorem that the Jacobian of any three circles is their orthotomic circle, will mean that the Jacobian of any three conics which each of them passes through the two given points is the orthotomic conic through the same two points, that is, the conic such that at each of its intersections with any one of the three conies, the two tangents are harmonically related to the pair of lines from this intersection to the two given points respectively. Such extended interpretation of any theorem is applicable even to the theorems which involve distances or angles; viz., the terms "distance" and "angle" have a determinate signification when interpreted in reference (not to the circular points at infinity, but instead thereof) to any two given points whatever (see as to this my "Sixth Memoir on Quantics," Nos. 220, *et seq.*, *Phil. Trans.*, vol. CXLIX. (1859), pp. 61–90; see p. 86 [158]). And this being so, the theorem can, without change in the statement thereof, be understood as referring to the two given points.

Article 58. I say then that any theorem (referring explicitly or implicitly) to the circular points at infinity I, J, may be understood as a theorem referring instead to any two given points. We might of course give the theorems in the first instance in terms explicitly referring to the two given points (viz., instead of a circle, speak of a conic through the two given points, and so in other instances); but, as just explained, this is not really more general, and the theorems would be given in a less concise and familiar form. It would not, on the face of the investigations, be apparent that in treating of the polyzomal curves

$$\sqrt{l(\Theta+\lambda\Phi)}+\sqrt{m(\Theta+\mu\Phi)}+\&c.\ =0,$$

($\Theta =$ a conic, $\Phi =$ a line, as above), that we were really treating of the curves the zomals whereof are circles, and therein of the theories of foci and foci-foci as about to be explained. And for these reasons I shall consider the two points $\Theta = 0$, $\Phi = 0$, to be the circular points at infinity I, J, and in the investigations, &c., make use of the terms circle, right angles, &c., which, in their ordinary significations, have implicit reference to these two points.

The present Part does not explicitly relate to the theory of polyzomal curves, but contains a series of researches, partly analytical and partly geometrical, which will be made use of in the following Parts III. and IV. of the Memoir.

59 to 62. The Circular Points at Infinity; Rectangular and Circular Coordinates.

Article 59. The coordinates made use of (except in the cases where the general trilinear coordinates (x, y, z), or any other coordinates, are explicitly referred to), will be either the ordinary rectangular coordinates x, y, or else, as we may term them, the circular coordinates ξ, η ($= x + iy$, $x - iy$ respectively, $i = \sqrt{-1}$ as usual), but in either case I shall introduce for homogeneity the coordinate z, it being understood that this coordinate is in fact $= 1$, and that it may be retained or replaced by this its value, in different investigations or stages of the same investigation, as may for the time being be most convenient. In more concise terms, we may say that the coordinates are either the rectangular coordinates x, y, and z ($= 1$), or else the circular coordinates ξ, η, and z ($= 1$). The equation of the line infinity is $z = 0$; the points I, J are given by the equations $(x + iy = 0, z = 0)$ and $(x - iy = 0, z = 0)$, or, what is the same thing, by the equations $(\xi = 0, z = 0)$ and $(\eta = 0, z = 0)$ respectively; or in the rectangular coordinates the coordinates of these points are $(-i, 1, 0)$ and $(i, 1, 0)$ respectively, and in the circular coordinates they are $(1, 0, 0)$ and $(0, 1, 0)$ respectively. It is, of course, only for points at infinity that the coordinate z is $= 0$ (and observe that for any such point the x and y or ξ and η coordinates may be regarded as finite); for every point whatever not at infinity the coordinate z is, as stated above, $= 1$.

Article 60. Consider a point A, whose coordinates (rectangular) are $(a, a', 1)$ and (circular) $(\alpha, \alpha', 1)$, viz., $\alpha = a + a'i$, $\alpha' = a - a'i$; then the equations of the lines through A to the points I, J, are

$$x - az + i(y - a'z) = 0, \quad x - az - i(y - a'z) = 0$$

respectively, or they are

$$\xi - \alpha z = 0, \quad \eta - \alpha' z = 0$$

respectively. These equations, if (a, a') or (α, α') are arbitrary, will, it is clear, be the equations of any two lines through the points I, J, respectively.

Article 61. We have from either of the equations in (x, y, z)

$$(x - az)^2 + (y - a'z)^2 = 0,$$

that is, the distance from each other of any two points $(x, y, 1)$, and $(a, a', 1)$ in a line through I or J is $= 0$. And in particular, if $z = 0$, then $x^2 + y^2 = 0$; that is, the distance of the point $(a, a', 1)$ from I or J is in each case $= 0$.

Article 62. Consider for a moment any three points P, Q, A; the perpendicular distance of P from QA is $= 2\triangle PQA \div$ distance QA; if Q be any point on the line through A to either of the points I, J, and in particular if Q be either of the points I, J, then the triangle PQA is finite, but the distance QA is $= 0$: that is, the perpendicular distance of P from the line through A to either of the points I, J, that is, from any line through either of these points, is $= \infty$. But, as just stated, the triangle PQA is finite, or say the triangles PIA, PJA are each finite; viz., the coordinates (rectangular) of P, A being $(x, y, z = 1)$, $(a, a', 1)$ or (circular) $(\xi, \eta, z = 1)$, $(\alpha, \alpha', 1)$, the expressions for the doubles of these triangles respectively are

$$\begin{vmatrix} x & y & z \\ -i & 1 & 0 \\ a & a' & 1 \end{vmatrix}, \quad \begin{vmatrix} x & y & z \\ i & 1 & 0 \\ a & a' & 1 \end{vmatrix},$$

that is, they are (rectangular coordinates) $x-az+i(y-a'z)$, $x-az-i(y-a'z)$, or (circular coordinates) $\xi-\alpha z$, $\eta-\alpha' z$.

Representing the double areas by $\triangle PIA$, $\triangle PJA$, respectively, and the squared distance of the points A, P, by $\mathfrak{A}$, we have

$$\mathfrak{A}=(x-az)^2+(y-a'z)^2=(\xi-\alpha z)(\eta-\alpha' z)=\triangle PIA\cdot\triangle PJA.$$

63. Antipoints; Definition and Fundamental Properties.

Article 63. Two pairs of points (A,B) and (A_1,B_1) which are such that the lines AB, A_1B_1 bisect each other at right angles in a point O in such wise that $OA=OB=iOA_1=iOB_1$, are said to be *antipoints*, each of the other. In rectangular coordinates, taking the coordinates of (A,B) to be $(a,0,1)$ and $(-a,0,1)$, those of (A_1,B_1) will be $(0,ai,1)$ and $(0,-ai,1)$ respectively, whence joining the points (A,B) with the points (I,J), the points A_1, B_1, are given as the intersections of the lines AI and BJ, and of the lines AJ and BI respectively. Or, what is the same thing, in any quadrilateral wherein I, J are opposite angles, the remaining pairs (A,B) and (A_1,B_1) are antipoints each of the other.

Article 64. In circular coordinates, if the coordinates of A are $(\alpha,\alpha',1)$, and those of B are $(\beta,\beta',1)$, then the equations of

$$\begin{aligned} AI,\quad AJ &\text{ are } \xi-\alpha z=0,\quad \eta-\alpha' z=0,\\ BI,\quad BJ &\text{ are } \xi-\beta z=0,\quad \eta-\beta' z=0,\end{aligned}$$

whence the equations of

$$\begin{aligned} A_1I,\quad A_1J &\text{ are } \xi-\alpha_1 z=0,\quad \eta-\beta' z=0,\\ B_1I,\quad B_1J &\text{ are } \xi-\beta z=0,\quad \eta-\alpha' z=0.\end{aligned}$$

Article 65. Considering any point P the coordinates of which are ξ, η, z $(=1)$, let $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{A}_1$, $\mathfrak{B}_1$ be its squared distances from the points A, B, A_1, B_1 respectively; then by what precedes

$$\begin{aligned}\mathfrak{A}&=(\xi-\alpha z)(\eta-\alpha' z),\\ \mathfrak{B}&=(\xi-\beta z)(\eta-\beta' z),\\ \mathfrak{A}_1&=(\xi-\alpha_1 z)(\eta-\beta' z),\\ \mathfrak{B}_1&=(\xi-\beta z)(\eta-\alpha' z),\end{aligned}$$

and thence

$$\mathfrak{A}\cdot\mathfrak{B}=\mathfrak{A}_1\cdot\mathfrak{B}_1;$$

that is, *the product of the squared distances of a point from any two points A, B, is equal to the product of the squared distances of the same point from the two antipoints* A_1, B_1. This theorem, which was, I believe, first given by me in the *Educational Times* (see reprint, vol. vi. 1866, p. 81), is an important one in the theory of foci. It is to be further noticed that we have

$$\mathfrak{A}+\mathfrak{B}-\mathfrak{A}_1-\mathfrak{B}_1=(\alpha-\beta)(\alpha'-\beta')z^2=Kz^2=K,$$

if $K=(\alpha-\alpha')(\beta-\beta')$, be the squared distance of the points A, B, = squared distance of points A_1, B_1.

66. Antipoints of a Circle.

Article 66. A similar notion to that of two pairs of antipoints is as follows, viz., if from the centre of a circle perpendicular to its plane and in opposite senses, we measure off two distances each $= i$ into the radius, the extremities of these distances are *antipoints of the circle.* It is clear that the antipoints of the circle and the extremities of any diameter thereof are (in the plane of these four points) pairs of antipoints. It is to be added that each antipoint is the centre of a sphere radius zero, or say of a cone-sphere, passing through the circle: the circle is thus the intersection of the two cone-spheres having their centres at the two antipoints respectively.

67. Antipoints in relation to a Pair of Orthotomic Circles.

Article 67. It is a well-known property that if any circle pass through the points (A, B), and any other circle through the antipoints (A_1, B_1), then these two circles cut at right angles. Conversely if a circle pass through the points A, B, then all the orthotomic circles which have their centres on the circle through the antipoints (A_1, B_1) will pass through the points A, B; and the like as to any circle through A_1, B_1.

72 to 77. On a System of Sixteen Points.

Article 68. Take (A, B, C, D) any four concyclic points, and let the antipoints of

$$\begin{array}{lll} (B, C), & (A, D) \text{ be } (B_1, C_1), & (A_1, D_1), \\ (C, A), & (B, D) \text{ be } (C_2, A_2), & (B_2, D_2), \\ (A, B), & (C, D) \text{ be } (A_3, B_3), & (C_3, D_3), \end{array}$$

then each of the three new sets (A_1, B_1, C_1, D_1), (A_2, B_2, C_2, D_2), (A_3, B_3, C_3, D_3) will be a set of four concyclic points.

Article 69. Let O be the centre of the circle through (A, B, C, D), say of the circle O, and then, the lines BC, AD meeting in R, the lines CA, BD in S, and the lines AB, CD in T, let each of these points be made the centre of a circle orthotomic to O, viz., let these new circles be called the circles R, S, T respectively.

As regards the circle R, since its centre lies in BC, the circle passes through (B_1, C_1); and since the centre lies in AD, the circle passes through (A_1, D_1); that is, the four points (A_1, B_1, C_1, D_1) lie in the circle R. Similarly (A_2, B_2, C_2, D_2) lie in the circle S, and (A_3, B_3, C_3, D_3) in the circle T.

Article 70. The points R, S, T are conjugate points in relation to the circle O; that is, ST, TR, RS are the polars of R, S, T respectively in regard to this circle; and they are, consequently, at right angles to the lines OR, OS, OT respectively; viz., the four centres O, R, S, T are such that the line joining any two of them cuts at right angles the line joining the other two of them, and we see that the relation between the four sets is in fact a symmetrical one; this is most easily seen by consideration of the circular points at infinity

I, J; the four sets of points may be arranged thus

$$\begin{array}{c|cccc}
 & I & I & I & I \\ \hline
J & A & A_1 & A_2 & A_3 \\
J & B & B_1 & B_2 & B_3 \\
J & C & C_1 & C_2 & C_3 \\
J & D & D_1 & D_2 & D_3
\end{array}$$

in such wise that any four of them in the same vertical line pass through I, and any four in the same horizontal line pass through J; and this being so, starting for instance with (A_1, B_1, C_1, D_1) we have antipoints

$$\begin{array}{lll}
\text{of } (A_1, C_1), \quad (A_1, B_1) & \text{are } (A, C), & (A_3, B_3), \\
\text{of } (A_3, B_3), \quad (A, D) & \text{are } (C_2, D_2), & (A_2, D_2), \\
\text{of } (A_2, D_2), \quad (C, D) & \text{are } (A, B), & (C, D),
\end{array}$$

and similarly if we start from (A_1, B_2, C_2, D_2) or (A_3, B_3, C_3, D_3).

Article 71. I return for a moment to the construction of (A_1, B_1, C_1, D_1); these are points on the circle R, and (A_1, D_1) are the antipoints of (B, C); that is, they are the intersections of the circle R by the line at right angles to BC from its middle point, or, what is the same thing, by the perpendicular on BC from O. Similarly (A_1, D_1) are the antipoints of (A, D); that is, they are the intersections of the circle R by the perpendicular on AD from O. And the like as to (A_2, B_2, C_2, D_2) and (A_3, B_3, C_3, D_3) respectively.

Article 72. Hence, starting with the points A, B, C, D on the circle O, and constructing as above the circles R, S, T, and constructing also the perpendiculars from O on the six chords AB, AC, &c.,

$$\begin{array}{rlcll}
\text{the perpendiculars on } BC, & \text{meet circle} & R \text{ in} & (B_1, C_1), & (A_1, D_1), \\
CA, & BD & S & (C_2, A_2), & (B_2, D_2), \\
AB, & CD & T & (A_3, B_3), & (C_3, D_3),
\end{array}$$

so that the whole system is given by means of the circles R, S, T, and the six perpendiculars.

Article 73. If to fix the ideas (A, B, C, D) are real points taken in order on the real circle O, then the points R, S, T are each of them real; but R and T lie outside, S inside the circle O. The circles R and T are consequently real, but the circle S imaginary, viz., its radius is $= i$ into a real quantity; the imaginary points (A_2, B_2, C_2, D_2) are thus given as the intersections of a real circle by a pair of real lines, and the like as to the imaginary points (A_3, B_3, C_3, D_3); but the imaginary points (A_1, B_1, C_1, D_1) are only given as the intersections of an imaginary circle (centre real and radius a pure imaginary) by a pair of real lines. The points (C_2, A_2) qua antipoints of (C, A) are easily constructed as the intersections of a real circle by a real line, and the like as to the points (B_2, D_2) qua antipoints of (B, D), but the construction for the two pairs of points cannot be effected by means of the same real circle.

78 to 80. Property in regard to Four Confocal Conics.

Article 74. All the conics which pass through the four concyclic points A, B, C, D, have their axes in fixed directions; but three such conics are the line-pairs (BC, AD), (CA, BD), and (AB, CD), whence the directions of the axes are those of the bisectors of the angles formed by any one of these pairs of lines; hence, in particular, considering either axis of a conic through the four points, the lines AB and CD are equally inclined on opposite sides to this axis, and this leads to the theorem that the antipoints (A_3, B_3), (C_3, D_3) are in a conic confocal to the given conic through (A, B, C, D); whence, also, considering any given conic whatever through (A, B, C, D), the points (A_1, B_1, C_1, D_1), (A_2, B_2, C_2, D_2), (A_3, B_3, C_3, D_3) lie severally in three conics, each of them confocal with the given conic.

Article 75. To prove this, consider any two confocal conics, say an ellipse and a hyperbola, and let F be one of their four intersections; join F with the common centre O, and let OT, ON be parallel to the tangent and normal respectively of the ellipse at the point F. OF, OT are in direction conjugate axes of the ellipse, and OF, ON are in direction conjugate axes of the hyperbola; and if they are also the axes in magnitude, that is, if the points T, N are the intersections of OT with the ellipse and of ON with the hyperbola respectively, then it is easy to show that $OT^2 + ON^2 = 0$. And this being so, imagine on the ellipse any two points A, B such that the chord AB is parallel to OT, that is conjugate to OF; AB is bisected by OF, say in a point K, or we have parallel to OT the semichords or ordinates $KA = KB$; and we may, perpendicularly to this or parallel to ON, draw through K in the hyperbola a chord A_3B_3, which chord will be bisected in K, or we shall have $KA_3 = KB_3$. Hence KA, KA_3 are in the ellipse and the hyperbola respectively ordinates conjugate to the same diameter OF, and the semi-diameters conjugate to OF being OT, ON respectively, we have $KA^2\,(= KB^2) : KA_3^2\,(= KB_3^2) = OT^2 : ON^2$; this is, $KA^2 = KB^2 = -KA_3^2 = -KB_3^2$; or (A_3, B_3) will be the antipoints of (A, B).

Article 76. Conversely, if in the ellipse we have the two points (A, B), then drawing the diameter OF conjugate to AB, and through its extremity F the confocal hyperbola, then the antipoints (A_3, B_3) will lie on the hyperbola. And similarly, if on the ellipse we have the two points (C, D), then drawing the diameter OG conjugate to CD, and through its extremity G a confocal hyperbola, the antipoints (C_3, D_3) will lie on the hyperbola. Suppose (A, B, C, D) are concyclic, then, as noticed, AB and CD will be equally inclined on opposite sides to the transverse axis of the ellipse; the conjugate diameters OF, OG will therefore be equally inclined on opposite sides of the transverse axis; and the points F and G will therefore be situate symmetrically on opposite sides of the transverse axis, that is, the points F and G will respectively determine the same confocal hyperbola, and we have thus the required theorem, viz., if (A, B, C, D) are any four concyclic points on an ellipse, or say on a conic, and if (A_3, B_3) are the antipoints of (A, B), and (C_3, D_3) the antipoints of (C, D), then (A_3, B_3, C_3, D_3) will lie on a conic confocal with the given conic.

81 to 85. System of the Sixteen Points, the Axial Case.

Article 77. The theorems hold good when the four points A, B, C, D are in a line; the antipoints (B_1, C_1) of (B, C), &c., are in this case situate symmetrically on opposite sides of the line, so that it is evident at sight that we have (A_1, B_1, C_1, D_1), (A_2, B_2, C_2, D_2), (A_3, B_3, C_3, D_3), each set in a circle; and that the centres R, S, T of these circles lie in the line. The construction for the general case becomes, however, indeterminate, and must therefore be varied. If in the general case we take any circle through (B, C), and any circle through (A, D), then the circle R cuts at right angles these two circles, and has, consequently, its centre in the radical axis of the two circles; whence, when the four points are in a line, taking any circle through (B, C), or in particular the circle on BC as diameter, and any circle through (A, D), or in particular the circle on AD as diameter, the radical axis of these two circles intersects the line in the required centre R, and the circle R is the circle with this centre cutting at right angles the two circles respectively; the circles S and T are, of course, obtained by the like construction in regard to the combinations $(C, A; B, D)$ and $(A, B; C, D)$, respectively. It may be added, that we have

$R\begin{cases}\text{centre}\\ \text{extremities}\end{cases}$	$S\begin{cases}\text{centre}\\ \text{sibiconjugate points of involutions}\end{cases}$	$T\begin{cases}\text{centre}\\ \text{of circles}\end{cases}$
BC; AD	CA; BD	AB; CD

and that (as in the general case) the circles R, S, T intersect each pair of them at right angles; and they are evidently each intersected at right angles by the line $ABCD$ (or axis of the figure), which replaces the circle O in the general case.

Article 78. If the points A, B, C, D are taken in order on the line, then the points R, S, T are all real, viz., the point R is situate, on one side or the other, outside AD, but the points S and T are each of them situate between B and C; the circles R and T are real, but the circle S has its radius a pure imaginary quantity.

Article 79. If one of the four points, suppose D, is at infinity on the line, then the antipoints of (A, D), of (B, D), and of (C, D) are each of them the two points (I, J).

It would at first sight appear that the only conditions for the circles R, S, T were the conditions of passing through the antipoints of (B, C), of (C, A), and of (A, B) respectively, and that these circles thus became indeterminate; but in fact the definition of the circles is then as follows, viz., R has its centre at A, and passes through the antipoints of (B, C): (whence square of radius $= AB\cdot AC$); S has its centre at B, and passes through the antipoints of (C, A) (whence square of radius $= BC \cdot BA$); T has its centre at C, and passes through the antipoints of (A, B) (whence square of radius $= CA\cdot CB$); and we have thus three circles each intersecting the other two at right angles.

Article 80. If two of the points, say C, D, are at infinity, then the circle R has its centre at the middle point of AB, and passes through I and J, that is, it is the circle on AB as diameter; similarly S is the line through B at right angles to AB, and T is the line through A at right angles to AB; the system of three circles is thus a system of a circle and two lines at right angles to each other and each passing through the extremity of a diameter of the circle.

Article 81. If three of the points, say B, C, D, are at infinity, then A is any point, R is any circle through A, S is the line through A at right angles to OA, and T is the line through A at right angles to OA; but this is a system which does not present any thing worthy of notice.

86 to 91. The Involution of Four Circles.

Article 82. Given any four circles, there are in general two circles which cut each of the four given circles at right angles; these two circles are called the *orthotomic circles* of the four given circles. If the four given circles have a common orthotomic circle, then they form a system of circles in involution.

Article 83. The condition that four circles may have a common orthotomic circle may be expressed analytically as follows: Let the equations of the four circles be

$$S_1 = x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0, \quad \&c.,$$

then the equation of a circle cutting all four at right angles is

$$S - \lambda_1S_1 - \lambda_2S_2 - \lambda_3S_3 - \lambda_4S_4 = 0,$$

where λ_1, λ_2, λ_3, λ_4 are determined by the conditions that the coefficients of x and y in S vanish, and that the radius of S is zero. These conditions give four equations, which in general cannot be satisfied; but if the four circles satisfy a certain condition, the equations become compatible, and there is a circle cutting all four at right angles.

Article 84. The condition may also be expressed geometrically: If four circles have a common orthotomic circle, then the centres of the four circles lie on a circle, and the squares of the radii are proportional to the squares of the distances of the centres from a fixed point.

Article 85. In the case where the four circles pass through two common points, they have in general a common orthotomic circle; for if A, B are the two common points, then any circle through the antipoints of (A, B) cuts all four circles at right angles.

Article 86. The system of four circles in involution is connected with the theory of the sixteen points as follows: Given four concyclic points (A, B, C, D), the circles through any three of these points form a system of four circles in involution; and the orthotomic circle of this system is the circle through the antipoints of any pair of the four points.

Article 87. The involution of four circles may be generalised as follows: Given any four curves of the second order passing through two fixed points, these curves form a system in involution, and there exists a curve of the second order cutting all four at right angles, that is, such that at each intersection the tangents to the two curves are harmonically related to the lines joining the intersection to the two fixed points.

92. On a Locus connected with the foregoing Properties.

Article 88. If, as above, A, B, C, D are any four points, and $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ are the squared distances of a current point P from the four points respectively, then the locus of the foci of the conics which pass through the four points is the tetrazomal curve

$$a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}} + d\sqrt{\mathfrak{D}} = 0.$$

In fact the sum $a\mathfrak{A} + b\mathfrak{B} + c\mathfrak{C} + d\mathfrak{D}$ has, it has been seen, a constant value for all positions of the point P; taking Q to be the other focus, its squared distances are $(k - \sqrt{\mathfrak{A}})^2$, &c., whence for the first-mentioned focus we have

$$a\mathfrak{A} + b\mathfrak{B} + c\mathfrak{C} + d\mathfrak{D} = a(k - \sqrt{\mathfrak{A}})^2 + b(k - \sqrt{\mathfrak{B}})^2 + c(k - \sqrt{\mathfrak{C}})^2 + d(k - \sqrt{\mathfrak{D}})^2,$$

or recollecting that $a + b + c + d = 0$, it follows that we have for the locus in question

$$a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}} + d\sqrt{\mathfrak{D}} = 0;$$

this locus will be discussed in the sequel. I remark here, that in the case where the four points are on a circle, then (as mentioned above), the axes of the several conics are in the same fixed directions; there are thus two sets of foci, those on the axis in one direction, and those on the axis in the other direction; it might therefore be anticipated, and it will appear, that in this case the tetrazomal breaks up into two trizomal curves.

93 to 98. Formulae as to the two Sets (A,B,C,D) and (A_1,B_1,C_1,D_1), each of four Concyclic Points.

Article 89. Consider the four points A, B, C, D on a circle, then taking, as before, their circular coordinates to be $(\alpha,\alpha',1)$, $(\beta,\beta',1)$, $(\gamma,\gamma',1)$, $(\delta,\delta',1)$, the condition that the points may be on a circle is

$$\begin{vmatrix} 1 & \alpha & \alpha' & \alpha\alpha' \\ 1 & \beta & \beta' & \beta\beta' \\ 1 & \gamma & \gamma' & \gamma\gamma' \\ 1 & \delta & \delta' & \delta\delta' \end{vmatrix} = 0,$$

viz., this equation may be written

$$(\beta-\gamma)(\alpha-\delta) : (\gamma-\alpha)(\beta-\delta) : (\alpha-\beta)(\gamma-\delta) = (\beta'-\gamma')(\alpha'-\delta') : (\gamma'-\alpha')(\beta'-\delta') : (\alpha'-\beta')(\gamma'-\delta')$$

or if, for shortness, we take

$$\begin{array}{llll} a=\beta-\gamma, & f=\alpha-\delta, & a'=\beta'-\gamma', & f'=\alpha'-\delta', \\ b=\gamma-\alpha, & g=\beta-\delta, & b'=\gamma'-\alpha', & g'=\beta'-\delta', \\ c=\alpha-\beta, & h=\gamma-\delta, & c'=\alpha'-\beta', & h'=\gamma'-\delta', \end{array}$$

and consequently

$$af+bg+ch=0,$$
$$a'f'+b'g'+c'h'=0,$$

$$\begin{array}{ll} a=g-h, & a'=g'-h', \\ b=h-f, & b'=h'-f', \\ c=f-g, & c'=f'-g', \end{array}$$

$$a+b+c=0, \quad a'+b'+c'=0,$$

then the equation is

$$af:bg:ch=a'f':b'g':c'h'.$$

Article 90. Let a, b, c, d denote as before $(a:b:c:d = BCD : -CDA : DAB : -ABC)$, then we have

$$a:b:c:d = \begin{vmatrix} \beta & \beta' & 1 \\ \gamma & \gamma' & 1 \\ \delta & \delta' & 1 \end{vmatrix} : - \begin{vmatrix} \gamma & \gamma' & 1 \\ \delta & \delta' & 1 \\ \alpha & \alpha' & 1 \end{vmatrix} : \begin{vmatrix} \delta & \delta' & 1 \\ \alpha & \alpha' & 1 \\ \beta & \beta' & 1 \end{vmatrix} : - \begin{vmatrix} \alpha & \alpha' & 1 \\ \beta & \beta' & 1 \\ \gamma & \gamma' & 1 \end{vmatrix},$$

and we may write

$$\begin{array}{lll} a=ah'-a'h, & a=a'g-ag', & g=gh'-g'h, \\ b=bh'-b'h, & b=b'f-bf', & h=hf'-h'f, \\ c=ch'-c'h, & c=cf'-c'f, & f=fg'-f'g, \\ d=-(cb'-c'b), & d=-(ac'-a'c), & d=-(ba'-b'a), \end{array}$$

viz., the expressions in the same horizontal line are equal, and a, b, c, d are proportional to the expressions in the four lines respectively.

Article 91. I say that we have

$$\frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'} = -\frac{d}{d'},$$

viz., this will be the case if

$$bc' = fg'd, \quad ac'h = hf'd, \quad a'bc = fg'd,$$

and selecting the convenient expressions for a, b, c, d, these equations become

$$bc'(gh'-g'h) = g'h(cb'-c'b), \quad ac'(hf'-h'f) = fh'(ac'-a'c), \quad a'b(fg'-f'g) = fg'(ba'-b'a),$$

viz., these equations are respectively $bgc'h' = b'g'ch$, $cha'f' = c'h'af$, $afb'g' = a'f'bg$, and are consequently satisfied. It thus appears that the equation

$$\frac{l}{a} + \frac{m}{b} + \frac{n}{c} + \frac{p}{d} = 0$$

is transformable into

$$\frac{l}{c'f'} + \frac{m}{c'g'} + \frac{n}{a'f'} + \frac{p}{f'g'} = 0,$$

which is of course one of a system of similar forms.

Article 92. Take (A_1, D_1) the antipoints of (A, D); (B_1, C_1) the antipoints of (B, C); or say that the circular coordinates of A_1, B_1, C_1, D_1 are $(\alpha, \delta', 1)$, $(\beta, \gamma', 1)$, $(\gamma, \beta', 1)$, $(\delta, \alpha', 1)$ respectively; the points A_1, B_1, C_1, D_1 are, as above mentioned, on a circle, the condition that this may be so being in fact

$$\begin{vmatrix} 1 & \alpha & \delta' & \alpha\delta' \\ 1 & \beta & \gamma' & \beta\gamma' \\ 1 & \gamma & \beta' & \gamma\beta' \\ 1 & \delta & \alpha' & \delta\alpha' \end{vmatrix} = 0,$$

equivalent to

$$af : bg : ch = a'f' : b'g' : c'h'.$$

Article 93. Let (a_1, b_1, c_1, d_1) be the corresponding quantities to (a, b, c, d), viz.,

$$a_1 : b_1 : c_1 : d_1 = B_1C_1D_1 : -C_1D_1A_1 : D_1A_1B_1 : -A_1B_1C_1;$$

we have

$$a_1 : b_1 : c_1 : d_1 = \begin{vmatrix} \beta & \gamma' & 1 \\ \gamma & \beta' & 1 \\ \delta & \alpha' & 1 \end{vmatrix} : - \begin{vmatrix} \gamma & \beta' & 1 \\ \delta & \alpha' & 1 \\ \alpha & \delta' & 1 \end{vmatrix} : \begin{vmatrix} \delta & \alpha' & 1 \\ \alpha & \delta' & 1 \\ \beta & \gamma' & 1 \end{vmatrix} : - \begin{vmatrix} \alpha & \delta' & 1 \\ \beta & \gamma' & 1 \\ \gamma & \beta' & 1 \end{vmatrix},$$

giving rise to a similar set of forms

$$\begin{array}{lcr} a_1 = \ldots, & -ac' + ba', a'g + b'a, & -c'g - b'h, \\ b_1 = -c'h - g'h, & \ldots, -f'b - g'f, & -f'h + c'f, \\ c_1 = b'c + h'g, & -f'c + h'f, \ldots, & f'g + g'f, \\ d_1 = g'c + h'h, & -h'a + a'c, -a'b - g'a, & \ldots, \end{array}$$

and leading to

$$\frac{a_1}{a'} = \frac{b_1}{c'g'} = \frac{c_1}{a'f'} = -\frac{d_1}{f'g'},$$

so that the equation

$$\frac{l_1}{a_1} + \frac{m_1}{b_1} + \frac{n_1}{c_1} + \frac{p_1}{d_1} = 0,$$

is transformable into

$$\frac{l_1}{c'f'} + \frac{m_1}{c'g'} + \frac{n_1}{a'f'} + \frac{p_1}{f'g'} = 0.$$

Article 94. Let A, B, C, D be, as above, points on a circle; (A_1, D_1) and (B_1, C_1) the antipoints of (A, D), (B, C) respectively. Write

$$\begin{aligned}
\mathfrak{A} &= (\xi - \alpha z)(\eta - \alpha' z), & \mathfrak{A}_1 &= (\xi - \alpha z)(\eta - \delta' z),\\
\mathfrak{B} &= (\xi - \beta z)(\eta - \beta' z), & \mathfrak{B}_1 &= (\xi - \beta z)(\eta - \gamma' z),\\
\mathfrak{C} &= (\xi - \gamma z)(\eta - \gamma' z), & \mathfrak{C}_1 &= (\xi - \gamma z)(\eta - \beta' z),\\
\mathfrak{D} &= (\xi - \delta z)(\eta - \delta' z), & \mathfrak{D}_1 &= (\xi - \delta z)(\eta - \alpha' z);
\end{aligned}$$

then we have identically

$$\begin{aligned}
f'\mathfrak{B} &= gg'\mathfrak{A} + cc'\mathfrak{D} + gc'\mathfrak{A}_1 + cg'\mathfrak{D}_1,\\
f'\mathfrak{C} &= hh'\mathfrak{A} + bb'\mathfrak{D} - hh'\mathfrak{A}_1 - bh'\mathfrak{D}_1,\\
g'\mathfrak{B}_1 &= gh'\mathfrak{A} - cb'\mathfrak{D} - gb'\mathfrak{A}_1 + ch'\mathfrak{D}_1,\\
g'\mathfrak{C}_1 &= hg'\mathfrak{A} - bc'\mathfrak{D} + hc'\mathfrak{A}_1 - bg'\mathfrak{D}_1,
\end{aligned}$$

or, in the foregoing notation,

$$\begin{aligned}
f'\mathfrak{B} &= gg'\mathfrak{A} + cc'\mathfrak{D} + gc'\mathfrak{A}_1 + cg'\mathfrak{D}_1,\\
f'\mathfrak{C} &= hh'\mathfrak{A} + bb'\mathfrak{D} - hh'\mathfrak{A}_1 - bh'\mathfrak{D}_1,\\
g'\mathfrak{B}_1 &= gh'\mathfrak{A} - cb'\mathfrak{D} - gb'\mathfrak{A}_1 + ch'\mathfrak{D}_1,\\
g'\mathfrak{C}_1 &= hg'\mathfrak{A} - bc'\mathfrak{D} + hc'\mathfrak{A}_1 - bg'\mathfrak{D}_1.
\end{aligned}$$

99 to 104. Further Properties in relation to the same Sets (A, B, C, D) and (A_1, B_1, C_1, D_1).

Article 95. It is to be shown that in virtue of these equations, and if moreover $\frac{a}{l} + \frac{b}{m} + \frac{c}{n} + \frac{d}{p} = 0$, then it is possible to find l_1, m_1, n_1, p_1, such that we have identically

$$-l\mathfrak{A} + m\mathfrak{B} + n\mathfrak{C} - p\mathfrak{D} + l_1\mathfrak{A}_1 - m_1\mathfrak{B}_1 - n_1\mathfrak{C}_1 + p_1\mathfrak{D}_1 = 0.$$

This equation will in fact be identically true if only

$$\begin{array}{rrr}
-ff'l + gg'm + hh'n & & - gh'm_1 - g'h'n_1 = 0,\\
cc'm + hh'n - ff'p & & + ci'n_1 + hc'p_1 = 0,\\
gc'm - hh'n & & + ff'l_1 + gh'n_1 - hc'p_1 = 0,\\
cg'm - hh'n & + ch'm_1 + hg'p_1 & + lf'p_1 = 0.
\end{array}$$

From the first and second equations eliminating m_1 or n_1, the other of these quantities disappears of itself, and we thus obtain two equations which must be equivalent to a single one, viz., we have

$$bc'ff'l + c'g'af'm + bha'f'n + g'h'f'p = 0,$$

$$b'cff'l + cga'f'm + h'h'af'n + gh'ff'p = 0;$$

which equations may also be written

$$\frac{l}{c'f'} + \frac{m}{c'g'} + \frac{n}{a'f'} + \frac{p}{f'g'} = 0,$$

$$\frac{l}{a'h'} + \frac{m}{b'h'} + \frac{n}{a'f'} + \frac{p}{a'b'} = 0,$$

and it thus appears that the equations are equivalent to each other, and to the assumed relation

$$\frac{a}{l} + \frac{b}{m} + \frac{c}{n} + \frac{d}{p} = 0.$$

Article 96. Similarly, from the third and fourth equations eliminating m_1 or n_1, the other of these quantities disappears of itself, and we find

$$b'c'ff'l_1 - cga'f'm_1 + afc'g'n_1 - dg'f'p_1 = 0,$$

$$bKff'l_1 - afb'h'm_1 + hha'f'n_1 - h'hff'p_1 = 0,$$

equations which may be written

$$\frac{l_1}{cf} + \frac{m_1}{cg} + \frac{n_1}{af} + \frac{p_1}{fg} = 0, \quad \frac{l_1}{ac} + \frac{m_1}{eg} + \frac{n_1}{af} + \frac{p_1}{ga} = 0,$$

and which are equivalent to each other, and to

$$\frac{a_1}{l_1} + \frac{b_1}{m_1} + \frac{c_1}{n_1} + \frac{d_1}{p_1} = 0.$$

Article 97. It thus appears that we have identically

$$l\mathfrak{A} - m\mathfrak{B} - n\mathfrak{C} + p\mathfrak{D} = l_1\mathfrak{A}_1 - m_1\mathfrak{B}_1 - n_1\mathfrak{C}_1 + p_1\mathfrak{D}_1,$$

and that the condition $\frac{a}{l} + \frac{b}{m} + \frac{c}{n} + \frac{d}{p} = 0$ is equivalent to the condition $\frac{a_1}{l_1} + \frac{b_1}{m_1} + \frac{c_1}{n_1} + \frac{d_1}{p_1} = 0$. The theorem is thus established.

Article 98. It follows from the foregoing that if we have the tetrazomal curve

$$\sqrt{l\mathfrak{A}} + \sqrt{m\mathfrak{B}} + \sqrt{n\mathfrak{C}} + \sqrt{p\mathfrak{D}} = 0,$$

where $\frac{a}{l} + \frac{b}{m} + \frac{c}{n} + \frac{d}{p} = 0$, then this curve is identical with the tetrazomal curve

$$\sqrt{l_1\mathfrak{A}_1} + \sqrt{m_1\mathfrak{B}_1} + \sqrt{n_1\mathfrak{C}_1} + \sqrt{p_1\mathfrak{D}_1} = 0,$$

where $\frac{a_1}{l_1} + \frac{b_1}{m_1} + \frac{c_1}{n_1} + \frac{d_1}{p_1} = 0$; and the two forms of the equation are equivalent to each other.

Article 99. In particular, if A, B, C, D are four concyclic points, and (A_1, D_1), (B_1, C_1) are the antipoints of (A, D), (B, C) respectively, then the tetrazomal curve

$$a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}} + d\sqrt{\mathfrak{D}} = 0$$

breaks up into two trizomal curves, viz., into

$$\sqrt{gg'\mathfrak{A}} + \sqrt{cc'\mathfrak{D}} + \sqrt{gc'\mathfrak{A}_1} + \sqrt{cg'\mathfrak{D}_1} = 0,$$

and

$$\sqrt{hh'\mathfrak{A}} + \sqrt{bb'\mathfrak{D}} - \sqrt{hh'\mathfrak{A}_1} - \sqrt{bh'\mathfrak{D}_1} = 0;$$

or, what is the same thing, the locus of the foci of conics through four concyclic points breaks up into two trizomal curves.

Part III. (Nos. 105 to 157).—On the Theory of Foci.

105 to 110. Explanation of the General Theory.

Article 100. If from a focus of a conic we draw two tangents to the curve, these pass respectively through the two circular points at infinity, and we have thence the generalised definition of a focus as established by Plücker, viz., in any curve a *focus* is a point such that the lines joining it with the two circular points at infinity are respectively tangents to the curve; or, what is the same thing, if from each of the circular points at infinity, say from the points I, J, tangents are drawn to the curve, the intersections of each tangent from the one point with each tangent from the other point are the foci of the curve. A curve of the class n has thus in general n^2 foci.

It is to be added that, as in the conic the line joining the points of contact of the two tangents from a focus is the directrix corresponding to that focus, so in general the line joining the points of contact of the tangents from the focus through the points I, J respectively is the directrix corresponding to the focus in question.

Article 101. A circular point at infinity I or J, may be an ordinary or a singular point on the curve, and the tangent at this point then counts, or, in the case of a multiple point, the tangents at this point count a certain number of times, say q times, among the tangents which can be drawn to the curve from the point; the number of the remaining tangents is thus $= n - q$. In particular, if the circular point at infinity be an ordinary point, then the tangent counts twice, or we have $q = 2$; if it be a node, each of the tangents counts twice, or $q = 4$; if it be a cusp, the tangent counts three times, or $q = 3$. Similarly, if the other circular point at infinity be an ordinary or a singular point on the curve, the tangent or tangents there count a certain number of times, say q' times, among the tangents to the curve from this point; the number of the remaining tangents is thus $= n - q'$. And if as usual we disregard the tangents at the two points I, J respectively, and attend only to the remaining tangents, the number of the foci is $= (n - q)(n - q')$.

Article 102. Among the tangents from the point I or J there may be a tangent which, either from its being a multiple tangent (that is, a tangent having ordinary contact at two or more distinct points), or from being an osculating tangent at one or more points, counts a certain number of times, say r, among the tangents from the point in question. Similarly, if among the tangents from the other point J or I, there is a tangent which counts r' times, then the foci are made up as follows, viz. we have

Intersections of the two singular tangents counting as	rr' foci.
Intersections of the first singular tangent with each of the ordinary tangents from the other circular point at infinity, as $(n - q' - r')r$	$(n - q - r)r'$
Do. for second singular tangent,	$(n - q' - r')r$
Intersections of the ordinary tangents	$(n - q - r)(n - q' - r')$
Giving together the $(n - q)(n - q')$ foci	

and the like observation applies to the more general case where the tangents from each of the points I, J include more than one singular tangent.

Article 103. There is yet another case to be considered; the line infinity may be an ordinary or a singular tangent to the curve: assuming that it counts s times among the tangents from either of the circular points at infinity, the numbers of the remaining tangents are $n-q-s$, $n-q'-s$ from the two points I, J respectively, and the number of foci is $=(n-q-s)(n-q'-s)$.

Article 104. In the case of a real curve the two points I, J are related in the same manner to the curve, and we have therefore $q=q'$; the singular tangents (if any) from the two points respectively being the same as well in character as in number. Writing $n-q-s=n-q'-s=p$, and not for the present attending to the case of singular tangents, I shall assume that the number of tangents to the curve from each of the two points is $=p$; the number of foci is thus $=p^2$; and to each focus there corresponds a directrix, viz., this is the line through the points of contact of the tangents from the focus to the two points I, J respectively.

Article 105. Consider any two foci A, B not in line with either of the points I, J, then joining these with the points I, J, and taking A_1, B_1 the intersections of AI, BJ and of AJ, BI (A_1, B_1 being therefore by a foregoing definition the antipoints of (A, B)), then A_1, B_1 are, it is clear, foci of the curve. We may out of the p^2 foci select, and that in $1\cdot 2\ldots p$ different ways, a system of p foci such that no two of them lie in line with either of the points I, J; and this being so, taking the antipoints of each of the $\frac{1}{2}p(p-1)$ pairs out of the p foci, we have, inclusively of the p foci, in all $p+2\cdot\frac{1}{2}p(p-1)$, that is p^2 foci, the entire system of foci.

111 to 117. On the Foci of Conics.

Article 106. A conic is a curve of the class 2, and the number of foci is thus $=4$. Taking as foci any two points A, B, the remaining two foci will be the antipoints A_1, B_1. In order that a given point A may be a focus, the conic must touch the lines AI, AJ; similarly, in order that a given point B may be a focus, the conic must touch the lines BI, BJ; the equation of a conic having the given points A, B for foci contains therefore a single arbitrary parameter.

Article 107. In the case, however, of the parabola the curve touches the line infinity; there is consequently from each of the points I, J only a single tangent to the curve, and consequently only one focus: the parabola having a given point A for its focus is a conic touching the line infinity and the lines AI, AJ, or say the three sides of the triangle AIJ; its equation contains therefore two arbitrary parameters.

Article 108. Returning to the general conic, there are certain trizomal forms of the focal equation, not of any great interest, but which may be mentioned. Using circular coordinates, and taking $(\alpha,\alpha',1)$ and $(\beta,\beta',1)$ for the coordinates of the given foci A, B respectively, the conic touches the lines $\xi-\alpha z=0$, $\eta-\alpha' z=0$, $\xi-\beta z=0$, $\eta-\beta' z=0$; the equation of a conic touching the first three lines is

$$\sqrt{l(\xi-\alpha z)}+\sqrt{m(\eta-\alpha' z)}+\sqrt{n(\xi-\beta z)}=0,$$

where l, m, n are arbitrary, and it is easy to obtain, in order that the conic may touch the fourth line $\eta-\beta' z=0$, the condition

$$n=-\frac{lm}{l+m}\cdot\frac{(\beta-\alpha)(\beta'-\alpha')}{(\beta-\alpha)+(\beta'-\alpha')}.$$

Article 109. In fact, n having this value, the equation gives

$$l(\xi-\alpha z)+m(\eta-\alpha' z)+2\sqrt{lm(\xi-\alpha z)(\eta-\alpha' z)} = -\frac{lm}{l+m}\frac{(\beta-\alpha)(\beta'-\alpha')}{(\beta-\alpha)+(\beta'-\alpha')}(\eta-\beta' z+(\beta'-\alpha')z),$$

and taking over the term

$$\frac{lm}{l+m}\frac{(\beta-\alpha)(\beta'-\alpha')}{(\beta-\alpha)+(\beta'-\alpha')}(\beta'-\alpha')z = \frac{lm}{l+m}(\beta-\alpha)z,$$

this gives

$$l(\xi-\beta z)+m(\eta-\alpha' z)+2\sqrt{lm(\xi-\alpha z)(\eta-\alpha' z)} = -\frac{lm}{l+m}\frac{(\beta-\alpha)(\beta'-\alpha')}{(\beta-\alpha)+(\beta'-\alpha')}(\eta-\beta' z).$$

It is easy to see that the equation may be written in any one of the four forms

$$\begin{aligned}
\sqrt{l(\xi-\alpha z)}+\sqrt{m(\eta-\beta' z)}+\sqrt{-\tfrac{lm}{l+m}\tfrac{(\beta-\alpha)(\beta'-\alpha')}{(\beta-\alpha)+(\beta'-\alpha')}(\eta-\alpha' z)} &= 0,\\
\sqrt{l(\xi-\alpha z)}+\sqrt{m(\eta-\alpha' z)}+\sqrt{-\tfrac{lm}{l+m}\tfrac{(\beta-\alpha)(\beta'-\alpha')}{(\beta-\alpha)+(\beta'-\alpha')}(\xi-\beta z)} &= 0,\\
\sqrt{l(\xi-\beta z)}+\sqrt{m(\eta-\alpha' z)}+\sqrt{-\tfrac{lm}{l+m}\tfrac{(\beta-\alpha)(\beta'-\alpha')}{(\beta-\alpha)+(\beta'-\alpha')}(\eta-\beta' z)} &= 0,\\
\sqrt{l(\xi-\beta z)}+\sqrt{m(\eta-\beta' z)}+\sqrt{-\tfrac{lm}{l+m}\tfrac{(\beta-\alpha)(\beta'-\alpha')}{(\beta-\alpha)+(\beta'-\alpha')}(\xi-\alpha z)} &= 0,
\end{aligned}$$

viz., in forms containing any three of the four radicals $\sqrt{\xi-\alpha z}$, $\sqrt{\xi-\beta z}$, $\sqrt{\eta-\alpha' z}$, $\sqrt{\eta-\beta' z}$. The conic is thus expressed as a trizomal curve, the zomals being each a line, viz., they are any three out of the four focal tangents; the order of the curve, as deduced from the general expression $2^{\nu-2}r$, is $= 2$; so that there is here no depression of order.

Article 110. But the ordinary form of the focal equation is a more interesting one; viz., $\mathfrak{A}$, $\mathfrak{B}$ being as usual the squared distances of the current point from the two given foci respectively, say

$$\mathfrak{A} = (x-az)^2+(y-a'z)^2, \quad \mathfrak{B} = (x-\beta z)^2+(y-\beta' z)^2,$$

then $2a$ being an arbitrary parameter, the equation is

$$2az+\sqrt{\mathfrak{A}}+\sqrt{\mathfrak{B}} = 0,$$

viz., the equation is here that of a trizomal curve, the zomals being curves of the second order, that is, the zomals are ($z^2 = 0$) the line infinity twice, and the line-pairs AI, AJ and BI, BJ respectively: the general expression $2^{\nu-2}r$ gives therefore the order $= 4$; but in the present case there are two branches, viz., the branches

$$2az+\sqrt{\mathfrak{A}}-\sqrt{\mathfrak{B}} = 0, \quad 2az-\sqrt{\mathfrak{A}}+\sqrt{\mathfrak{B}} = 0,$$

each ideally containing ($z = 0$) the line infinity; the curve contains therefore ($2z^2 = 0$) the line infinity twice, and omitting this factor the order is $= 2$, as it should be.

Article 111. To express the equation by means of the other two foci A_1, B_1, writing the equation under the form

$$2\mathfrak{A}+2\mathfrak{B}+2\sqrt{\mathfrak{A}\mathfrak{B}}-4a^2z^2 = 0,$$

and then if $\mathfrak{A}_1$, $\mathfrak{B}_1$ are the squared distances of the current point from A_1, B_1 respectively, we have (ante, No. 65),

$$\mathfrak{A}\mathfrak{B} = \mathfrak{A}_1\mathfrak{B}_1,$$

$$\mathfrak{A} + \mathfrak{B} - 2\mathfrak{A}_1 - 2\mathfrak{B}_1 = Kz^2 = K,$$

where K is the squared distance of the foci A, B, $= 4a^2e^2$ suppose: whence putting $a^2(1 - e^2) = b^2$, the equation becomes

$$2\mathfrak{A}_1 + 2\mathfrak{B}_1 + 2\sqrt{\mathfrak{A}_1\mathfrak{B}_1} - 4b^2z^2 = 0,$$

that is

$$2bz + \sqrt{\mathfrak{A}_1} + \sqrt{\mathfrak{B}_1} = 0,$$

which is the required new form. It is hardly necessary to remark that the equation $2az+\sqrt{\mathfrak{A}}+\sqrt{\mathfrak{B}} = 0$, putting therein $z = 1$, and expressing $\mathfrak{A}$, $\mathfrak{B}$ in rectangular coordinates measured along the axes, is the ordinary focal equation $2a = \sqrt{(x - ae)^2 + y^2} + \sqrt{(x + ae)^2 + y^2}$.

Article 112. I remark that the equation $2az + \sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} = 0$ gives rise to $4a^2z^2 + \mathfrak{A} + \mathfrak{B} + 4a^2z\sqrt{\mathfrak{A}} = 0$, but here $\mathfrak{A} - \mathfrak{B} = -4aexz$, so that the equation contains $z = 0$, and omitting this it becomes $(az - ex) + \sqrt{\mathfrak{A}} = 0$, a bizomal form, being a curve of the order $= 2$, as it should be; this is in fact the ordinary equation in regard to a focus and its directrix.

118 to 123. Theorem of the Variable Zomal as applied to a Conic.

Article 113. The equation $2kz + \sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} = 0$ is in like manner that of a conic; in fact, this would be a curve of the order $= 4$, but there are as before the two branches

$$2kz + \sqrt{\mathfrak{A}} - \sqrt{\mathfrak{B}} = 0, \quad 2kz - \sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} = 0,$$

each ideally containing $(z = 0)$ the line infinity, and the order is thus reduced to be $= 2$. Each of the circles $\mathfrak{A} = 0$, $\mathfrak{B} = 0$ is a circle having double contact with the conic (this of course implies that the centre of the circle is on an axis of the conic). We may if we please start from the form $2az + \sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} = 0$, and then by means of the theorem of the variable zomal introduce into the equation one, two, or three such circles.

Article 114. It is in this point of view that I will consider the question, viz., adapting the formula to the case of the ellipse, and starting from the form

$$2az + \sqrt{(x - aez)^2 + y^2} + \sqrt{(x + aez)^2 + y^2} = 0,$$

the equation of the variable zomal or circle of double contact may be taken to be

$$4\tfrac{c}{a}\sqrt{(x - qaez)^2 + y^2} + (x - \tfrac{c}{a}qaez)^2 + y^2 + (x + \tfrac{c}{a}qaez)^2 + y^2,$$

where q is an arbitrary parameter; writing for greater simplicity $z = 1$, and reducing, the equation is

$$(x - qae)^2 + y^2 = b^2(1 - q^2).$$

Article 115. If $q < 1$, then writing $q = \sin\theta$, we obtain the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

as the envelope of the variable circle

$$(x - ae\sin\theta)^2 + y^2 = b^2\cos^2\theta,$$

viz., of a circle having its centre on the major axis at a distance $= ae\sin\theta$ from the centre, and its radius $= b\cos\theta$. (I notice, in passing, that this gives in practice a very convenient graphical construction of the ellipse.) It may be remarked that for $\theta = \pm\sin^{-1} e$, the circle becomes

$$(x - \pm(a \mp b))^2 + y^2 = b^2,$$

viz., this is the circle of curvature at one or other of the extremities of the major axis; as θ passes from 0 to $\pm\sin^{-1} e$ we have a series of real circles, which, by their continued intersection, generate the ellipse; as θ increases from $\pm\sin^{-1} e$ to ±90°, the circles continue real, but the consecutive circles no longer intersect in any real point, and ultimately for $\theta = \pm 90^\circ$, the circles become evanescent at the two foci respectively.

Article 116. In the case $q > 1$, we have a real representation of

$$(x - qae)^2 + y^2 + b^2(q^2 - 1),$$

as the squared distance of the point (x, y) from a point $(X, 0, Z)$ out of the plane of the figure, viz., putting this

$$= (x - X)^2 + y^2 + Z^2,$$

we have

$$qae = X, \quad Z^2 = b^2(q^2 - 1),$$

whence

$$\frac{X^2}{a^2e^2} - \frac{Z^2}{b^2} = 1;$$

or, what is the same thing,

$$\frac{X^2}{a^2} - \frac{Z^2}{b^2} = e^2,$$

that is, the locus is the focal hyperbola, viz., a hyperbola in the plane of zx, having its vertices at the foci, and its foci at the vertices of the ellipse.

Article 117. If instead of the form first considered, we start from the trizomal form

$$2bz + \sqrt{ax^2 + (y - aeiz)^2} + \sqrt{ax^2 + (y + aeiz)^2} = 0,$$

then we have the zomal or circle of double contact under the form

$$\tfrac{b^2}{a^2} + (y - qaei)^2 = a^2(1 - q^2);$$

or putting herein $q = -i\tan\phi$, this is

$$x^2 + (y - ae\tan\phi)^2 = a^2\sec^2\phi,$$

so that we have the ellipse as the envelope of a variable circle having its centre on the minor axis of the ellipse, distance from the centre $= ae\tan\phi$, and radius $= a\sec\phi$. This is, in fact, Gergonne's theorem, according to which the ellipse is the secondary caustic or orthogonal trajectory of rays issuing from a point and refracted at a right line into a rarer medium. It is to be remarked that for $\tan\phi = \pm\frac{b}{a}$, the equation of the circle is

$$x^2 + (y \pm (a-b))^2 = b^2,$$

viz., this is the circle of curvature at one or other extremity of the minor axis; from $\phi = 0$ to $\phi = \pm\tan^{-1}\frac{b}{a}$, the intersections of the consecutive circles are real, and give the entire real ellipse; from $\phi = \pm\tan^{-1}\frac{b}{a}$ to $\phi = \pm 90$, the circles are still real, but the intersections of consecutive circles are imaginary.

Article 118. If in the equation of the generating circle we interchange x, y, a, b, the equation becomes

$$(x - ae\tan\phi)^2 + y^2 = b^2\sec^2\phi,$$

which is (as it should be) equivalent to the former equation

$$(x - ae\sin\theta)^2 + y^2 = b^2\cos^2\theta,$$

the identity being established by means of the equation

$$\cos\theta = \frac{\cos\phi}{\sqrt{1-e^2\sin^2\phi}}, \text{ and therefore } \sin\theta = i\tan\phi\sqrt{1-\frac{\cos^2\phi}{1-e^2\sin^2\phi}}, \tan\theta = i\sin\phi\sqrt{1-e^2\sin^2}$$

which is Jacobi's imaginary transformation in the theory of Elliptic Functions.

124 to 126. Foci of the Circular Cubic and the Bicircular Quartic.

Article 119. For a cubic curve, the class is in general $= 6$, and the number of the foci is $= 36$. But a specially interesting case is that of a circular cubic, viz., a cubic passing through each of the circular points at infinity. Here, at each of the circular points at infinity, the tangent at this point reckons twice among the tangents to the curve from the point; the number of the remaining tangents is thus $= 4$, and the number of the foci is $= 16$. If from any two points whatever on the curve tangents be drawn to the curve, then the two pencils of tangents are, and that in four different ways, homologous to each other, viz., if the tangents of the first pencil are $(1, 2, 3, 4)$, and those of the second pencil, taken in a proper order, are $(1', 2', 3', 4')$, then we have $(1, 2, 3, 4)$ homologous with each of the arrangements $(1', 2', 3', 4')$, $(2', 1', 4', 3')$, $(3', 4', 1', 2')$, $(4', 3', 2', 1')$. And in each case the intersections of the four corresponding tangents lie on a conic passing through the two given points on the curve.

Article 120. Hence taking the points on the curve to be the circular points at infinity, we have the sixteen foci lying in fours upon four different circles—that is, we have four tetrads of concyclic foci. Let any one of these tetrads be A, B, C, D, then if

Antipoints of $(B, C), (A, D)$ are $(B_1, C_1), (A_1, D_1)$,
$(C, A), (B, D)$ are $(C_2, A_2), (B_2, D_2)$,
$(A, B), (C, D)$ are $(A_3, B_3), (C_3, D_3)$,

the four tetrads of concyclic foci are

$$\begin{array}{llll} A, & B, & C, & D; \\ A_1, & B_1, & C_1, & D_1; \\ A_2, & B_2, & C_2, & D_2; \\ A_3, & B_3, & C_3, & D_3. \end{array}$$

It is to be observed that if A, B, C, D are any four points on a circle, then if, as above, we pair these in any manner, and take the antipoints of each pair, the four antipoints lie on a circle, and thus the original system A, B, C, D, of four points on a circle, leads to the remaining three systems of four points on a circle. The theory is in fact that already discussed ante, No. 72 *et seq.*

Article 121. The preceding theory applies without alteration to the bicircular quartic, viz., the quartic curve which has a node at each of the circular points at infinity. The class is here $= 8$, but among the tangents from a node each of the two tangents at the node is to be reckoned twice, and the number of the remaining tangents is $= 4$: the number of foci is $= 16$. And, by the general theorem that in a binodal quartic the pencils of tangents from the two nodes respectively are homologous, the sixteen foci are related to each other precisely in the manner of the foci of the circular cubic. The latter is in fact a particular case of the former, viz., the bicircular quartic may break up into the line infinity, and a circular cubic.

127 to 129. Centre of the Circular Cubic, and Nodo-Foci, &c. of the Bicircular Quartic.

Article 122. The tangents at I, J have not been recognised as tangents from I, J, giving by their intersection a focus, but it is necessary in the theory to pay attention to the tangents in question. It is clear that these tangents are in fact asymptotes; viz., in the case of the circular cubic they are the two imaginary asymptotes of the curve, and in the case of a bicircular quartic, the two pairs of imaginary parallel asymptotes; but it is convenient to speak of them as the tangents at I, J.

Article 123. In the case of a circular cubic, the tangents at I and J meet in a point which I call the *centre* of the curve, viz., this is the intersection of the two imaginary asymptotes. In the case of a bicircular quartic, the tangents at I consist of a pair of lines meeting in a point which I call a *nodo-focus*, and the tangents at J of a pair of lines meeting in another point which I call the other nodo-focus; the two nodo-foci are antipoints of each other, and the line joining them passes through the centre of the curve.

Article 124. The nodo-foci of a bicircular quartic are to be distinguished from the ordinary foci; the nodo-foci are the intersections of the tangents at the circular points at infinity, while the ordinary foci are the intersections of the tangents from the circular points at infinity to the curve. The nodo-foci are, however, included among the foci in the wider sense, viz., they are the points which correspond to the case where the tangents from I and J are the tangents at these points respectively.

130 to 157. Additional Investigations on the Theory of Foci.

Article 125. The theory of foci may be further developed by considering the properties of the polyzomal curves in relation to the circular points at infinity. I have already shown that the general trizomal curve $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$ may be expressed by means of variable zomals, and that when the zomals are circles, the curve belongs to the theory of foci. I proceed to consider some further developments of this theory.

Article 126. If the zomals of a trizomal curve are circles, then the curve is in general a bicircular quartic; for the order of the trizomal is $= 2r$, and if $r = 2$, the order is $= 4$; and since each zomal is a circle, the curve passes through the circular points at infinity, and has a node at each of these points. The trizomal curve $\sqrt{lU} + \sqrt{mV} + \sqrt{nW} = 0$, where $U = 0$, $V = 0$, $W = 0$ are circles, is thus a bicircular quartic.

Article 127. The foci of a bicircular quartic may be determined as follows: The tangents from the circular points at infinity to the curve are the lines joining these points to the foci; and since the curve is a trizomal with circular zomals, the tangents from I and J may be determined by means of the equations of the zomals. In particular, if the zomals are circles passing through two fixed points, then the foci are the antipoints of these fixed points.

Article 128. The case where the zomals are circles having a common radical axis is particularly interesting; for in this case the trizomal curve degenerates into a pair of circles, or into a circular cubic and the line infinity. The condition for this degeneration is that the Jacobian of the three circles passes through the circular points at infinity.

Article 129. I remark that the focal equation of a conic may be generalised to the case of a bicircular quartic. If $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ are the squared distances of the current point from four foci, then the equation

$$a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}} + d\sqrt{\mathfrak{D}} = 0,$$

where $\frac{a}{l} + \frac{b}{m} + \frac{c}{n} + \frac{d}{p} = 0$, represents a bicircular quartic having the four given points for foci. This is the generalisation of the focal equation of a conic to the case of a quartic curve.

Article 130. The theory of polyzomal curves is thus seen to be intimately connected with the theory of foci, and the results obtained in the present memoir furnish a new method of investigating the properties of curves defined by their foci. The method is particularly applicable to the circular cubic and the bicircular quartic, and it is probable that further developments of the theory will lead to important results in regard to these and other curves.

Article 131. I conclude this part of the memoir by remarking that the theory of the variable zomal, which has been established for the trizomal curve, may be extended to the general ν-zomal curve. Given the ν-zomal curve $\sqrt{U_1} + \sqrt{U_2} + \ldots + \sqrt{U_\nu} = 0$, if the zomals are connected by an identical relation $a_1U_1 + a_2U_2 + \ldots + a_\nu U_\nu = 0$, then the curve may be expressed by means of any $\nu - 1$ of the zomals, and the equation will contain a variable parameter. This is the general theorem of the variable zomal, of which the theorem for the trizomal is a particular case.

Article 132. In the case of the tetrazomal curve, the theorem of the variable zomal does not in general hold; for, as has been shown, the tetrazomal curve does not in general admit of being expressed by means of a new zomal in addition to three of the original zomals. There are, however, particular cases in which the theorem does hold, and these cases are of considerable interest in the theory of curves.

Article 133. The present memoir contains the foundations of a new theory of curves, which I have called the theory of polyzomal curves. The theory is far from being complete, and there are many questions which remain to be investigated. I hope, however, that the results which have been obtained will be found to be of sufficient interest to justify the publication of this preliminary account of the theory.

Collected Mathematical Papers of Arthur Cayley

Volume VI

414. ON POLYZOMAL CURVES.

(From the Proceedings of the London Mathematical Society, Vol. IV., 1871–3, pp. 347–352; 423–526; 527–572.)

[Continued from page 522.]

129. In the case of a bicircular quartic, the two tangents at I and the two tangents at J meet in four points, which (although not recognising them as foci) I call the nodo-foci; these lie in pairs on two lines, diagonals of the quadrilateral formed by the four tangents (the third diagonal is of course the line IJ), which diagonals I call the "nodal axes;" and the point of intersection of the two nodal axes is the "centre" of the curve. The nodo-foci are four points, two of them real, the other two imaginary, viz., they are two pairs of antipoints, the lines through the two pairs respectively being, of course, the nodal axes; these are consequently real lines bisecting each other at right angles in the centre (with the relation $1 : i$ between the distances). The centre may also be defined as the intersection of the harmonic of IJ in regard to the tangents at I, and the harmonic of this same line in regard to the tangents at J. Speaking of the tangents as asymptotes, the nodo-foci are the angles of the rhombus formed by the two pairs of parallel asymptotes; the nodal axes are the diagonals of this rhombus, and the centre is the point of intersection of the two diagonals; as such it is also the intersection of the two lines drawn parallel to and midway between the lines forming each pair of parallel asymptotes.

Article No. 130. Circular Cubic and Bicircular Quartic; the Nodal or Symmetrical Case.

130. In a circular cubic or bicircular quartic, the pencil of the tangents from I and that of the tangents through J, considered as corresponding to each other in some one of the four arrangements, may be such that the line IJ considered as belonging to the two pencils respectively shall correspond to itself, and when this is so, the four foci, A, B, C, D, which are the intersections of the corresponding tangents in question, will lie in a line (viz., the conic which exists in the general case will break up into a line-pair consisting of the line IJ and another line). The line in question may be called the focal axis; it will presently be shown that in the case of the circular cubic it passes through the centre, and that in the case of the bicircular quartic it not only passes through the centre, but coincides with one or other of the nodal axes, viz., with that passing through the real or the imaginary nodo-foci; that is, the curve may have on the focal axis two real or else two imaginary nodo-foci. The focal axis contains, as has been mentioned, four foci; the remaining twelve foci are situate symmetrically, six on each side of the focal axis, the arrangement of the sixteen foci being as mentioned ante, No. 81 *et seq.*; the focal axis is in fact an axis of symmetry of the curve, and if preferred it may be named the axis of symmetry, transverse axis, or simply the axis. And the curve (circular cubic, or bicircular quartic) is in this case a "symmetrical" or "axial" curve.

Article Nos. 131 to 140. Circular Cubic and Bicircular Quartic: Singular Forms.

131. The circular cubic may have a node or a cusp. If this were at one of the points I, J the curve would be imaginary, and I do not attend to the case; and for the same reason, for the bicircular quartic I do not attend to the case where one of

132. I consider first the case of the bicircular quartic where each of the points I, J is a cusp. The curve is in this case of necessity symmetrical[1]; it is in fact a Cartesian; viz., the Cartesian may be taken by definition to be a quartic curve having a cusp at each of the circular points at infinity. But in this case, as distinguished from the general case of the bicircular quartic, there is an essential degeneration of all the focal properties, and it is necessary to explain what these become. The centre is evidently the intersection of the cuspidal tangents; the nodo-foci (so far as they can be said to exist) coalesce with the centre, and they do not in so coalescing determine any definite directions for the nodal axes; that is, there are no nodal axes, and the only theorem in regard to the focal axis or axis of symmetry is, that it passes through the centre. Of the four tangents through the point I, one has come to coincide with the line IJ; and similarly, of the four tangents through the point J one has come to coincide with the line JI: there remain only three tangents through I and three tangents through J, and these by their intersections determine nine foci, viz., three foci A, B, C on the axis, and besides (B_1, C_1) the antipoints of (B, C); (C_1, A_1) the antipoints of (C, A); and (A_1, B_1) the antipoints of (A, B).

133. The remaining seven foci have disappeared, viz., we may consider that one of them has gone off to infinity on the focal axis, and that three pairs of foci have come to coincide with the points I, J respectively. The circle (as in the general case of a symmetrical quartic) has become a line, the focal axis; the circles R, S, T (contrary to what might at first sight appear) continue to be determinate circles, viz., these have their centres at A, B, C respectively, and pass through the points (B_1, C_1), (C_1, A_1), and (A_1, B_1) respectively, see ante, No. 83. But on each of these circles we have not more than two proper foci, and it is only on the axis as representing the circle that we have three proper foci, the axial foci A, B, C: in regard hereto it is to be remarked that the equation of the curve can be expressed not only by means of these three foci in the form $\sqrt{\mathfrak{A}}+\sqrt{\mathfrak{B}}+\sqrt{\mathfrak{C}}=0$; but by means of any two of them in the form $\sqrt{\mathfrak{A}}+\sqrt{\mathfrak{B}}+K=0$, where K is a constant, or, what is the same thing (z being introduced for homogeneity in the expressions of $\mathfrak{A}$ and $\mathfrak{B}$ respectively), in the form $\sqrt{\mathfrak{A}}+\sqrt{\mathfrak{B}}+Kz^2=0$.

134. Using for the moment the expression "twisted" as opposed to symmetrical (viz., the curve is twisted when there is not any axis of symmetry, but the foci lie only on circles) then the classification is

$$\begin{array}{l}
\text{Circular Cubics, twisted,}\\
\qquad\qquad\text{symmetrical,}\\
\text{Bicircular Quartics, twisted,}\\
\qquad\qquad\text{symmetrical,}\left\{\begin{array}{l}\text{Ordinary,}\\ \text{Bicuspidal} = \text{Cartesian,}\end{array}\right.
\end{array}$$

and each of these kinds may be general, nodal, or cuspidal; viz., for the two last mentioned kinds there may be a node or a cusp at a real point of the curve.

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[1] It will appear, post Nos. 161—164, that if starting with three given points as the foci of a bicircular quartic, we impose the condition that the nodes at I, J shall be each of them a cusp, then either the quartic will be the circle through the three points taken twice, in which case the assumed focal property of the given three points disappears altogether, or else the three points must be in lined, and thus the curve be symmetrical, that is, a Cartesian.

135. In the case of a node,—say the point N; first if the curve (circular cubic or bicircular quartic) be twisted then of the four foci A, B, C, D we have two, suppose B and C, coinciding with N; and the sixteen foci are as follows, viz.

B, C, A, D are N, N, A, D;		
B_1, C_1, A_1, D_1	„	$N, N,$ Antipoints of (A, D);
C_1, A_1, B_1, D_1	„	Antipoints of (N, A), Antipoints of (N, B);
A_3, B_3, C_3, D_3	do.	

viz., we have the points (A, D) each once, the node four times, the antipoints of (A, D) once, and the antipoints of (N, A) and of (N, D), each pair twice. But properly there are only four foci, viz., the points A, D and their antipoints. The circle subsists as in the general case, and so does the circle $R(BC, AD)$, viz., this has for centre the intersection of the line AB by the tangent at N to the circle C, and it passes through the point N, of course cutting the circle at right angles; the circles S and T each reduce themselves each to the point N considered as an evanescent circle, or what is the same thing to the line-pair NI, NJ.

136. The case is nearly the same if the curve be symmetrical, but in the case of the bicircular quartic excluding the Cartesian: viz., we have on the axis the foci B, C coinciding at N, and the other two foci A, D; the sixteen foci are as above and the circle R is determined by the proper construction as applied to the case in hand, viz., the centre is the intersection of the axis by the radical axis of the point N (considered as an evanescent circle) and the circle on AD as diameter; that is $RN^2 = RA \cdot RB$. And the circles S and T reduce themselves each to the point N considered as an evanescent circle.

137. Next if we have a cusp, say the point K: first if the curve (circular cubic or bicircular quartic) be twisted then of the four foci A, B, C, D, three, suppose A, B, C, coincide with K; and the sixteen foci are as follows, viz.,

B, C, A, D are K, K, K, D;		
B_1, C_1, A_1, D_1	„	$K, K,$ Antipoints of (K, D),
C_1, A_1, B_1, D_1	do.	
A_3, A_2, C_1, D_3	do.	

viz., we have the point D once, the point K nine times, and the antipoints of K, D three times. But properly the point D is the only focus. The circle is, it would appear, any circle through K, D, but possibly the particular circle which touches the cuspidal tangent may be a better representative of the circle of the general case; the circles R, S, T reduce themselves each to the point K considered as an evanescent point.

138. The like is the case if the curve be symmetrical, but in the case of the bicircular quartic excluding the Cartesian; the circle is here the axis, which is in fact the cuspidal tangent.

139. For the Cartesian, if there is a node N; then of the three foci A, B, C, two, suppose B and C, coincide with N; the nine foci are A once, N four times, and the antipoints of N, A twice; but properly the point A is the only focus. And if there be a cusp K; then all the three foci A, B, C coincide with K; and the nine foci are K nine times; but properly the point K is the only focus.

140. The conclusion is, that in the singular forms the focal properties undergo an essential degradation, and that we have in the several cases only a limited or even no proper set of foci.

Article Nos. 141 to 149. Analytical Theory for the Circular Cubic.

141. The equation for the circular cubic may be taken to be

$$\xi(\eta^2 + \theta\eta z + ez^2) + z^2(a\xi + b\eta + cz) = 0,$$

viz. (ξ, η, z) being any coordinates whatever, this is the general equation of a cubic passing through the points $(\xi = 0, z = 0)$, $(\eta = 0, z = 0)$, and at these points touched by the lines $\xi = 0$, $\eta = 0$ respectively. And if $(\xi, \eta, z = 1)$ be circular coordinates, then we have the general equation of a circular cubic having the lines $\xi = 0$, $\eta = 0$ for its asymptotes, or say the point $\xi = 0$, $\eta = 0$ for its centre; the equation of the remaining asymptote is evidently $p\xi + q\eta + ez = 0$; to make the curve real we must have (p, q) and (a, b) conjugate imaginaries, e and c real.

142. Taking in any case the points I, J to be the points $\xi = 0, z = 0$ and $\eta = 0, z = 0$ respectively, for the equation of a tangent from I write $p\xi = \theta z$; then we have

$$\theta\eta(\theta z + q\eta + ez) + z(a\theta z + bp\eta + cpz) = 0,$$

that is

$$z^2(a\theta + cp) + \eta z(\theta^2 + e\theta + bp) + \eta^2 \cdot q\theta = 0,$$

and the line will be a tangent if only

$$(\theta^2 + e\theta + bp)^2 - 4q\theta(a\theta + cp) = 0,$$

that is

$$\theta^4 + 2e\theta^3 + \theta^2(e^2 + 2bp - 4aq) + 2\theta(bp - 2cq) + b^2p^2 = 0,$$

which is the equation for the determination of the tangents from I.

143. The first and second cases are those in which two roots are equal, giving a node at I or a cusp at I respectively; and similarly the third and fourth cases are those in which two roots are equal, giving a node at J or a cusp at J respectively. For a node at I we have $bp = 0$ and $e = 0$; but if $p = 0$, then the cubic is $(\xi\eta + z^2)(b\eta + cz) + e\eta^2 z = 0$, which has a cusp at J; hence for a node at I we must have $b = 0$, $e = 0$, and the equation is

$$\xi(\eta^2 + \theta\eta z) + z^2(a\xi + cz) = 0.$$

For a cusp at I we have the same conditions $b = 0$, $e = 0$, and the further condition $aq = 0$; but if $a = 0$, then the cubic is $(\xi\eta^2 + q\xi\eta z + cz^3) = 0$, which is a line and a conic; hence we must have $q = 0$, and the equation is

$$\xi\eta^2 + z^2(a\xi + cz) = 0.$$

Similarly for a node or cusp at J we have $a = 0$, $e = 0$, with $q = 0$ for the cusp, and the equations are

$$\eta(\xi^2 + p\xi z) + z^2(b\eta + cz) = 0 \quad \text{and} \quad \eta\xi^2 + z^2(b\eta + cz) = 0.$$

144. First if $p = q$, $a = b$: the equation becomes

$$\xi\eta(\eta + pz) + z^2(a\xi + a\eta + cz) = 0,$$

which has a node at the point $(\xi = 0, \eta = 0)$, that is, the centre.

145. The equation for the bicircular quartic may be taken to be

$$k(\xi^2 - \alpha z^2)(\eta^2 - \beta z^2) + z^2\{a\xi + b\eta + cz^2 + \xi\eta(a'\xi + b'\eta) + c'z^2\} = 0,$$

viz. (ξ, η, z) being any coordinates whatever, this is the equation of a quartic curve having the points $(\xi = 0, z = 0)$ and $(\eta = 0, z = 0)$ for nodes. And if $(\xi, \eta, z = 1)$ be circular coordinates, then we have the equation of a bicircular quartic, the points I, J being the nodes.

146. The equation may be written

$$k(\xi^2 - \alpha z^2)(\eta^2 - \beta z^2) + z^2 \cdot u = 0,$$

where $u = 0$ represents a conic. For the tangents from I we have $\xi^2 = \alpha z^2$; the equation gives $\xi^2 - \alpha z^2 = 0$, and thence $k\alpha(\eta^2 - \beta z^2) + u = 0$; that is,

$$\eta^2(k\alpha + b') + \eta(b\xi + ez) + k\alpha(a - \beta)z^2 + a\xi^2 + c\xi z + c'z^2 = 0.$$

But writing herein $\xi^2 = \alpha z^2$, $\xi = \sqrt{\alpha} \cdot z$, this becomes

$$\eta^2(k\alpha + b') + \eta(b\sqrt{\alpha} \cdot z + ez) + z^2(k\alpha^2 + a\alpha + c\sqrt{\alpha} + c') = 0,$$

which is a quadratic for the determination of the tangents from I; viz., these are $\xi = \sqrt{\alpha} \cdot z$, where $\sqrt{\alpha}$ is a root of

$$k\alpha^2 + (a + b'\beta)\alpha + c\sqrt{\alpha} + c' + e\beta = 0.$$

Similarly for the tangents from J.

147. The two equations may be written

$$
\begin{gathered}
24k\alpha^2\theta = 8\beta\theta^3 + 8k\alpha\beta\cdot\theta - Gka\alpha\theta - 8k^2\alpha\beta\cdot\theta - 4kc\theta + e^2,\\
-8k^2\alpha\beta\cdot\theta - 4kc\theta + e^2,\\
10\theta^2 - 6\alpha\alpha + 3e - 6kb'\beta + 3e'\\
24k\alpha^2\beta^2 + 24kc + 6
\end{gathered}
$$

which equations have the same invariants; in fact for the first equation the invariants are found to be as follows, viz., if for shortness $\Theta = -8k^2\alpha\beta\cdot 8k^2 - 4kc\theta + e^2$, then

$$
I = \Theta\{576k^2\alpha^2\beta^2 + 576k^2c\alpha\beta^2 + 144k^2(a^2\alpha + b'^2\beta) + 72k\cdot e\alpha\beta + 3c^2\}.
$$

$$
\begin{gathered}
J = \Theta\{576k^2\alpha^2\beta^2 + 576k^2c\alpha\beta^2 + 144k^2(a^2\alpha + b'^2\beta) + 36k\cdot e\alpha\beta - c^3\}\\
-c^2\{576k^2\alpha\beta\cdot c\alpha\beta^2 + 144k^2c(a^2\alpha + b'^2\beta) + 36k\cdot e\alpha\beta^2\}
\end{gathered}
$$

and then by symmetry the other equation has the same invariants. The absolute invariant $I^3 \div J^2$ has thus the same value in the two equations, that is, the equations are linearly transformable the one into the other.

148. In the first case, if θ be determined by the foregoing equation, we may take as corresponding tangents through the two nodes respectively $\xi = \sqrt{\alpha}\cdot z$, $\eta = \sqrt{\beta}\cdot z$; the foci (A, B, C, D), which are the intersections of the pairs of lines $(\xi = \sqrt{\alpha}\cdot z, \eta = \sqrt{\beta}\cdot z)$, &c., lie, it is clear, in the line $\beta\xi - \alpha\eta = 0$, which is one of the nodal axes of the curve. Similarly, in the second case, if θ be determined by the foregoing equation, we may take as corresponding tangents through the two nodes respectively $\xi = \sqrt{\alpha}\cdot z$, $\eta = -\sqrt{\beta}\cdot z$; the foci (A, B, C, D), which are the intersections of the pairs of lines $(\xi = \sqrt{\alpha}\cdot z, \eta = -\sqrt{\beta}\cdot z)$, &c., lie in the line $\beta\xi + \alpha\eta = 0$, which is the other of the nodal axes of the curve. In either case the foci A, B, C, D lie in a line, that is, we have the curve symmetrical; and, as we have just seen, the focal axis, or axis of symmetry, is one or other of the nodal axes.

149. In the case of the Cartesian, or when $\alpha = 0$, $\beta = 0$, viz., the equation $a\alpha = b\beta$ is satisfied identically, and this seems to show that the Cartesian is symmetrical; it is to be observed, however, that for $\alpha = 0$, $\beta = 0$ the foregoing formulæ fail, and it is proper to repeat the investigation for the special case.

Article Nos. 150 to 153. Casey's Theorem for the Bicircular Quartic.

150. The theorem that, for a bicircular quartic, if $(\xi - \alpha z = 0, \eta - \alpha' z = 0)$, $(\xi - \beta z = 0, \eta - \beta' z = 0)$, $(\xi - \gamma z = 0, \eta - \gamma' z = 0)$, $(\xi - \delta z = 0, \eta - \delta' z = 0)$ are the equations of the pairs of tangents from the four foci A, B, C, D respectively, then the equation of the curve is

$$\sqrt{(\xi - \alpha z)(\eta - \alpha' z)} + \sqrt{(\xi - \beta z)(\eta - \beta' z)} + \sqrt{(\xi - \gamma z)(\eta - \gamma' z)} + \sqrt{(\xi - \delta z)(\eta - \delta' z)} = 0,$$

where the radicals are connected by a linear equation, may be established as follows.

151. Consider the case of the bicircular quartic, and take as before $(\xi = 0, z = 0)$ and $(\eta = 0, z = 0)$ for the coordinates of the points I, J respectively. Let the two tangents from the focus A be $\xi - \alpha z = 0$, $\eta - \alpha' z = 0$, say for shortness $p = 0$, $p' = 0$, then the equation of the curve is expressible in the form $pp'\Omega = \mho^2$, where $\Omega = 0$, $\mho = 0$ are each of them a circle, viz., Ω and $\mho$ are each of them a quadric function containing the terms z^2, $z\eta$, $z\xi$, and $\xi\eta$. Taking an indeterminate coefficient λ, the equation may be written

$$pp'(\Omega + 2\lambda\mho + \lambda^2 pp') = (\mho + \lambda pp')^2,$$

and then λ may be so determined that $\Omega + 2\lambda\mho + \lambda^2 pp' = 0$, shall be a 0-circle, that is, shall have its centre at I; viz. this will be the case if the coefficients of η^2, ηz, z^2 are each $= 0$; or, what is the same thing, if the circle reduces itself to the line-pair $\xi = 0$, $z = 0$; that is, if the coefficients of ξ^2, $\xi\eta$, ξz, η^2, ηz, z^2 are proportional to $0, 0, 0, 0, 0, 1$ respectively.

152. We have to show that the four foci $(p=0, p'=0)$, $(q=0, q'=0)$, $(r=0, r'=0)$, $(s=0, s'=0)$, lie in a line; that is, that the line in question is an axis of the curve. But this is a particular case of the theorem that, if $(p=0, p'=0)$, $(q=0, q'=0)$, $(r=0, r'=0)$ are any three foci, then the circle through the three foci (which, it has been shown, exists for any three foci) is the circle R. In fact, if the three foci lie in a line, then the circle R is the line, or say the circle on the line at infinity between the two circular points at infinity; that is, the circle is the line at infinity, and the axis is thus shown to be a line containing the four foci.

153. The analytical investigation of the focal properties of the bicircular quartic may be presented under a somewhat different form. Writing the equation as

$$(\xi^2-\alpha z^2)(\eta^2-\beta z^2)+(a\xi+b\eta+cz^2+a'\xi\eta+b'\xi^2+c'\eta^2)z^2=0,$$

the condition that this shall represent a bicircular quartic is that α and β shall each be determinate, so that the nodes at I and J shall be proper nodes; and this requires that k shall not $=0$, and that a', b' shall not be both of them $=0$. If $k=0$, the curve degenerates into a conic twice; if $a'=0$, $b'=0$, the curve degenerates into a pair of circles.

154. The foregoing conclusions may be verified by means of the expressions for the foci in terms of the coefficients of the equation. For if we write

$$\Omega = a\xi + b\eta + cz^2 + a'\xi\eta + b'\xi^2 + c'\eta^2,$$

then the equation is

$$k(\xi^2 - \alpha z^2)(\eta^2 - \beta z^2) + z^2\Omega = 0,$$

and for the tangents at $(\xi = 0, z = 0)$ we have $\xi^2 = \alpha z^2$, and thence

$$k\alpha(\eta^2 - \beta z^2) + \Omega = 0;$$

similarly for finding the tangents at $(\eta = 0, z = 0)$ we have only to to interchange ξ and η, α and β, and write Ω' in place of Ω.

155. The condition for the determination of the foci is that the equation

$$\sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} + \sqrt{\mathfrak{C}} + \sqrt{\mathfrak{D}} = 0$$

shall be expressible as a determinant; viz. this is

$$\begin{vmatrix} \sqrt{\mathfrak{A}} & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} & \sqrt{\mathfrak{D}} \\ 1 & 1 & 1 & 1 \end{vmatrix} = 0,$$

where the equation is satisfied by writing

$$\sqrt{\mathfrak{A}} : \sqrt{\mathfrak{B}} : \sqrt{\mathfrak{C}} : \sqrt{\mathfrak{D}} = \lambda : \mu : \nu : \rho,$$

where $\lambda + \mu + \nu + \rho = 0$.

156. Calculating $AB' - A'B$, $CA' - C'A$, $CD' - CD$, $BD' - B'D$, these are at once found to be proportional to a, b, c, d respectively; and hence we have the theorem that for a bicircular quartic if $(\xi - \alpha z = 0, \eta - \alpha' z = 0)$ are the equations of the tangents from any focus, then the equation of the curve may be written in the form

$$a\sqrt{(\xi - \alpha z)(\eta - \alpha' z)} + b\sqrt{(\xi - \beta z)(\eta - \beta' z)} + c\sqrt{(\xi - \gamma z)(\eta - \gamma' z)} + d\sqrt{(\xi - \delta z)(\eta - \delta' z)} = 0,$$

where a, b, c, d are proportional to the distances of the foci from any line whatever.

Article Nos. 157 to 160. The Focal Conic.

157. I attend in the first place to the focal conic of the bicircular quartic; the theorem is that given any three circles A, B, C, there exists a conic passing through the centres of these three circles, and having the property that, taking on the conic any point D as the centre of a fourth circle D, the four circles have a common orthogonal circle; or, what is the same thing, that the four circles are tangential to a fifth circle. The proof may be presented as follows.

158. Let the equations of the three circles be

$$x^2+y^2+2gx+2fy+c = 0, \quad (A), x^2+y^2+2g'x+2f'y+c' = 0, \quad (B), x^2+y^2+2g''x+2f''y+c'' = 0, \quad (C),$$

then the equation of any circle orthogonal to these three is

$$\begin{vmatrix} x^2+y^2 & x & y & 1 \\ -c & g & f & -1 \\ -c' & g' & f' & -1 \\ -c'' & g'' & f'' & -1 \end{vmatrix} = 0;$$

and the condition that a fourth circle

$$x^2 + y^2 + 2G\xi + 2F\eta + C = 0,$$

shall be orthogonal to this, is found to be

$$\lambda(aG + bF + C + d) + \mu(a'G + b'F + C + d') + \nu(a''G + b''F + C + d'') = 0,$$

which is the equation of a conic passing through the centres of the three circles A, B, C.

159. To find the envelope of a circle whose centre lies on a given conic, and which cuts at right angles a given circle. Let the conic be

$$(a, b, c, f, g, h | x, y, z)^2 = 0,$$

and the circle be $(x-\alpha)^2+(y-\beta)^2=r^2$; then for the circle through the centre (x, y, z) we have

$$(X-x)^2+(Y-y)^2=\rho^2,$$

where $\rho^2=(x-\alpha)^2+(y-\beta)^2-r^2$. The condition of orthogonality gives

$$x^2+y^2-\rho^2=0, \quad \text{or} \quad x^2+y^2-(x-\alpha)^2-(y-\beta)^2+r^2=0,$$

that is

$$2\alpha x+2\beta y-(\alpha^2+\beta^2-r^2)=0,$$

which is a line; and the envelope is found by combining this with the conic.

160. In particular, if the conic is a conic passing through the points I, J, then the envelope is a bicircular quartic. For if the conic passes through I and J, then the condition that the centre of the variable circle shall lie on the conic is

$$l\xi+m\eta+n+\lambda\sqrt{(\xi\eta)}=0,$$

and the envelope of the circle is found to be

$$(l^2, m^2, n^2, -mn, -nl, -lm | \xi^2, \eta^2, z^2, \eta z, z\xi, \xi\eta)=0,$$

which is the equation of a bicircular quartic.

Article Nos. 161 to 164. Condition that a Bicircular Quartic shall be a Cartesian.

161. The condition that a bicircular quartic shall be a Cartesian, or shall have each of its nodes at I and J a cusp, may be investigated as follows. The equation is

$$k(\xi^2-\alpha z^2)(\eta^2-\beta z^2)+z^2\Omega=0,$$

and for a cusp at I we must have $\alpha=0$ and the tangent $\xi=0$ a cuspidal tangent; that is, writing $\xi=0$ in the equation, the resulting equation in (η, z) must have a cusp at $(\eta=0, z=0)$, which requires that the terms in η^4, $\eta^3 z$ shall vanish, and that the term $\eta^2 z^2$ shall not vanish. The conditions are thus found to be $k\beta+c'=0$ and $e=0$; and similarly for a cusp at J, $k\alpha+b'=0$ and $e'=0$.

162. We have thus the conditions for the Cartesian

$$b' = -k\alpha, \quad c' = -k\beta, \quad e = 0, \quad e' = 0,$$

and the equation becomes

$$k(\xi^2 - \alpha z^2)(\eta^2 - \beta z^2) + z^2(a\xi + b\eta + cz^2 - k\alpha\eta^2 - k\beta\xi^2) = 0,$$

which may be written

$$k(\xi^2 - \alpha z^2)(\eta^2 - \beta z^2) - k\alpha z^2\eta^2 - k\beta z^2\xi^2 + z^2(a\xi + b\eta + cz^2) = 0,$$

or, what is the same thing,

$$k(\xi^2\eta^2 - \alpha\eta^2 z^2 - \beta\xi^2 z^2 + \alpha\beta z^4) - k\alpha z^2\eta^2 - k\beta z^2\xi^2 + z^2(a\xi + b\eta + cz^2) = 0,$$

that is

$$k\xi^2\eta^2 - 2k\alpha\eta^2 z^2 - 2k\beta\xi^2 z^2 + k\alpha\beta z^4 + z^2(a\xi + b\eta + cz^2) = 0.$$

But this may be reduced to the standard form of the Cartesian.

163. The conditions may also be expressed in terms of the foci. If the foci are given as the intersections of the pairs of lines $(\xi = \alpha_r z, \eta = \beta_r z)$, then the condition is that the four pairs of lines shall be such that the product equation

$$\prod_{r=1}^{4}(\eta - \beta_r z) = 0$$

shall have the coefficients of η^4, $\eta^3 z$ each $= 0$; and similarly for the other pair of tangents.

164. In particular, if three foci A, B, C are given, then for the Cartesian the fourth focus D is determined as the intersection of two known lines; and it appears that the three foci must lie on a line, that is, the Cartesian is necessarily symmetrical.

165. There is not, in general, any identical equation

$$a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}} + d\sqrt{\mathfrak{D}} = 0,$$

but if the four foci lie on a circle, then such an equation exists. In fact, if the foci lie on a circle, then we may write

$$\sqrt{\mathfrak{A}} : \sqrt{\mathfrak{B}} : \sqrt{\mathfrak{C}} : \sqrt{\mathfrak{D}} = \lambda : \mu : \nu : \rho,$$

where $\lambda + \mu + \nu + \rho = 0$, and the condition is satisfied.

Article Nos. 166 to 170. Applications of the Focal Theory.

166. The focal theory of the bicircular quartic leads to many interesting applications. In particular, it gives a construction for the quartic when the four foci and the constant k are given; and it leads to the investigation of the bitangents and of the nodes and cusps of the curve.

167. If the four foci are given, then the equation of the curve is at once written down in the form

$$\sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} + \sqrt{\mathfrak{C}} + \sqrt{\mathfrak{D}} = 0,$$

where $\mathfrak{A} = 0$, $\mathfrak{B} = 0$, $\mathfrak{C} = 0$, $\mathfrak{D} = 0$ are the equations of the pairs of tangents from the four foci respectively.

168. The equation may be rationalized by writing it in the form

$$\sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} = -\sqrt{\mathfrak{C}} - \sqrt{\mathfrak{D}},$$

and squaring; we thus obtain

$$\mathfrak{A} + \mathfrak{B} + 2\sqrt{\mathfrak{A}\mathfrak{B}} = \mathfrak{C} + \mathfrak{D} + 2\sqrt{\mathfrak{C}\mathfrak{D}},$$

that is

$$\mathfrak{A} + \mathfrak{B} - \mathfrak{C} - \mathfrak{D} = 2(\sqrt{\mathfrak{C}\mathfrak{D}} - \sqrt{\mathfrak{A}\mathfrak{B}}),$$

and squaring again,

$$(\mathfrak{A} + \mathfrak{B} - \mathfrak{C} - \mathfrak{D})^2 = 4(\mathfrak{C}\mathfrak{D} + \mathfrak{A}\mathfrak{B} - 2\sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}\mathfrak{D}}).$$

The rationalized equation is thus of the fourth order in (ξ, η, z), and represents a bicircular quartic.

169. The bitangents of the bicircular quartic may be investigated by means of the focal equation. A bitangent is a line which touches the curve in two points; and if the line be taken as $lx + my + nz = 0$, then the condition of tangency gives a relation between the coefficients (l, m, n). It appears that there are, in general, four bitangents, and that these correspond to the four foci in such wise that each bitangent is the polar of a focus with respect to a certain conic.

170. The investigation leads to the consideration of the Jacobian of four circles, and to the theorem that the four foci of a bicircular quartic are the centres of the four circles which are orthogonal to three given circles. The theory is intimately connected with that of the "focal conic" already referred to.

Article Nos. 171 and 172. Locus of Centre of Variation.

171. The equation of the bicircular quartic may be written in the form

$$\frac{\xi^2}{a} + \frac{\eta^2}{b} + \frac{z^2}{c} + \frac{2\eta z}{a'} + \frac{2z\xi}{b'} + \frac{2\xi\eta}{c'} = 0,$$

and the centre of the curve is then determined as the intersection of the lines

$$\frac{\xi}{a} + \frac{\eta}{c'} + \frac{z}{b'} = 0, \quad \frac{\xi}{c'} + \frac{\eta}{b} + \frac{z}{a'} = 0.$$

172. If the curve varies, the locus of the centre is found by eliminating the variable parameters between the equations; and it appears that the locus is, in general, a conic. This conic is the "focal conic" of the system of bicircular quartics.

173. The line of centres is the axis of the curve, but the centres A, B, C, D are not in general on a line; they are the centres of the circles A, B, C, D respectively, and the condition that the four circles shall have a common orthogonal circle is the condition already referred to in connexion with the focal conic.

Article Nos. 174 to 177. Casey's Equation.

174. Casey's well-known equation for four circles touching a fifth circle may be established by means of the focal properties of the bicircular quartic. If four circles touch a fifth circle, then the distances of their centres from any line satisfy a certain linear relation; and conversely, if such a relation exists, the four circles touch a fifth circle.

175. Let the four circles be A, B, C, D, and let their equations be

$$x^2 + y^2 + 2g_r x + 2f_r y + c_r = 0, \quad (r = 1, 2, 3, 4).$$

If these touch a fifth circle, then writing the equation of the fifth circle as

$$x^2 + y^2 + 2Gx + 2Fy + C = 0,$$

the condition of tangency gives

$$(G - g_r)^2 + (F - f_r)^2 = (C - c_r)^2,$$

for $r = 1, 2, 3, 4$. These are four equations which must be consistent, and the condition of consistency is Casey's equation.

176. Casey's equation may be written in the form

$$\begin{vmatrix} d_{12} & d_{13} & d_{14} \\ d_{21} & d_{23} & d_{24} \\ d_{31} & d_{32} & d_{34} \end{vmatrix} = 0,$$

where d_{rs} is the distance between the centres of the circles r and s, measured with the proper sign. The equation is symmetrical in regard to the four circles, and it is the condition that the four circles shall be tangential to a fifth circle.

177. The equation may also be presented in a form which brings into view the connexion with the bicircular quartic. If the four circles have a common orthogonal circle, then the four foci lie on a conic; and conversely, if the four foci lie on a conic, then the four circles have a common orthogonal circle. This is the theorem of the focal conic already referred to.

Article Nos. 178 to 181. Case of Double Contact; Casey's Equation in the Problem of the Tangent Circles.

178. In the problem of the tangent circles,—given three circles, to draw the circles touching all three,—there are, in general, eight solutions. The investigation leads to Casey's equation, which gives the condition that four circles shall touch a fifth. The eight circles may be divided into four sets of two, each set touching a different one of the four pairs of circles which are the inverses of the given three circles.

179. The equation for the tangent circles may be obtained as follows. Let the three given circles be A, B, C, and let the required circle be S. Let d, d', d'' be the distances of the centre of S from the centres of A, B, C respectively, and let r, r', r'', ρ be the radii of A, B, C, S respectively. Then for external contact we have

$$d = r + \rho, \quad d' = r' + \rho, \quad d'' = r'' + \rho,$$

and for internal contact the corresponding equations with the signs of r, r', r'' changed. The elimination of ρ between these equations leads to a relation between d, d', d'' which is the condition that the circle S shall touch the three circles A, B, C.

180. The condition may be expressed as the vanishing of a determinant. Writing the equations

$$d - r - \rho = 0, \quad d' - r' - \rho = 0, \quad d'' - r'' - \rho = 0,$$

and eliminating ρ, we obtain

$$\begin{vmatrix} 1 & 1 & 0 & 0 \\ r & 0 & d & -1 \\ r' & d' & 0 & -1 \\ r'' & 0 & d'' & -1 \end{vmatrix} = 0,$$

which is the condition for the tangent circle. This condition is equivalent to Casey's equation for the four circles A, B, C, S.

181. The investigation shows that the problem of the tangent circles is intimately connected with the theory of the bicircular quartic. The eight tangent circles correspond to the eight foci of a certain bicircular quartic, and the conditions of tangency are given by the focal equation of this quartic.

Article Nos. 182 and 183. Focal Formulæ for the General Curve.

182. Comparing the formula of No. 120 with that of No. 50, we see that the focal formulae for the general curve are as follows:—If the coordinates of a point on the curve are proportional to (ξ, η, ζ), and if (α, β, γ), $(\alpha', \beta', \gamma')$, &c. are the coordinates of the foci, then the distances of the point from the foci are proportional to

$$\sqrt{(a\xi + b\eta + c\zeta)(a'\xi + b'\eta + c'\zeta)},$$

and the equation of the curve is obtained by equating to zero a linear function of the square roots of these expressions.

183. In the case of a curve of the nth order, there are n^2 foci, and the equation of the curve may be written in the form

$$\sum_{r=1}^{n^2} \sqrt{\mathfrak{A}_r} = 0,$$

where $\mathfrak{A}_r = 0$ is the equation of the pair of tangents from the rth focus. This is the generalization of the focal property of the conic, and it applies to all algebraic curves.

Article Nos. 184 and 185. Case of the Circular Cubic.

184. In the case of the circular cubic, the equation may be written in the form

$$\sqrt{\mathfrak{A}} + \sqrt{\mathfrak{B}} + \sqrt{\mathfrak{C}} = 0,$$

where $\mathfrak{A} = 0$, $\mathfrak{B} = 0$, $\mathfrak{C} = 0$ are the equations of the pairs of tangents from the three foci respectively. The curve has three foci, and the equation is the condition that the sum of the distances from the three foci, measured in a certain way, shall be zero.

185. The rationalized equation of the circular cubic is obtained by squaring twice, and it is found to be of the form

$$(\mathfrak{A} + \mathfrak{B} + \mathfrak{C})^2 - 4(\mathfrak{A}\mathfrak{B} + \mathfrak{B}\mathfrak{C} + \mathfrak{C}\mathfrak{A}) = 0,$$

which is a particular case of the general equation of the third order. The curve has a node at each of the circular points at infinity, and the three foci lie on a line, which is the axis of the curve.

186. For the circular cubic, the three foci A, B, C lie on a line, and the equation of the curve may be written in the form

$$a\sqrt{\mathfrak{A}}+b\sqrt{\mathfrak{B}}+c\sqrt{\mathfrak{C}}=0,$$

where a, b, c are constants. The rationalized equation is

$$a^2\mathfrak{A}+b^2\mathfrak{B}+c^2\mathfrak{C}+2bc\sqrt{\mathfrak{B}\mathfrak{C}}+2ca\sqrt{\mathfrak{C}\mathfrak{A}}+2ab\sqrt{\mathfrak{A}\mathfrak{B}}=0,$$

and squaring again we obtain the equation of the cubic.

187. The focal circles of the circular cubic are three circles, each passing through two of the foci and through the node at infinity. The centres of these circles lie on the axis of the curve, and they are the centres of the three pairs of antipoints of the foci.

188. The classification of circular cubics is as follows:—

(1) General circular cubic: three distinct foci, no singularities.

(2) Nodal circular cubic: a node at a finite point, two of the foci coinciding at the node.

(3) Cuspidal circular cubic: a cusp at a finite point, all three foci coinciding at the cusp.

(4) Symmetrical circular cubic: the three foci on a line, the curve having an axis of symmetry.

Each of these may be real or imaginary, and the curve may have one or two branches.

189. These values of $\sqrt{l}$, $\sqrt{m}$, $\sqrt{n}$ give the equations of the two circles through the three foci; and the envelope of the line joining the points of contact of tangent circles is the focal axis of the curve.

190. The theorem that the four foci of a bicircular quartic lie on a circle may be proved directly from the focal equation. If the equation is

$$\sqrt{\mathfrak{A}}+\sqrt{\mathfrak{B}}+\sqrt{\mathfrak{C}}+\sqrt{\mathfrak{D}}=0,$$

then the condition that the four foci shall lie on a circle is that there shall exist a linear relation between the square roots; and this condition is satisfied identically when the four foci lie on a circle.

191. The converse theorem is also true: if the four foci lie on a circle, then the curve is a bicircular quartic. This follows from the fact that the focal equation determines the curve uniquely when the foci and the constant k are given.

192. The focal circles of the bicircular quartic are four circles, each passing through three of the foci. The centres of these circles lie on the focal conic, and the circles are orthogonal to the common orthogonal circle of the four foci.

193. The eight foci of a bicircular quartic consist of four real and four imaginary points, or of two real and six imaginary points, or of eight imaginary points, according to the nature of the curve. The real foci lie on the real focal circles, and the imaginary foci on the imaginary focal circles.

194. The sixteen foci of the general bicircular quartic are arranged in four tetrads of concyclic foci. Each tetrad gives rise to a focal circle, and the four focal circles have a common orthogonal circle. The centres of the four focal circles lie on the focal conic.

195. The focal properties of the bicircular quartic are closely connected with the theory of the bitangents. Each bitangent of the quartic corresponds to a pair of foci, and the four bitangents correspond to the four pairs of foci which can be formed from the four foci.

196. The nodes and cusps of the bicircular quartic may be investigated by means of the focal equation. A node corresponds to the coincidence of two foci, and a cusp to the coincidence of three foci. The conditions for a node or cusp may thus be expressed in terms of the focal coordinates.

197. The case of the Cartesian, where each of the circular points at infinity is a cusp, is the limiting case of the bicircular quartic when three of the foci coincide at each cusp. The focal equation then degenerates, and the nine foci reduce to three axial foci and their antipoints.

198. The focal theory may be extended to curves of higher order. A curve of order n has n^2 foci, and its equation may be written in the form

$$\sum_{r=1}^{n^2} \sqrt{\mathfrak{A}_r} = 0,$$

where $\mathfrak{A}_r = 0$ is the equation of the pair of tangents from the rth focus. The theory of the focal circles and of the focal conic may be generalized accordingly.

Article Nos. 199 to 201. Tetrazomal Curves.

199. A tetrazomal curve is defined as the locus of a point such that the sum of the squares of the distances from four fixed points, each multiplied by a constant coefficient, is zero. If the four fixed points are A, B, C, D, and the coefficients are a, b, c, d, then the equation of the curve is

$$a \cdot PA^2 + b \cdot PB^2 + c \cdot PC^2 + d \cdot PD^2 = 0,$$

where P is any point on the curve.

200. The general equation of a tetrazomal curve may be written in the form

$$a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}} + d\sqrt{\mathfrak{D}} = 0,$$

where $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$ are quadric functions of the coordinates. The curve is of the fourth order, and it has the four fixed points as foci.

201. The tetrazomal curve includes as particular cases the bicircular quartic (when the four fixed points are concyclic) and the circular cubic (when one of the coefficients is zero). The general theory of tetrazomal curves is thus a generalization of the focal theory of the bicircular quartic.

202. The classification of tetrazomal curves is as follows:—

I. General tetrazomal: four distinct foci, no singularities; order = 4.

II. Nodal tetrazomal: a node at one of the foci, or at another point; order = 4.

III. Cuspidal tetrazomal: a cusp at one of the foci, or at another point; order = 4.

IV. Degenerate tetrazomal: the curve breaks up into two conics, or into other simpler forms.

Each of these kinds may be real or imaginary, and the curve may have one or more branches.

203. The focal circles of a tetrazomal curve are circles passing through three of the four foci. There are four such circles, and they have a common orthogonal circle. The centres of the focal circles lie on a conic, which is the focal conic of the curve.

204. IV. $\sqrt{l}+\sqrt{m}+\sqrt{n}+\sqrt{p}=0$, $a\sqrt{l}+b\sqrt{m}+c\sqrt{n}+d\sqrt{p}=0$; order of tetrazomal = 4; orders of trizomals = 3 each; corresponding to IV. supra.

Article Nos. 205 to 207. Pentazomal and Higher Curves.

205. V. $\sqrt{l}=\sqrt{m}$; order of tetrazomal = 5; orders of trizomals = 3 each.

206. The theory may be extended to pentazomal curves, defined by the equation

$$a\sqrt{\mathfrak{A}}+b\sqrt{\mathfrak{B}}+c\sqrt{\mathfrak{C}}+d\sqrt{\mathfrak{D}}+e\sqrt{\mathfrak{E}}=0,$$

where $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$, $\mathfrak{D}$, $\mathfrak{E}$ are quadric functions of the coordinates. The curve is of the fifth order, and it has five foci.

207. VII. If we have $1,1,1,1=0$, and $(1,1,1,1)(\sqrt{l},\sqrt{m},\sqrt{n},\sqrt{p})=0$, the condition is satisfied identically; and conversely, if the condition is satisfied, the curve is a tetrazomal. This is the generalization of the theorem that four points lie on a circle if and only if the sum of the opposite angles of the quadrilateral is equal to two right angles.

208. The general theory of polyazomal curves includes all curves defined by an equation of the form

$$\sum_{r=1}^{n} a_r \sqrt{\mathfrak{A}_r} = 0,$$

where $\mathfrak{A}_r$ are quadric functions of the coordinates. The curve is of order $2n$, and it has n foci. The theory of the focal circles, the focal conic, and the bitangents may be generalized to these curves.

209. The polyazomal curve of order $2n$ has n focal circles, each passing through $n - 1$ of the n foci. The n focal circles have a common orthogonal circle, and their centres lie on a curve of order $n - 1$, which is the focal curve of the polyazomal.

210. The bitangents of a polyazomal curve correspond to the pairs of foci, and their number is $\frac{1}{2}n(n-1)$. Each bitangent is the polar of a pair of foci with respect to a certain conic, and the envelope of the bitangents is a curve of order $2n - 2$.

211. The conditions for a node or cusp of a polyazomal curve may be expressed in terms of the focal coordinates. A node corresponds to the coincidence of two foci, and a cusp to the coincidence of three foci. The conditions are analogous to those for the bicircular quartic.

212. The degenerate forms of polyazomal curves correspond to special relations between the foci and the coefficients. If two coefficients are equal, the curve has a node; if three are equal, it has a cusp; and if all are equal, it breaks up into simpler components.

213. The analytical theory of polyazomal curves may be developed by means of the method of symmetric functions. If the foci are given, and the coefficients are regarded as variables, then the equation of the curve may be written in the form

$$\sum_{r=1}^{n} a_r \sqrt{(\xi - \alpha_r z)(\eta - \beta_r z)} = 0,$$

and the rationalized equation is obtained by eliminating the radicals. This leads to an equation of order $2n$ in the coordinates.

214. The condition that a polyazomal curve shall have a node is that the discriminant of the rationalized equation shall vanish. This condition may be expressed as a determinant of order $2n$, whose elements are symmetric functions of the focal coordinates and the coefficients.

215. The condition that a polyazomal curve shall have a cusp is more complicated. It requires that the discriminant shall vanish, and also that certain subdeterminants shall vanish. The conditions may be expressed as the vanishing of the invariants of the rationalized equation.

216. The focal properties of polyazomal curves are closely connected with the theory of transformations. If a polyazomal curve is transformed by inversion, the foci are transformed into the foci of the transformed curve, and the coefficients are multiplied by certain factors. The focal circles and the focal curve are also transformed into the corresponding loci of the transformed curve.

217. The theory of polyazomal curves has many applications in geometry and physics. In geometry, it furnishes a general method for the investigation of curves with given focal properties. In physics, it is connected with the theory of potential, and with the problem of determining the equipotential surfaces of a system of charges.

218. The equation of a polyazomal curve may be interpreted as the condition that the sum of the distances from a variable point to a system of fixed points, each multiplied by a constant coefficient, shall be zero. This is a generalization of the definition of the ellipse and hyperbola as the locus of a point the sum or difference of whose distances from two fixed points is constant.

219. The investigation of the singularities of polyazomal curves is of great importance for the theory of algebraic curves. The results obtained for these curves may be extended, by means of birational transformation, to curves of the most general kind.

220. The memoir concludes with the indication of certain directions in which the theory may be extended. The theory of polyazomal curves is capable of generalization in many ways; and it is hoped that the present investigation may serve as a foundation for future researches on this interesting subject.

184. In the case of a circular cubic, we must have

$$\rho(\beta-\gamma)(\alpha-\delta)+\sigma(\gamma-\alpha)(\beta-\delta)+\tau(\alpha-\beta)(\gamma-\delta)=0,$$

$$\sqrt{\alpha\rho}(\beta-\gamma)+\sqrt{\beta\sigma}(\gamma-\alpha)+\sqrt{\gamma\tau}(\alpha-\beta)=0,$$

which, when the foci A, B, C, D are given, determine the values of $\rho:\sigma:\tau$ in order that the curve may be a circular cubic. We see at once that there are two sets of values, and consequently two circular cubics having each of them the given points A, B, C, D for a set of concyclic foci. The two systems may be written

$$\sqrt{\rho}:\sqrt{\sigma}:\sqrt{\tau}=\sqrt{\alpha\delta}-\sqrt{\beta\gamma}:\sqrt{\beta\delta}-\sqrt{\gamma\alpha}:\sqrt{\gamma\delta}-\sqrt{\alpha\beta},$$

viz., it being understood that $\sqrt{\alpha\delta}$ means $\sqrt{\alpha}\sqrt{\delta}$, &c., then, according as $\sqrt{\delta}$ has one or other of its two opposite values, we have one or other of the two systems of values of $\rho:\sigma:\tau$. To verify this, observe that the product

$$(\sqrt{\alpha\delta}-\sqrt{\beta\gamma})(\sqrt{\beta\delta}-\sqrt{\gamma\alpha})(\sqrt{\gamma\delta}-\sqrt{\alpha\beta})$$

is symmetrical in regard to α, β, γ, δ.

and for either set of values the verification of the relation

$$\sqrt{\alpha\rho}(\beta-\gamma)+\sqrt{\beta\sigma}(\gamma-\alpha)+\sqrt{\gamma\tau}(\alpha-\beta)=0,$$

will depend on the two identical equations

$$\sin A\sin(B-C)+\sin B\sin(C-A)+\sin C\sin(A-B)=0,$$

$$\cos A\sin(B-C)+\cos B\sin(C-A)+\cos C\sin(A-B)=0;$$

although the foregoing solution for the case of a circular cubic is the most elegant one, I will presently return to the question and give the solution in a different form.

Article No. 186. Focal Formulæ for the Symmetrical Curve.

186. In the symmetrical case, where the foci A, B, C, are on a line, then if, as usual, a, b, c, d denote the distances from a fixed point, we have the expressions of (a,b,c,d) in a form adapted to the formulae of No. 49, viz.,

$$a:b:c:d=(b-c)(c-d)(d-b):(c-d)(d-a)(a-c):(d-a)(a-b)(b-d):(a-b)(b-c)(c-a).$$

187. In fact, these values obviously satisfy the second equation; and to see that they satisfy the first equation, we have only to write them under the form

$$\rho : \sigma : \tau = M - \tfrac{1}{4}(b+c)(a+d) : M - \tfrac{1}{4}(c+a)(b+d) : M - \tfrac{1}{4}(a+b)(c+d),$$

where $M = (a+b+c+d)^2$. The first set gives for the curve

$$(b-c)\sqrt{\mathfrak{A}} + (c-a)\sqrt{\mathfrak{B}} + (a-b)\sqrt{\mathfrak{C}} = 0,$$

but this contains the line $z = 0$ not once only, but twice; it in fact is $(y^2 = 0)$, the axis taken twice; the only proper cubic with the foci A, B, C, in lined is therefore

$$(b-c)(a+d-b-c)\sqrt{\mathfrak{A}} + (c-a)(b+d-c-a)\sqrt{\mathfrak{B}} + (a-b)(c+d-a-b)\sqrt{\mathfrak{C}} = 0,$$

the equation of which is, of course, expressible in each of the other three forms.

Article Nos. 188 to 192. Case of the General Circular Cubic.

188. Returning to the general case of the circular cubic, the lines AD, BC meet in R, and if we denote by a_1, b_1, c_1, d_1 the distances from R of the four points respectively, so that $b_1c_1 = a_1d_1 = RA \cdot RD$ (observe that if as usual A, B, C, D are taken in order on the circle O, then A, C are on opposite sides of R, and B, D on opposite sides of R), then the two sets of values are

$$\sqrt{\rho} : \sqrt{\sigma} : \sqrt{\tau} = \sqrt{a_1d_1} - \sqrt{b_1c_1} : \sqrt{b_1d_1} - \sqrt{c_1a_1} : \sqrt{c_1d_1} - \sqrt{a_1b_1},$$

and

$$\sqrt{\rho} : \sqrt{\sigma} : \sqrt{\tau} = \sqrt{a_1d_1} + \sqrt{b_1c_1} : \sqrt{b_1d_1} + \sqrt{c_1a_1} : \sqrt{c_1d_1} + \sqrt{a_1b_1}.$$

189. These values of $\sqrt{l}$, $\sqrt{m}$, $\sqrt{n}$ give the equations of the two circular cubics with the foci (A, B, C, D), the equation of each of them under a fourfold form, viz., we have

$$(\cdot, d_1 - c_1, b_1 - d_1, c_1 - b_1)(\sqrt{\mathfrak{A}}, \sqrt{\mathfrak{B}}, \sqrt{\mathfrak{C}}, \sqrt{\mathfrak{D}}) = 0,$$

$$(-d_1 + c_1, \cdot, a_1 - d_1, b_1 - a_1)(\sqrt{\mathfrak{A}}, \sqrt{\mathfrak{B}}, \sqrt{\mathfrak{C}}, \sqrt{\mathfrak{D}}) = 0,$$
$$(-b_1 + d_1, -a_1 + d_1, \cdot, a_1 - b_1)(\sqrt{\mathfrak{A}}, \sqrt{\mathfrak{B}}, \sqrt{\mathfrak{C}}, \sqrt{\mathfrak{D}}) = 0,$$
$$(-c_1 + b_1, -b_1 + a_1, -a_1 + c_1, \cdot)(\sqrt{\mathfrak{A}}, \sqrt{\mathfrak{B}}, \sqrt{\mathfrak{C}}, \sqrt{\mathfrak{D}}) = 0,$$

(first curve), and

$$(\cdot, -c_1 - d_1, d_1 + b_1, -b_1 + c_1)(\sqrt{\mathfrak{A}}, \sqrt{\mathfrak{B}}, \sqrt{\mathfrak{C}}, \sqrt{\mathfrak{D}}) = 0,$$

etc. (second curve).

190. Similarly AC and BD meet in S, and if we denote by a_2, b_2, c_2, d_2 the distances from S of the four points respectively, so that $c_2a_2 = b_2d_2 = SA \cdot SC$ (observe that if as usual A, B, C, D are taken in order on the circle O, then A, C are on opposite sides of S, and B, D on opposite sides of S), then we obtain the same two cubics, each equation under a fourfold form, with the values (a_2, b_2, c_2, d_2).

191. And similarly AB and CD meet in T, and if we denote by a_3, b_3, c_3, d_3 the distances from T of the four points respectively, so that $a_3b_3 = c_3d_3 = TA \cdot TB$, then we again obtain the same two cubics, each equation under a fourfold form, with the values (a_3, b_3, c_3, d_3).

192. The three systems have been obtained independently, but they may of course be derived each from any other of them: to show how this is, recollecting that we have

$$RA \cdot RD = RB \cdot RC, \quad SA \cdot SC = SB \cdot SD, \quad TA \cdot TB = TC \cdot TD,$$

these are the equations which connect the distances (a_1, b_1, c_1, d_1), (a_2, b_2, c_2, d_2), (a_3, b_3, c_3, d_3) respectively.

193. And hence

$$b_1(-b_1c_1+c_1^2+a_1d_1-d_1^2)+c_1(-b_1d_1+c_1d_1+a_1d_1-c_1d_1)=0,$$

that is

$$b_1:c_1=b_1:-(c_1),\text{ and this with } a_1:b_1:c_1:d_1 \text{ gives all the ratios, or we have}$$

$$a_2:b_2:c_2:d_2=a_1(d_1-c_1):b_1(b_1-c_1):c_1(c_1-d_1):d_1(c_1-a_1),\ \&\text{c.}$$

We have then for example

$$b_2-c_2:c_2-a_2:a_2-b_2=b_1-c_1:c_1-a_1:a_1-b_1;\ \&\text{c.},$$

showing the identity of the forms in (a_1,b_1,c_1,d_1) and (a_2,b_2,c_2,d_2).

Article No. 194. Transformation to a New Set of Concyclic Foci.

194. Consider the equation

$$\sqrt{l\mathfrak{A}}+\sqrt{m\mathfrak{B}}+\sqrt{n\mathfrak{C}}+\sqrt{p\mathfrak{D}}=0,$$

which refers to the foci A, B, C, D; and taking the fourth concyclic focus, let (A_1,D_1) be the antipoints of (A,D) and (B_1,C_1) the antipoints of (B,C); so that (A_1,B_1,C_1,D_1) are another set of concyclic foci. Then the equation may be transformed so as to refer to the new set of foci; and the transformed equation is found to be

$$\sqrt{l_1\mathfrak{A}_1}+\sqrt{m_1\mathfrak{B}_1}+\sqrt{n_1\mathfrak{C}_1}+\sqrt{p_1\mathfrak{D}_1}=0,$$

where l_1, m_1, n_1, p_1 are linear functions of l, m, n, p, with coefficients depending on the distances between the old and new foci.

195. I. The general case; l, m, n, p not subjected to any condition. The curve is here of the order $= 8$; it has a quadruple point at each of the points I, J (and there is consequently no other point at infinity); the number of nodes is $= 6$. Hence dps. $= 6 + 2 \cdot 10 = 26$; class is $= 12$; deficiency $= 3$.

196. II. $\sqrt{l} + \sqrt{m} + \sqrt{n} + \sqrt{p} = 0$; order of tetrazomal $= 6$; and hence order of each of the trizomals is $= 3$. There is here a single branch containing ($z^2 = 0$) the line infinity twice; the order is $= 6$. Each of the points I, J is a double point, and there are therefore two more points at infinity, that is (besides the asymptotes at I, J), there are two (real or imaginary) asymptotes. The number of nodes, as in the general case, is $= 6$. Hence dps. $= 6 + 2 \cdot 1 = 8$; class is $= 14$; deficiency $= 2$. I notice the included particular case where the circles reduce themselves to their centres; viz., we have here the curve

$$a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}} + d\sqrt{\mathfrak{D}} = 0,$$

which (see ante No. 93) is in fact the curve which is the locus of the foci of the conics which pass through the four points A, B, C, D. It is at present assumed that the four points are not in a circle; this case will be considered post No. 200.

197. III. $\sqrt{l}+\sqrt{m}=0$, $\sqrt{n}+\sqrt{p}=0$; order of the tetrazomal is $=6$; and hence order of each of the trizomals is $=3$. To verify this, observe that here

$$(\sqrt{l},\sqrt{m},\sqrt{n},\sqrt{p})=(\sqrt{l},-\sqrt{l},\sqrt{n},-\sqrt{n}),$$

which since $a+b+c+d=0$ gives $l:m:n:p=a:b:c:d$; and the equation becomes

$$\sqrt{l}(\sqrt{\mathfrak{A}}-\sqrt{\mathfrak{B}})+\sqrt{n}(\sqrt{\mathfrak{C}}-\sqrt{\mathfrak{D}})=0,$$

which is a pair of trizomals, each of order 3.

198. IV. $\sqrt{l}+\sqrt{m}+\sqrt{n}+\sqrt{p}=0$, $a\sqrt{l}+b\sqrt{m}+c\sqrt{n}+d\sqrt{p}=0$; corresponds to IV. supra, viz., there is a branch ideally containing ($z^2=0$) the line infinity twice. But, observe that whereas in IV. supra, in order that this might be so, it was necessary to impose on l, m, n, p three conditions giving the definite systems of values $\sqrt{l}:\sqrt{m}:\sqrt{n}:\sqrt{p}=a:b:c:d$, in the present case only two conditions are imposed, so that a single arbitrary parameter is left.

199. V. $\sqrt{l}=\sqrt{m}=\sqrt{n}=\sqrt{p}$; corresponds to V. supra.

200. VI. $\sqrt{l}+\sqrt{m}=0$, $\sqrt{n}+\sqrt{p}=0$, $a\sqrt{l}+b\sqrt{m}+c\sqrt{n}+d\sqrt{p}=0$, or what is the same thing, $\sqrt{l}:\sqrt{m}:\sqrt{n}:\sqrt{p}=c-d:d-c:b-a:a-b$; the equation is thus $(c-d)(\sqrt{\mathfrak{A}}-\sqrt{\mathfrak{B}})-(a-b)(\sqrt{\mathfrak{C}}-\sqrt{\mathfrak{D}})=0$. There is here one branch ideally containing ($z^2=0$) the line infinity twice.

201. VII. If we have $l, m, n, p = 0$, and $(1,1,1,1)(\sqrt{l}, \sqrt{m}, \sqrt{n}, \sqrt{p}) = 0$, see ante, Nos. 87 *et seq.* it there appears that (unless the centres A, B, C, are in a line) the condition signifies that the four circles have a common orthotomic circle; and when we have also

$$\frac{ab-cd}{l} + \frac{cd-ab}{m} + \frac{ac-bd}{n} + \frac{bd-ac}{p} = 0.$$

The formulæ for the decomposition are given ante, Nos. 42 *et seq.* Writing therein $\mathfrak{A}^\circ$, $\mathfrak{B}^\circ$, $\mathfrak{C}^\circ$, $\mathfrak{D}^\circ$ in place of $\mathfrak{U}$, $\mathfrak{V}$, $\mathfrak{W}$, $\mathfrak{T}$ respectively, it thereby appears that the tetrazomal curve $\sqrt{l\mathfrak{A}^\circ} + \sqrt{m\mathfrak{B}^\circ} + \sqrt{n\mathfrak{C}^\circ} + \sqrt{p\mathfrak{D}^\circ} = 0$, breaks up into the two trizomal curves

$$\sqrt{\lambda\mathfrak{A}^\circ} + \sqrt{\mu\mathfrak{B}^\circ} + \sqrt{\nu\mathfrak{C}^\circ} = 0, \quad \sqrt{\lambda'\mathfrak{A}^\circ} + \sqrt{\mu'\mathfrak{B}^\circ} + \sqrt{\nu'\mathfrak{C}^\circ} = 0,$$

where

$$\lambda = l + m, \quad \mu = m + n, \quad \nu = n + l,$$

and where we have

$$\lambda' = -l + m, \quad \mu' = -m + n, \quad \nu' = -n + l.$$

Article Nos. 198 to 203. Cases of the Decomposable Curve, Centres not in a line.

198. I assume, in the first instance, that the centres of the circles are not in a line. The tetrazomal curve $\sqrt{l\mathfrak{A}} + \sqrt{m\mathfrak{B}} + \sqrt{n\mathfrak{C}} + \sqrt{p\mathfrak{D}} = 0$ breaks up into two trizomals when the coefficients (l, m, n, p) satisfy certain conditions; and the forms of the trizomals are as follows.

199. The several cases of the decomposable curve are as follows:
I. The general case; l, m, n, p not subjected to any condition. The curve is here of the order $= 8$; it has a quadruple point at each of the points I, J (and there is consequently no other point at infinity); the number of nodes is $= 6$. Hence dps. $= 6 + 2 \cdot 10 = 26$; class is $= 12$; deficiency $= 3$.
II. $\sqrt{l} + \sqrt{m} + \sqrt{n} + \sqrt{p} = 0$; order of tetrazomal $= 6$; and hence order of each of the trizomals is $= 3$. There is here a single branch containing ($z^2 = 0$) the line infinity twice; the order is $= 6$. Each of the points I, J is a double point, and there are therefore two more points at infinity. The number of nodes is $= 6$. Hence dps. $= 6 + 2 \cdot 1 = 8$; class is $= 14$; deficiency $= 2$.
III. $\sqrt{l} + \sqrt{m} = 0$, $\sqrt{n} + \sqrt{p} = 0$; order of the tetrazomal is $= 6$; and hence order of each of the trizomals is $= 3$.
IV. $\sqrt{l} + \sqrt{m} + \sqrt{n} + \sqrt{p} = 0$, $a\sqrt{l} + b\sqrt{m} + c\sqrt{n} + d\sqrt{p} = 0$; corresponds to IV. supra, viz., there is a branch ideally containing ($z^2 = 0$) the line infinity twice. But in the present case only two conditions are imposed, so that a single arbitrary parameter is left.
V. $\sqrt{l} = \sqrt{m} = \sqrt{n} = \sqrt{p}$; corresponds to V. supra.
VI. $\sqrt{l} + \sqrt{m} = 0$, $\sqrt{n} + \sqrt{p} = 0$, $a\sqrt{l} + b\sqrt{m} + c\sqrt{n} + d\sqrt{p} = 0$; the equation is $(c - d)(\sqrt{\mathfrak{A}} - \sqrt{\mathfrak{B}}) - (a - b)(\sqrt{\mathfrak{C}} - \sqrt{\mathfrak{D}}) = 0$. There is here one branch ideally containing ($z^2 = 0$) the line infinity twice.

202. VII. If we have l, m, n, p satisfying the two conditions, and $(1,1,1,1)$ with $(\sqrt{l}, \sqrt{m}, \sqrt{n}, \sqrt{p})$, see ante, Nos. 87 *et seq.* it there appears that (unless the centres A, B, C, are in a line) the condition signifies that the four circles have a common orthotomic circle; and when we have also

$$\frac{ab-cd}{l}+\frac{cd-ab}{m}+\frac{ac-bd}{n}+\frac{bd-ac}{p}=0.$$

The formulæ for the decomposition are given ante, Nos. 42 *et seq.*

203. Writing therein $\mathfrak{A}^\circ$, $\mathfrak{B}^\circ$, $\mathfrak{C}^\circ$, $\mathfrak{D}^\circ$ in place of $\mathfrak{U}$, $\mathfrak{V}$, $\mathfrak{W}$, $\mathfrak{T}$ respectively, it thereby appears that the tetrazomal curve $\sqrt{l\mathfrak{A}^\circ}+\sqrt{m\mathfrak{B}^\circ}+\sqrt{n\mathfrak{C}^\circ}+\sqrt{p\mathfrak{D}^\circ}=0$, breaks up into the two trizomal curves

$$\sqrt{\lambda\mathfrak{A}^\circ}+\sqrt{\mu\mathfrak{B}^\circ}+\sqrt{\nu\mathfrak{C}^\circ}=0, \quad \sqrt{\lambda'\mathfrak{A}^\circ}+\sqrt{\mu'\mathfrak{B}^\circ}+\sqrt{\nu'\mathfrak{C}^\circ}=0,$$

where

$$\lambda=l+m, \quad \mu=m+n, \quad \nu=n+l,$$

and

$$\lambda'=-l+m, \quad \mu'=-m+n, \quad \nu'=-n+l.$$

Article Nos. 204 to 207. Further Cases of the Decomposable Curve.

204. IV. $\sqrt{l}+\sqrt{m}+\sqrt{n}+\sqrt{p}=0$, $a\sqrt{l}+b\sqrt{m}+c\sqrt{n}+d\sqrt{p}=0$; order of tetrazomal is $=4$; orders of trizomals $=3$ each; corresponding to IV. supra.

205. V. $\sqrt{l}=\sqrt{m}=\sqrt{n}=\sqrt{p}$; order of tetrazomal is $=5$; orders of trizomals $=3$ each.

206. And hence one or other of the two functions

$$\sqrt{l\mathfrak{A}}+\sqrt{m\mathfrak{B}}+\sqrt{n\mathfrak{C}}, \quad \sqrt{l\mathfrak{A}}+\sqrt{m\mathfrak{B}}+\sqrt{n\mathfrak{C}}$$

is $=0$; that is, one of the trizomal curves is a cubic.

207. III. $\sqrt{l}+\sqrt{p}=0$, $\sqrt{m}+\sqrt{n}=0$; order of the tetrazomal is $=6$; and hence order of each of the trizomals is $=3$. To verify this, observe that here

$$(\sqrt{l}, \sqrt{m}, \sqrt{n}, \sqrt{p})=(\sqrt{l}, \sqrt{m}, -\sqrt{m}, -\sqrt{l}),$$

which since $a+b+c+d=0$ gives the required ratios.

208. In the general case where the points A, B, C, D are not on a circle, the tetrazomal curve is indecomposable; and the order is $= 8$ in the general case, $= 6$ in the special case where $\sqrt{l}+\sqrt{m}+\sqrt{n}+\sqrt{p}=0$, and so on as above.

209. The decomposable cases arise when there exists a linear relation between the square roots of the focal distances; and this is the condition that the four foci shall lie on a circle. When this condition is satisfied, the tetrazomal curve breaks up into two trizomal curves, each of which is a cubic.

210. The several cases may be tabulated as follows:

Case	**Order of Tetrazomal**	**Order of Trizomals**
I. General	8	4, 4
II. $\sum\sqrt{l}=0$	6	3, 3
III. $\sqrt{l}+\sqrt{m}=0,\ \sqrt{n}+\sqrt{p}=0$	6	3, 3
IV. $\sum\sqrt{l}=0,\ \sum al=0$	4	3, 3
V. $\sqrt{l}=\sqrt{m}=\sqrt{n}=\sqrt{p}$	5	3, 3
VI. $\sqrt{l}+\sqrt{m}=0,\ \sqrt{n}+\sqrt{p}=0,\ \sum a\sqrt{l}=0$	6	3, 3

211. The condition that the tetrazomal curve shall be decomposable is that the four foci shall lie on a circle. This condition may be expressed analytically as follows. Let the coordinates of the foci be (x_r, y_r), $r = 1, 2, 3, 4$; then the condition that they shall lie on a circle is

$$\begin{vmatrix} x_1^2 + y_1^2 & x_1 & y_1 & 1 \\ x_2^2 + y_2^2 & x_2 & y_2 & 1 \\ x_3^2 + y_3^2 & x_3 & y_3 & 1 \\ x_4^2 + y_4^2 & x_4 & y_4 & 1 \end{vmatrix} = 0.$$

When this condition is satisfied, the tetrazomal curve breaks up into two trizomals.

212. The condition may also be expressed in terms of the distances between the foci. If d_{12}, d_{13}, &c. are the distances between the pairs of foci, then the condition is

$$d_{12}d_{34} + d_{13}d_{24} + d_{14}d_{23} = 0,$$

which is the condition that four points shall lie on a circle.

213. When the four foci lie on a circle, the tetrazomal curve is said to be "symmetrical;" and the two trizomals into which it breaks up are called the "components" of the tetrazomal. The components are each of them a circular cubic, and they have the same foci as the original tetrazomal.

214. The equation $\sqrt{l}+\sqrt{m}+\sqrt{n}+\sqrt{p}=0$ gives $\sqrt{l}+\sqrt{m}+\sqrt{n}+\sqrt{p}=0$, and combining with $a+b+c+d=0$, we obtain the ratios of $l:m:n:p$ in terms of $a:b:c:d$. The resulting equation is

$$(b-c)\sqrt{\mathfrak{A}}+(c-a)\sqrt{\mathfrak{B}}+(a-b)\sqrt{\mathfrak{C}}=0,$$

which is the equation of a circular cubic. The tetrazomal curve thus breaks up into two trizomals, each of which is a circular cubic.

215. IV. $\sqrt{l}+\sqrt{m}+\sqrt{n}+\sqrt{p}=0$, $a\sqrt{l}+b\sqrt{m}+c\sqrt{n}+d\sqrt{p}=0$; order of tetrazomal is $=4$; orders of trizomals $=3$ each; corresponding to IV. supra. The two conditions determine the ratios $l:m:n:p$ uniquely, and the tetrazomal is completely determined.

216. The several cases of the decomposable curve may be distinguished by the number of conditions imposed on the coefficients (l,m,n,p). In Case I. there are no conditions; in Case II. there is one condition; in Case III. there are two conditions; and in Case IV. there are three conditions. The greater the number of conditions, the lower the order of the tetrazomal.

217. The general theory of polyazomal curves includes the theory of the tetrazomal as a particular case. A polyazomal curve of order n is defined by the equation

$$\sum_{r=1}^{n} a_r \sqrt{\mathfrak{A}_r} = 0,$$

where $\mathfrak{A}_r = 0$ is the equation of a pair of lines. The curve is of order $2n$, and it has n foci. The theory of the focal circles, the focal conic, and the bitangents may be generalized to these curves.

218. The condition that a polyazomal curve shall break up into curves of lower order is that the foci shall lie on a conic. When this condition is satisfied, the curve breaks up into two curves of order n, each having the same foci as the original curve.

219. The number of conditions required for the decomposition of a polyazomal curve is $n-1$. Thus for a tetrazomal ($n = 4$) the number of conditions is 3; for a pentazomal ($n = 5$) it is 4; and so on. The conditions are linear in the coefficients $(a_1, a_2, \ldots, a_n)$.

220. The general theory of polyazomal curves is thus seen to be a natural generalization of the theory of the bicircular quartic. The methods employed in the investigation of the bicircular quartic may be applied, with suitable modifications, to the investigation of polyazomal curves of any order.

221. The memoir concludes with a summary of the principal results obtained. The theory of polyazomal curves has been developed from the fundamental conception of the focus of an algebraic curve; and it has been shown that the focal properties of these curves are intimately connected with the theory of transformations, and with the general theory of algebraic curves.
222. The principal results are as follows:

(1) Every algebraic curve of order n has n^2 foci, which are the intersections of the tangents from the circular points at infinity.

(2) The equation of the curve may be written in the form $\sum \sqrt{\mathfrak{A}_r} = 0$, where $\mathfrak{A}_r = 0$ is the equation of the pair of tangents from the rth focus.

(3) The focal circles of the curve are circles passing through $n-1$ of the n^2 foci; and they have a common orthogonal circle.

(4) The condition that the curve shall have a node or a cusp is that two or three of the foci shall coincide.

(5) The condition that the curve shall break up into curves of lower order is that the foci shall lie on a conic.

223. The theory of polyazomal curves is capable of further extension; and it is hoped that the present investigation may serve as a foundation for future researches on this interesting subject.

[The memoir continues with additional notes and references to previous work.]

224. And then

$$(d-a)\sqrt{l} = (b-d)\sqrt{m} + (c-d)\sqrt{n},$$
$$(d-a)\sqrt{p} = (a-b)\sqrt{m} + (a-c)\sqrt{n},$$

and substituting these values in the equation $a\sqrt{l} + b\sqrt{m} + c\sqrt{n} + d\sqrt{p} = 0$, we obtain the condition for the decomposition.

225. $\sqrt{l} = \sqrt{m} = \sqrt{n} = \sqrt{p}$; order of tetrazomal is $= 5$; orders of trizomals $= 3$ each. In this case the four foci are concyclic, and the tetrazomal breaks up into two circular cubics.

Article Nos. 226 to 230. Cases of the Decomposable Curve, Centres in a line.

226. I proceed to consider the case where the centres of the circles are in a line. In this case the tetrazomal curve is necessarily decomposable; and the components are of lower order than in the general case.

227. If the centres A, B, C, D are in a line, then the four circles have a common orthogonal circle only if the line of centres passes through the centre of the orthogonal circle. In this case the tetrazomal breaks up into two trizomals, each of order 3.

228. If the line of centres does not pass through the centre of the orthogonal circle, then the tetrazomal is indecomposable; but it may have a node or a cusp at one of the points at infinity.

229. The case where the centres are in a line is of importance for the theory of the symmetrical bicircular quartic. In this case the focal axis is the line of centres; and the four foci lie on this line. The curve has an axis of symmetry, and the focal circles are symmetrically situated with respect to this axis.

230. The investigation of this case leads to the consideration of the Cartesian curve as a limiting case of the bicircular quartic. When the four foci are in a line, the bicircular quartic degenerates into a Cartesian; and the focal properties undergo the degradation already referred to.

231. The case where the centres are in a line may be further subdivided according to the relative positions of the centres. If the centres are in order A, B, C, D on the line, then the distances satisfy certain inequalities; and the nature of the curve depends on these inequalities.

232. If the centres are such that $AB \cdot CD = AC \cdot BD = AD \cdot BC$, then the four points form a harmonic range; and the tetrazomal curve has special properties. In this case the curve is symmetrical, and the focal circles are equal.

233. The condition that four points on a line shall form a harmonic range may be expressed in various ways. If the distances from a fixed point on the line are a, b, c, d, then the condition is

$$\frac{2}{b-c} = \frac{1}{a-b} + \frac{1}{d-c},$$

or, what is the same thing,

$$(a+d)(b+c) = 2(ad+bc).$$

When this condition is satisfied, the tetrazomal curve has a node at each of the circular points at infinity.

234. The case of harmonic centres is of interest in connexion with the theory of the bicircular quartic. The harmonic relation between the foci leads to simplifications in the equation of the curve; and the focal circles have properties analogous to those of the circles of Apollonius.

235. The memoir concludes with the indication of certain directions in which the theory may be extended. The theory of polyazomal curves is capable of generalization in many ways; and it is hoped that the present investigation may serve as a foundation for future researches on this interesting subject.

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236. *Note on the Focal Properties of the Bicircular Quartic.*—The foregoing theory of the focal properties of the bicircular quartic may be presented under a somewhat different form, which brings into clearer view the connexion with the theory of the conic.

237. If we regard the bicircular quartic as the envelope of a circle which cuts three given circles at given angles, then the centre of the variable circle describes a conic; and the foci of the bicircular quartic are the centres of the three given circles. This is the fundamental theorem of the focal theory.

238. The theorem may be proved analytically as follows. Let the three given circles be

$$x^2 + y^2 + 2g_r x + 2f_r y + c_r = 0, \quad (r = 1, 2, 3),$$

and let the variable circle be

$$(x - \xi)^2 + (y - \eta)^2 = \rho^2.$$

The condition that the variable circle shall cut the rth given circle at a given angle θ_r is

$$\rho^2 + r_r^2 - d_r^2 = 2\rho r_r \cos\theta_r,$$

where r_r is the radius of the rth circle, and d_r is the distance between the centres. Eliminating ρ between these three equations, we obtain the locus of (ξ, η), which is a conic.

239. The conic which is the locus of the centre of the variable circle is the "focal conic" of the system of circles. It passes through the centres of the three given circles; and its properties are analogous to those of the director circle of a conic.

240. The focal conic may be used to investigate the properties of the bicircular quartic. If four circles have a common orthogonal circle, then the four centres lie on a conic; and conversely, if four points lie on a conic, then the four circles having these points as centres and cutting a given circle orthogonally have a common orthogonal circle.

241. The theorem of the focal conic leads to a simple proof of Casey's equation for four circles touching a fifth. If four circles touch a fifth circle, then their centres lie on a conic; and the condition that four points shall lie on a conic is equivalent to Casey's equation.

242. The focal properties of the bicircular quartic are thus seen to be intimately connected with the theory of conics and circles. The theory provides a unified point of view from which many apparently disconnected theorems may be regarded as corollaries of a single fundamental principle.

Article Nos. 243 to 245. Applications to the Theory of Curves.

243. The theory of polyazomal curves has many applications to the theory of algebraic curves. In particular, it furnishes a method for the investigation of curves with given singularities, and for the classification of curves of a given order.

244. The method consists in regarding the curve as the locus of a point whose distances from fixed points satisfy a given relation. The fixed points are the foci of the curve; and the given relation is the focal equation. By varying the relation, we obtain a system of curves with the same foci but different singularities.

245. The theory may also be applied to the investigation of the bitangents and inflexions of a curve. The bitangents correspond to the double tangents from the foci; and the inflexions correspond to the points where the focal circles touch the curve.

246. The classification of curves by means of their focal properties is as follows. A curve of order n has n^2 foci; and the arrangement of these foci determines the nature of the curve. If the foci are distinct, the curve is general; if some of the foci coincide, the curve has singularities; and if the foci lie on a conic, the curve is degenerate.

247. The number of foci which must coincide in order that the curve shall have a node or a cusp may be determined as follows. For a node, two foci must coincide; for a cusp, three foci must coincide; and for a tacnode, four foci must coincide. The greater the number of coincident foci, the higher the singularity.

248. The condition that a curve shall have a given number of nodes and cusps may be expressed as a system of equations between the focal coordinates. These equations are algebraic; and their solution gives the focal conditions for the given singularities.

249. The theory of focal singularities is thus seen to be a powerful instrument for the investigation of algebraic curves. It provides a method for the systematic discussion of curves with given singularities; and it leads to results which are not easily obtainable by other methods.

Article Nos. 250 to 252. Conclusion.

250. The present memoir has developed the theory of polyazomal curves from the fundamental conception of the focus of an algebraic curve. It has been shown that the focal properties of these curves are intimately connected with the theory of transformations, and with the general theory of algebraic curves.

251. The principal results of the investigation are: the theorem that every algebraic curve of order n has n^2 foci; the theorem that the equation of the curve may be written in the form $\sum \sqrt{\mathfrak{A}_r} = 0$; the theorem of the focal circles; and the theorem of the decomposition of polyazomal curves when the foci lie on a conic.

252. The theory is capable of further extension; and it is hoped that it may serve as a foundation for future researches on the focal properties of algebraic curves.

253. The condition that the four circles shall have a common orthogonal circle is that the distances of their centres from any line shall satisfy a certain linear relation. This condition may be written in the form

$$\begin{vmatrix} a^2 & a & 1 \\ b^2 & b & 1 \\ c^2 & c & 1 \end{vmatrix} = 0,$$

where a, b, c are the distances of the three centres from any line. The condition is equivalent to the condition that the three centres shall lie on a conic.

254. When the four circles have a common orthogonal circle, the tetrazomal curve breaks up into two trizomals; and each of these is a circular cubic. The four foci of the tetrazomal are the centres of the four circles; and the three foci of each trizomal are three of these four centres.

255. The condition for the decomposition may also be expressed as follows. If the four circles are A, B, C, D, and if S is their common orthogonal circle, then the radical axis of any two of the circles passes through the centre of S. The four radical axes thus meet in a point, which is the centre of the common orthogonal circle.

256. The theorem that four circles have a common orthogonal circle if and only if their centres lie on a conic is a fundamental theorem of the theory. It is the generalization of the theorem that three circles have a common orthogonal circle if and only if their centres are collinear.

257. The theorem may be proved analytically as follows. Let the equations of the four circles be

$$x^2 + y^2 + 2g_r x + 2f_r y + c_r = 0, \quad (r = 1, 2, 3, 4).$$

The condition that these shall have a common orthogonal circle is that there shall exist values of G, F, C such that

$$(G - g_r)^2 + (F - f_r)^2 = C - c_r, \quad (r = 1, 2, 3, 4).$$

Eliminating G, F, C, we obtain the condition that the four circles shall have a common orthogonal circle.

258. The condition thus obtained is equivalent to the condition that the four points (g_r, f_r) shall lie on a conic. For the condition that four points shall lie on a conic is

$$\begin{vmatrix} g_1^2+f_1^2 & g_1 & f_1 & 1 \\ g_2^2+f_2^2 & g_2 & f_2 & 1 \\ g_3^2+f_3^2 & g_3 & f_3 & 1 \\ g_4^2+f_4^2 & g_4 & f_4 & 1 \end{vmatrix} = 0;$$

and this is precisely the condition that the four circles shall have a common orthogonal circle.

259. The theorem is thus established; and it forms the foundation of the theory of the focal properties of polyazomal curves. The theorem that the tetrazomal curve breaks up into two trizomals when the four foci lie on a circle is a particular case of this general theorem.

260. The investigation of the present memoir has been confined to the consideration of curves of the fourth order; but the methods employed are applicable to curves of any order. The theory of polyazomal curves of order n presents many interesting problems, which it is hoped to investigate in a future memoir.

Article Nos. 261 to 263. Note on the Focal Conic.

261. The focal conic of a system of circles has been defined as the locus of the centre of a circle which cuts the given circles at given angles. The focal conic passes through the centres of the given circles; and its properties are analogous to those of the director circle of a conic.

262. The equation of the focal conic may be obtained as follows. Let the given circles be

$$x^2+y^2+2g_rx+2f_ry+c_r=0, \quad (r=1,2,3).$$

The condition that the variable circle $(x-\xi)^2+(y-\eta)^2=\rho^2$ shall cut the rth circle at an angle θ_r is

$$\rho^2+r_r^2-d_r^2=2\rho r_r\cos\theta_r.$$

Eliminating ρ between these three equations, we obtain the locus of (ξ, η), which is the focal conic.

263. The focal conic is thus seen to be a conic passing through the centres of the three given circles. Its equation is of the second degree in (ξ, η); and its coefficients depend on the radii and the angles of intersection.

264. The focal conic has many interesting properties. In particular, it is the locus of the centre of a circle which cuts three given circles orthogonally; and it is the envelope of the radical axis of two circles which cut the three given circles at given angles.

265. The focal conic is also the locus of a point whose distances from the centres of the three given circles are proportional to the radii of those circles. This property is analogous to the property of the director circle of a conic, which is the locus of a point whose distances from the foci are equal.

266. The focal conic may be used to investigate the properties of the system of circles. If four circles have a common orthogonal circle, then the focal conics of any three of them pass through the centre of the fourth; and conversely, if the focal conics of three circles pass through a point, then the four circles have a common orthogonal circle.

267. The theory of the focal conic is thus seen to be an important part of the theory of circles and their orthogonal trajectories. It provides a method for the investigation of the properties of systems of circles; and it leads to many elegant theorems in geometry.

Article Nos. 268 to 270. Applications to the Problem of Apollonius.

268. The problem of Apollonius,—to describe a circle touching three given circles,—is one of the most famous problems in geometry. The solution by means of the focal conic is as follows.

269. Let the three given circles be A, B, C. The focal conic of these three circles passes through their centres; and the centre of the required circle is a point on the focal conic. The radius of the required circle is determined by the condition that it shall touch the three given circles.

270. The condition of tangency gives a quadratic equation for the radius; and the two values of the radius correspond to the two solutions of the problem. The eight solutions of the problem of Apollonius correspond to the eight points in which the focal conic is met by the three circles which are the loci of the centres of circles cutting the given circles at the supplementary angles.

271. The theorem that the eight solutions of the problem of Apollonius correspond to the eight foci of a bicircular quartic is a beautiful example of the connexion between the theory of circles and the theory of curves. The bicircular quartic which has the three given circles for its focal circles has the required touching circles for its bitangents; and the eight foci of the quartic are the centres of the eight touching circles.

272. The investigation of the problem of Apollonius by means of the focal conic leads to many generalizations. The problem may be extended to the case of four or more circles; and the solution may be obtained by means of the focal conic of the system.

273. The theory of the focal conic thus provides a unified method for the investigation of problems relating to circles and their contacts. It is a powerful instrument for the solution of geometrical problems; and it leads to results which are not easily obtainable by other methods.

Article Nos. 274 to 276. Generalization of the Focal Theory.

274. The theory of foci may be generalized as follows. Instead of considering the intersections of the tangents from the circular points at infinity, we may consider the intersections of the tangents from any two fixed points. These intersections may be called the "generalized foci" of the curve; and the theory of these foci is analogous to the theory of the ordinary foci.

275. If the two fixed points are P and Q, then the generalized foci of a curve of order n are the intersections of the tangents from P and Q to the curve. The number of generalized foci is n^2; and their arrangement determines the nature of the curve.

276. The theory of generalized foci may be applied to the investigation of curves with given properties. In particular, it leads to a generalization of the theorem that the sum of the distances from any point of a conic to the two foci is constant. The generalized theorem is that the sum of the distances from any point of a curve to the two fixed points P and Q, measured along the tangents to the curve, is constant.

277. The condition that four circles shall have a common orthogonal circle may be expressed in terms of the distances of their centres. If the centres are A, B, C, D, and the distances are $AB = c$, $AC = b$, $AD = d$, $BC = a$, $BD = f$, $CD = e$, then the condition is

$$\begin{vmatrix} 0 & c^2 & b^2 & d^2 & 1 \\ c^2 & 0 & a^2 & f^2 & 1 \\ b^2 & a^2 & 0 & e^2 & 1 \\ d^2 & f^2 & e^2 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{vmatrix} = 0.$$

This is the condition that the four points shall lie on a conic; and it is equivalent to the condition that the four circles shall have a common orthogonal circle.

278. The condition may also be written in the form

$$(ab + cd)(ac + bd)(ad + bc) = 0,$$

which shows that the condition is satisfied if $ab + cd = 0$, or $ac + bd = 0$, or $ad + bc = 0$. These are the conditions that the four points shall form a harmonic range.

279. The general theory of polyazomal curves includes the theory of the bicircular quartic, the circular cubic, and the conic as particular cases. The methods of the theory are applicable to curves of any order; and they lead to results which are of great importance for the theory of algebraic curves.

280. The memoir concludes with the hope that the theory of polyazomal curves may serve as a foundation for future researches on the focal properties of algebraic curves. The theory is capable of extension in many directions; and it is believed that it will lead to important discoveries in the theory of curves and their applications.

End of the Memoir on Polyazomal Curves.

COLLECTED MATHEMATICAL PAPERS

OF

ARTHUR CAYLEY, Sc.D., F.R.S.,

SADLERIAN PROFESSOR OF PURE MATHEMATICS
IN THE UNIVERSITY OF CAMBRIDGE.

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414. On Polyzomal Curves (continued)

ANNEX III. ON THE NORM OF

$$(b-c)\sqrt{a^2+\alpha}+(c-a)\sqrt{b^2+\beta}+(a-b)\sqrt{c^2+\gamma},$$

WHEN THE CENTRES ARE IN A LINE.

The norm of $\sqrt{U}+\sqrt{V}+\sqrt{W}$ is

$$(1,1,1,-1,-1,-1 \mid U,V,W),$$

whence that of $\sqrt{U}+\sqrt{U'}+\sqrt{V}+\sqrt{V'}+\sqrt{W}+\sqrt{W'}$ is

$$\begin{aligned}&(1,1,1,-1,-1,-1 \mid U,V,W)\\ &\qquad+(1,1,1,-1,-1,-1 \mid U',V',W')\\ &\qquad\qquad+2(1,1,1,-1,-1,-1 \mid U,V,W \,|\, U',V',W'),\end{aligned}$$

where the last term is = 2 into

$$U'(-U-V+W)+V'(-U+V-W)+W'(-U-V+W);$$

and the norm of $\sqrt{U+U'+U''}+\sqrt{V+V'+V''}+\sqrt{W+W'+W''}$ is obviously composed in a similar manner.

Now, applying the formula to obtain the norm of

$$(b-c)\sqrt{a^2+\alpha}+(c-a)\sqrt{b^2+\beta}+(a-b)\sqrt{c^2+\gamma},$$

the expression contains six terms, two of which are at once seen to vanish; and writing for shortness ( ) in place of $(1,1,1,-1,-1,-1)$ the remaining terms are

$$\begin{aligned}&(\;)\{(b-c)^2\alpha,(c-a)^2\beta,(a-b)^2\gamma\}\\ &\qquad+2(\;)\{(b-c)^2\alpha,(c-a)^2\beta,(a-b)^2\gamma \,|\, (b-c)^2\alpha',(c-a)^2\beta',(a-b)^2\gamma'\}\\ &\qquad+2\theta(\;)\{(b-c)^2\alpha,(c-a)^2\beta,(a-b)^2\gamma \,|\, (b-c)^2,(c-a)^2,(a-b)^2\}\\ &\qquad\qquad+2\theta(\;)\{(b-c)^2\alpha',(c-a)^2\beta',(a-b)^2\gamma' \,|\, (b-c)^2,(c-a)^2,(a-b)^2\}\end{aligned}$$

the first of these terms requires no reduction; the second, omitting the factor 2, is

$$\begin{aligned}&(b-c)^2\alpha\,[(b-c)^2\alpha'-(c-a)^2\beta'-(a-b)^2\gamma']\\ &\qquad+(c-a)^2\beta\,[-(b-c)^2\alpha'+(c-a)^2\beta'-(a-b)^2\gamma']\\ &\qquad\qquad+(a-b)^2\gamma\,[-(b-c)^2\alpha'-(c-a)^2\beta'+(a-b)^2\gamma'],\end{aligned}$$

which is

$$=2(a-b)(b-c)(c-a)\,[bc(b-c)\alpha+ca(c-a)\beta+ab(a-b)\gamma].$$

Similarly the third term, omitting the factor 2θ, is

$$\begin{aligned}&(b-c)^2\alpha\,[(b-c)^2-(c-a)^2-(a-b)^2]\\ &\qquad+(c-a)^2\beta\,[-(b-c)^2+(c-a)^2-(a-b)^2]\\ &\qquad\qquad+(a-b)^2\gamma\,[-(b-c)^2-(c-a)^2+(a-b)^2],\end{aligned}$$

which is

$$=2(a-b)(b-c)(c-a)\,[(b-c)\alpha+(c-a)\beta+(a-b)\gamma],$$

and for the last term, omitting the factor 2θ, this may be deduced therefrom by writing (α',β',γ') in place of (α,β,γ), viz., it is

$$=-2(a-b)^2(b-c)^2(c-a)^2.$$

Hence, restoring the omitted factors, and collecting, we find

$$\begin{aligned}
&\text{Norm}\{(b-c)\sqrt{a^2+\alpha}+(c-a)\sqrt{b^2+\beta}+(a-b)\sqrt{c^2+\gamma}\}\\
&= (b-c)^4\alpha^2+(c-a)^4\beta^2+(a-b)^4\gamma^2-2(c-a)^2(a-b)^2\beta\gamma-2(a-b)^2(b-c)^2\gamma\alpha-2(b-c)^2(c-a)^2\alpha\beta\\
&\qquad + 4\theta(a-b)(b-c)(c-a)\,[(b-c)\alpha+(c-a)\beta+(a-b)\gamma]\\
&\qquad + 4(a-b)(b-c)(c-a)\,[bc(b-c)\alpha+ca(c-a)\beta+ab(a-b)\gamma]\\
&\qquad - 4\theta^2(a-b)^2(b-c)^2(c-a)^2.
\end{aligned}$$

Hence, first writing $a-x$, $b-x$, $c-x$ in place of a, b, c; then y^2 for θ, and $(-\alpha'', -\beta'', -\gamma'')$ for (α, β, γ); and finally introducing z for homogeneity, we find

$$\begin{gathered}
\text{Norm}\{(b-c)\sqrt{(x-az)^2+\alpha'' z^2}+(c-a)\sqrt{(x-bz)^2+\beta''^2 z^2}+(a-b)\sqrt{(x-cz)^2+\gamma''^2 z^2}\} = z^8 \text{ into} \\
z^{-4}\{(b-c)^4\alpha''^2+(c-a)^4\beta''^2+(a-b)^4\gamma''^2 \\
-2(c-a)^2(a-b)^2\beta''\gamma''-2(a-b)^2(b-c)^2\gamma''\alpha''-2(b-c)^2(c-a)^2\alpha''\beta''\} \\
-4y^2(b-c)(c-a)(a-b)\,[(b-c)\alpha''+(c-a)\beta''+(a-b)\gamma''] \\
-4(b-c)(c-a)(a-b)\{(b-c)\alpha''[2bc-2x(b+c)+x^2] \\
+(c-a)\beta''[2ca-2x(c+a)+x^2]+(a-b)\gamma''[2ab-2x(a+b)+x^2]\} \\
\hfill -4y^4(b-c)^2(c-a)^2(a-b)^2,
\end{gathered}$$

so that the equation $(b-c)\sqrt{S}+(c-a)\sqrt{S'}+(a-b)\sqrt{S''}=0$, in its rationalised form, contains $(z^4=0)$ the line infinity twice, and the curve is thus a conic. If $\alpha''=\beta''=\gamma''=k''^2$, then the expression of the norm is

$$= z^4 \text{ into } -4(a-b)^2(b-c)^2(c-a)^2(y^4-k''^4z^4),$$

viz., when the three circles have each of them the same radius k'', the curve is the pair of parallel lines $y^4-k''^4z^4=0$; and in particular when $k''=0$, or the circles reduce themselves each to a point, then the curve is $y^4=0$, the axis twice.

ANNEX IV. ON THE TRIZOMAL CURVES

$$\sqrt{lU}+\sqrt{mV}+\sqrt{nW}=0, \text{ which have a Cusp, or two Nodes.}$$

The trizomal curve $\sqrt{lU}+\sqrt{mV}+\sqrt{nW}=0$, has not in general any nodes or cusps: in the particular case where the zomal curves are circles, we have however seen how the ratios $l:m:n$ may be determined so that the curve shall acquire a node, two nodes, or a cusp; viz., regarding a, b, c as current areal coordinates, we have here a conic $\alpha+\beta+\gamma=0$, the locus of the centres of the variable circle, and the solution depends on establishing a relation between this conic and the orthotomic circle or Jacobian of the three given circles. I have in my paper "Investigations in connection with Casey's Equation," *Quart. Math. Jour.* vol. viii. (1867), pp. 334—342, [395] given, after Professor Cremona, a solution of the general question to find the number of the curves $\sqrt{lU}+\sqrt{mV}+\sqrt{nW}=0$, which have a cusp, or which have two nodes, and I will here reproduce the leading points of the investigation. I remark, that although one of the loci involved in it is the same as that occurring in the case of the three circles (viz., we have in each case the Jacobian of the given curves), the other two loci Σ and Λ, which present themselves, seem to have no relation to the conic of centres which is made use of in the particular case.

We have the curves $U=0$, $V=0$, $W=0$, each of the same order r; and considering a point the coordinates whereof are (l,m,n), we regard as corresponding to this point the curve $\sqrt{lU}+\sqrt{mV}+\sqrt{nW}=0$, say for shortness, the curve Ω, being as above a curve of the order $2r$, having r^2 contacts with each of the given curves $U=0$, $V=0$, $W=0$. As long as the point (l,m,n) is arbitrary, the curve Ω has not any node, and in order that this curve may have a node, it is necessary that the point (l,m,n) shall lie on a certain curve Λ; this being so, the node will, it is easy to see, lie on the curve J, the Jacobian of the three given curves; and the curves J and Λ will correspond to each other point to point, viz., taking for (l,m,n) any point whatever on the curve Λ, the curve Ω will have a node at some one point of J; and conversely, in order that the curve Ω may be a curve having a node at a given point of J, the point (l,m,n) must be at some one point of the curve Λ.

The curve Λ has, however, nodes and cusps; each node of Λ corresponds to two points of J, viz., for (l,m,n) at a node of Λ, the curve Ω is a binodal curve having a node at each of the corresponding points of J; each cusp of Λ corresponds to two coincident points of J, viz. for (l,m,n) at a cusp of Λ, the curve Ω has a node at the corresponding point of J. The number of the binodal curves Ω is thus equal to the number of the nodes of Λ, and the number of the cuspidal curves Ω is equal to the number of the cusps of Λ; and the question is to find the Plückerian numbers of the curve Λ.

This Professor Cremona accomplished in a very ingenious manner, by bringing the curve Λ into connexion with another curve Σ (viz., Σ is the locus of the nodes of those curves $lU+mV+nW=0$ which have a node), and the result arrived at is that for the curve Λ

$$\begin{aligned}
\text{Order} &= 3(r-1)(3r-2),\\
\text{Class} &= 6(r-1)^2,\\
\text{Nodes} &= \tfrac{1}{2}(r-1)(27r^3-63r^2+22r+16),\\
\text{Cusps} &= 3(r-1)(7r-8),\\
\text{Double tangents} &= \tfrac{1}{2}(r-1)(12r^3-36r^2+19r+16),\\
\text{Inflexions} &= 12(r-1)(r-2);
\end{aligned}$$

so that, finally, the number of the cuspidal curves $\sqrt{lU}+\sqrt{mV}+\sqrt{nW}=0$, is found to be $=3(r-1)(7r-8)$, and the number of the binodal curves of the same form is found to be $=\tfrac{1}{2}(r-1)(27r^3-63r^2+22r+16)$. When the given curves are conics, or for $r=2$, these numbers are $=18$ and 36 respectively; but the formulae are not applicable to the case where the conics have a point or points of intersection in common; nor, consequently, to the case of the three circles.

415. Corrections and Additions to the Memoir on the Theory of Reciprocal Surfaces (Phil. Trans. vol. clix. 1869, [411]).

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLXII. (for the year 1872), pp. 83—87. Received July 22,—Read November 16, 1871.]

1. I am indebted to Dr Zeuthen for the remark that although the "off-points" and "off-planes," as explained in the memoir, are real singularities, they are not the singularities to which the θ, θ' of the formulae refer. The most convenient way of correcting this is to retain all the formulae with θ, θ' as they stand, but to write ω, ω' for the number of "off-points" and "off-planes" respectively; viz. we thus have

ω, off-points,
θ, unexplained singular points,

and

ω', off-planes,
θ', unexplained singular planes,

the formulae as they stand, taking account of the unexplained singularities θ and θ', but not taking any account at all of the off-points and off-planes ω, ω'. The extended formulae in which these are taken into account are:

$$a(n-2) = \kappa - \beta + \rho + 2\sigma + 3\omega, \tag{1}$$
$$b(n-2) = \rho + 2\beta + 3\gamma + 3i, \tag{2}$$
$$c(n-2) = 2\sigma + 4\beta + \gamma + i + \omega, \tag{3}$$

$$a(n-2)(n-3) = 2(\delta - C - \beta\omega) + 3(ac - 3\sigma - \chi - 3\omega) + 2(ab - 2p - j), \tag{4}$$
$$b(n-2)(n-3) = 4k + (ab - 2p - j) + 3(bc - 3\beta - 2\gamma - i), \tag{5}$$
$$c(n-2)(n-3) = 6h + (ac - 3\sigma - \chi - 3\omega) + 2(bc - 3\beta - 2\gamma - i), \tag{6}$$

which replace Salmon's original formulae (A) and (B).

2. In the reciprocal formulae (A$'$) and (B$'$), for ω, ω' we must of course write ω', ω respectively: viz. these are

$$a'(n'-2) = \kappa' - B + \rho' + 2\sigma' + 3\omega', \tag{7}$$
$$b'(n'-2) = \rho' + 2\beta' + 3\gamma' + 3i', \tag{8}$$
$$c'(n'-2) = 2\sigma' + 4\beta' + \gamma' + i' + \omega', \tag{9}$$

$$a'(n'-2)(n'-3) = 2(B - C - 3\omega') + 3(a'c' - 3\sigma' - \chi' - 3\omega') + 2(a'b' - 2p' - j'), \tag{10}$$

$$b'(n'-2)(n'-3) = 4k' + (a'b' - 2p' - j') + 3(b'c' - 3\beta' - 2\gamma' - i'), \tag{11}$$
$$c'(n'-2)(n'-3) = 6h' + (a'c' - 3\sigma' - \chi' - 3\omega') + 2(b'c' - 3\beta' - 2\gamma' - i'). \tag{12}$$

The equations involving the points χ, ω, and the planes χ', ω' respectively are

$$c(-5n+6) + 3\omega = 6r + 18\beta + 5\gamma + 4i + 2\chi + 3\omega, \tag{13}$$
$$c'(5n-12) - 3\omega' = 6r' + 18\beta' + 5\gamma' + 4i' + 2\chi' + 3\omega', \tag{14}$$

(compare Salmon, p. 586, equations (C) and (D) near the bottom). The first of these is obtained from the value of $c(n-2)(n-3)$ by means of the equation

$$c' - c = 2h + 3\beta \quad \text{or} \quad c' = c + 2h + 3\beta,$$

which gives

$$(c + 2h + 3\beta)(n-2)(n-3) = 6h + (ac - 3\sigma - \chi - 3\omega) + 2(bc - 3\beta - 2\gamma - i);$$

substituting herein for ac and bc their values and reducing, we find

$$c\{(n-2)(n-3) - a - 2b\} = 6h - 6\beta - 6\sigma - 2\chi - 3\omega - 4\gamma - 2i,$$

where

$$(n-2)(n-3) - a - 2b = -5n + 6,$$

and the equation thus becomes

$$c(-5n+6) = 6h - 6\beta - 6\sigma - 2\chi - 3\omega - 4\gamma - 2i,$$

or, since $6h - 6\sigma = 6r$,

$$c(-5n+6) = 6r + 18\beta + 5\gamma + 4i + 2\chi + 3\omega.$$

The reciprocal of the first of these is

$$c'(-5n'+6) = 6h' - 6\beta' - 6\sigma' - 2\chi' - 3\omega' - 4\gamma' - 2i',$$

viz. writing $a = n(n-1) - 2b - 3c$, and $\kappa = 3n(n-2) - 6b - 8c$, this is

$$= 4n(n-2) - 8b - 11c - 2j - 3\chi' - 2C - 4B - 3\omega';$$

and it thus appears that the order c' of the spinode curve is reduced by 3 for each off-plane ω'.

4. As to the other two equations, writing for ρ, σ their values, these become

$$j + 6t + 3i + 5\beta + 6\gamma = b(2n-4) - 2q,$$
$$2\chi + 3\omega + 4i + 18\beta + 6\gamma = c(5n-12) - 6r + 3\omega,$$

equations which admit of a geometrical interpretation. In fact, when there is only a nodal curve, the first equation is

$$j + 6t = b(2n-4) - 2q,$$

which we may verify when the nodal curve is a complete intersection, $P = 0$, $Q = 0$; for if the equation of the surface is $(A, B, C \mid P, Q)^2 = 0$, where the degrees of A, B, C, P, Q are respectively $n-2f$, $n-f-g$, $n-2g$, f, $g = 0$ (so that $b = fg$), then the number of pinch-points is thus $= fg(2n-2f-2g)$, $= (2n-4)fg - 2fg(f+g-2)$; but for the curve $P = 0$, $Q = 0$ we have $t = 0$, and its order and class are $b = fg$, $q = fg(f+g-2)$, or the formula is thus verified.

Similarly, when there is only a cuspidal curve, the second equation is

$$2\chi + 3\omega = c(5n-12) - 6r + 3\omega,$$

which equation may be verified when the cuspidal curve is a complete intersection, $P = 0$, $Q = 0$, and the points χ, ω are given as the intersections of the curve with the surface $(A, B, C \mid N, -M)^2 = 0$.

Now $AC - B^2$ vanishing for $P = 0$, $Q = 0$ we must have $A = \Lambda\alpha^2 + A'$, $B = \Lambda\alpha\beta + B'$, $C = \Lambda\beta^2 + C'$, where A', B', C' vanish for $P = 0$, $Q = 0$; and thence $M = \Lambda M' + M''$, $N = \Lambda N' + N''$, where M'', N'' vanish for $P = 0$, $Q = 0$. The equation $(A, B, C \mid N, -M)^2 = 0$, writing therein $P = 0$, $Q = 0$, thus becomes $\Lambda'(N'\alpha - M'\beta)^2 = 0$; and its intersections with the curve $P = 0$, $Q = 0$ are the points $P = 0$, $Q = 0$, Λ' each three times, and the points $P = 0$, $Q = 0$, $N'\alpha - M'\beta = 0$ each twice; viz. they are the points $2\chi + 3\omega$. But if the degree of Λ is $= \lambda$, then the degrees of N', M', α, β are $2n - 3f - 2g - \lambda$, $2n - 2f - 3g - \lambda$, $n - 2f - \lambda$, $n - 2g - \lambda$, whence the degree of $\Lambda'(N'\alpha - M'\beta)^2$ is $= 5n - 6f - 6g$, and the number of points is $= fg(5n - 6f - 6g)$, viz. this is $= fg(5n-12) - 6fg(f+g-2)$, or it is $= c(5n-12) - 6r$; so that β being $= 0$, the equation is verified.

5. It was also pointed out to me by Dr Zeuthen that in the value of $24t$ given in No. 10 the term involving χ should be -6χ instead of $+6\chi$, and that in consequence the coefficients of β' are erroneous in several others of the formulae. Correcting these, and at the same time introducing the terms in ω, ω', and writing down also the terms in θ' as they stand, we have

$$\begin{aligned}
4\rho &= \ldots - 2\chi + 3\theta - 3\omega,\\
24t &= \ldots - 6\chi + 9\theta - 9\omega,\\
2\sigma &= \ldots - \theta - \omega,\\
8\beta &= \ldots + 6\chi - 9\theta + 9\omega,\\
8k &= \ldots - 6\chi + 17\theta - 25\omega,\\
2j &= \ldots + 6\chi - 9\theta + 15\omega,\\
8\omega' &= \ldots - 30\chi + 21\theta - 45\omega,\\
c' &= \ldots - 12\chi + 10\theta - 20\omega.
\end{aligned}$$

The equations of No. 11, used afterwards, No. 53, should thus be

$$\begin{aligned}
&4i + 6r = (5n - 12)c - 18\beta - 5\gamma - 2\chi + 3\theta - 3\omega,\\
&\qquad - 24t - 8\beta + 18r = (-8n + 16)b + (15n - 36)c - 34\beta + 6\gamma + 4j - 6\chi + 9\theta - 9\omega;
\end{aligned}$$

and from these I deduce

$$44q' + 6r = (44n - 88)b + (-45n + 63)c - 4\tfrac{2}{3}\beta - 3\tfrac{1}{3}\gamma - 13i - 8j - 22\chi + \ldots$$

6. In No. 32 we have (without alteration) $\theta = 16$; but in the application (Nos. 40 and 41) to the surface $FP^2 + GR - Q^2 = 0$ we have $b = 0$, and there are $t = fpq$ off-points, $F = 0$, $P = 0$, $Q = 0$, and $\chi = 9Pl$ close-points, $G = 0$, $P = 0$, $Q = 0$. The new equations involving ω are thus satisfied.

7. I have ascertained that the value of β' obtained, Nos. 51 to 64 of the memoir, is inconsistent with that obtained in the "Addition" by consideration of the deficiency, and that it is in fact incorrect. The reason is that, although, as stated No. 53, the values of two of the coefficients D, E may be assumed at pleasure, they cannot, in conjunction with a given system of values of A, B, C, be thus assumed at pleasure; viz. A, B, C being $= 110$, 272, 44 respectively, the values of D, E are really determinate. I have no direct investigation, but by working back from the formula in the Addition I find that we must have $D = 271$, $E = 315$; the values of the remaining coefficients then are

$$F = \tfrac{7}{2}, \quad G = -\tfrac{7}{2}, \quad H = -\tfrac{15}{2}, \quad I = -198;$$

or the formula is

$$\begin{aligned}
\beta' = 2n(n-2)(11n-24) &- (110n - 272)b + 44j\\
&+ \tfrac{1}{2}(271n - 315)c + \tfrac{7}{2}i - \tfrac{7}{2}\beta - 4\tfrac{1}{2}\gamma - 198\,\epsilon\\
&- hC - gB - i\,\iota - \chi\,j - \theta\,\beta - \ldots\omega\\
&+ h'C + g'B + i'\,\iota' + \chi'\,j' + \theta'\,\beta' - f'\,\omega';
\end{aligned}$$

but I have not as yet any means of determining the coefficients f, f' of the terms in ω, ω'.

From the several cases of a cubic surface we obtain as in the memoir; but applying to the same surfaces the reciprocal equation for β, instead of the results of the memoir, we find

$$\begin{array}{rlrl} h = & 24, & h' = & 4, \\ g + 2\chi = & 45, & g' + 2\chi' = & -198, \\ g + g' = & 9, & & \\ \theta + \theta' = & 518, & & \\ \lambda + \lambda' = & -2, & & \end{array}$$

(so that now $\lambda + \lambda' = -2$, as is also given by the cubic scroll). And combining the two sets of results, we have

$$\begin{array}{lcl} h = 24, & & h' = 4, \\ g = 45 - 2\chi, & & g' = -198 - 2\chi', \\ \theta = 9, & & \\ \lambda = -1 + \tfrac{1}{2}\theta, & & \end{array}$$

but the coefficients g, χ, χ', f, f' are still undetermined. To make the result agree with that of the Addition, I assume $\chi = -86$, $\chi' = -1$, $\theta = +28$; whence we have

$$\begin{aligned} \beta' = 2n(n-2)(11n-24) - (110n-272)b + 44j \\ - (271n-315)c + \tfrac{7}{2}i - \tfrac{7}{2}\beta + 198\,\epsilon \\ - 24C - 28B + 86\,\iota - \omega - 24\theta + \ldots\omega \\ + 4C' + 10B' + \iota' + \chi' j' + 8\chi' - 4\omega'; \end{aligned}$$

and if we substitute herein the foregoing value of $44q' + 6r$, we obtain

$$\begin{aligned} \beta' = 2n(n-2)(11n-24) + (-66n+184)b + (-93n+252)c \\ + 15\tfrac{1}{3}\beta + 9\tfrac{2}{3}\gamma + 6\iota - 24\theta - 28B - \iota - 2\tfrac{1}{3}j - 3\tfrac{1}{3}\chi + \theta - f\,\omega \\ + 4C' + 10B' + \iota' + \chi' j' + 8\chi' - 4\omega', \end{aligned}$$

which, except as to the terms in ω, ω', the coefficients of which are not determined, agrees with the value given in the Addition.

Dr Zeuthen considers that in general $i' = i$; I presume this is so, but have not verified it.

416. On the Theory of Reciprocal Surfaces.

[Published as an Addition by Prof. Cayley in Dr Salmon's *Treatise on the Analytic Geometry of Three Dimensions*, 4th Ed. (8vo. Dublin, 1882), pp. 592—604.]

620. In further developing the theory of reciprocal surfaces it has been found necessary to take account of other singularities, some of which are as yet only imperfectly understood. It will be convenient to give the following complete list of the quantities which present themselves:

n, order of the surface.
a, order of the tangent cone drawn from any point to the surface.
B, number of nodal edges of the cone.
K, number of its cuspidal edges.
ρ, class of nodal torse.
σ, class of cuspidal torse.
b, order of nodal curve.
k, number of its apparent double points.
f, number of its actual double points.
t, number of its triple points.
j, number of its pinch-points.
q, its class.
c, order of cuspidal curve.
h, number of its apparent double points.

θ, number of its points of an unexplained singularity.
χ, number of its close-points.
ω, number of its off-points.
r, its class.
β, number of intersections of nodal and cuspidal curves, stationary points on cuspidal curve.
γ, number of intersections, stationary points on nodal curve.
i, number of intersections, not stationary points on either curve.
C, number of cnicnodes of surface.
B, number of binodes.

And corresponding reciprocally to these:

n', class of surface.
a', class of section by arbitrary plane.
δ', number of double tangents of section.
κ', number of its inflexions.
ρ', order of node-couple curve.
σ', order of spinode curve.
b', class of node-couple torse.
k', number of its apparent double planes.
f', number of its actual double planes.
t', number of its triple planes.
j', number of its pinch-planes.
q', its order.
c', class of spinode torse.
h', number of its apparent double planes.
θ', number of its planes of a certain unexplained singularity.
χ', number of its close-planes.
ω', number of its off-planes.
r', its order.
β', number of common planes of node-couple and spinode torse, stationary planes of spinode torse.
γ', number of common planes, stationary planes of node-couple torse.
i', number of common planes, not stationary planes of either torse.
C', number of cnictropes of surface.
B', number of its bitropes.

In all 46 quantities.

621. In part explanation, observe that the definitions of ρ and σ agree with those given, Art. 609: the nodal torse is the torse enveloped by the tangent planes along the nodal curve; if the nodal curve meets the curve of contact α, then a tangent plane of the nodal torse passes through the arbitrary point, that is, ρ will be the number of these planes which pass through the arbitrary point, viz. the class of the torse. So also the cuspidal torse is the torse enveloped by the tangent planes along the cuspidal curve; and σ will be the number of these tangent planes which pass through the arbitrary point, viz. it will be the class of the torse.

Again, as regards ρ' and σ': the node-couple torse is the envelope of the bitangent planes of the surface, and the node-couple curve is the locus of the points of contact of these planes; similarly, the spinode torse is the envelope of the parabolic planes of the surface, and the spinode curve is the locus of the points of contact of these planes.

622. The several singularities may arise as follows:

Nodal curve.—The surface may meet a plane in a curve having a node; the locus of the node is the nodal curve; each point of this curve is a point on two sheets of the surface. The pinch-points are the points of the nodal curve, each of which is also a point on the cuspidal curve; they are points where the two tangent planes of the nodal curve coincide with each other and with the tangent plane of the cuspidal curve.

Cuspidal curve.—The surface may meet a plane in a curve having a cusp; the locus of the cusp is the cuspidal curve; each point of this curve is a point on two consecutive sheets of the surface.

The several quantities b, k, f, t, j; c, h, θ, χ, ω; β, γ, i refer to the nodal and cuspidal curves respectively and to their intersections; the significations of q and r will appear presently.

623. The section by any plane is a curve; a line in the plane meets this curve in n points, and the envelope of the line is the tangential of the curve. When the plane passes through a point on the nodal curve, two of the n points unite together at the point on the nodal curve; the line has thus a double point and the tangential has a corresponding cusp. When the plane passes through a pinch-point, the two branches of the nodal curve have the same tangent, and the tangential acquires a certain singularity, say the line has there two double points; this is not a point on the cuspidal curve. The several points of the nodal curve give rise to the same singularity of the tangential, and the class of the tangential (which is the class of section a') is thus reduced; the reduction is $2q$ and we have

$$q = b' - b - 2k - 2f - 3j - 6t, \quad \text{say } q = b' - b - 2k - 2f - 3j - 6t.$$

624. So when the plane passes through a point on the cuspidal curve, two of the n points unite together at the point on the cuspidal curve; the line has thus a cusp and the tangential has a corresponding node. The several points of the cuspidal curve give rise to the same singularity of the tangential, and the class of the tangential is thus further reduced; the reduction is $3r$ and we have

$$r = c' - c - 2h - 3\beta.$$

625. At a point i the nodal curve crosses the cuspidal curve, being on the side away from the two half-sheets of the surface acnodal, and on the side of the two half-sheets crunodal, viz. the two half-sheets intersect each other along this portion of the nodal curve. There is at the point a single tangent plane, which is a plane i'; and we thus have $i = i'$.

626. As already mentioned, a cnicnode C is a point where, instead of a tangent plane, we have a tangent quadricone; and at a binode B the quadricone degenerates into a pair of planes. A cnictrope C' is a plane touching the surface along a conic; in the case of a bitrope B', the conic degenerates into a flat conic or pair of points.

627. In the original formulae for $a(n-2)$, $b(n-2)$, $c(n-2)$, we have to write $\kappa - B$ instead of κ, and the formulae are further modified by reason of the singularities θ and ω. So in the original formulae for $a(n-2)(n-3)$, $b(n-2)(n-3)$, $c(n-2)(n-3)$, we have instead of δ to write $B - C - 3\omega$; and to substitute new expressions for $[ab]$, $[ac]$, $[bc]$, viz. these are

$$\begin{aligned}
[ab] &= ab - 2\rho - j, \\
[ac] &= ac - 3\sigma - \chi - \omega, \\
[bc] &= bc - 3\beta - 2\gamma - i.
\end{aligned}$$

The whole series of equations thus is:

$$a' = n(n-1) - 2b - 3c, \tag{15}$$

$$\kappa' = 3n(n-2) - 6b - 8c, \tag{16}$$

$$\delta' = \tfrac{1}{2}n(n-2)(n^2-9) - (n^2-n-6)(2b+3c) + 2b(b-1) + 6bc + \tfrac{9}{2}c(c-1). \tag{17}$$

$$a(n-2) = \kappa - B + \rho + 2\sigma + 3\omega, \tag{7}$$

$$b(n-2) = \rho + 2\beta + 3\gamma + 3i, \tag{8}$$

$$c(n-2) = 2\sigma + 4\beta + \gamma + i + \omega, \tag{9}$$

$$a(n-2)(n-3) = 2(\delta - C - 3\omega) + 3(ac - 3\sigma - \chi - 3\omega) + 2(ab - 2\rho - j), \tag{10}$$

$$b(n-2)(n-3) = 4k + (ab - 2\rho - j) + 3(bc - 3\beta - 2\gamma - i), \tag{11}$$

$$c(n-2)(n-3) = 6h + (ac - 3\sigma - \chi - 3\omega) + 2(bc - 3\beta - 2\gamma - i), \tag{12}$$

$$q = b' - b - 2k - 2f - 3j - 6t, \tag{13}$$

$$r = c' - c - 2h - 3\beta. \tag{14}$$

Also, reciprocal to these:

$$\begin{aligned} a' &= n'(n'-1) - 2b' - 3c', && (15)\\ \kappa' &= 3n'(n'-2) - 6b' - 8c', && (16)\\ \delta' &= \tfrac{1}{2}n'(n'-2)(n'^2-9) - (n'^2-n'-6)(2b'+3c') + 2b'(b'-1) + 6b'c' + \tfrac{9}{2}c'(c'-1). && (17) \end{aligned}$$

$$\begin{aligned} a'(n'-2) &= \kappa' - B + \rho' + 2\sigma' + 3\omega', && (18)\\ b'(n'-2) &= \rho' + 2\beta' + 3\gamma' + 3i', && (19)\\ c'(n'-2) &= 2\sigma' + 4\beta' + \gamma' + i' + \omega', && (20) \end{aligned}$$

$$\begin{aligned} a'(n'-2)(n'-3) = 2(B' - C' - 3\omega') &+ 3(a'c' - 3\sigma' - \chi' - 3\omega')\\ &+ 2(a'b' - 2\rho' - j'), \quad (21) \end{aligned}$$

$$\begin{aligned} b'(n'-2)(n'-3) &= 4k' + (a'b' - 2\rho' - j') + 3(b'c' - 3\beta' - 2\gamma' - i'), && (22)\\ c'(n'-2)(n'-3) &= 6h' + (a'c' - 3\sigma' - \chi' - 3\omega') + 2(b'c' - 3\beta' - 2\gamma' - i'), && (23) \end{aligned}$$

$$\begin{aligned} q' &= b' - b - 2k' - 2f' - 3j' - 6t', && (24)\\ r' &= c' - c' - 2h' - 3\beta'. && (25) \end{aligned}$$

together with one other independent relation, in all 26 relations between the 46 quantities.

628. The new relation may be presented under several different forms, equivalent to each other in virtue of the foregoing 25 relations; these are

$$\begin{aligned} &2(n-1)(n-2)(n-3) - 12(n-3)(b+c) + 6q + 6r + 24i + 42\beta + 30\gamma - \chi\\ &\quad = 2(n'-1)(n'-2)(n'-3) - 12(n'-3)(b'+c') + 6q' + 6r' + 24i' + 42\beta' + 30\gamma' - \chi', \quad (26) \end{aligned}$$

$$\begin{aligned} &26n - 12c - 4C - 10\beta + \beta - 7j - 8\chi + 16i - 4\omega\\ &\quad = 26n' - 12c' - 4C' - 10\beta' + \beta' - 7j' - 8\chi' + 16i' - 4\omega', \quad (27) \end{aligned}$$

in each of which two equations Σ is used to denote the same function of the accented letters that the left-hand side is of the unaccented letters.

$$\begin{aligned} \beta' + \beta = 2n(n-2)(11n-24) &+ (-66n+184)b + (-93n+252)c\\ &+ 22(2\beta + 3\gamma + 3i) + 27(4\beta + \gamma + i)\\ &+ i + \beta - 24C - 28B - \iota - 2\tfrac{1}{3}j - 3\tfrac{1}{3}\chi + \theta\beta - f\omega\\ &+ 4C' + 10B' + \iota' + 7j' + 8\chi' - 4\omega'. \quad (28) \end{aligned}$$

Or, reciprocally,

$$\begin{aligned} \beta' + \beta = 2n'(n'-2)(11n'-24) &+ (-66n'+184)b' + (-93n'+252)c'\\ &+ 22(2\beta' + 3\gamma' + 3i') + 27(4\beta' + \gamma' + i')\\ &+ i' + \beta' - 24C' - 28B' - \iota' - 2\tfrac{1}{3}j' - 3\tfrac{1}{3}\chi' + \theta'\beta' - f'\omega'\\ &+ 4C + 10B + \iota + 7j + 8\chi - 4\omega. \quad (29) \end{aligned}$$

The foregoing equation (26) in fact expresses that the surface and its reciprocal have the same deficiency; viz. the expression for the deficiency is

$$\begin{aligned}\text{Deficiency} &= \tfrac{1}{6}(n-1)(n-2)(n-3)-(n-3)(b+c)+\tfrac{1}{2}(q+r)+2t+i+\beta+\gamma+i-\theta,\\ &= \tfrac{1}{6}(n'-1)(n'-2)(n'-3)-\text{\&c.} \qquad (30)\end{aligned}$$

629. The equation (28) (due to Prof. Cayley) is the correct form of an expression for β', first obtained by him (with some errors in the numerical coefficients) from independent considerations, but which is best obtained by means of the equation (26); and (27) is a relation presenting itself in the investigation. In fact, considering a as standing for its value $n(n-1)-2b-3c$, we have from the first 25 equations

$$\begin{aligned} &0 = 2(n-1)(n-2)(n-3)-12(n-3)(b+c)+\dots\\ -2\sigma(n-2):\quad &\kappa-B-\rho-2\sigma-3\omega=0,\\ -4b(n-2):\quad &-\rho-2\beta-3\gamma-3i=0,\\ -6c(n-2):\quad &-2\sigma-4\beta-\gamma-i-\omega=0,\\ +2:\quad &2q+2r+4i+14\beta+10\gamma+\dots\end{aligned}$$

and multiplying these equations by the numbers set opposite to them respectively, and adding, we find

$$\begin{aligned}&-2n^3+12n^2+4n+b(12n-36)+c(12n-48)\\ &\qquad -6q-6r-4i-10\beta-4\gamma-24i-7j-8\chi+2\theta-4\omega=\Sigma,\end{aligned}$$

and adding thereto (26) we have the equation (27); and from this (28), or by a like process, (29), is obtained without much difficulty. As to the 8 Σ-equations or symmetries, observe that the first, third, fourth, and fifth are in fact included among the original equations (for an expression which vanishes is in fact $=\Sigma$); we have from them moreover $3n-c=3a'-\kappa'$, and thence $3n-c-\kappa=3a'-\kappa'-\kappa$, which is $=\Sigma$, or we have thus the second equation; but the sixth, seventh, and eighth equations have yet to be obtained.

630. The equations (15), (16), (17) give

$$\begin{aligned}
n' &= a(a-1) - 2\delta' - 3\kappa', \\
c' &= 3a(a-2) - 6\delta' - 8\kappa', \\
\delta' &= \tfrac{1}{2}a(a-2)(a^2-9) - (a^2-a-6)(2\delta'+3\kappa') + 2\delta'(\delta'-1) + 6\delta'\kappa' + \tfrac{9}{2}\kappa'(\kappa'-1);
\end{aligned}$$

from (7), (8), (9) we have

$$(a-b-c)(n-2) = \kappa - B - 6\beta - 4\gamma - 3i - \theta + 2\omega,$$
$$(a-2b-3c)(n-2)(n-3) = 2(\delta - C) - 8k - 18h - 6c + 18\beta + 12\gamma + 9i - 6\omega),$$

and substituting these values for κ and δ, and for a its value $= n(n-1) - 2b - 3c$, we obtain the values of n', c', b'; viz. the value of n' is

$$\begin{aligned}
n' = n(n-1)^2 - n(7b+12c) + 4b^2 + 8b + 9c^2 + 15c \\
- 2k - 18f + 18\beta + 12\gamma + 12t - 9i - 2\theta - 3j - 3\chi.
\end{aligned}$$

Observe that the effect of a cnicnode C is to reduce the class by 2, and that of a binode B to reduce it by 3.

631. We have

$$(n-2)(n-3) = n^2 - n + (-4n+6) = a + 2b + 3c + (-4n+6),$$

and making this substitution in the equations (10), (11), (12), which contain $(n-2)(n-3)$, these become

$$\begin{aligned}
a(-4n+6) &= 2(\delta - C) - a^2 - 4\rho - 9\sigma - 2j - 3\chi - 1\omega, \\
b(-4n+6) &= 4k - 2b^2 - 9\beta - 6\gamma - 3i - 2\rho - j, \\
c(-4n+6) &= 6h - 3c^2 - 6\beta - 4\gamma - 2i - 3\sigma - \chi + \omega,
\end{aligned}$$

(the foregoing equations (C) Salmon p. 586); and adding to each equation four times the corresponding equation with the factor $(n-2)$, these become

$$\begin{aligned}
a^2 - 2a &= 2(\delta - C) + 4(\kappa - B) - \rho - 2j - 3\chi - 3\omega, \\
2b^2 - 2b &= 4k - \rho + 6\gamma + 12t - 3i + 2\rho - j, \\
3c^2 - 2c &= 6h + 10\beta + 4\rho - 2i + 5\sigma - \chi + \omega.
\end{aligned}$$

Writing in the first of these $a^2 - 2a = n' + 2\delta' + 3\kappa' - a$, and reducing the other two by means of the values of q, r, the equations become

$$\begin{aligned} \sigma' &= a^2 - 2C - 4\rho + \kappa - \sigma - 2j - 3\chi - 3\omega, \\ 2q + \beta + 3i + j &= 2\rho, \\ 3r + c + 2i + \chi &= 5\sigma + \beta + 4\rho + \omega, \end{aligned}$$

which give at once the last three of the 8 Σ-equations.

The reciprocal of the first of these is

$$\sigma' = a - n + \kappa' - 2j' - 3\chi' - 2C' - 4B' - 3\omega',$$

or writing herein $a = n(n-1) - 2b - 3c$ and $\kappa' = 3n(n-2) - 6b - 8c$, this is

$$\sigma' = 4n(n-2) - 8b - 11c - 2j - 3\chi' - 2C' - 4B' - 3\omega',$$

giving the order of the spinode curve; viz. for a surface of the order n without singularities this is $= 4n(n-2)$, the product of the orders of the surface and its Hessian.

632. Instead of obtaining the second and third equations as above, we may to the value of $b(-4n+6)$ add twice the value of $b(n-2)$; and to twice the value of $c(-4n+6)$ add three times the value of $c(n-2)$, thus obtaining equations free from ρ and σ respectively; these equations are

$$\begin{aligned} b(-2n+2) &= 4k - 2b' - 5\beta - 3i + 6t - j, \\ c(-5n+6) &= 12h - 6c' - 5\gamma - 4i - 2\chi + 3\sigma - 3\omega, \end{aligned}$$

equations which, introducing therein the values of q and r, may also be written

$$\begin{aligned} b(2n-4) &= 2q + 5\beta + 6\gamma + 6t + 3i + j + 4f, \\ c(5n-12) + 3\omega &= 6r + 18\beta + 5\gamma + 4i + 2\chi + 3\omega. \end{aligned}$$

Considering as given, n the order of the surface; the nodal curve with its singularities b, k, f, t; the cuspidal curve and its singularities c, h; and the quantities β, γ, i which relate to the intersections of the nodal and cuspidal curves; the first of the two equations gives j, the number of pinch-points, being singularities of the nodal curve *quoad* the surface; and the second equation establishes a relation between θ, χ, ω, the numbers of singular points of the cuspidal curve *quoad* the surface.

In the case of a nodal curve only, if this be a complete intersection $P = 0$, $Q = 0$, the equation of the surface is $(A, B, C \mid P, Q)^2 = 0$, and the first equation is

$$b(-2n+2) = 4k - 2b^2 + 6t - j;$$

or, assuming $t = 0$, say $j = 2(n-1)b - 2b^2 + 4k$, which may be verified; and so in the case of a cuspidal curve only, when this is a complete intersection $P = 0$, $Q = 0$, the equation of the surface is $(A, B, C \mid P, Q)^2 = 0$, where $AC - B^2 = MP + NQ$; and the second equation is

$$c(-5n+6) = 12h - 6c^2 - 2\chi + 3\theta - 3\omega,$$

or, say

$$2\chi + 3\omega = (5n-6)c - 6c^2 + 12h + 3\theta,$$

which may also be verified.

633. We may in the first instance out of the 46 quantities consider as given the 14 quantities

$$n;\quad b,k,f,t;\quad c,h,\theta,\chi;\quad \beta,\gamma,i;\quad C,B,$$

then of the 26 relations, 17 determine the 17 quantities

$$a,\delta,\kappa,\rho,\sigma;\quad j,q;\quad r,\omega;\quad n',a',\delta',\kappa';\quad \rho',\sigma',c',\iota,$$

and there remain the 9 equations (18), (19), (20), (21), (22), (23), (24), (25), (28), connecting the 15 quantities

$$\rho',\sigma';\quad k',f',t',j',c',h',\theta',\chi',\omega',r';\quad \beta',\gamma';\quad C',B'.$$

634. The formulae of course give a, δ, κ, q, ρ, σ in terms of n, b, c, and the other quantities respectively; but they do not directly give the curve-singularities k', f', t', j' of the node-couple torse, nor the curve-singularities h', θ', χ', ω' of the spinode torse, nor do they give β', γ', i', C', B': and until these can be obtained, the theory of reciprocal surfaces must be considered as incomplete.

635. The question of singularities has been considered under a more general point of view by Zeuthen, in the memoir "Recherche des singularités qui ont rapport à une droite multiple d'une surface," *Math. Annalen*, t. IV. pp. 1—20, 1871. He attributes to the surface:

A number of singular points, viz. points at any one of which the tangents form a cone of the order μ, and class ν, with $y+\eta$ double lines, of which y are tangents to branches of the nodal curve through the point, and $z+\zeta$ stationary lines, whereof z are tangents to branches of the cuspidal curve through the point, and with u double planes and v stationary planes; moreover, these points have only the properties which are the most general in the case of a surface regarded as a locus of points; and Σ denotes a sum extending to all such points. (The foregoing general definition includes the cnicnodes ($\mu=\nu=2$, $y=\eta=z=\zeta=u=v=0$); and [also, but not properly] the binodes ($\mu=2$, $\nu=1$, $y=\eta=\zeta=$ &c. $=0$), [it includes also the off-points ($\mu=\nu=3$, $z=\zeta=1$, $y=\eta=\xi=0$)].)

And, further, a number of singular planes, viz. planes any one of which touches along a curve of the class μ' and order ν', with $y'+\eta'$ double tangents, of which y' are generating lines of the node-couple torse, $z'+\zeta'$ stationary tangents, of which z' are generating lines of the spinode torse, u' double points and v' cusps; it is, moreover, supposed that these planes have only the properties which are the most general in the case of a surface regarded as an envelope of its tangent planes; and Σ' denotes a sum extending to all such planes. [The definition includes the cnictropes ($\mu'=\nu'=2$, $y'=\eta'=z'=\zeta'=u'=v'=0$), and [also, but not properly] the bitropes ($\mu'=2$, $\nu'=1$, $\eta'=\zeta'=$ &c. $=0$), [it includes also the off-planes ($\mu'=\nu'=3$, $z'=\zeta'=1$, $y'=\eta'=\xi'=0$)].)

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Notes and References.

384. The conclusion arrived at Nos. 27—30 that the transformed curve of the order $D+1$ depends upon $4D-6$ parameters is at variance with Riemann's theorem according to which the number of parameters is $3p-3$, (p Riemann $= D$ Cayley), $= 3D-3$, and this last is the correct value. My erroneous conclusion is referred to in the preface to Clebsch and Gordan's *Theorie der Abel'schen Functionen* (Leipzig, 1866),

> "Unter den von Riemann behandelten Theilen der Theorie haben wir die Frage nach der Anzahl der Moduln einer Klasse von Abel'schen Functionen ausschliessen zu müssen geglaubt. Diese Frage ist durch die scharfsinnigen Betrachtungen des Herrn Cayley Gegenstand der Controverse geworden: sie ist überhaupt wohl zunächst nur durch tiefe algebraische Untersuchungen endgültig zu entscheiden, für deren Schwierigkeiten die gegenwärtig bekannten Methoden nicht mehr auszureichen scheinen."

In the case D (or p) $= 3$, my value is 10, Riemann's is 9: that the latter is correct was shown by a direct proof in the paper Brill, "Note bezüglich der Zahl der Moduln einer Klasse von algebraischen Gleichungen," *Math. Ann.*, t. i. (1869), pp. 401—406: the explanation of my error is given in the paper, Cayley, "Note on the Theory of Invariants," *Math. Ann.*, t. III. (1871), pp. 268—271.

400. The question here considered, viz., the expression of a binary sextic f in the form $v^3 - u^2$, v and u a cubic and a quadric respectively, forms the basis of the very interesting investigations contained in the Memoir, Clebsch "Zur Theorie der binären Formen sechster Ordnung und zur Dreitheilung der hyperelliptischen Functionen," *Gott. Abh.*, t. xiv. (1869), pp. 1 &c. Considering f as a given sextic it is remarked that the number of solutions, or what is the same thing the number of the functions u or v, although at first sight $= 45$, is really $= 40$; supposing that there is a given solution u, v, or that the sextic function is in the first instance given in the form $v^3 - u^2$, then if any other solution is u', v', we have $v'^3 - v^3 = u'^2 - u^2$, where v', u' are functions to be determined: there are in all 39 solutions, a set of 27 and a set of 12 solutions: viz. writing the equation in the form $(v+v')(v-v') = (u-u')(u-\epsilon u')(u-\epsilon^2 u')$, ϵ an imaginary cube root of unity, then either the $v+v'$ and the $v-v'$ contain each of them as a factor one of the quadric functions $u-u'$, $u-\epsilon u'$, $u-\epsilon^2 u'$ (which gives the set of 27 solutions) or else the $v+v'$ and the $v-v'$ are each of them the product of three linear factors of the quadric functions respectively (which gives the set of 12 solutions). It may be added that the 27 solutions form 9 groups of 3 each and that these 9 groups depend upon Hesse's equation of the order 9 for the determination of the inflexions of a cubic curve; and that the 12 solutions are determined by an equation of the order 12 which is the known resolvent of this order arising from Hesse's equation and is solved by means of a quartic equation with a quadrinvariant $= 0$. As appears by the title of the memoir, the question is connected with that of the trisection of the hyperelliptic functions.

401, 403. On the subject of Pascal's theorem, see Veronese, "Nuove teoremi sull' hexagrammum mysticum," *R. Accad. dei Lincei* (1876—77), pp. 7—61; Miss Christine Ladd (Mrs Franklin), "The Pascal Hexagram," *Amer. Math. Jour.*, t. ii. (1879), pp. 1—12, and Veronese, "Interpretations géométriques de la théorie des substitutions de n lettres, particulièrement pour $n = 3$, 4, 5, en relation avec les groupes de l'Hexagramme Mystique," *Ann. di Matem.*, t. xi. 1882—83, pp. 93—236. See also Richmond, "A Symmetrical System of Equations of the Lines on a Cubic Surface which has a Conical Point," *Quart. Math. Jour.*, t. xxii. (1889), pp. 170—179, where the author discusses a perfectly symmetrical system of the lines on the cubic surface and deduces from them equations of the lines relating to a Pascal's hexagon; there are of course through the conical point 6 lines lying on a quadric cone and these by their intersections with the plane give the six points of the hexagon; the interest of the paper consists as well in the connexion established between the two theories as in the perfectly symmetrical form given to the equations.

406, 407. A correction was made by Halphen to the fundamental theorem of Chasles that the number of the conics (X, R) is $= a\mu + b\nu$, he finds that a diminution is in some cases required, and thus that the general form is, Number of conics $(X, R) = a\mu + b\nu - r$: see Halphen's two Notes, *Comptes Rendus*, 4 Sep. and 13 Nov., 1876, t. LXXXIII. pp. 537 and 886, and his papers "Sur la théorie des caractéristiques pour les coniques," *Proc. Lond. Math. Soc.*, t. IX. (1877—1878), pp. 149—170, and "Sur les nombres des coniques qui dans un plan satisfont à cinq conditions projectives et indépendantes entre elles," *Proc. Lond. Math. Soc.*, t. X. (1878—79), pp. 76—87: also Zeuthen's paper "Sur la revision de la théorie des caractéristiques de M. Study," *Math. Ann.*, t. XXXVII. (1890), pp. 461—464, where the point is brought out very clearly and tersely.

The correction rests upon a more complete development of the notion of the line-pair-point, viz. this degenerate form of conic seems at first sight to depend upon three parameters only, the two parameters which determine the position of the coincident lines, and a third parameter which determines the position therein of the coincident points: but there is really a fourth parameter. [Compare herewith the point-pair, or indefinitely thin conic, which working with point-coordinates presents itself in the first instance as a coincident line-pair depending on two parameters only, but which really depends also on the two parameters which determine the position therein of the vertices.]

As to the fourth parameter of the line-pair-point the most simple definition is a metrical one; taking the semiaxes of the degenerate conic to be a and b ($a = 0$, $b = 0$) then we have two positive integers p and q prime to each other such that the ratio $a^q : b^p$ is finite; and this being so the fractional or it may be integer number $p : q$ is the fourth parameter in question. But it is preferable to adopt Halphen's purely descriptive definition, viz. we consider a conic 1° in reference to three given points y, z, t on a given line, and take x, x' for the remaining intersections of the conic with the lines yzt, ytz respectively; the fourth parameter is the anharmonic ratio of the pencil x, x', y, z.

408, 409. As to the correction, see Halphen, “Sur une série de courbes analogues aux courbes de Lamé,” *Comptes Rendus*, t. LXXXIV. (1877), pp. 365—367.

411. The “Addition” referred to in the text is given as [415] of the present volume.

412. For the curve $x^3 + y^3 + z^3 + 6lxyz = 0$, the sextactic points are given as the intersections of the curve by the Hessian $l(x^3 + y^3 + z^3) - (1 + 2l^3)xyz = 0$. But these are the points of contact of the tangents from the 9 flexions; in fact the equation of a tangent is $x(\alpha^2 + 2l\beta\gamma) + \ldots = 0$, and for the point of contact we have $x : y : z = \alpha(\alpha^3 + 2l\beta\gamma) : \ldots$; substituting in the equation of the curve, this becomes $(\alpha^3 + \beta^3 + \gamma^3 + 6l\alpha\beta\gamma)(\alpha^6 + \ldots - 10l\alpha^3\beta^3\gamma^3 + 2l^2\beta^3\gamma^3 \cdot \alpha^3 + \ldots) = 0$; the first factor vanishes for a point on the curve, and the second factor is $= (\alpha^3 + \beta^3 + \gamma^3)^2 - 4(1 + 2l^3)^2\alpha^3\beta^3\gamma^3$ which $= 0$ gives the 9 points of contact. But these are the intersections of the curve by the Hessian; and hence the theorem.

413. The theorem that a cubic curve has 27 sextactic points was first given in my paper “Note on the Inflections of a Cubic Curve,” *Quart. Math. Jour.*, t. XII. (1873), pp. 365—366 [417]; see also the paper “On the Sextactic Points of a Plane Curve,” *Phil. Trans.*, vol. CLXV. (for the year 1875), pp. 545—578 [428].

414. The formulae for the class of the envelope of the osculating conics were verified by Mr. T. Cotterill in the particular case of a cubic curve; see his paper “On the Number and Nature of the Sextactic Points of a Plane Curve,” *Quart. Math. Jour.*, t. XII. (1873), pp. 132—143.

415. The corrections and additions here made are given in a more complete form in [416].

416. The formulae given in this Addition require some explanation in reference to the singularities χ, χ' and ω, ω' which present themselves. In Salmon's original formulae (A), (B), the terms in χ were omitted; the terms in ω were subsequently introduced by Dr Zeuthen. The singularities in question are:

χ, close-points: these are points on the cuspidal curve, not ordinary points thereof, but points where the cuspidal curve has a certain singularity analogous to that of the point of contact of a stationary tangent; or say, they are points where the cuspidal curve is met by its envelope. The number of them is χ.

ω, off-points: these are isolated singular points of the surface, not situate on either the nodal or the cuspidal curve.

And reciprocally:

χ', close-planes; ω', off-planes.

417. In the present state of the theory of surfaces, the only case in which we can be certain of the complete system of singularities is that of the general surface of the order n without any singularities; for this case $n' = n(n-1)^2$.

418. The expression for the deficiency given in the text is not universally applicable; the deficiency of a surface is not in general determined by the order of the surface and the orders of its nodal and cuspidal curves only, but depends also on the manner in which these curves are situate upon the surface. The question is considered by Zeuthen, "Note sur la déficience des surfaces algébriques," *Comptes Rendus*, t. LXXXIV. (1877), pp. 365—367.

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419. The theory of reciprocal surfaces is considered under a more general point of view by Zeuthen in the memoir "Recherche des singularités qui ont rapport à une droite multiple d'une surface," *Math. Annalen*, t. IV. (1871), pp. 1—30; and also in the memoir "Sur la déficience des surfaces algébriques," *Comptes Rendus*, t. LXXXIV. (1877), pp. 365—367.

420. With Zeuthen, the nodal curve has

$$3t+f+3\theta'+\chi' \quad \text{double points}$$

($k=i+3t+f+3\theta'+2\chi'$, if k denotes, as with me, the number of apparent double points of the curve), and it has

$$\gamma+d+3\chi'+\xi' \quad \text{stationary points.}$$

The cuspidal curve has

$$g+6\chi'+12\xi'+U'+4\theta'+2\Sigma+2\Sigma' \quad \text{double points}$$

($h=h+g+6\chi'+12\xi'+U'+4\theta'+2\Sigma+2\Sigma'$, if h denotes, as with me, the number of apparent double points of the curve), and it has

$$\beta+6+2\theta'+\Sigma \quad \text{stationary points}$$

and the nodal and cuspidal curves intersect in

$$3\beta+2\gamma+i+12\theta'+2\Sigma+2\Sigma' \quad \text{points;}$$

where I have written Σ and Σ' to denote sums (different in the different equations) determined by Zeuthen, and depending on the singularities K and S' respectively.

For comparison of my formulae with Zeuthen's it is thus proper in my formulae to write $\theta=0$, $\omega=0$, $\chi=0$ (but in the first instance I retain θ) and in his formulae to write $\theta'=0$, $\chi'=0$, $d=0$, $g=0$, $e=0$, $\Sigma=0$, $\Sigma'=0$. Doing this the last mentioned formulae give as with me $3t+f$ double points and γ stationary points for the nodal curve, but they give for the cuspidal curve $6\chi'+12\xi'$ (instead of 0) double points and β stationary points; and the two curves intersect (as with me) in $3\beta+2\gamma+i$ points.

There is a real discrepancy in the number $6\chi'+12\xi'$ of double points on the cuspidal curve.

I compare his $(6 + 2 \times 6 + 1 = 33)$ relations

$$\begin{array}{ll} a = a', & d = d', \\ i = i', & g = g', \\ \kappa = \kappa', & h = h', \end{array}$$

$$n(n-1) = a + 2b + 3c. \tag{6}$$

$$a(a-1) = n' + 2\delta' + 3\kappa'. \tag{7}$$

$$c - \kappa' = 3(n-a). \tag{8}$$

$$b(b-1) = g + 2k + 3j + 3\omega' + \chi'. \tag{9}$$

$$\tfrac{1}{2}(b-q) = \gamma + d - s + \Sigma', \quad \text{[determines } s\text{]}. \tag{10}$$

$$c(c-1) = r + 2h + 3\beta + 6\theta' + 3e. \tag{11}$$

$$\tfrac{1}{2}(c-r) = \beta + e - m + 2\theta' + 2\Sigma, \quad \text{[determines } m\text{]}. \tag{12}$$

$$a(n-2) = \kappa - \beta + \rho + 2\sigma + \Sigma. \tag{13}$$

$$b(n-2) = \rho + 2\beta + 3\gamma + 3i + 9\theta' + \chi'. \tag{14}$$

$$c(n-2) = 2\sigma + 4\beta + \gamma + 8\chi' + 16\xi' + 12\theta' + \Sigma. \tag{15}$$

$$a(n-2)(n-3) = 2(\delta - 3C) + 3(ac - 3\sigma - \chi) + 2(ab - 2\rho - j). \tag{16}$$

$$b(n-2)(n-3) = 4(k - 3\xi') + (ab - 2\rho - j) + 3(bc - 3\beta - 2\gamma - i) + 3\chi' + 2\Sigma'. \tag{17}$$

$$\begin{aligned} c(n-2)(n-3) = {} & 6(h - 6\chi' - 12\xi' - U' - 4\theta' - \Sigma) \\ & + (ac - 3\sigma - \chi) + 2(bc - 3\beta - 2\gamma - i) \\ & - 30\theta' + 2\Sigma + 2\Sigma'. \end{aligned} \tag{18}$$

with the like reciprocal equations $(6')$ to $(18')$

$$\begin{aligned} & a + m - r - \theta - \chi' - 4\eta' + \Sigma' = 0, \\ & \qquad 4\mu' - 3\xi' - 14\eta' + \Sigma = \theta' + m' - f' - \beta' - 4j - 3\chi - 14\eta + \Sigma', \end{aligned}$$

—where κ, &c.

and my $(3+22+1=26)$ relations as follows:

$$a=a', \qquad f=f'. \tag{1}$$

$$i=i'. \tag{2}$$

$$\kappa=\kappa'. \tag{3}$$

$$a=n(n-1)-2b-3c. \tag{4}$$

$$\kappa'=3n(n-2)-6b-8c. \tag{5}$$

$$\delta'=\tfrac{1}{2}n(n-2)(n^2-9)-\&\text{c}. \tag{6}$$

$$\text{(A)}\quad q=b'-b-2k-2f-3j-6t. \tag{13}$$

$$\text{(B)}\quad r=c'-c-2h-3\beta. \tag{14}$$

$$\text{(C)}\quad a(n-2)=\kappa-B+\rho+2\sigma+3\omega. \tag{7}$$

$$\text{(D)}\quad b(n-2)=\rho+2\beta+3\gamma+3i. \tag{8}$$

$$\text{(E)}\quad c(n-2)=2\sigma+4\beta+\gamma+i+\omega. \tag{9}$$

$$\text{(F)}\quad a(n-2)(n-3)=2(\delta-C-3\omega)+3(ac-3\sigma-\chi-3\omega)+2(ab-2\rho-j). \tag{10}$$

$$\text{(G)}\quad b(n-2)(n-3)=4k+(ab-2\rho-j)+3(bc-3\beta-2\gamma-i). \tag{11}$$

$$\text{(H)}\quad c(n-2)(n-3)=6h+(ac-3\sigma-\chi-3\omega)+2(bc-3\beta-2\gamma-i), \tag{12}$$

with the like reciprocal equations (4′) to (14′);

$$\begin{aligned}\text{(I)}\quad &2(n-1)(n-2)(n-3)-12(n-3)(b+c)+6q+6r+24i+42\beta+30\gamma-\chi\\ &=2(n'-1)(n'-2)(n'-3)-12(n'-3)(b'+c')+6q'+6r'+24i'+42\beta'+30\gamma'-\chi'.\end{aligned} \tag{26}$$

$$\begin{gathered}k+\xi+3t+3\theta'+\chi'-E,\\ h+\theta'+6\chi'+12\xi'+U'+4\theta'+\Sigma+2\Sigma'.\end{gathered}$$

Substituting for κ; h their values we have instead of (A), (B), (C), (D) the equations

$$\begin{aligned}
&\text{(A')}\quad b'-b=q+2k+2f+6t+6\theta'+3j+3\chi'+\Sigma+\Sigma',\\
&\text{(B')}\quad c'-c=r+2h+2g+12\chi'+2U'+2\xi'+W'+3\theta'+3e+\Sigma+\Sigma',\\
&\text{(C')}\quad b(n-2)(n-3)=4k+2\chi'\theta'+(ab-2\rho-j)+3(bc-3\beta-2\gamma-i)+\Sigma+\Sigma',\\
&\text{(D')}\quad c(n-2)(n-3)=6h-30\theta'+(ac-3\sigma-\chi)+2(bc-3\beta-2\gamma-i)+\Sigma+\Sigma'.
\end{aligned}$$

Writing as before $C=0$, $\omega=0$; $U=0$, $\chi'=0$, $d=0$, $g=0$, $e=0$, and neglecting the terms in Σ, Σ', the two equations (E) become

$$\begin{aligned}
\text{Zeuthen}\quad & c(n-2)=2\sigma+4\beta+\gamma+8\chi'+16\xi',\\
\text{Cayley}\quad & c(n-2)=2\sigma+4\beta+\gamma+\theta,
\end{aligned}$$

which can be made to agree by writing $\theta=8\chi'+16\xi'$. But we have

$$\begin{aligned}
\text{Zeuthen}\quad \text{(F)}\quad & c'-c=r+2h+3\beta+12\chi'+24\xi',\\
\text{Cayley}\quad \text{(5)}\quad & c'-c=r+2h+3\beta,
\end{aligned}$$

values which differ by the terms $12\chi'+24\xi'$, or if θ has the value just written down, the term $\frac{1}{2}\beta$.

I refrain from a comparison of the two equations (I), — and of the expressions for the deficiency given by these two equations respectively; but I notice here the expression for the deficiency obtained by Zeuthen in the last section (XIV.) of his Memoir, viz. this is

$$\begin{aligned}
24(D+1)=c'-12n+24a+\Sigma\delta+3\kappa-15c+2\sigma+6\rho+12\chi'+6g+9e&\\
+8B+2U'+18\theta+6U'+6\theta'&\\
+\Sigma(3\kappa'+3\xi'+8\xi+13\eta)+\Sigma'(6\eta').&
\end{aligned}$$

The problem is a very difficult one, and it cannot be held that as yet a complete solution has been obtained. Take in plane geometry the question of reciprocal curves; here, using throughout point-coordinates, we start with a curve represented by the general equation $(x, y, z)^n=0$, such a curve has only isolated singularities, viz. the line-singularities of the inflexion and the double tangent, we know the expression in point-coordinates of any such singularity (inflexion or double tangent as the case may be), viz. we can at once write down the equation of a curve of the order n having a given stationary tangent and point of contact therewith, or a given double tangent and two points of contact therewith. Returning to the general curve $(x, y, z)^n=0$, we know that the reciprocal curve has other isolated singularities, viz. the point-singularities which correspond to these, the double point (or node) and the stationary point (or cusp), and we know the expression of any such singularity (node or cusp as the case may be), viz. we can at once write down the equation of a curve of the order n having at a given point a node with given tangents, or a cusp with given tangent. And then starting afresh with a curve of the order n having a node or a cusp we obtain the effect thereof as regards the line-singularities of the inflexion and the double tangent. We are thus led to consider as ordinary singularities in the theory the above-mentioned four singularities of the inflexion, the double tangent, the node and the cusp: and we know further that any other singularity whatever of a plane curve is compounded in a definite manner of a certain number of some or all of these singularities.

But in the theory of surfaces, starting in like manner with the general equation $(x, y, z, w)^n = 0$, such a surface has torse-singularities, the node-couple torse, and the spinode-torse; each of these is in general an indecomposable torse of a certain kind (but there is the new cause of complication that it may break into two or more separate torses), but we do not know the analytical expression of these singularities, nor consequently the analytical expression of the curve-singularities which correspond to them, the nodal curve and the cuspidal curve. Thus if we attempt to start with a surface $(x, y, z, w)^n = 0$ having a nodal curve, we can indeed write down the equation in its most general form, viz. if the nodal curve has for its complete expression the k equations $P = 0$, $Q = 0$, $R = 0$, &c. (viz. if the curve is such that every surface whatever through the curve is of the form $\Omega = AP+BQ+CR+\ldots = 0$) then the most general equation of the surface having this curve for a nodal curve is $(A, B, C, \ldots \mid P, Q, R, \ldots)^2 = 0$, but this form is far too complicated to be worked with; and if for simplicity we take the nodal curve to be a complete intersection $P = 0$, $Q = 0$, and consequently the equation of the surface to be $(A, B, C \mid P, Q)^2 = 0$, then it is by no means clear that we do not in this way introduce limitations extraneous to the general theory. The same difficulty applies of course, and with yet greater force, to the cuspidal curve; and even if we could deal separately with the cases of a surface having a given nodal curve, and a given cuspidal curve, this would in no wise solve the problem for the more general case of a surface having a given nodal curve and a given cuspidal curve. It is to be added that the general surface of the order n has no plane- or point-singularities, and thus that such singularities (which correspond most nearly to the singularities considered in the theory of reciprocal curves) present themselves in the theory of reciprocal surfaces as extraordinary singularities.

END OF VOL. VI.

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